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Nicolas Privault

Understanding Markov Chains

Examples and Applications

Third Edition

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Nicolas Privault

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Examples and Applications

Third Edition

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Preface

Stochastic and Markovian modeling are of importance to many areas of science including physics, biology, engineering, as well as economics, finance, and social sciences. This text is an undergraduate-level introduction to the Markovian modeling of time-dependent randomness in discrete and continuous time, mostly on discrete state spaces. The emphasis is put on the understanding of concepts by examples and elementary derivations. Some more advanced topics on Poisson stochastic integrals are also covered.

The book is mostly self-contained except for its main prerequisites, which consist in a knowledge of basic probabilistic concepts. This includes random variables, discrete distributions (essentially binomial, geometric, and Poisson), continuous distributions (Gaussian and gamma) and their probability density functions, expectation, independence, and conditional probabilities, some of which are recalled in the first chapter. Such basic topics can be regarded as belonging to the field of "static" probability, i.e. probability without time dependence, as opposed to the contents of this text which is dealing with random evolution over time.

Our treatment of time-dependent randomness revolves around the important technique of first step analysis for random walks, branching processes, and more generally for Markov chains in discrete and continuous time, with application to the computation of ruin probabilities and mean hitting times. In addition to the treatment of Markov chains, a brief introduction to martingales is given in discrete time. This provides a different way to recover the computations of ruin probabilities and mean hitting times which have been presented in the Markovian framework. Spatial Poisson processes on abstract spaces are also considered without any time ordering, with the inclusion of some recent results on deviation inequalities and moment identities for Poisson stochastic integrals.

There already exist many textbooks on stochastic processes and Markov chains, including, e.g., Çinlar (1975), Bosq and Nguyen (1996), Karlin and Taylor (1998), Ross (1996), Norris (1998), Durrett (1999), Grimmett and Stirzaker (2001), Jones and Smith (2001), Steele (2001), Medhi (2010). In comparison with the existing literature, which is sometimes dealing with structural properties of stochastic processes via a more compact and abstract treatment, the present book tends to emphasize elementary

and explicit calculations instead of quicker arguments that may shorten the path to the solution, while being sometimes difficult to reproduce by undergraduate students.

The text also includes 151 exercises and 21 longer problems whose solutions are completely worked out in the solutions manual provided as supplementary material. Some of the exercises have been influenced by Çinlar (1975), Karlin and Taylor (1998), Ross (1996), Jones and Smith (2001), Medhi (2010), and other references, while a large number of them are original, and their solutions have been derived independently in the solutions manual available as supplementary material.

The *problems*, which are longer than the exercises, are based on various topics of application, such as cookie-excited random walks (Problem 3.10), phase-type distributions (Problem 4.13), the Wright-Fisher model in population genetics (Problem 5.20), probabilistic automata (Problem 5.22), time reversibility (Problem 7.25), the PageRank algorithm (Problem 7.26), and the Ising model (Problem 7.27). This book also contains 63 figures and three tables, as well as two Python codes and 12 R codes, also available as supplementary material.

Some theorems whose proofs are technical, as in Chaps. 7 and 9, have been quoted from Karlin and Taylor (1998) and Bosq and Nguyen (1996). The first (2013) and second (2018) editions have benefited from numerous questions, comments and suggestions from undergraduate students in Stochastic Processes at the Nanyang Technological University (NTU) in Singapore. I also thank Yung-Hsiang Huang and Moti Ben-Ari for many corrections and suggestions to the 2018 second edition.

A pdf solution manual for end-of-chapter problems are available for download by instructors adopting this book for use in their courses from https://sites.google.com/springernature.com/extramaterial/lecturer-material.

Singapore
October 2025

Nicolas Privault

Introduction

A stochastic[1] process is a mathematical tool used for the modeling of time-dependent random phenomena. Here, the term "stochastic" means random and "process" refers to the time-evolving status of a given system. Stochastic processes have applications to multiple fields, and can be useful anytime one recognizes the role of randomness and unpredictability of events that can occur at random times in an e.g. physical, biological, or financial system.

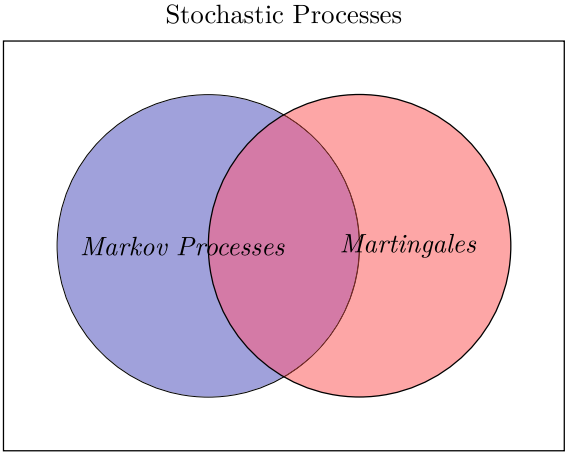
For example, in applications to physics one can mention phase transitions, atomic emission phenomena, etc. In biology the time behavior of live beings is often subject to randomness, at least when the observer has only access to partial information. This latter point is of importance, as it links that the notion of randomness to the concept of information: what appears random to an observer may not be random to another observer equipped with more information. Think, for example, of the observation of the apparent random behavior of cars turning at a crossroad versus the point of view of car drivers, each of whom are acting according to their own decisions. In finance the importance of modeling time-dependent random phenomena is quite clear, as no one can make definite predictions for the future moves of risky assets. The concrete outcome of random modeling lies in the computation of *expectations* or *expected values*, which often turn out to be more useful than the probability values themselves. An average or expected lifetime, for example, can be easier to interpret than a (small) probability of default. The long term statistical behavior of random systems, which also involves the estimation of expectations, is a related issue of interest.

Basically, a stochastic process is a time-dependent family $(X_t)_{t\in T}$ of random variables, where t is a time index belonging to a parameter set or time scale T. In other words, instead of considering a single random variable X, one considers a whole family of random variables $(X_t)_{t\in T}$, with the addition of another level of technical difficulty. The time scale T can be finite (e.g. $T = \{1, 2, \ldots, N\}$) or countably infinite (e.g. $T = \mathbb{N} = \{0, 1, 2, \ldots\}$) or even uncountable (e.g. $T = [0, 1]$, $T = \mathbb{R}_+$). The case of uncountable T corresponds to continuous-time stochastic processes, and

[1] From the Greek "στόχος" (stokhos), meaning "guess", or "conjecture".

this setting is the most theoretically difficult. A serious treatment of continuous-time processes would actually require additional background in *measure theory*, which is outside of the scope of this text. Measure theory is the general study of measures on abstract spaces, including probability measures as a particular case, and allows for a rigorous treatment of integrals via integration in the Lebesgue sense. The Lebesgue integral is a powerful tool that allows one to integrate functions and random variables under minimal technical conditions. Here we mainly work in a discrete-time framework that mostly does not require the use of measure theory.

That being said, the definition of a stochastic process $(X_t)_{t\in T}$ remains vague at this stage since virtually *any* family of random variables could be called a stochastic process. In addition, working at such a level of generality without imposing any structure or properties on the processes under consideration could be of little practical use. As we will see later on, stochastic processes can be classified into two main families:



– *Markov Processes*
  Roughly speaking, a stochastic process $(X_s)_{s\geq 0}$ is a Markov process when its statistical behavior after time t can be recovered from the value X_t of the process at time t. In particular, the values X_s of the process at times $s \in [0, t)$ have no influence on this behavior as long as the value of X_t is known.
– *Martingales*
  Originally, a martingale is a strategy designed to win repeatedly in a game of chance. In mathematics, a stochastic process $(X_t)_{t\in\mathbb{R}_+}$ is a martingale if the best possible estimate at time s of its future value X_t at time $t > s$ is simply given by X_s. This requires the careful definition of a "best possible estimate", and for this we need the tool of conditional expectation which relies on estimation in the mean-square sense. Martingale are useful in physics and finance, where they are linked to the notion of equilibrium.

Time series of order greater than one form another class of stochastic processes that may have neither the Markov property nor the martingale property in general.

The outline of this text is as follows. After reviewing in Chap. 1 the probabilistic tools required in the analysis of Markov chains, we consider simple gambling problems in Chap. 2, due to their practical usefulness and to the fact that they only require a minimal theoretical background. Next, in Chap. 3 we turn to the study of discrete-time random walks with infinite state space, which can be defined as stochastic processes with independent increments, without requiring much abstract formalism. In Chap. 4 we introduce the general framework of Markov chains in discrete time, which includes the gambling process and the simple random walk of Chapters 2 and 3 as particular cases. In the subsequent Chaps. 5–7, Markov chains are considered from the point of view of first step analysis, which is introduced in detail in Chap. 5. The classification of states is reviewed in Chap. 6, with application to the long-run behavior of Markov chains in Chap. 7, which also includes a short introduction to the Markov chain Monte Carlo method. Branching processes are other examples of discrete-time Markov processes which have important applications in life sciences, e.g. for population dynamics or the control of disease outbreak, and they are considered in Chap. 8. Then in Chap. 9, we deal with Markov chains in continuous time, including Poisson and birth and death processes. Martingales are considered in Chap. 10, where they are used to recover in a simple and elegant way the main results of Chap. 2 on ruin probabilities and mean exit times for gambling processes. Spatial Poisson point processes, which can be defined on an abstract space without requiring an ordered time index, are presented in Chap. 11. The end of that chapter covers additional results on moments and deviation inequalities for Poisson stochastic integrals, whose proofs should be accessible to advanced undergraduates. Reliability theory is an important engineering application of Markov chains, and it is reviewed in Chap. 12. The stochastic processes considered in this text have a discrete state space and discontinuous trajectories, see also Privault (2024) for more applications of such processes.

Contents

1 Probability Background 1
1.1 Probability Sample Space and Events 1
1.2 Probability Measures 4
1.3 Conditional Probabilities and Independence 6
1.4 Random Variables 8
1.5 Probability Distributions 10
1.6 Expectation of Random Variables 18
1.7 Moment and Probability Generating Functions 28
1.8 Conditional Expectation 35

2 Gambling Problems 43
2.1 Constrained Random Walk 43
2.2 Ruin Probabilities 45
2.3 Mean Game Duration 58

3 Random Walks 75
3.1 Unrestricted Random Walk 75
3.2 Mean and Variance 76
3.3 Distribution 77
3.4 First Return to Zero 79

4 Discrete-Time Markov Chains 101
4.1 Markov Property 101
4.2 Transition Matrix 104
4.3 Examples of Markov Chains 107
4.4 Higher-Order Transition Probabilities 110
4.5 The Two-State Discrete-Time Markov Chain 114
References 134

5 First Step Analysis 135
5.1 Hitting Probabilities 135
5.2 Mean Hitting and Absorption Times 143
5.3 First Return Times 149
5.4 Mean Number of Returns 155

6 Classification of States 177
6.1 Communicating States 177
6.2 Recurrent States 180
6.3 Transient States 181
6.4 Positive Versus Null Recurrence 186
6.5 Periodicity and Aperiodicity 188

7 Long-Run Behavior of Markov Chains 195
7.1 Limiting Distributions 195
7.2 Stationary Distributions 198
7.3 Markov Chain Monte Carlo (MCMC) 210

8 Branching Processes 229
8.1 Construction and Examples 229
8.2 Probability Generating Functions 232
8.3 Extinction Probabilities 236

9 Continuous-Time Markov Chains 251
9.1 The Poisson Process 251
9.2 Continuous-Time Markov Chains 257
9.3 Transition Semigroup and Infinitesimal Generator 263
9.4 Two-State Continuous-Time Markov Chain 274
9.5 Limiting and Stationary Distributions 277
9.6 Discrete-Time Embedded Chain 285
9.7 Mean Absorption Time and Probabilities 290

10 Discrete-Time Martingales 301
10.1 Filtrations and Conditional Expectations 301
10.2 Martingales–Definition and Properties 303
10.3 Stopping Times 305
10.4 Ruin Probabilities 309
10.5 Mean Game Duration 314
References 326

11 Spatial Poisson Processes 327
11.1 Spatial Poisson Processes 327
11.2 Poisson Stochastic Integrals 330
11.3 Transformations of Poisson Measures 333
11.4 Moments of Poisson Stochastic Integrals 335
11.5 Deviation Inequalities 339

12 Reliability Theory 345
12.1 Survival Probabilities 345
12.2 Poisson Process with Time-Dependent Intensity 346
12.3 Mean Time to Failure 348

Some Useful Identities 351
References 355
Subject Index 359
Author Index 365

List of Figures

Fig. 1.1 Probability density function φ_X 11
Fig. 1.2 Exponential CDF and PDF 13
Fig. 1.3 Probability $\mathbb{P}((X, Y) \in [-0.5, 1] \times [-0.5, 1])$ computed as a volume integral 14
Fig. 2.1 Sample path of a gambling process $(X_n)_{n\geq 0}$ 44
Fig. 2.2 Ruin probability $f_{20}(k)$ function of $X_0 = k \in [0, 20]$ for $S = 20$ and $p = 0.45$ 50
Fig. 2.3 Expected terminal wealth of Player A starting from state $k = 1$ 56
Fig. 2.4 Ruin probability as a function of $p \in [0, 1]$ for $S = 20$ and $k = 10$ 57
Fig. 2.5 Two sample paths of a gambling process $(X_n)_{n\geq 0}$ 58
Fig. 2.6 Mean game duration $h_{20}(k)$ as a function of $X_0 = k \in [0, 20]$ for $p = 0.45$ 64
Fig. 2.7 Mean game duration as a function of $p \in [0, 1]$ for $S = 20$ and $k = 10$ 66
Fig. 3.1 Sample path of the random walk $(S_n)_{n\geq 0}$ 76
Fig. 3.2 Graph of $120 = \binom{10}{7} = \binom{10}{3}$ paths linking $(0, 0)$ to $(10, 4)$ 78
Fig. 3.3 Sample path of the random walk $(S_n)_{n\geq 0}$ with return to state 0 79
Fig. 3.4 Last return to state 0 at time $k = 10$ 81
Fig. 3.5 Probability generating functions of T_0^r for p = 0.35 and p = 0.5 87
Fig. 3.6 Illustration of the range process 95
Fig. 4.1 Graph of a five-state Markov chain 110
Fig. 4.2 Sample path of a two-state chain in continuous time with $a = 0.2$ and $b = 0.4$ 119
Fig. 4.3 The proportion of chain values at state 1 tends to $1/3 = a/(a + b)$ 120

Fig. 4.4 Convergence graph for the two-state Markov chain with $a = 0.2$ and $b = 0.4$ 121
Fig. 4.5 Sample path of a five-state Markov chain 121
Fig. 5.1 Mean return times to state 0 on the maze (5.3.7) 155
Fig. 7.1 Global balance condition (discrete time) 199
Fig. 7.2 Stationary distribution on the maze (5.3.7) 208
Fig. 7.3 Stationary distribution by state numbering 208
Fig. 7.4 Simulated stationary distribution 209
Fig. 7.5 Detailed balance condition (discrete time) 210
Fig. 7.6 Simulation of the voter model with $N = 199$, $p = 0.98$, and $z_0 = z_{N+1} = +1$ 225
Fig. 7.7 Simulation of the voter model with $N = 3$, $p = \sqrt{0.75}$, and $z_0 = z_4 = +1$ 228
Fig. 8.1 Sample of a branching process 230
Fig. 8.2 Sample graph of a branching process 231
Fig. 8.3 Generating function of Y_1 with $p = 0.65$ 240
Fig. 8.4 Generating function of Y_1 with $p = 3/8 < 1/2$ and $\alpha_1 = 1$ 242
Fig. 8.5 Generating function of Y_1 with $p = 1/2$ and $\alpha_1 = 1$ 243
Fig. 8.6 Generating function of Y_1 with $p = 3/4 > 1/2$ and $\alpha_1 = q/p = 1/3$ 243
Fig. 8.7 Probability generating function of Y_1 244
Fig. 8.8 Sample path of a branching process $(X_n)_{n\geq 0}$ 244
Fig. 9.1 Sample path of a Poisson process $(N_t)_{t\in \mathbf{R}_+}$ 252
Fig. 9.2 Sample path of a birth process $(X_t^b)_{t\in \mathbf{R}_+}$ 259
Fig. 9.3 Sample path of a death process $(X_t^d)_{t\in \mathbf{R}_+}$ 261
Fig. 9.4 Sample path of a birth and death process $(X_t)_{t\in \mathbf{R}_+}$ 262
Fig. 9.5 Sample path of a continuous-time two-state chain with $\alpha = 20$ and $\beta = 40$ 278
Fig. 9.6 The proportion of process values in the state 1 converges to $1/3 = \alpha/(\alpha + \beta)$ 278
Fig. 9.7 Global balance condition (continuous time) 279
Fig. 9.8 Birth and death process $(X_t)_{t\in \mathbf{R}_+}$ with its embedded chain $(Z_n)_{n\geq 0}$ 285
Fig. 9.9 Discrete-time embedded chain $(Z_n)_{n\geq 0}$ based on the path of Fig. 9.8 286
Fig. 10.1 Sample path of the random walk $(S_n)_{n\geq 0}$ 302
Fig. 10.2 Stopped process 307
Fig. 10.3 Two sample paths of the random walk $(S_n)_{n\geq 0}$ 310
Fig. 10.4 Stopped process 318
Fig. 10.5 Possible paths of the process $(M_n)_{n\geq 0}$ 319
Fig. 10.6 Possible paths of the stopped process $(M_{\tau\wedge n})_{n\geq 0}$ 320
Fig. 10.7 Sample path of the random walk $(S_n)_{n\geq 0}$ 320
Fig. 10.8 Convex function 322
Fig. 10.9 Random walk supremum above $x = 1.5$ 325

Fig. 11.1 Poisson random samples with $\omega(A) = 2$, $\omega(B) = 4$, $\omega(C) = 3$ 328
Fig. 11.2 Two Poisson point process samples 329
Fig. 11.3 Poisson point process sample on the plane 329
Fig. 11.4 Gamma Lévy density $\rho(x) = \lambda e^{-xt}/x$ 333
Fig. 11.5 Boolean model in dimension three 344

List of Tables

Table 7.1 Summary of Markov chain properties 214
Table 8.1 Summary of computing methods 245
Table 9.1 Summary of computing methods 294
Table 10.1 List of martingales 316

Chapter 1
Probability Background



In this chapter, we review a number of basic probabilistic tools that will needed for the study of stochastic processes in the subsequent chapters. We refer the reader to e.g. Pitman (1999), Jacod and Protter (2000), Devore (2003), for additional background on probability theory.

1.1 Probability Sample Space and Events

We will need the following notation coming from set theory. Given A and B to abstract sets, "$A \subset B$" means that A is contained in B, and in this case, $B \setminus A$ denotes the set of elements of B which do not belong to A. The property that the element ω belongs to the set A is denoted by "$\omega \in A$", and given two sets A and Ω such that $A \subset \Omega$, we let $A^c = \Omega \setminus A$ denote the *complement* of A in Ω. The finite set made of n elements $\omega_1, \ldots, \omega_n$ is denoted by $\{\omega_1, \ldots, \omega_n\}$, and we will usually distinguish between the element ω and its associated singleton set $\{\omega\}$.

A probability sample space is an abstract set Ω that contains the possible outcomes of a random experiment.

Examples

(i) Coin tossing: $\Omega = \{H, T\}$.

(ii) Rolling one die: $\Omega = \{1, 2, 3, 4, 5, 6\}$.

(iii) Picking one card at random in a pack of 52: $\Omega = \{1, 2, 3, \ldots, 52\}$.

(iv) An integer-valued random outcome: $\Omega = \mathbb{N} = \{0, 1, 2, \ldots\}$.

In this case the outcome $\omega \in \mathbb{N}$ can be the random number of trials needed until some event occurs.

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(v) A nonnegative, real-valued outcome: $\Omega = \mathbb{R}_+$.

In this case, the outcome $\omega \in \mathbb{R}_+$ may represent the (nonnegative) value of a continuous random time.

(vi) A random continuous parameter (such as time, weather, price or wealth, temperature, ...): $\Omega = \mathbb{R}$.

(vii) Random choice of a continuous path in the space $\Omega = \mathcal{C}(\mathbb{R}_+)$ of all continuous functions on $\mathbb{R}_+$.

In this case, $\omega \in \Omega$ is a function $\omega : \mathbb{R}_+ \longrightarrow \mathbb{R}$ and a typical example is the graph $t \longmapsto \omega(t)$ of a stock price over time.

Product spaces:

Probability sample spaces can be built as product spaces and used for the modeling of repeated random experiments.

(i) Rolling two dice: $\Omega = \{1, 2, 3, 4, 5, 6\} \times \{1, 2, 3, 4, 5, 6\}$. In this case, a typical element of Ω is written as $\omega = (k, l)$ with $k, l \in \{1, 2, 3, 4, 5, 6\}$.

(ii) A finite number n of real-valued samples: $\Omega = \mathbb{R}^n$. In this case, the outcome ω is a vector $\omega = (x_1, \ldots, x_n) \in \mathbb{R}^n$ with n components.

Note that to some extent, the more complex Ω is, the better it fits a practical and useful situation, e.g. $\Omega = \{H, T\}$ corresponds to a simple coin tossing experiment while $\Omega = \mathcal{C}(\mathbb{R}_+)$ the space of continuous functions on $\mathbb{R}_+$ can be applied to the modeling of stock markets. On the other hand, in many cases and especially in the most complex situations, we will *not* attempt to specify Ω explicitly.

Events

An event is a collection of outcomes, which is represented by a subset of Ω. In what follows we consider collections of events, called *σ-algebras* (or *σ-fields*), according to the following definition.

Definition 1.1 A collection $\mathcal{G}$ of events is a *σ-algebra* on Ω provided that it satisfies the following conditions:

(i) $\emptyset \in \mathcal{G}$,

(ii) For all countable sequences $(A_n)_{n\geq 1}$ such that $A_n \in \mathcal{G}$, $n \geq 1$, we have $\bigcup_{n\geq 1} A_n \in \mathcal{G}$,

(iii) $A \in \mathcal{G} \Longrightarrow (\Omega \setminus A) \in \mathcal{G}$,

where $\Omega \setminus A := \{\omega \in \Omega \, : \, \omega \notin A\}$.

Note that Properties (ii) and (iii) above also imply the stability of σ-algebras under intersections, as

$$\bigcap_{n\geq 1} A_n = \left(\bigcup_{n\geq 1} A_n^c \right)^c \in \mathcal{G}, \tag{1.1.1}$$

for all countable sequences $A_n \in \mathcal{G}$, $n \geq 1$.

The collection of all events in Ω will often be denoted by $\mathcal{F}$. The empty set $\emptyset$ and the full space Ω are considered as events but they are of less importance because Ω corresponds to "any outcome may occur" while $\emptyset$ corresponds to an absence of outcome, or no experiment.

In the context of stochastic processes, two σ-algebras $\mathcal{F}$ and $\mathcal{G}$ such that $\mathcal{F} \subset \mathcal{G}$ will refer to two different amounts of information, the amount of information associated to $\mathcal{F}$ being here lower than the one associated to $\mathcal{G}$.

The formalism of σ-algebras helps in describing events in a short and precise way.

Examples

(i) Let $\Omega = \{1, 2, 3, 4, 5, 6\}$.
The event $A = \{2, 4, 6\}$ corresponds to
"the result of the experiment is an even number".

(ii) Taking again $\Omega = \{1, 2, 3, 4, 5, 6\}$,

$$\mathcal{F} := \{\Omega, \emptyset, \{2, 4, 6\}, \{1, 3, 5\}\}$$

defines a σ-algebra on Ω which corresponds to the knowledge of parity of an integer picked at random from 1 to 6.

Note that in the set-theoretic notation, an event A is a subset of Ω, *i.e.* $A \subset \Omega$, while it is an element of $\mathcal{F}$, i.e. $A \in \mathcal{F}$. For example, we have $\Omega \supset \{2, 4, 6\} \in \mathcal{F}$, while $\{\{2, 4, 6\}, \{1, 3, 5\}\} \subset \mathcal{F}$.

(iii) Taking

$$\mathcal{G} := \{\Omega, \emptyset, \{2, 4, 6\}, \{2, 4\}, \{6\}, \{1, 2, 3, 4, 5\}, \{1, 3, 5, 6\}, \{1, 3, 5\}\} \supset \mathcal{F},$$

defines a σ-algebra on Ω which is bigger than $\mathcal{F}$ and includes the parity information contained in $\mathcal{F}$, in addition to information on whether the outcome of the experiment is equal to 6 or not.

(iv) Take

$$\Omega = \{H, T\} \times \{H, T\} = \{(H, H), (H, T), (T, H), (T, T)\}.$$

In this case, the collection $\mathcal{F}$ of all possible events is given by

$$\begin{aligned}\mathcal{F} = \{&\emptyset, \{(H, H)\}, \{(T, T)\}, \{(H, T)\}, \{(T, H)\}, \\ &\{(T, T), (H, H)\}, \{(H, T), (T, H)\}, \{(H, T), (T, T)\}, \\ &\{(T, H), (T, T)\}, \{(H, T), (H, H)\}, \{(T, H), (H, H)\}, \\ &\{(H, H), (T, T), (T, H)\}, \{(H, H), (T, T), (H, T)\}, \\ &\{(H, T), (T, H), (H, H)\}, \{(H, T), (T, H), (T, T)\}, \Omega\}\,.\end{aligned} \tag{1.1.2}$$

Note that the set $\mathcal{F}$ of all events considered in (1.1.2) above has altogether

$$1 = \binom{n}{0} \text{ event of cardinality 0,}$$
$$4 = \binom{n}{1} \text{ events of cardinality 1,}$$
$$6 = \binom{n}{2} \text{ events of cardinality 2,}$$
$$4 = \binom{n}{3} \text{ events of cardinality 3,}$$
$$1 = \binom{n}{4} \text{ event of cardinality 4,}$$

with $n = 4$, for a total of

$$16 = 2^n = \sum_{k=0}^{4} \binom{4}{k} = 1 + 4 + 6 + 4 + 1$$

events. The collection of events

$$\mathcal{G} := \{\emptyset, \{(T, T), (H, H)\}, \{(H, T), (T, H)\}, \Omega\}$$

defines a sub σ-algebra of $\mathcal{F}$, which corresponds to the restricted information "the results of two coin tossings are different".

Exercise: Write down the set of all events on $\Omega = \{H, T\}$.

Note also that (H, T) is different from (T, H), whereas $\{(H, T), (T, H)\}$ is equal to $\{(T, H), (H, T)\}$.

In addition, we will distinguish between the *outcome* $\omega \in \Omega$ and its associated *event* $\{\omega\} \in \mathcal{F}$, which satisfies $\{\omega\} \subset \Omega$.

1.2 Probability Measures

Definition 1.2 A *probability measure* is a mapping $\mathbb{P} : \mathcal{F} \longrightarrow [0, 1]$ that assigns a probability $\mathbb{P}(A) \in [0, 1]$ to any event $A \in \mathcal{F}$, with the properties

(a) $\mathbb{P}(\Omega) = 1$, and

(b) $\mathbb{P}\left(\bigcup_{n\geq 1} A_n\right) = \sum_{n\geq 1} \mathbb{P}(A_n)$, whenever $A_k \cap A_l = \emptyset, k \neq l$.

Property (b) above is named the *law of total probability*. It states in particular that we have

$$\mathbb{P}(A_1 \cup \cdots \cup A_n) = \mathbb{P}(A_1) + \cdots + \mathbb{P}(A_n)$$

when the subsets $A_1, \ldots, A_n$ of Ω are disjoint, and

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) \tag{1.2.1}$$

if $A \cap B = \emptyset$. We also have the *complement rule*

$$\mathbb{P}(A^c) = \mathbb{P}(\Omega \setminus A) = \mathbb{P}(\Omega) - \mathbb{P}(A) = 1 - \mathbb{P}(A).$$

When A and B are not necessarily disjoint, we can write

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B),$$

which extends to arbitrary families of events $(A_i)_{i \in I}$ indexed by a finite set I as the inclusion-exclusion principle

$$\mathbb{P}\left(\bigcup_{i \in I} A_i\right) = \sum_{J \subset I} (-1)^{|J|+1} \mathbb{P}\left(\bigcap_{j \in J} A_j\right), \tag{1.2.2}$$

and

$$\mathbb{P}\left(\bigcap_{j \in J} A_j\right) = \sum_{I \subset J} (-1)^{|I|+1} \mathbb{P}\left(\bigcup_{i \in I} A_i\right). \tag{1.2.3}$$

The triple

$$(\Omega, \mathcal{F}, \mathbb{P}) \tag{1.2.4}$$

is called a *probability space*, and was introduced by A.N. Kolmogorov (1903–1987). This setting is generally referred to as the *Kolmogorov framework*.

A property or event is said to hold $\mathbb{P}$-*almost surely* (also written $\mathbb{P}$-*a.s.*) if it holds with probability equal to one.

Example

Take

$$\Omega = \big\{(T, T), (H, H), (H, T), (T, H)\big\}$$

and

$$\mathcal{F} = \{\emptyset, \{(T, T), (H, H)\}, \{(H, T), (T, H)\}, \Omega\}.$$

The *uniform* probability measure $\mathbb{P}$ on $(\Omega, \mathcal{F})$ is given by setting

$$\mathbb{P}(\{(T, T), (H, H)\}) := \frac{1}{2} \quad \text{and} \quad \mathbb{P}(\{(H, T), (T, H)\}) := \frac{1}{2}.$$

In addition, we have the following convergence properties.

1. Let $(A_n)_{n\in\mathbb{N}}$ be a *non-decreasing* sequence of events, i.e. $A_n \subset A_{n+1}, n \geq 0$. Then we have
$$\mathbb{P}\left(\bigcup_{n\in\mathbb{N}} A_n\right) = \lim_{n\to\infty} \mathbb{P}(A_n). \tag{1.2.5}$$

2. Let $(A_n)_{n\in\mathbb{N}}$ be a *non-increasing* sequence of events, i.e. $A_{n+1} \subset A_n, n \geq 0$. Then we have
$$\mathbb{P}\left(\bigcap_{n\in\mathbb{N}} A_n\right) = \lim_{n\to\infty} \mathbb{P}(A_n). \tag{1.2.6}$$

Theorem 1.3 *Borel-Cantelli Lemma. Let $(A_n)_{n\geq 1}$ denote a sequence of events on $(\Omega, \mathcal{F}, \mathbb{P})$, such that*
$$\sum_{n\geq 1} \mathbb{P}(A_n) < \infty.$$

Then we have
$$\mathbb{P}\left(\bigcap_{n\geq 1}\bigcup_{k\geq n} A_k\right) = 0,$$

i.e. the probability that A_n occurs infinitely many times occur is zero.

1.3 Conditional Probabilities and Independence

We start with examples.

Consider a population $\Omega = M \cup W$ made of a set M of men and a set W of women. Here, the σ-algebra $\mathcal{F} = \{\Omega, \emptyset, W, M\}$ corresponds to the information given by gender. After polling the population, e.g. for a market survey, it turns out that a proportion $p \in [0, 1]$ of the population declares to like apples, while a proportion $1 - p$ declares to dislike apples. Let $A \subset \Omega$ denote the subset of individuals who like apples, while $A^c \subset \Omega$ denotes the subset individuals who dislike apples, with

$$p = \mathbb{P}(A) \quad \text{and} \quad 1 - p = \mathbb{P}(A^c),$$

e.g. $p = 60\%$ of the population likes apples. It may be interesting to get a more precise information and to determine

– the relative proportion $\dfrac{\mathbb{P}(A \cap W)}{\mathbb{P}(W)}$ of women who like apples, and
– the relative proportion $\dfrac{\mathbb{P}(A \cap M)}{\mathbb{P}(M)}$ of men who like apples.

Here, $\mathbb{P}(A \cap W)/\mathbb{P}(W)$ represents the probability that a randomly chosen woman in W likes apples, and $\mathbb{P}(A \cap M)/\mathbb{P}(M)$ represents the probability that a randomly chosen man in M likes apples. Those two ratios are interpreted as *conditional probabilities*, for example $\mathbb{P}(A \cap M)/\mathbb{P}(M)$ denotes the probability that a given individual likes apples *given that* he is a man.

For another example, suppose that the population Ω is split as $\Omega = Y \cup O$ into a set Y of "young" people and another set O of "old" people, and denote by $A \subset \Omega$ the set of people who voted for candidate A in an election. Here it can be of interest to find out the relative proportion

$$\mathbb{P}(A \mid Y) = \frac{\mathbb{P}(Y \cap A)}{\mathbb{P}(Y)}$$

of young people who voted for candidate A.

Definition 1.4 Given any two events $A, B \subset \Omega$ with $\mathbb{P}(B) \neq 0$, we call

$$\mathbb{P}(A \mid B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

the probability of A *given* B, or *conditionally to* B.

Remark 1.5 We note that if $\mathbb{P}(B) = 1$ we have $\mathbb{P}(A \cap B^c) \leq \mathbb{P}(B^c) = 0$, hence $\mathbb{P}(A \cap B^c) = 0$, which implies

$$\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c) = \mathbb{P}(A \cap B),$$

and $\mathbb{P}(A \mid B) = \mathbb{P}(A)$.

We also recall the following property:

$$\begin{aligned}
\mathbb{P}\left(B \cap \bigcup_{n\geq 1} A_n\right) &= \mathbb{P}\left(\bigcup_{n\geq 1}(B \cap A_n)\right) \\
&= \sum_{n\geq 1} \mathbb{P}(B \cap A_n) \\
&= \sum_{n\geq 1} \mathbb{P}(B \mid A_n)\mathbb{P}(A_n) \\
&= \sum_{n\geq 1} \mathbb{P}(A_n \mid B)\mathbb{P}(B),
\end{aligned}$$

for any family of disjoint events $(A_n)_{n\geq 1}$ with $A_i \cap A_j = \emptyset$, $i \neq j$, and $\mathbb{P}(B) > 0$, $n \geq 1$. This also shows that conditional probability measures are probability measures, in the sense that whenever $\mathbb{P}(B) > 0$, we have

(a) $\mathbb{P}(\Omega \mid B) = 1$, and

(b) $\mathbb{P}\left(\bigcup_{n\geq 1} A_n \,\middle|\, B\right) = \sum_{n\geq 1} \mathbb{P}(A_n \mid B)$, whenever $A_k \cap A_l = \emptyset$, $k \neq l$.

In particular, if $\bigcup_{n\geq 1} A_n = \Omega$, $(A_n)_{n\geq 1}$ becomes a *partition* of Ω and we get the *law of total probability*

$$\mathbb{P}(B) = \sum_{n\geq 1} \mathbb{P}(B \cap A_n) = \sum_{n\geq 1} \mathbb{P}(A_n \mid B)\mathbb{P}(B) = \sum_{n\geq 1} \mathbb{P}(B \mid A_n)\mathbb{P}(A_n), \quad (1.3.1)$$

provided that $A_i \cap A_j = \emptyset$, $i \neq j$, and $\mathbb{P}(B) > 0$, $n \geq 1$.

Remark. In general we have

$$\mathbb{P}\left(A \,\middle|\, \bigcup_{n\geq 1} B_n\right) \neq \sum_{n\geq 1} \mathbb{P}(A \mid B_n),$$

even when $B_k \cap B_l = \emptyset$, $k \neq l$. Indeed, taking for example $A = \Omega = B_1 \cup B_2$ with $B_1 \cap B_2 = \emptyset$ and $\mathbb{P}(B_1) = \mathbb{P}(B_2) = 1/2$, we have

$$1 = \mathbb{P}(\Omega \mid B_1 \cup B_2) \neq \mathbb{P}(\Omega \mid B_1) + \mathbb{P}(\Omega \mid B_2) = 2.$$

Independent events

Definition 1.6 Two events A and B such that $\mathbb{P}(A), \mathbb{P}(B) > 0$ are said to be *independent* if

$$\mathbb{P}(A \mid B) = \mathbb{P}(A). \quad (1.3.2)$$

We note that the independence condition (1.3.2) is equivalent to

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

1.4 Random Variables

A real-valued random variable is a mapping

$$\begin{aligned} X : \Omega &\longrightarrow \mathbb{R} \\ \omega &\longmapsto X(\omega) \end{aligned}$$

from a probability sample space Ω into the state space $\mathbb{R}$. Given

$$X : \Omega \longrightarrow \mathbb{R}$$

a random variable and a (measurable)[1] subset A of $\mathbb{R}$, we denote by $\{X \in A\}$ the event

$$\{X \in A\} := \{\omega \in \Omega \, : \, X(\omega) \in A\}.$$

Examples

(i) Let $\Omega := \{1, 2, 3, 4, 5, 6\} \times \{1, 2, 3, 4, 5, 6\}$, and consider the mapping

$$\begin{aligned} X : \Omega &\longrightarrow \mathbb{R} \\ (k, l) &\longmapsto k + l. \end{aligned}$$

Then X is a random variable giving the sum of the two numbers appearing on each die.

(ii) the time needed everyday to travel from home to work or school is a random variable, as the precise value of this time may change from day to day under unexpected circumstances.

(iii) the price of a risky asset can be modeled using a random variable.

In the sequel we will often use the notion of *indicator function* $\mathbb{1}_A$ of an event $A \subset \Omega$.

Definition 1.7 For any $A \subset \Omega$, the indicator function $\mathbb{1}_A$ is the random variable

$$\begin{aligned} \mathbb{1}_A : \Omega &\longrightarrow \{0, 1\} \\ \omega &\longmapsto \mathbb{1}_A(\omega) \end{aligned}$$

defined by

$$\mathbb{1}_A(\omega) = \begin{cases} 1 \ \text{ if } \omega \in A, \\ 0 \ \text{ if } \omega \notin A. \end{cases}$$

Indicator functions satisfy the property

$$\mathbb{1}_{A \cap B}(\omega) = \mathbb{1}_A(\omega) \mathbb{1}_B(\omega), \tag{1.4.1}$$

since

$$\begin{aligned} \mathbb{1}_{A \cap B}(\omega) = 1 &\iff \omega \in A \cap B \\ &\iff \omega \in A \text{ and } \omega \in B \\ &\iff \mathbb{1}_A(\omega) = 1 \text{ and } \mathbb{1}_B(\omega) = 1 \\ &\iff \mathbb{1}_A(\omega) \mathbb{1}_B(\omega) = 1. \end{aligned}$$

We also have

$$\mathbb{1}_{A \cup B} = \mathbb{1}_A + \mathbb{1}_B - \mathbb{1}_{A \cap B} = \mathbb{1}_A + \mathbb{1}_B - \mathbb{1}_A \mathbb{1}_B,$$

[1] Measurability of subsets of $\mathbb{R}$ refers to *Borel measurability*, a concept which will not be defined in this text.

and

$$\mathbb{1}_{A\cup B} = \mathbb{1}_A + \mathbb{1}_B, \tag{1.4.2}$$

if $A \cap B = \emptyset$.

For example, if $\Omega = \mathbb{N}$ and $A = \{k\}$, for all $l \geq 0$ we have

$$\mathbb{1}_{\{k\}}(l) = \begin{cases} 1 & \text{if } k = l, \\ 0 & \text{if } k \neq l. \end{cases}$$

Given X a random variable, we also let

$$\mathbb{1}_{\{X=n\}} = \begin{cases} 1 & \text{if } X = n, \\ 0 & \text{if } X \neq n, \end{cases}$$

and

$$\mathbb{1}_{\{X<n\}} = \begin{cases} 1 & \text{if } X < n, \\ 0 & \text{if } X \geq n. \end{cases}$$

1.5 Probability Distributions

The *probability distribution* of a random variable $X : \Omega \longrightarrow \mathbb{R}$ is the collection

$$\{\mathbb{P}(X \in A) \ : \ A \text{ is a measurable subset of } \mathbb{R}\}.$$

As the collection of *measurable* subsets of $\mathbb{R}$ coincides with the σ-algebra generated by the intervals in $\mathbb{R}$, the distribution of X can be reduced to the knowledge of the probabilities

$$\{\mathbb{P}(a < X \leq b) = \mathbb{P}(X \leq b) - \mathbb{P}(X \leq a) \ : \ a < b \in \mathbb{R}\},$$

or of the cumulative distribution functions

$$\{\mathbb{P}(X \leq a) \ : \ a \in \mathbb{R}\}, \quad \text{or} \quad \{\mathbb{P}(X \geq a) \ : \ a \in \mathbb{R}\},$$

see e.g. Corollary 3.8 in Çınlar (2011).

Two random variables X and Y are said to be independent under the probability $\mathbb{P}$ if their probability distributions satisfy

$$\mathbb{P}(X \in A \ , \ Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$$

for all (measurable) subsets A and B of $\mathbb{R}$.

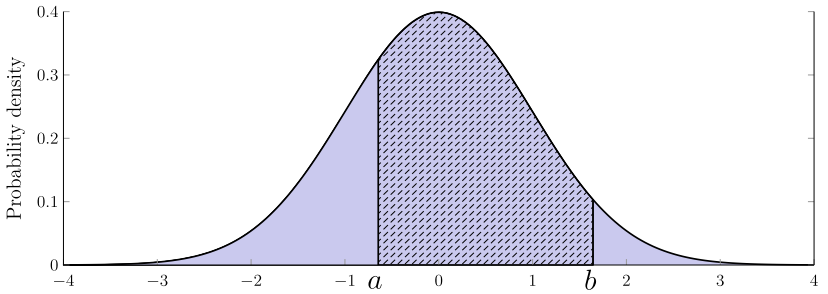


Fig. 1.1 Probability density function φ_X

Distributions admitting a density

We say that the distribution of X admits a probability *density* distribution function $\varphi_X : \mathbb{R} \longrightarrow \mathbb{R}_+$ if, for all $a \leq b$, the probability $\mathbb{P}(a \leq X \leq b)$ can be written as

$$\mathbb{P}(a \leq X \leq b) = \int_a^b \varphi_X(x)dx,$$

see Fig. 1.1. We also say that the distribution of X is absolutely continuous, or that X is an absolutely continuous random variable. This, however, does *not* imply that the density function $\varphi_X : \mathbb{R} \longrightarrow \mathbb{R}_+$ is continuous.

In particular, we always have

$$\int_{-\infty}^{\infty} \varphi_X(x)dx = \mathbb{P}(-\infty \leq X \leq \infty) = 1$$

for any probability density functions $\varphi_X : \mathbb{R} \longrightarrow \mathbb{R}_+$.

Remark 1.8 Note that if the distribution of X admits a probability density function φ_X, then for all $a \in \mathbb{R}$ we have

$$\mathbb{P}(X = a) = \int_a^a \varphi_X(x)dx = 0, \tag{1.5.1}$$

and this is not a contradiction.

In particular, Remark 1.8 shows that

$$\mathbb{P}(a \leq X \leq b) = \mathbb{P}(X = a) + \mathbb{P}(a < X \leq b) = \mathbb{P}(a < X \leq b) = \mathbb{P}(a < X < b),$$

for $a \leq b$. Property (1.5.1) appears for example in the framework of lottery games with a large number of participants, in which a given number "a" selected in advance has a very low (almost zero) probability to be chosen.

The probability density function φ_X can be recovered from the Cumulative Distribution Functions (CDFs)

$$x \longmapsto F_X(x) := \mathbb{P}(X \leq x) = \int_{-\infty}^{x} \varphi_X(s)ds,$$

and

$$x \longmapsto 1 - F_X(x) = \mathbb{P}(X \geq x) = \int_{x}^{\infty} \varphi_X(s)ds,$$

as

$$\varphi_X(x) = \frac{\partial F_X}{\partial x}(x) = \frac{\partial}{\partial x}\int_{-\infty}^{x} \varphi_X(s)ds = -\frac{\partial}{\partial x}\int_{x}^{\infty} \varphi_X(s)ds, \qquad x \in \mathbb{R}.$$

Examples

(i) The *uniform* distribution on an interval.

The probability density function of the uniform distribution on the interval $[a, b]$, $a < b$, is given by

$$\varphi(x) = \frac{1}{b-a}\mathbb{1}_{[a,b]}(x), \qquad x \in \mathbb{R}.$$

(ii) The *Gaussian* distribution.

The probability density function of the standard normal distribution is given by

$$\varphi(x) = \frac{1}{\sqrt{2\pi}}\mathrm{e}^{-x^2/2}, \qquad x \in \mathbb{R}.$$

More generally, the probability density function of the Gaussian distribution with mean $\mu \in \mathbb{R}$ and variance $\sigma^2 > 0$ is given by

$$\varphi(x) := \frac{1}{\sqrt{2\pi\sigma^2}}\mathrm{e}^{-(x-\mu)^2/(2\sigma^2)}, \qquad x \in \mathbb{R}.$$

In this case, we write $X \simeq \mathcal{N}(\mu, \sigma^2)$.

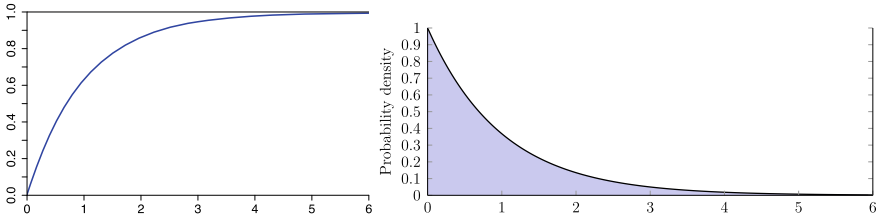
(iii) The *exponential* distribution (Fig. 1.2).

The probability density function of the exponential distribution with parameter $\lambda > 0$ is given by

$$\varphi(x) := \lambda \mathbb{1}_{[0,\infty)}(x)\mathrm{e}^{-\lambda x} = \begin{cases} \lambda \mathrm{e}^{-\lambda x}, & x \geq 0 \\ 0, & x < 0. \end{cases} \tag{1.5.2}$$

We also have

$$\mathbb{P}(X > t) = \mathrm{e}^{-\lambda t}, \qquad t \geq 0. \tag{1.5.3}$$



(a) Exponential CDF. (b) Exponential PDF.

Fig. 1.2 Exponential CDF and PDF

(iv) The *gamma* distribution.

The probability density function of the gamma distribution is given by

$$\varphi(x) := \frac{a^\lambda}{\Gamma(\lambda)} \mathbb{1}_{[0,\infty)}(x) x^{\lambda-1} \mathrm{e}^{-ax} = \begin{cases} \dfrac{a^\lambda}{\Gamma(\lambda)} x^{\lambda-1} \mathrm{e}^{-ax}, & x \geq 0 \\ 0, & x < 0, \end{cases}$$

where $a > 0$ and $\lambda > 0$ are scale and shape parameters, and

$$\Gamma(\lambda) := \int_0^\infty x^{\lambda-1} \mathrm{e}^{-x} dx, \qquad \lambda > 0,$$

is the gamma function.

(v) The *Cauchy* distribution.

The probability density function of the Cauchy distribution is given by

$$\varphi(x) := \frac{1}{\pi(1+x^2)}, \qquad x \in \mathbb{R}.$$

(vi) The *lognormal* distribution.

The probability density function of the lognormal distribution is given by

$$\varphi(x) := \mathbb{1}_{[0,\infty)}(x) \frac{1}{x\sigma\sqrt{2\pi}} \mathrm{e}^{-(\mu-\log x)^2/(2\sigma^2)} = \begin{cases} \dfrac{1}{x\sigma\sqrt{2\pi}} \mathrm{e}^{-(\mu-\log x)^2/(2\sigma^2)}, & x \geq 0 \\ 0, & x < 0. \end{cases}$$

Exercise: For each of the above probability density functions φ, check that the condition

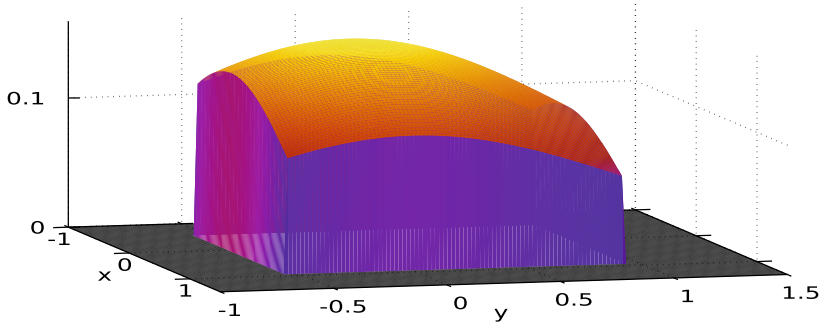


Fig. 1.3 Probability $\mathbb{P}((X, Y) \in [-0.5, 1] \times [-0.5, 1])$ computed as a volume integral

$$\int_{-\infty}^{\infty} \varphi(x)dx = 1$$

is satisfied.

Joint densities

Given two absolutely continuous random variables $X : \Omega \longrightarrow \mathbb{R}$ and $Y : \Omega \longrightarrow \mathbb{R}$, we can form the $\mathbb{R}^2$-valued random variable (X, Y) defined by

$$\begin{aligned}(X, Y) : \Omega &\longrightarrow \mathbb{R}^2 \\ \omega &\longmapsto (X(\omega), Y(\omega)).\end{aligned}$$

We say that (X, Y) admits a joint probability density

$$\varphi_{(X,Y)} : \mathbb{R}^2 \longrightarrow \mathbb{R}_+$$

when

$$\mathbb{P}((X, Y) \in A \times B) = \mathbb{P}(X \in A \text{ and } Y \in B) = \int_B \int_A \varphi_{(X,Y)}(x, y)dxdy$$

for all *measurable* subsets A, B of $\mathbb{R}$, see Fig. 1.3.

The probability density function $\varphi_{(X,Y)}$ can be recovered from the joint cumulative distribution function

$$(x, y) \longmapsto F_{(X,Y)}(x, y) := \mathbb{P}(X \leq x \text{ and } Y \leq y) = \int_{-\infty}^{x} \int_{-\infty}^{y} \varphi_{(X,Y)}(s, t)dsdt,$$

and

$$(x, y) \longmapsto \mathbb{P}(X \geq x \text{ and } Y \geq y) = \int_{x}^{\infty} \int_{y}^{\infty} \varphi_{(X,Y)}(s, t)dsdt,$$

as

$$\varphi_{(X,Y)}(x,y) = \frac{\partial^2}{\partial x \partial y} F_{(X,Y)}(x,y) \tag{1.5.4}$$

$$= \frac{\partial^2}{\partial x \partial y} \int_{-\infty}^{x} \int_{-\infty}^{y} \varphi_{(X,Y)}(s,t) ds dt \tag{1.5.5}$$

$$= \frac{\partial^2}{\partial x \partial y} \int_{x}^{\infty} \int_{y}^{\infty} \varphi_{(X,Y)}(s,t) ds dt,$$

$x, y \in \mathbb{R}$.

The probability densities $\varphi_X : \mathbb{R} \longrightarrow \mathbb{R}_+$ and $\varphi_Y : \mathbb{R} \longrightarrow \mathbb{R}_+$ of $X : \Omega \longrightarrow \mathbb{R}$ and $Y : \Omega \longrightarrow \mathbb{R}$ are called the *marginal densities* of (X, Y), and are given by

$$\varphi_X(x) = \int_{-\infty}^{\infty} \varphi_{(X,Y)}(x,y) dy, \qquad x \in \mathbb{R}, \tag{1.5.6}$$

and

$$\varphi_Y(y) = \int_{-\infty}^{\infty} \varphi_{(X,Y)}(x,y) dx, \qquad y \in \mathbb{R}.$$

The conditional probability density $\varphi_{X|Y=y} : \mathbb{R} \longrightarrow \mathbb{R}_+$ of X given $Y = y$ is defined by

$$\varphi_{X|Y=y}(x) := \frac{\varphi_{(X,Y)}(x,y)}{\varphi_Y(y)}, \qquad x, y \in \mathbb{R}, \tag{1.5.7}$$

provided that $\varphi_Y(y) > 0$. In particular, X and Y are independent if and only if

$$\varphi_{X|Y=y}(x) = \varphi_X(x), \quad i.e., \quad \varphi_{(X,Y)}(x,y) = \varphi_X(x)\varphi_Y(y), \qquad x, y \in \mathbb{R}.$$

Example

- If $X_1, \ldots, X_n$ are independent exponentially distributed random variables with parameters $\lambda_1, \ldots, \lambda_n$ we have

$$\begin{aligned} \mathbb{P}(\min(X_1, \ldots, X_n) > t) &= \mathbb{P}(X_1 > t, \ldots, X_n > t) \\ &= \mathbb{P}(X_1 > t) \cdots \mathbb{P}(X_n > t) \\ &= \mathrm{e}^{-(\lambda_1 + \cdots + \lambda_n)t}, \qquad t \geq 0, \end{aligned} \tag{1.5.8}$$

  hence $\min(X_1, \ldots, X_n)$ is an exponentially distributed random variable with parameter $\lambda_1 + \cdots + \lambda_n$.

  From the probability density function of (X_1, X_2) given by

$$\varphi_{(X_1,X_2)}(x,y) = \varphi_{X_1}(x)\varphi_{X_2}(y) = \lambda_1 \lambda_2 \mathrm{e}^{-\lambda_1 x - \lambda_2 y}, \qquad x, y \geq 0,$$

we can write

$$\begin{aligned}\mathbb{P}(X_1 < X_2) &= \mathbb{P}(X_1 \leq X_2)\\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{y} \varphi_{(X_1,X_2)}(x,y)dxdy\\ &= \lambda_1\lambda_2 \int_0^{\infty}\int_0^{y} \mathrm{e}^{-\lambda_1 x-\lambda_2 y}dxdy\\ &= \frac{\lambda_1}{\lambda_1+\lambda_2},\end{aligned} \tag{1.5.9}$$

and we note that

$$\mathbb{P}(X_1 = X_2) = \lambda_1\lambda_2 \int_{\{(x,y)\in\mathbb{R}_+^2\,:\,x=y\}} \mathrm{e}^{-\lambda_1 x-\lambda_2 y}dxdy = 0.$$

Discrete distributions

We only consider integer-valued random variables, i.e. the distribution of X is given by the values of $\mathbb{P}(X = k)$, $k \geq 0$.

Examples

(i) The *Bernoulli* distribution.

We have

$$\mathbb{P}(X = 1) = p \qquad \text{and} \qquad \mathbb{P}(X = 0) = 1 - p, \tag{1.5.10}$$

where $p \in [0, 1]$ is a parameter.

Note that any Bernoulli random variable $X : \Omega \longrightarrow \{0, 1\}$ can be written as the indicator function

$$X = \mathbb{1}_A$$

on Ω with $A = \{X = 1\} = \{\omega \in \Omega \,:\, X(\omega) = 1\}$.

(ii) The *binomial* distribution.

We have

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, \ldots, n,$$

where $n \geq 1$ and $p \in [0, 1]$ are parameters and $\binom{n}{k} = n!/(k!(n-k)!)$, $0 \leq k \leq n$.

(iii) The *geometric* distribution.

In this case, we have

$$\mathbb{P}(X = k) = (1 - p)p^k, \qquad k \geq 0, \tag{1.5.11}$$

where $p \in (0, 1)$ is a parameter. For example, if $(X_k)_{k\in\mathbb{N}}$ is a sequence of independent Bernoulli random variables with distribution (1.5.10), then the random variable,[2]

$$T_0 := \inf\{k \geq 0 \ : \ X_k = 0\}$$

can denote the duration of a game until the time that the wealth X_k of a player reaches 0. The random variable T_0 has the geometric distribution (1.5.11) with parameter $p \in (0, 1)$.

(iv) The *negative binomial* (or *Pascal*) distribution.

We have

$$\mathbb{P}(X = k) = \binom{k + r - 1}{r - 1}(1 - p)^r p^k, \qquad k \geq 0, \tag{1.5.12}$$

where $p \in (0, 1)$ and $r \geq 1$ are parameters. Note that the sum of $r \geq 1$ independent geometric random variables with parameter p has a negative binomial distribution with parameter (r, p). In particular, the negative binomial distribution recovers the geometric distribution when $r = 1$.

(v) The *Poisson* distribution.

We have

$$\mathbb{P}(X = k) = \frac{\lambda^k}{k!}\mathrm{e}^{-\lambda}, \qquad k \geq 0,$$

where $\lambda > 0$ is a parameter.

The probability that a discrete nonnegative random variable $X : \Omega \longrightarrow \mathbb{N} \cup \{+\infty\}$ is finite is given by

$$\mathbb{P}(X < \infty) = \sum_{k\geq 0} \mathbb{P}(X = k), \tag{1.5.13}$$

and we have

$$1 = \mathbb{P}(X = \infty) + \mathbb{P}(X < \infty) = \mathbb{P}(X = \infty) + \sum_{k\geq 0} \mathbb{P}(X = k).$$

[2] The notation "inf" stands for "infimum", meaning the smallest $n \geq 0$ such that $X_n = 0$, if such an n exists.

Remark 1.9 The distribution of a discrete random variable cannot admit a probability density. If this were the case, by Remark 1.8 we would have $\mathbb{P}(X = k) = 0$ for all $k \geq 0$ and

$$1 = \mathbb{P}(X \in \mathbb{R}) = \mathbb{P}(X \in \mathbb{N}) = \sum_{k \geq 0} \mathbb{P}(X = k) = 0,$$

which is a contradiction.

Given two discrete random variables X and Y, the conditional distribution of X given $Y = k$ is given by

$$\mathbb{P}(X = n \mid Y = k) = \frac{\mathbb{P}(X = n \text{ and } Y = k)}{\mathbb{P}(Y = k)}, \qquad n \geq 0,$$

provided that $\mathbb{P}(Y = k) > 0$, $k \geq 0$.

1.6 Expectation of Random Variables

The *expectation*, or *expected value*, of a random variable X is the mean, or average value, of X. In practice, expectations can be even more useful than probabilities. For example, knowing that a given equipment (such as a bridge) has a failure probability of 1.78493 out of a billion can be of less practical use than knowing the expected lifetime (e.g. 200000 years) of that equipment.

For example, the time $T(\omega)$ to travel from home to work/school can be a random variable with a new outcome and value every day, however, we usually refer to its expectation $\mathbb{E}[T]$ rather than to its sample values that may change from day to day.

Expected value of a Bernoulli random variable

Any Bernoulli random variable $X : \Omega \longrightarrow \{0, 1\}$ can be written as the indicator function $X := \mathbb{1}_A$ where A is the event $A = \{X = 1\}$, and the parameter $p \in [0, 1]$ of X is given by

$$p = \mathbb{P}(X = 1) = \mathbb{P}(A) = \mathbb{E}[\mathbb{1}_A] = \mathbb{E}[X].$$

The expectation of a Bernoulli random variable with parameter p is defined as

$$\mathbb{E}[\mathbb{1}_A] := 1 \times \mathbb{P}(A) + 0 \times \mathbb{P}(A^c) = \mathbb{P}(A). \tag{1.6.1}$$

Expected value of a discrete random variable

Next, let $X : \Omega \longrightarrow \mathbb{N}$ be a discrete random variable. The expectation $\mathbb{E}[X]$ of X is defined as the sum

$$\mathbb{E}[X] = \sum_{k\geq 0} k\mathbb{P}(X = k), \tag{1.6.2}$$

in which the possible values $k \geq 0$ of X are weighted by their probabilities. More generally we have

$$\mathbb{E}[\phi(X)] = \sum_{k\geq 0} \phi(k)\mathbb{P}(X = k),$$

for all sufficiently summable functions $\phi : \mathbb{N} \longrightarrow \mathbb{R}$.

The expectation of the indicator function $X = \mathbb{1}_A = \mathbb{1}_{\{X=1\}}$ can be recovered from (1.6.2) as

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{1}_A] = 0 \times \mathbb{P}(\Omega \setminus A) + 1 \times \mathbb{P}(A) = 0 + \mathbb{P}(A) = \mathbb{P}(A).$$

Note that the expectation is a *linear* operation, i.e. we have

$$\mathbb{E}[aX + bY] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y], \qquad a, b \in \mathbb{R}, \tag{1.6.3}$$

provided that

$$\mathbb{E}[|X|] + \mathbb{E}[|Y|] < \infty.$$

Examples

(i) Expected value of a Poisson random variable with parameter $\lambda > 0$:

$$\mathbb{E}[X] = \sum_{k\geq 0} k\mathbb{P}(X = k) = \mathrm{e}^{-\lambda} \sum_{k\geq 1} k\frac{\lambda^k}{k!} = \lambda \mathrm{e}^{-\lambda} \sum_{k\geq 0} \frac{\lambda^k}{k!} = \lambda, \tag{1.6.4}$$

where we used the exponential series (A.1).

(ii) Estimating the expected value of a Poisson random variable using R: Taking $\lambda := 2$, we can use the following R code:

```
poisson_samples <- rpois(100000, lambda = 2)
poisson_samples
mean(poisson_samples)
```

Given $X : \Omega \longrightarrow \mathbb{N} \cup \{+\infty\}$ a discrete nonnegative random variable X, we have

$$\mathbb{P}(X < \infty) = \sum_{k\geq 0} \mathbb{P}(X = k),$$

and

$$1 = \mathbb{P}(X = \infty) + \mathbb{P}(X < \infty) = \mathbb{P}(X = \infty) + \sum_{k \geq 0} \mathbb{P}(X = k),$$

and in general

$$\mathbb{E}[X] = +\infty \times \mathbb{P}(X = \infty) + \sum_{k \geq 0} k\mathbb{P}(X = k).$$

In particular, $\mathbb{P}(X = \infty) > 0$ implies $\mathbb{E}[X] = \infty$, and the finiteness condition $\mathbb{E}[X] < \infty$ implies $\mathbb{P}(X < \infty) = 1$, however, the converse is *not true*. For example, assume that X has the geometric distribution

$$\mathbb{P}(X = k) := \frac{1}{2^{k+1}}, \qquad k \geq 0, \tag{1.6.5}$$

with parameter $p = 1/2$, and

$$\mathbb{E}[X] = \sum_{k \geq 0} \frac{k}{2^{k+1}} = \frac{1}{4} \sum_{k \geq 1} \frac{k}{2^{k-1}} = \frac{1}{4} \frac{1}{(1 - 1/2)^2} = 1 < \infty,$$

by (A.4). Letting $\phi(X) := 2^X$, we have

$$\mathbb{P}(\phi(X) < \infty) = \mathbb{P}(X < \infty) = \sum_{k \geq 0} \frac{1}{2^{k+1}} = 1,$$

and

$$\mathbb{E}[\phi(X)] = \sum_{k \geq 0} \phi(k)\mathbb{P}(X = k) = \sum_{k \geq 0} \frac{2^k}{2^{k+1}} = \sum_{k \geq 0} \frac{1}{2} = +\infty,$$

hence the expectation $\mathbb{E}[\phi(X)]$ is *infinite* although $\phi(X)$ is *finite* with probability one.[3]

Conditional expectation

The notion of expectation takes its full meaning under conditioning. For example, the expected return of a random asset usually depends on information such as economic data, location, etc. In this case, replacing the expectation by a conditional expectation will provide a better estimate of the expected value.

For instance, life expectancy is a natural example of a conditional expectation since it typically depends on location, gender, and other parameters.

The *conditional expectation* of a finite discrete random variable $X : \Omega \longrightarrow \mathbb{N}$ given an event A is defined by

[3] This is the St. Petersburg paradox.

$$\mathbb{E}[X \mid A] = \sum_{k\geq 0} k\mathbb{P}(X = k \mid A) = \sum_{k\geq 1} k\frac{\mathbb{P}(X = k \text{ and } A)}{\mathbb{P}(A)}.$$

Lemma 1.10 *Given an event A such that $\mathbb{P}(A) > 0$, we have*

$$\mathbb{E}[X \mid A] = \frac{1}{\mathbb{P}(A)}\mathbb{E}\left[X\mathbb{1}_A\right]. \tag{1.6.6}$$

Proof The proof is done only for $X : \Omega \longrightarrow \mathbb{N}$ a discrete random variable, however, (1.6.6) is valid for general real-valued random variables. By Relation (1.4.1) we have

$$\begin{aligned}
\mathbb{E}[X \mid A] &= \sum_{k\geq 0} k\mathbb{P}(X = k \mid A) \\
&= \frac{1}{\mathbb{P}(A)} \sum_{k\geq 0} k\mathbb{P}(\{X = k\} \cap A) = \frac{1}{\mathbb{P}(A)} \sum_{k\geq 0} k\,\mathbb{E}\left[\mathbb{1}_{\{X=k\}\cap A}\right] \\
&= \frac{1}{\mathbb{P}(A)} \sum_{k\geq 0} k\,\mathbb{E}\left[\mathbb{1}_{\{X=k\}}\mathbb{1}_A\right] = \frac{1}{\mathbb{P}(A)}\,\mathbb{E}\left[\mathbb{1}_A \sum_{k\geq 0} k\mathbb{1}_{\{X=k\}}\right] \\
&= \frac{1}{\mathbb{P}(A)}\,\mathbb{E}\left[\mathbb{1}_A X\right],
\end{aligned}$$

where we used the relation

$$X = \sum_{k\geq 0} k\mathbb{1}_{\{X=k\}}$$

which holds since X takes only integer values. □

Example

(i) For example, consider $\Omega = \{1, 3, -1, -2, 5, 7\}$ with the non-uniform probability measure given by

$$\mathbb{P}(\{-1\}) = \mathbb{P}(\{-2\}) = \mathbb{P}(\{1\}) = \mathbb{P}(\{3\}) = \frac{1}{7},\ \mathbb{P}(\{5\}) = \frac{2}{7},\ \mathbb{P}(\{7\}) = \frac{1}{7},$$

and the random variable

$$X : \Omega \longrightarrow \mathbb{Z}$$

given by

$$X(k) = k, \qquad k = 1, 3, -1, -2, 5, 7.$$

Here, $\mathbb{E}[X \mid X > 1]$ denotes the expected value of X given

$$A = \{X > 1\} = \{3, 5, 7\} \subset \Omega,$$

i.e. the mean value of X given that X is strictly positive. This conditional expectation can be computed as

$$\begin{aligned}
&\mathbb{E}[X \mid X > 1] \\
&= 3 \times \mathbb{P}(X = 3 \mid X > 1) + 5 \times \mathbb{P}(X = 5 \mid X > 1) + 7 \times \mathbb{P}(X = 7 \mid X > 1) \\
&= \frac{3 + 2 \times 5 + 7}{4} \\
&= \frac{3 + 5 + 5 + 7}{7 \times 4/7} \\
&= \frac{1}{\mathbb{P}(X > 1)} \mathbb{E}\left[X \mathbb{1}_{\{X>1\}}\right],
\end{aligned}$$

where $\mathbb{P}(X > 1) = 4/7$ and the truncated expectation $\mathbb{E}\left[X \mathbb{1}_{\{X>1\}}\right]$ is given by $\mathbb{E}\left[X \mathbb{1}_{\{X>1\}}\right] = (3 + 2 \times 5 + 7)/7$.

(ii) Estimating a conditional expectation using *R*:

```
geo_samples <- rgeom(100000, prob = 1/4)
mean(geo_samples)
mean(geo_samples[geo_samples<10])
```

Taking $p := 3/4$, by (A.4) we have

$$\mathbb{E}[X] = (1 - p) \sum_{k \geq 1} k p^k = \frac{p}{1 - p} = 3,$$

and

$$\begin{aligned}
\mathbb{E}[X \mid X < 10] &= \frac{1}{\mathbb{P}(X < 10)} \mathbb{E}\left[X \mathbb{1}_{\{X<10\}}\right] \\
&= \frac{1}{\mathbb{P}(X < 10)} \sum_{k=0}^{9} k \mathbb{P}(X = k) \\
&= \frac{1}{\sum_{k=0}^{9} p^k} \sum_{k=1}^{9} k p^k \\
&= \frac{p(1 - p)}{1 - p^{10}} \frac{\partial}{\partial p} \sum_{k=0}^{9} p^k \\
&= \frac{p(1 - p)}{1 - p^{10}} \frac{\partial}{\partial p} \left(\frac{1 - p^{10}}{1 - p} \right) \\
&= \frac{p(1 - p^{10} - 10(1 - p)p^9)}{(1 - p)(1 - p^{10})} \\
&\simeq 2.4032603455.
\end{aligned}$$

If the random variable $X : \Omega \longrightarrow \mathbb{N}$ is independent[4] of the event A, we have

$$\mathbb{E}[X\mathbb{1}_A] = \mathbb{E}[X]\,\mathbb{E}[\mathbb{1}_A] = \mathbb{E}[X]\mathbb{P}(A),$$

and we naturally find

$$\mathbb{E}[X \mid A] = \mathbb{E}[X]. \tag{1.6.7}$$

Taking $X = \mathbb{1}_A$ with

$$\begin{aligned} \mathbb{1}_A : \Omega &\longrightarrow \{0,1\} \\ \omega &\longmapsto \mathbb{1}_A := \begin{cases} 1 & \text{if } \omega \in A, \\ 0 & \text{if } \omega \notin A, \end{cases} \end{aligned}$$

shows that, in particular,

$$\begin{aligned} \mathbb{E}[\mathbb{1}_A \mid A] &= 0 \times \mathbb{P}(X = 0 \mid A) + 1 \times \mathbb{P}(X = 1 \mid A) \\ &= \mathbb{P}(X = 1 \mid A) \\ &= \mathbb{P}(A \mid A) \\ &= 1. \end{aligned}$$

One can also define the conditional expectation of X given $A = \{Y = k\}$, as

$$\mathbb{E}[X \mid Y = k] = \sum_{n \geq 0} n\mathbb{P}(X = n \mid Y = k),$$

where $Y : \Omega \longrightarrow \mathbb{N}$ is a discrete random variable.

Proposition 1.11 *Given X a discrete random variable such that $\mathbb{E}[|X|] < \infty$, we have the relation*

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]], \tag{1.6.8}$$

which is sometimes referred to as the tower property.

Proof We have

$$\begin{aligned} \mathbb{E}[\mathbb{E}[X \mid Y]] &= \sum_{k \geq 0} \mathbb{E}[X \mid Y = k]\mathbb{P}(Y = k) \\ &= \sum_{k \geq 0} \sum_{n \geq 0} n\mathbb{P}(X = n \mid Y = k)\mathbb{P}(Y = k) \\ &= \sum_{n \geq 0} n \sum_{k \geq 0} \mathbb{P}(X = n \text{ and } Y = k) \\ &= \sum_{n \geq 0} n\mathbb{P}(X = n) = \mathbb{E}[X], \end{aligned}$$

[4] *i.e.*, $\mathbb{P}(\{X = k\} \cap A) = \mathbb{P}(\{X = k\})\mathbb{P}(A)$ for all $k \geq 0$.

where we used the marginal distribution

$$\mathbb{P}(X = n) = \sum_{k \geq 0} \mathbb{P}(X = n \text{ and } Y = k), \qquad n \geq 0,$$

that follows from the *law of total probability* (1.3.1) with $A_k = \{Y = k\}$, $k \geq 0$.

Taking

$$Y = \sum_{k \geq 0} k \mathbb{1}_{A_k},$$

with $A_k := \{Y = k\}$, $k \geq 0$, from (1.6.8) we also get the *law of total expectation*

$$\begin{aligned} \mathbb{E}[X] &= \mathbb{E}[\mathbb{E}[X \mid Y]] \\ &= \sum_{k \geq 0} \mathbb{E}[X \mid Y = k] \mathbb{P}(Y = k) \\ &= \sum_{k \geq 0} \mathbb{E}[X \mid A_k] \mathbb{P}(A_k). \end{aligned} \tag{1.6.9}$$

Example.

Life expectancy in Singapore is $\mathbb{E}[T] = 80$ years overall, where T denotes the lifetime of a given individual chosen at random. Let $G \in \{m, w\}$ denote the gender of that individual. The statistics show that

$$\mathbb{E}[T \mid G = m] = 78 \quad \text{and} \quad \mathbb{E}[T \mid G = w] = 81.9,$$

and we have

$$\begin{aligned} 80 &= \mathbb{E}[T] \\ &= \mathbb{E}[\mathbb{E}[T|G]] \\ &= \mathbb{P}(G = w)\, \mathbb{E}[T \mid G = w] + \mathbb{P}(G = m)\, \mathbb{E}[T \mid G = m] \\ &= 81.9 \times \mathbb{P}(G = w) + 78 \times \mathbb{P}(G = m) \\ &= 81.9 \times (1 - \mathbb{P}(G = m)) + 78 \times \mathbb{P}(G = m), \end{aligned}$$

showing that

$$80 = 81.9 \times (1 - \mathbb{P}(G = m)) + 78 \times \mathbb{P}(G = m),$$

i.e.

$$\mathbb{P}(G = m) = \frac{81.9 - 80}{81.9 - 78} = \frac{1.9}{3.9} = 0.487.$$

Variance

The *variance* of a random variable X is defined by

$$\mathrm{Var}[X] := \mathbb{E}\left[X^2\right] - (\mathbb{E}[X])^2,$$

provided that $\mathbb{E}\left[|X|^2\right] < \infty$. If $(X_k)_{k=1,\ldots,n}$ is a sequence of independent random variables, we have

$$\begin{aligned}
\mathrm{Var}\left[\sum_{k=1}^n X_k\right] &= \mathbb{E}\left[\left(\sum_{k=1}^n X_k\right)^2\right] - \left(\mathbb{E}\left[\sum_{k=1}^n X_k\right]\right)^2 \\
&= \mathbb{E}\left[\sum_{k=1}^n X_k \sum_{l=1}^n X_l\right] - \mathbb{E}\left[\sum_{k=1}^n X_k\right]\mathbb{E}\left[\sum_{l=1}^n X_l\right] \\
&= \mathbb{E}\left[\sum_{k=1}^n \sum_{l=1}^n X_k X_l\right] - \sum_{k=1}^n \sum_{l=1}^n \mathbb{E}[X_k]\,\mathbb{E}[X_l] \\
&= \sum_{k=1}^n \mathbb{E}\left[X_k^2\right] + \sum_{1\le k\neq l\le n} \mathbb{E}[X_k X_l] - \sum_{k=1}^n (\mathbb{E}[X_k])^2 - \sum_{1\le k\neq l\le n} \mathbb{E}[X_k]\,\mathbb{E}[X_l] \\
&= \sum_{k=1}^n \left(\mathbb{E}\left[X_k^2\right] - (\mathbb{E}[X_k])^2\right) \\
&= \sum_{k=1}^n \mathrm{Var}[X_k]. \qquad (1.6.10)
\end{aligned}$$

Random sums

In the sequel we consider $Y : \Omega \longrightarrow \mathbb{N}$ an *a.s.* finite, integer-valued random variable, i.e. we have $\mathbb{P}(Y < \infty) = 1$ and $\mathbb{P}(Y = \infty) = 0$. Based on the tower property of conditional expectations (1.6.8) or ordinary conditioning, the expectation of a random sum $\sum_{k=1}^Y X_k$, where $(X_k)_{k\in\mathbb{N}}$ is a sequence of random variables, can be computed from the *tower property* (1.6.8) or from the *law of total expectation* (1.6.9) as

$$\begin{aligned}
\mathbb{E}\left[\sum_{k=1}^Y X_k\right] &= \mathbb{E}\left[\mathbb{E}\left[\sum_{k=1}^Y X_k \,\middle|\, Y\right]\right] \\
&= \sum_{n\ge 0} \mathbb{E}\left[\sum_{k=1}^Y X_k \,\middle|\, Y = n\right] \mathbb{P}(Y = n) \\
&= \sum_{n\ge 0} \mathbb{E}\left[\sum_{k=1}^n X_k \,\middle|\, Y = n\right] \mathbb{P}(Y = n),
\end{aligned}$$

and if the sequence $(X_k)_{k\in\mathbb{N}}$ is (mutually) independent of Y, this yields

$$\mathbb{E}\left[\sum_{k=1}^{Y} X_k\right] = \sum_{n\geq 0} \mathbb{E}\left[\sum_{k=1}^{n} X_k\right] \mathbb{P}(Y=n)$$
$$= \sum_{n\geq 0} \mathbb{P}(Y=n) \sum_{k=1}^{n} \mathbb{E}[X_k].$$

Random products

Similarly, for a random product we will have, using the independence of Y with $(X_k)_{k\in\mathbb{N}}$,

$$\mathbb{E}\left[\prod_{k=1}^{Y} X_k\right] = \sum_{n\geq 0} \mathbb{E}\left[\prod_{k=1}^{n} X_k\right] \mathbb{P}(Y=n) \tag{1.6.11}$$
$$= \sum_{n\geq 0} \mathbb{P}(Y=n) \prod_{k=1}^{n} \mathbb{E}[X_k],$$

where the last equality requires the (mutual) independence of the random variables in the sequence $(X_k)_{k\geq 1}$.

Distributions admitting a density

Given a random variable X whose distribution admits a probability density $\varphi_X : \mathbb{R} \longrightarrow \mathbb{R}_+$ we have

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x\varphi_X(x)dx,$$

and more generally,

$$\mathbb{E}[\phi(X)] = \int_{-\infty}^{\infty} \phi(x)\varphi_X(x)dx, \tag{1.6.12}$$

for all sufficiently integrable function ϕ on $\mathbb{R}$. For example, if X has a standard normal distribution we have

$$\mathbb{E}[\phi(X)] = \int_{-\infty}^{\infty} \phi(x)\mathrm{e}^{-x^2/2} \frac{dx}{\sqrt{2\pi}}.$$

Examples

(a) In case X has a Gaussian distribution with mean $\mu \in \mathbb{R}$ and variance $\sigma^2 > 0$, we have

$$\mathbb{E}[\phi(X)] = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} \phi(x)\mathrm{e}^{-(x-\mu)^2/(2\sigma^2)}dx. \tag{1.6.13}$$

(b) The uniform random variable U on $[0, 1]$ satisfies $\mathbb{E}[U] = 1/2 < \infty$ and

$$\mathbb{P}(1/U < \infty) = \mathbb{P}(U > 0) = \mathbb{P}(U \in (0, 1]) = 1,$$

however, we have

$$\mathbb{E}[1/U] = \int_0^1 \frac{dx}{x} = +\infty,$$

and $\mathbb{P}(1/U = +\infty) = \mathbb{P}(U = 0) = 0$.

(c) If the random variable X has an exponential distribution with parameter $\mu > 0$ we have

$$\mathbb{E}\left[\mathrm{e}^{\lambda X}\right] = \mu \int_0^\infty \mathrm{e}^{\lambda x} \mathrm{e}^{-\mu x} dx = \begin{cases} \dfrac{\mu}{\mu - \lambda} < \infty & \text{if } \mu > \lambda, \\ +\infty, & \text{if } \mu \leq \lambda. \end{cases}$$

Exercise: In case $X \simeq \mathcal{N}(\mu, \sigma^2)$ has a Gaussian distribution with mean $\mu \in \mathbb{R}$ and variance $\sigma^2 > 0$, check that

$$\mu = \mathbb{E}[X] \quad \text{and} \quad \sigma^2 = \mathbb{E}\left[X^2\right] - (\mathbb{E}[X])^2.$$

When $(X, Y) : \Omega \longrightarrow \mathbb{R}^2$ is a $\mathbb{R}^2$-valued couple of random variables whose distribution admits a probability density $\varphi_{X,Y} : \mathbb{R}^2 \longrightarrow \mathbb{R}_+$ we have

$$\mathbb{E}[\phi(X, Y)] = \int_{-\infty}^\infty \int_{-\infty}^\infty \phi(x, y) \varphi_{X,Y}(x, y) dx dy,$$

for all sufficiently integrable function ϕ on $\mathbb{R}^2$.

The expectation of an absolutely continuous random variable satisfies the same linearity property (1.6.3) as in the discrete case.

The conditional expectation of an absolutely continuous random variable can be defined as

$$\mathbb{E}[X \mid Y = y] = \int_{-\infty}^\infty x \varphi_{X|Y=y}(x) dx$$

where the conditional probability density $\varphi_{X|Y=y}(x)$ is defined in (1.5.7), with the relation

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]] \tag{1.6.14}$$

which is called the *tower property* and holds as in the discrete case, since

$$\begin{aligned}\mathbb{E}[\mathbb{E}[X \mid Y]] &= \int_{-\infty}^{\infty} \mathbb{E}[X \mid Y = y]\varphi_Y(y)dy \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x\varphi_{X|Y=y}(x)\varphi_Y(y)dxdy \\ &= \int_{-\infty}^{\infty} x\int_{-\infty}^{\infty} \varphi_{(X,Y)}(x,y)dydx \\ &= \int_{-\infty}^{\infty} x\varphi_X(x)dx = \mathbb{E}[X],\end{aligned}$$

where we used Relation (1.5.6) between the probability density of (X, Y) and its marginal X.

For example, an exponentially distributed random variable X with probability density function (1.5.2) has the expected value

$$\mathbb{E}[X] = \lambda \int_0^{\infty} x\mathrm{e}^{-\lambda x}dx = \frac{1}{\lambda}.$$

1.7 Moment and Probability Generating Functions

Characteristic functions

The *characteristic function* of a random variable X is the function

$$\Psi_X : \mathbb{R} \longrightarrow \mathbb{C}$$

defined by

$$\Psi_X(t) = \mathbb{E}\left[\mathrm{e}^{itX}\right], \qquad t \in \mathbb{R}.$$

The characteristic function Ψ_X of a random variable X with probability density function $f : \mathbb{R} \longrightarrow \mathbb{R}_+$ satisfies

$$\Psi_X(t) = \int_{-\infty}^{\infty} \mathrm{e}^{ixt}\varphi(x)dx, \qquad t \in \mathbb{R}.$$

On the other hand, if $X : \Omega \longrightarrow \mathbb{N}$ is a discrete random variable we have

$$\Psi_X(t) = \sum_{n\geq 0} \mathrm{e}^{itn}\mathbb{P}(X = n), \qquad t \in \mathbb{R}.$$

One of the main applications of characteristic functions is to provide a characterization of probability distributions, as in the following theorem.

Theorem 1.12 *Two random variables $X : \Omega \longrightarrow \mathbb{R}$ and $Y : \Omega \longrightarrow \mathbb{R}$ have same distribution if and only if*

$$\Psi_X(t) = \Psi_Y(t), \qquad t \in \mathbb{R}.$$

Theorem 1.12 is used to identify or to determine the probability distribution of a random variable X, by comparison with the characteristic function Ψ_Y of a random variable Y whose distribution is known.

The characteristic function of a random vector (X, Y) is the function $\Psi_{X,Y} : \mathbb{R}^2 \longrightarrow \mathbb{C}$ defined by

$$\Psi_{X,Y}(s,t) = \mathbb{E}\left[\mathrm{e}^{isX+itY}\right], \qquad s,t \in \mathbb{R}.$$

Theorem 1.13 *The random variables $X : \Omega \longrightarrow \mathbb{R}$ and $Y : \Omega \longrightarrow \mathbb{R}$ are* independent *if and only if*

$$\Psi_{X,Y}(s,t) = \Psi_X(s)\Psi_Y(t), \qquad s,t \in \mathbb{R}.$$

A random variable X has a Gaussian distribution with mean μ and variance σ^2 if and only if its characteristic function satisfies

$$\mathbb{E}\left[\mathrm{e}^{i\alpha X}\right] = \mathrm{e}^{i\alpha\mu-\alpha^2\sigma^2/2}, \qquad \alpha \in \mathbb{R}. \tag{1.7.1}$$

From Theorems 1.12 and 1.13, we deduce the following proposition.

Proposition 1.14 *Let $X \simeq \mathcal{N}(\mu, \sigma_X^2)$ and $Y \simeq \mathcal{N}(\nu, \sigma_Y^2)$ be independent Gaussian random variables. Then $X + Y$ also has a Gaussian distribution*

$$X + Y \simeq \mathcal{N}(\mu + \nu, \sigma_X^2 + \sigma_Y^2).$$

Proof Since X and Y are independent, by Theorem 1.13 the characteristic function Ψ_{X+Y} of $X + Y$ is given by

$$\begin{aligned}\Psi_{X+Y}(t) &= \Psi_X(t)\Psi_Y(t)\\ &= \mathrm{e}^{it\mu-t^2\sigma_X^2/2}\mathrm{e}^{it\nu-t^2\sigma_Y^2/2}\\ &= \mathrm{e}^{it(\mu+\nu)-t^2(\sigma_X^2+\sigma_Y^2)/2}, \qquad t \in \mathbb{R},\end{aligned}$$

where we used (1.7.1). Consequently, the characteristic function of $X + Y$ is that of a Gaussian random variable with mean $\mu + \nu$ and variance $\sigma_X^2 + \sigma_Y^2$ and we conclude by Theorem 1.12. □

Moment generating functions

The *moment generating function* of a random variable X is the function $M_X : \mathbb{R} \longrightarrow \mathbb{R}$ defined by

$$M_X(t) := \mathbb{E}\left[\mathrm{e}^{tX}\right],$$

for t in a neighborhood of 0. In particular, we have

$$\mathbb{E}[X^n] = \frac{\partial^n}{\partial t^n} M_X(0), \qquad n \geq 1,$$

provided that $\mathbb{E}[|X|^n] < \infty$, and

$$M_X(t) = \mathbb{E}\left[\mathrm{e}^{tX}\right] = \sum_{n\geq 0} \frac{t^n}{n!} \mathbb{E}[X^n],$$

provided that $\mathbb{E}\left[\mathrm{e}^{t|X|}\right] < \infty$, $t \in \mathbb{R}$, and for this reason the moment generating function G_X characterizes the *moments* $\mathbb{E}[X^n]$ of $X : \Omega \longrightarrow \mathbb{N}$, $n \geq 0$.

The moment generating function M_X of a random variable X with probability density function $f : \mathbb{R} \longrightarrow \mathbb{R}_+$ satisfies

$$M_X(t) = \int_{-\infty}^{\infty} \mathrm{e}^{xt} \varphi(x) dx, \qquad t \in \mathbb{R}.$$

For example, the moment generating functions (MGF) of a Gaussian random variable X with mean μ and variance σ^2 is given by

$$\mathbb{E}\left[\mathrm{e}^{\alpha X}\right] = \mathrm{e}^{\alpha\mu+\alpha^2\sigma^2/2}, \qquad \alpha \in \mathbb{R}. \tag{1.7.2}$$

Note that in probability, the moment generating function is written as a *bilateral* transform defined using an integral from $-\infty$ to $+\infty$.

Probability generating functions

Consider

$$X : \Omega \longrightarrow \mathbb{N} \cup \{+\infty\}$$

a *discrete* random variable possibly taking infinite values. The *probability generating function* of X is the *function*

$$\begin{aligned} G_X :[-1, 1] &\longrightarrow \mathbb{R} \\ s &\longmapsto G_X(s) \end{aligned}$$

defined by

$$G_X(s) := \mathbb{E}\left[s^X \mathbb{1}_{\{X<\infty\}}\right] = \sum_{n\geq 0} s^n \mathbb{P}(X = n), \qquad -1 \leq s \leq 1. \tag{1.7.3}$$

Note that the series summation in (1.7.3) is over the *finite* integers, which explains the presence of the truncating indicator function $\mathbb{1}_{\{X<\infty\}}$ inside the expectation in (1.7.3). If the random variable $X : \Omega \longrightarrow \mathbb{N}$ is almost surely finite, i.e. $\mathbb{P}(X < \infty) = 1$, we simply have

$$G_X(s) = \mathbb{E}\left[s^X\right] = \sum_{n\geq 0} s^n \mathbb{P}(X = n), \qquad -1 \leq s \leq 1,$$

and for this reason the probability generating function G_X characterizes the *probability distribution* $\mathbb{P}(X = n)$, $n \geq 0$, of $X : \Omega \longrightarrow \mathbb{N}$.

Examples

(i) Poisson distribution. Consider a random variable X with probability generating function

$$G_X(s) = \mathrm{e}^{\lambda(s-1)}, \qquad -1 \leq s \leq 1,$$

for some $\lambda > 0$. What is the distribution of X?

Using the exponential series (A.1) we have

$$G_X(s) = \mathrm{e}^{\lambda(s-1)} = \mathrm{e}^{-\lambda} \sum_{n\geq 0} s^n \frac{\lambda^n}{n!}, \qquad -1 \leq s \leq 1, \tag{1.7.4}$$

hence by identification with (1.7.3) we find

$$\mathbb{P}(X = n) = \mathrm{e}^{-\lambda} \frac{\lambda^n}{n!}, \qquad n \geq 0,$$

i.e. X has the Poisson distribution with parameter λ.

(ii) Geometric distribution. Given X a random variable with geometric distribution $\mathbb{P}(X = n) = (1 - p)p^n$, $n \geq 0$, we have

$$G_X(s) = \sum_{n\geq 0} s^n \mathbb{P}(X = n) = (1 - p) \sum_{n\geq 0} s^n p^n = \frac{1 - p}{1 - ps}, \qquad -1 < s < 1,$$

where we applied the geometric series (A.3).

We note that from (1.7.3) we can write

$$G_X(s) = \mathbb{E}[s^X], \qquad -1 < s < 1,$$

since $s^X = s^X \mathbb{1}_{\{X<\infty\}}$ when $-1 < s < 1$.

Properties of probability generating functions

(i) Taking $s = 1$, we have

$$G_X(1) = \sum_{n\geq 0} \mathbb{P}(X = n) = \mathbb{P}(X < \infty) = \mathbb{E}\left[\mathbb{1}_{\{X<\infty\}}\right],$$

hence

$$G_X(1) = \mathbb{P}(X < \infty).$$

(ii) Taking $s = 0$, we have

$$G_X(0) = \mathbb{E}\left[0^X\right] = \mathbb{E}\left[\mathbb{1}_{\{X=0\}}\right] = \mathbb{P}(X = 0),$$

since $0^0 = 1$ and $0^X = \mathbb{1}_{\{X=0\}}$, hence

$$G_X(0) = \mathbb{P}(X = 0). \tag{1.7.5}$$

(iii) The *derivative*[5] $G'_X(s)$ of $G_X(s)$ with respect to s satisfies

$$G'_X(s) = \sum_{n\geq 1} n s^{n-1} \mathbb{P}(X = n), \qquad -1 < s < 1,$$

hence if $\mathbb{P}(X < \infty) = 1$ we have provided that $\mathbb{E}[X] < \infty$.

$$G'_X(1^-) = \mathbb{E}[X] = \sum_{k\geq 0} k\mathbb{P}(X = k), \tag{1.7.6}$$

(iv) By computing the second derivative

$$\begin{aligned} G''_X(s) &= \sum_{k\geq 2} k(k-1)s^{k-2}\mathbb{P}(X = k) \\ &= \sum_{k\geq 0} k(k-1)s^{k-2}\mathbb{P}(X = k) \\ &= \sum_{k\geq 0} k^2 s^{k-2}\mathbb{P}(X = k) - \sum_{k\geq 0} k s^{k-2}\mathbb{P}(X = k), \qquad -1 < s < 1, \end{aligned}$$

we similarly find

[5] Here $G'_X(1^-)$ denotes the derivative on the left at the point $s = 1$.

$$\begin{aligned} G_X''(1^-) &= \sum_{k\geq 0} k(k-1)\mathbb{P}(X=k) \\ &= \sum_{k\geq 0} k^2\mathbb{P}(X=k) - \sum_{k\geq 0} k\mathbb{P}(X=k) \\ &= \mathbb{E}\left[X^2\right] - \mathbb{E}[X] \\ &= \mathbb{E}[X(X-1)], \end{aligned}$$

provided that $\mathbb{E}\left[X^2\right] < \infty$.
More generally, using the k-th derivative of G_X we can compute the *factorial moment*

$$G_X^{(k)}(1^-) = \mathbb{E}[X(X-1)\cdots(X-k+1)], \qquad k \geq 1, \tag{1.7.7}$$

provided that $\mathbb{E}[|X^k|] < \infty$. In particular, we have provided that $\mathbb{E}\left[X^2\right] < \infty$.

$$\mathrm{Var}[X] = G_X''(1^-) + G_X'(1^-)(1 - G_X'(1^-)), \tag{1.7.8}$$

(v) When $X : \Omega \longrightarrow \mathbb{N}$ and $Y : \Omega \longrightarrow \mathbb{N}$ are two finite independent random variables we have $-1 \leq s \leq 1$.

$$G_{X+Y}(s) = \mathbb{E}\left[s^{X+Y}\right] = \mathbb{E}\left[s^X s^Y\right] = \mathbb{E}\left[s^X\right]\mathbb{E}\left[s^Y\right] = G_X(s)G_Y(s), \tag{1.7.9}$$

(vi) The probability generating function can also be used from (1.7.3) to recover the distribution of the discrete random variable X as extending (1.7.5) to all $n \geq 0$.

$$\mathbb{P}(X=n) = \frac{1}{n!}\frac{\partial^n}{\partial s^n} G_X(s)_{|s=0}, \qquad n \geq 0, \tag{1.7.10}$$

Exercise: Show that the probability generating function of a Poisson random variable X with parameter $\lambda > 0$ is given by

$$G_X(s) = \mathrm{e}^{\lambda(s-1)}, \qquad -1 \leq s \leq 1.$$

From the generating function we also recover the mean

$$\mathbb{E}[X] = G'_X(1^-) = \lambda \mathrm{e}^{\lambda(s-1)}_{|s=1} = \lambda,$$

of the Poisson random variable X with parameter λ, and its variance

$$\begin{aligned}\mathrm{Var}[X] &= G''_X(1^-) + G'_X(1^-) - (G'_X(1^-))^2 \\ &= \lambda^2 \mathrm{e}^{\lambda(s-1)}_{|s=1} + \lambda s \mathrm{e}^{\lambda(s-1)}_{|s=1} - \lambda^2 \\ &= \lambda^2 + \lambda - \lambda^2 = \lambda,\end{aligned}$$

by (1.7.8).

Moments and cumulants

The *cumulants* of a random variable X are the sequence $(\kappa_n^X)_{n\in\mathbb{N}}$ of numbers defined by the logarithmic generating function

$$\log\left(\mathbb{E}\left[\mathrm{e}^{tX}\right]\right) = \sum_{n\geq 1} \kappa_n^X \frac{t^n}{n!}, \qquad t \in \mathbb{R}.$$

The *n-th* moment of a random variable X can be written in terms of its cumulants as

$$\begin{aligned}\mathbb{E}[X^n] &= \sum_{k=0}^{n} \frac{1}{k!} \sum_{\substack{d_1+\cdots+d_k=n \\ d_1,\ldots,d_k\geq 1}} \frac{n!}{d_1!\cdots d_k!} \kappa_{d_1}^X \cdots \kappa_{d_k}^X \qquad (1.7.11) \\ &= \sum_{k=1}^{n} \sum_{B_1,\ldots,B_k} \kappa_{|B_1|}^X \cdots \kappa_{|B_k|}^X,\end{aligned}$$

where the above sum runs over the partitions $B_1, \ldots, B_k$ of $\{1, \ldots, n\}$ with cardinality $|B_i|$ (Faà di Bruno formula). This also shows that

$$\mathbb{E}[X^n] = A_n(\kappa_1^X, \kappa_2^X, \ldots, \kappa_n^X),$$

where

$$A_n(x_1, \ldots, x_n) = n! \sum_{\substack{r_1+2r_2+\cdots+nr_n=n \\ r_1,\ldots,r_n\geq 0}} \prod_{l=1}^{n} \left(\frac{1}{r_l!}\left(\frac{x_l}{l!}\right)^{r_l}\right)$$

is the Bell polynomial of degree n. These relations follow from the Faà di Bruno formula, cf. e.g. Sect. 2.4 and Relation (2.4.4) p. 27 of Lukacs (1970). Indeed we have

$$\begin{aligned}
&\sum_{n\geq 0}\frac{t^n}{n!}\,\mathbb{E}[X^n]=\mathbb{E}\left[\mathrm{e}^{tX}\right]\\
&=\exp\left(\log\left(\mathbb{E}\left[\mathrm{e}^{tX}\right]\right)\right)=\sum_{k\geq 0}\frac{1}{k!}\left(\log\left(\mathbb{E}\left[\mathrm{e}^{tX}\right]\right)\right)^k\\
&=\sum_{k\geq 0}\frac{1}{k!}\left(\sum_{n\geq 1}\kappa_n^X\frac{t^n}{n!}\right)^k\\
&=\sum_{k\geq 0}\frac{1}{k!}\sum_{n\geq 1}t^n\sum_{\substack{d_1+\cdots+d_k=n\\ d_1\geq 1,\ldots,d_k\geq 1}}\frac{\kappa_{d_1}^X}{d_1!}\cdots\frac{\kappa_{d_1}^X}{d_k!}\\
&=\sum_{n\geq 0}\frac{t^n}{n!}\sum_{k=0}^n\frac{1}{k!}\sum_{\substack{d_1+\cdots+d_k=n\\ d_1\geq 1,\ldots,d_k\geq 1}}\frac{n!}{d_1!\cdots d_k!}\frac{\kappa_{d_1}^X}{d_1!}\cdots\frac{\kappa_{d_1}^X}{d_k!},
\end{aligned}$$

which shows (1.7.11).

1.8 Conditional Expectation

The construction of conditional expectations of the form $\mathbb{E}[X \mid Y]$ given above for discrete and absolutely continuous random variables can be generalized to σ-algebras.

Definition 1.15 Given $\mathcal{F}$ a σ-algebra on Ω, a random variable $X:\Omega\longrightarrow\mathbb{R}$ is said to be $\mathcal{F}$-measurable if

$$\{X\leq x\}:=\{\omega\in\Omega\ :\ X(\omega)\leq x\}\in\mathcal{F},$$

for all $x\in\mathbb{R}$.

Intuitively, when X is $\mathcal{F}$-measurable, the knowledge of the values of X depends only on the information contained in $\mathcal{F}$. For example, when $\mathcal{F}=\sigma(A_1,\ldots,A_n)$ where $(A_n)_{n\geq 1}$ is a *partition* of Ω with $\bigcup_{n\geq 1}A_n=\Omega$, any $\mathcal{F}$-measurable random variable X can be written as

$$X(\omega)=\sum_{k=1}^n c_k\mathbb{1}_{A_k}(\omega),\qquad \omega\in\Omega,$$

for some $c_1,\ldots,c_n\in\mathbb{R}$.

Definition 1.16 Given $(\Omega,\mathcal{F},\mathbb{P})$ a probability space we let $L^2(\Omega,\mathcal{F})$ denote the space of $\mathcal{F}$-measurable and square-integrable random variables, i.e.

$$L^2(\Omega,\mathcal{F}):=\left\{X:\Omega\longrightarrow\mathbb{R}\ :\ \mathbb{E}\left[|X|^2\right]<\infty\right\}.$$

More generally, for $p \geq 1$ one can define the space $L^p(\Omega, \mathcal{F})$ of $\mathcal{F}$-measurable and p-integrable random variables as

$$L^p(\Omega, \mathcal{F}) := \big\{X : \Omega \longrightarrow \mathbb{R} \ : \ \mathbb{E}[|X|^p] < \infty\big\}.$$

We define an *inner product* $\langle \cdot, \cdot \rangle_{L^2(\Omega,\mathcal{F})}$ between elements of $L^2(\Omega, \mathcal{F})$, as

$$\langle X, Y \rangle_{L^2(\Omega,\mathcal{F})} := \mathbb{E}[XY], \quad X, Y \in L^2(\Omega, \mathcal{F}). \tag{1.8.1}$$

This inner product is associated to the norm $\| \cdot \|_{L^2(\Omega)}$ by the relation

$$\|X\|_{L^2(\Omega)} = \sqrt{\mathbb{E}\left[X^2\right]} = \sqrt{\langle X, X \rangle_{L^2(\Omega,\mathcal{F})}}, \qquad X \in L^2(\Omega, \mathcal{F}).$$

The norm $\| \cdot \|_{L^2(\Omega)}$ also defines the *mean-square* distance

$$\|X - Y\|_{L^2(\Omega)} = \sqrt{\mathbb{E}\left[(X - Y)^2\right]}$$

between random variables $X, Y \in L^2(\Omega, \mathcal{F})$, and it induces a notion of *orthogonality*, namely X is *orthogonal* to Y in $L^2(\Omega, \mathcal{F})$ if and only if

$$\langle X, Y \rangle_{L^2(\Omega,\mathcal{F})} = 0.$$

Proposition 1.17 *The ordinary expectation $\mathbb{E}[X]$ achieves the minimum distance*

$$\big\|X - \mathbb{E}[X]\big\|^2_{L^2(\Omega)} = \min_{c \in \mathbb{R}} \|X - c\|^2_{L^2(\Omega)}. \tag{1.8.2}$$

Proof It suffices to differentiate

$$\frac{\partial}{\partial c} \mathbb{E}\left[(X - c)^2\right] = -2\,\mathbb{E}[X - c] = 0,$$

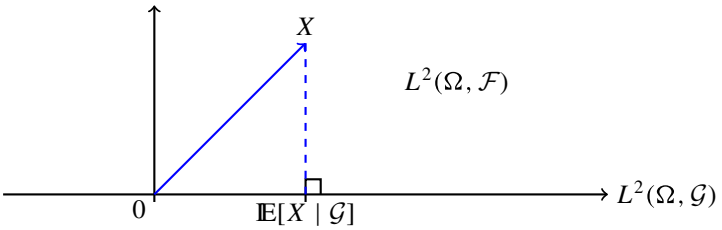
showing that the minimum in (1.8.2) is reached when $\mathbb{E}[X - c] = 0$, i.e. $c = \mathbb{E}[X]$. □

Similarly to Proposition 1.17, the conditional expectation will be defined by a distance minimizing procedure.

Definition 1.18 Given $\mathcal{G} \subset \mathcal{F}$ a sub σ-algebra of $\mathcal{F}$ and $X \in L^2(\Omega, \mathcal{F})$, the conditional expectation of X given $\mathcal{G}$, and denoted

$$\mathbb{E}[X \mid \mathcal{G}],$$

is defined as the *orthogonal projection* of X onto $L^2(\Omega, \mathcal{G})$.



As a consequence of the uniqueness of the orthogonal projection onto the subspace $L^2(\Omega, \mathcal{G})$ of $L^2(\Omega, \mathcal{F})$, the conditional expectation $\mathbb{E}[X \mid \mathcal{G}]$ is characterized by the relation

$$\langle Y, X - \mathbb{E}[X \mid \mathcal{G}] \rangle_{L^2(\Omega,\mathcal{F})} = 0,$$

which rewrites as

$$\mathbb{E}[Y(X - \mathbb{E}[X \mid \mathcal{G}])] = 0,$$

i.e.

$$\mathbb{E}[YX] = \mathbb{E}[Y\,\mathbb{E}[X \mid \mathcal{G}]],$$

for all bounded and $\mathcal{G}$-measurable random variables Y, where $\langle \cdot, \cdot \rangle_{L^2(\Omega,\mathcal{F})}$ denotes the inner product (1.8.1) in $L^2(\Omega, \mathcal{F})$. The next proposition extends Proposition 1.17 as a consequence of Definition 1.18. See Theorem 5.1.4 p. 197 of Stroock (2011) for an extension of the construction of conditional expectation to the space $L^1(\Omega, \mathcal{F})$ of integrable random variable.

Proposition 1.19 *The conditional expectation $I\!E[X \mid \mathcal{G}]$ realizes the minimum in mean-square distance between $X \in L^2(\Omega, \mathcal{F})$ and $L^2(\Omega, \mathcal{G})$, i.e. we have*

$$\|X - I\!E[X \mid \mathcal{G}]\|_{L^2(\Omega)} = \min_{Y \in L^2(\Omega,\mathcal{G})} \|X - Y\|_{L^2(\Omega)}. \tag{1.8.3}$$

Proof This is a consequence of the Pythagorean theorem written as

$$\|X - Y\|_{L^2(\Omega)} = \|X - \mathbb{E}[X \mid \mathcal{G}]\|_{L^2(\Omega)} + \|\,\mathbb{E}[X \mid \mathcal{G}] - Y\|_{L^2(\Omega)},$$

for any $Y \in L^2(\Omega, \mathcal{G})$.

The following proposition will often be used as a characterization of $\mathbb{E}[X \mid \mathcal{G}]$.

Proposition 1.20 *Given $X \in L^2(\Omega, \mathcal{F})$, $Z := I\!E[X \mid \mathcal{G}]$ is the unique random variable Z in $L^2(\Omega, \mathcal{G})$ that satisfies the relation*

$$I\!E[YX] = I\!E[YZ] \tag{1.8.4}$$

for all bounded and $\mathcal{G}$-measurable random variables Y.

We note that taking $Y = \mathbf{1}$ in (1.8.4) yields

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[X]. \tag{1.8.5}$$

In particular, when $\mathcal{G} = \{\emptyset, \Omega\}$ we have $\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X \mid \{\emptyset, \Omega\}]$, and

$$\mathbb{E}[X \mid \{\emptyset, \Omega\}] = \mathbb{E}[\mathbb{E}[X \mid \{\emptyset, \Omega\}]] = \mathbb{E}[X], \tag{1.8.6}$$

because $\mathbb{E}[X \mid \{\emptyset, \Omega\}]$ is in $L^2(\Omega, \{\emptyset, \Omega\})$ and is *a.s.* constant. In addition, the conditional expectation operator has the following properties.

(i) $\mathbb{E}[XY \mid \mathcal{G}] = Y\,\mathbb{E}[X \mid \mathcal{G}]$ if Y depends only on the information contained in $\mathcal{G}$.
Proof. By the characterization (1.8.4) it suffices to show that

$$\mathbb{E}[H(XY)] = \mathbb{E}[H(Y\,\mathbb{E}[X|\mathcal{G}])], \tag{1.8.7}$$

for all bounded and $\mathcal{G}$-measurable random variables H, which implies $\mathbb{E}[XY \mid \mathcal{G}] = Y\,\mathbb{E}[X \mid \mathcal{G}]$.

Relation (1.8.7) holds from (1.8.4) because the product HY is $\mathcal{G}$-measurable hence Y in (1.8.4) can be replaced with HY.

(ii) $\mathbb{E}[Y|\mathcal{G}] = Y$ when Y depends only on the information contained in $\mathcal{G}$.

Proof. This is a consequence of point (i) above by taking $X := \mathbf{1}$.

(iii) $\mathbb{E}[\mathbb{E}[X|\mathcal{G}] \mid \mathcal{H}] = \mathbb{E}[X|\mathcal{H}]$ if $\mathcal{H} \subset \mathcal{G}$, called the *tower property*.
Proof. First, we note that by (1.8.5), (iii) holds when $\mathcal{H} = \{\emptyset, \Omega\}$. Next, by the characterization (1.8.4) it suffices to show that

$$\mathbb{E}[H\,\mathbb{E}[X|\mathcal{G}]] = \mathbb{E}[H\,\mathbb{E}[X|\mathcal{H}]], \tag{1.8.8}$$

for all bounded and $\mathcal{H}$-measurable random variables H, which will imply (iii) from (1.8.4).

In order to prove (1.8.8) we check that by point (i) above and (1.8.5) we have

$$\begin{aligned}\mathbb{E}[H\,\mathbb{E}[X|\mathcal{G}]] &= \mathbb{E}[\mathbb{E}[HX|\mathcal{G}]] = \mathbb{E}[HX] \\ &= \mathbb{E}[\mathbb{E}[HX|\mathcal{H}]] = \mathbb{E}[H\,\mathbb{E}[X|\mathcal{H}]],\end{aligned}$$

and we conclude by the characterization (1.8.4).

(iv) $\mathbb{E}[X|\mathcal{G}] = \mathbb{E}[X]$ when X "does not depend" on the information contained in $\mathcal{G}$ or, more precisely stated, when the random variable X is *independent* of the σ-algebra $\mathcal{G}$.

Proof. It suffices to note that for all bounded $\mathcal{G}$-measurable Y we have

$$\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y] = \mathbb{E}[Y\,\mathbb{E}[X]],$$

and we conclude again by (1.8.4).

(v) If Y depends only on $\mathcal{G}$ and X is independent of $\mathcal{G}$, then

$$\mathbb{E}[h(X,Y)|\mathcal{G}] = \mathbb{E}[h(X,x)]_{x=Y}. \tag{1.8.9}$$

Proof. This relation can be proved using the tower property, by noting that for any bounded $Z \in L^2(\Omega,\mathcal{G})$ we have

$$\begin{aligned}\mathbb{E}[Z\,\mathbb{E}[h(x,X)]_{x=Y}] &= \mathbb{E}[Z\,\mathbb{E}[h(x,X) \mid \mathcal{G}]_{x=Y}] \\ &= \mathbb{E}[Z\,\mathbb{E}[h(Y,X) \mid \mathcal{G}]] \\ &= \mathbb{E}[\mathbb{E}[Zh(Y,X) \mid \mathcal{G}]] \\ &= \mathbb{E}[Zh(Y,X)],\end{aligned}$$

which yields (1.8.9) by the characterization (1.8.4).

The notion of conditional expectation can be extended from square-integrable random variables in $L^2(\Omega,\mathcal{F})$ to integrable random variables in $L^1(\Omega,\mathcal{F})$, see e.g. Theorem 5.1 in Kallenberg (2002).

Proposition 1.21 *When the σ-algebra $\mathcal{G} := \sigma(A_1, A_2, \ldots, A_n)$ is generated by n disjoint events $A_1, A_2, \ldots, A_n \in \mathcal{F}$, we have*

$$\mathbb{E}[X \mid \mathcal{G}] = \sum_{k=1}^{n} \mathbb{1}_{A_k}\,\mathbb{E}[X \mid A_k] = \sum_{k=1}^{n} \mathbb{1}_{A_k}\,\frac{\mathbb{E}[X\mathbb{1}_{A_k}]}{\mathbb{P}(A_k)}.$$

Proof It suffices to note that the $\mathcal{G}$-measurable random variables can be generated by indicator functions of the form $\mathbb{1}_{A_l}$, and that

$$\begin{aligned}\mathbb{E}\left[\mathbb{1}_{A_l}\sum_{k=1}^{n}\mathbb{1}_{A_k}\frac{\mathbb{E}[X\mathbb{1}_{A_k}]}{\mathbb{P}(A_k)}\right] &= \mathbb{E}\left[\mathbb{1}_{A_l}\frac{\mathbb{E}[X\mathbb{1}_{A_l}]}{\mathbb{P}(A_l)}\right] \\ &= \frac{\mathbb{E}[X\mathbb{1}_{A_l}]}{\mathbb{P}(A_l)}\,\mathbb{E}\left[\mathbb{1}_{A_l}\right] \\ &= \mathbb{E}\left[X\mathbb{1}_{A_l}\right], \qquad l = 1,2,\ldots,n,\end{aligned}$$

showing (1.8.4). The relation

$$\mathbb{E}[X \mid A_k] = \frac{\mathbb{E}[X\mathbb{1}_{A_k}]}{\mathbb{P}(A_k)}, \qquad k = 1,2,\ldots,n,$$

follows from Lemma 1.10. □

For example, in case $\Omega = \{a,b,c,d\}$ and $\mathcal{G} = \big\{\emptyset, \Omega, \{a,b\}, \{c\}, \{d\}\big\}$, we have

$$\begin{aligned}\mathbb{E}[X \mid \mathcal{G}] &= \mathbb{1}_{\{a,b\}}\,\mathbb{E}[X \mid \{a,b\}] + \mathbb{1}_{\{c\}}\,\mathbb{E}[X \mid \{c\}] + \mathbb{1}_{\{d\}}\,\mathbb{E}[X \mid \{d\}] \\ &= \mathbb{1}_{\{a,b\}}\frac{\mathbb{E}\left[X\mathbb{1}_{\{a,b\}}\right]}{\mathbb{P}(\{a,b\})} + \mathbb{1}_{\{c\}}\frac{\mathbb{E}\left[X\mathbb{1}_{\{c\}}\right]}{\mathbb{P}(\{c\})} + \mathbb{1}_{\{d\}}\frac{\mathbb{E}\left[X\mathbb{1}_{\{d\}}\right]}{\mathbb{P}(\{d\})}.\end{aligned}$$

Regarding conditional probabilities we have similarly, for $A \subset \Omega = \{a, b, c, d\}$,

$$\begin{aligned}\mathbb{P}(A \mid \mathcal{G}) &= \mathbb{1}_{\{a,b\}}\frac{\mathbb{P}\big(A \cap \{a,b\}\big)}{\mathbb{P}(\{a,b\})} + \mathbb{1}_{\{c\}}\frac{\mathbb{P}\big(A \cap \{c\}\big)}{\mathbb{P}(\{c\})} + \mathbb{1}_{\{d\}}\frac{\mathbb{P}\big(A \cap \{d\}\big)}{\mathbb{P}(\{d\})} \\ &= \mathbb{1}_{\{a,b\}}\mathbb{P}(A \mid \{a,b\}) + \mathbb{1}_{\{c\}}\mathbb{P}(A \mid \{c\}) + \mathbb{1}_{\{d\}}\mathbb{P}(A \mid \{d\}).\end{aligned}$$

In particular, if $A = \{a\} \subset \Omega = \{a, b, c, d\}$ we find

$$\begin{aligned}\mathbb{P}(\{a\} \mid \mathcal{G}) &= \mathbb{1}_{\{a,b\}}\mathbb{P}(\{a\} \mid \{a,b\}) \\ &= \mathbb{1}_{\{a,b\}}\frac{\mathbb{P}\big(\{a\} \cap \{a,b\}\big)}{\mathbb{P}(\{a,b\})} \\ &= \mathbb{1}_{\{a,b\}}\frac{\mathbb{P}(\{a\})}{\mathbb{P}(\{a,b\})}.\end{aligned}$$

In other words, the probability of getting the outcome a is $\mathbb{P}(\{a\})/\mathbb{P}(\{a,b\})$ knowing that the outcome is either a or b, otherwise it is zero.

Exercises

Exercise 1.1 Prove the equivalence between the two versions (1.2.2)–(1.2.3) of the inclusion-exclusion principle.

Exercise 1.2 Consider the random variable $X : \Omega \longrightarrow \mathbb{N} \cup \{\infty\}$ with distribution

$$\mathbb{P}(X = k) = qp^k, \qquad k \in \mathbb{N} = \{0, 1, 2, \ldots\},$$

where $q \in [0, 1-p]$ and $0 \leq p < 1$.

(a) Compute $\mathbb{P}(X < \infty)$ and $\mathbb{P}(X = \infty)$ by considering two cases, and give the value of $\mathbb{E}[X]$ when $0 \leq q < 1 - p$.
(b) Assume that $q = 1 - p$ and consider the random variable $Y := r^X$ for some $r > 0$. Explain why $\mathbb{P}(Y < \infty) = 1$ and compute $\mathbb{E}[Y]$ by considering two cases depending on the value of $r > 0$.

Exercise 1.3 Let $N \in \{1, 2, 3, 4, 5, 6\}$ denote the integer random variable obtained by tossing a six faced die and by noting the number on the upper side of the die. Given the value of N, an *independent*, unbiased coin is thrown N times. We denote by Z the total number of heads that appear in the process of throwing the coin N times.

(a) Using conditioning on the value of $N \in \{1, 2, 3, 4, 5, 6\}$, compute the mean and the variance of the random variable Z.
(b) Determine the probability distribution of Z.
(c) Recover the result of Question a from the data of the probability distribution computed in Question b.

Exercise 1.4 Thinning of Poisson random variables. Given a random sample N of a Poisson random variable with parameter λ, we perform a number N of *independent* $\{0, 1\}$-valued Bernoulli experiments *independent* of N, each of them with parameter $p \in (0, 1)$. We let Z denote the total number of $+1$ outcomes occurring in the N Bernoulli trials.

(a) Express Z as a random sum, and use this expression to compute the mean and variance of Z.
(b) Compute the probability distribution of Z.
(c) Recover the result of Question a from the data of the probability distribution computed in Question b.

Exercise 1.5 Given X and Y two independent exponentially distributed random variables with parameters λ and μ, show the relation

$$\mathbb{E}[\min(X, Y) \mid X < Y] = \frac{1}{\lambda + \mu} = \mathbb{E}[\min(X, Y)]. \tag{1.8.10}$$

Exercise 1.6 Given a random sample L of a gamma random variable with probability density function

$$\varphi_L(x) = \mathbb{1}_{[0,\infty)} x \mathrm{e}^{-x},$$

consider U a uniform random variable taking values in the interval $[0, L]$ and let $V = L - U$.

Compute the joint probability density function of the couple (U, V) of random variables.

Exercise 1.7 Let X and Y denote two independent Poisson random variables with parameters λ and μ.

(a) Show that the random variable $X + Y$ has the Poisson distribution with parameter $\lambda + \mu$.
(b) Compute the conditional distribution $\mathbb{P}(X = k \mid X + Y = n)$ given that $X + Y = n$, for all $k, n \in \mathbb{N}$.
(c) Assume that respective parameters of the distributions of X and Y are random, independent, and chosen according to an exponential distribution with parameter $\theta > 0$.
Give the probability distributions of X and Y, and compute the conditional distribution $\mathbb{P}(X = k \mid X + Y = n)$ given that $X + Y = n$, for all $k, n \in \mathbb{N}$.

(d) Assume now that X and Y have same random parameter represented by a single exponentially distributed random variable Λ with parameter $\theta > 0$, independent of X and Y.
Compute the conditional distribution $\mathbb{P}(X = k \mid X + Y = n)$ given that $X + Y = n$, for all $k, n \in \mathbb{N}$.

Exercise 1.8 A red pen and a green pen are put in a hat. A pen is chosen at random in the hat, and replaced inside after it color has been noted.

– In case the pen is of red color, then a supplementary red pen is placed in the hat.
– On the other hand if the pen color is green, then another green pen is added.

After this first part of the experiment is completed, a second pen is chosen at random.

Determine the probability that the first drawn pen was red, given that the color of the second pen chosen was red.

Exercise 1.9 A machine relies on the functioning of three parts, each of which having a probability $1 - p$ of being under failure, and a probability p of functioning correctly. All three parts are functioning independently of the others, and the machine is working if and only if two at least of the parts are operating.

(a) Compute the probability that the machine is functioning.
(b) Suppose that the machine itself is set in a random environment in which the value of the probability p becomes random. Precisely we assume that p is a uniform random variable taking real values between 0 and 1, independently of the state of the system. Compute the probability that the machine operates in this random environment.

Chapter 2
Gambling Problems



This chapter consists in a detailed study of a fundamental example of random walk that can only evolve by going up of down by one unit within the finite state space $\mathbb{S} = \{0, 1, \ldots, S\}$. This allows us in particular to have a first look at the first step analysis technique that will be repeatedly used in the general framework of Markov chains, particularly in Chap. 5.

2.1 Constrained Random Walk

To begin, let us repeat that this chapter on "gambling problems" is not primarily designed to help a reader dealing with problem gambling, although some comments on this topic are made of the end of Sect. 2.3.

We consider an amount $\$S$ of S dollars which is to be shared between two players A and B. At each round, Player A may earn \$1 with probability $p \in (0, 1)$, and in this case Player B loses \$1. Conversely, Player A may lose \$1 with probability $q := 1 - p$, in which case Player B gains \$1, and the successive rounds are independent.

We let X_n represent the wealth of Player A at time $n \in \mathbb{N}$, while $S - X_n$ represents the wealth of Player B at time $n \in \mathbb{N}$.

The initial wealth X_0 of Player A could be negative, but for simplicity we will assume that it is comprised between 0 and S. Assuming that the value of X_n, $n \geq 0$, belongs to $\{1, 2, \ldots, S-1\}$ at the time step n, at the next step $n+1$ we will have

$$X_{n+1} = \begin{cases} X_n + 1 & \text{if Player A wins round } n+1, \\ X_n - 1 & \text{if Player B wins round } n+1. \end{cases}$$

Moreover, as soon as X_n hits one of the boundary points $\{0, S\}$, the process remains frozen at that state over time, i.e.

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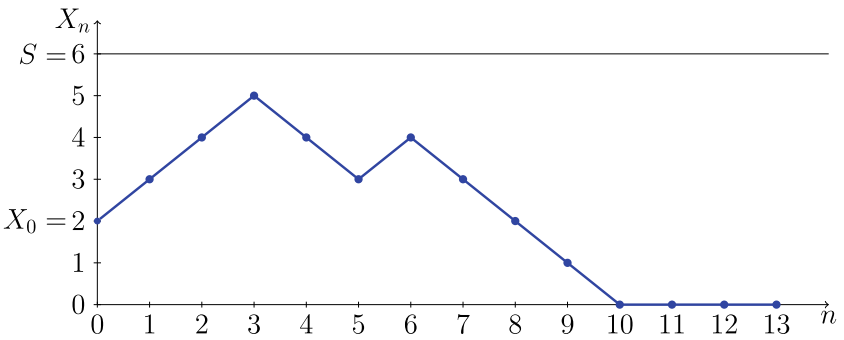


Fig. 2.1 Sample path of a gambling process $(X_n)_{n\geq 0}$

$$(X_n = 0) \Longrightarrow (X_{n+1} = 0) \quad \text{and} \quad (X_n = S) \Longrightarrow (X_{n+1} = S),$$

or

$$\mathbb{P}(X_{n+1} = 0 \mid X_n = 0) = 1 \quad \text{and} \quad \mathbb{P}(X_{n+1} = S \mid X_n = S) = 1, \quad n \geq 0.$$

In other words, the game ends whenever the wealth of any of the two players reaches \$0, in which case the other player's account contains \$$S$, see Fig. 2.1. Among the main issues of interest are:

- the probability that Player A (or B) gets eventually ruined,
- the mean duration of the game.

We will also be interested in the probability distribution of the random game duration T, i.e. in the knowledge of $\mathbb{P}(T = n)$, $n \geq 0$.

According to the above problem description, for all $n \in \mathbb{N}$ we have

$$\mathbb{P}(X_{n+1} = k+1 \mid X_n = k) = p \quad \text{and} \quad \mathbb{P}(X_{n+1} = k-1 \mid X_n = k) = q,$$

$k = 1, 2, \ldots, S-1$, and in this case the chain is said to be *time homogeneous* since the transition probabilities do not depend on the time index n.

Since we do not focus on the behavior of the chain after it hits states 0 or S, the probability distribution of X_{n+1} given $\{X_n = 0\}$ or $\{X_n = S\}$ can be left unspecified.

The probability sample space Ω corresponding to this experiment could be taken as the (uncountable) set

$$\Omega := \{-1, +1\}^{\mathbb{N}} = \Big\{\omega = (\omega_0, \omega_1, \ldots) \ : \ \omega_i = \pm 1, \ n \in \mathbb{N}\Big\},$$

with any element $\omega \in \Omega$ represented by a countable sequence of $+1$ or -1, depending whether the process goes up or down at each time step. However, in what follows we will not focus on this particular expression of Ω.

2.2 Ruin Probabilities

We are interested in the event

$$R_A = \text{“Player } A \text{ loses all his capital at some time”} = \bigcup_{n\in\mathbb{N}} \{X_n = 0\}, \tag{2.2.1}$$

and in computing the conditional probability

$$f_S(k) := \mathbb{P}(R_A \mid X_0 = k), \qquad k = 0, 1, \ldots, S. \tag{2.2.2}$$

Pathwise Analysis

First, let us note that the problem is easy to solve in the cases $S = 1$, $S = 2$ and $S = 3$.

(i) $\underline{S = 1}$.

In this case, the boundary $\{0, 1\}$ is reached from time 0 and we find

$$\begin{cases} f_1(0) = \mathbb{P}(R_A \mid X_0 = 0) = 1, \\ f_1(1) = \mathbb{P}(R_A \mid X_0 = 1) = 0. \end{cases} \tag{2.2.3}$$

(ii) $\underline{S = 2}$.

In this case, we find

$$\begin{cases} f_2(0) = \mathbb{P}(R_A \mid X_0 = 0) = 1, \\ f_2(1) = \mathbb{P}(R_A \mid X_0 = 1) = q, \\ f_2(2) = \mathbb{P}(R_A \mid X_0 = 2) = 0. \end{cases} \tag{2.2.4}$$

(iii) $\underline{S = 3}$.

The value of $f_3(1) = \mathbb{P}(R_A \mid X_0 = 1)$ is computed by noting that starting from state $\textcircled{1}$, one can reach state $\textcircled{0}$ only by an odd number $2n + 1$ of steps, $n \in \mathbb{N}$, and that every such path decomposes into $n + 1$ independent downwards steps, each of them having probability q, and n upwards steps, each of them with probability p. By summation over n using the geometric series identity (A.3), this yields

$$\begin{cases} f_3(0) = \mathbb{P}(R_A \mid X_0 = 0) = 1, \\ f_3(1) = \mathbb{P}(R_A \mid X_0 = 1) = q \sum_{n\geq 0} (pq)^n = \dfrac{q}{1-pq}, \\ f_3(2) = \mathbb{P}(R_A \mid X_0 = 2) = q^2 \sum_{n\geq 0} (pq)^n = \dfrac{q^2}{1-pq}, \\ f_3(3) = \mathbb{P}(R_A \mid X_0 = 3) = 0. \end{cases} \tag{2.2.5}$$

The value of $f_3(2)$ is computed similarly by considering $n+2$ independent downwards steps, each of them with probability q, and n upwards steps, each of them with probability p. Clearly, things become quite complicated for $S \geq 4$, and increasingly difficult as S gets larger.

First Step Analysis

The general case will be solved by the method of *first step analysis*, which will be repeatedly applied to other Markov processes in Chaps. 3 and 5 and elsewhere.

Lemma 2.1 *For all $k = 1, 2, \ldots, S-1$ we have*

$$\mathbb{P}(R_A \mid X_0 = k) = p\mathbb{P}(R_A \mid X_0 = k+1) + q\mathbb{P}(R_A \mid X_0 = k-1).$$

Proof The idea is to apply conditioning given the first transition from X_0 to X_1. For all $k = 1, 2, \ldots, S-1$, by (1.2.1) we have

$$\begin{aligned} &\mathbb{P}(R_A \mid X_0 = k) \\ &= \mathbb{P}(R_A \text{ and } X_1 = k+1 \mid X_0 = k) + \mathbb{P}(R_A \text{ and } X_1 = k-1 \mid X_0 = k) \\ &= \frac{\mathbb{P}(R_A \text{ and } X_1 = k+1 \text{ and } X_0 = k)}{\mathbb{P}(X_0 = k)} + \frac{\mathbb{P}(R_A \text{ and } X_1 = k-1 \text{ and } X_0 = k)}{\mathbb{P}(X_0 = k)} \\ &= \frac{\mathbb{P}(R_A \text{ and } X_1 = k+1 \text{ and } X_0 = k)}{\mathbb{P}(X_1 = k+1 \text{ and } X_0 = k)} \times \frac{\mathbb{P}(X_1 = k+1 \text{ and } X_0 = k)}{\mathbb{P}(X_0 = k)} \\ &\quad + \frac{\mathbb{P}(R_A \text{ and } X_1 = k-1 \text{ and } X_0 = k)}{\mathbb{P}(X_1 = k-1 \text{ and } X_0 = k)} \times \frac{\mathbb{P}(X_1 = k-1 \text{ and } X_0 = k)}{\mathbb{P}(X_0 = k)} \\ &= \mathbb{P}(R_A \mid X_1 = k+1 \text{ and } X_0 = k)\mathbb{P}(X_1 = k+1 \mid X_0 = k) \\ &\quad + \mathbb{P}(R_A \mid X_1 = k-1 \text{ and } X_0 = k)\mathbb{P}(X_1 = k-1 \mid X_0 = k) \\ &= p\mathbb{P}(R_A \mid X_1 = k+1 \text{ and } X_0 = k) + q\mathbb{P}(R_A \mid X_1 = k-1 \text{ and } X_0 = k) \\ &= p\mathbb{P}(R_A \mid X_0 = k+1) + q\mathbb{P}(R_A \mid X_0 = k-1), \end{aligned}$$

where we used Lemma 2.2 below on the last step. □

In the case $S = 3$, Lemma 2.1 shows that

$$\begin{cases} f_3(0) = \mathbb{P}(R_A \mid X_0 = 0) = 1, \\ f_3(1) = pf_3(2) + qf_3(0) = pf_3(2) + q = pqf_3(1) + q, \\ f_3(2) = pf_3(3) + qf_3(1) = qf_3(1) = pqf_3(2) + q^2, \\ f_3(3) = \mathbb{P}(R_A \mid X_0 = 3) = 0, \end{cases}$$

whose solution can be checked to be given by (2.2.5).

More generally, Lemma 2.1 shows that the function

$$f_S : \{0, 1, \ldots, S\} \longrightarrow [0, 1]$$

defined by (2.2.2) satisfies the linear equation[1]

$$f_S(k) = pf_S(k+1) + qf_S(k-1), \qquad k = 1, 2, \ldots, S-1, \quad (2.2.6)$$

subject to the *boundary conditions*

$$f_S(0) = \mathbb{P}(R_A \mid X_0 = 0) = 1, \quad (2.2.7)$$

and

$$f_S(S) = \mathbb{P}(R_A \mid X_0 = S) = 0, \quad (2.2.8)$$

for $k \in \{0, S\}$. It can be easily checked that the expressions (2.2.3), (2.2.4) and (2.2.5) do satisfy the above Eq. (2.2.6) and the boundary conditions (2.2.7) and (2.2.8).

Note that Lemma 2.1 is frequently stated without proof. The last step of the proof stated above relies on the following lemma, which shows that the data of X_1 entirely determines the probability of the ruin event R_A. In other words, the probability of ruin depends only on the initial amount k owned by the gambler when he enters the casino. Whether he enters the casino at time 1 with $X_1 = k \pm 1$ or at time 0 with $X_0 = k \pm 1$ makes no difference on the ruin probability.

[1] Due to the relation $(f + g)(k) = f(k) + g(k)$ we can check that if f and g are two solutions of (2.2.6) then $f + g$ is also a solution of (2.2.6), hence the equation is *linear*.

Lemma 2.2 *For all $k = 1, 2, \ldots, S-1$ we have*

$$\mathbb{P}(R_A \mid X_1 = k \pm 1 \text{ and } X_0 = k) = \mathbb{P}(R_A \mid X_1 = k \pm 1) = \mathbb{P}(R_A \mid X_0 = k \pm 1).$$

In other words, the ruin probability depends on the data of the starting point and not on the starting time.

Proof This relation can be shown in various ways:

1. Descriptive proof (*preferred*): we note that given $X_1 = k+1$, the transition from X_0 to X_1 has no influence on the future of the process after time 1, and the probability of ruin starting at time 1 is the same as if the process is started at time 0.
2. Algebraic proof. First, for $1 < k \leq S-1$ and $k \pm 1 \geq 1$, letting $\widetilde{X}_0 := X_1 - Z$ where $Z \simeq X_1 - X_0$ has same distribution as $X_1 - X_0$ and is independent of X_1, by (2.2.1) we have

$$\begin{aligned}
&\mathbb{P}(R_A \mid X_1 = k \pm 1 \text{ and } X_0 = k) \\
&= \mathbb{P}\Big(\bigcup_{n\geq 0} \{X_n = 0\} \,\Big|\, X_1 = k \pm 1,\ X_0 = k \Big) \\
&= \frac{\mathbb{P}\Big(\Big(\bigcup_{n\geq 0} \{X_n = 0\} \Big) \cap \{X_1 = k \pm 1\} \cap \{X_0 = k\} \Big)}{\mathbb{P}\left(\{X_1 = k \pm 1\} \cap \{X_0 = k\} \right)} \\
&= \frac{\mathbb{P}\Big(\bigcup_{n\geq 0} \left(\{X_n = 0\} \cap \{X_1 = k \pm 1\} \cap \{X_0 = k\} \right) \Big)}{\mathbb{P}\left(\{X_1 = k \pm 1\} \cap \{X_0 = k\} \right)} \\
&= \frac{\mathbb{P}\Big(\bigcup_{n\geq 2} \left(\{X_n = 0\} \cap \{X_1 = k \pm 1\} \cap \{X_0 = k\} \right) \Big)}{\mathbb{P}\Big(\{X_1 = k \pm 1\} \cap \{X_0 = k\} \Big)} \\
&= \frac{\mathbb{P}\Big(\Big(\bigcup_{n\geq 2} \{X_n = 0\} \Big) \cap \{X_1 = k \pm 1\} \cap \{X_0 = k\} \Big)}{\mathbb{P}\left(\{X_1 = k \pm 1\} \cap \{X_0 = k\} \right)} \qquad (2.2.9) \\
&= \frac{\mathbb{P}\Big(\Big(\bigcup_{n\geq 2} \{X_n = 0\} \Big) \cap \{X_1 = k \pm 1\} \cap \{X_1 - \widetilde{X}_0 = \pm 1\} \Big)}{\mathbb{P}\big(\{X_1 = k \pm 1\} \cap \{X_1 - \widetilde{X}_0 = \pm 1\} \big)} \qquad (2.2.10) \\
&= \frac{\mathbb{P}\Big(\Big(\bigcup_{n\geq 2} \{X_n = 0\} \Big) \cap \{X_1 = k \pm 1\} \Big) \mathbb{P}\big(\{X_1 - \widetilde{X}_0 = \pm 1\} \big)}{\mathbb{P}(\{X_1 = k \pm 1\}) \mathbb{P}(\{X_1 - \widetilde{X}_0 = \pm 1\})} \\
&= \mathbb{P}\Big(\bigcup_{n\geq 2} \{X_n = 0\} \,\Big|\, \{X_1 = k \pm 1\} \Big) \\
&= \mathbb{P}\Big(\bigcup_{n\geq 1} \{X_n = 0\} \,\Big|\, \{X_0 = k \pm 1\} \Big) \\
&= \mathbb{P}(R_A \mid X_0 = k \pm 1). \qquad (2.2.11)
\end{aligned}$$

Note that when switching from (2.2.9) to (2.2.10) and then to (2.2.11) using the relation $\widetilde{X}_0 := X_1 - Z$ we regard the process increment starting from X_1 as run backward in time. In case $k = 1$ we easily find that

$$\mathbb{P}(R_A \mid X_1 = 0 \text{ and } X_0 = 1) = 1 = \mathbb{P}(R_A \mid X_0 = 0),$$

since $\{X_1 = 0\} \subset R_A = \bigcup_{n\in\mathbb{N}} \{X_n = 0\}$.

□

In the remaining of this section we will prove that in the non-symmetric case $p \neq q$ the solution of (2.2.6) is given by

$$f_S(k) = \mathbb{P}(R_A \mid X_0 = k) = \frac{(q/p)^k - (q/p)^S}{1 - (q/p)^S} = \frac{1 - (p/q)^{S-k}}{1 - (p/q)^S}, \tag{2.2.12}$$

$k = 0, 1, \ldots, S$, and that in the symmetric case $p = q = 1/2$ the solution of (2.2.6) is given by

$$f_S(k) = \mathbb{P}(R_A \mid X_0 = k) = \frac{S - k}{S} = 1 - \frac{k}{S}, \tag{2.2.13}$$

$k = 0, 1, \ldots, S$, cf. also Exercise 2.3 for a different derivation.

Remark that (2.2.12) and (2.2.13) do satisfy both boundary conditions (2.2.7) and (2.2.8). When the number S of states becomes large we find that, for all $k \geq 0$,

$$f_\infty(k) := \lim_{S\to\infty} \mathbb{P}(R_A \mid X_0 = k) = \begin{cases} 1 & \text{if } q \geq p, \\ \left(\dfrac{q}{p}\right)^k & \text{if } p > q, \end{cases} \tag{2.2.14}$$

which represents the probability of hitting the origin starting from state $\textcircled{k}$, cf. (2.3.14) and (3.4.16) below, and also Exercise 3.2-(c) for a different derivation of this statement in the framework of random walks.

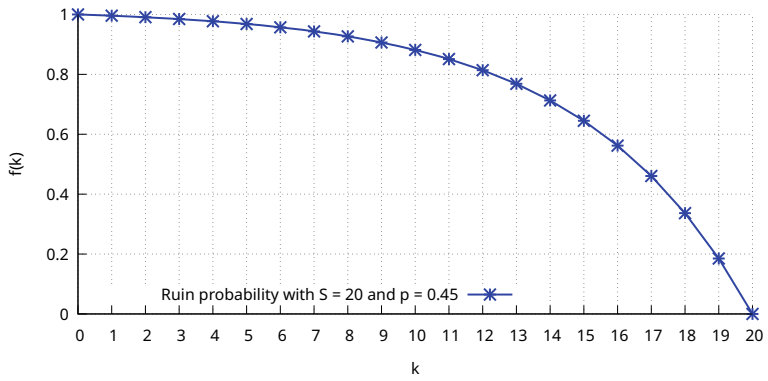


Fig. 2.2 Ruin probability $f_{20}(k)$ function of $X_0 = k \in [0, 20]$ for $S = 20$ and $p = 0.45$

Exercise: Check that (2.2.12) agrees with (2.2.4) and (2.2.5) when $S = 2$ and $S = 3$.

In the graph of Fig. 2.2 the ruin probability (2.2.12) is plotted as a function of k for $p = 0.45$ and $q = 0.55$.
We now turn to the solution of (2.2.6), for which we develop two different approaches (called here the "standard solution" and the "direct solution") that both recover (2.2.12) and (2.2.13).

Standard solution method

We decide to look for a solution of (2.2.6) of the form[2]

$$k \longmapsto f_S(k) = Ca^k, \tag{2.2.15}$$

where C and a are constants which will be determined from the boundary conditions and from the Eq. (2.2.6), respectively.

Substituting (2.2.15) into (2.2.6) when C is non-zero yields the *characteristic equation*

$$pa^2 - a + q = p(a-1)(a-q/p) = 0, \tag{2.2.16}$$

of degree 2 in the unknown a, and this equation admits in general two solutions a_1 and a_2 given by

$$\{a_1, a_2\} = \left(\frac{1+\sqrt{1-4pq}}{2p}, \frac{1-\sqrt{1-4pq}}{2p}\right) = \left(1, \frac{q}{p}\right) = (1, r),$$

[2] Where did we get this idea? From intuition, experience, or empirically by multiple trials and errors.

for all $p \in (0, 1]$, with

$$a_1 = 1 \quad \text{and} \quad a_2 = r = \frac{q}{p}.$$

Note that we have $a_1 = a_2 = 1$ in case $p = q = 1/2$.

Non-symmetric case : $p \neq q$ − **Proof of** (2.2.12)

In this case we have $p \neq q$, i.e.[3] $r \neq 1$, and

$$f_S(k) = C_1 a_1^k = C_1 \qquad \text{and} \qquad f_S(k) = C_2 r^k$$

are both solutions of (2.2.6). Since (2.2.6) is linear, the sum of two solutions remains a solution, hence the general solution of (2.2.6) is given by

$$f_S(k) = C_1 a_1^k + C_2 a_2^k = C_1 + C_2 r^k, \qquad k = 0, 1, \ldots, S, \tag{2.2.17}$$

where $r = q/p$ and C_1, C_2 are two constants to be determined from the boundary conditions. From (2.2.7), (2.2.8) and (2.2.17) we have

$$\begin{cases} f_S(0) = 1 = C_1 + C_2, \\ f_S(S) = 0 = C_1 + C_2 r^S, \end{cases} \tag{2.2.18}$$

and solving the system (2.2.18) of two equations we find

$$C_1 = -\frac{r^S}{1 - r^S} \quad \text{and} \quad C_2 = \frac{1}{1 - r^S},$$

which yields (2.2.12) as by (2.2.17) we have

$$f_S(k) = C_1 + C_2 r^k = \frac{r^k - r^S}{1 - r^S} = \frac{(q/p)^k - (q/p)^S}{1 - (q/p)^S}, \quad k = 0, 1, \ldots, S.$$

Symmetric case : $p = q = 1/2$ − **Proof of** (2.2.13)

In this case, Equation (2.2.6) rewrites as

$$f(k) = \frac{1}{2} f(k + 1) + \frac{1}{2} f(k - 1), \qquad k = 1, 2, \ldots, S - 1, \tag{2.2.19}$$

and we have $r = 1$ (fair game) and (2.2.16) reads

$$a^2 - 2a + 1 = (a - 1)^2 = 0,$$

[3] From the Latin "id est" meaning "that is".

which has the unique solution $a = 1$, since the constant function $f(k) = C$ is solution of (2.2.6).
However, this is not enough and we need to combine $f(k) = C_1$ with a second solution. Noting that $g(k) = C_2 k$ is also solution of (2.2.6), the general solution is found to have the form

$$f_S(k) = f(k) + g(k) = C_1 + C_2 k. \tag{2.2.20}$$

From (2.2.7), (2.2.8) and (2.2.20) we have

$$\begin{cases} f_S(0) = 1 = C_1, \\ f_S(S) = 0 = C_1 + C_2 S, \end{cases} \tag{2.2.21}$$

and solving the system (2.2.21) of two equations we find

$$C_1 = 1 \text{ and } C_2 = -1/S,$$

which yields the quite intuitive solution

$$f_S(k) = \mathbb{P}(R_A \mid X_0 = k) = \frac{S-k}{S} = 1 - \frac{k}{S}, \quad k = 0, 1, \ldots, S. \tag{2.2.22}$$

Direct solution method

Noting that $p + q = 1$ and due to its special form, we can rewrite (2.2.6) as

$$(p+q) f_S(k) = p f_S(k+1) + q f_S(k-1),$$

$k = 1, 2, \ldots, S-1$, i.e. as the *difference equation*

$$p(f_S(k+1) - f_S(k)) - q(f_S(k) - f_S(k-1)) = 0, \tag{2.2.23}$$

$k = 1, 2, \ldots, S-1$, which rewrites as

$$f_S(k+1) - f_S(k) = \frac{q}{p}(f_S(k) - f_S(k-1)), \qquad k = 1, 2, \ldots, S-1,$$

hence for $k = 1$ we have

$$f_S(2) - f_S(1) = \frac{q}{p}(f_S(1) - f_S(0)),$$

and for $k = 2$ we find

$$f_S(3) - f_S(2) = \frac{q}{p}(f_S(2) - f_S(1)) = \left(\frac{q}{p}\right)^2 (f_S(1) - f_S(0)).$$

Following by induction on $k \geq 2$, we can show that

$$f_S(k+1) - f_S(k) = \left(\frac{q}{p}\right)^k (f_S(1) - f_S(0)), \tag{2.2.24}$$

$k = 0, 1, \ldots, S-1$. Next, by the telescoping sum

$$f_S(n) = f_S(0) + \sum_{k=0}^{n-1} (f_S(k+1) - f_S(k)),$$

Relation (2.2.24) implies

$$f_S(n) = f_S(0) + (f_S(1) - f_S(0)) \sum_{k=0}^{n-1} \left(\frac{q}{p}\right)^k, \tag{2.2.25}$$

$n = 1, 2, \ldots, S-1$. The remaining question is how to find $f_S(1) - f_S(0)$, knowing that $f_S(0) = 1$ by (2.2.7).

Non-symmetric case : $p \neq q$

In this case, we have $r = q/p \neq 1$ and we get

$$\begin{aligned} f_S(n) &= f_S(0) + (f_S(1) - f_S(0)) \sum_{k=0}^{n-1} r^k \\ &= f_S(0) + \frac{1-r^n}{1-r} (f_S(1) - f_S(0)), \end{aligned} \tag{2.2.26}$$

$n = 1, 2, \ldots, S-1$, where we used (A.2).

Conditions (2.2.7) and (2.2.8) show that

$$0 = f_S(S) = 1 + \frac{1-r^S}{1-r} (f_S(1) - f_S(0)),$$

hence

$$f_S(1) - f_S(0) = -\frac{1-r}{1-r^S},$$

and combining this relation with (2.2.26) yields

$$f_S(n) = f_S(0) - \frac{1-r^n}{1-r^S} = 1 - \frac{1-r^n}{1-r^S} = \frac{r^n - r^S}{1-r^S},$$

$n = 0, 1, \ldots, S$, which recovers (2.2.12).

Symmetric case : $p = q = 1/2$

In this case, we have $r = 1$ and in order to solve (2.2.19) we note that (2.2.23) simply becomes

$$f_S(k+1) - f_S(k) = f_S(1) - f_S(0), \qquad k = 0, 1, \ldots, S-1,$$

and (2.2.25) reads

$$f_S(n) = f_S(0) + n(f_S(1) - f_S(0)), \qquad n = 1, 2, \ldots, S-1,$$

which has the form (2.2.20). Then the conditions $f_S(0) = 1$ and $f_S(S) = 0$, cf. (2.2.7) and (2.2.8), yield

$$0 = f_S(S) = 1 + S(f_S(1) - f_S(0)), \quad \text{hence} \quad f_S(1) - f_S(0) = -\frac{1}{S},$$

and

$$f_S(n) = f_S(0) - \frac{n}{S} = 1 - \frac{n}{S} = \frac{S-n}{S},$$

$n = 0, 1, \ldots, S$, which coincides with (2.2.22).

Remark 2.3 Note that when $p = q = 1/2$, (2.2.23) can be read as a discretization of a continuous Laplace equation as

$$\begin{aligned} \frac{\partial^2 f}{\partial x^2}(x) &\simeq \frac{\partial f}{\partial x}(x+1/2) - \frac{\partial f}{\partial x}(x-1/2) \\ &\simeq (f(x+1) - f(x) - (f(x) - f(x-1))) \\ &= 0, \qquad x \in \mathbb{R}, \end{aligned} \tag{2.2.27}$$

which admits a solution of the form

$$f(x) = f(0) + xf'(0) = f(0) + x(f(1) - f(0)), \qquad x \in \mathbb{R},$$

showing the intuition behind the linear form of (2.2.20).

In order to compute the probability of ruin of Player B given that $X_0 = k$ we only need to swap k to $S - k$ and to exchange p and q in (2.2.12). In other words, when $X_0 = k$ then Player B starts with an initial amount $S - k$ and a probability q of winning each round, which by (2.2.12) yields

$$\mathbb{P}(R_B \mid X_0 = k) = \frac{(p/q)^{S-k} - (p/q)^S}{1 - (p/q)^S} = \frac{1 - (q/p)^k}{1 - (q/p)^S} \quad \text{if } p \neq q, \quad (2.2.28)$$

where

$$R_B := \text{“Player } B \text{ loses all his capital at some time”} = \bigcup_{n \in \mathbb{N}} \{X_n = S\}.$$

In the symmetric case $p = q = 1/2$ we similarly find

$$\mathbb{P}(R_B \mid X_0 = k) = \frac{k}{S}, \quad k = 0, 1, \ldots, S, \quad \text{if } p = q = \frac{1}{2}, \quad (2.2.29)$$

see also Exercise 2.3 below.

Note that (2.2.28) and (2.2.29) satisfy the expected boundary conditions

$$\mathbb{P}(R_B \mid X_0 = 0) = 0 \quad \text{and} \quad \mathbb{P}(R_B \mid X_0 = S) = 1,$$

since X_0 represents the wealth of Player A at time 0.

By (2.2.12) and (2.2.28) we can check that

$$\begin{aligned} \mathbb{P}(R_A \cup R_B \mid X_0 = k) &= \mathbb{P}(R_A \mid X_0 = k) + \mathbb{P}(R_B \mid X_0 = k) \\ &= \frac{(q/p)^k - (q/p)^S}{1 - (q/p)^S} + \frac{1 - (q/p)^k}{1 - (q/p)^S} \\ &= 1, \qquad k = 0, 1, \ldots, S, \end{aligned} \quad (2.2.30)$$

which means that eventually one of the two players has to lose the game, see also Exercise 10.5. This means in particular that, with probability one, the game cannot continue endlessly.

In other words, we have

$$\mathbb{P}\big(R_A^c \cap R_B^c \mid X_0 = k\big) = 0, \qquad k = 0, 1, \ldots, S.$$

In the particular case $S = 3$ we can indeed check that, taking

$$A_n := \bigcap_{k=1}^{n} \big\{X_{2k-1} = 1 \text{ and } X_{2k} = 2\big\}, \qquad n \geq 1,$$

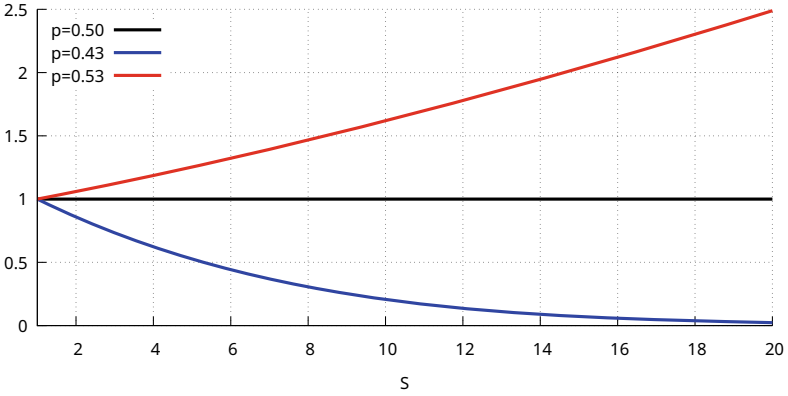


Fig. 2.3 Expected terminal wealth of Player A starting from state $k = 1$

the sequence $(A_n)_{n\geq 1}$ is non-increasing, hence by (1.2.6) we have

$$\begin{aligned}
&\mathbb{P}\left(\bigcap_{n\geq 1}\left\{X_{2n-1}=1 \text{ and } X_{2n}=2\right\} \,\middle|\, X_0=2\right)\\
&\quad=\mathbb{P}\left(\bigcap_{n\geq 1}\bigcap_{k=1}^{n}\left\{X_{2k-1}=1 \text{ and } X_{2k}=2\right\} \,\middle|\, X_0=2\right)\\
&\quad=\mathbb{P}\left(\bigcap_{n\geq 1} A_n \,\middle|\, X_0=2\right)\\
&\quad=\lim_{n\to\infty}\mathbb{P}(A_n \mid X_0=2)\\
&\quad=\lim_{n\to\infty}\mathbb{P}\left(\bigcap_{k=1}^{n}\left\{X_{2k-1}=1 \text{ and } X_{2k}=2\right\} \,\middle|\, X_0=2\right)\\
&\quad=q\lim_{n\to\infty}(pq)^n=0,
\end{aligned}$$

since we always have $0 \leq pq < 1$. However, this is *a priori* not completely obvious when $S \geq 4$.

The expected terminal wealth of Player A starting from state $\textcircled{k}$ is given by

$$0\times\mathbb{P}(R_A \mid X_0=k)+S\times\mathbb{P}(R_B \mid X_0=k)=\begin{cases} S\dfrac{1-(q/p)^k}{1-(q/p)^S} & \text{if } p\neq q,\\ k & \text{if } p=q=\dfrac{1}{2}. \end{cases}$$

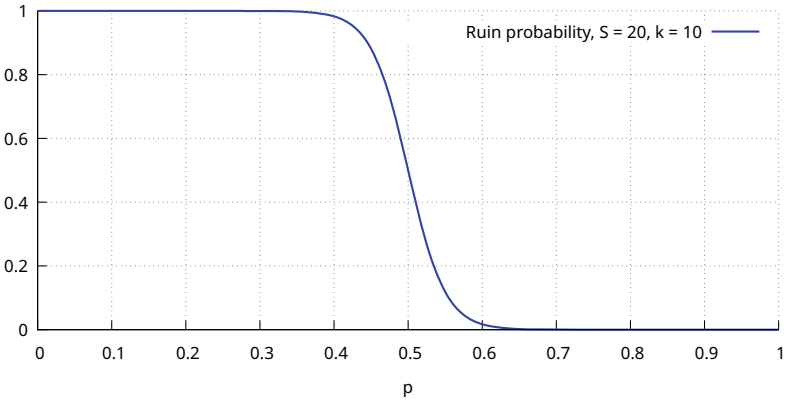


Fig. 2.4 Ruin probability as a function of $p \in [0, 1]$ for $S = 20$ and $k = 10$

When the number S of states becomes large, (2.2.28) also shows that for all $k \geq 0$ we have

$$\lim_{S\to\infty} \mathbb{P}(R_B \mid X_0 = k) = \begin{cases} 0 & \text{if } p \leq q, \\ 1 - \left(\dfrac{q}{p}\right)^k & \text{if } p > q, \end{cases}$$

which represents the complement of the probability (2.2.14) of hitting the origin starting from state $\textcircled{k}$, and is the probability that the process $(X_n)_{n\geq 0}$ "escapes to infinity" (Fig. 2.3).

In Fig. 2.4 below the ruin probability (2.2.12) is plotted as a function of p for $S = 20$ and $k = 10$.

Gambling machines in casinos are computer-controlled, and most countries permit by law a certain degree of "unfairness" (see the notions of "payout percentage" or "return to player") by taking $p < 1/2$ in order to allow the house to make an income.[4] Interestingly, we can note that taking *e.g.* $p = 0.45 < 1/2$ gives a ruin probability

$$\mathbb{P}(R_A \mid X_0 = 10) = 0.8815,$$

almost equal to 90%, which means that the slightly unfair probability $p = 0.45$ at the level of each round translates into a probability of only $0.1185 \simeq \%12$ of finally winning the game, i.e. a division by 4 from 0.45, although the average proportion of winning rounds is still 45%.

Consequently, a "slightly unfair" game at each round may become devastatingly unfair in the long run.[5] Most (but not all) gamblers are aware that gambling machines

[4] In this game, the payout is \$2 and the payout percentage is $2p$.

[5] "You can't lose with house odds and time. You can't win with gambler odds and time." Peter Kaufman *via* Dennis Hong.

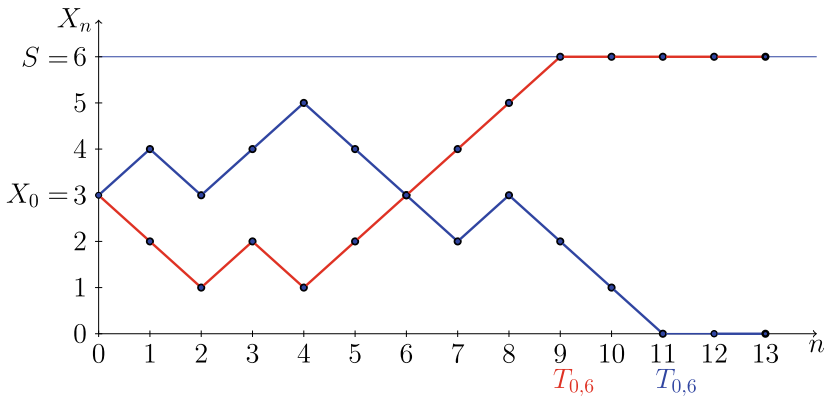


Fig. 2.5 Two sample paths of a gambling process $(X_n)_{n\geq 0}$

can be slightly unfair, however, most people would intuitively believe that a small degree of unfairness on each round should only translate into a reasonably low degree of unfairness in the long run.

2.3 Mean Game Duration

Let now

$$T_{0,S} = \inf\{n \geq 0 \ : \ X_n = 0 \text{ or } X_n = S\}$$

denote the time[6] until any of the states ⓪ or Ⓢ is reached by $(X_n)_{n\geq 0}$, with $T_{0,S} = +\infty$ in case neither states are ever reached, i.e. when there exists no integer $n \geq 0$ such that $X_n = 0$ or $X_n = S$, see Fig. 2.5.
Note that by (2.2.30) we have

$$\mathbb{P}(T_{0,S} < \infty \mid X_0 = k) = \mathbb{P}(R_A \cup R_B \mid X_0 = k) = 1, \quad k = 0, 1, \ldots, S.$$

and therefore

$$\mathbb{P}(T_{0,S} = \infty \mid X_0 = k) = 0, \qquad k = 0, 1, \ldots, S.$$

We are now interested in computing the expected duration

$$h_S(k) := \mathbb{E}[T_{0,S} \mid X_0 = k]$$

[6] The notation "inf" stands for "infimum", meaning the smallest $n \geq 0$ such that $X_n = 0$ or $X_n = S$, if such an n exists.

of the game given that Player A starts with a wealth equal to $X_0 = k \in \{0, 1, \ldots, S\}$. Clearly, we have the boundary conditions

$$\begin{cases} h_S(0) = \mathbb{E}[T_{0,S} \mid X_0 = 0] = 0, & \text{(2.3.1a)} \\ h_S(S) = \mathbb{E}[T_{0,S} \mid X_0 = S] = 0. & \text{(2.3.1b)} \end{cases}$$

We start by considering the particular cases $S = 2$ and $S = 3$.

(i) $\underline{S = 2}$.

We have

$$T_{0,2} = \begin{cases} 0 \text{ if } X_0 = 0, \\ 1 \text{ if } X_0 = 1, \\ 0 \text{ if } X_0 = 2, \end{cases}$$

thus $T_{0,2}$ is *deterministic* given the value of X_0 and we simply have $h_2(1) = T_{0,2} = 1$ when $X_0 = 1$.

(ii) $\underline{S = 3}$.

In this case, the probability distribution of $T_{0,3}$ given $X_0 \in \{0, 1, 2, 3\}$ can be determined explicitly and we find, when $X_0 = 1$,

$$\begin{cases} \mathbb{P}(T_{0,3} = 2k \mid X_0 = 1) = p^2(pq)^{k-1}, & k \geq 1, \\ \mathbb{P}(T_{0,3} = 2k+1 \mid X_0 = 1) = q(pq)^k, & k \geq 0, \end{cases}$$

since in an even number $2k$ of steps we can only exit through state $\textcircled{3}$ after starting from $\textcircled{1}$, while in an odd number $2k+1$ of steps we can only exit through state $\textcircled{0}$. By exchanging p with q in the above formulas we get, when $X_0 = 2$,

$$\begin{cases} \mathbb{P}(T_{0,3} = 2k \mid X_0 = 2) = q^2(pq)^{k-1}, & k \geq 1, \\ \mathbb{P}(T_{0,3} = 2k+1 \mid X_0 = 2) = p(pq)^k, & k \geq 0, \end{cases}$$

whereas $T_{0,3} = 0$ whenever $X_0 = 0$ or $X_0 = 3$.

As a consequence, we can directly compute

$$h_3(2) = \mathbb{E}[T_{0,3} \mid X_0 = 2] = 2\sum_{k\geq 1} k\mathbb{P}(T_{0,3} = 2k \mid X_0 = 2) \tag{2.3.2}$$

$$+ \sum_{k\geq 0}(2k+1)\mathbb{P}(T_{0,3} = 2k+1 \mid X_0 = 2)$$

$$= 2q^2 \sum_{k\geq 1} k(pq)^{k-1} + p\sum_{k\geq 0}(2k+1)(pq)^k$$

$$= \frac{2q^2}{(1-pq)^2} + \frac{2p^2q}{(1-pq)^2} + \frac{p}{1-pq}$$

$$= \frac{2q^2+p+qp^2}{(1-pq)^2}$$

$$= \frac{2(1-p)q+1-q+q(1-q)p}{(1-pq)^2}$$

$$= \frac{1+q-pq-q^2p}{(1-pq)^2}$$

$$= \frac{1+q}{1-pq}, \tag{2.3.3}$$

where we applied (A.4), and by exchanging p and q we get

$$h_3(1) = \mathbb{E}[T_{0,3} \mid X_0 = 1] = \frac{2p^2+q+pq^2}{(1-pq)^2} = \frac{1+p}{1-pq}. \tag{2.3.4}$$

Again, things can become quite complicated for $S \geq 4$, and increasingly difficult when S becomes larger.

In the general case $S \geq 4$ we will only compute the conditional expectation of $T_{0,S}$ and not its probability distribution. For this we rely again on *first step analysis*, as stated in the following lemma.

Lemma 2.4 *For all $k = 1, 2, \ldots, S-1$ we have*

$$\mathbb{E}[T_{0,S} \mid X_0 = k] = 1 + p\,\mathbb{E}[T_{0,S} \mid X_0 = k+1] + q\,\mathbb{E}[T_{0,S} \mid X_0 = k-1].$$

Proof We condition on the first transition from X_0 to X_1. Using the equality $\mathbb{1}_A = \mathbb{1}_{A\cap B} + \mathbb{1}_{A\cap B^c}$ under the form

$$\mathbb{1}_{\{X_0=k\}} = \mathbb{1}_{\{X_1=k+1, X_0=k\}} + \mathbb{1}_{\{X_1=k-1, X_0=k\}},$$

cf. (1.4.2), and conditional expectations we show, by first step analysis, that for all $k = 1, 2, \ldots, S-1$, applying Lemma 1.10 and (1.6.6) successively to $A = \{X_0 = k\}$, $A = \{X_0 = k-1\}$ and $A = \{X_0 = k+1\}$, we have

$$
\begin{aligned}
&\mathbb{E}[T_{0,S} \mid X_0 = k] = \frac{1}{\mathbb{P}(X_0 = k)} \mathbb{E}\left[T_{0,S} \mathbb{1}_{\{X_0 = k\}}\right] \\
&= \frac{1}{\mathbb{P}(X_0 = k)} \left(\mathbb{E}\left[T_{0,S} \mathbb{1}_{\{X_1 = k+1, X_0 = k\}}\right] + \mathbb{E}\left[T_{0,S} \mathbb{1}_{\{X_1 = k-1, X_0 = k\}}\right]\right) \\
&= \frac{\mathbb{P}(X_1 = k+1 \text{ and } X_0 = k)}{\mathbb{P}(X_0 = k)} \mathbb{E}[T_{0,S} \mid X_1 = k+1, X_0 = k] \\
&\quad + \frac{\mathbb{P}(X_1 = k-1 \text{ and } X_0 = k)}{\mathbb{P}(X_0 = k)} \mathbb{E}[T_{0,S} \mid X_1 = k-1, X_0 = k] \\
&= \mathbb{P}(X_1 = k+1 \mid X_0 = k)\, \mathbb{E}[T_{0,S} \mid X_1 = k+1, X_0 = k] \\
&\quad + \mathbb{P}(X_1 = k-1 \mid X_0 = k)\, \mathbb{E}[T_{0,S} \mid X_1 = k-1, X_0 = k] \\
&= p\, \mathbb{E}[T_{0,S} \mid X_1 = k+1, X_0 = k] + q\, \mathbb{E}[T_{0,S} \mid X_1 = k-1, X_0 = k] \qquad (2.3.5) \\
&= p\, \mathbb{E}[T_{0,S} + 1 \mid X_0 = k+1, X_{-1} = k] + q\, \mathbb{E}[T_{0,S} + 1 \mid X_0 = k-1, X_{-1} = k] \qquad (2.3.6) \\
&= p\, \mathbb{E}[T_{0,S} + 1 \mid X_0 = k+1] + q\, \mathbb{E}[T_{0,S} + 1 \mid X_0 = k-1] \\
&= p(1 + \mathbb{E}[T_{0,S} \mid X_0 = k+1]) + q(1 + \mathbb{E}[T_{0,S} \mid X_0 = k-1]) \\
&= p + q + p\, \mathbb{E}[T_{0,S} \mid X_0 = k+1] + q\, \mathbb{E}[T_{0,S} \mid X_0 = k-1] \\
&= 1 + p\, \mathbb{E}[T_{0,S} \mid X_0 = k+1] + q\, \mathbb{E}[T_{0,S} \mid X_0 = k-1].
\end{aligned}
$$

From (2.3.5) to (2.3.6) we relabelled X_1 as X_0, which amounts to changing $T_{0,S} - 1$ into $T_{0,S}$, or equivalently changing $T_{0,S}$ into $T_{0,S} + 1$. □

In the case $S = 3$, Lemma 2.4 shows that

$$
\begin{cases}
h_3(0) = \mathbb{E}[T_{0,S} \mid X_0 = 0] = 0, \\
h_3(1) = 1 + p h_3(2) + q h_3(0) = 1 + p h_3(2) = 1 + p(1 + q h_3(1)) = 1 + p + pq h_3(1), \\
h_3(2) = 1 + p h_3(3) + q h_3(1) = 1 + q h_3(1) = 1 + q(1 + p h_3(2)) = 1 + q + pq h_3(2), \\
h_3(3) = \mathbb{E}[T_{0,S} \mid X_0 = 3] = 0,
\end{cases}
$$

whose solution can be checked to be given by (2.3.3)–(2.3.4).

More generally, defining the function $h_S : \{0, 1, \ldots, S\} \longrightarrow \mathbb{R}_+$ by

$$
h_S(k) := \mathbb{E}[T_{0,S} \mid X_0 = k], \qquad k = 0, 1, \ldots, S,
$$

Lemma 2.4 shows that

$$
h_S(k) = p(1 + h_S(k+1)) + q(1 + h_S(k-1)) = 1 + p h_S(k+1) + q h_S(k-1),
$$

$k = 1, 2, \ldots, S-1$, i.e. we have to solve the equation

$$\begin{cases} h_S(k) = 1 + ph_S(k+1) + qh_S(k-1), & 1 \le k < S, \quad (2.3.7a) \\ h_S(0) = h_S(S) = 0, & (2.3.7b) \end{cases}$$

for the function $h_S(k)$. Using the fact that $p + q = 1$, we can rewrite (2.3.7a) as

$$(p+q)h_S(k) = 1 + ph_S(k+1) + qh_S(k-1), \qquad k = 1, 2, \ldots, S-1,$$

or as the *difference equation*

$$p(h_S(k+1) - h_S(k)) - q(h_S(k) - h_S(k-1)) = -1, \qquad k = 1, 2, \ldots, S-1, \tag{2.3.8}$$

under the *boundary conditions* (2.3.7b).

The equation

$$p(f(k+1) - f(k)) - q(f(k) - f(k-1)) = 0, \qquad k = 1, 2, \ldots, S-1, \tag{2.3.9}$$

cf. (2.2.23), is called the *homogeneous equation* associated to (2.3.8).

We will use the following fact:

The general solution to (2.3.8) can be written as the sum of a *homogeneous solution* of (2.3.9) and a *particular solution* of (2.3.8).

Non-symmetric case : $p \neq q$

By (2.2.17) we know that the *homogeneous solution* of (2.3.9) is of the form $C_1 + C_2 r^k$. Next, searching for a *particular solution* of (2.3.8) of the form $k \longmapsto Ck$ shows that C has to be equal to $C = 1/(q-p)$. Therefore, when $r = q/p \neq 1$, the general solution of (2.3.8) has the form

$$h_S(k) = C_1 + C_2 r^k + \frac{1}{q-p} k. \tag{2.3.10}$$

From the boundary conditions (2.3.1) and (2.3.1b) and from (2.3.10) we have

$$\begin{cases} h_S(0) = 0 = C_1 + C_2, \\ h_S(S) = 0 = C_1 + C_2 r^S + \dfrac{1}{q-p} S, \end{cases} \tag{2.3.11}$$

and solving the system (2.3.11) of two equations we find

$$C_1 = -\frac{S}{(q-p)(1-r^S)} \quad \text{and} \quad C_2 = \frac{S}{(q-p)(1-r^S)},$$

hence from (2.3.10) we get

$$h_S(k) = \mathbb{E}[T_{0,S} \mid X_0 = k] = \frac{1}{q-p}\left(k - S\frac{1-(q/p)^k}{1-(q/p)^S}\right) \tag{2.3.12}$$
$$= \frac{1}{q-p}\left(k - S\mathbb{P}(R_B \mid X_0 = k)\right), \quad k = 0, 1, 2, \ldots, S,$$

which does satisfy the boundary conditions (2.3.1) and (2.3.1b). Note that changing k to $S-k$ and p to q does not modify (2.3.12), as it also represents the mean game duration for Player B. In particular, we note that

$$\mathbb{E}[T_{0,S} \mid X_0 = k] < +\infty, \qquad k = 0, 1, 2, \ldots, S.$$

When $p = 1$, i.e. $r = 0$, we can check easily that *e.g.* $h_S(k) = S - k, k = 0, 1, 2, \ldots, S$. On the other hand, when the number S of states becomes large, we find that for all $k \geq 1$,

$$h_\infty(k) := \lim_{S\to\infty} h_S(k) = \lim_{S\to\infty} \mathbb{E}[T_{0,S} \mid X_0 = k] = \begin{cases} \infty & \text{if } q \leq p, \\ \dfrac{k}{q-p} & \text{if } q > p, \end{cases} \tag{2.3.13}$$

with $h_\infty(0) = 0$, cf. also the symmetric case treated in (2.3.19) below when $p = q = 1/2$. In particular, for $k \geq 1$ we have, see (2.2.14),

$$\begin{cases} \mathbb{P}(T_0 < \infty \mid X_0 = k) = \left(\dfrac{q}{p}\right)^k < 1 \text{ and } \mathbb{E}[T_0 \mid X_0 = k] = \infty, \text{ for } p > q, \\ \mathbb{P}(T_0 < \infty \mid X_0 = k) = 1 \text{ and } \mathbb{E}[T_0 \mid X_0 = k] = \infty, \text{ for } p = q = \dfrac{1}{2}, \\ \mathbb{P}(T_0 < \infty \mid X_0 = k) = 1 \text{ and } \mathbb{E}[T_0 \mid X_0 = k] < \infty, \text{ for } p < q. \end{cases} \tag{2.3.14}$$

We note in particular that the mean game duration $\mathbb{E}[T_0 \mid X_0 = k]$ is infinite in the fair game case $p = q = 1/2$. When $r = q/p = 1$, this yields an example of a random variable T_0 which is (almost surely[7]) finite, while its expectation is infinite.[8] This

[7] "Almost surely" means "with probability 1".

[8] Recall that an infinite set of finite data values may have an infinite average. Similarly, there exist finite surfaces that enclose an infinite volume.

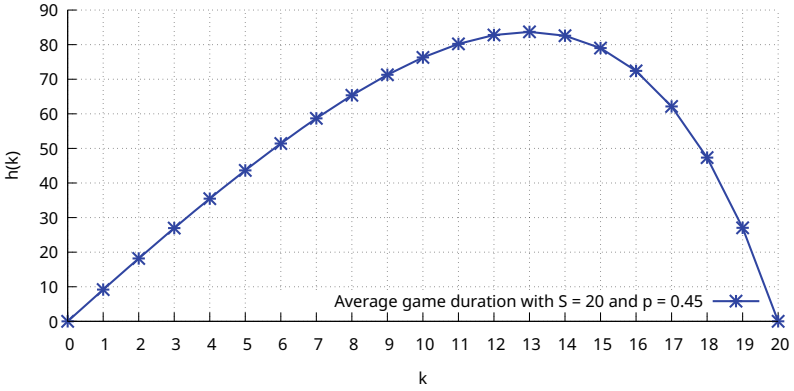


Fig. 2.6 Mean game duration $h_{20}(k)$ as a function of $X_0 = k \in [0, 20]$ for $p = 0.45$

situation is similar to that of the St. Petersburg paradox as in (1.6.5). Similarly, one can find sequences $(X_n)_{n\geq 0}$ of random variables such that $X_n \to 0$ with probability one, while $\mathbb{E}[X_n] \to \infty$ as n tends to infinity.

When $S = 2$ it is easy to show that (2.3.12) yields $h_2(1) = 1$. When $S = 3$, (2.3.12) shows that, using the relation $p + q = 1$,[9]

$$\mathbb{E}[T_{0,3} \mid X_0 = 1] = \frac{1}{q-p}\left(1 - 3\frac{1-q/p}{1-(q/p)^3}\right) = \frac{1+p}{1-pq}, \tag{2.3.15}$$

and

$$\mathbb{E}[T_{0,3} \mid X_0 = 2] = \frac{1}{q-p}\left(2 - 3\frac{1-(q/p)^2}{1-(q/p)^3}\right) = \frac{1+q}{1-pq}, \tag{2.3.16}$$

however, it takes more time to show that (2.3.15) and (2.3.16) agree respectively with (2.3.2) and (2.3.4). In Fig. 2.6 the mean game duration (2.3.12) is plotted as a function of k for $p = 0.45$.

Symmetric case : $p = q = 1/2$
In this case (fair game), the *homogeneous solution* of (2.3.9) is $C_1 + C_2 k$, given by (2.2.20).

Since $r = 1$ we see that $k \longmapsto Ck$ can no longer be a particular solution of (2.3.8). However, we can search for a particular solution of the form $k \longmapsto Ck^2$, in which case we find that C has to be equal to $C = -1$.

Therefore when $r = q/p = 1$ the general solution of (2.3.8) has the form

$$h_S(k) = C_1 + C_2 k - k^2, \qquad k = 0, 1, 2, \ldots, S. \tag{2.3.17}$$

[9] This point is left as exercise.

From the boundary conditions (2.3.1) and (2.3.1b) and from (2.3.17) we have

$$\begin{cases} h_S(0) = 0 = C_1, \\ h_S(S) = 0 = C_1 + C_2 S - S^2, \end{cases} \tag{2.3.18}$$

and solving the above system (2.3.18) of two equations yields

$$C_1 = 0 \quad \text{and} \quad C_2 = S,$$

hence from (2.3.17) we get

$$h_S(k) = \mathbb{E}[T_{0,S} \mid X_0 = k] = k(S-k), \qquad k = 0, 1, 2, \ldots, S, \tag{2.3.19}$$

which does satisfy the boundary conditions (2.3.1) and (2.3.1b) and coincides with (2.3.13) when S goes to infinity. In particular, we have

$$\mathbb{E}[T_{0,S} \mid X_0 = k] < +\infty, \qquad k = 0, 1, 2, \ldots, S.$$

We also note that for all values of $p \in [0, 1]$ the expectation $\mathbb{E}[T_{0,S} \mid X_0 = k]$ has a *finite* value, which recovers the fact that the game duration $T_{0,S}$ is finite with probability one for all $k = 0, 1, \ldots, S$, i.e. $\mathbb{P}(T_{0,S} = \infty \mid X_0 = k) = 0$ for all $k = 0, 1, \ldots, S$, see Exercise 10.5.

Remark 2.5 When $r = 1$, by the same argument as in (2.2.27) we find that (2.3.8) is a discretization of the continuous Poisson equation

$$\frac{1}{2} \frac{\partial^2 f}{\partial x^2}(x) = -1, \qquad x \in \mathbb{R},$$

which has for solution

$$f(x) = f(0) + x f'(0) - x^2, \qquad x \in \mathbb{R}.$$

Relation (2.3.19) can also be recovered from (2.3.12) by letting p go to $1/2$. In Fig. 2.7, the expected game duration (2.3.12) is plotted as a function of p for $S = 20$ and $k = 10$.

As expected, the duration will be maximal in a fair game for $p = q = 1/2$. On the other hand, it always takes exactly $10 = S - k = k$ steps to end the game in case $p = 0$ or $p = 1$, in which case there is no randomness. When $p = 0.45$ the expected duration of the game becomes 76.3, which represents only a drop of 24% from the

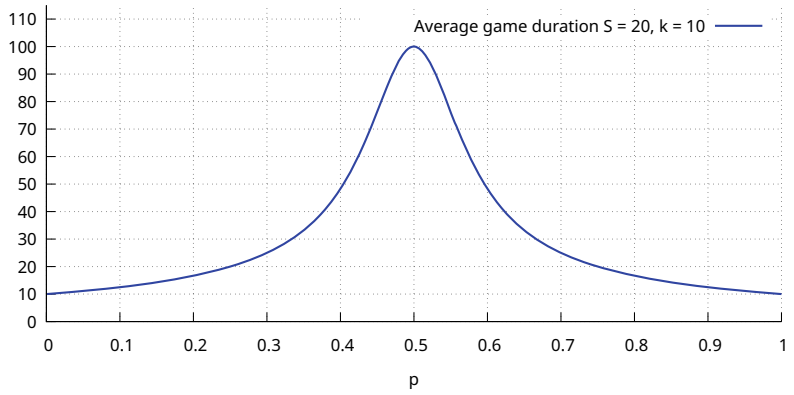


Fig. 2.7 Mean game duration as a function of $p \in [0, 1]$ for $S = 20$ and $k = 10$

"fair" value 100, as opposed to the 73% drop noticed above in terms of winning probabilities. Thus, a game with $p = 0.45$ is only slightly shorter than a fair game, whereas the probability of winning the game drops down to 0.12.

Remark 2.6 In this Chap. 2 we have noticed an interesting connection between analysis and probability. That is, a probabilistic quantity such as $k \longmapsto \mathbb{P}(R_A \mid X_0 = k)$ or $k \longmapsto \mathbb{E}[T_{0,S} \mid X_0 = k]$ can be shown to satisfy a difference equation which is solved by analytic methods. This fact actually extends beyond the present simple framework, and in continuous time it yields other connections between probability and partial differential equations.

In Chap. 3 we will consider a family of simple random walks which can be seen as "unrestricted" gambling processes.

Exercises

Exercise 2.1 We consider a gambling problem with the possibility of a draw,[10] i.e. at time n the gain X_n of Player A can increase by one unit with probability $r \in (0, 1/2]$, decrease by one unit with probability r, or remain stable with probability $1 - 2r$. We let

$$f(k) := \mathbb{P}(R_A \mid X_0 = k)$$

denote the probability of ruin of Player A, and let

[10] Also called "lazy random walk".

$$h(k) := \mathbb{E}[T_{0,S} \mid X_0 = k]$$

denote the expectation of the game duration $T_{0,S}$ starting from $X_0 = k, k = 0, 1, \ldots, S$.

(a) Using first step analysis, write down the difference equation satisfied by $f(k)$ and its boundary conditions, $k = 0, 1, \ldots, S$. We refer to this equation as the *homogeneous equation*.
(b) Solve the *homogeneous equation* of Question (a) by your preferred method. Is this solution compatible with your intuition of the problem? Why?
(c) Using first step analysis, write down the difference equation satisfied by $h(k)$ and its boundary conditions, $k = 0, 1, \ldots, S$.
(d) Find a particular solution of the equation of Question (c).
(e) Solve the equation of Question (c).
Hint: recall that the general solution of the equation is the sum of a particular solution and a solution of the *homogeneous equation*.
(f) How does the mean duration $h(k)$ behave as r goes to zero? Is this solution compatible with your intuition of the problem? Why?

Exercise 2.2 Recall that for any standard gambling process $(Z_k)_{k\geq 0}$ on a state space $\{a, a+1, \ldots, b-1, b\}$ with absorption at states $\textcircled{a}$ and $\textcircled{b}$ and probabilities $p \neq q$ of moving by ± 1, the probability of hitting state $\textcircled{a}$ before hitting state $\textcircled{b}$ after starting from state $Z_0 = k \in \{a, a+1, \ldots, b-1, b\}$ is given by

$$\frac{1-(p/q)^{b-k}}{1-(p/q)^{b-a}}. \tag{2.3.20}$$

In questions (a)-(b)-(c) below we consider a gambling process $(X_k)_{k\geq 0}$ on the state space $\{0, 1, \ldots, S\}$ with absorption at $\textcircled{0}$ and $\textcircled{S}$ and probabilities $p \neq q$ of moving by ± 1.

(a) Using Relation (2.3.20), give the probability of coming back in finite time to a given state $m \in \{1, 2, \ldots, S-1\}$ after starting from $X_0 = k \in \{m+1, \ldots, S\}$.
(b) Using Relation (2.3.20), give the probability of coming back in finite time to the given state $m \in \{1, 2, \ldots, S-1\}$ after starting from $X_0 = k \in \{0, 1, \ldots, m-1\}$.
(c) Using first step analysis, give the probability of coming back to state $\textcircled{m}$ in finite time after starting from $X_0 = m$.
(d) Using first step analysis, compute the mean time to either come back to m of reach any of the two boundaries $\{0, S\}$, whichever comes first?
(e) Repeat the above questions (c)–(d) with equal probabilities $p = q = 1/2$, in which case the probability of hitting state $\textcircled{a}$ before hitting state $\textcircled{b}$ after starting from state $Z_0 = k$ is given by

$$\frac{b-k}{b-a}, \qquad k = a, a+1, \ldots, b-1, b. \tag{2.3.21}$$

Exercise 2.3 Consider the gambling process $(X_n)_{n\geq 0}$ on the state space $\mathbb{S} = \{0, 1, \ldots, S\}$, with probability p, resp. q, of moving up, resp. down, at each time step. For $k = 0, 1, \ldots, S$, let τ_k denote the first hitting time

$$\tau_k := \inf\{n \geq 0 \ : X_n = k\}.$$

of state $\textcircled{k}$ by the process $(X_n)_{n\geq 0}$, and let

$$p_k := \mathbb{P}(\tau_{k+1} < \tau_0 \mid X_0 = k), \qquad k = 0, 1, \ldots, S-1,$$

denote the probability of hitting state $\boxed{k+1}$ before hitting state $\textcircled{0}$.

(a) Show that $p_k = \mathbb{P}(\tau_{k+1} < \tau_0 \mid X_0 = k)$ satisfies the recurrence equation

$$p_k = p + qp_{k-1}p_k, \qquad k = 1, 2, \ldots, S-1, \tag{2.3.22}$$

i.e.

$$p_k = \frac{p}{1 - qp_{k-1}}, \qquad k = 1, 2, \ldots, S-1.$$

(b) Check by substitution that the solution of (2.3.22) is given by

$$p_k = \frac{1 - (q/p)^k}{1 - (q/p)^{k+1}}, \qquad k = 0, 1, \ldots, S-1. \tag{2.3.23}$$

(c) Compute $\mathbb{P}(\tau_S < \tau_0 \mid X_0 = k)$ by a product formula and recover (2.2.12) and (2.2.28) based on the result of part (2.3.23).

(d) Show that (2.2.13) and (2.2.29) can be recovered in a similar way in the symmetric case $p = q = 1/2$ by trying the solution $p_k = k/(k+1)$, $k = 0, 1, \ldots, S-1$.

Exercise 2.4 Consider a gambling process $(X_n)_{n\geq 0}$ on the state space $\{0, 1, 2\}$, with transition probabilities given by

$$\begin{bmatrix} \mathbb{P}(X_1 = 0 \mid X_0 = 0) & \mathbb{P}(X_1 = 1 \mid X_0 = 0) & \mathbb{P}(X_1 = 2 \mid X_0 = 0) \\ \mathbb{P}(X_1 = 0 \mid X_0 = 1) & \mathbb{P}(X_1 = 1 \mid X_0 = 1) & \mathbb{P}(X_1 = 2 \mid X_0 = 1) \\ \mathbb{P}(X_1 = 0 \mid X_0 = 2) & \mathbb{P}(X_1 = 1 \mid X_0 = 2) & \mathbb{P}(X_1 = 2 \mid X_0 = 2) \end{bmatrix}$$

$$= \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{bmatrix} q & p & 0 \\ q & 0 & p \\ 0 & 0 & 1 \end{bmatrix} \end{array},$$

where $0 < p < 1$ and $q = 1 - p$. In this game, Player A is allowed to "rebound" from state $\textcircled{0}$ to state $\textcircled{1}$ with probability p, and state $\textcircled{2}$ is absorbing.

In order to be ruined, Player A has to visit state $\textcircled{0}$ *twice*. Let

$$f(k) := \mathbb{P}(R_A \mid X_0 = k), \qquad k = 0, 1, 2,$$

denote the probability of ruin of Player A starting from $k = 0, 1, 2$. Starting from $\textcircled{0}$ counts as one visit to $\textcircled{0}$.

(a) Compute the boundary condition $f(0)$ using pathwise analysis.
(b) Give the value of the boundary condition $f(2)$, and compute $f(1)$ by first step analysis.

Exercise 2.5 (a) Recover (2.3.19) from (2.3.12) by letting p go to $1/2$, i.e. when $r = q/p$ goes to 1.
(b) Recover (2.2.22) from (2.2.12) by letting p go to $1/2$, i.e. when $r = q/p$ goes to 1.

Exercise 2.6 Extend the setting of Exercise 2.1 to a non-symmetric gambling process with draw and respective probabilities $\alpha > 0$, $\beta > 0$, and $1 - \alpha - \beta > 0$ of increase, decrease, and draw. Compute the ruin probability $f(k)$ and the mean game duration $h(k)$ in this extended framework. Check that when $\alpha = \beta \in (0, 1/2)$ we recover the result of Exercise 2.1.

Problem 2.7 We consider a discrete-time process $(X_n)_{n\geq 0}$ that models the wealth of a gambler within $\{0, 1, \ldots, S\}$, with the transition probabilities

$$\begin{cases} \mathbb{P}(X_{n+1} = k + 1 \mid X_n = k) = p, & k = 0, 1, \ldots, S - 1, \\ \mathbb{P}(X_{n+1} = k - 1 \mid X_n = k) = q, & k = 1, 2, \ldots, S - 1, \end{cases}$$

with

$$\mathbb{P}(X_{n+1} = 0 \mid X_n = 0) = q \quad \text{and} \quad \mathbb{P}(X_{n+1} = S \mid X_n = S) = 1,$$

for all $n \in \mathbb{N} = \{0, 1, 2, \ldots\}$, where $q = 1 - p$ and $p \in (0, 1]$. In this model the gambler is given a second chance, and may be allowed to "rebound" to state $\textcircled{1}$ after reaching $\textcircled{0}$. Let

$$W = \bigcup_{n\in\mathbb{N}} \{X_n = S\}$$

denote the event that the player eventually wins the game.

(a) Let

$$g(k) := \mathbb{P}(W \mid X_0 = k)$$

denote the probability that the player eventually wins after starting from state $k \in \{0, 1, \ldots, S\}$. Using first step analysis, write down the difference equations satisfied by $g(k)$, $k = 0, 1, \ldots, S-1$, and their boundary condition(s), which may not be given in explicit form. This question is standard, however, one has to pay attention to the special behavior of the process at state ⓪.

(b) Obtain $\mathbb{P}(W \mid X_0 = k)$ for all $k = 0, 1, \ldots, S$ as the unique solution to the system of equations stated in Question (a).

The answer to this question is very simple and can be obtained through intuition. However, a (mathematical) proof is required.

(c) Let

$$T_S = \inf\{n \geq 0 \ : \ X_n = S\}$$

denote the first hitting time of S by the process $(X_n)_{n\geq 0}$. Let

$$h(k) := \mathbb{E}[T_S \mid X_0 = k]$$

denote the expected time until the gambler wins after starting from state $k \in \{0, 1, \ldots, S\}$. Using first step analysis, write down the difference equations satisfied by $h(k)$ for $k = 0, 1, \ldots, S-1$, and state the corresponding boundary condition(s).

Again, one has to pay attention to the special behavior of the process at state ⓪, as the equation obtained by first step analysis for $h(0)$ will take a particular form and can be viewed as a second boundary condition.

(d) Compute $\mathbb{E}[T_S \mid X_0 = k]$ for all $k = 0, 1, \ldots, S$ by solving the equations of Question (c).

This question is more difficult than Question (b), and it could be skipped at first reading since its result is not used in the sequel. One can solve the associated *homogeneous equation* for $k = 1, 2, \ldots, S-1$ using the results of Sect. 2.3, and a particular solution can be found by observing that here we consider the time until Player A (not B) wins. As usual, the cases $p \neq q$ and $p = q = 1/2$ have to be considered separately at some point. The formula obtained for $p = 1$ should be quite intuitive and may help you check your result.

(e) Let now

$$T_0 = \inf\{n \geq 0 \ : \ X_n = 0\}$$

denote the first hitting time of 0 by the process $(X_n)_{n\geq 0}$. Using the results of Sect. 2.2 for the ruin of Player B, write down the value of

$$p_k := \mathbb{P}(T_S < T_0 \mid X_0 = k)$$

as a function of p, S, and $k = 0, 1, \ldots, S$.

Note that according to the notation of Chap. 2, $\{T_S < T_0\}$ denotes the event "Player A wins the game".

(f) Explain why the equality

$$\begin{aligned}\mathbb{P}(T_S < T_0 \mid X_1 = k+1 \text{ and } X_0 = k) &= \mathbb{P}(T_S < T_0 \mid X_1 = k+1) \\ &= \mathbb{P}(T_S < T_0 \mid X_0 = k+1). \end{aligned} \tag{2.3.24}$$

holds for $k \in \{0, 1, \ldots, S-1\}$ (an explanation in words will be sufficient here).

(g) Using Relation (2.3.24), show that the probability

$$\mathbb{P}(X_1 = k+1 \mid X_0 = k \text{ and } T_S < T_0)$$

of an upward step given that state (S) is reached first, is equal to

$$\begin{aligned}\mathbb{P}(X_1 = k+1 \mid X_0 = k \text{ and } T_S < T_0) &= p\frac{\mathbb{P}(T_S < T_0 \mid X_0 = k+1)}{\mathbb{P}(T_S < T_0 \mid X_0 = k)} \\ &= p\frac{p_{k+1}}{p_k}, \end{aligned} \tag{2.3.25}$$

$k = 1, 2, \ldots, S-1$, to be computed explicitly from the result of Question (e). How does this probability compare to the value of p?

No particular difficulty here, the proof should be a straightforward application of the definition of conditional probabilities.

(h) Compute the probability

$$\mathbb{P}(X_1 = k-1 \mid X_0 = k \text{ and } T_0 < T_S), \qquad k = 1, 2, \ldots, S,$$

of a downward step given that state (0) is reached first, using similar arguments to Question (g).

(i) Let

$$h(k) = \mathbb{E}[T_S \mid X_0 = k,\ T_S < T_0], \qquad k = 1, 2, \ldots, S,$$

denote the expected time until the player wins, given that state (0) is never reached. Using the transition probabilities (2.3.25), state the finite difference equations satisfied by $h(k)$, $k = 1, 2, \ldots, S-1$, and their boundary condition(s).

The derivation of the equation is standard, but you have to make a careful use of conditional transition probabilities given $\{T_S < T_0\}$. There is an issue on whether and how $h(0)$ should appear in the system of equations, but this point can actually be solved.

(j) Solve the equation of Question (i) when $p = 1/2$ and compute $h(k)$ for $k = 1, 2, \ldots, S$. What can be said of $h(0)$?

There is actually a way to transform this equation using an homogeneous equation already solved in Sect. 2.3.

Problem 2.8 Let $S \geq 1$. We consider a discrete-time process $(X_n)_{n\geq 0}$ that models the wealth of a gambler within $\{0, 1, \ldots, S\}$, with the transition probabilities

$$\mathbb{P}(X_{n+1} = k+2 \mid X_n = k) = p, \quad \mathbb{P}(X_{n+1} = k-1 \mid X_n = k) = 2p,$$

and

$$\mathbb{P}(X_{n+1} = k \mid X_n = k) = r, \quad k \in \mathbb{Z},$$

for all $n \in \mathbb{N} = \{0, 1, 2, \ldots\}$, where $p > 0$, $r \geq 0$, and $3p + r = 1$. We let

$$\tau := \inf\{n \geq 0 \ : \ X_n \leq 0 \text{ or } X_n \geq S\}.$$

(a) Consider the probability

$$g(k) := \mathbb{P}(X_\tau \geq S \mid X_0 = k)$$

that the game ends with Player A winning the game, starting from $X_0 = k$. Give the values of $g(0)$, $g(S)$ and $g(S+1)$.

(b) Using first step analysis, write down the difference equation satisfied by $g(k), k = 1, 2, \ldots, S-1$, and its boundary conditions, by taking *overshoot* into account. We refer to this equation as the *homogeneous equation*.

(c) Solve the equation of Question (b) from its characteristic equation as in (2.2.16).

(d) Does the answer to Question (c) depend on p? Why?

(e) Consider the expected time

$$h(k) := \mathbb{E}[\tau \mid X_0 = k], \qquad k = 0, 1, \ldots, S+1,$$

spent until the end of the game. Give the values of $h(0)$, $h(S)$ and $h(S+1)$.

(f) Using first step analysis, write down the difference equation satisfied by $h(k)$, $k = 1, 2, \ldots, S-1$, and its boundary conditions.

(g) Find a particular solution of the equation of Question (e).

(h) Solve the equation of Question (c).

Hint: the general solution of the equation is the sum of a particular solution and a solution of the associated *homogeneous equation*.

(i) How does the mean duration $h(k)$ behave as p goes to zero? Is this compatible with your intuition of the problem? Why?

(j) How do the values of $g(k)$ and $h(k)$ behave for fixed $k \in \{1, 2, \ldots, S-1\}$ as S tends to infinity?

Problem 2.9 (Anděl and Hudecová (2012)). Consider a gambling process $(X_n)_{n\geq 0}$ on the state space $\mathbb{S} = \{0, 1, \ldots, S\}$, with transition probabilities

$$\mathbb{P}(X_{n+1} = k+1 \mid X_n = k) = p, \qquad \mathbb{P}(X_{n+1} = k-1 \mid X_n = k) = q,$$

$k = 1, 2, \ldots, S-1$, with $p + q = 1$. Let

$$\tau := \inf\{n \geq 0 \ : \ X_n = 0 \text{ or } X_n = S\}$$

denote the time until the process hits either state ⓪ or state Ⓢ, and consider the second moment

$$h(k) := \mathbb{E}\left[\tau^2 \mid X_0 = k\right],$$

of τ after starting from $k = 0, 1, 2, \ldots, S$.

(a) Give the values of $h(0)$ and $h(S)$.
(b) Using first step analysis, find an equation satisfied by $h(k)$ and involving $\mathbb{E}[\tau \mid X_0 = k+1]$ and $\mathbb{E}[\tau \mid X_0 = k-1]$, $k = 1, 2, \ldots, S-1$.
(c) From now on, we take $p = q = 1/2$. Recall that in this case we have

$$\mathbb{E}[\tau \mid X_0 = k] = (S-k)k, \qquad k = 0, 1, \ldots, S.$$

Show that the function $h(k)$ satisfies the finite difference equation

$$h(k) = -1 + 2(S-k)k + \frac{1}{2}h(k+1) + \frac{1}{2}h(k-1), \qquad k = 1, 2, \ldots, S-1. \tag{2.3.26}$$

(d) Knowing that

$$k \longmapsto \frac{2}{3}k^2 - \frac{2S}{3}k^3 + \frac{k^4}{3}$$

is a particular solution of the equation (2.3.26) of Question (c), and that the solution of the *homogeneous equation*

$$f(k) = \frac{1}{2}f(k+1) + \frac{1}{2}f(k-1), \qquad k = 1, 2, \ldots, S-1,$$

takes the form

$$f(k) = C_1 + C_2 k,$$

compute the value of the expectation $h(k)$ solution of (2.3.26) for all $k = 0, 1, \ldots, S$.
(e) Compute the variance

$$v(k) = \mathbb{E}[\tau^2 \mid X_0 = k] - (\mathbb{E}[\tau \mid X_0 = k])^2$$

of the game duration starting from $k = 0, 1, \ldots, S$.
(f) Compute $v(1)$ when $S = 2$ and explain why the result makes pathwise sense.

Chapter 3
Random Walks



In this chapter we consider our second important example of discrete-time stochastic process, which is a random walk allowed to evolve over the set $\mathbb{Z}$ of signed integers without any boundary restriction. Of particular importance are the probabilities of return to a given state in finite time, as well as the corresponding mean return time.

3.1 Unrestricted Random Walk

The simple unrestricted random walk $(S_n)_{n\geq 0}$, also called the *Bernoulli* random walk, is defined by $S_0 = 0$ and

$$S_n = \sum_{k=1}^{n} X_k = X_1 + \cdots + X_n, \qquad n \geq 1,$$

where the random walk *increments* $(X_k)_{k\geq 1}$ form a family of independent, $\{-1, +1\}$-valued random variables (Fig. 3.1).

We will assume in addition that the family $(X_k)_{k\geq 1}$ is *i.i.d.*, i.e. it is made of *independent and identically distributed* Bernoulli random variables, with distribution

$$\begin{cases} \mathbb{P}(X_k = +1) = p, \\ \mathbb{P}(X_k = -1) = q, \qquad k \geq 1, \end{cases}$$

with $p + q = 1$.

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N. Privault, *Understanding Markov Chains*, Springer Undergraduate Mathematics Series,
https://doi.org/10.1007/978-981-95-4057-0_3

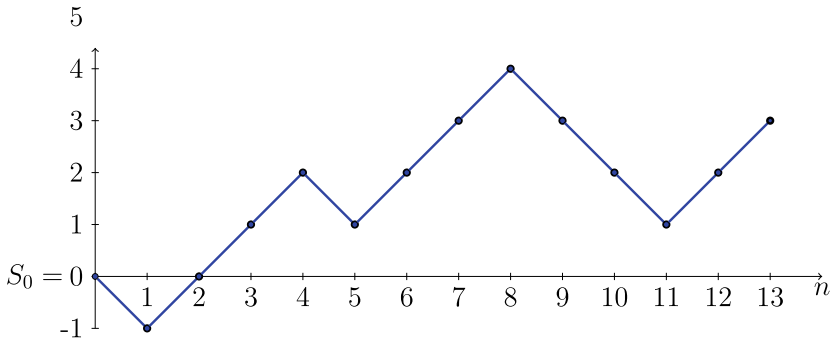


Fig. 3.1 Sample path of the random walk $(S_n)_{n\geq 0}$

3.2 Mean and Variance

In this case, the mean and variance of X_n are given by

$$\mathbb{E}[X_n] = -1 \times q + 1 \times p = 2p - 1 = p - q,$$

and

$$\begin{aligned} \mathrm{Var}[X_n] &= \mathbb{E}[X_n^2] - (\mathbb{E}[X_n])^2 \\ &= 1 \times q + 1 \times p - (2p-1)^2 \\ &= 4p(1-p) = 4pq. \end{aligned}$$

As a consequence, we find that

$$\mathbb{E}[S_n \mid S_0 = 0] = \mathbb{E}\left[\sum_{k=1}^{n} X_k\right] = \sum_{k=1}^{n} \mathbb{E}\left[X_k\right] = n(2p-1) = n(p-q),$$

and the variance can be computed by (1.6.10) as

$$\mathrm{Var}[S_n \mid S_0 = 0] = \mathrm{Var}\left[\sum_{k=1}^{n} X_k\right] = \sum_{k=1}^{n} \mathrm{Var}[X_k] = 4npq,$$

where we used (1.6.10).

3.3 Distribution

First we note that in an even number of time steps, $(S_n)_{n\geq 0}$ can only reach an even state in $\mathbb{Z}$ starting from $\textcircled{0}$. Similarly, in an odd number of time steps, $(S_n)_{n\geq 0}$ can only reach an odd state in $\mathbb{Z}$ starting from $\textcircled{0}$. Indeed, starting from $S_n = k$ the value of S_{n+2} after two time steps can only belong to $\{k-2, k, k+2\}$. Consequently, we have

$$\begin{cases} \mathbb{P}(S_{2n} = 2k+1 \mid S_0 = 0) = 0, & k \in \mathbb{Z},\ n \geq 0, \\ \mathbb{P}(S_{2n+1} = 2k \mid S_0 = 0) = 0, & k \in \mathbb{Z},\ n \geq 0, \end{cases} \tag{3.3.1}$$

and

$$\mathbb{P}(S_n = k \mid S_0 = 0) = 0, \qquad \text{for } k < -n \text{ or } k > n, \tag{3.3.2}$$

since $S_0 = 0$. Next, let l denote the number of upwards steps between time 0 and time $2n$, whereas $2n - l$ will denote the number of downwards steps. If $S_{2n} = 2k$ we have

$$2k = l - (2n - l) = 2l - 2n,$$

hence there are $l = n + k$ upwards steps and $2n - l = n - k$ downwards steps, $-n \leq k \leq n$. The probability of a given paths having $l = n + k$ upwards steps and $2n - l = n - k$ downwards steps is

$$p^{n+k} q^{n-k}$$

and in order to find $\mathbb{P}(S_{2n} = 2k \mid S_0 = 0)$ we need to multiply this probability by the total number of paths leading from $\textcircled{0}$ to $\boxed{2k}$ in $2n$ steps. We find that this number of paths is

$$\binom{2n}{n+k} = \binom{2n}{n-k}$$

which represents the number of ways to arrange $n + k$ upwards steps (or $n - k$ downwards steps) within $2n$ time steps.

Hence, we have

$$\mathbb{P}(S_{2n} = 2k \mid S_0 = 0) = \binom{2n}{n+k} p^{n+k} q^{n-k}, \quad -n \leq k \leq n, \tag{3.3.3}$$

in addition to (3.3.1) and (3.3.2). In Fig. 3.2 we enumerate the $120 = \binom{10}{7} = \binom{10}{3}$ possible paths corresponding to $n = 5$ and $k = 2$.

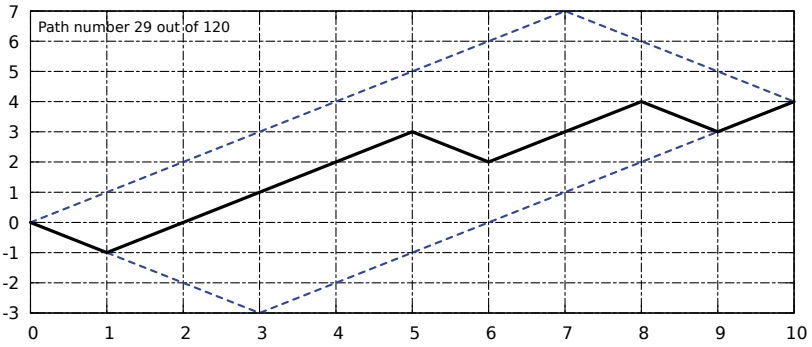


Fig. 3.2 Graph of $120 = \binom{10}{7} = \binom{10}{3}$ paths linking $(0, 0)$ to $(10, 4)$

Exercises:

(i) Show by a similar analysis that

$$\mathbb{P}(S_{2n+1} = 2k+1 \mid S_0 = 0) = \binom{2n+1}{n+k+1} p^{n+k+1} q^{n-k}, \quad -n \le k \le n, \tag{3.3.4}$$

i.e. $(2n + 1 + S_{2n+1})/2$ is a binomial random variable with parameter $(2n + 1, p)$, and

$$\mathbb{P}\left(\frac{2n+1+S_{2n+1}}{2} = k \,\middle|\, S_0 = 0\right) = \mathbb{P}\left(S_{2n+1} = 2k - 2n - 1 \mid S_0 = 0\right)$$
$$= \binom{2n+1}{k} p^k q^{2n+1-k},$$

$k = 0, 1, \ldots, 2n + 1$.

(ii) Show that $n + S_{2n}/2$ is a binomial[1] random variable with parameter $(2n, p)$, i.e., show that

$$\mathbb{P}\left(n + \frac{S_{2n}}{2} = k \,\middle|\, S_0 = 0\right) = \mathbb{P}\left(S_{2n} = 2k - 2n \mid S_0 = 0\right)$$
$$= \binom{2n}{k} p^k q^{2n-k}, \qquad k = 0, 1, \ldots, 2n.$$

[1] Note that S_{2n} is always an even number after we start from $S_0 = 0$.

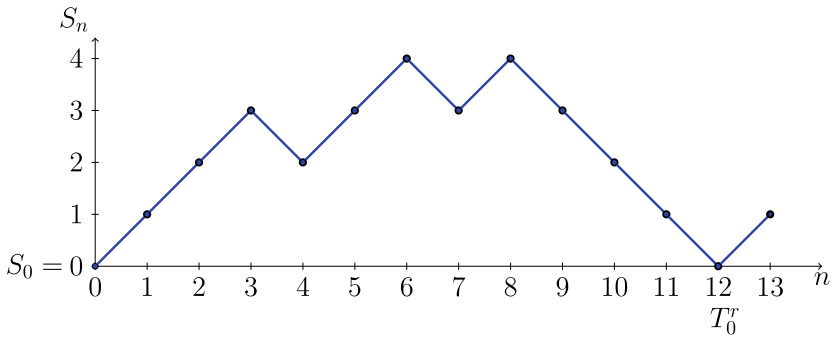


Fig. 3.3 Sample path of the random walk $(S_n)_{n\geq 0}$ with return to state 0

3.4 First Return to Zero

Let

$$T_0^r := \inf\{n \geq 1 \ : \ S_n = 0\}$$

denote the time of first return to $\textcircled{0}$ of the random walk started at $\textcircled{0}$, with the convention $\inf \emptyset = +\infty$.[2] We are interested in particular in computing the mean time $\mathbb{E}[T_0^r \mid S_0 = 0]$ it takes to return to state $\textcircled{0}$ after starting from state $\textcircled{0}$, see Fig. 3.3.

We are interested in computing the distribution

$$g(n) = \mathbb{P}(T_0^r = n \mid S_0 = 0), \qquad n \geq 1,$$

of the first return time T_0^r to $\textcircled{0}$. It is easy to show by pathwise analysis that T_0^r can only be even-valued starting from $\textcircled{0}$, hence $g(2k+1) = 0$ for all $k \in \mathbf{N}$, and in particular we have

$$\mathbb{P}(T_0^r = 1 \mid S_0 = 0) = 0, \quad \mathbb{P}(T_0^r = 2 \mid S_0 = 0) = 2pq, \tag{3.4.1}$$

and

$$\mathbb{P}(T_0^r = 4 \mid S_0 = 0) = 2p^2q^2, \tag{3.4.2}$$

by considering the two paths leading from $\textcircled{0}$ to $\textcircled{0}$ in two steps and the only two paths leading from $\textcircled{0}$ to $\textcircled{0}$ in four steps without hitting $\textcircled{0}$. However, the computation of $\mathbb{P}(T_0^r = 2n \mid S_0 = 0)$ by this method is difficult to extend to all $n \geq 3$.

[2] Recall that the notation "inf" stands for "infimum", meaning the smallest $n \geq 0$ such that $S_n = 0$, with $T_0^r = +\infty$ if no such $n \geq 0$ exists.

In order to completely solve this problem we will rely on the computation of the probability generating function $G_{T_0^r}$ of T_0^r, cf. (3.4.9) below.

This computation will use the following tools:

– convolution equation, see Relation (3.4.3) below,
– Taylor expansions, see Relation (3.4.22) below,
– probability generating functions.

First, we will need the following Lemma 3.1 which will be used in the proof of Lemma 3.2 below.

Lemma 3.1 (Convolution equation) *The function*

$$\begin{aligned} g : \{1, 2, 3, \ldots\} &\longrightarrow [0, 1] \\ n &\longmapsto g(n) \end{aligned}$$

defined by

$$g(n) := \mathbb{P}(T_0^r = n \mid S_0 = 0), \qquad n \geq 1,$$

satisfies the convolution equation

$$h(n) = \sum_{k=0}^{n-2} g(n-k)h(k), \qquad n \geq 1, \tag{3.4.3}$$

with the initial condition $g(1) = 0$*, where* $h(n) := \mathbb{P}(S_n = 0 \mid S_0 = 0)$ *is given from* (3.3.3) *by*

$$h(2n) = \binom{2n}{n} p^n q^n, \quad \textit{and} \quad h(2n+1) = 0, \qquad n \in \mathbf{N}. \tag{3.4.4}$$

Proof We first partition the event $\{S_n = 0\}$ into

$$\{S_n = 0\} = \bigcup_{k=0}^{n-2} \big\{S_k = 0,\ S_{k+1} \neq 0, \ldots, S_{n-1} \neq 0,\ S_n = 0\big\}, \qquad n \geq 1,$$

according to all possible times $k = 0, 1, \ldots, n-2$ of *last* return to state ⓪ *before* time n, with $\{S_1 = 0\} = \emptyset$ since we are starting from $S_0 = 0$, see Fig. 3.4.

Then, we have

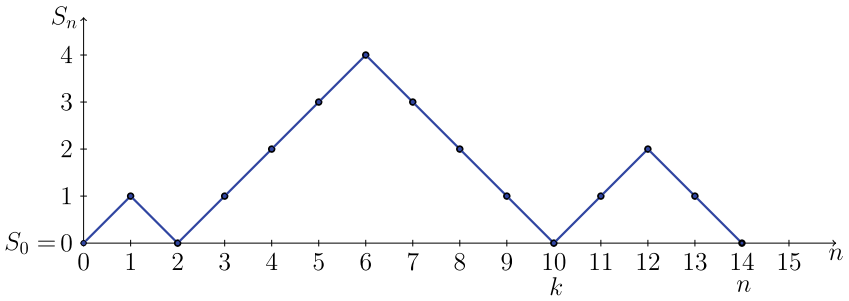


Fig. 3.4 Last return to state 0 at time $k = 10$

$$
\begin{aligned}
h(n) :&= \mathbb{P}(S_n = 0 \mid S_0 = 0) \\
&= \sum_{k=0}^{n-2} \mathbb{P}\big(S_k = 0,\ S_{k+1} \neq 0, \ldots, S_{n-1} \neq 0,\ S_n = 0 \mid S_0 = 0\big) \\
&= \sum_{k=0}^{n-2} \mathbb{P}\big(S_{k+1} \neq 0, \ldots, S_{n-1} \neq 0,\ S_n = 0 \mid S_k = 0,\ S_0 = 0\big)\mathbb{P}(S_k = 0 \mid S_0 = 0) \\
&= \sum_{k=0}^{n-2} \mathbb{P}\big(S_{k+1} \neq 0, \ldots, S_{n-1} \neq 0,\ S_n = 0 \mid S_k = 0\big)\mathbb{P}(S_k = 0 \mid S_0 = 0) \qquad (3.4.5) \\
&= \sum_{k=0}^{n-2} \mathbb{P}\big(S_1 \neq 0, \ldots, S_{n-k-1} \neq 0,\ S_{n-k} = 0 \mid S_0 = 0\big)\mathbb{P}(S_k = 0 \mid S_0 = 0) \qquad (3.4.6) \\
&= \sum_{k=0}^{n-2} \mathbb{P}(T_0^r = n-k \mid S_0 = 0)\mathbb{P}(S_k = 0 \mid S_0 = 0) \\
&= \sum_{k=0}^{n-2} h(k)g(n-k), \qquad n \geq 1, \qquad (3.4.7)
\end{aligned}
$$

where from (3.4.5) to (3.4.6) we applied a backward shift of k steps in time. □

We now need to solve the convolution equation (3.4.3) for $g(n) = \mathbb{P}(T_0^r = n \mid S_0 = 0)$, $n \geq 1$, knowing that $g(1) = 0$. For this we will derive a simple equation for the probability generating function

$$
\begin{aligned}
G_{T_0^r} : [-1, 1] &\longrightarrow \mathbf{R} \\
s &\longmapsto G_{T_0^r}(s)
\end{aligned}
$$

of the random variable T_0^r, defined by

$$G_{T_0^r}(s) := \mathbb{E}\left[s^{T_0^r}\mathbb{1}_{\{T_0^r<\infty\}}\,\middle|\,S_0=0\right] = \sum_{n\geq 0} s^n \mathbb{P}(T_0^r = n \mid S_0 = 0) = \sum_{n\geq 0} s^n g(n),$$

$-1 \leq s \leq 1$, cf. (1.7.3).

Recall that the knowledge of $G_{T_0^r}(s)$ provides certain information on the distribution of T_0^r, such as the probability

$$\mathbb{P}(T_0^r < \infty \mid S_0 = 0) = \mathbb{E}\left[\mathbb{1}_{\{T_0^r<\infty\}}\,\middle|\,S_0=0\right] = G_{T_0^r}(1)$$

and the expected value

$$\mathbb{E}\left[T_0^r\mathbb{1}_{\{T_0^r<\infty\}}\,\middle|\,S_0=0\right] = \sum_{n\geq 1} n\mathbb{P}(T_0^r = n \mid S_0 = 0) = G'_{T_0^r}(1^-).$$

In Lemma 3.2 below we will compute $G_{T_0^r}(s)$ for all $s \in [-1, 1]$. First, let the function

$$\begin{aligned} H : \mathbf{R} &\longrightarrow \mathbf{R} \\ s &\longmapsto H(s) \end{aligned}$$

be defined by

$$H(s) := \sum_{k\geq 0} h(k)s^k = \sum_{k\geq 0} s^k \mathbb{P}(S_k = 0 \mid S_0 = 0), \qquad -1 \leq s \leq 1.$$

In the following lemma we show that the function $H(s)$ can be computed in closed form.

Proposition 3.1 *We have*

$$H(s) = (1 - 4pqs^2)^{-1/2}, \qquad |s| < \frac{1}{2\sqrt{pq}}.$$

Proof By (3.4.4) and the fact that $\mathbb{P}(S_{2k+1} = 0 \mid S_0 = 0) = 0$, $k \in \mathbf{N}$, we have

$$
\begin{aligned}
H(s) &= \sum_{k\geq 0} s^k \mathbb{P}(S_k = 0 \mid S_0 = 0) \qquad (3.4.8)\\
&= \sum_{k\geq 0} s^{2k} \mathbb{P}(S_{2k} = 0 \mid S_0 = 0) = \sum_{k\geq 0} s^{2k} \binom{2k}{k} p^k q^k \\
&= \sum_{k\geq 0} (pq)^k s^{2k} \frac{(2k)(2k-1)(2k-2)(2k-3)\times\cdots\times 4\times 3\times 2\times 1}{(k(k-1)\times\cdots\times 2\times 1)^2} \\
&= \sum_{k\geq 0} (4pq)^k s^{2k} \frac{k(k-1/2)(k-2/2)(k-3/2)\times\cdots\times (4/2)\times (3/2)\times (2/2)\times (1/2)}{(k(k-1)\times\cdots\times 2\times 1)^2} \\
&= \sum_{k\geq 0} (4pq)^k s^{2k} \frac{(k-1/2)(k-3/2)\times\cdots\times (3/2)\times (1/2)}{k(k-1)\times\cdots\times 2\times 1} \\
&= \sum_{k\geq 0} (-1)^k (4pq)^k s^{2k} \frac{(-1/2-(k-1))(3/2-k)\times\cdots\times (-3/2)\times (-1/2)}{k(k-1)\times\cdots\times 2\times 1} \\
&= \sum_{k\geq 0} (-4pqs^2)^k \frac{(-1/2)\times (-3/2)\times\cdots\times (3/2-k)(-1/2-(k-1))}{k!} \\
&= (1-4pqs^2)^{-1/2},
\end{aligned}
$$

for $|4pqs^2| < 1$.[3]

Remark 3.1 We note that, taking $s = 1$, by (1.6.1) we have

$$
\begin{aligned}
H(1) &= \sum_{k\geq 0} \mathbb{P}(S_k = 0 \mid S_0 = 0) \\
&= \sum_{k\geq 0} \mathbb{E}[\mathbb{1}_{\{S_k=0\}} \mid S_0 = 0] \\
&= \mathbb{E}\left[\sum_{k\geq 0} \mathbb{1}_{\{S_k=0\}} \,\middle|\, S_0 = 0 \right],
\end{aligned}
$$

hence $H(1) = 1/\sqrt{1-4pq}$ represents the mean number of visits of the random walk $(S_n)_{n\geq 0}$ to state $\textcircled{0}$.

Next, based on the convolution equation (3.4.3) of Lemma 3.1 we compute $G_{T_0^r}(s)$ in Lemma 3.2 by deriving and solving an Eq. (3.4.13) for $G_{T_0^r}(s)$. This method has some similarities with the z-transform method used in electrical engineering.

Lemma 3.2 *The probability generating function $G_{T_0^r}$ of the first return time T_0^r to $\textcircled{0}$ is given by*

[3] We used the Taylor expansion $(1+x)^\alpha = \sum_{k\geq 0} \frac{x^k}{k!} \alpha(\alpha-1)\times\cdots\times(\alpha-(k-1))$, cf. Relation (A.8).

$$G_{T_0^r}(s) = 1 - \frac{1}{H(s)} = 1 - \sqrt{1 - 4pqs^2}, \qquad 4pqs^2 < 1. \quad (3.4.9)$$

Proof We have, taking into account the relation $g(1) = \mathbb{P}(T_0^r = 1 \mid S_0 = 0) = 0$,

$$\begin{aligned} G_{T_0^r}(s)H(s) &= \left(\sum_{n\geq 1} s^n g(n)\right)\left(\sum_{k\geq 0} s^k h(k)\right) \\ &= \sum_{n\geq 2}\sum_{k\geq 0} s^{n+k} g(n)h(k) = \sum_{k\geq 0}\sum_{n\geq 2} s^{n+k} g(n)h(k) \quad (3.4.10) \\ &= \sum_{l\geq 2} s^l \sum_{k=0}^{l-2} g(l-k)h(k) \quad (3.4.11) \\ &= \sum_{l\geq 1} s^l h(l) \\ &= \sum_{l\geq 1} s^l \mathbb{P}(S_l = 0 \mid S_0 = 0) \\ &= -1 + \sum_{l\geq 0} s^l \mathbb{P}(S_l = 0 \mid S_0 = 0) = H(s) - 1, \quad (3.4.12) \end{aligned}$$

where from line (3.4.10) to line (3.4.11) we have applied the change of variable $(k, n) \longmapsto (k, l)$ with $l = n + k$, and from line (3.4.11) to line (3.4.12) we have used the convolution equation (3.4.3) of Lemma 3.1. This shows that $G_{T_0^r}(s)$ satisfies the equation

$$G_{T_0^r}(s)H(s) = H(s) - 1, \qquad 4pqs^2 < 1. \quad (3.4.13)$$

Solving (3.4.13) yields the value of $G_{T_0^r}(s)$ for all s such that $4pqs^2 < 1$.

See Exercise 3.4-(e) for another derivation of (3.4.9) based on first step analysis.[4] We will apply our knowledge of $G_{T_0^r}(s)$ to the computation of the first return time distribution of T_0^r, the probability of return to ⓪ in finite time, and the mean return time to ⓪.

[4] "Any good theorem should have several proofs, the more the better. For two reasons: usually, different proofs have different strengths and weaknesses, and they generalise in different directions—they are not just repetitions of each other. Some of them are good for this application, some are good for that application. They all shed light on the area. If you cannot look at a problem from different directions, it is probably not very interesting; the more perspectives, the better !"—Sir Michael Atiyah.

Probability of return to zero in finite time
The probability that the first return to $\textcircled{0}$ occurs within a finite time is

$$\begin{aligned}\mathbb{P}\big(T_0^r < \infty \mid S_0 = 0\big) &= \mathbb{E}\left[\mathbb{1}_{\{T_0^r<\infty\}} \,\middle|\, S_0 = 0\right] \\ &= \mathbb{E}\left[1^{T_0^r}\mathbb{1}_{\{T_0^r<\infty\}} \,\middle|\, S_0 = 0\right] \\ &= G_{T_0^r}(1) = 1 - \sqrt{1-4pq} \\ &= 1 - |2p-1| = 1 - |p-q| = \begin{cases} 2q, & p \geq 1/2, \\ 2p, & p \leq 1/2, \end{cases} \\ &= 2\min(p,q), \end{aligned} \tag{3.4.14}$$

hence

$$\mathbb{P}(T_0^r = \infty \mid S_0 = 0) = |2p-1| = |p-q|. \tag{3.4.15}$$

see also the following Python code:

```
from sympy import *
import sympy as sp
k = sp.Symbol("k");p = sp.Symbol("p"); q = sp.Symbol("q")
prob=summation(p**k*q**k*factorial(2*k)/factorial(k)**2/(2*k-1), (k, 1, oo))
simplify(prob.args[0][0])
```

Note that in (2.2.14) above we have shown that the probability of hitting state $\textcircled{0}$ in finite time starting from any state $\textcircled{k}$ with $k \geq 1$ is given by

$$\mathbb{P}\big(T_0^r < \infty \mid S_0 = k\big) = \min\left(1, \left(\frac{q}{p}\right)^k\right), \qquad k \geq 1, \tag{3.4.16}$$

i.e.

$$\mathbb{P}(T_0^r = \infty \mid S_0 = k) = \max\left(0, 1 - \left(\frac{q}{p}\right)^k\right), \qquad k \geq 1,$$

see Exercise 3.2-(c), Exercise 3.4-(c) and Exercises 3.4-(b) and 5.9-(a) for other derivations of (3.4.16).

In the non-symmetric case $p \neq q$, Relation (3.4.14) shows that

$$\mathbb{P}(T_0^r < \infty \mid S_0 = 0) < 1 \quad \text{and} \quad \mathbb{P}(T_0^r = \infty \mid S_0 = 0) > 0,$$

whereas in the symmetric case (or fair game) $p = q = 1/2$ we find that

$$\mathbb{P}(T_0^r < \infty \mid S_0 = 0) = 1 \qquad \text{and} \qquad \mathbb{P}(T_0^r = \infty \mid S_0 = 0) = 0,$$

i.e. the random walk returns to $\textcircled{0}$ with probability one.

Mean return time to zero

(i) In the non-symmetric case $p \neq q$, by (3.4.15), the time T_0^r needed to return to state $\textcircled{0}$ is infinite with probability

$$\mathbb{P}(T_0^r = \infty \mid S_0 = 0) = |p - q| > 0,$$

hence the expected value[5]

$$\begin{aligned} \mathbb{E}[T_0^r \mid S_0 = 0] &= \infty \times \mathbb{P}(T_0^r = \infty \mid S_0 = 0) + \sum_{k \geq 1} k \mathbb{P}(T_0^r = k \mid S_0 = 0) \\ &= \infty \end{aligned} \tag{3.4.17}$$

is infinite in that case.
Note that starting from $S_0 = k \geq 1$, by (2.3.13) we have found that the mean hitting time of state $\textcircled{0}$ equals

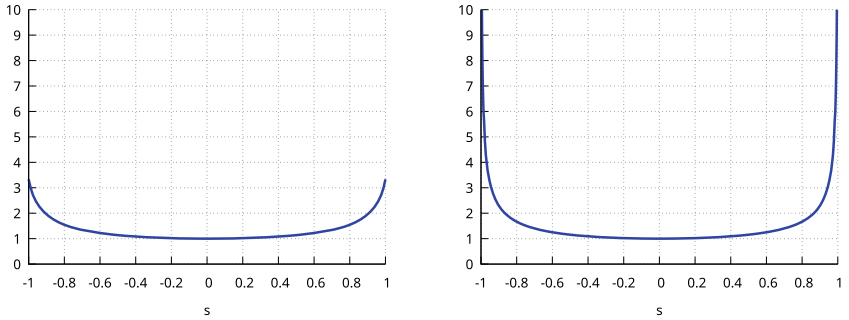
$$\mathbb{E}[T_0^r \mid S_0 = k] = \begin{cases} \infty & \text{if } q \leq p, \\ \dfrac{k}{q - p} & \text{if } q > p. \end{cases} \tag{3.4.18}$$

In particular, we have $\mathbb{P}(T_0^r < \infty \mid S_0 = k) = 1$ when $q > p$ and $k \geq 1$, which is consistent with (3.4.16). See Exercise 5.9-(b) for other derivations of (2.3.13)–(3.4.18) using the probability generating function $s \mapsto G_{T_0^r}(s)$.

Remark 3.2 By (3.4.9), the truncated expectation $\mathbb{E}\left[T_0^r \mathbb{1}_{\{T_0^r < \infty\}} \mid S_0 = 0\right]$ satisfies

$$\begin{aligned} \mathbb{E}\left[T_0^r \mathbb{1}_{\{T_0^r < \infty\}} \mid S_0 = 0\right] &= \sum_{n \geq 1} n \mathbb{P}(T_0^r = n \mid S_0 = 0) \\ &= G'_{T_0^r}(1^-) \\ &= \frac{4pqs}{\sqrt{1 - 4pqs^2}}\bigg|_{s=1} \\ &= \frac{4pq}{\sqrt{1 - 4pq}} \\ &= \frac{4pq}{|p - q|}, \end{aligned} \tag{3.4.19}$$

[5] Note that the summation $\sum_{k \geq 1} = \sum_{1 \leq k < \infty}$ actually *excludes* the value $k = \infty$.



(a) Graph of $G_{T_0^r}(s)$ with $p = 0.35$. (b) Graph of $G_{T_0^r}(s)$ with $p = 0.5$.

Fig. 3.5 Probability generating functions of T_0^r for p = 0.35 and p = 0.5

when $p \neq q$, see for example Fig. 3.5a and the following Python code:

```
from sympy import *
import sympy as sp
n = sp.Symbol("n");p = sp.Symbol("p"); q = sp.Symbol("q")
expectation=summation(2*n*p**n*q**n*factorial(2*n)/factorial(n)**2/(2*n-1),
     (n, 1, oo))
expectation.args[1].args[0][0]
```

which shows in particular from (3.4.14) and Lemma 1.10 that

$$\begin{aligned}\mathbb{E}[T_0^r \mid T_0^r < \infty,\ S_0 = 0] &= \frac{1}{\mathbb{P}(T_0^r < \infty \mid S_0 = 0)}\,\mathbb{E}[T_0^r \mathbb{1}_{\{T_0^r<\infty\}} \mid S_0 = 0] \\ &= \frac{1}{\min(p,q)} \times \frac{2pq}{|p-q|} = 2\frac{\max(p,q)}{|p-q|},\end{aligned}$$

see (3.4.14).

(ii) In the symmetric case $p = q = 1/2$ we have $\mathbb{P}(T_0^r < \infty \mid S_0 = 0) = 1$ and

$$\mathbb{E}[T_0^r \mid S_0 = 0] = \mathbb{E}\left[T_0^r \mathbb{1}_{\{T_0^r<\infty\}} \middle| S_0 = 0\right] = G'_{T_0^r}(1^-) = \infty \tag{3.4.20}$$

as the slope of $s \mapsto G_{T_0^r}(s)$ in Fig. 3.5b is infinite at $s = 1$, or by taking the limit as $p, q \to 1/2$ in (3.4.18) or (3.4.19).

When $p = q = 1/2$ the random walk returns to state ⓪ with probability *one* within a *finite* (random) time, while the average of this random time is *infinite*. This yields another example of a random variable T_0^r which is almost surely finite, while its expectation is infinite as in the St. Petersburg paradox.

```
nsim <- 10000; N=1000000; T<-2.0; t <- 0:(N-1); dt <- 1; mean=0.0;
for (i in 1:nsim){signal=0;colour="blue";
Z <- 2*(rbinom(N,1,0.5)-0.5);X <- c(1,N+1);X[1]=0;j=1;
while (j<N && signal==0){j=j+1;X[j]=X[j-1]+Z[j];if (X[j]==0)
     {signal=1;mean=mean+j-1}}
plot(t[1:j], X[1:j], xlab = "t", ylab = "", type = "p", ylim =
     c(min(X[1:j])-max(X
[1:j]),-min(X[1:j])+max(X[1:j])), col = colour,main=paste("Time=",j-1,", Mean=
",mean,"/",i,"=",round(mean/i, digits=1)), xaxs="i", xaxt="n",lwd=3)
lines(t[1:j], X[1:j], type = "l",col="blue",lwd=2)
axis(side=1, at=c(0:j), c(0:j));axis(side=1, pos=0, at=c(0:j), c(0:j))
text((j-1)/2,0.5,paste(j-1),cex=5);
readline(prompt = "Pause. Press <Enter>...")}
```

This shows how even a fair game can be risky when the player's wealth is negative, as it will take on average an infinite time to recover the losses.

First return time distribution

Proposition 3.2 can also be obtained from the path counting result of Exercise 3.7, which shows that the number of path linking to ⓪ over $2k$ time steps without hitting ⓪ in between is given by $\dfrac{1}{2k-1}\binom{2k}{k}$, $k \geq 1$.

Proposition 3.2 *The probability distribution* $\mathbb{P}\big(T_0^r = n \mid S_0 = 0\big)$ *of the first return time* T_0^r *to* ⓪ *is given by*

$$\mathbb{P}(T_0^r = 2k \mid S_0 = 0) = \frac{1}{2k-1}\binom{2k}{k}(pq)^k, \qquad k \geq 1, \tag{3.4.21}$$

with $\mathbb{P}\big(T_0^r = 2k+1 \mid S_0 = 0\big) = 0$, $k \in \mathbf{N}$.

Proof By applying a Taylor expansion to $s \longmapsto 1-(1-4pqs^2)^{1/2}$ in (3.4.9), we get

$$\begin{aligned} G_{T_0^r}(s) &= 1-(1-4pqs^2)^{1/2} \\ &= 1-\sum_{k\geq 0}\frac{1}{k!}(-4pqs^2)^k\left(\frac{1}{2}-0\right)\left(\frac{1}{2}-1\right)\times\cdots\times\left(\frac{1}{2}-(k-1)\right) \\ &= \frac{1}{2}\sum_{k\geq 1}s^{2k}\frac{(4pq)^k}{k!}\left(1-\frac{1}{2}\right)\times\cdots\times\left(k-1-\frac{1}{2}\right), \end{aligned} \tag{3.4.22}$$

where we used (A.8) for $\alpha = 1/2$. By identification of (3.4.22) with the expansion

$$G_{T_0^r}(s) = \sum_{n\geq 0} s^n \mathbb{P}(T_0^r = n \mid S_0 = 0), \qquad -1 \leq s \leq 1,$$

we obtain

$$\begin{aligned}\mathbb{P}(T_0^r = 2k \mid S_0 = 0) &= g(2k)\\ &= \frac{(4pq)^k}{k!}\frac{1}{2}\left(1-\frac{1}{2}\right)\times\cdots\times\left(k-1-\frac{1}{2}\right)\\ &= \frac{(4pq)^k}{2k!}\prod_{m=1}^{k-1}\left(m-\frac{1}{2}\right)\\ &= \frac{1}{2k-1}\binom{2k}{k}(pq)^k, \qquad k \geq 1,\end{aligned}$$

while $\mathbb{P}\big(T_0^r = 2k+1 \mid S_0 = 0\big) = g(2k+1) = 0$, $k \in \mathbf{N}$. This conclusion could also be obtained using (1.7.10) from the relation

$$\mathbb{P}(T_0^r = n \mid S_0 = 0) = \frac{1}{n!}\frac{\partial^n}{\partial s^n} G_{T_0^r}(s)_{|s=0}, \qquad n \geq 0.$$

□

Exercise: Check that the formula (3.4.21) recovers (3.4.1) and (3.4.2) when $k = 0, 1, 2$.

Using the independence of increments of the random walk $(S_n)_{n\geq 0}$, one can also show that the probability generating function of the first passage time

$$T_k := \inf\{n \geq 0 \ : \ S_n = k\}$$

to any level $k \geq 1$ is given by

$$G_{T_k}(s) = \left(\frac{1-\sqrt{1-4pqs^2}}{2qs}\right)^k, \qquad 4pqs^2 < 1, \quad q \leq p, \tag{3.4.23}$$

from which the distribution of T_k can be computed given the series expansion of $G_{T_k}(s)$, cf. Exercise 3.4-(b) with $i := -k$.

The gambling process of Chap. 2 and the standard random walk $(S_n)_{n\geq 0}$ will later be reconsidered as particular cases in the general framework of Markov chains of Chaps. 4 and 5.

Exercises

Exercise 3.1 We consider the simple random walk $(S_n)_{n\geq 0}$ of Sect. 3.1 with independent increments and started at $S_0 = 0$, in which the probability of advance is p and the probability of retreat is $1 - p$.

(a) Enumerate all possible sample paths that conduct to $S_4 = 2$ starting from $S_0 = 0$.

(b) Show that

$$\mathbb{P}(S_4 = 2 \mid S_0 = 0) = \binom{4}{3} p^3(1-p) = \binom{4}{1} p^3(1-p).$$

(c) Show that we have

$$\begin{aligned} &\mathbb{P}(S_n = k \mid S_0 = 0) \qquad (3.4.24)\\ &= \begin{cases} \binom{n}{(n+k)/2} p^{(n+k)/2}(1-p)^{(n-k)/2}, & n+k \text{ even and } |k| \leq n, \\ 0, & n+k \text{ odd or } |k| > n. \end{cases} \end{aligned}$$

(d) Show, by a direct argument using a "last step" analysis at time $n+1$ on random walks, that $p_{n,k} := \mathbb{P}(S_n = k \mid S_0 = 0)$ satisfies the difference equation

$$p_{n+1,k} = p p_{n,k-1} + q p_{n,k+1}, \qquad (3.4.25)$$

subject to the boundary conditions $p_{0,0} = 1$ and $p_{0,k} = 0$, $k \neq 0$.

(e) Confirm that $p_{n,k} = \mathbb{P}(S_n = k \mid S_0 = 0)$ given by (3.4.24) satisfies the equation (3.4.25) and its boundary conditions.

Exercise 3.2 Consider a random walk $(S_n)_{n\geq 0}$ on $\mathbb{Z}$ with independent increments and probabilities p, resp. $q = 1-p$ of moving up by one step, resp. down by one step. Let

$$T_0 = \inf\{n \geq 0 \ : \ S_n = 0\}$$

denote the hitting time of state $\textcircled{0}$.

(a) Explain why for any $k \geq 1$ we have

$$\mathbb{E}[T_0 \mid S_0 = k] = k\, \mathbb{E}[T_0 \mid S_0 = 1],$$

and compute $\mathbb{E}[T_0 \mid S_0 = 1]$ using first step analysis when $q > p$. What can we conclude when $p \geq q$?

(b) Explain why, by the Markov property, we have

$$\mathbb{P}(T_0 < \infty \mid S_0 = k) = (\mathbb{P}(T_0 < \infty \mid S_0 = 1))^k, \qquad k \geq 1.$$

(c) Using first step analysis for random walks, show that $\alpha := \mathbb{P}(T_0 < \infty \mid S_0 = 1)$ satisfies the quadratic equation

$$p\alpha^2 - \alpha + q = p(\alpha - q/p)(\alpha - 1) = 0,$$

and give the values of $\mathbb{P}(T_0 < \infty \mid S_0 = 1)$ and $\mathbb{P}(T_0 = \infty \mid S_0 = 1)$ in the cases $p < q$ and $p \geq q$ respectively.

Exercise 3.3 Consider the random walk

$$S_n := X_1 + \cdots + X_n, \qquad n \geq 1,$$

with $S_0 = 0$, where $(X_k)_{k\geq 1}$ is a sequence of Bernoulli random variables with

$$\mathbb{P}(X_k = 1) = p \in (0,1), \qquad \mathbb{P}(X_k = -1) = q \in (0,1),$$

and $p + q = 1$. Recall that the probability generating function (PGF)

$$G_{T_0^r}(s) = \sum_{k\geq 0} s^k \mathbb{P}(T_0^r = k), \qquad s \in [-1,1], \tag{3.4.26}$$

of the first return time $T_0^r := \inf\{S_n = 0 \ : \ n \geq 1\}$ to state ⓪ is given by

$$G_{T_0^r}(s) = 1 - \sqrt{1 - 4pqs^2}, \qquad s \in [-1,1]. \tag{3.4.27}$$

(a) Compute $\mathbb{P}\big(T_0^r = 0\big)$ and $\mathbb{P}\big(T_0^r < \infty\big)$ from $G_{T_0^r}$.
(b) By differentiation of (3.4.26) and (3.4.27), compute $\mathbb{P}\big(T_0^r = 1\big)$, $\mathbb{P}\big(T_0^r = 2\big)$, $\mathbb{P}\big(T_0^r = 3\big)$ and $\mathbb{P}\big(T_0^r = 4\big)$ using the probability generating function $G_{T_0^r}$.
(c) Compute $\mathbb{E}\left[T_0^r \mid T_0^r < \infty\right]$ using the probability generating function $G_{T_0^r}$.

Exercise 3.4 Consider a simple random walk $(X_n)_{n\geq 0}$ on $\mathbb{Z}$ with respective probabilities p and q of increment and decrement. Let

$$T_0 := \inf\{n \geq 0 \ : \ X_n = 0\}$$

denote the first hitting time of state ⓪, and consider the probability generating function

$$G_i(s) := \mathbb{E}\left[s^{T_0} \mid X_0 = i\right], \qquad -1 < s < 1, \quad i \in \mathbb{Z}.$$

(a) By a first step analysis argument, find the finite difference equation satisfied by $G_i(s)$, and its boundary condition(s) at $i = 0$ and $i = \pm\infty$.
(b) Find the value of $G_i(s)$ for all $i \in \mathbb{Z}$ and $s \in (0,1)$, and recover the result of (3.4.23) on the probability generating function of the hitting time T_0 of ⓪ starting from state ⓘ.
(c) Recover Relations (2.2.14)–(3.4.16) using $G_i(s)$.
(d) Recover Relations (2.3.13)–(3.4.18) by differentiation of $s \mapsto G_i(s)$.
(e) Recover the result of (3.4.9) on the probability generating function of the return time T_0^r to ⓪.

Exercise 3.5 Using the probability distribution (3.4.21) of T_0^r, recover the fact that $\mathbb{E}[T_0^r \mathbb{1}_{\{T_0^r < \infty\}} \mid S_0 = 0] = \infty$, when $p = q = 1/2$.

Exercise 3.6 Consider a sequence $(X_k)_{k\geq 1}$ of independent Bernoulli random variables with

$$\mathbb{P}(X_k = 1) = p, \quad \text{and} \quad \mathbb{P}(X_k = -1) = q, \qquad k \geq 1,$$

where $p + q = 1$, and let the process $(M_n)_{n\geq 0}$ be defined by $M_0 := 0$ and

$$M_n := \sum_{k=1}^{n} 2^{k-1} X_k, \qquad n \geq 1.$$

(a) Compute $\mathbb{E}[M_n]$ for all $n \geq 0$.
(b) Consider the hitting time $\tau := \inf\{n \geq 1 \ : \ M_n = 1\}$ and the stopped process

$$M_{\min(n,\tau)} = M_n \mathbb{1}_{\{n<\tau\}} + \mathbb{1}_{\{\tau\leq n\}}, \qquad n \geq 0.$$

Determine the possible values of $M_{\min(n,\tau)}$, and the probability distribution of $M_{\min(n,\tau)}$ at any time $n \geq 1$.
(c) Give an interpretation of the stopped process $(M_{\min(n,\tau)})_{n\geq 0}$ in terms of strategy in a game started at $M_0 = 0$.
(d) Based on the result of Question (b), compute $\mathbb{E}[M_{\min(n,\tau)}]$ for all $n \geq 1$.

Exercise 3.7 Consider a random walk $(S_n)_{n\geq 0}$ on $\mathbb{Z}$ with increments ± 1, started at $S_0 = 0$. Recall that the number of paths joining states (0) and $\boxed{2k}$ over $2m$ time steps is

$$\binom{2m}{m+k}. \tag{3.4.28}$$

(a) Compute the total number of paths joining $S_1 = 1$ to $S_{2n-1} = 1$.

Hint: Apply the formula (3.4.28).
(b) Compute the total number of paths joining $S_1 = 1$ to $S_{2n-1} = -1$.

Hint: Apply the formula (3.4.28).
(c) Show that to every one path joining $S_1 = 1$ to $S_{2n-1} = 1$ by *crossing or hitting* (0) we can associate one path joining $S_1 = 1$ to $S_{2n-1} = -1$, in a one-to-one correspondence.

Hint: Draw a sample path joining $S_1 = 1$ to $S_{2n-1} = 1$, and reflect it in such a way that the reflected path then joins $S_1 = 1$ to $S_{2n-1} = -1$.
(d) Compute the total number of paths joining $S_1 = 1$ to $S_{2n-1} = 1$ by *crossing or hitting* (0).

Hint: Combine the answers to part (b) and part (c).
(e) Compute the total number of paths joining $S_1 = 1$ to $S_{2n-1} = 1$ *without* crossing or hitting (0).

Hint: Combine the answers to part (a) and (d).

(f) Give the total number of paths joining $S_0 = 0$ to $S_{2n} = 0$ *without* crossing or hitting $\textcircled{0}$ between time 1 and time $2n - 1$.

Hint: Apply two times the answer to part (e). A drawing is recommended.

Problem 3.8 Range process. Consider the random walk $(S_n)_{n\geq 0}$ defined by $S_0 = 0$ and

$$S_n := X_1 + \cdots + X_n, \qquad n \geq 1,$$

where $(X_k)_{k\geq 1}$ is an *i.i.d.*[6] family of $\{-1, +1\}$-valued random variables with distribution

$$\begin{cases} \mathbb{P}(X_k = +1) = p, \\ \mathbb{P}(X_k = -1) = q, \end{cases}$$

$k \geq 1$, where $p + q = 1$. We let R_n denote the *range* of $(S_0, S_1, \ldots, S_n)$, i.e. the (random) number of distinct values appearing in the sequence $(S_0, S_1, \ldots, S_n)$.

(a) Explain why

$$R_n = 1 + \left(\sup_{k=0,1,\ldots,n} S_k\right) - \left(\inf_{k=0,1,\ldots,n} S_k\right),$$

and give the value of R_0 and R_1.

(b) Show that for all $k \geq 1$, $R_k - R_{k-1}$ is a Bernoulli random variable, and that

$$\mathbb{P}(R_k - R_{k-1} = 1) = \mathbb{P}(S_k - S_0 \neq 0, S_k - S_1 \neq 0, \ldots, S_k - S_{k-1} \neq 0).$$

(c) Show that for all $k \geq 1$ we have

$$\mathbb{P}(R_k - R_{k-1} = 1) = \mathbb{P}(X_1 \neq 0, X_1 + X_2 \neq 0, \ldots, X_1 + \cdots + X_k \neq 0).$$

(d) Show why the telescoping identity $R_n = R_0 + \sum_{k=1}^{n} (R_k - R_{k-1})$ holds for all $n \in \mathbf{N}$.

(e) Show that $\mathbb{P}(T_0^r = \infty) = \lim_{k\to\infty} \mathbb{P}(T_0^r > k)$.

(f) From the results of Questions (c) and (d), show that

$$\mathbb{E}[R_n] = \sum_{k=0}^{n} \mathbb{P}(T_0^r > k), \qquad n \geq 0,$$

where $T_0^r = \inf\{n \geq 1 \ : \ S_n = 0\}$ is the time of first return to $\textcircled{0}$ of the random walk.

[6] independent and identically distributed.

(g) From the results of Questions (e) and (f), show that

$$\mathbb{P}(T_0^r = \infty) = \lim_{n\to\infty} \frac{1}{n}\, \mathbb{E}[R_n].$$

(h) Show that

$$\lim_{n\to\infty} \frac{1}{n}\, \mathbb{E}[R_n] = 0.$$

when $p = 1/2$, and that $\mathbb{E}[R_n] \simeq_{n\to\infty} n|p-q|$, when $p \neq 1/2$.[7]

Hints and comments

(a) No mathematical computation is needed here, a credible explanation (in words) is sufficient. It may be of interest to also compute $\mathbb{E}[R_2]$.
(b) Show first that the two events

$$\{R_k - R_{k-1} = 1\} \quad \text{and} \quad \{S_k - S_0 \neq 0, S_k - S_1 \neq 0, \ldots, S_k - S_{k-1} \neq 0\}$$

are the same.
(c) Here, the events

$$\{R_k - R_{k-1} = 1\} \quad \text{and} \quad \{X_1 \neq 0, X_1 + X_2 \neq 0, \ldots, X_1 + \cdots + X_k \neq 0\}$$

are not the same, however, we can show the equality between the probabilities.
(d) Telescoping identities can be useful in many situations, not restricted to probability or stochastic processes.
(e) Use point 2 on Sect. 1.3, after identifying what the events A_k are.
(f) The basic identity $\mathbb{E}[\mathbb{1}_A] = \mathbb{P}(A)$ can be used.
(g) A mathematically rigorous proof is asked here. The following definition of limit may be used:

A real number a is said to be the limit of the sequence $(x_n)_{n\geq 1}$, written $a = \lim_{n\to\infty} x_n$, if and only if for every real number $\varepsilon > 0$, there exists a natural number N such that for every $n > N$ we have $|x_n - a| < \varepsilon$.

Alternatively, we may use the notion of Cesàro mean and state and apply the relevant theorem.
(h) Use the formula (3.4.15) giving $\mathbb{P}(T_0^r = \infty)$.

In Fig. 3.6 the height at time n of the colored area coincides with $R_n - 1$.

Problem 3.9 Recurrent random walk in d dimensions. Let $\{e_1, e_2, \ldots, e_d\}$ denote the canonical basis of $\mathbf{R}^d$, i.e.

$$e_k = (0, \ldots, \underset{\substack{\uparrow \\ \mathbf{k}}}{1}, 0, \ldots, 0), \qquad k = 1, 2, \ldots, d.$$

[7] The meaning of $f(n) \simeq_{n\to\infty} g(n)$ is $\lim_{n\to\infty} f(n)/g(n) = 1$, provided that $g(n) \neq 0, n \geq 1$.

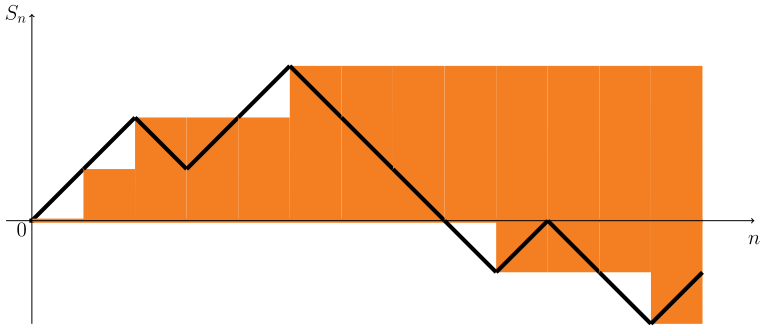


Fig. 3.6 Illustration of the range process

Consider the symmetric $\mathbb{Z}^d$-valued random walk

$$S_n = X_1 + \cdots + X_n, \qquad n \geq 0,$$

started at $S_0 = \mathbf{0} = (0, 0, \ldots, 0)$, where $(X_n)_{n\geq 1}$ is a sequence of (mutually) independent uniformly distributed random variables

$$X_n \in \{e_1, e_2, \ldots, e_d, -e_1, -e_2, \ldots, -e_d\}, \qquad n \geq 1,$$

with distribution

$$\mathbb{P}(X_n = e_k) = \mathbb{P}(X_n = -e_k) = \frac{1}{2d}, \qquad k = 1, 2, \ldots, d.$$

Let

$$T_{\mathbf{0}}^r := \inf\{n \geq 1 \ : \ S_n = \mathbf{0}\}$$

denote the first return time of $(S_n)_{n\geq 0}$ to $\mathbf{0} = (0, 0, \ldots, 0)$. The random walk is said to be *recurrent* if $\mathbb{P}(T_{\mathbf{0}}^r < \infty) = 1$.

(a) Show that the probability distribution $\mathbb{P}\big(T_{\mathbf{0}}^r = n\big)$, $n \geq 1$, satisfies the equation

$$\mathbb{P}\big(S_n = \mathbf{0}\big) = \sum_{k=2}^{n} \mathbb{P}\big(T_{\mathbf{0}}^r = k\big)\mathbb{P}\big(S_{n-k} = \mathbf{0}\big), \qquad n \geq 1.$$

(b) Show that

$$\sum_{n=2}^{m} \mathbb{P}\big(T_{\mathbf{0}}^r = n\big) \geq 1 - \frac{1}{\displaystyle\sum_{n=0}^{m} \mathbb{P}\big(S_n = \mathbf{0}\big)}, \qquad m \geq 1. \tag{3.4.29}$$

Hint: Start by showing that

$$\sum_{n=1}^{m} \mathbb{P}\big(S_n = \mathbf{0}\big) = \sum_{k=2}^{m} \mathbb{P}\big(T_{\mathbf{0}}^r = k\big) \sum_{l=0}^{m-k} \mathbb{P}\big(S_l = \mathbf{0}\big).$$

(c) Show that under the condition

$$\sum_{n\geq 0} \mathbb{P}\big(S_n = \mathbf{0}\big) = \infty,$$

we have $\mathbb{P}\big(T_{\mathbf{0}}^r < \infty\big) = 1$.

(d) Show that

$$\sum_{n=2}^{m} \mathbb{P}\big(T_{\mathbf{0}}^r = n\big) \leq \frac{\displaystyle\sum_{n=2}^{2m} \mathbb{P}\big(S_n = \mathbf{0}\big)}{\displaystyle\sum_{n=0}^{m} \mathbb{P}\big(S_n = \mathbf{0}\big)}, \qquad m \geq 1. \tag{3.4.30}$$

Hint: Start by showing that

$$\sum_{n=1}^{2m} \mathbb{P}\big(S_n = \mathbf{0}\big) = \sum_{n=2}^{2m} \mathbb{P}\big(T_{\mathbf{0}}^r = n\big) \sum_{l=0}^{2m-n} \mathbb{P}\big(S_l = \mathbf{0}\big).$$

(e) Show that under the condition

$$\sum_{n\geq 0} \mathbb{P}\big(S_n = \mathbf{0}\big) < \infty,$$

we have $\mathbb{P}\big(T_{\mathbf{0}}^r = \infty\big) > 0$.

(f) When $d = 1$, compute $\mathbb{P}(S_{2n} = 0)$, $n \geq 1$, and show that the one-dimensional random walk is *recurrent*, i.e. we have $\mathbb{P}(T_0^r < \infty) = 1$.

Hint: Use Stirling's approximation $n! \simeq (n/e)^n \sqrt{2\pi n}$ as n tends to ∞.

(g) When $d = 2$, show that we have

$$\mathbb{P}\big(S_{2n} = \mathbf{0}\big) = \frac{(2n)!}{4^{2n}(n!)^2} \sum_{k=0}^{n} \binom{n}{k}^2, \qquad n \geq 1, \tag{3.4.31}$$

and that the two-dimensional random walk is *recurrent*, i.e. we have $\mathbb{P}\big(T_{\mathbf{0}}^r < \infty\big) = 1$.

Hint: Use the combinatorial identity

$$\binom{2n}{n} = \sum_{k=0}^{n} \binom{n}{k}^2$$

and Stirling's approximation.

(h) Given $i_1, i_2, \ldots, i_d \in \mathbf{N}$, count all paths starting from $\mathbf{0}$ and returning to $\mathbf{0}$ via i_k "forward" steps in the direction e_k and i_k "backward" steps in the direction $-e_k$, $k = 1, 2, \ldots, d$. Deduce an expression for $\mathbb{P}\big(S_{2n} = \mathbf{0}\big)$ that generalizes (3.4.31) to all $d \geq 2$.

Hint: Use multinomial coefficients.

(i) Based on the Euclidean division $n = a_n d + b_n$ where $b_n \in \{0, 1, \ldots, d-1\}$, show that we have

$$\sum_{n\geq 1} \mathbb{P}\big(S_{2n} = \mathbf{0}\big) \leq \sum_{n\geq 1} \frac{(2n)!}{2^{2n} d^n n! (a_n!)^d a_n^{b_n}}$$

Hint: Use the bound

$$i_1! i_2! \cdots i_d! \geq (a_n!)^d (a_n + 1)^{b_n}$$

which is valid for any $n = i_1 + \cdots + i_d$, and the identity

$$d^n = \sum_{\substack{i_1+\cdots+i_d=n \\ i_1, i_2, \ldots, i_d \geq 0}} \frac{n!}{i_1! \cdots i_d!}.$$

(j) Applying Stirling's approximation to $n!$, $(2n)!$ and $a_n!$, show that there exists a constant $C > 0$ such that for all n sufficiently large we have

$$\frac{(2n)!}{2^{2n} d^n n! (a_n!)^d} \leq \frac{C}{n^{d/2}}.$$

Hint: We have $\lim_{m\to\infty}(1 + x/m)^m = \mathrm{e}^x$ for all $x \in \mathbf{R}$.

(k) Is the random walk recurrent when $d \geq 3$?

Problem 3.10 (Benjamini and Wilson 2003, Antal and Redner 2005). Random walks in a cookie environment, also called excited random walks, can be used to model the behavior of primitive organisms.

In the absence of cookies the random walk is symmetric, with probabilities $1/2$ of going up and down, and it can *rebound* to $\textcircled{1}$ with probability $1/2$ after hitting state $\textcircled{0}$.

When the random walk encounters a cookie, its behavior becomes modified and it restarts with probabilities p and $q = 1 - p$ of moving up, resp. down, where $p \in [0, 1]$. Every encountered cookie is eaten by the organism, and when the random walk reaches an empty spot it restarts with equal probabilities $1/2$ of moving up or

down. In this case, the organism wanders without a preferred direction. The random walk is *attracted* by the cookies when $p > 1/2$, and *repulsed* when $p < 1/2$.

(a) Does the cookie random walk $(S_n)_{n\geq 0}$ have the Markov property? Explain your answer.

(b) Suppose that at time $n \geq 1$ the random walk has just eaten a cookie at state $x \geq 1$, after eating all cookies at states $1, 2, \ldots, x-1$. Show that in this case, the probability of reaching $\boxed{x+1}$ before reaching $\textcircled{0}$ is given by

$$p + (1-p)\frac{x-1}{x+1} = 1 - \frac{2q}{x+1}, \qquad x \geq 1.$$

Hint: Use first step analysis together with formula (2.2.13) Lemma 2.2, see also Remark 2.3.

(c) For any $x \geq 1$, let τ_x denote the first hitting time

$$\tau_x := \inf\{n \geq 1 \ : \ S_n = x\}.$$

Give the value of $\mathbb{P}(\tau_1 < \tau_0 \mid S_0 = 0)$, and show that for all $x \geq 1$ we have[8]

$$\mathbb{P}(\tau_x < \tau_0 \mid S_0 = 0) = \frac{\mathbf{1}}{\mathbf{2}} \exp\left(\sum_{l=2}^{x} \log\left(1 - \frac{2q}{l}\right)\right), \qquad x \geq 1,$$

where "log" denotes the *natural logarithm* "ln".

(d) Show the bound $\mathbb{P}(\tau_x < \tau_0 \mid S_0 = 0) \leq (x/2)^{-2q}$ for $x \geq 2$.

Hint: Use comparison arguments between integrals and series.

(e) Show that $\mathbb{P}(\tau_0 < \infty \mid S_0 = 0) = 1$, provided that $p < 1$.[9]

Hint: Consider $\mathbb{P}(\tau_0 \leq \tau_x \mid S_0 = 0)$ and use Relation (1.2.5) page 6.

(f) Starting from state $\textcircled{0}$, compute the mean time needed by the random walk to reach state $\textcircled{1}$.

Hint: Apply first step analysis, or locate and use the relevant result in Problem 2.7.

(g) Suppose that a cookie has just been eaten at state $x \geq 1$, after eating all cookies at states $1, 2, \ldots, x-1$. Show that the mean time to reach the next cookie at state $\boxed{x+1}$ is $1 + q(4x+2)$.

Hint: Locate and apply the relevant results in Problem 2.7 together with first step analysis.

(h) Show that the mean time to reach state $\textcircled{x}$ starting from $\textcircled{0}$ is given by

$$\mathbb{E}[\tau_x \mid S_0 = 0] = 1 - 2q + x + 2qx^2, \qquad x \geq 1.$$

[8] We use the convention $\sum_{k=2}^{1} a_k = 0$ for any sequence (a_k).

[9] In this case, the cookie random walk is said to be *recurrent*.

Hint: Perform a summation on the results of Questions (3.10) and (3.10).

(i) Suppose that a cookie has just been eaten at state $x \geq 1$, after eating all cookies at states $1, 2, \ldots, x-1$. Show that, *given one does not hit* $\textcircled{0}$, the probabilities of moving up to $\boxed{x+1}$, resp. down to $\boxed{x-1}$, are

$$\frac{p}{1-2q/(x+1)} \quad \text{and} \quad \frac{q(x-1)/(x+1)}{1-2q/(x+1)}, \qquad x \geq 1.$$

Hint: Proceed similarly to Question (g) of Problem 2.7.

(j) Suppose that a cookie has just been eaten at state $x \geq 1$, after eating all cookies at states $1, 2, \ldots, x-1$. Show that the mean time to reach the next cookie at state $\boxed{x+1}$ *given one does not hit* $\textcircled{0}$ is

$$1 + \frac{q-2q/(x+1)}{1-2q/(x+1)} \frac{4x}{3}, \qquad x \geq 1.$$

Hint: Locate and use the relevant result in Problem 2.7.

Chapter 4
Discrete-Time Markov Chains



In this chapter we start the general study of discrete-time Markov chains by focusing on the Markov property and on the role played by transition probability matrices. We also include a complete study of the time evolution of the two-state chain, which represents the simplest example of Markov chain.

4.1 Markov Property

We consider a discrete-time stochastic process $(Z_n)_{n\geq 0}$ taking values in a discrete state space $\mathbb{S}$, typically $\mathbb{S} = \mathbb{Z}$. The $\mathbb{S}$-valued process $(Z_n)_{n\geq 0}$ is said to be Markov, or to have the *Markov property* if, for all $n \geq 1$, the probability distribution of Z_{n+1} is determined by the state Z_n of the process at time n, and does not depend on the past values of Z_k for $k = 0, 1, \ldots, n-1$.

In other words, for all $n \geq 1$ and all $i_0, i_1, \ldots, i_n, j \in \mathbb{S}$ we have

$$\mathbb{P}(Z_{n+1} = j \mid Z_n = i_n,\ Z_{n-1} = i_{n-1}, \ldots, Z_0 = i_0) = \mathbb{P}(Z_{n+1} = j \mid Z_n = i_n).$$

In particular, we have

$$\mathbb{P}(Z_{n+1} = j \mid Z_n = i_n,\ Z_{n-1} = i_{n-1}) = \mathbb{P}(Z_{n+1} = j \mid Z_n = i_n),$$

and

$$\mathbb{P}(Z_2 = j \mid Z_1 = i_1, Z_0 = i_0) = \mathbb{P}(Z_2 = j \mid Z_1 = i_1).$$

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https://doi.org/10.1007/978-981-95-4057-0_4

Note that this feature is apparent in the statement of Lemma 2.2. In addition, we have the following facts.

1. *Chain rule.* The first order transition probabilities can be used for the complete computation of the probability distribution of the process as

$$\begin{aligned}
&\mathbb{P}(Z_n = i_n, Z_{n-1} = i_{n-1}, \ldots, Z_0 = i_0) \\
&= \mathbb{P}(Z_n = i_n \mid Z_{n-1} = i_{n-1}, \ldots, Z_0 = i_0)\mathbb{P}(Z_{n-1} = i_{n-1}, \ldots, Z_0 = i_0) \\
&= \mathbb{P}(Z_n = i_n \mid Z_{n-1} = i_{n-1})\mathbb{P}(Z_{n-1} = i_{n-1} \mid Z_{n-2} = i_{n-2}, \ldots, Z_0 = i_0) \\
&\times \mathbb{P}(Z_{n-2} = i_{n-2}, \ldots, Z_0 = i_0) \\
&= \mathbb{P}(Z_n = i_n \mid Z_{n-1} = i_{n-1})\mathbb{P}(Z_{n-1} = i_{n-1} \mid Z_{n-2} = i_{n-2}) \\
&\times \mathbb{P}(Z_{n-2} = i_{n-2} \mid Z_{n-3} = i_{n-3}, \ldots, Z_0 = i_0)\mathbb{P}(Z_{n-3} = i_{n-3}, \ldots, Z_0 = i_0) \\
&= \mathbb{P}(Z_n = i_n \mid Z_{n-1} = i_{n-1})\mathbb{P}(Z_{n-1} = i_{n-1} \mid Z_{n-2} = i_{n-2}) \\
&\times \mathbb{P}(Z_{n-2} = i_{n-2} \mid Z_{n-3} = i_{n-3})\mathbb{P}(Z_{n-3} = i_{n-3}, \ldots, Z_0 = i_0),
\end{aligned}$$

which shows, reasoning by induction, that

$$\begin{aligned}
&\mathbb{P}(Z_n = i_n, Z_{n-1} = i_{n-1}, \ldots, Z_0 = i_0) \\
&\qquad = \mathbb{P}(Z_n = i_n \mid Z_{n-1} = i_{n-1}) \cdots \mathbb{P}(Z_1 = i_1 \mid Z_0 = i_0)\mathbb{P}(Z_0 = i_0),
\end{aligned} \tag{4.1.1}$$

or

$$\begin{aligned}
&\mathbb{P}(Z_n = i_n, Z_{n-1} = i_{n-1}, \ldots, Z_1 = i_1 \mid Z_0 = i_0) \\
&\qquad = \mathbb{P}(Z_n = i_n \mid Z_{n-1} = i_{n-1}) \cdots \mathbb{P}(Z_1 = i_1 \mid Z_0 = i_0),
\end{aligned} \tag{4.1.2}$$

$i_0, i_1, \ldots, i_n \in \mathbb{S}$.

2. By the *law of total probability* (1.3.1) applied under $\mathbb{P}$ to the events $A_{i_0} = \{Z_1 = i_1 \text{ and } Z_0 = i_0\}$, $i_0 \in \mathbb{S}$, we also have

$$\begin{aligned}
\mathbb{P}(Z_1 = i_1) &= \sum_{i_0 \in \mathbb{S}} \mathbb{P}(Z_1 = i_1, Z_0 = i_0) \\
&= \sum_{i_0 \in \mathbb{S}} \mathbb{P}(Z_1 = i_1 \mid Z_0 = i_0)\mathbb{P}(Z_0 = i_0), \quad i_1 \in \mathbb{S},
\end{aligned} \tag{4.1.3}$$

and similarly, under the probability measure $\mathbb{P}(\cdot \mid Z_0 = i_0)$,

$$\begin{aligned}
\mathbb{P}(Z_2 = i_2 \mid Z_0 = i_0) &= \sum_{i_1 \in \mathbb{S}} \mathbb{P}(Z_2 = i_2 \text{ and } Z_1 = i_1 \mid Z_0 = i_0) \\
&= \sum_{i_1 \in \mathbb{S}} \mathbb{P}(Z_2 = i_2 \mid Z_1 = i_1)\mathbb{P}(Z_1 = i_1 \mid Z_0 = i_0),
\end{aligned} \tag{4.1.4}$$

$i_0, i_2 \in \mathbb{S}$.

Example

The random walk

$$S_n := X_1 + X_2 + \cdots + X_n, \qquad n \geq 0, \tag{4.1.5}$$

considered in Chap. 3, where $(X_n)_{n\geq 1}$ is a sequence of (mutually) independent $\mathbb{Z}$-valued random increments, is a discrete-time Markov chain with $\mathbb{S} = \mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$.

Indeed, the value of S_{n+1} depends only on S_n and on the value of the next increment X_{n+1}. In other words, for all $j, i_n, \ldots, i_1 \in \mathbb{Z}$ we have (note that $S_0 = 0$ here)

$$\begin{aligned}
&\mathbb{P}(S_{n+1} = j \mid S_n = i_n,\ S_{n-1} = i_{n-1}, \ldots, S_1 = i_1) \qquad (4.1.6)\\
&= \frac{\mathbb{P}(S_{n+1} = j, S_n = i_n, S_{n-1} = i_{n-1}, \ldots, S_1 = i_1)}{\mathbb{P}(S_n = i_n,\ S_{n-1} = i_{n-1}, \ldots, S_1 = i_1)}\\
&= \frac{\mathbb{P}(S_{n+1} - S_n = j - i_n, S_n - S_{n-1} = i_n - i_{n-1}, \ldots, S_2 - S_1 = i_2 - i_1, S_1 = i_1)}{\mathbb{P}(S_n - S_{n-1} = i_n - i_{n-1}, \ldots, S_2 - S_1 = i_2 - i_1, S_1 = i_1)}\\
&= \frac{\mathbb{P}(X_{n+1} = j - i_n, X_n = i_n - i_{n-1}, \ldots, X_2 = i_2 - i_1, X_1 = i_1)}{\mathbb{P}(X_n = i_n - i_{n-1}, \ldots, X_2 = i_2 - i_1, X_1 = i_1)}\\
&= \frac{\mathbb{P}(X_{n+1} = j - i_n)\mathbb{P}(X_n = i_n - i_{n-1}, \ldots, X_2 = i_2 - i_1, X_1 = i_1)}{\mathbb{P}(X_n = i_n - i_{n-1}, \ldots, X_2 = i_2 - i_1, X_1 = i_1)}\\
&= \mathbb{P}(X_{n+1} = j - i_n)\\
&= \frac{\mathbb{P}(X_{n+1} = j - i_n)\mathbb{P}(X_n + \cdots + X_1 = i_n)}{\mathbb{P}(X_1 + \cdots + X_n = i_n)}\\
&= \frac{\mathbb{P}(X_{n+1} = j - i_n, X_n + \cdots + X_1 = i_n)}{\mathbb{P}(X_1 + \cdots + X_n = i_n)}\\
&= \frac{\mathbb{P}(X_{n+1} = j - i_n \text{ and } S_n = i_n)}{\mathbb{P}(S_n = i_n)} = \frac{\mathbb{P}(S_{n+1} = j \text{ and } S_n = i_n)}{\mathbb{P}(S_n = i_n)}\\
&= \mathbb{P}(S_{n+1} = j \mid S_n = i_n).
\end{aligned}$$

In addition, the Markov chain $(S_n)_{n\geq 0}$ is *time homogeneous* if the random sequence $(X_n)_{n\geq 1}$ is identically distributed.

In particular, we have

$$\mathbb{P}(S_{n+1} = j \mid S_n = i) = \mathbb{P}(X_{n+1} = j - i),$$

hence the transition probability from state $\textcircled{i}$ to state $\textcircled{j}$ of a random walk with independent increments depends only on the difference $j - i$ and on the distribution of X_{n+1}.

More generally, all processes with independent increments are Markov processes. However, *not all Markov chains have independent increments*. In fact, the Markov chains of interest in this chapter do not have independent increments.

4.2 Transition Matrix

In what follows we will assume that the Markov chain $(Z_n)_{n\geq 0}$ is *time homogeneous*, i.e. the probability $\mathbb{P}(Z_{n+1} = j \mid Z_n = i)$ is independent of $n \in \mathbb{N}$. In this case, the random evolution of a Markov chain $(Z_n)_{n\geq 0}$ is determined by the data of

$$P_{i,j} := \mathbb{P}(Z_1 = j \mid Z_0 = i), \qquad i, j \in \mathbb{S}, \tag{4.2.1}$$

which coincides with the probability $\mathbb{P}(Z_{n+1} = j \mid Z_n = i)$ for all $n \in \mathbb{N}$. These data can be encoded into a matrix indexed by $\mathbb{S}^2 = \mathbb{S} \times \mathbb{S}$, called the *transition matrix* of the Markov chain:

$$\big[\, P_{i,j} \,\big]_{i,j\in\mathbb{S}} = [\, \mathbb{P}(Z_1 = j \mid Z_0 = i) \,]_{i,j\in\mathbb{S}},$$

also written on $\mathbb{S} := \mathbb{Z}$ as

$$P = \big[\, P_{i,j} \,\big]_{i,j\in\mathbb{S}} = \begin{bmatrix} \ddots & \vdots & \vdots & \vdots & \vdots & \vdots & \iddots \\ \cdots & P_{-2,-2} & P_{-2,-1} & P_{-2,0} & P_{-2,1} & P_{-2,2} & \cdots \\ \cdots & P_{-1,-2} & P_{-1,-1} & P_{-1,0} & P_{-1,1} & P_{-1,2} & \cdots \\ \cdots & P_{0,-2} & P_{0,-1} & P_{0,0} & P_{0,1} & P_{0,2} & \cdots \\ \cdots & P_{1,-2} & P_{1,-1} & P_{1,0} & P_{1,1} & P_{1,2} & \cdots \\ \cdots & P_{2,-2} & P_{2,-1} & P_{2,0} & P_{2,1} & P_{2,2} & \cdots \\ \iddots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}.$$

The notion of transition matrix is related to that of (weighted) adjacency matrix in graph theory.

Note the inversion of the order of indices (i, j) between $\mathbb{P}(Z_{n+1} = j \mid Z_n = i)$ and $P_{i,j}$. In particular, the initial state $\textcircled{i}$ is a *row number* in the matrix, while the final state $\textcircled{j}$ corresponds to a *column number*.

By the *law of total probability* (1.3.1) applied to the probability measure $\mathbb{P}(\cdot \mid Z_0 = i)$, we also have the equality

$$\sum_{j\in\mathbb{S}} \mathbb{P}(Z_1 = j \mid Z_0 = i) = \mathbb{P}\left(\bigcup_{j\in\mathbb{S}} \{Z_1 = j\} \mid Z_0 = i\right) = \mathbb{P}(\Omega) = 1, \quad i \in \mathbb{S}, \tag{4.2.2}$$

i.e. the *rows* of the transition matrix satisfy the condition

$$\sum_{j\in\mathbb{S}} P_{i,j} = 1,$$

for every row index $i \in \mathbb{S}$.

Using the matrix notation $P = (P_{i,j})_{i,j\in\mathbb{S}}$, and Relation (4.1.1) we find

$$\mathbb{P}(Z_n = i_n, Z_{n-1} = i_{n-1}, \ldots, Z_0 = i_0) = P_{i_{n-1},i_n} \cdots P_{i_0,i_1}\mathbb{P}(Z_0 = i_0),$$

$i_0, i_1, \ldots, i_n \in \mathbb{S}$, and we rewrite (4.1.3) as

$$\mathbb{P}(Z_1 = i) = \sum_{j\in\mathbb{S}} \mathbb{P}(Z_1 = i \mid Z_0 = j)\mathbb{P}(Z_0 = j) = \sum_{j\in\mathbb{S}} P_{j,i}\mathbb{P}(Z_0 = j), \quad i \in \mathbb{S}. \tag{4.2.3}$$

A state $k \in \mathbb{S}$ is said to be *absorbing* if $P_{k,k} = 1$.

In what follows we will often consider $\mathbb{S} = \mathbb{N} = \{0, 1, 2, \ldots\}$ and $\mathbb{N}$-valued Markov chains, in which case the *transition matrix* $\left[\mathbb{P}(Z_{n+1} = j \mid Z_n = i) \right]_{i,j\in\mathbb{N}}$ of the chain is written as

$$\left[P_{i,j} \right]_{i,j\in\mathbb{N}} = \begin{bmatrix} P_{0,0} & P_{0,1} & P_{0,2} & \cdots \\ P_{1,0} & P_{1,1} & P_{1,2} & \cdots \\ P_{2,0} & P_{2,1} & P_{2,2} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}.$$

From (4.2.2) we have

$$\sum_{j\geq 0} P_{i,j} = 1,$$

for all $i \in \mathbb{N}$.

In case the Markov chain $(Z_k)_{k\geq 0}$ takes values in the finite state space $\mathbb{S} = \{0, 1, \ldots, N\}$ its $(N + 1) \times (N + 1)$ transition matrix will simply have the form

$$\left[P_{i,j} \right]_{0\leq i,j\leq N} = \begin{bmatrix} P_{0,0} & P_{0,1} & P_{0,2} & \cdots & P_{0,N} \\ P_{1,0} & P_{1,1} & P_{1,2} & \cdots & P_{1,N} \\ P_{2,0} & P_{2,1} & P_{2,2} & \cdots & P_{2,N} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ P_{N,0} & P_{N,1} & P_{N,2} & \cdots & P_{N,N} \end{bmatrix}.$$

Still on the finite state space $\mathbb{S} = \{0, 1, \ldots, N\}$, Relation (4.2.3) can be restated in the language of matrix and vector products using the shorthand notation:

$$\eta = \pi P, \tag{4.2.4}$$

where

$$\eta := [\mathbb{P}(Z_1 = 0), \ldots, \mathbb{P}(Z_1 = N)] = [\eta_0, \eta_1, \ldots, \eta_N] \in \mathbb{R}^{N+1}$$

is the row vector "distribution of Z_1",

$$\pi := [\mathbb{P}(Z_0 = 0), \ldots, \mathbb{P}(Z_0 = N)] = [\pi_0, \ldots, \pi_N] \in \mathbb{R}^{N+1}$$

is the row vector representing the probability distribution of Z_0, and

$$[\eta_0, \eta_1, \ldots, \eta_N] = [\pi_0, \ldots, \pi_N] \times \begin{bmatrix} P_{0,0} & P_{0,1} & P_{0,2} & \cdots & P_{0,N} \\ P_{1,0} & P_{1,1} & P_{1,2} & \cdots & P_{1,N} \\ P_{2,0} & P_{2,1} & P_{2,2} & \cdots & P_{2,N} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ P_{N,0} & P_{N,1} & P_{N,2} & \cdots & P_{N,N} \end{bmatrix}. \tag{4.2.5}$$

Invariant vectors

A row vector π such that $\pi = \pi P$ is said to be *invariant* or *stationary* by the transition matrix P.

For example, in case the matrix P takes the form

$$P = \left[P_{i,j}\right]_{0 \le i,j \le N} = \begin{bmatrix} \pi_0 & \pi_1 & \pi_2 & \pi_3 & \cdots & \pi_N \\ \pi_0 & \pi_1 & \pi_2 & \pi_3 & \cdots & \pi_N \\ \pi_0 & \pi_1 & \pi_2 & \pi_3 & \cdots & \pi_N \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \pi_0 & \pi_1 & \pi_2 & \pi_3 & \cdots & \pi_N \end{bmatrix},$$

with all rows equal and $\pi_0 + \pi_1 + \cdots + \pi_N = 1$, then we have $\pi = \pi P$, i.e. π is an invariant (or stationary) distribution for P.

4.3 Examples of Markov Chains

The wide range of applications of Markov chains to engineering, physics and biology has already been mentioned in the introduction. Here we consider some more specific examples.

(i) Random walk.

The transition matrix $[\, P_{i,j} \,]_{i,j\in\mathbb{S}}$ of the unrestricted random walk (4.1.5) is given by

$$[\, P_{i,j} \,]_{i,j\in\mathbb{S}} = \begin{array}{c} \\ i-2 \\ i-1 \\ i \\ i+1 \\ i+2 \\ \\ \end{array} \begin{array}{c} \begin{array}{ccccc} \quad & i-1 & i & i+1 & \quad \end{array} \\ \begin{bmatrix} \ddots & \vdots & \vdots & \vdots & \vdots & \iddots \\ \cdots & 0 & p & 0 & 0 & 0 \cdots \\ \cdots & q & 0 & p & 0 & 0 \cdots \\ \cdots & 0 & q & 0 & p & 0 \cdots \\ \cdots & 0 & 0 & q & 0 & p \cdots \\ \cdots & 0 & 0 & 0 & q & 0 \cdots \\ \iddots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix} \end{array} \tag{4.3.1}$$

(ii) Gambling process.

The transition matrix $[\, P_{i,j} \,]_{0\le i,j\le S}$ of the gambling process on $\{0, 1, \ldots, S\}$ with absorbing states (0) and (S), see Chap. 2, is given by

$$P = [\, P_{i,j} \,]_{0\le i,j\le S} = \begin{bmatrix} 1 & 0 & 0 & 0 & \cdots & \cdots & 0 & 0 & 0 & 0 \\ q & 0 & p & 0 & \cdots & \cdots & 0 & 0 & 0 & 0 \\ 0 & q & 0 & p & \cdots & \cdots & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \ddots & \ddots & \ddots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \ddots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & \cdots & q & 0 & p & 0 \\ 0 & 0 & 0 & 0 & \cdots & \cdots & 0 & q & 0 & p \\ 0 & 0 & 0 & 0 & \cdots & \cdots & 0 & 0 & 0 & 1 \end{bmatrix}.$$

(iii) Credit rating.

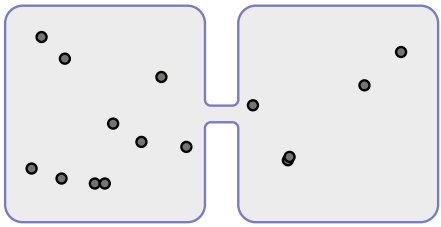
[Transition probabilities are expressed in %].

Rating at the start of a year	Rating at the end of the year									
	AAA	AA	A	BBB	BB	B	CCC	D	N.R.	Total
AAA	90.34	5.62	0.39	0.08	0.03	0	0	0	3.5	100
AA	0.64	88.78	6.72	0.47	0.06	0.09	0.02	0.01	3.21	100
A	0.07	2.16	87.94	4.97	0.47	0.19	0.01	0.04	4.16	100
BBB	0.03	0.24	4.56	84.26	4.19	0.76	0.15	0.22	5.59	100
BB	0.03	0.06	0.4	6.09	76.09	6.82	0.96	0.98	8.58	100
B	0	0.09	0.29	0.41	5.11	74.62	3.43	5.3	10.76	100
CCC	0.13	0	0.26	0.77	1.66	8.93	53.19	21.94	13.14	100
D	0	0	0	0	1.0	3.1	9.29	51.29	37.32	100
N.R.	0	0	0	0	0	0.1	8.55	74.06	17.07	100

We note that higher ratings are more stable since the diagonal coefficients of the matrix go decreasing. On the other hand, starting from an AA rating it is easier to be downgraded to A (probability 6.72%) than to be upgraded to AAA (probability 0.64%).

(iv) Ehrenfest chain.

Two volumes of air (left and right), containing a total of N balls, are connected by a pipe.



At each time step, one picks a ball at random and moves it to the other side. Let $Z_n \in \{0, 1, \ldots, N\}$ denote the number of balls in the left side at time n. The transition probabilities $\mathbb{P}(Z_{n+1} = j \mid Z_n = i)$, $0 \le i, j \le N$, are given by

$$\mathbb{P}(Z_{n+1} = k+1 \mid Z_n = k) = \frac{N-k}{N}, \qquad k = 0, 1, \ldots, N-1, \tag{4.3.2}$$

and

$$\mathbb{P}(Z_{n+1} = k-1 \mid Z_n = k) = \frac{k}{N}, \qquad k = 1, 2, \ldots, N, \tag{4.3.3}$$

with

$$P = \begin{bmatrix} 0 & 1 & 0 & \cdots & \cdots & 0 & 0 \\ 1/N & 0 & (N-1)/N & \cdots & \cdots & 0 & 0 \\ 0 & 2/N & 0 & \cdots & \cdots & 0 & 0 \\ 0 & 0 & 3/N & \cdots & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & \ddots & 3/N & 0 & 0 \\ 0 & 0 & \cdots & \ddots & 0 & 2/N & 0 \\ 0 & 0 & \cdots & \cdots & (N-1)/N & 0 & 1/N \\ 0 & 0 & \cdots & \cdots & 0 & 1 & 0 \end{bmatrix},$$

cf. Exercises 6.7, 7.4 and Problem 7.25 on the (modified) Ehrenfest chain, and Exercise 4.9 on the Bernoulli-Laplace chain.

v) Markov chains in music.

By a statistical analysis of note transitions, every type of music can be encoded into a Markov chain. An example of such an analysis is presented in the next transition matrix.

	A	A♯	B	C	D	E	F	G	G♯
A	4/19	0	3/19	0	2/19	1/19	0	6/19	3/19
A♯	1	0	0	0	0	0	0	0	0
B	7/15	0	1/15	4/15	0	3/15	0	0	0
C	0	0	6/15	3/15	6/15	0	0	0	0
D	0	0	0	3/11	3/11	5/11	0	0	0
E	4/19	1/19	0	3/19	0	5/19	4/19	1/19	1/19
F	0	0	0	1/5	0	0	1/5	0	3/5
G	1/5	0	1/5	2/5	0	0	0	1/5	0
G♯	0	0	3/4	0	0	1/4	0	0	0

(vi) Text generation.

Markov chains can be used to generate sentences in a given language, based on a statistical analysis on the transition between words in a sample text. The state space of the Markov chain can be made of different word sequences as in e.g. Problem 4.14.

Other applications of Markov chains include:

– Memory management in computer science,

– Logistics, supply chain management, and waiting queues,

– Modeling of insurance claims (Problem 4.13),

– Board games, e.g. Snakes and Ladders,

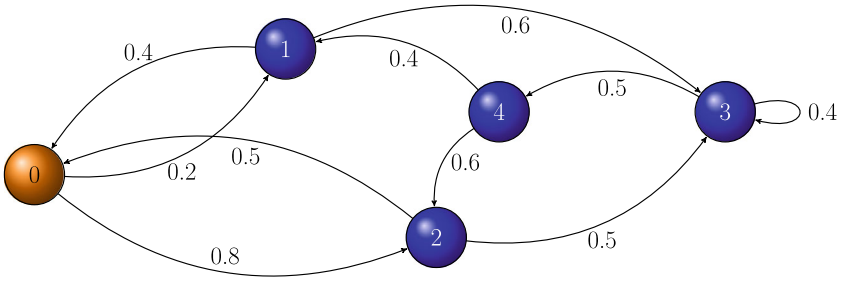


Fig. 4.1 Graph of a five-state Markov chain

– Genetics, cf. the Wright-Fisher model (Problem 5.20).

– Random fields in imaging,

– Artificial intelligence, learning theory and machine learning (Problems 4.14 and 5.19.)

Graph representation

Whenever possible, we will represent a Markov chain using a graph, as in the following example with transition matrix

$$P = \begin{bmatrix} 0 & 0.2 & 0.8 & 0 & 0 \\ 0.4 & 0 & 0 & 0.6 & 0 \\ 0.5 & 0 & 0 & 0.5 & 0 \\ 0 & 0 & 0 & 0.4 & 0.6 \\ 0 & 0.4 & 0.6 & 0 & 0 \end{bmatrix}, \tag{4.3.4}$$

see Fig. 4.1.

4.4 Higher-Order Transition Probabilities

As noted above, the transition matrix P is a convenient way to record $\mathbb{P}(Z_{n+1} = j \mid Z_n = i)$, $i, j \in \mathbb{S}$, into an array of data.

However, it is *much more than that*, as already hinted at in Relation (4.2.4). Suppose for example that we are interested in the two-step transition probability

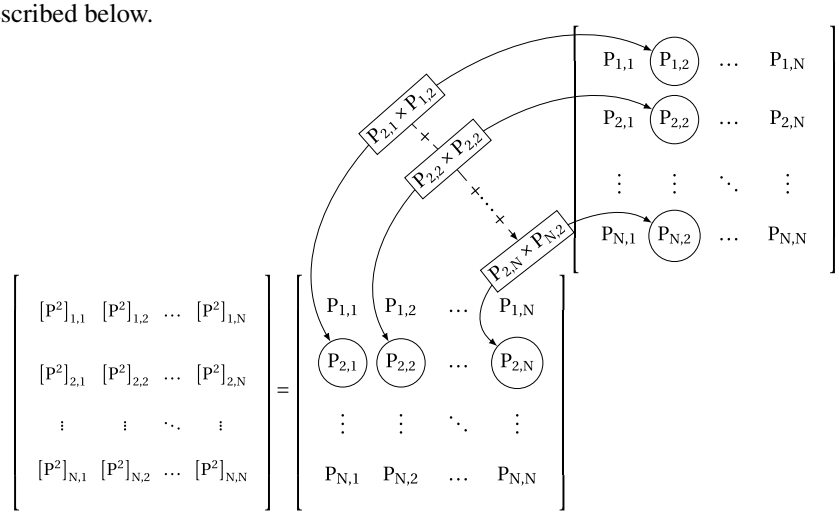
$$\mathbb{P}(Z_{n+2} = j \mid Z_n = i).$$

This probability does not appear in the transition matrix P, but it can be computed by first step analysis, applying the *law of total probability* (1.3.1) to the probability measure $\mathbb{P}(\cdot \mid Z_n = i)$ as follows.

(i) 2-step transitions. Denoting by $\mathbb{S}$ the state space of the process we have, using (4.1.4),

$$\begin{aligned}
\mathbb{P}(Z_{n+2} = j \mid Z_n = i) &= \sum_{l \in \mathbb{S}} \mathbb{P}(Z_{n+2} = j \text{ and } Z_{n+1} = l \mid Z_n = i) \\
&= \sum_{l \in \mathbb{S}} \frac{\mathbb{P}(Z_{n+2} = j,\ Z_{n+1} = l,\ Z_n = i)}{\mathbb{P}(Z_n = i)} \\
&= \sum_{l \in \mathbb{S}} \frac{\mathbb{P}(Z_{n+2} = j,\ Z_{n+1} = l,\ Z_n = i)}{\mathbb{P}(Z_{n+1} = l \text{ and } Z_n = i)} \frac{\mathbb{P}(Z_{n+1} = l \text{ and } Z_n = i)}{\mathbb{P}(Z_n = i)} \\
&= \sum_{l \in \mathbb{S}} \mathbb{P}(Z_{n+2} = j \mid Z_{n+1} = l \text{ and } Z_n = i) \mathbb{P}(Z_{n+1} = l \mid Z_n = i) \\
&= \sum_{l \in \mathbb{S}} \mathbb{P}(Z_{n+2} = j \mid Z_{n+1} = l) \mathbb{P}(Z_{n+1} = l \mid Z_n = i) \\
&= \sum_{l \in \mathbb{S}} P_{i,l} P_{l,j} \\
&= [P^2]_{i,j}, \qquad i, j \in \mathbb{S},
\end{aligned}$$

where we used (4.2.1), in agreement with the matrix multiplication mechanism described below.



Hence, using matrix product notation, we find

$$(\mathbb{P}(Z_{n+2} = j \mid Z_n = i))_{0 \le i, j \le N}$$

$$= \begin{bmatrix} P_{0,0} & P_{0,1} & P_{0,2} & \cdots & P_{0,N} \\ P_{1,0} & P_{1,1} & P_{1,2} & \cdots & P_{1,N} \\ P_{2,0} & P_{2,1} & P_{2,2} & \cdots & P_{2,N} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ P_{N,0} & P_{N,1} & P_{N,2} & \cdots & P_{N,N} \end{bmatrix} \times \begin{bmatrix} P_{0,0} & P_{0,1} & P_{0,2} & \cdots & P_{0,N} \\ P_{1,0} & P_{1,1} & P_{1,2} & \cdots & P_{1,N} \\ P_{2,0} & P_{2,1} & P_{2,2} & \cdots & P_{2,N} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ P_{N,0} & P_{N,1} & P_{N,2} & \cdots & P_{N,N} \end{bmatrix}.$$

(ii) k-step transitions. More generally, we have the following result.

Proposition 4.1 *For all $k \in \mathbb{N}$ we have the relation*

$$\left[\mathbb{P}(Z_{n+k} = j \mid Z_n = i) \right]_{i,j \in \mathbb{S}} = \left[[P^k]_{i,j} \right]_{i,j \in \mathbb{S}} = P^k. \tag{4.4.1}$$

Proof We prove (4.4.1) by induction. Clearly, the statement holds for $k = 0$ and $k = 1$. Next, for all $k \in \mathbb{N}$, we have

$$\begin{aligned} \mathbb{P}(Z_{n+k+1} = j \mid Z_n = i) &= \sum_{l \in \mathbb{S}} \mathbb{P}(Z_{n+k+1} = j \text{ and } Z_{n+k} = l \mid Z_n = i) \\ &= \sum_{l \in \mathbb{S}} \frac{\mathbb{P}(Z_{n+k+1} = j,\ Z_{n+k} = l,\ Z_n = i)}{\mathbb{P}(Z_n = i)} \\ &= \sum_{l \in \mathbb{S}} \frac{\mathbb{P}(Z_{n+k+1} = j,\ Z_{n+k} = l,\ Z_n = i)}{\mathbb{P}(Z_{n+k} = l \text{ and } Z_n = i)} \frac{\mathbb{P}(Z_{n+k} = l \text{ and } Z_n = i)}{\mathbb{P}(Z_n = i)} \\ &= \sum_{l \in \mathbb{S}} \mathbb{P}(Z_{n+k+1} = j \mid Z_{n+k} = l \text{ and } Z_n = i) \mathbb{P}(Z_{n+k} = l \mid Z_n = i) \\ &= \sum_{l \in \mathbb{S}} \mathbb{P}(Z_{n+k+1} = j \mid Z_{n+k} = l) \mathbb{P}(Z_{n+k} = l \mid Z_n = i) \\ &= \sum_{l \in \mathbb{S}} \mathbb{P}(Z_{n+k} = l \mid Z_n = i) P_{l,j}. \end{aligned}$$

We have just checked that the family of matrices

$$\left[\mathbb{P}(Z_{n+k} = j \mid Z_n = i) \right]_{i,j \in \mathbb{S}}, \qquad k \geq 1,$$

satisfies the same induction relation as the *matrix power* P^k, i.e.

$$[P^{k+1}]_{i,j} = \sum_{l \in \mathbb{S}} [P^k]_{i,l} P_{l,j},$$

and the same initial condition, hence by induction on $k \geq 0$ the equality

$$\left[\mathbb{P}(Z_{n+k} = j \mid Z_n = i) \right]_{i,j \in \mathbb{S}} = \left[[P^k]_{i,j} \right]_{i,j \in \mathbb{S}} = P^k$$

holds not only for $k = 0$ and $k = 1$, but also for all $k \in \mathbb{N}$. □

The matrix product relation

$$P^{m+n} = P^m P^n = P^n P^m,$$

which reads

$$[P^{m+n}]_{i,j} = \sum_{l \in \mathbb{S}} [P^m]_{i,l} [P^n]_{l,j} = \sum_{l \in \mathbb{S}} [P^n]_{i,l} [P^m]_{l,j}, \qquad i, j \in \mathbb{S},$$

can now be interpreted as

$$\begin{aligned} \mathbb{P}(Z_{n+m} = j \mid Z_0 = i) &= \sum_{l \in \mathbb{S}} \mathbb{P}(Z_m = j \mid Z_0 = l) \mathbb{P}(Z_n = l \mid Z_0 = i) \\ &= \sum_{l \in \mathbb{S}} \mathbb{P}(Z_n = j \mid Z_0 = l) \mathbb{P}(Z_m = l \mid Z_0 = i), \end{aligned}$$

$i, j \in \mathbb{S}$, which is called the *Chapman-Kolmogorov* equation, cf. also the triple (1.2.4).

Example The gambling process $(Z_n)_{n \geq 0}$.

– Taking $S = 4$ and $p = 40\%$, the transition matrix of the gambling process on $\mathbb{S} = \{0, 1, \ldots, 4\}$ of Chap. 2 reads

$$P = \left[P_{i,j} \right]_{0 \leq i,j \leq 4} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0.6 & 0 & 0.4 & 0 & 0 \\ 0 & 0.6 & 0 & 0.4 & 0 \\ 0 & 0 & 0.6 & 0 & 0.4 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \tag{4.4.2}$$

and we can check by hand calculation that:

$$P^2 = P \times P$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0.6 & 0 & 0.4 & 0 & 0 \\ 0 & 0.6 & 0 & 0.4 & 0 \\ 0 & 0 & 0.6 & 0 & 0.4 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0.6 & 0 & 0.4 & 0 & 0 \\ 0 & 0.6 & 0 & 0.4 & 0 \\ 0 & 0 & 0.6 & 0 & 0.4 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0.6 & 0.24 & 0 & 0.16 & 0 \\ 0.36 & 0 & 0.48 & 0 & 0.16 \\ 0 & 0.36 & 0 & 0.24 & 0.4 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

Exercise: From the above matrix (4.4.2), check that

$$\mathbb{P}(Z_2 = 4 \mid Z_0 = 2) = [P^2]_{2,4} = 0.16,$$

$$\mathbb{P}(Z_2 = 1 \mid Z_0 = 2) = [P^2]_{2,1} = 0, \qquad \text{and}$$
$$\mathbb{P}(Z_2 = 2 \mid Z_0 = 2) = [P^2]_{2,2} = 0.48.$$

Example. The fifth order transitions of the chain with Markov matrix (4.3.4) can be computed from the fifth matrix power

$$P^5 = \begin{bmatrix} 0.14352 & 0.09600 & 0.25920 & 0.30160 & 0.19968 \\ 0.15840 & 0.10608 & 0.24192 & 0.30400 & 0.18960 \\ 0.17040 & 0.10920 & 0.23280 & 0.30880 & 0.17880 \\ 0.17664 & 0.11520 & 0.22800 & 0.30928 & 0.17088 \\ 0.14904 & 0.09600 & 0.25440 & 0.30520 & 0.19536 \end{bmatrix}.$$

Note that for large transition orders (*e.g.* for 1000 time steps) we get

$$P^{1000} = \begin{bmatrix} 0.16273 & 0.10613 & 0.24056 & 0.30660 & 0.18396 \\ 0.16273 & 0.10613 & 0.24056 & 0.30660 & 0.18396 \\ 0.16273 & 0.10613 & 0.24056 & 0.30660 & 0.18396 \\ 0.16273 & 0.10613 & 0.24056 & 0.30660 & 0.18396 \\ 0.16273 & 0.10613 & 0.24056 & 0.30660 & 0.18396 \end{bmatrix},$$

which suggests a *convergence phenomenon* in large time for the Markov chain, see Chap. 7 for details.

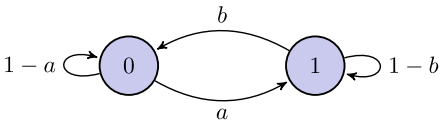
Example For the simple random walk of Chap. 3, computing the probability to travel from ⓪ to $\boxed{2k} = \boxed{10}$ in $2n = 20$ time steps involves a summation over $\binom{20}{10+5} = \binom{2n}{n+k} = 15504$ paths, which can be evaluated by computing $\left[P^{20}\right]_{0,10}$, cf. also Fig. 3.2.

4.5 The Two-State Discrete-Time Markov Chain

The above discussion shows that there is some interest in computing the n-th order transition matrix P^n. Although this is generally difficult, this is actually possible when the number of states equals two, i.e. $\mathbb{S} = \{0, 1\}$.

We close this chapter with a complete study of the two-state Markov chain, whose transition matrix has the form

$$P = \begin{bmatrix} 1-a & a \\ b & 1-b \end{bmatrix}, \tag{4.5.1}$$



where $a \in [0,1]$ and $b \in [0,1]$.
We also have

$$\mathbb{P}(Z_{n+1} = 1 \mid Z_n = 0) = a, \qquad \mathbb{P}(Z_{n+1} = 0 \mid Z_n = 0) = 1 - a,$$

and

$$\mathbb{P}(Z_{n+1} = 0 \mid Z_n = 1) = b, \qquad \mathbb{P}(Z_{n+1} = 1 \mid Z_n = 1) = 1 - b.$$

The matrix power

$$P^n = \begin{bmatrix} 1-a & a \\ b & 1-b \end{bmatrix}^n = \underbrace{\begin{bmatrix} 1-a & a \\ b & 1-b \end{bmatrix} \times \cdots \times \begin{bmatrix} 1-a & a \\ b & 1-b \end{bmatrix}}_{n \text{ times}}$$

of the transition matrix P is computed for all $n \geq 0$ in Proposition 4.2. We always exclude the case $a = b = 0$ since it corresponds to the trivial case where $P = \mathrm{I}$ is the identity matrix (constant chain).

Proposition 4.2 *We have*

$$P^n = \frac{1}{a+b} \begin{bmatrix} b + a(1-a-b)^n & a(1-(1-a-b)^n) \\ b(1-(1-a-b)^n) & a + b(1-a-b)^n \end{bmatrix}, \qquad n \geq 0.$$

Proof This result will be proved by a *diagonalization* argument. The matrix P has two eigenvectors

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} -a \\ b \end{bmatrix},$$

with respective eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 1 - a - b$.

Hence P can be written in the *diagonal form*

$$P = M \times D \times M^{-1}, \tag{4.5.2}$$

i.e.

$$P = \begin{bmatrix} 1 & -a \\ 1 & b \end{bmatrix} \times \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \times \begin{bmatrix} \dfrac{b}{a+b} & \dfrac{a}{a+b} \\ -\dfrac{1}{a+b} & \dfrac{1}{a+b} \end{bmatrix}.$$

As a consequence of (4.5.2), we have

$$P^n = (M \times D \times M^{-1})^n = (M \times D \times M^{-1}) \cdots (M \times D \times M^{-1})$$
$$= M \times D \times \cdots \times D \times M^{-1} = M \times D^n \times M^{-1},$$

where

$$D^n = \begin{bmatrix} 1 & 0 \\ 0 & \lambda_2^n \end{bmatrix}, \qquad n \geq 0.$$

hence

$$P^n = \begin{bmatrix} 1 & -a \\ 1 & b \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 0 & \lambda_2^n \end{bmatrix} \times \begin{bmatrix} \dfrac{b}{a+b} & \dfrac{a}{a+b} \\ -\dfrac{1}{a+b} & \dfrac{1}{a+b} \end{bmatrix}$$
$$= \tfrac{1}{a+b} \begin{bmatrix} b & a \\ b & a \end{bmatrix} + \tfrac{\lambda_2^n}{a+b} \begin{bmatrix} a & -a \\ -b & b \end{bmatrix}$$
$$= \tfrac{1}{a+b} \begin{bmatrix} b + a\lambda_2^n & a(1-\lambda_2^n) \\ b(1-\lambda_2^n) & a + b\lambda_2^n \end{bmatrix}. \tag{4.5.3}$$

□

For an alternative proof of Proposition 4.2, see also Exercise 1.4.1 page 5 of Norris (1998) in which P^n is written as

$$P^n = \begin{bmatrix} 1 - a_n & -a_n \\ b_n & 1 - b_n \end{bmatrix}$$

and the relation $P^{n+1} = P \times P^n$ is used to find induction relations for a_n and b_n, cf. the solution of Exercise 7.19 for a similar analysis.

From the result of Proposition 4.2 we may now compute the probabilities

$$\mathbb{P}(Z_n = 0 \mid Z_0 = 0) = \frac{b + a\lambda_2^n}{a+b}, \quad \mathbb{P}(Z_n = 1 \mid Z_0 = 0) = \frac{a(1-\lambda_2^n)}{a+b} \tag{4.5.4}$$

and

$$\mathbb{P}(Z_n = 0 \mid Z_0 = 1) = \frac{b(1-\lambda_2^n)}{a+b}, \quad \mathbb{P}(Z_n = 1 \mid Z_0 = 1) = \frac{a + b\lambda_2^n}{a+b}. \tag{4.5.5}$$

As an example, the value of $\mathbb{P}(Z_3 = 0 \mid Z_0 = 0)$ could also be computed using pathwise analysis as

$$\mathbb{P}(Z_3 = 0 \mid Z_0 = 0) = (1-a)^3 + ab(1-b) + 2(1-a)ab,$$

which coincides with (4.5.4), i.e.

$$\mathbb{P}(Z_3 = 0 \mid Z_0 = 0) = \frac{b + a(1-a-b)^3}{a+b},$$

for $n = 3$. Under the condition

$$-1 < \lambda_2 = 1 - a - b < 1,$$

which is equivalent to $(a, b) \neq (0, 0)$ and $(a, b) \neq (1, 1)$, we can let n go to infinity in (4.5.3) to derive the large time behavior, or limiting distribution, of the Markov chain:

$$\lim_{n\to\infty} P^n = \lim_{n\to\infty} \begin{bmatrix} \mathbb{P}(Z_n = 0 \mid Z_0 = 0) & \mathbb{P}(Z_n = 1 \mid Z_0 = 0) \\ \mathbb{P}(Z_n = 0 \mid Z_0 = 1) & \mathbb{P}(Z_n = 1 \mid Z_0 = 1) \end{bmatrix} = \frac{1}{a+b} \begin{bmatrix} b & a \\ b & a \end{bmatrix}.$$

Note that convergence will be faster when $a + b$ is closer to 1.

Hence we have

$$\lim_{n\to\infty} \mathbb{P}(Z_n = 1 \mid Z_0 = 0) = \lim_{n\to\infty} \mathbb{P}(Z_n = 1 \mid Z_0 = 1) = \frac{a}{a+b} \tag{4.5.6}$$

and

$$\lim_{n\to\infty} \mathbb{P}(Z_n = 0 \mid Z_0 = 0) = \lim_{n\to\infty} \mathbb{P}(Z_n = 0 \mid Z_0 = 1) = \frac{b}{a+b}. \tag{4.5.7}$$

Consequently,

$$\pi = [\pi_0, \pi_1] := \left[\frac{b}{a+b}, \frac{a}{a+b}\right] \tag{4.5.8}$$

is a *limiting distribution* as n goes to infinity, provided that $(a, b) \neq (1, 1)$. In other words, whatever the initial state Z_0, the probability of being at $\textcircled{1}$ after a "large" time becomes close to $a/(a + b)$, while the probability of being at $\textcircled{0}$ becomes close to $b/(a + b)$.

In case $a = b = 0$, we have

$$P = I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

the chain is constant and it clearly admits its initial distribution as limiting distribution. In case $a = b = 1$, we have

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

and there is no limiting distribution as the chain switches indefinitely between state ⓪ and ①.

The notions of limiting and invariant (or stationary) distributions will be treated in Chap. 7 in the general framework of Markov chains, see for example Proposition 7.7.

Remarks

(i) The limiting distribution π in (4.5.8) is *invariant* (or stationary) by P in the sense that

$$\pi P = \frac{1}{a+b}[b,a]\begin{bmatrix} 1-a & a \\ b & 1-b \end{bmatrix} = \frac{1}{a+b}\begin{bmatrix} b(1-a)+ab \\ ab+a(1-b) \end{bmatrix}^\top$$
$$= \frac{1}{a+b}[b,a] = \pi,$$

i.e. π is invariant (or stationary) with respect to P, and the invariance relation (4.2.4):

$$\pi = \pi P,$$

which means that $\mathbb{P}(Z_1 = k) = \pi_k$ if $\mathbb{P}(Z_0 = k) = \pi_k$, $k = 0, 1$. For example, the distribution $\pi = [1/2, 1/2]$ is clearly invariant (or stationary) for the swapping chain with $a = b = 1$ and transition matrix

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

while $\pi := [1/3, 2/3]$ will *not* be invariant (or stationary) for this chain. This is a two-state particular case of the circular chain of Example (7.2.4).

(ii) If $a + b = 1$, one sees that

$$P^n = \begin{bmatrix} b & a \\ b & a \end{bmatrix} = P$$

for all $n \geq 0$, and we find

$$\mathbb{P}(Z_n = 1 \mid Z_k = 0) = \mathbb{P}(Z_n = 1 \mid Z_k = 1) = \mathbb{P}(Z_n = 1) = a$$

and

$$\mathbb{P}(Z_n = 0 \mid Z_k = 0) = \mathbb{P}(Z_n = 0 \mid Z_k = 1) = \mathbb{P}(Z_n = 0) = b$$

for all $k = 0, 1, \ldots, n-1$, regardless of the initial distribution $[\mathbb{P}(Z_0 = 0), \mathbb{P}(Z_0 = 1)]$. In this case, Z_n is independent of Z_k, as we have

$$\mathbb{P}(Z_n = i,\ Z_k = j) = \mathbb{P}(Z_n = i \mid Z_k = j)\mathbb{P}(Z_k = j) = \mathbb{P}(Z_n = i)\mathbb{P}(Z_k = j),$$

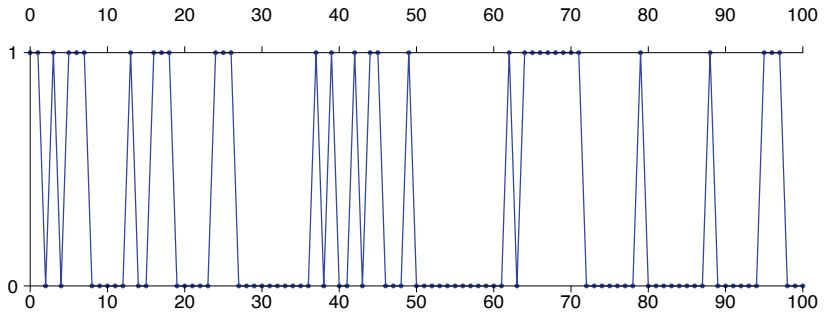


Fig. 4.2 Sample path of a two-state chain in continuous time with $a = 0.2$ and $b = 0.4$

$i, j = 0, 1, 0 \le k < n$, and $(Z_n)_{n\ge 0}$ is an *i.i.d* sequence of random variables with distribution $(1-a, a) = (b, a)$ over $\{0, 1\}$, cf. Exercise 4.3 in the general case.

(iii) A given proportion $p = a/(a+b) \in (0, 1)$ of visits to state (1) in the long run can be reached by any $a \in (0, p]$ and $b \in (0, 1-p]$ satisfying $a = bp/(1-p)$. Smaller values of a and b will lead to increased *stickiness*. The case $(a, b) = (p, 1-p)$ satisfies $a + b = 1$ and corresponds to minimal stickiness, i.e. to the independence of the sequence $(Z_n)_{n\ge 0}$.

(iv) When $a = b = 1$ in (4.5.1) the limit $\lim_{n\to\infty} P^n$ does not exist as we have

$$P^n = \begin{cases} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, & n = 2k, \\ \\ \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, & n = 2k+1, \end{cases}$$

and the chain is indefinitely switching at each time step from one state to the other.

In Figs. 4.2 and 4.3 we consider a simulation of the two-state random walk with transition matrix

$$P = \begin{bmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{bmatrix},$$

i.e. $a = 0.2$ and $b = 0.4$. Figure 4.2 represents a sample path $(x_n)_{n=0,1,\ldots,100}$ of the chain, while Fig. 4.3 represents the sample average

$$y_n = \frac{1}{n+1}(x_0 + x_1 + \cdots + x_n), \qquad n = 0, 1, \ldots, 100,$$

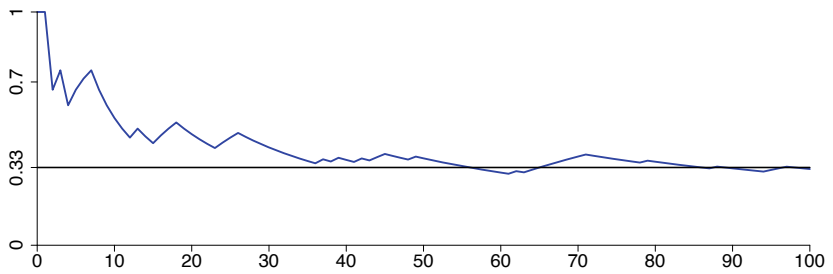


Fig. 4.3 The proportion of chain values at state 1 tends to $1/3 = a/(a+b)$

which counts the proportion of values of the chain in the state (1). This proportion is found to converge to $a/(a+b) = 1/3$. This is actually a consequence of the Ergodic Theorem, cf. Theorem 7.12 in Chap. 7.
The code of the above program, written in R is given below.

```
dim=2; a=0.2; b=0.4; # Parameter definitions
P=matrix(c(1-a,a,b,1-b),nrow=dim,ncol=dim,byrow=TRUE) # Transition matrix
N=100; Z=array(N+1); # Number of time steps
for(ll in seq(1,N)) {Z[1]=sample(dim,size=1,prob=P[2,])
# Random simulation of Z[j+1] given Z[j]
for (j in seq(1,N))
     Z[j+1]=sample(dim,size=1,prob=P[Z[j],]);Y=array(N+1);S=0;
# Computation of the average over the l first steps
for(l in seq(1,N+1)) { Z[l]=Z[l]-1; S=S+Z[l]; Y[l]=S/l;}
X=array(N+1); for(l in seq(1,N+1)) {X[l]=l-1;};par(mfrow=c(2,1))
plot(X,Y,type="l",yaxt="n",xaxt="n",xlim=c(0,N),xlab="",
     ylim=c(0,1),ylab="",xaxs="i", col="black", main="",bty="n")
segments(0,a/(a+b),N,a/(a+b))
axis(2,pos=0,at=c(0.0,0.1,0.2,0.3,0.4,0.5,0.6,0.7,0.8,0.9,1.0))
axis(1,pos=0,at=seq(0,N,10),outer=TRUE)
plot(X,Z,type="o",xlab="",ylab="",xlim=c(0,N),yaxt="n",xaxt="n",xaxs="i",
col="black",main="",pch=20,bty="n")
axis(1,pos=1,at=seq(0,N+1,10),outer=TRUE,padj=-4,tcl=0.5)
axis(1,pos=0,at=seq(0,N+1,10),outer=TRUE);axis(2,las=2,at=0:1)
readline(prompt = "Pause. Press <Enter> to continue...")}
```

We close this chapter with two other Markov chain sample paths presented in Figs. 4.4 and 4.5. In Fig. 4.4, we check again that the proportion of chain values in the state (1) converges to $1/3$ for a two-state Markov chain.
In Fig. 4.5 we draw a sample path of a five-state Markov chain.

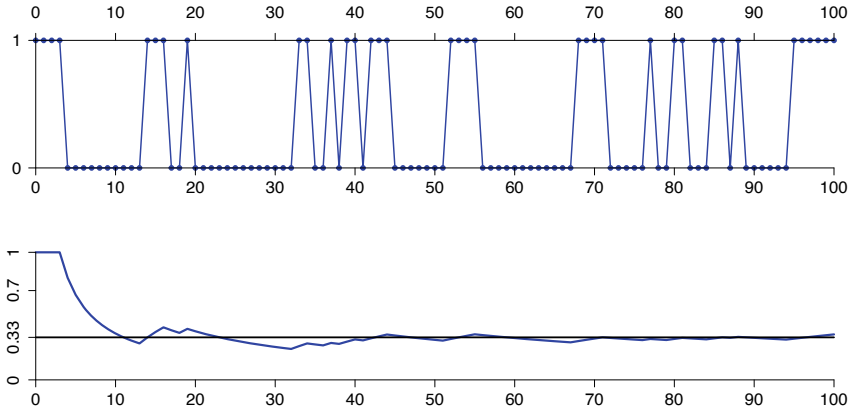


Fig. 4.4 Convergence graph for the two-state Markov chain with $a = 0.2$ and $b = 0.4$

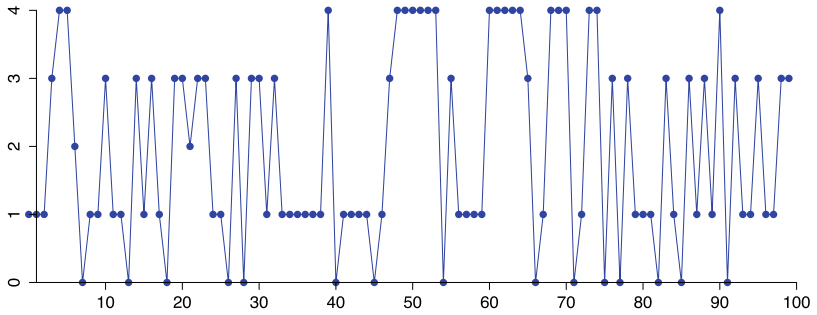


Fig. 4.5 Sample path of a five-state Markov chain

Exercises

Exercise 4.1 Consider the *symmetric* random walk $(S_n)_{n\geq 0}$ on $\mathbb{S} = \mathbb{Z}$ with independent increments ± 1 chosen with equal probability $1/2$, started at $S_0 = 0$.

(a) Is the process $Z_n := 2S_n + 1$ a Markov chain?
(b) Is the process $Z_n := (S_n)^2$ a Markov chain?

Exercise 4.2 Consider the Markov chain $(Z_n)_{n\geq 0}$ with state space $\mathbb{S} = \{1, 2\}$ and transition matrix

$$P = \begin{matrix} \\ 1 \\ 2 \end{matrix} \begin{matrix} \begin{matrix} 1 & 2 \end{matrix} \\ \begin{bmatrix} 0.4 & 0.6 \\ 0.8 & 0.2 \end{bmatrix} \end{matrix}.$$

(a) Compute $\mathbb{P}(Z_7 = 1 \text{ and } Z_5 = 2 \mid Z_4 = 1 \text{ and } Z_3 = 2)$.
(b) Compute $\mathbb{E}[Z_2 \mid Z_1 = 1]$.

Exercise 4.3 Consider a transition probability matrix P of the form

$$P = \left[P_{i,j}\right]_{0\le i,j\le N} = \begin{bmatrix} \pi_0 & \pi_1 & \pi_2 & \pi_3 & \cdots & \pi_N \\ \pi_0 & \pi_1 & \pi_2 & \pi_3 & \cdots & \pi_N \\ \pi_0 & \pi_1 & \pi_2 & \pi_3 & \cdots & \pi_N \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \pi_0 & \pi_1 & \pi_2 & \pi_3 & \cdots & \pi_N \end{bmatrix},$$

where $\pi = [\pi_0, \pi_1, \ldots, \pi_N] \in [0,1]^{N+1}$ is a vector such that

$$\pi_0 + \pi_1 + \cdots + \pi_N = 1.$$

(a) Compute the matrix power P^n for all $n \ge 2$.
(b) Show that the vector π is an invariant (or stationary) distribution for P.
(c) Show that if $\mathbb{P}(Z_0 = i) = \pi_i$, $i = 0, 1, \ldots, N$, then Z_n is independent of Z_k for all $0 \le k < n$, and $(Z_n)_{n\ge 0}$ is an *i.i.d* sequence of random variables with distribution $\pi = [\pi_0, \pi_1, \ldots, \pi_N]$ over $\{0, 1, \ldots, N\}$.

Exercise 4.4 Consider a $\{0,1\}$-valued "hidden" two-state Markov chain $(X_n)_{n\ge 0}$ with *transition probability matrix*

$$P = \begin{bmatrix} P_{0,0} & P_{0,1} \\ P_{1,0} & P_{1,1} \end{bmatrix} = \begin{bmatrix} \mathbb{P}(X_1 = 0 \mid X_0 = 0) & \mathbb{P}(X_1 = 1 \mid X_0 = 0) \\ \mathbb{P}(X_1 = 0 \mid X_0 = 1) & \mathbb{P}(X_1 = 1 \mid X_0 = 1) \end{bmatrix},$$

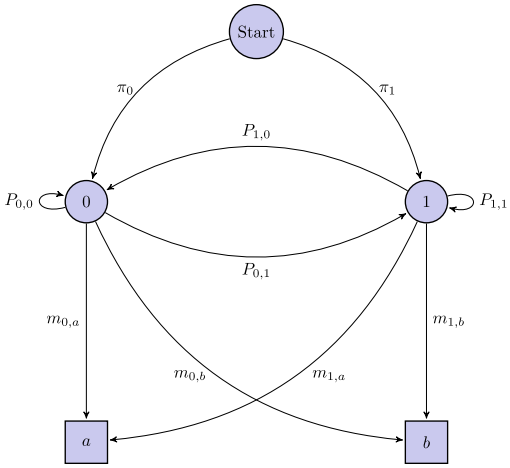
and initial distribution

$$\pi = [\pi_0, \pi_1] = [\mathbb{P}(X_0 = 0), \mathbb{P}(X_0 = 1)].$$

We observe a two-state process $(O_k)_{k\ge 0}$ whose state $O_k \in \{a, b\}$ at every time $k \in \mathbb{N}$ has a conditional distribution given $X_k \in \{0, 1\}$ denoted by

$$M = \begin{bmatrix} m_{0,a} & m_{0,b} \\ m_{1,a} & m_{1,b} \end{bmatrix} = \begin{bmatrix} \mathbb{P}(O_k = a \mid X_k = 0) & \mathbb{P}(O_k = b \mid X_k = 0) \\ \mathbb{P}(O_k = a \mid X_k = 1) & \mathbb{P}(O_k = b \mid X_k = 1) \end{bmatrix},$$

called the *emission probability* matrix.



(a) Using the matrix entries of π, P and M, compute the four conditional probabilities

$$\begin{cases} \mathbb{P}\big((O_0, O_1) = (a, b) \mid (X_0, X_1) = (0, 0)\big), \\ \mathbb{P}\big((O_0, O_1) = (a, b) \mid (X_0, X_1) = (0, 1)\big), \\ \mathbb{P}\big((O_0, O_1) = (a, b) \mid (X_0, X_1) = (1, 0)\big), \\ \mathbb{P}\big((O_0, O_1) = (a, b) \mid (X_0, X_1) = (1, 1)\big). \end{cases}$$

Hint: By independence, the conditional probability of observing $(O_0, O_1) = (a, b)$ given that $(X_0, X_1) = (x, y)$ splits as

$$\begin{aligned} &\mathbb{P}((O_0, O_1) = (a, b) \mid (X_0, X_1) = (x, y)) \\ &\qquad = \mathbb{P}(O_0 = a \mid X_0 = x)\mathbb{P}(O_1 = b \mid X_1 = y). \end{aligned}$$

(b) Using π, P and M, compute $\mathbb{P}(X_0 = 1, X_1 = 1)$ and the four probabilities

$$\begin{cases} \mathbb{P}\big((O_0, O_1) = (a, b) \text{ and } (X_0, X_1) = (0, 0)\big), \\ \mathbb{P}\big((O_0, O_1) = (a, b) \text{ and } (X_0, X_1) = (0, 1)\big), \\ \mathbb{P}\big((O_0, O_1) = (a, b) \text{ and } (X_0, X_1) = (1, 0)\big), \\ \mathbb{P}\big((O_0, O_1) = (a, b) \text{ and } (X_0, X_1) = (1, 1)\big). \end{cases}$$

(c) Find the probability $\mathbb{P}((O_0, O_1) = (a, b))$ that the observed sequence is (a, b).

Hint: Use the law of total probability based on all possible values of (X_0, X_1).

(d) Compute the hidden state probabilities

$$\begin{cases} \mathbb{P}((X_0, X_1) = (0, 1) \mid (O_0, O_1) = (a, b)), \\ \mathbb{P}((X_0, X_1) = (1, 0) \mid (O_0, O_1) = (a, b)), \end{cases}$$

and

$$\mathbb{P}(X_1 = 1 \mid (O_0, O_1) = (a, b)) \text{ and } \mathbb{P}(X_1 = 0 \mid (O_0, O_1) = (a, b)),$$

given that the observed sequence is (a, b).

Exercise 4.5 Consider the two-dimensional random walk $(S_n)_{n\geq 0}$ started at $S_0 = (0, 0)$ on $\mathbb{S} = \mathbb{Z}^2$, where, starting from a location $S_n = (i, j)$ the chain can move to any of the points $(i + 1, j + 1)$, $(i + 1, j - 1)$, $(i - 1, j + 1)$, $(i - 1, j - 1)$ with equal probability $1/4$. The next Questions (a) and (b) are independent.

(a) We denote by $S_n = (X_n, Y_n)$ the coordinates of S_n at time $n \geq 0$ and let $Z_n := X_n^2 + Y_n^2$. Does the process $(Z_n)_{n\geq 0}$ have the Markov property? Justify your answer.

Hint: Use the fact that a same value of Z_n may correspond to different locations of (X_n, Y_n) on the circle, for example $(X_n, Y_n) = (5, 0)$ and $(X_n, Y_n) = (4, 3)$ when $Z_n = 25$.

(b) Suppose in addition that the random walk cannot visit any site more than once, as in a *snake game*. In this case, does the random walk $(S_n)_{n\geq 0}$ have the Markov property? Justify your answer.



Questions (a) and (b) above are independent.

Exercise 4.6 The Elephant Random Walk $(S_n)_{n\geq 0}$ (Schütz and Trimper 2008) is a discrete-time $\mathbb{Z}$-valued random walk

$$S_n := X_1 + \cdots + X_n, \qquad n \in \mathbb{N},$$

whose increments $X_k = S_k - S_{k-1}$, $k \geq 1$, are recursively defined as follows:

– At time $n = 1$, X_1 is a Bernoulli $\{-1, +1\}$-valued random variable with

$$\mathbb{P}(X_1 = +1) = p \quad \text{and} \quad \mathbb{P}(X_1 = -1) = q = 1 - p \in (0, 1).$$

– At any subsequent time $n \geq 2$, one randomly draws an integer time index $k \in \{1, 2, \ldots, n-1\}$ with uniform probability, and lets $X_n := X_k$ with probability p, and $X_n := -X_k$ with probability $q := 1 - p$.

Does the Elephant Random Walk $(S_n)_{n\geq 0}$ have the Markov property?

Exercise 4.7 Consider a two-state Markov chain $(X_n)_{n\geq 0}$ with state space $\mathbb{S} = \{0, 1\}$ and transition matrix

$$P = \begin{matrix} \\ 0 \\ 1 \end{matrix} \begin{matrix} \begin{matrix} 0 & \quad 1 \end{matrix} \\ \begin{bmatrix} 1-a & a \\ b & 1-b \end{bmatrix} \end{matrix},$$

where $a, b \in (0, 1)$, and define a new stochastic process $(Z_n)_{n\geq 1}$ by $Z_n = (X_{n-1}, X_n)$, $n \geq 1$. Argue that $(Z_n)_{n\geq 1}$ is a Markov chain, and write down its transition matrix. Start by determining the state space of $(Z_n)_{n\geq 1}$.

Exercise 4.8 Given $p \in [0, 1)$, consider the Markov chain $(X_n)_{n\geq 0}$ on the state space $\{0, 1, 2\}$ having the transition matrix

$$P = \begin{matrix} \\ 0 \\ 1 \\ 2 \end{matrix} \begin{matrix} \begin{matrix} 0 & 1 & 2 \end{matrix} \\ \begin{bmatrix} p & q & 0 \\ 0 & p & q \\ 0 & 0 & 1 \end{bmatrix} \end{matrix},$$

with $q := 1 - p$.

(a) Give the probability distribution of the first hitting time

$$T_2 := \inf\big\{n \geq 0 \ : \ X_n = 2\big\}.$$

of state (2) starting from $X_0 = (0)$.

Hint: The sum $Z = X_1 + \cdots + X_n$ of n independent geometrically distributed random variables $X_1, \ldots X_n$ on $\{1, 2, \ldots\}$ has the negative binomial distribution

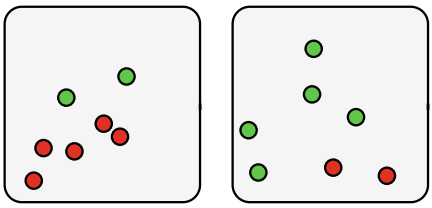
$$\mathbb{P}(Z = k \mid X_0 = 1) = \binom{k-1}{k-d}(1-p)^d p^{k-d}, \qquad k \geq d.$$

(b) Compute the mean hitting time $\mathbb{E}[T_2 \mid X_0 = 0]$ of state (2) starting from $X_0 = 0$.

Hint:+ We have

$$\sum_{k\geq 1} kp^{k-1} = \frac{1}{(1-p)^2} \quad \text{and} \quad \sum_{k\geq 2} k(k-1)p^{k-2} = \frac{2}{(1-p)^3}, \qquad 0 \leq p < 1.$$

Exercise 4.9 Bernoulli-Laplace chain. Consider two boxes and a total of $2N$ balls made of N red balls and N green balls. At time 0, a number $k = X_0$ of red balls and a number $N - k$ of green balls are placed in the first box, while the remaining $N - k$ red balls and k green balls are placed in the second box.



At each unit of time, one ball is chosen randomly out of N in each box, and the two balls are interchanged. Write down the transition matrix of the Markov chain $(X_n)_{n\geq 0}$ with state space $\mathbb{S} = \{0, 1, 2, \ldots, N\}$, representing the number of red balls in the first box. Start for example from $N = 5$.

Exercise 4.10 (a) After winning k dollars, a gambler either receives $k + 1$ dollars with probability p, or has to *quit* the game and lose everything with probability $q = 1 - p$. Starting from *one* dollar, find a model for the time evolution of the wealth of the player using a Markov chain whose transition probability matrix P will be described explicitly along with its powers P^n of all orders $n \geq 1$.

(b) (Success runs Markov chain). We modify the model of Question (a) by allowing the gambler to start playing again and win with probability p after reaching state $\textcircled{0}$. Write down the corresponding transition probability matrix P, and compute P^n for all $n \geq 2$.

Exercise 4.11 Let $(X_k)_{k\geq 0}$ be the Markov chain with transition matrix

$$P = \begin{bmatrix} 1/4 & 0 & 1/2 & 1/4 \\ 0 & 1/5 & 0 & 4/5 \\ 0 & 1 & 0 & 0 \\ 1/3 & 1/3 & 0 & 1/3 \end{bmatrix}.$$

A new process is defined by letting

$$Z_n := \begin{cases} 0 & \text{if } X_n = 0 \text{ or } X_n = 1, \\ X_n & \text{if } X_n = 2 \text{ or } X_n = 3, \end{cases}$$

i.e.

$$Z_n = X_n \mathbb{1}_{\{X_n \in \{2,3\}\}}, \qquad n \geq 0.$$

(a) Compute

$$\mathbb{P}(Z_{n+1} = 2 \mid Z_n = 0 \text{ and } Z_{n-1} = 2)$$

and

$$\mathbb{P}(Z_{n+1} = 2 \mid Z_n = 0 \text{ and } Z_{n-1} = 3),$$

$n \geq 1$.

(b) Is $(Z_n)_{n \geq 0}$ a Markov chain?

Exercise 4.12 (Olaleye et al. 2009) Abeokuta, one of the major towns of the defunct Western Region of Nigeria, has recently seen an astronomic increase in vehicular activities. The intensity of vehicle traffic at the Lafenwa intersection which consists of Ayetoro, Old Bridge and Ita-Oshin routes, is modeled according to three states L/M/H = { Low / Moderate / High }.

(a) During year 2005, low intensity incoming traffic has been observed at Lafenwa intersection for $\eta_L = 50\%$ of the time, moderate traffic has been observed for $\eta_M = 40\%$ of the time, while high traffic has been observed during $\eta_H = 10\%$ of the time.

Given the correspondence table

Incoming traffic	Vehicles per hour
L (low intensity)	360
M (medium intensity)	505
H (high intensity)	640

compute the average incoming traffic per hour in year 2005.

(b) The analysis of incoming daily traffic volumes at Lafenwa intersection between years 2004 and 2005 shows that the probability of switching states within {L, M, H} is given by the Markov transition probability matrix

$$P = \begin{bmatrix} 2/3 & 1/6 & 1/6 \\ 1/3 & 1/2 & 1/6 \\ 1/6 & 2/3 & 1/6 \end{bmatrix}.$$

Based on the knowledge of P and $\eta = [\eta_L, \eta_M, \eta_H]$, give a projection of the respective proportions of traffic in the states L/M/H for year 2006.

(c) Based on the result of Question (b), give a projected estimate for the average incoming traffic per hour in year 2006.
(d) By solving the equation $\pi = \pi P$ for the invariant (or stationary) probability distribution $\pi = [\pi_L, \pi_M, \pi_H]$, give a long term projection of steady traffic at Lafenwa intersection. *Hint*: we have $\pi_L = 11/24$.

Problem 4.13 Phase-type distributions are used in insurance to model the heavy-tailed random claim sizes appearing in reserve and surplus processes.

(a) Given $p \in [0, 1]$, construct the transition matrix of a two-state Markov chain on the state space $\{0, 1\}$ satisfying the following two conditions:

(i) State $\textcircled{0}$ is absorbing, i.e. $\mathbb{P}(X_{n+1} = 0 \mid X_n = 0) = 1$, *and*
(ii) The first hitting time

$$T_0 := \inf\{n \geq 0 \ : \ X_n = 0\}$$

of state $\textcircled{0}$ starting from state $\textcircled{1}$ has the geometric distribution p given by

$$\mathbb{P}(T_0 = k \mid X_0 = 1) = (1 - p)p^{k-1}, \qquad k \geq 1.$$

(b) Given $p \in [0, 1]$ and $d \geq 1$, construct the transition matrix of a $d + 1$-state Markov chain on the state space $\{0, 1, \ldots, d\}$ satisfying the following two conditions:

(i) State $\textcircled{0}$ is absorbing, i.e. $\mathbb{P}(X_{k+1} = 0 \mid X_k = 0) = 1$, *and*
(ii) The first hitting time T_0 of state $\textcircled{0}$ starting from state d has the negative binomial distribution

$$\mathbb{P}(T_0 = k \mid X_0 = d) = \binom{k-1}{k-d}(1 - p)^d p^{k-d}, \qquad k \geq d.$$

(c) In what follows we consider a discrete-time Markov chain on $\{0, 1, \ldots, d\}$ having d transient[1] states $\{1, 2, \ldots, d\}$, and 0 as absorbing state. Show that its transition matrix P takes the form

$$P = \big[P_{i,j}\big]_{0 \leq i,j \leq d} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ \alpha_1 & Q_{1,1} & \cdots & Q_{1,d} \\ \alpha_2 & Q_{2,1} & \cdots & Q_{2,d} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_d & Q_{d,1} & \cdots & Q_{d,d} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \alpha & Q \end{bmatrix},$$

[1] Here, the transience condition implies that $\mathbb{P}(T_0 < \infty \mid X_0 = i) = 1$ for all $i = 1, 2, \ldots, d$, it will be ensured by assuming that $I - Q$ is invertible, cf. Sects. 5.4 and 6.3 for details.

where α is the column vector $\alpha = [\alpha_1, \alpha_2, \ldots, \alpha_d]^\top$ and Q is the $d \times d$ matrix

$$Q = \begin{bmatrix} Q_{1,1} & \cdots & Q_{1,d} \\ \vdots & \ddots & \vdots \\ Q_{d,1} & \cdots & Q_{d,d} \end{bmatrix},$$

and give the relation between the vector α and the matrix Q using the column vector $e := [1, 1, \ldots, 1]^\top$.

(d) Show by induction on $n \geq 0$ that we have

$$P^n = \begin{bmatrix} 1 & 0 \\ (I - Q^n)e & Q^n \end{bmatrix}, \tag{4.5.9}$$

where I is the $d \times d$ identity matrix

$$I := \begin{bmatrix} 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \ddots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix}.$$

(e) Compute the probability distribution of T_0 using the vector α and the matrix Q^{n-1}, where T_0 is the first hitting time of state $\textcircled{0}$ starting from state $i \geq 1$.

Hint: Partition the event $\{T_0 = n\}$ as

$$\{T_0 = n\} = \bigcup_{k=1}^{d} \{X_n = 0 \text{ and } X_{n-1} = k\}.$$

(f) From now on, we assume that the initial distribution of X_0 is given by the d-dimensional vector

$$\beta = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_d \end{bmatrix},$$

i.e. $\beta_i = \mathbb{P}(X_0 = i)$, $i = 1, 2, \ldots, d$, with $\mathbb{P}(X_0 = 0) = 0$. Compute the probability distribution of T_0 using the vectors α, β, and the matrix Q^{n-1}.

(g) Compute the cumulative distribution function $P(T_0 \leq n)$ of T_0 the vectors β, e, and the matrix Q^n.

(h) Recover the result of Question (g) by rewriting $\mathbb{P}(T_0 \leq n)$ as the probability of not being in any state $i = 1, 2, \ldots, d$ at time n.

(i) Compute the probability generating function

$$G_{T_0}(s) := \sum_{k\geq 0} s^k \mathbb{P}(T_0 = k)$$

of T_0, assuming the existence of the matrix inverse $(I - sQ)^{-1}$ given by the series

$$(I - sQ)^{-1} = \sum_{k\geq 0} s^k Q^k, \qquad s \in (-1, 1].$$

(j) Using the probability generating function $s \mapsto G_{T_0}(s)$, compute the first and second moments $E[T_0]$ and $E[T_0^2]$ of T_0.

Problem 4.14 Hidden Markov models have applications to speech recognition, face recognition, emotion recognition, genomics and biological sequence analysis, sentiment analysis, unsupervised learning, climate studies, etc.

Consider a $\{0, 1\}$-valued "hidden" two-state Markov chain $(X_n)_{n\geq 0}$ with *transition probability matrix*

$$P = \begin{bmatrix} P_{0,0} & P_{0,1} \\ P_{1,0} & P_{1,1} \end{bmatrix} = \begin{bmatrix} \mathbb{P}(X_1 = 0 \mid X_0 = 0) & \mathbb{P}(X_1 = 1 \mid X_0 = 0) \\ \mathbb{P}(X_1 = 0 \mid X_0 = 1) & \mathbb{P}(X_1 = 1 \mid X_0 = 1) \end{bmatrix},$$

and initial distribution

$$\pi = [\pi_0, \pi_1] = [\mathbb{P}(X_0 = 0), \mathbb{P}(X_0 = 1)].$$

We observe a stochastic process $(O_k)_{k\geq 0}$ whose state $O_k \in \{a, b, c\}$ at every time $k \in \mathbb{N}$ has a conditional distribution given $X_k \in \{0, 1\}$ denoted by

$$\begin{aligned} M &= \begin{bmatrix} m_{0,a} & m_{0,b} & m_{0,c} \\ m_{1,a} & m_{1,b} & m_{1,c} \end{bmatrix} \\ &= \begin{bmatrix} \mathbb{P}(O_k = a \mid X_k = 0) & \mathbb{P}(O_k = b \mid X_k = 0) & \mathbb{P}(O_k = c \mid X_k = 0) \\ \mathbb{P}(O_k = a \mid X_k = 1) & \mathbb{P}(O_k = b \mid X_k = 1) & \mathbb{P}(O_k = c \mid X_k = 1) \end{bmatrix}, \end{aligned}$$

called the *emission probability* matrix.

(a) Using the matrix entries of π, P and M, compute $\mathbb{P}(X_0 = 1, X_1 = 1, X_2 = 0)$ and the probability

$$\mathbb{P}((O_0, O_1, O_2) = (c, a, b) \text{ and } (X_0, X_1, X_2) = (1, 1, 0))$$

of observing the sequence $(O_0, O_1, O_2) = (c, a, b)$ when $(X_0, X_1, X_2) = (1, 1, 0)$.

Hint: By independence, the conditional probability of observing $(O_0, O_1, O_2) = (c, a, b)$ given that $(X_0, X_1, X_2) = (1, 1, 0)$ splits as

$$\begin{aligned}\mathbb{P}((O_0, O_1, O_2) = (c, a, b) \mid (X_0, X_1, X_2) = (1, 1, 0)) \\ = \mathbb{P}(O_0 = c \mid X_0 = 1)\mathbb{P}(O_1 = a \mid X_1 = 1)\mathbb{P}(O_2 = b \mid X_2 = 0).\end{aligned}$$

(b) Find the probability $\mathbb{P}((O_0, O_1, O_2) = (c, a, b))$ that the observed sequence is (c, a, b).

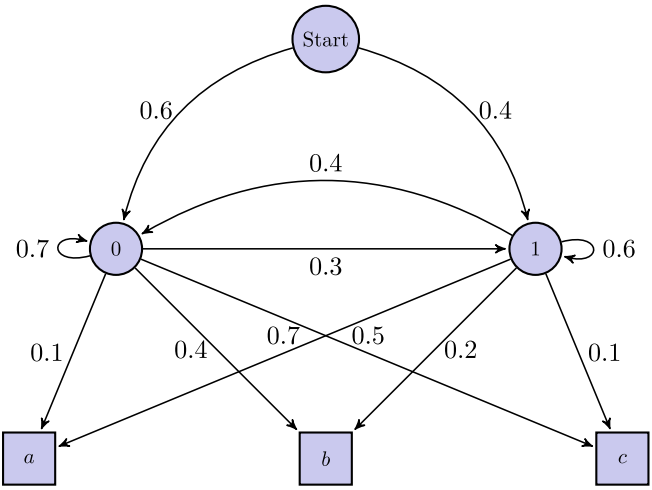
Hint: Use the law of total probability based on all possible values of (X_0, X_1, X_2).

(c) Compute the probabilities

$$\mathbb{P}(X_1 = 1 \mid (O_0, O_1, O_2) = (c, a, b)), \quad \text{and} \quad \mathbb{P}(X_1 = 0 \mid (O_0, O_1, O_2) = (c, a, b)).$$

From Questions (d) to (i) below we take $\pi = [\pi_0, \pi_1] := [0.6, 0.4]$ and

$$P := \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}, \quad M := \begin{bmatrix} m_{0,a} & m_{0,b} & m_{0,c} \\ m_{1,a} & m_{1,b} & m_{1,c} \end{bmatrix} = \begin{bmatrix} 0.1 & 0.4 & 0.5 \\ 0.7 & 0.2 & 0.1 \end{bmatrix}. \quad (4.5.10)$$



(d) Compute[2] the eight probabilities

$$\mathbb{P}((X_0, X_1, X_2) = (x, y, z) \text{ and } (O_0, O_1, O_2) = (c, a, b))$$

for all $x, y, z \in \{0, 1\}$, and identify the most likely sample sequence of values for (X_0, X_1, X_2).

(e) Compute[1] the probability $\mathbb{P}((O_0, O_1, O_2) = (c, a, b))$ that the observed sequence is (c, a, b).

[2] Give a numerical value with exactly three significant digits.

(f) Compute[1] the six probabilities

$$\mathbb{P}(X_k = 1 \mid (O_0, O_1, O_2) = (c, a, b)), \quad \mathbb{P}(X_k = 0 \mid (O_0, O_1, O_2) = (c, a, b)),$$

$k = 0, 1, 2$. What is the most likely sequence for (X_0, X_1, X_2) according to this computation?

Based on the knowledge of $\mathbb{P}((X_0, X_1, X_2) = (1, 1, 0) \mid (O_0, O_1, O_2) = (c, a, b))$, one can build an estimator $\widehat{P}$, $\hat{\pi}$, $\widehat{M}$ of the model parameters P, π and M, as

$$\begin{cases} \hat{\pi}_i := \mathbb{P}(X_0 = i \mid (O_0, O_1, O_2) = (c, a, b)), & \text{(4.5.11a)} \\ \widehat{P}_{i,j} := \dfrac{\sum_{t=0}^{N-1} \mathbb{P}(X_t = i, X_{t+1} = j \mid (O_0, O_1, O_2) = (c, a, b))}{\sum_{t=0}^{N-1} \mathbb{P}(X_t = i \mid (O_0, O_1, O_2) = (c, a, b))} & \text{(4.5.11b)} \\ \widehat{M}_{i,k} := \dfrac{\sum_{t=0}^{N} \mathbb{1}_{\{O_t = k\}} \mathbb{P}(X_t = i \mid (O_0, O_1, O_2) = (c, a, b))}{\sum_{t=0}^{N} \mathbb{P}(X_t = i \mid (O_0, O_1, O_2) = (c, a, b))}, & \text{(4.5.11c)} \end{cases}$$

with here $N = 2$. Note that (4.5.11b) averages the number of times the observed state is "k" given that the hidden state is "i", which gives an estimate of the conditional probability $M_{i,k}$.

(g) Compute[3] the vector estimate $\hat{\pi} = [\hat{\pi}_0, \hat{\pi}_1]$ using (4.5.11) and the data of Equation (4.5.10).

(h) Compute[1] the matrix estimate $\widehat{P}$ using (4.5.11a) and the data of Equation (4.5.10).

(i) Compute[1] the matrix estimate $\widehat{M}$ using (4.5.11b) and the data of Equation (4.5.10).

(j) Iterating the estimates (4.5.11)–(4.5.11b) is computationally intensive, however, this procedure admits an efficient recursive implementation via the Baum-Welch algorithm which is based on the Expectation-Maximization (EM) algorithm.

Imagine an alien trying to analyse an English manuscript without any prior knowledge of English. Using a simple two-state hidden chain $(X_n)_{n \geq 0}$ he will try to uncover some *features* of the language, starting with a *binary classification* of the alphabet.

[3] Give a numerical value with exactly three significant digits.

```
install.packages("HMM")
library (HMM)
library (lattice)
text = readChar("my_own_text_file.txt",nchars=10000)
data <- unlist (strsplit (gsub ("[^a-z]", "_", tolower (text)), ""))
pi=c(0.4,0.6)
P=t(matrix(c(c(0.6177499,0.3822501),c(0.8826096,0.1173904)),nrow=2,ncol=2))
M=t(matrix(c(c(0.037192964,0.009902360,0.032833978,0.044882670,0.057331132,
0.052143890,0.013665015,0.036187536,0.072293323,0.044793972,0.060008388,
0.004256270,0.024770706,0.053520546,0.014232306,0.046981769,0.053733382,
0.066355203,0.046817817,0.006912535,0.016201697,0.013425499,0.024694447,
0.064902148,0.046170421,0.033586536,0.022203489),
c(0.0389931197,0.0697183142,0.0239154174,0.0512772632,0.0404732634,0.0059687348,
0.0211687193,0.0625229746,0.0039632091,0.0567828864,0.0468108656,0.0168355418,
0.0627882213,0.0286478204,0.0389215263,0.0064318198,0.0001698078,0.0493758725,
0.0652709152,0.0069580806,0.0093043072,0.0028807932,0.0521827110,0.0608822385,
0.0645417465,0.0555249876,0.0576888424)),nrow=27,ncol=2))
model <- initHMM (c("0", "1"),c("_", letters), pi, P, M)
system.time (estimate <- baumWelch (model, data, 100)) # 100 iterations
xyplot(estimate$hmm$emissionProbs[1,] ~ c(1:27), scales=list(x=list(at=1:27,
labels=c("_", letters))),type="h", lwd=5, xlab="", ylab="")
```

Using a text of your choice and a $\{0, 1\}$-valued hidden Markov chain $(X_n)_{n\geq 0}$, estimate the corresponding emission probability matrix M by running the above R code which relies on the HMM (Hidden Markov Model) package. A text length of $N \simeq 10,000$ characters can be a minimum. The initial values of π, P and M have been set randomly.

What do you conclude?

Instructions:

Specify your choice of document used.
Attach a graph with a discussion of your output.

Suggestions and possible variations:

Try a language different from English.

Increase the state space of $(X_n)_{n\geq 0}$ in order to uncover more features of the chosen language.

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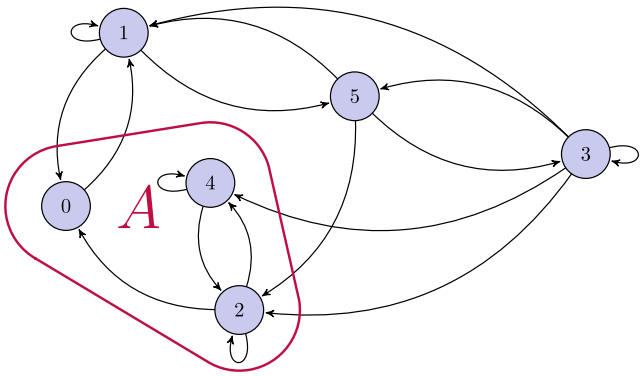
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Chapter 5
First Step Analysis

Starting with this chapter we introduce the systematic use of the first step analysis technique, in a general framework that covers the examples of random walks already treated in Chaps. 2 and 3. The main applications of first step analysis are the computation of hitting probabilities, mean hitting and absorption times, mean first return times, and average number of returns to a given state.

5.1 Hitting Probabilities

Let us consider a Markov chain $(Z_n)_{n\geq 0}$ with state space $\mathbb{S}$, and let $A \subset \mathbb{S}$ denote a subset of $\mathbb{S}$, as in the following example with $\mathbb{S} = \{0, 1, 2, 3, 4, 5\}$ and $A := \{0, 2, 4\}$.



We are interested in the first time T_A the chain hits the subset A, where

$$T_A := \inf\{n \geq 0 \ : \ Z_n \in A\}, \tag{5.1.1}$$

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N. Privault, *Understanding Markov Chains*, Springer Undergraduate Mathematics Series,
https://doi.org/10.1007/978-981-95-4057-0_5

with $T_A = 0$ if $Z_0 \in A$ and

$$T_A = +\infty \quad \text{if} \quad \{n \geq 0 \, : \, Z_n \in A\} = \emptyset,$$

i.e. if $Z_n \notin A$ for all $n \in \mathbb{N}$. In case the transition matrix P satisfies

$$P_{k,l} = \mathbb{1}_{\{k=l\}} \quad \text{for all} \quad k, \, l \in A, \tag{5.1.2}$$

the set $A \subset \mathbb{S}$ is said to be *absorbing*.

Similarly to the gambling problem of Chap. 2, we would like to compute the probabilities

$$g_l(k) = \mathbb{P}(Z_{T_A} = l \text{ and } T_A < \infty \mid Z_0 = k)$$

of hitting the set $A \subset \mathbb{S}$ through state $l \in A$ starting from $k \in \mathbb{S}$, where Z_{T_A} represents the location of the chain $(Z_n)_{n \geq 0}$ at the hitting time T_A.

This computation can be achieved by first step analysis, using the *law of total probability* (1.3.1) for the probability measure $\mathbb{P}(\cdot \mid Z_0 = k)$ and the Markov property, as follows.

Proposition 5.1 *Assume that* (5.1.2) *holds. The hitting probabilities*

$$g_l(k) := \mathbb{P}(Z_{T_A} = l \text{ and } T_A < \infty \mid Z_0 = k), \qquad k \in \mathbb{S}, \; l \in A,$$

satisfy the equation

$$g_l(k) = \sum_{m \in \mathbb{S}} P_{k,m} g_l(m) = P_{k,l} + \sum_{m \in \mathbb{S} \setminus A} P_{k,m} g_l(m), \tag{5.1.3}$$

$k \in \mathbb{S} \setminus A$, $l \in A$, *subject to the boundary conditions*

$$g_l(k) = \mathbb{P}(Z_{T_A} = l \text{ and } T_A < \infty \mid Z_0 = k) = \mathbb{1}_{\{k=l\}} = \begin{cases} 1 \text{ if } k = l, \\ 0 \text{ if } k \neq l, \end{cases}$$

$k, l \in A$.

which hold since $T_A = 0$ *whenever one starts from* $Z_0 \in A$.

Proof For all $k \in \mathbb{S} \setminus A$ we have $T_A \geq 1$ given that $Z_0 = k$, hence we can write

$$
\begin{aligned}
g_l(k) &= \mathbb{P}(Z_{T_A} = l \text{ and } T_A < \infty \mid Z_0 = k) \\
&= \sum_{m \in \mathbb{S}} \mathbb{P}(Z_{T_A} = l \text{ and } T_A < \infty \mid Z_1 = m \text{ and } Z_0 = k)\mathbb{P}(Z_1 = m \mid Z_0 = k) \\
&= \sum_{m \in \mathbb{S}} \mathbb{P}(Z_{T_A} = l \text{ and } T_A < \infty \mid Z_1 = m)\mathbb{P}(Z_1 = m \mid Z_0 = k) \\
&= \sum_{m \in \mathbb{S}} P_{k,m}\mathbb{P}(Z_{T_A} = l \text{ and } T_A < \infty \mid Z_1 = m) \\
&= \sum_{m \in \mathbb{S}} P_{k,m}\mathbb{P}(Z_{T_A} = l \text{ and } T_A < \infty \mid Z_0 = m) \\
&= \sum_{m \in \mathbb{S}} P_{k,m} g_l(m), \qquad k \in \mathbb{S} \setminus A, \quad l \in A,
\end{aligned}
$$

where the relation

$$
\mathbb{P}(Z_{T_A} = l \text{ and } T_A < \infty \mid Z_1 = m) = \mathbb{P}(Z_{T_A} = l \text{ and } T_A < \infty \mid Z_0 = m)
$$

follows from the fact that the probability of ruin does not depend on the initial time the counter is started. □

Equation (5.1.3) can be rewritten in matrix form as

$$
g_l = P g_l, \qquad l \in A, \tag{5.1.4}
$$

where g_l is a column vector, i.e.

$$
\begin{bmatrix} g_l(0) \\ \vdots \\ g_l(N) \end{bmatrix} = \begin{bmatrix} P_{0,0} & P_{0,1} & P_{0,2} & \cdots & P_{0,N} \\ P_{1,0} & P_{1,1} & P_{1,2} & \cdots & P_{1,N} \\ P_{2,0} & P_{2,1} & P_{2,2} & \cdots & P_{2,N} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ P_{N,0} & P_{N,1} & P_{N,2} & \cdots & P_{N,N} \end{bmatrix} \times \begin{bmatrix} g_l(0) \\ \vdots \\ g_l(N) \end{bmatrix}, \qquad l \in A,
$$

subject to the boundary condition

$$
g_l(k) = \mathbb{P}(Z_{T_A} = l \text{ and } T_A < \infty \mid Z_0 = k) = \mathbb{1}_{\{l\}}(k) = \begin{cases} 1, & k = l, \\ 0, & k \neq l, \end{cases}
$$

for all $k, l \in A$. See e.g. Theorem 3.4 page 40 of Karlin and Taylor (1981) for a uniqueness result for the solution of such equations, and Theorem 2.1 in Goldberg (1986) for the uniqueness of solutions to difference equations in general.

In addition, the hitting probabilities

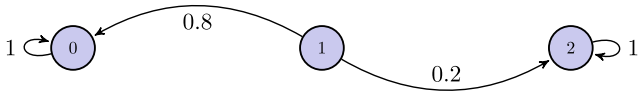
$$g_l(k) = \mathbb{P}(Z_{T_A} = l \text{ and } T_A < \infty \mid Z_0 = k), \qquad k \in \mathbb{S},$$

satisfy the condition

$$\begin{aligned} 1 &= \mathbb{P}(T_A = \infty \mid Z_0 = k) + \sum_{l \in A} \mathbb{P}(Z_{T_A} = l \text{ and } T_A < \infty \mid Z_0 = k) \\ &= \mathbb{P}(T_A = \infty \mid Z_0 = k) + \sum_{l \in A} g_l(k), \end{aligned} \tag{5.1.5}$$

for all $k \in \mathbb{S}$. Note that we may have $\mathbb{P}(T_A = \infty \mid Z_0 = k) > 0$, for example in the following chain with $A = \{0\}$ and $k = 1$ we have

$$\mathbb{P}(T_0 = \infty \mid Z_0 = 1) = 0.2.$$



More generally, if $f : A \longrightarrow \mathbb{R}$ is a function on the domain A and $\mathbb{P}(T_A < \infty \mid Z_0 = k) = 1$, letting

$$\begin{aligned} g_A(k) &:= \mathbb{E}\left[f(Z_{T_A}) \middle| Z_0 = k \right] \\ &= \sum_{l \in A} f(l) \mathbb{P}(Z_{T_A} = l \mid Z_0 = k), \qquad k \in \mathbb{S}, \end{aligned}$$

by linearity we find the *Dirichlet problem*

$$(I - P) g_A = 0,$$

subject to the boundary condition

$$g_A(k) = f(k), \qquad k \in A,$$

see, e.g., Theorem 5.3 in Privault (2008) for a continuous-time analog.

The next lemma will be used in Chap. 8 on branching Processes.

Lemma 5.2 *Assume that state* $\textcircled{j} \in \mathbb{S}$ *is absorbing. Then for all* $\textcircled{i} \in \mathbb{S}$ *we have*

$$\mathbb{P}(T_j < \infty \mid Z_0 = i) = \lim_{n \to \infty} \mathbb{P}(Z_n = j \mid Z_0 = i).$$

Proof We have

$$\{T_j < \infty\} = \bigcup_{n \geq 1} \{Z_n = j\},$$

because the finiteness of T_j means that Z_n becomes equal to $\textcircled{j}$ for some $n \in \mathbb{N}$. In addition, since $\textcircled{j} \in \mathbb{S}$ is absorbing it holds that

$$\{Z_n = j\} \subset \{Z_{n+1} = j\}, \qquad n \geq 0,$$

hence given that $\{Z_0 = i\}$, by (1.2.5) we have

$$\begin{aligned} \alpha_1 &= \mathbb{P}(T_j < \infty \mid Z_0 = i) = \mathbb{P}\left(\bigcup_{n \geq 1} \{Z_n = j\} \,\middle|\, Z_0 = i \right) \\ &= \mathbb{P}\left(\lim_{n \to \infty} \{Z_n = j\} \,\middle|\, Z_0 = i \right) = \lim_{n \to \infty} \mathbb{P}(Z_n = j \mid Z_0 = i). \end{aligned} \tag{5.1.6}$$

□

Block Triangular Transition Matrices

Assume now that the state space is $\mathbb{S} = \{0, 1, \ldots, N\}$ and the transition matrix P has the form

$$P = \begin{bmatrix} Q & R \\ 0 & I \end{bmatrix}, \tag{5.1.7}$$

where Q is a square $(r+1) \times (r+1)$ matrix, R is a $(r+1) \times (N-r)$ matrix, and

$$\mathrm{I} = \begin{bmatrix} 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix}$$

is the $(N-r) \times (N-r)$ identity matrix, in which case the states in $\{r+1, r+2, \ldots, N\}$ are *absorbing*.

If the set $A := \{r+1, r+2, \ldots, N\}$ is made of the absorbing states of the chain, we have the boundary conditions

$$g_l(m) = \mathbb{1}_{\{m=l\}}, \qquad l = 0, 1, \ldots, N, \quad m = r+1, r+2, \ldots, N, \tag{5.1.8}$$

hence the equation (5.1.3) can be rewritten as

$$\begin{aligned} g_l(k) &= \sum_{m=0}^{N} P_{k,m} g_l(m) \\ &= \sum_{m=0}^{r} P_{k,m} g_l(m) + \sum_{m=r+1}^{N} P_{k,m} g_l(m) \\ &= \sum_{m=0}^{r} P_{k,m} g_l(m) + P_{k,l} \\ &= \sum_{m=0}^{r} Q_{k,m} g_l(m) + R_{k,l}, \qquad k = 0, 1, \ldots, r, \quad l = r+1, \ldots, N, \end{aligned}$$

from (5.1.8) and since $P_{k,l} = R_{k,l}$, $k = 0, 1, \ldots, r$, $l = r+1, \ldots, N$. Hence we have

$$g_l(k) = \sum_{m=0}^{r} Q_{k,m} g_l(m) + R_{k,l}, \qquad k = 0, 1, \ldots, r, \quad l = r+1, \ldots, N.$$

Remark 5.3 In the case of the two-state Markov chain with transition matrix (4.5.1) and $A = \{0\}$, we simply find $g_0(0) = 1$ and

$$g_0(1) = b + (1 - b) \times g_0(1),$$

hence $g_0(1) = 1$ if $b > 0$ and $g_0(1) = 0$ if $b = 0$.

Example Consider a Markov chain on the state space $\mathbb{S} = \{0, 1, 2, 3\}$ with transition matrix of the form

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ a & b & c & d \\ \alpha & \beta & \gamma & \eta \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{5.1.9}$$

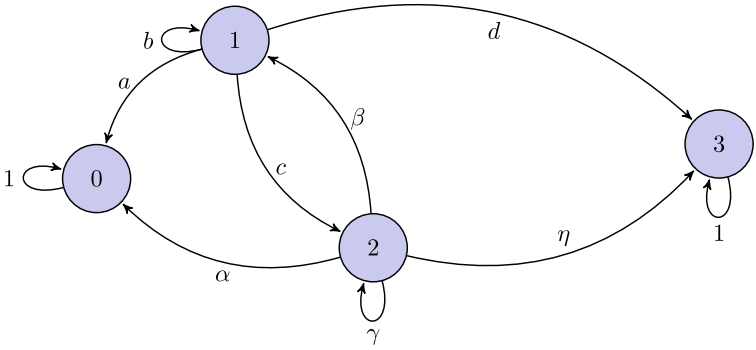
Let $A = \{0, 3\}$ denote the absorbing states of the chain, and let

$$T_{0,3} = \inf\{n \geq 0 \ : \ Z_n = 0 \text{ or } Z_n = 3\}$$

and compute the probabilities

$$g_0(k) = \mathbb{P}(Z_{T_{0,3}} = 0 \mid Z_0 = k)$$

of hitting state $\textcircled{0}$ first within $\{0, 3\}$ starting from $k = 0, 1, 2, 3$. The chain has the following graph:



(5.1.10)

Noting that ⓪ and ③ are absorbing states, and writing the relevant rows of the first step analysis matrix equation $g = Pg$, we have

$$\begin{cases} g_0(0) = 1 \\ g_0(1) = a \times 1 + b g_0(1) + c g_0(2) + d \times 0 \\ g_0(2) = \alpha \times 1 + \beta g_0(1) + \gamma g_0(2) + \eta \times 0 \\ g_0(3) = 0, \end{cases}$$

i.e.

$$\begin{cases} g_0(0) = 1 \\ g_0(1) = a + b g_0(1) + c g_0(2) \\ g_0(2) = \alpha + \beta g_0(1) + \gamma g_0(2) \\ g_0(3) = 0, \end{cases}$$

which has for solution

$$\begin{cases} g_0(0) = 1 \\ g_0(1) = \dfrac{c\alpha + a(1-\gamma)}{(1-b)(1-\gamma) - c\beta} \\ g_0(2) = \dfrac{a\beta + \alpha(1-b)}{(1-b)(1-\gamma) - c\beta} \\ g_0(3) = 0. \end{cases} \tag{5.1.11}$$

We have $g_l(0) = g_l(3) = 0$ for $l = 1, 2$, and by a similar analysis, letting

$$g_3(k) := \mathbb{P}(Z_{T_{0,3}} = 3 \mid Z_0 = k), \qquad k = 0, 1, 2, 3,$$

we find

$$\begin{cases} g_3(0) = 0 \\ g_3(1) = \dfrac{c\eta + d(1-\gamma)}{(1-b)(1-\gamma) - c\beta} \\ g_3(2) = \dfrac{\beta d + \eta(1-b)}{(1-b)(1-\gamma) - c\beta} \\ g_3(3) = 1, \end{cases}$$

and we note that

$$g_0(1) + g_3(1) = \frac{c\alpha + a(1-\gamma)}{(1-b)(1-\gamma) - c\beta} + \frac{c\eta + d(1-\gamma)}{(1-b)(1-\gamma) - c\beta} = 1,$$

since $\alpha + \eta = 1 - \gamma - \beta$ and $a + d = 1 - b - c$, and similarly

$$g_0(2) + g_3(2) = \frac{a\beta + \alpha(1-b)}{(1-b)(1-\gamma) - c\beta} + \frac{\beta d + \eta(1-b)}{(1-b)(1-\gamma) - c\beta} = 1.$$

We also check that in case $a = d$ and $\alpha = \eta$ we have

$$g_0(1) = \frac{c\alpha + a(\beta + 2\alpha)}{(c+2a)(\beta+2\alpha) - c\beta} = \frac{c\alpha + a\beta + 2a\alpha}{2c\alpha + 2a\beta + 4a\alpha} = g_0(2) = \frac{1}{2}, \quad (5.1.12)$$

and

$$g_0(1) = g_3(1) = g_0(2) = g_3(2) = \frac{1}{2}.$$

Note that, letting

$$T_0 := \inf\{n \geq 0 \ : \ Z_n = 0\} \quad \text{and} \quad T_3 := \inf\{n \geq 0 \ : \ Z_n = 0\},$$

we also have

$$g_0(k) = \mathbb{P}(Z_{T_{0,3}} = 0 \mid Z_0 = k) = \mathbb{P}(T_0 < \infty \mid Z_0 = k)$$

and

$$g_3(k) = \mathbb{P}(Z_{T_{0,3}} = 3 \mid Z_0 = k) = \mathbb{P}(T_3 < \infty \mid Z_0 = k)$$

$k = 0, 1, 2, 3$.

5.2 Mean Hitting and Absorption Times

We are now interested in the mean hitting time

$$h_A(k) := \mathbb{E}[T_A \mid Z_0 = k]$$

it takes for the chain to hit the set $A \subset \mathbb{S}$ starting from a state $k \in \mathbb{S}$. In case the set A is absorbing we refer to $h_A(k)$ as the *mean absorption time* into A starting from the state $\textcircled{k}$. Clearly, since $T_A = 0$ whenever $Z_0 = k \in A$, we have

$$h_A(k) = 0, \qquad \text{for all } k \in A.$$

Proposition 5.4 *The mean hitting times*

$$h_A(k) := \mathbb{E}[T_A \mid Z_0 = k], \qquad k \in \mathbb{S},$$

satisfy the equations

$$h_A(k) = 1 + \sum_{l \in \mathbb{S}} P_{k,l} h_A(l) = 1 + \sum_{l \in \mathbb{S} \setminus A} P_{k,l} h_A(l), \quad k \in \mathbb{S} \setminus A, \tag{5.2.1}$$

subject to the boundary conditions

$$h_A(k) = \mathbb{E}[T_A \mid Z_0 = k] = 0, \qquad k \in A.$$

Proof For all $k \in \mathbb{S} \setminus A$, by first step analysis using the *law of total expectation* applied to the probability measure $\mathbb{P}(\cdot \mid Z_0 = l)$, and the Markov property we have

$$
\begin{aligned}
h_A(k) &= \mathbb{E}[T_A \mid Z_0 = k] \\
&= \sum_{l \in \mathbb{S}} \mathbb{E}\left[T_A \mathbb{1}_{\{Z_1 = l\}} \mid Z_0 = k\right] \\
&= \frac{1}{\mathbb{P}(Z_0 = k)} \sum_{l \in \mathbb{S}} \mathbb{E}\left[T_A \mathbb{1}_{\{Z_1 = l\}} \mathbb{1}_{\{Z_0 = k\}}\right] \\
&= \frac{1}{\mathbb{P}(Z_0 = k)} \sum_{l \in \mathbb{S}} \mathbb{E}\left[T_A \mathbb{1}_{\{Z_1 = l \text{ and } Z_0 = k\}}\right] \\
&= \sum_{l \in \mathbb{S}} \mathbb{E}\left[T_A \mid Z_1 = l \text{ and } Z_0 = k\right] \frac{\mathbb{P}(Z_1 = l \text{ and } Z_0 = k)}{\mathbb{P}(Z_0 = k)} \\
&= \sum_{l \in \mathbb{S}} \mathbb{E}[T_A \mid Z_1 = l \text{ and } Z_0 = k] \mathbb{P}(Z_1 = l \mid Z_0 = k) \\
&= \sum_{l \in \mathbb{S}} \mathbb{E}[1 + T_A \mid Z_0 = l] \mathbb{P}(Z_1 = l \mid Z_0 = k) \\
&= \sum_{l \in \mathbb{S}} (1 + \mathbb{E}[T_A \mid Z_0 = l]) \mathbb{P}(Z_1 = l \mid Z_0 = k) \\
&= \sum_{l \in \mathbb{S}} \mathbb{P}(Z_1 = l \mid Z_0 = k) + \sum_{l \in \mathbb{S}} \mathbb{P}(Z_1 = l \mid Z_0 = k) \, \mathbb{E}[T_A \mid Z_0 = l] \\
&= 1 + \sum_{l \in \mathbb{S}} \mathbb{P}(Z_1 = l \mid Z_0 = k) \, \mathbb{E}[T_A \mid Z_0 = l] \\
&= 1 + \sum_{l \in \mathbb{S}} P_{k,l} h_A(l), \qquad k \in \mathbb{S} \setminus A,
\end{aligned}
$$

with the relation

$$
\mathbb{E}[T_A \mid Z_1 = l, \ Z_0 = k] = 1 + \mathbb{E}[T_A \mid Z_0 = l].
$$

Hence we have

$$
h_A(k) = 1 + \sum_{l \in \mathbb{S}} P_{k,l} h_A(l), \quad k \in \mathbb{S} \setminus A, \tag{5.2.2}
$$

subject to the boundary conditions

$$
h_A(k) = \mathbb{E}[T_A \mid Z_0 = k] = 0, \qquad k \in A, \tag{5.2.3}
$$

the Condition (5.2.3) implies that (5.2.2) becomes

$$
h_A(k) = 1 + \sum_{l \in \mathbb{S} \setminus A} P_{k,l} h_A(l), \qquad k \in \mathbb{S} \setminus A.
$$

The Eq. (5.2.1) can be rewritten in matrix form as

$$h_A = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} + Ph_A,$$

by considering only the rows with index $k \in A^c = \mathbb{S} \setminus A$, subject to the boundary conditions

$$h_A(k) = 0, \qquad k \in A.$$

Block Triangular Transition Matrices

When the transition matrix P has the form (5.1.7) and $A = \{r+1, r+2, \ldots, N\}$, Eq. (5.2.2) rewrites as

$$\begin{aligned} h_A(k) &= 1 + \sum_{l=0}^{N} P_{k,l} h_A(l) \\ &= 1 + \sum_{l=0}^{r} P_{k,l} h_A(l) + \sum_{l=r+1}^{N} P_{k,l} h_A(l) \\ &= 1 + \sum_{l=0}^{r} P_{k,l} h_A(l), \qquad 0 \le k \le r, \end{aligned}$$

since $h_A(l) = 0, l = r+1, r+2, \ldots, N$, i.e.

$$h_A(k) = 1 + \sum_{l=0}^{r} P_{k,l} h_A(l), \qquad 0 \le k \le r,$$

with $h_A(k) = 0, k = r+1, \ldots, n$.

Two-State Chain

In the case of the two-state Markov chain with transition matrix (4.5.1) and $A = \{0\}$, we simply find $h_{\{0\}}(0) = h_{\{1\}}(1) = 0$ and

$$h_{\{0\}}(1) = b \times 1 + (1-b)(1 + h_{\{0\}}(1)) = 1 + (1-b)h_{\{0\}}(1), \tag{5.2.4}$$

with solution

$$h_{\{0\}}(1) = b \sum_{k\geq 1} k(1-b)^{k-1} = \frac{1}{b},$$

and similarly we find

$$h_{\{1\}}(0) = a \sum_{k\geq 1} k(1-a)^{k-1} = \frac{1}{a},$$

cf. also (5.3.3) below.

Utility Functionals

The above can be generalized to derive an equation for an expectation of the form

$$h_A(k) := \mathbb{E}\left[\sum_{i=0}^{T_A} f(Z_i) \,\middle|\, Z_0 = k \right], \qquad k = 0, 1, \ldots, N,$$

where $f(\cdot)$ is a given utility function, as follows:

$$\begin{aligned}
h_A(k) &= \mathbb{E}\left[\sum_{i=0}^{T_A} f(Z_i) \,\middle|\, Z_0 = k \right] \\
&= \sum_{l=0}^{N} P_{k,l} \left(f(k) + \mathbb{E}\left[\sum_{i=1}^{T_A} f(Z_i) \,\middle|\, Z_1 = l \right] \right) \\
&= \sum_{l=0}^{N} P_{k,l} f(k) + \sum_{l=0}^{N} P_{k,l}\, \mathbb{E}\left[\sum_{i=1}^{T_A} f(Z_i) \,\middle|\, Z_1 = l \right] \\
&= f(k) \sum_{l=0}^{N} P_{k,l} + \sum_{l=0}^{r} P_{k,l}\, \mathbb{E}\left[\sum_{i=0}^{T_A} f(Z_i) \,\middle|\, Z_0 = l \right] \\
&= f(k) + \sum_{l=0}^{r} P_{k,l} h_A(l), \qquad k \in A^c := \{0, 1, \ldots, r\},
\end{aligned}$$

with $A := \{r+1, \ldots, N\}$, hence

$$h_A(k) = f(k) + \sum_{l=0}^{r} P_{k,l} h_A(l), \qquad k \in A^c = \{0, 1, \ldots, r\},$$

which can be rewritten as

$$\begin{bmatrix} h_A(0) \\ \vdots \\ h_A(r) \end{bmatrix} = \begin{bmatrix} f(0) \\ \vdots \\ f(r) \end{bmatrix} + \begin{bmatrix} P_{0,0} & P_{0,1} & P_{0,2} & \cdots & P_{0,r} \\ P_{1,0} & P_{1,1} & P_{1,2} & \cdots & P_{1,r} \\ P_{2,0} & P_{2,1} & P_{2,2} & \cdots & P_{2,r} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ P_{r,0} & P_{r,1} & P_{r,2} & \cdots & P_{r,r} \end{bmatrix} \times \begin{bmatrix} h_A(0) \\ \vdots \\ h_A(r) \end{bmatrix},$$

with the boundary condition

$$h_A(k) = f(k), \qquad k \in A = \{r+1, \ldots, n\},$$

see also Exercise 5.19 and Problem 5.23.

For example:

– When $f = \mathbb{1}_{A^c} = \mathbb{1}_{\{0,1,\ldots,r\}}$ is the indicator function over the set A^c, i.e.

$$f(Z_i) = \mathbb{1}_{A^c}(Z_i) = \begin{cases} 1 & \text{if } Z_i \notin A, \\ 0 & \text{if } Z_i \in A, \end{cases}$$

the quantity $h_A(k)$ coincides with the mean hitting time of the set A starting from the state $\textcircled{k}$. In particular, when $A = \{m\}$ this recovers the equation

$$h_{\{m\}}(k) = 1 + \sum_{\substack{l \in \mathbb{S} \\ l \neq m}} P_{k,l} h_{\{m\}}(l), \qquad k \in \mathbb{S} \setminus \{m\}, \tag{5.2.5}$$

with $h_{\{m\}}(m) = 0$.

– When f is the indicator function $f = \mathbb{1}_{\{l\}}$, i.e.

$$f(Z_i) = \mathbb{1}_{\{l\}}(Z_i) = \begin{cases} 1 & \text{if } Z_i = l, \\ 0 & \text{if } Z_i \neq l, \end{cases}$$

with $l \in A^c$, the quantity $h_A(k)$ counts the mean number of visits to state $\textcircled{l}$ starting from $\textcircled{k}$ before hitting the set A.

– See Exercise 5.19, and Problem 5.21 for a complete solution in case $f(k) := k$ and $(Z_k)_{k \geq 0}$ is the gambling process of Chap. 2.

Poisson Equation

Taking

$$h_A(k) := \mathbb{E}\left[\sum_{i=0}^{T_A-1} f(Z_i) \,\middle|\, Z_0 = k\right], \qquad k = 0, 1, \ldots, N,$$

leads to the same first step analysis equation

$$h_A(k) = f(k) + \sum_{m=0}^{r} P_{k,m} h_A(m), \qquad k \in A^c := \{0, 1, \ldots, r\},$$

or

$$(I - [P_{k,m}]_{k,m=0,1,\ldots,r}) \times h_A = f,$$

with the boundary condition

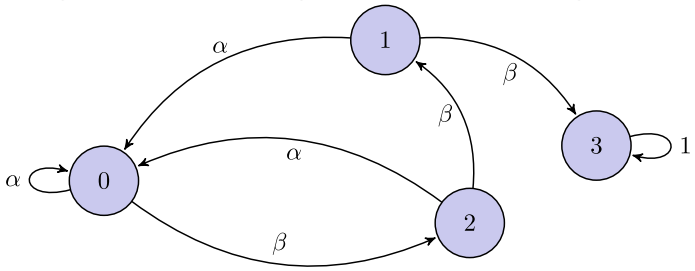
$$h_A(k) = 0, \qquad k \in A = \{r+1, \ldots, n\},$$

see e.g. Theorem 5.5 in Privault (2008) for a continuous-time analog.

Example Consider a Markov chain whose transition probability matrix is given by

$$P = \left[P_{i,j}\right]_{0 \le i,j \le 3} = \begin{bmatrix} \alpha & 0 & \beta & 0 \\ \alpha & 0 & 0 & \beta \\ \alpha & \beta & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

where $\alpha, \beta \ge 0$ and $\alpha + \beta = 1$. Taking $A := \{3\}$, determine the mean time it takes to reach state (3) starting from state (0). We observe that state (3) is absorbing:



Let

$$h_3(k) := \mathbb{E}[T_3 \mid Z_0 = k]$$

denote the mean (hitting) time needed to reach state (3), after starting from state $k = 0, 1, 2, 3$. We get

$$\begin{cases} h_3(0) = \alpha(1 + h_3(0)) + \beta(1 + h_3(2)) = 1 + \alpha h_3(0) + \beta h_3(2) \\ h_3(1) = \alpha(1 + h_3(0)) + \beta(1 + h_3(3)) = 1 + \alpha h_3(0) \\ h_3(2) = \alpha(1 + h_3(0)) + \beta(1 + h_3(1)) = 1 + \alpha h_3(0) + \beta h_3(1) \\ h_3(3) = 0, \end{cases}$$

which, using the relation $\alpha = 1 - \beta$, yields

$$h_3(3) = 0, \qquad h_3(1) = \frac{1}{\beta^3}, \qquad h_3(2) = \frac{1+\beta}{\beta^3}, \qquad h_3(0) = \frac{1+\beta+\beta^2}{\beta^3}.$$

Since state (3) can only be reached from state (1) with probability β, it is natural that the hitting times go to infinity as β goes to zero. We also check that $h_3(3) < h_3(1) < h_3(2) < h_3(0)$, as can be expected from the above graph. In addition, $(h_3(1), h_3(2), h_3(0))$ converge to $(1, 2, 3)$ as β goes to 1, as can be expected.

5.3 First Return Times

Consider now the first *return* time T_j^r to state $j \in \mathbb{S}$, defined by

$$T_j^r := \inf\{n \geq 1 \ : \ Z_n = j\},$$

with

$$T_j^r = +\infty \quad \text{if } Z_n \neq j \text{ for all } n \geq 1.$$

Note that in contrast with the definition (5.1.1) of the hitting time T_j, the infimum is taken here for $n \geq 1$ as it takes at least one step out of the initial state in order to *return* to state (j). Nevertheless we have $T_j = T_j^r$ if the chain is started from a state (i) different from (j).

We denote by

$$\mu_j(i) := \mathbb{E}[T_j^r \mid Z_0 = i] \geq 1$$

the *mean return time* to state $j \in \mathbb{S}$ after starting from state $i \in \mathbb{S}$.

Mean return times can also be computed by first step analysis. We have

$$\begin{aligned}
\mu_j(i) &= \mathbb{E}[T_j^r \mid Z_0 = i] \\
&= 1 \times \mathbb{P}(Z_1 = j \mid Z_0 = i) \\
&\quad + \sum_{\substack{l \in \mathbb{S} \\ l \neq j}} \mathbb{P}(Z_1 = l \mid Z_0 = i)(1 + \mathbb{E}[T_j^r \mid Z_0 = l]) \\
&= P_{i,j} + \sum_{\substack{l \in \mathbb{S} \\ l \neq j}} P_{i,l}(1 + \mu_j(l)) \\
&= P_{i,j} + \sum_{\substack{l \in \mathbb{S} \\ l \neq j}} P_{i,l} + \sum_{\substack{l \in \mathbb{S} \\ l \neq j}} P_{i,l}\mu_j(l) \\
&= \sum_{l \in \mathbb{S}} P_{i,l} + \sum_{\substack{l \in \mathbb{S} \\ l \neq j}} P_{i,l}\mu_j(l) \\
&= 1 + \sum_{\substack{l \in \mathbb{S} \\ l \neq j}} P_{i,l}\mu_j(l),
\end{aligned}$$

hence

$$\mu_j(i) = 1 + \sum_{\substack{l \in \mathbb{S} \\ l \neq j}} P_{i,l}\mu_j(l), \qquad i, j \in \mathbb{S}. \tag{5.3.1}$$

See e.g. Theorem 5.9 page 49 of Karlin and Taylor (1981) for a uniqueness result for the solution of such equations.

Hitting Times Versus Return Times

Note that the mean return time equation in (5.3.1) does not include any boundary condition, in contrast with the mean hitting time Eq. (5.2.5) in Sect. 5.2. In addition, the time T_i^r to return to state $\textcircled{i}$ is always at least one by construction, hence $\mu_i(i) \geq 1$ and cannot vanish, while we always have $h_i(i) = 0$ as boundary condition, $i \in \mathbb{S}$. On the other hand, for $i \neq j$ we have by definition

$$\mu_i(j) = \mathbb{E}[T_i^r \mid Z_0 = j] = \mathbb{E}[T_i \mid Z_0 = j] = h_i(j),$$

and for $i = j$ the mean return time $\mu_j(j)$ can be computed from the hitting times $h_j(l)$, $l \neq j$, by first step analysis as

$$\begin{aligned}\mu_j(j) &= \sum_{l\in S} P_{j,l}(1+h_j(l)) \\ &= P_{j,j} + \sum_{l\neq j} P_{j,l}(1+h_j(l)) \\ &= \sum_{l\in S} P_{j,l} + \sum_{l\neq j} P_{j,l}h_j(l) \\ &= 1 + \sum_{l\neq j} P_{j,l}h_j(l), \qquad j \in \mathbb{S}, \end{aligned} \tag{5.3.2}$$

which is in agreement with (5.3.1) when $i = j$.

In practice we may prefer to compute first the hitting times $h_i(j) = 0$ under the boundary conditions $h_i(i) = 0$, and then to recover the return time $\mu_i(i)$ from (5.3.2), $i, j \in \mathbb{S}$.

Examples

(i) Mean return times for the two-state Markov chain.

The mean return time $\mu_0(i) = \mathbb{E}[T_0^r \mid Z_0 = i]$ to state ⓪ starting from state ⓘ $\in \{0, 1\}$ satisfies

$$\begin{cases} \mu_0(0) = (1-a) \times 1 + a(1+\mu_0(1)) = 1 + a\mu_0(1) \\ \mu_0(1) = b \times 1 + (1-b)(1+\mu_0(1)) = 1 + (1-b)\mu_0(1) \end{cases}$$

which yields

$$\mu_0(0) = 1 + \frac{a}{b} \quad \text{and} \quad \mu_0(1) = h_0(1) = \frac{1}{b}, \tag{5.3.3}$$

cf. also (5.2.4) above for the computation of $\mu_0(1) = h_0(1) = 1/b$ as a mean hitting time. In the two-state case, the distribution of T_0^r given $Z_0 = 0$ is given by

$$f_{0,0}^{(n)} := \mathbb{P}(T_0^r = n \mid Z_0 = 0) = \begin{cases} 0 & \text{if } n = 0, \\ 1-a & \text{if } n = 1, \\ ab(1-b)^{n-2} & \text{if } n \geq 2, \end{cases} \tag{5.3.4}$$

hence (5.3.3) can be directly recovered as[1]

[1] We are using the identities $\sum_{k\geq 0} r^k = (1-r)^{-1}$ and $\sum_{k\geq 1} kr^{k-1} = (1-r)^{-2}$, cf. (A.3) and (A.4).

$$\begin{aligned}
\mu_0(0) &= \mathbb{E}[T_0^r \mid Z_0 = 0] \\
&= \sum_{n\geq 0} n\mathbb{P}(T_0^r = n \mid Z_0 = 0) \\
&= \sum_{n\geq 0} n f_{0,0}^{(n)} \\
&= 1 - a + ab \sum_{n\geq 2} n(1-b)^{n-2} \\
&= 1 - a + ab \sum_{n\geq 0} (n+2)(1-b)^n \\
&= 1 - a + ab(1-b) \sum_{n\geq 0} n(1-b)^{n-1} + 2ab \sum_{n\geq 0} (1-b)^n \\
&= \frac{a+b}{b} = 1 + \frac{a}{b}, \qquad (5.3.5)
\end{aligned}$$

where we used the identity (A.4). Similarly, we check that

$$\begin{cases} \mu_1(0) = 1 + (1-a)\mu_1(0) \\ \mu_1(1) = 1 + b\mu_1(0), \end{cases}$$

which yields

$$\mu_1(0) = h_1(0) = \frac{1}{a} \quad \text{and} \quad \mu_1(1) = 1 + \frac{b}{a},$$

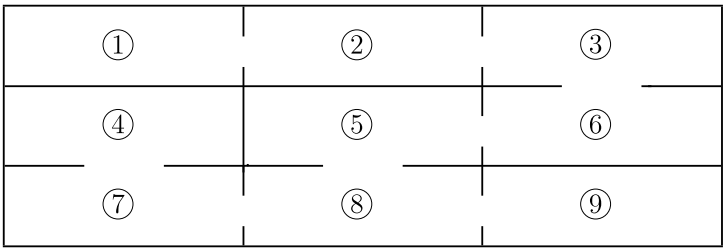
and can be directly recovered by

$$\mu_1(1) = 1 - b + ab \sum_{n\geq 0} (n+2)(1-a)^n = \frac{a+b}{a} = 1 + \frac{b}{a}, \qquad (5.3.6)$$

as in (5.3.3) and (5.3.5) above, by swapping a with b and state ⓪ with state ①.

(ii) Maze problem.

Mazes provide natural examples of Markovian systems as their users tend rely on their current positions and to forget past information. More generally, Markovian systems can be used as an approximation of a non-Markovian reality.
Consider a fish placed in an aquarium with 9 compartments:



(5.3.7)

The fish moves randomly: at each time step it changes compartments and if it finds $k \geq 1$ exit doors from one compartment, it will choose one of them with probability $1/k$, i.e. the transition matrix is

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 0 & 0 & 1/2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 1/2 & 0 & 1/2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 0 & 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 0 & 1/3 & 0 & 1/3 & 0 & 1/3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}.$$

Question: Find the average time to come back to state ① after starting from state ①.

Answer: Letting

$$T_l^r = \inf\{n \geq 1 \ : \ Z_n = l\}$$

denote the first return time to state l, and defining

$$\mu_1(k) := \mathbb{E}[T_1^r \mid Z_0 = k]$$

the mean return time to state ① starting from k, we have

$$\begin{cases} \mu_1(1) = 1 + \mu_1(2) \\ \mu_1(2) = \dfrac{1}{2}(1+0) + \dfrac{1}{2}(1+\mu_1(3)) = 1 + \dfrac{1}{2}\mu_1(3) \\ \mu_1(3) = \dfrac{1}{2}(1+\mu_1(2)) + \dfrac{1}{2}(1+\mu_1(6)) = 1 + \dfrac{1}{2}\mu_1(2) + \dfrac{1}{2}\mu_1(6) \\ \mu_1(4) = 1 + \mu_1(7) \\ \mu_1(5) = \dfrac{1}{2}(1+\mu_1(8)) + \dfrac{1}{2}(1+\mu_1(6)) = 1 + \dfrac{1}{2}\mu_1(8) + \dfrac{1}{2}\mu_1(6) \\ \mu_1(6) = \dfrac{1}{2}(1+\mu_1(3)) + \dfrac{1}{2}(1+\mu_1(5)) = 1 + \dfrac{1}{2}\mu_1(3) + \dfrac{1}{2}\mu_1(5) \\ \mu_1(7) = 1 + \dfrac{1}{2}\mu_1(4) + \dfrac{1}{2}\mu_1(8) = \dfrac{1}{2}(1+\mu_1(4)) + \dfrac{1}{2}(1+\mu_1(8)) \\ \mu_1(8) = \dfrac{1}{3}(1+\mu_1(7)) + \dfrac{1}{3}(1+\mu_1(5)) + \dfrac{1}{3}(1+\mu_1(9)) \\ \qquad = 1 + \dfrac{1}{3}\left(\mu_1(7) + \mu_1(5) + \mu_1(9)\right) \\ \mu_1(9) = 1 + \mu_1(8), \end{cases}$$

or

$$\begin{cases} \mu_1(1) = 1 + \mu_1(2), \quad \mu_1(2) = 1 + \dfrac{1}{2}\mu_1(3), \quad \mu_1(3) = 2 + \dfrac{2}{3}\mu_1(6), \\ \mu_1(4) = 1 + \mu_1(7), \quad \mu_1(5) = 1 + \dfrac{1}{2}\mu_1(8) + \dfrac{1}{2}\mu_1(6), \quad 0 = 30 + 3\mu_1(8) - 5\mu_1(6), \\ \mu_1(7) = 3 + \mu_1(8), \quad 0 = 80 + 5\mu_1(6) - 5\mu_1(8), \quad \mu_1(9) = 1 + \mu_1(8), \end{cases}$$

which yields

$$\begin{aligned} &\mu_1(1) = 16,\ \mu_1(2) = 15,\ \mu_1(3) = 28,\ \mu_1(4) = 59,\ \mu_1(5) = 48, \\ &\mu_1(6) = 39,\ \mu_1(7) = 58,\ \mu_1(8) = 55,\ \mu_1(9) = 56. \end{aligned} \tag{5.3.8}$$

Consequently, it takes on average 16 steps to come back to ① starting from ①, and 59 steps to reach ① starting from ④. These data are illustrated in the following picture in which the numbers represent the average time it takes to return to ① starting from a given state.

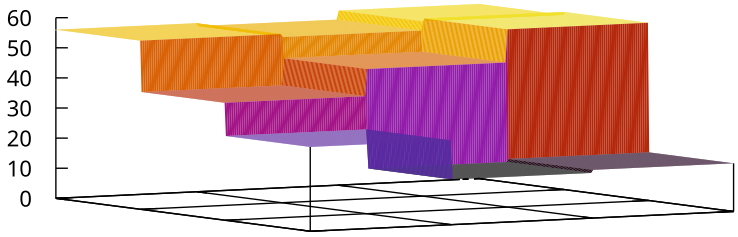


Fig. 5.1 Mean return times to state 0 on the maze (5.3.7)

$\mu_1(1) = 16$	$\mu_1(2) = 15$	$\mu_1(3) = 28$
$\mu_1(4) = 59$	$\mu_1(5) = 48$	$\mu_1(6) = 39$
$\mu_1(7) = 58$	$\mu_1(8) = 55$	$\mu_1(9) = 56$

Figure 5.1 represents the mean return times to state $\textcircled{0}$ according to the initial state on the maze (5.3.7).

5.4 Mean Number of Returns

Return Probabilities

In what follows, we let

$$p_{ij} = \mathbb{P}(T_j^r < \infty \mid Z_0 = i) = \mathbb{P}(Z_n = j \text{ for some } n \geq 1 \mid Z_0 = i), \quad i, j \in \mathbb{S},$$

denote the probability of return to state $\textcircled{j}$ in finite time[2] The probability p_{ii} of return to state $\textcircled{i}$ within a finite time starting from state $\textcircled{i}$ can be computed as follows:

[2] When $\textcircled{i} \neq \textcircled{j}$, p_{ij} is the probability of *visiting* state $\textcircled{j}$ in finite time after starting from state $\textcircled{i}$. starting from state $\textcircled{i}$.

$$\begin{aligned} p_{ii} &= \mathbb{P}(Z_n = i \text{ for some } n \geq 1 \mid Z_0 = i) \\ &= \mathbb{P}\Big(\bigcup_{n\geq 1} \{Z_n = i\} \,\Big|\, Z_0 = i \Big) \\ &= \sum_{n\geq 1} \mathbb{P}(Z_n = i, Z_{n-1} \neq i, \ldots, Z_1 \neq i \mid Z_0 = i) \\ &= \sum_{n\geq 1} f_{i,i}^{(n)}, \end{aligned} \tag{5.4.1}$$

where

$$f_{i,j}^{(n)} := \mathbb{P}(T_j^r = n \mid Z_0 = i) = \mathbb{P}(Z_n = j, Z_{n-1} \neq j, \ldots, Z_1 \neq j \mid Z_0 = i),$$

$i, j \in \mathbb{S}$, is the probability distribution of T_j^r given that $Z_0 = i$, with

$$f_{i,i}^{(0)} = \mathbb{P}(T_i^r = 0 \mid Z_0 = i) = 0.$$

Note that we have

$$f_{i,i}^{(1)} = \mathbb{P}(Z_1 = i \mid Z_0 = i) = P_{i,i}, \qquad i \in \mathbb{S}.$$

Convolution Equation

By conditioning on the first return time $k \geq 1$, the return time probability distribution $f_{i,i}^{(k)} = \mathbb{P}(T_i^r = k \mid Z_0 = i)$ satisfies the convolution equation

$$\begin{aligned} [P^n]_{i,i} &= \mathbb{P}(Z_n = i \mid Z_0 = i) \\ &= \sum_{k=1}^{n} \mathbb{P}(Z_k = i, Z_{k-1} \neq i, \ldots, Z_1 \neq i \mid Z_0 = i) \mathbb{P}(Z_n = i \mid Z_k = i) \\ &= \sum_{k=1}^{n} \mathbb{P}(Z_k = i, Z_{k-1} \neq i, \ldots, Z_1 \neq i \mid Z_0 = i) \mathbb{P}(Z_{n-k} = i \mid Z_0 = i) \\ &= \sum_{k=1}^{n} f_{i,i}^{(k)} [P^{n-k}]_{i,i}, \end{aligned}$$

which extends the convolution equation (3.4.7) from random walks to the more general setting of Markov chains.

The return probabilities p_{ij} will be used below to compute the average number of returns to a given state, and the distribution $f_{i,j}^{(k)}$, $k \geq 1$, of T_j^r given that $Z_0 = i$ will be useful in Section 6.4 on positive and null recurrence.

Number of Returns

Let

$$R_j := \sum_{n\geq 1} \mathbb{1}_{\{Z_n=j\}} \tag{5.4.2}$$

denote the number of returns[3] to state $\boxed{j}$ by the chain $(Z_n)_{n\geq 0}$. The next proposition shows that, given $\{Z_0 = i\}$, R_j has a zero-modified geometric distribution with initial mass $1 - p_{ij}$.

Proposition 5.5 *The probability distribution of the number of returns R_j to state j given that $\{Z_0 = i\}$ is given by*

$$\mathbb{P}(R_j = m \mid Z_0 = i) = \begin{cases} 1 - p_{ij}, & m = 0, \\ p_{ij} \times (p_{jj})^{m-1} \times (1 - p_{jj}), & m \geq 1, \end{cases}$$

Proof When the chain never visits state $\boxed{j}$ starting from $Z_0 = i$ we have $R_j = 0$, and this happens with probability

$$\begin{aligned} \mathbb{P}(R_j = 0 \mid Z_0 = i) &= \mathbb{P}(T_j^r = \infty \mid Z_0 = i) \\ &= 1 - \mathbb{P}(T_j^r < \infty \mid Z_0 = i) \\ &= 1 - p_{ij}. \end{aligned}$$

Next, when the chain $(Z_n)_{n\geq 0}$ makes a number $R_j = m \geq 1$ of visits to state $\boxed{j}$ starting from state $\boxed{i}$, it makes a first visit to state $\boxed{j}$ with probability p_{ij} and then makes $m - 1$ returns to state $\boxed{j}$, each with probability p_{jj}. After those m visits, it never returns to state $\boxed{j}$, and this event occurs with probability $1 - p_{jj}$. Hence, given that $\{Z_0 = i\}$ we have

$$\mathbb{P}(R_j = m \mid Z_0 = i) = \begin{cases} p_{ij} \times (p_{jj})^{m-1} \times (1 - p_{jj}), & m \geq 1, \\ 1 - p_{ij}, & m = 0, \end{cases}$$

by the same argument as in (5.3.4) above.

In case $i = j$, R_i is simply the number of returns to state $\boxed{i}$ starting from state $\boxed{i}$, and it has the geometric distribution

$$\mathbb{P}(R_i = m \mid Z_0 = i) = (1 - p_{ii})(p_{ii})^m, \qquad m \geq 0. \tag{5.4.3}$$

[3] Here, R_j is called a number of returns because the time counter is started at $n = 1$ and excludes the initial state.

Proposition 5.6 *We have*

$$\mathbb{P}(R_j < \infty \mid Z_0 = i) = \begin{cases} 1 - p_{ij}, & \text{if } p_{jj} = 1, \\ 1, & \text{if } p_{jj} < 1. \end{cases}$$

Proof We note that

$$\begin{aligned} \mathbb{P}(R_j < \infty \mid Z_0 = i) &= \mathbb{P}(R_j = 0 \mid Z_0 = i) + \sum_{m \geq 1} \mathbb{P}(R_j = m \mid Z_0 = i) \\ &= 1 - p_{ij} + p_{ij}(1 - p_{jj}) \sum_{m \geq 1} (p_{jj})^{m-1} \\ &= \begin{cases} 1 - p_{ij}, & \text{if } p_{jj} = 1, \\ 1, & \text{if } p_{jj} < 1. \end{cases} \end{aligned}$$

□

We also have

$$\mathbb{P}(R_j = \infty \mid Z_0 = i) = \begin{cases} p_{ij}, & \text{if } p_{jj} = 1, \\ 0, & \text{if } p_{jj} < 1. \end{cases}$$

In particular, if $p_{jj} = 1$, i.e. state $\textcircled{j}$ is recurrent, we have

$$\mathbb{P}(R_j = m \mid Z_0 = i) = 0, \qquad m \geq 1,$$

and in this case,

$$\begin{cases} \mathbb{P}(R_j < \infty \mid Z_0 = i) = \mathbb{P}(R_j = 0 \mid Z_0 = i) = 1 - p_{ij}, \\ \mathbb{P}(R_j = \infty \mid Z_0 = i) = 1 - \mathbb{P}(R_j < \infty \mid Z_0 = i) = p_{ij}. \end{cases}$$

On the other hand, when $i = j$, by (1.5.13) we find

$$\begin{aligned} \mathbb{P}(R_i < \infty \mid Z_0 = i) &= \sum_{m \geq 0} \mathbb{P}(R_i = m \mid Z_0 = i) \\ &= (1 - p_{ii}) \sum_{m \geq 0} (p_{ii})^m \\ &= \begin{cases} 0, & \text{if } p_{ii} = 1, \\ 1, & \text{if } p_{ii} < 1, \end{cases} \end{aligned} \tag{5.4.4}$$

hence

$$\mathbb{P}(R_i = \infty \mid Z_0 = i) = \begin{cases} 1, & \text{if } p_{ii} = 1, \\ 0, & \text{if } p_{ii} < 1, \end{cases} \tag{5.4.5}$$

i.e. the number of returns to a recurrent state is infinite with probability one. This is an example of a 0–1 law.

Mean Number of Returns

The notion of *mean number of returns* will be needed for the classification of states of Markov chains in Chap. 6.

Proposition 5.7 *The mean number of returns to state* $\textcircled{j}$ *is given by*

$$\mathbb{E}[R_j \mid X_0 = i] = \frac{p_{ij}}{1 - p_{jj}},$$

and it is finite, i.e. $\mathbb{E}[R_j \mid X_0 = i] < \infty$, *if and only if* $p_{jj} < 1$, $i, j \in \mathbb{S}$.

Proof By (A.4), when $p_{jj} < 1$ we have $\mathbb{P}(R_j < \infty \mid X_0 = i) = 1$ and

$$\begin{aligned} \mathbb{E}[R_j \mid X_0 = i] &= \sum_{m \geq 1} m \mathbb{P}(R_j = m \mid X_0 = i) && (5.4.6) \\ &= (1 - p_{jj}) p_{ij} \sum_{m \geq 1} m (p_{jj})^{m-1} \\ &= \frac{p_{ij}}{1 - p_{jj}}, && (5.4.7) \end{aligned}$$

hence

$$\mathbb{E}[R_j \mid X_0 = i] < \infty \quad \text{if} \quad p_{jj} < 1.$$

If $p_{jj} = 1$ then $\mathbb{E}[R_j \mid X_0 = i] = \infty$ unless $p_{ij} = 0$, in which case $\mathbb{P}(R_j = 0 \mid X_0 = i) = 1$ and $\mathbb{E}[R_j \mid X_0 = i] = 0$.

By analogy with (A.3), the matrix inverse

$$G := (I - P)^{-1} = \sum_{n \geq 0} P^n = I + \sum_{n \geq 1} P^n \tag{5.4.8}$$

of $I - P$ is called the *potential kernel*, or the *resolvent* of P. See Althoen et al. (1993) for an application of the next proposition to the Snakes and Ladders game.

Proposition 5.8 *If* $\textcircled{m}$ *is the* only *absorbing state in* $\mathbb{S}$, *we have the expression*

$$\mathbb{E}[T_m \mid Z_0 = i] = \sum_{\substack{j \in \mathbb{S} \\ j \neq m}} \left[(I - P)^{-1} \right]_{i,j}, \quad i \neq m. \tag{5.4.9}$$

Proof By (5.4.2) and (5.4.8) we have

$$\begin{aligned}\mathbb{E}[R_j \mid Z_0 = i] &= \mathbb{E}\left[\sum_{n\geq 1} \mathbb{1}_{\{Z_n=j\}} \,\middle|\, Z_0 = i\right] \\ &= \sum_{n\geq 1} \mathbb{E}[\mathbb{1}_{\{Z_n=j\}} \mid Z_0 = i] \\ &= \sum_{n\geq 1} \mathbb{P}(Z_n = j \mid Z_0 = i) \\ &= \sum_{n\geq 1} \left[P^n\right]_{i,j} \\ &= -\mathbb{1}_{\{i=j\}} + \left[(I-P)^{-1}\right]_{i,j}.\end{aligned}$$

On the other hand, after starting from $i \neq m$ we have

$$T_m = 1 + \sum_{\substack{j\in\mathbb{S}\\ j\neq m}} R_j,$$

hence

$$\begin{aligned}\mathbb{E}[T_m \mid Z_0 = i] &= 1 + \sum_{\substack{j\in\mathbb{S}\\ j\neq m}} \mathbb{E}[R_j \mid Z_0 = i] \\ &= 1 + \sum_{\substack{j\in\mathbb{S}\\ j\neq m}} \left(-\mathbb{1}_{\{i=j\}} + \left[(I-P)^{-1}\right]_{i,j}\right) \\ &= \sum_{\substack{j\in\mathbb{S}\\ j\neq m}} \left[(I-P)^{-1}\right]_{i,j}.\end{aligned}$$

□

Exercises

Exercise 5.1 Consider the Markov chain $(X_n)_{n\geq 0}$ with state space $\mathbb{S} = \{0, 1, 2, 3\}$ and transition probabilities

$$\begin{aligned}&\mathbb{P}(X_1 = 0 \mid X_0 = 0) = 1, && \mathbb{P}(X_1 = 3 \mid X_0 = 3) = 1,\\ &\mathbb{P}(X_1 = 0 \mid X_0 = 1) = 1/2, && \mathbb{P}(X_1 = 2 \mid X_0 = 1) = 1/2,\\ &\mathbb{P}(X_1 = 1 \mid X_0 = 2) = 1/3, && \mathbb{P}(X_1 = 3 \mid X_0 = 2) = 2/3.\end{aligned}$$

(a) Draw the graph of the chain and write down its transition matrix.

(b) Compute $\alpha := \mathbb{P}(T_3 < \infty \mid X_0 = 1)$ and $\beta := \mathbb{P}(T_3 < \infty \mid X_0 = 2)$, where

$$T_3 := \inf\{n \geq 0 \ : \ X_n = 3\}.$$

(c) Letting

$$T_{0,3} := \inf\{n \geq 0 \ : \ X_n = 0 \text{ or } X_n = 3\},$$

compute $\mathbb{E}[T_{0,3} \mid X_0 = 1]$ and $\mathbb{E}[T_{0,3} \mid X_0 = 2]$.

Exercise 5.2 Consider the two-state Markov chain $(X_n)_{n\geq 0}$ with state space $\mathbb{S} = \{0, 1\}$ and transition matrix

$$P = \begin{bmatrix} 0.4 & 0.6 \\ 0.8 & 0.2 \end{bmatrix}.$$

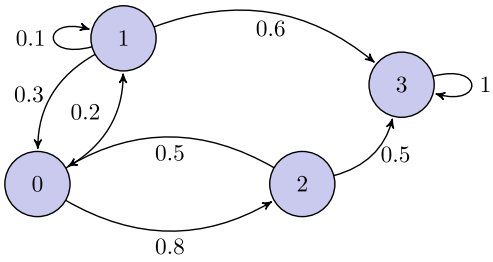
Compute the mean duration between two visits to state $\textcircled{1}$.

Exercise 5.3 Consider the Markov chain $(X_n)_{n\geq 0}$ on $\mathbb{S} = \{0, 1, 2\}$ whose transition probability matrix P is given by

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 \\ 1/3 & 0 & 2/3 \\ 0 & 1 & 0 \end{bmatrix} \end{array}.$$

(a) Draw a graph of the chain and find the probability $g_0(k)$ that the chain is absorbed into state $\textcircled{0}$ given that it started from states $k = 0, 1, 2$.
(b) Determine the mean time $h_0(k)$ it takes until the chain is absorbed into state $\textcircled{0}$, after starting from $k = 0, 1, 2$.

Exercise 5.4 Consider the Markov chain with graph



(5.4.10)

and let

$$T_k^r := \inf\{n \geq 1 \ : \ X_n = k\}$$

denote the *return* time to state $k = 0, 1, 2, 3$.

(a) Find the probabilities

$$p_2(k) := \mathbb{P}\big(T_2^r < \infty \mid X_0 = k\big),$$

of returning to state $\textcircled{2}$ in finite time after starting from $k = 0, 1, 2, 3$.

(b) Find the probabilities

$$p_1(k) := \mathbb{P}\big(T_1^r < \infty \mid X_0 = k\big), \qquad k = 0, 1, 2, 3.$$

of *returning* to state $\textcircled{1}$ in finite time after starting from $k = 0, 1, 2, 3$.

(c) Compute the mean return times $\mu_2(k) := \mathbb{E}\left[T_2^r \mid X_0 = k\right]$ to state $\textcircled{2}$ after starting from $k = 0, 1, 2, 3$.

Exercise 5.5 Consider the Markov chain $(X_n)_{n\geq 0}$ with state space $\mathbb{S} = \{0, 1, 2, 3\}$ and transition probability matrix given by

$$\left[\ P_{i,j}\ \right]_{0\leq i,j\leq 3} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0.3 & 0 & 0.7 & 0 \\ 0 & 0.3 & 0 & 0.7 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

(a) What are the absorbing states of the chain $(X_n)_{n\geq 0}$?

(b) Given that the chain starts at $\textcircled{1}$, find its probability of absorption $g_0(k) = \mathbb{P}(T_0 < \infty \mid X_0 = k)$ into state $\textcircled{0}$ for $k = 0, 1, 2, 3$.

(c) Find the mean hitting times $h_1(k) = \mathbb{E}[T_1 \mid X_0 = k]$ of state $\textcircled{1}$ starting from state $\textcircled{k}$, for $k = 0, 1, 2, 3$.

Exercise 5.6 Consider the random walk with Markov transition matrix given by

$$\left[\ P_{i,j}\ \right]_{0\leq i,j\leq 3} = \begin{bmatrix} 0.5 & 0 & 0.5 & 0 \\ 0.5 & 0 & 0 & 0.5 \\ 0.5 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Compute the average time it takes to reach state $\textcircled{3}$ given that the chain is started at state $\textcircled{0}$.

Exercise 5.7 Consider the Markov chain $(X_n)_{n\geq 0}$ on $\{0, 1, 2, 3\}$ whose transition probability matrix P is given by

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 0.5 & 0 & 0.5 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

(a) Draw the graph of this chain.
(b) Find the probability $g_0(k)$ that the chain is absorbed into state $\textcircled{0}$ given that it started from state $k = 0, 1, 2, 3$.
(c) Determine the mean time $h(k)$ it takes until the chain hits an absorbing state, after starting from $k = 0, 1, 2, 3$.

Exercise 5.8 Consider a discrete-time, time-homogeneous Markov chain $(X_n)_{n\geq 0}$ on a state space $\mathbb{S}$, and the first hitting time

$$T_A = \inf\{n \geq 0 \, : \, X_n \in A\},$$

of a subset A of $\mathbb{S}$. Show that $(X_n)_{n\geq 0}$ has the *strong Markov property* with respect to T_A, i.e. show that for all $n, m \geq 0$, $j \in \mathbb{S}$, and $(i_k)_{k\geq 0} \subset \mathbb{S}$ we have

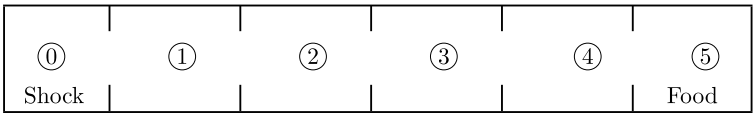
$$\mathbb{P}(X_{T_A+n} = j \mid X_{T_A} = i_0, \ldots, X_0 = i_{T_A} \text{ and } T_A < +\infty) = \mathbb{P}(X_n = j \mid X_0 = i_0).$$

Exercise 5.9 We consider the simple random walk $(S_n)_{n\geq 0}$ of Chap. 3.

(a) Using first step analysis, recover the formula (3.4.16) for the probability $\mathbb{P}(T_0 < \infty \mid S_0 = k)$ of hitting state $\textcircled{0}$ in finite time starting from any state $\textcircled{k} \geq 0$ when $q < p$.
(b) Using first step analysis, recover the formula (3.4.18) giving the mean hitting time $\mathbb{E}[T_0 \mid S_0 = k]$ of state $\textcircled{0}$ from any state $\textcircled{k} \geq 0$ when $q > p$.

Exercise 5.10 A player tosses a fair six-sided die and records the number appearing on the uppermost face. The die is then tossed again, and the second result is added to the first one. This procedure is repeated until the sum of all results becomes strictly greater than 10. Compute the probability that the game finishes with a cumulative sum equal to 13.

Exercise 5.11 A fish is placed into the linear maze as shown, and its state at time n is denoted by $X_n \in \{0, 1, 2, 3, 4, 5\}$:



Starting from any state $k \in \{1, 2, 3, 4\}$, the fish moves to the right with probability p and to the left with probability q such that $p + q = 1$ and $p \in (0, 1)$. Consider the hitting times

$$T_0 = \inf\{n \geq 0 \ : \ X_n = 0\}, \quad \text{and} \quad T_5 = \inf\{n \geq 0 \ : \ X_n = 5\},$$

and $g(k) = \mathbb{P}(T_5 < T_0 \mid X_0 = k)$, $k = 0, 1, \ldots, 5$.

(a) Using first step analysis, write down the equation satisfied by $g(k)$, $k = 0, 1, \ldots, 5$, and give the values of $g(0)$ and $g(5)$.

(b) Assume that the fish is equally likely to move right or left at each step. Compute the probability that starting from state (k) it finds the food before getting shocked, for $k = 0, 1, \ldots, 5$.

Exercise 5.12 Starting from a state $m \geq 1$ at time k, the next state of a random device at time $k + 1$ is uniformly distributed among $\{0, 1, \ldots, m - 1\}$, with (0) as an absorbing state.

(a) Model the time evolution of this system using a Markov chain whose transition probability matrix will be given explicitly.

(b) Let $h_0(m)$ denote the mean time until the system reaches the state zero for the first time after starting from state (m). Using first step analysis, write down the equation satisfied by $h_0(m)$, $m \geq 1$ and give the values of $h_0(0)$ and $h_0(1)$.

(c) Show that $h_0(m)$ is given by $h_0(m) = h_0(m - 1) + \dfrac{1}{m}$, $m \geq 1$, and that

$$h_0(m) = \sum_{k=1}^{m} \frac{1}{k},$$

for all $m \geq 0$.

Exercise 5.13 An individual is placed in a castle tower having three exits. Exit A leads to a tunnel that returns to the tower after three days of walk. Exit B leads to a tunnel that returns to the tower after one day of walk. Exit C leads to the outside. Since the inside of the tower is dark, each exit is chosen at random with probability $1/3$. The individual decides to remain outside after exiting the tower, and you may choose the number of steps it takes from Exit C to the outside of the tower, e.g. take it equal to 0 for simplicity.

(a) Show that this problem can be modeled using a Markov chain $(X_n)_{n \geq 0}$ with four states. Draw the graph of the chain $(X_n)_{n \geq 0}$.

(b) Write down the transition matrix of the chain $(X_n)_{n\geq 0}$.
(c) Starting from inside the tower, find the average time it takes to exit the tower.

Exercise 5.14 A mouse is trapped in a maze. Initially, it has to choose one of two directions. If it goes to the right, then it will wander around in the maze for three minutes and will then return to its initial position. If it goes to the left, then with probability $1/3$ it will depart the maze after two minutes of travelling, and with probability $2/3$ it will return to its initial position after five minutes of travelling. Assuming that the mouse is at all times equally likely to go to the left or to the right, what is the expected number of minutes that it will remain trapped in the maze?

Exercise 5.15 This exercise is a particular case of (5.1.9). Consider the Markov chain whose transition probability matrix P is given by

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0.1 & 0.6 & 0.1 & 0.2 \\ 0.2 & 0.3 & 0.4 & 0.1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

(a) Find the probability that the chain finishes at $\textcircled{0}$ given that it was started at state $\textcircled{1}$.
(b) Determine the mean time it takes until the chain reaches an absorbing state.

Exercise 5.16 This exercise is stated in the framework of (5.1.9). Consider the Markov chain $(X_n)_{n\geq 0}$ with state space $\mathbb{S} = \{0, 1, 2, 3\}$ and transition probability matrix given by

$$\left[\, P_{i,j} \,\right]_{0\leq i,j\leq 3} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0.3 & 0 & 0.4 & 0.3 \\ 0.3 & 0.4 & 0 & 0.3 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

(a) What are the absorbing states of the chain $(X_n)_{n\geq 0}$?
(b) Denoting by $T_k := \inf\{n \geq 0 \,:\, X_n = k\}$ the first hitting time of state $\textcircled{k}$, find the probabilities $g_1(k) = \mathbb{P}(T_1 < \infty \mid X_0 = k)$ of hitting state $\textcircled{1}$ in finite time after starting from state $\textcircled{k}$, for $k = 0, 1, 2, 3$.
(c) Denoting by $T_1^r := \inf\{n \geq 1 \,:\, X_n = 1\}$ the first *return time* to state $\textcircled{1}$, find the probabilities $p_1(k) = \mathbb{P}(T_1^r < \infty \mid X_0 = k)$ of returning to state $\textcircled{1}$ in finite time after starting from state $\textcircled{k}$, for $k = 0, 1, 2, 3$.
(d) Find the mean hitting times $h_1(k) = \mathbb{E}[T_1 \mid X_0 = k]$ of state $\textcircled{1}$ and the mean return times $\mu_1(k) = \mathbb{E}[T_1 \mid X_0 = k]$ to state $\textcircled{1}$ after starting from state $\textcircled{k}$, for $k = 0, 1, 2, 3$.

Exercise 5.17 Consider the Markov chain on $\{0, 1, 2\}$ with transition matrix

$$\begin{bmatrix} 1/3 & 1/3 & 1/3 \\ 1/4 & 3/4 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

(a) Compute the probability $\mathbb{P}(T_2 < \infty \mid X_0 = 1)$ of *hitting* state ② in finite time starting from state ①, and the probability $\mathbb{P}\big(T_1^r < \infty \mid X_0 = 1\big)$ of *returning* to state ① in finite time.
(b) Compute the mean *return time* $\mu_1(1) = \mathbb{E}[T_1^r \mid X_0 = 1]$ to state ① and the mean *hitting time* $h_2(1) = \mathbb{E}[T_2 \mid X_0 = 1]$ of state ② starting from state ①.

Exercise 5.18 Consider a sequence $(X_n)_{n\geq 1}$ of independent Bernoulli random variables with the distribution

$$\mathbb{P}(X_n = a) = p, \qquad \mathbb{P}(X_n = b) = q, \qquad n \geq 1,$$

where $p \in (0, 1]$ and $q := 1 - p$. Let $T^{(m)}$ denote the time of the first appearance of m consecutive a's in $(X_n)_{n\geq 1}$, with e.g. $T^{(4)} = 8$ in the following sequence:

$$(b, a, a, b, \overbrace{a, a, a, a}^{4 \text{ times}}, b, a, a, b, \ldots).$$
$$\begin{matrix} & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & & & & \\ & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & & & & \end{matrix}$$

(a) By first step analysis, find an equation satisfied by $\mathbb{E}[T^{(m)}]$.
(b) Compute the mean time $\mathbb{E}[T^{(m)}]$ until we encounter an m-winning streak, for all $m \geq 1$.
Hint. We have

$$\sum_{k=1}^{m} k p^{k-1} = \frac{\partial}{\partial p} \sum_{k=0}^{m} p^k = \frac{\partial}{\partial p} \left(\frac{1 - p^{m+1}}{1 - p} \right) = \frac{1 - (m+1)p^m + m p^{m+1}}{(1-p)^2}.$$

Exercise 5.19 Consider a *Markov Decision Process* (MDP)[4] on a state space $\mathbb{S}$, with set of actions $\mathbb{A}$ and family $(P^a)_{a\in\mathbb{A}}$ of transition probability matrices

$$P : \mathbb{S} \times \mathbb{S} \times \mathbb{A} \to [0, 1],$$
$$(k, l, a) \longmapsto P_{k,l}^a.$$

Using first step analysis, derive the *Bellman equation*

$$V^*(k) = R(k) + \gamma \max_{a\in\mathcal{A}} \sum_{l\in\mathbb{S}} P_{k,l}^a V^*(l), \qquad k \in \mathbb{S},$$

[4] See Problem 5.23 for details.

satisfied by the optimal value function

$$V^*(k) = \max_{\pi} \mathbb{E}\left[\sum_{n\geq 0} \gamma^n R(X_n) \;\middle|\; X_0 = k \right],$$

over all *policies* $\pi : \mathbb{S} \to \mathbb{A}$ giving the action chosen at every given state in $\mathbb{S}$, where $\gamma \in (0, 1)$ is a discount factor and $R : \mathbb{S} \to \mathbb{R}$ is a reward function.

Problem 5.20 Let $(X_n)_{n\geq 0}$ be a Markov chain on $\{0, 1, \ldots, N\}$, $N \geq 1$, with transition matrix $P = \big[P_{i,j} \; big \big]_{0\leq i,j\leq N}$.

(a) Consider the hitting times

$$T_0 = \inf\{n \geq 0 \; : \; X_n = 0\}, \qquad T_N = \inf\{n \geq 0 \; : \; X_n = N\},$$

and

$$g(k) = \mathbb{P}(T_0 < T_N \mid X_0 = k), \qquad k = 0, 1, \ldots, N.$$

What are the values of $g(0)$ and $g(N)$?

(b) Show, using first step analysis, that the function g satisfies the relation

$$g(k) = \sum_{l=0}^{N} P_{k,l} g(l), \qquad k = 1, \ldots, N-1. \tag{5.4.11}$$

(c) In this question and the following ones we consider the Wright-Fisher stochastic model in population genetics, in which the state X_n denotes the number of individuals in the population at time n, and

$$P_{k,l} = \mathbb{P}(X_{n+1} = l \mid X_n = k) = \binom{N}{l} \left(\frac{k}{N}\right)^l \left(1 - \frac{k}{N}\right)^{N-l},$$

$k, l = 0, 1, \ldots, N$. Write down the transition matrix P when $N = 3$.

(d) Show, from Question (b), that given that the solution to (5.4.11) is unique, we have

$$\mathbb{P}(T_0 < T_N \mid X_0 = k) = \frac{N-k}{N}, \qquad k = 0, 1, \ldots, N.$$

(e) Let

$$T_{0,N} = \inf\{n \geq 0 \; : \; X_n = 0 \text{ or } X_n = N\},$$

and

$$h(k) = \mathbb{E}[T_{0,N} \mid X_0 = k], \qquad k = 0, 1, \ldots, N.$$

What are the values of $h(0)$ and $h(N)$?

(f) Show, using first step analysis, that the function h satisfies the relation

$$h(k) = 1 + \sum_{l=0}^{N} P_{k,l} h(l), \qquad k = 1, 2, \ldots, N-1.$$

(g) Assuming that $N = 3$, compute

$$h(k) = \mathbb{E}[T_{0,3} \mid X_0 = k], \qquad k = 0, 1, 2, 3.$$

Problem 5.21 Consider a gambling process $(X_n)_{n\geq 0}$ on the state space $\{0, 1, \ldots, N\}$, with transition probabilities

$$\mathbb{P}(X_{n+1} = k+1 \mid X_n = k) = p, \qquad \mathbb{P}(X_{n+1} = k-1 \mid X_n = k) = q,$$

$k = 1, 2, \ldots, N-1$, with $p + q = 1$. Let

$$\tau := \inf\{n \geq 0 \ : \ X_n = 0 \text{ or } X_n = N\}$$

denote the time until the process hits either state $\textcircled{0}$ or state N, and consider the expectation

$$h(k) := \mathbb{E}\left[\sum_{i=0}^{\tau-1} X_i \ \middle| \ X_0 = k \right],$$

of the random sum $\displaystyle\sum_{0\leq i<\tau} X_i$ of all chain values visited *before* the process hits 0 or N after starting from $k = 0, 1, 2, \ldots, N$.

(a) Give the values of $h(0)$ and $h(N)$.[5]

(b) Show, by first step analysis, that $h(k)$ satisfies the equations

$$h(k) = k + ph(k+1) + qh(k-1), \quad k = 1, 2, \ldots, N-1. \tag{5.4.12}$$

From now on, we take $p = q = 1/2$.

(c) Find a particular solution of Eq. (5.4.12).

(d) Knowing that the solution of the associated *homogeneous equation*

$$f(k) = \frac{1}{2} f(k+1) + \frac{1}{2} f(k-1), \qquad k = 1, 2, \ldots, N-1,$$

takes the form

[5] We apply the convention $\displaystyle\sum_{i=0}^{-1} = \sum_{0\leq i<0} = 0.$

$$f(k) = C_1 + C_2 k,$$

show that the expected value $h(k)$ solution of (5.4.12) is given by

$$h(k) = k\frac{N^2 - k^2}{3}, \qquad k = 0, 1, \ldots, N.$$

(e) Compute $h(1)$ when $N = 2$ and explain why this result makes pathwise sense.
(f) Suppose that you start a business with initial monthly income of \$4K. Every month the income you receive from that business may *increase* or *decrease* by \$1K with equal probabilities $(1/2, 1/2)$. You decide to stop that business as soon as your monthly income hits the levels 0 or \$70K, whichever comes first.
  (i) Compute the expected duration of your business (in number of months).[6]
  (ii) Compute your expected accumulated wealth until the month before you stop your business.
  (iii) Compute your expected accumulated wealth under the assumption that your monthly income remains constant equal to \$4K over the same mean duration as in Question (f-*i*) above.
  (iv) Any comment?

Problem 5.22 Probabilistic automata (Gusev (2014)). Given an alphabet $\Sigma = \{a, b\}$ made of two letters we denote by Σ^* the set of all words over $\{a, b\}$, i.e. Σ^* is made of all finite sequences of symbols in $\{a, b\}$.

A *language* $\mathcal{L}$ over $\{a, b\}$ is a collection of (finite) words in Σ^*.

In computer science, an *automaton* given by a function

$$f : \{a, b\} \times \{0, 1, \ldots, n\} \longrightarrow \{0, 1, \ldots, n\}$$

is reading words of the form $a_1 a_2 \cdots a_m \in \mathcal{L}$ by producing a sequence $x_1, x_2, \ldots, x_m$ of integers *via* the recursion

$$x_1 := f(a_1, x_0),\ x_2 := f(a_2, x_1),\ x_3 := f(a_3, x_2), \ldots, x_m := f(a_m, x_{m-1}).$$

A word $a_1 a_2 \cdots a_m \in \mathcal{L}$, $m \geq 1$, is said to *synchronize* the automaton f *to state* $\textcircled{n}$ if we have $x_m = \textcircled{n}$. Here, $\textcircled{n}$ is regarded as a *sink state*, also called *accepting state*.

One says that the automaton f *recognizes* the language $\mathcal{L}$ if every word $a_1 a_2 \cdots a_m \in \mathcal{L}$ synchronizes the automaton f *to state* $\textcircled{n}$, i.e. satisfies $x_m = \textcircled{n}$, starting from *any* initial state $\textcircled{x}_0$.

[6] Recall that $\mathbb{E}[T_{0,N} \mid X_0 = k] = k(N - k)$, $k = 0, 1, \ldots, N$, see (2.3.19).

Synchronizing automata are connected to algebra and combinatorics, and they have applications in many areas including robotics, coding theory, network security management, chip design, industrial automation, biocomputing, etc.

(a) Let $n = 5$ and consider the automaton given by the function

f	0	1	2	3	4	5
a	1	2	3	3	1	5
b	0	0	0	4	5	5

Draw a graph for this automaton and find the smallest integers $l, m \geq 1$ such that f can recognize the word $a^l b^m$ starting from *any* state of the automaton.

(b) Determine the language recognized by the automaton and give an example of a five-letter word that cannot synchronize the automaton when it starts from state ④.
Hint: The set of words recognized by the automaton can be written using the notation $\Sigma^* xxxxx \Sigma^*$ which represents the concatenations of a word in Σ^* followed by a certain word $xxxxx$ (always the same, to be determined), followed by another word in Σ^*.

(c) Let $(X_k)_{k\geq 1}$ be a sequence of *i.i.d.* random variables taking values in $\{a, b\}$, with $\mathbb{P}(X_k = a) = p \in (0, 1)$ and $\mathbb{P}(X_k = b) = q = 1 - p$. Consider the random process $(Y_k)_{k\geq 0}$ started at Y_0, with

$$Y_1 = f(X_1, Y_0),\ Y_2 = f(X_2, Y_1), \ldots, Y_k = f(X_k, Y_{k-1}), \ldots$$

Show that the process $(Y_k)_{k\geq 0}$ is a Markov chain on the state space $\{0, 1, 2, 3, 4, 5\}$, and write down its graph and its transition matrix.

(d) Find the average time it takes until the automaton f becomes synchronized by the random words generated from $(X_k)_{k\geq 1}$, starting from the initial state $Y_0 = ⓪$.
Hint: Call $h_5(k)$ the average time it takes to reach state ⑤ starting from state $k = 0, 1, 2, 3, 4, 5$, and check first that $h_5(4) = ph_5(0)$.

(e) (10 points) Let now $n = 4$ and consider the automaton g defined by

g	0	1	2	3	4
a	0	2	2	1	4
b	0	0	3	4	4

Draw a graph for this automaton.

(f) Find the unique shortest word that synchronizes the automation to state ⓪ after starting from all states 1, 2, 3, and the unique shortest word that synchronizes the automaton to state ④ after starting from all states 1, 2, 3.

(g) Consider the random process $(Z_k)_{k\geq 0}$ started at Z_0, with

$$Z_1 = g(X_1, Z_0),\ Z_2 = g(X_2, Z_1), \ldots, Z_k = g(X_k, Z_{k-1}), \ldots$$

Show that the process $(Z_k)_{k\geq 0}$ is a Markov chain on the state space $\{0, 1, 2, 3, 4\}$, draw the graph of the chain and write down its transition matrix.

(h) Find the probability that the first synchronized word is "abab" when the automaton is started from state (1). Note that according to Question (f), synchronization may here occur through state (0) or through state (4).

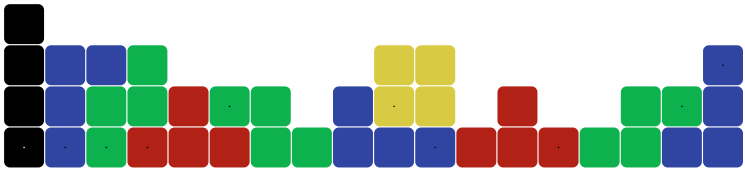
Problem 5.23 *Markov Decision Processes* (MDPs) have applications in game theory, robotics, automated control, operations research, information theory, multi-agent systems, swarm intelligence, genetic algorithms, etc., through their use in *reinforcement learning*, and *Q-learning*.

A *Markov Decision Process* (MDP) with state space $\mathbb{S}$ consists in:

– a *finite* set $\mathbb{A}$ of possible *actions*,
– a family $\left(P^{(a)}\right)_{a\in\mathbb{A}}$ of *transition probability matrices* $\left(P^{(a)}_{i,j}\right)_{i,j\in\mathbb{S}}$, and
– a *policy* $\pi : \mathbb{S} \to \mathbb{A}$ giving the action $\pi(k) \in \mathbb{A}$ taken at every given state in $k \in \mathbb{S}$.

When a MDP is in state $X_n = k$ at time n we look up the action $\pi(k) \in \mathbb{A}$ given by the policy π, and we generate the new value X_{n+1} using the transition probabilities $P^{(\pi(k))}_{k,\cdot} = \left(P^{(\pi(k))}_{k,l}\right)_{l\in\mathbb{S}}$.

The Pacman game is a natural application of Markov Decision Processes. The Tetris game can also be modeled as a Markov decision process whose (random) state consists in a given board configuration together with one of seven tile shapes. The actions are naturally chosen among the 40 placement choices for the falling tile, and a new tile shape is chosen with uniform probability $1/7$ at each time step.



The following ten questions are interdependent, and should be treated in sequence. The R package MDPtoolbox can be used to check your results, however, explanations are required.

(a) Consider a Markov chain $(X_n)_{n\geq 0}$ with state space $\mathbb{S}$ and transition matrix $P = (P_{i,j})_{i,j\in\mathbb{S}}$. Derive the first step analysis equation for the value function

$$V(k) := \mathbb{E}\left[\sum_{n\geq 0} R(X_n) \,\middle|\, X_0 = k\right], \qquad k \in \mathbb{S}, \tag{5.4.13}$$

defined as the total accumulated reward obtained after starting from state (k), where $R : \mathbb{S} \to \mathbb{R}$ is a reward function.[7]

[7] We always assume that $R(\cdot)$ and $(X_n)_{n\geq 0}$ are such that the series in (5.4.13) are convergent.

(a) In **Questions (b1)**, **(b2)** and **(b3)** below we consider the deterministic MDP on the state space $\mathbb{S} = \{1, 2, 3, 4, 5, 6, 7\}$ with actions $\mathbb{A} = \{\downarrow, \rightarrow\}$ and transition probability matrices

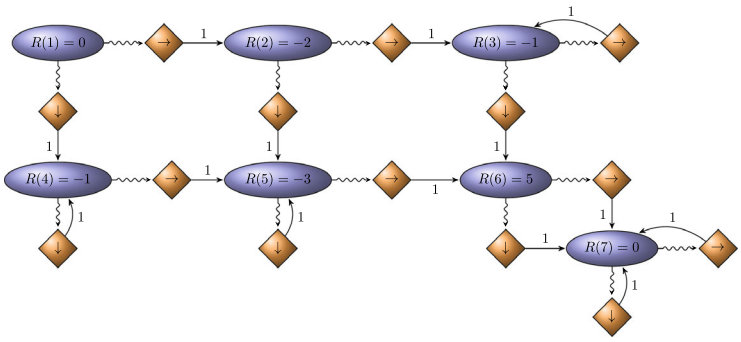
$$P^{(\downarrow)} := \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \qquad P^{(\rightarrow)} := \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix},$$

and the reward function $R : \mathbb{S} \to \mathbb{R}$ given by

$$R(1) = 0,\ R(2) = -2,\ R(3) = -1,\ R(4) = -1,\ R(5) = -3,\ R(6) = 5. \tag{5.4.14}$$

① $R(1) = 0$	② $R(2) = -2$	③ $R(3) = -1$
④ $R(4) = -1$	⑤ $R(5) = -3$	⑥ $R(6) = +5$

This MDP can be represented by the following graph, in which the $\rightsquigarrow$ arrows represent the policy choices, while the straight arrows denote Markov transitions.



(b1) Compute the *optimal action-value* function[8]

$$Q^*(k, \downarrow) := \max_{\pi\,:\,\pi(k)=\downarrow} \mathbb{E}\left[\sum_{n\geq 0} R(X_n) \,\middle|\, X_0 = k\right] \tag{5.4.15}$$

[8] In the maxima (5.4.15) the action is taken equal to "↓", resp. "→" at the first step only.

and

$$Q^*(k,\rightarrow) := \max_{\pi\,:\,\pi(k)=\rightarrow} \mathbb{E}\left[\sum_{n\geq 0} R(X_n) \,\middle|\, X_0 = k\right], \tag{5.4.16}$$

starting from state $X_0 = k \in \mathbb{S}$, in the following order: $Q^*(7,\downarrow)$, $Q^*(7,\rightarrow)$, $Q^*(6,\downarrow)$, $Q^*(6,\rightarrow)$, $Q^*(3,\downarrow)$, $Q^*(3,\rightarrow)$, $Q^*(5,\rightarrow)$, $Q^*(5,\downarrow)$, $Q^*(2,\downarrow)$, $Q^*(2,\rightarrow)$, $Q^*(4,\rightarrow)$, $Q^*(4,\downarrow)$, $Q^*(1,\downarrow)$, $Q^*(1,\rightarrow)$.

(b2) Find the *optimal value* function

$$V^*(k) := \max_{\pi} \mathbb{E}\left[\sum_{n\geq 0} R(X_n) \,\middle|\, X_0 = k\right],$$

at all states $k = 1, 2, \ldots, 7$.

(b3) Find the optimal policy $\pi^* = (\pi^*(1), \pi^*(2), \pi^*(3), \pi^*(4), \pi^*(5))$ of actions leading to the optimal gain starting from any state.[9]

(c) **Questions (c1), (c2), (c3)** below are for a general MDP.

(c1) Using first step analysis, show that for a general MDP the *optimal action-value* function[10]

$$Q^*(k,a) := \max_{\pi\,:\,\pi(k)=a} \mathbb{E}\left[\sum_{n\geq 0} R(X_n) \,\middle|\, X_0 = k\right], \qquad k \in \mathbb{S}, \quad a \in \mathbb{A}, \tag{5.4.17}$$

can be written using the transition probability matrix $P^{(a)}$ and the optimal value function $V^*(\cdot)$.[11]

(c2) How can the optimal policy $\pi^*(k)$ at state $k \in \mathbb{S}$ be computed from the optimal action-value function $Q^*(k,a)$?

(c3) By applying first step analysis to the *optimal value* function

$$V^*(k) := \max_{\pi} \mathbb{E}\left[\sum_{n\geq 0} R(X_n) \,\middle|\, X_0 = k\right], \qquad k \in \mathbb{S}, \tag{5.4.18}$$

derive the *Bellman equation* satisfied by the value function $V(k)$ using the *optimal action-value* function $Q^*(k,a)$.

(d) In **Questions (d1), (d2)** and **(d3)** below we let $p \in [0,1]$ and consider the stochastic MDP on the state space $\mathbb{S} = \{1,2,3,4,5,6,7\}$, with actions $\mathbb{A} = \{\downarrow, \rightarrow\}$ and transition probability matrices

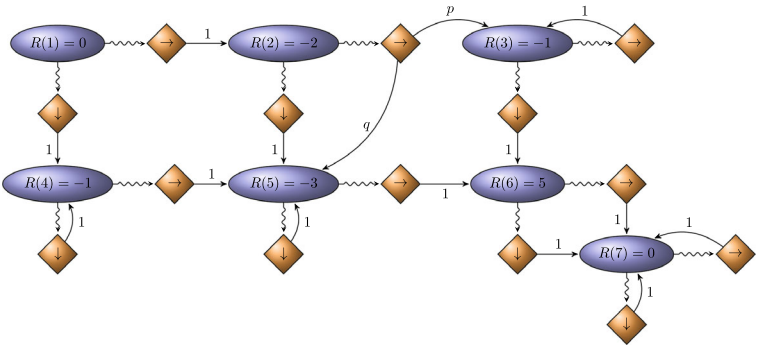
[9] The values of $\pi^*(6)$ and $\pi^*(7)$ are not considered here because they do not affect the total reward.

[10] In the maxima (5.4.17) the action is taken equal to a at the first step only. After moving to a new state we maximize the future reward according to the best policy choice.

[11] We always assume that $R(\cdot)$ and $(X_n)_{n\geq 0}$ are such that the series in (5.4.17) and (5.4.18) is convergent.

$$P^{(\downarrow)} := \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad P^{(\rightarrow)} := \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & p & 0 & q & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix},$$

and the reward function (5.4.14). This MDP can be represented by the following graph, in which the $\rightsquigarrow$ arrows represent policy choices, while the straight arrows denote Markov transitions.



(d1) Using the argument of Question (c1), compute the *optimal action-value* function[12]

$$Q^*(k, \downarrow) := \max_{\pi \,:\, \pi(k)=\downarrow} \mathbb{E}\left[\sum_{n\geq 0} R(X_n) \,\middle|\, X_0 = k \right] \tag{5.4.19}$$

and

$$Q^*(k, \rightarrow) := \max_{\pi \,:\, \pi(k)=\rightarrow} \mathbb{E}\left[\sum_{n\geq 0} R(X_n) \,\middle|\, X_0 = k \right], \tag{5.4.20}$$

starting from state $X_0 = k \in \mathbb{S}$, in the following order: $Q^*(7, \downarrow)$, $Q^*(7, \rightarrow)$, $Q^*(6, \downarrow)$, $Q^*(6, \rightarrow)$, $Q^*(3, \downarrow)$, $Q^*(3, \rightarrow)$, $Q^*(5, \rightarrow)$, $Q^*(5, \downarrow)$, $Q^*(2, \downarrow)$, $Q^*(2, \rightarrow)$, $Q^*(4, \rightarrow)$, $Q^*(4, \downarrow)$, $Q^*(1, \downarrow)$, $Q^*(1, \rightarrow)$.
Remark: Some values of $Q^*(k, \downarrow)$, $Q^*(k, \rightarrow)$ may now depend on p.

(d2) Using the result of Question (d1), find the *optimal value* function

$$V^*(k) := \max_{\pi} \mathbb{E}\left[\sum_{n\geq 0} R(X_n) \,\middle|\, X_0 = k \right],$$

at all states $k = 1, 2, \ldots, 7$, depending on the value of $p \in [0, 1]$.

[12] In the maxima (5.4.19) the action is taken equal to "$\downarrow$", resp. "$\rightarrow$" at the first step only.

(d3) Find the optimal policy $\pi^* = (\pi^*(1), \pi^*(2), \pi^*(3), \pi^*(4), \pi^*(5))$ of actions leading to the optimal gain starting from any state, depending on the value of $p \in [0, 1]$.[13]

```
install.packages("MDPtoolbox")
library(MDPtoolbox)
P <- array(0,c(7,7,2))
P[,,1] <- matrix(c(.,...,.,.,...,.), nrow=7, ncol=7, byrow=TRUE)
P[,,2] <- matrix(c(.,...,.,.,...,.), nrow=7, ncol=7, byrow=TRUE)
R <- array(0,c(7,2))
R[,1] <- matrix(c(.,...,.), nrow=1, ncol=7, byrow=TRUE)
R[,2] <- R[,1]
mdp_check(P, R)
mdp_value_iteration(P,R,discount=1)
```

[13] The values of $\pi^*(6)$ and $\pi^*(7)$ are not considered because they do not affect the total reward.

Chapter 6
Classification of States



In this chapter we present the notions of communicating, transient and recurrent states, as well as the concept of irreducibility of a Markov chain. We also examine the notions of positive and null recurrence, periodicity, and aperiodicity of such chains. Those topics will be important when analysing the long-run behavior of Markov chains in Chap. 7.

6.1 Communicating States

Definition 6.1 A state $\textcircled{j} \in \mathbb{S}$ is to be *accessible* from another state $\textcircled{i} \in \mathbb{S}$, and we write $\textcircled{i} \longmapsto \textcircled{j}$, if there exists a *finite* integer $n \geq 0$ such that

$$[P^n]_{i,j} = \mathbb{P}(X_n = j \mid X_0 = i) > 0.$$

In other words, it is possible to travel from $\textcircled{i}$ to $\textcircled{j}$ with non-zero probability in a certain (random) number of steps. We also say that state $\textcircled{i}$ leads to state $\textcircled{j}$, and when $i \neq j$ we have

$$\mathbb{P}(T_j^r < \infty \mid X_0 = i) \geq \mathbb{P}(T_j^r \leq n \mid X_0 = i) \geq \mathbb{P}(X_n = j \mid X_0 = i) > 0.$$

Remark 6.2 Since $P^0 = \mathrm{I}$ and $[P^0]_{i,j} = \mathbb{P}(X_0 = j \mid X_0 = i) = \mathbb{1}_{\{i=j\}}$ the definition of accessibility states implicitly that any state $\textcircled{i}$ is always accessible from itself (in zero time steps) even if $P_{i,i} = 0$.

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N. Privault, *Understanding Markov Chains*, Springer Undergraduate Mathematics Series,
https://doi.org/10.1007/978-981-95-4057-0_6

In case $\textcircled{i} \longmapsto \textcircled{j}$ and $\textcircled{j} \longmapsto \textcircled{i}$ we say that $\textcircled{i}$ and $\textcircled{j}$ *communicate*[1] and we write $\textcircled{i} \longleftrightarrow \textcircled{j}$.

The binary relation "$\longleftrightarrow$" is called an equivalence relation as it satisfies the following properties:

(a) Reflexivity: For all $i \in \mathbb{S}$ we have $\textcircled{i} \longleftrightarrow \textcircled{i}$

(b) Symmetry: For all $i, j \in \mathbb{S}$ we have that $\textcircled{i} \longleftrightarrow \textcircled{j}$ is equivalent to $\textcircled{j} \longleftrightarrow \textcircled{i}$.

(c) Transitivity: For all $i, j, k \in \mathbb{S}$ such that $\textcircled{i} \longleftrightarrow \textcircled{j}$ and $\textcircled{j} \longleftrightarrow \textcircled{k}$, we have $\textcircled{i} \longleftrightarrow \textcircled{k}$.

Proof It is clear that the relation $\longleftrightarrow$ is reflexive and symmetric. The proof of transitivity can be stated as follows. If $\textcircled{i} \longmapsto \textcircled{j}$ and $\textcircled{j} \longmapsto \textcircled{k}$, there exists $a \geq 1$ and $b \geq 1$ such that

$$[P^a]_{i,j} > 0, \qquad [P^b]_{j,k} > 0.$$

Next, by the Markov property and (4.1.2), for all $n \geq a + b$ we have

$$\begin{aligned}
&\mathbb{P}(X_n = k \mid X_0 = i) \\
&= \sum_{l,m \in \mathbb{S}} \mathbb{P}(X_n = k,\ X_{n-b} = l,\ X_a = m \mid X_0 = i) \qquad (6.1.1) \\
&= \sum_{l,m \in \mathbb{S}} \mathbb{P}(X_n = k \mid X_{n-b} = l)\mathbb{P}(X_{n-b} = l \mid X_a = m)\mathbb{P}(X_a = m \mid X_0 = i) \\
&\geq \mathbb{P}(X_n = k \mid X_{n-b} = j)\mathbb{P}(X_{n-b} = j \mid X_a = j)\mathbb{P}(X_a = j \mid X_0 = i) \\
&= [P^a]_{i,j}[P^{n-a-b}]_{j,j}[P^b]_{j,k} \\
&\geq 0.
\end{aligned}$$

The conclusion follows by taking $n = a + b$, in which case we have

$$\mathbb{P}(X_n = k \mid X_0 = i) \geq [P^a]_{i,j}[P^b]_{j,k} > 0.$$

□

The equivalence relation "$\longleftrightarrow$" induces a *partition* of $\mathbb{S}$ into disjoint classes $A_1, A_2, \ldots, A_m$ such that $\mathbb{S} = A_1 \cup \cdots \cup A_m$, and

(a) we have $\textcircled{i} \longleftrightarrow \textcircled{j}$ for all $i, j \in A_q$, and

(b) we have $\textcircled{i} \nleftrightarrow \textcircled{j}$ whenever $i \in A_p$ and $j \in A_q$ with $p \neq q$.

The sets $A_1, A_2, \ldots, A_m$ are called the *communicating classes* of the chain.

Definition 6.3 A Markov chain whose state space is made of a unique communicating class is said to be *irreducible*, otherwise the chain is said to be *reducible*.

[1] In graph theory, one says that $\textcircled{i}$ and $\textcircled{i}$ are *strongly connected*.

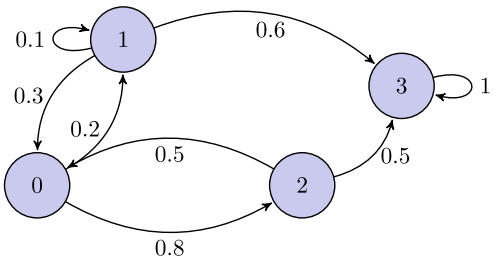
In particular, a Markov chain is irreducible if its transition matrix P is *regular*, *i.e.* if there exists $n \geq 1$ such that all entries of the matrix power P^n are strictly positive.

The R package "markovchain" can be used to test the irreducibility of a given chain.

```
install.packages("devtools"); library(devtools) # Install RTools as well
devtools::install_github('spedygiorgio/markovchain') # Choose option 2 -
     CRAN packages only
library (markovchain)
statesNames <- c("0", "1")
mcA <- new("markovchain",transitionMatrix = matrix(c(0.7,0.3,0.1,0.9),
byrow = TRUE, nrow = 2,dimnames = list(statesNames, statesNames)))
is.irreducible(mcA)
```

Clearly, all states in $\mathbb{S}$ communicate when $(X_n)_{n\geq 0}$ is irreducible. In case the i-th column of a transition matrix P vanishes, i.e. $P_{k,i} = 0, k \in \mathbb{S}$, then state $\textcircled{i}$ cannot be reached from any other state and $\textcircled{i}$ becomes a communicating class on its own, as is the case of state $\textcircled{1}$ in Exercise 4.10 for $n \geq 2$, or in Exercise 7.13. The same is true of absorbing states. However, having a returning loop with probability strictly lower than one is not sufficient to turn a given state into a communicating class on its own. Clearly, the existence of at least one absorbing state $\textcircled{i}$ with $P_{i,i} = 1$ makes a chain reducible.

Exercise: Find the communicating classes of the Markov chain with transition matrix (5.4.10) for the equivalence relation "$\longleftrightarrow$".



$$P = \begin{bmatrix} 0 & 0.2 & 0.8 & 0 \\ 0.3 & 0.1 & 0 & 0.6 \\ 0.5 & 0 & 0 & 0.5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The above state space $\mathbb{S} = \{0, 1, 2, 3\}$ is partitioned into two communicating classes which are $\{0, 1, 2\}$ and $\{3\}$.

6.2 Recurrent States

Definition 6.4 A state $\textcircled{i} \in \mathbb{S}$ is said to be *recurrent* if, starting from state $\textcircled{i}$, the chain $(X_n)_{n\geq 0}$ will return to state $\textcircled{i}$ within a finite (random) time, with probability 1, i.e.,

$$p_{ii} := \mathbb{P}(T_i^r < \infty \mid X_0 = i) = \mathbb{P}(X_n = i \text{ for some } n \geq 1 \mid X_0 = i) = 1. \quad (6.2.1)$$

Proposition 6.5 uses the mean number of returns R_i to state $\textcircled{i}$ defined in (5.4.2), and relies on the geometric distribution (5.4.3) of R_i given that $X_0 = i$.

Proposition 6.5 *For any state $\textcircled{i} \in \mathbb{S}$, the following statements are equivalent:*

(i) the state $\textcircled{i} \in \mathbb{S}$ is recurrent, i.e. $p_{ii} = 1$,
(ii) the number of returns to $\textcircled{i} \in \mathbb{S}$ is a.s.[2] *infinite, i.e.*

$$\mathbb{P}(R_i = \infty \mid X_0 = i) = 1, \text{ i.e. } \mathbb{P}(R_i < \infty \mid X_0 = i) = 0, \quad (6.2.2)$$

(iii) the mean number of returns to $\textcircled{i} \in \mathbb{S}$ is infinite, i.e.

$$\mathbb{E}[R_i \mid X_0 = i] = \infty, \quad (6.2.3)$$

(iv) we have

$$\sum_{n\geq 1} f_{i,i}^{(n)} = 1, \quad (6.2.4)$$

where $f_{i,i}^{(n)} := \mathbb{P}(T_i^r = n \mid X_0 = i)$, $n \geq 1$, is the distribution of T_i^r.

Proof Question (6.5) follows by the definition (6.2.1) of recurrent states.

(i) Relation (6.2.2) is equivalent to (6.2.1) by (5.4.4) and (5.4.5).
(ii) Relation (6.2.3) is equivalent to (6.2.1) by (5.4.6).
(iii) Relation (6.2.4) is equivalent to (6.2.1) by (5.4.1).

□

For example, state $\textcircled{0}$ is recurrent for the random walk of Chap. 3 when $p = q = 1/2$, while it is not recurrent if $p \neq q$ as by (3.4.14) we have

$$p_{0,0} = \mathbb{P}(T_0 < \infty) = 2\min(p, q). \quad (6.2.5)$$

As a consequence of (6.2.3), we have the following result.

Corollary 6.6 *A state $i \in \mathbb{S}$ is recurrent if and only if*

[2] "Almost surely", i.e. with probability one.

$$\sum_{n\geq 1}[P^n]_{i,i}=\infty,$$

i.e. the above series diverges.

Proof For all $i, j \in \mathbb{S}$, by (5.4.9) we have

$$\begin{aligned}\mathbb{E}[R_j \mid X_0=i] &= \mathbb{E}\left[\sum_{n\geq 1}\mathbb{1}_{\{X_n=j\}} \,\middle|\, X_0=i\right] = \sum_{n\geq 1}\mathbb{E}[\mathbb{1}_{\{X_n=j\}} \mid X_0=i]\\ &= \sum_{n\geq 1}\mathbb{P}(X_n=j \mid X_0=i) = \sum_{n\geq 1}[P^n]_{i,j}, \qquad (6.2.6)\end{aligned}$$

as in (5.4.9). To conclude, we let $j=i$ and apply (6.2.3). □

Corollary 6.6 admits the following consequence, which shows that any state communicating with a recurrent state is itself recurrent. In other words, recurrence is a *class property*, as all states in a given communicating class are recurrent as soon as one of them is recurrent.

Corollary 6.7 (Class property) *Let* $\textcircled{j} \in \mathbb{S}$ *be a recurrent state. Then any state* $\textcircled{i} \in \mathbb{S}$ *that communicates with state* $\textcircled{j}$ *is also recurrent.*

Proof By definition, since $\textcircled{i} \longmapsto \textcircled{j}$ and $\textcircled{j} \longmapsto \textcircled{i}$, there exists $a \geq 1$ and $b \geq 1$ such that

$$[P^a]_{i,j} > 0 \qquad [P^b]_{j,i} > 0,$$

and from (6.1.1) applied with $k=i$ we find

$$\begin{aligned}\sum_{n\geq a+b}[P^n]_{i,i} &= \sum_{n\geq a+b}\mathbb{P}(X_n=i \mid X_0=i)\\ &\geq [P^a]_{i,j}[P^b]_{j,i}\sum_{n\geq a+b}[P^{n-a-b}]_{j,j}\\ &= [P^a]_{i,j}[P^b]_{j,i}\sum_{n\geq 0}[P^n]_{j,j}\\ &= \infty,\end{aligned}$$

which shows that state $\textcircled{i}$ is recurrent from Corollary 6.6 and the assumption that state $\textcircled{j}$ is recurrent. □

A communicating class $A \subset \mathbb{S}$ is therefore recurrent if any of its states is recurrent.

6.3 Transient States

A state $\textcircled{i} \in \mathbb{S}$ is said to be *transient* when it is not recurrent, i.e., by (6.2.1),

$$p_{ii} = \mathbb{P}(T_i^r < \infty \mid X_0 = i) = \mathbb{P}(X_n = i \text{ for some } n \geq 1 \mid X_0 = i) < 1, \quad (6.3.1)$$

or

$$\mathbb{P}(T_i^r = \infty \mid X_0 = i) > 0.$$

Proposition 6.8 *For any state* $\textcircled{i} \in \mathbb{S}$*, the following statements are equivalent:*

(i) the state $\textcircled{i} \in \mathbb{S}$ *is transient, i.e.* $p_{ii} < 1$,
(ii) the number of returns to $\textcircled{i} \in \mathbb{S}$ *is* a.s.[3] *finite, i.e.*

$$\mathbb{P}(R_i = \infty \mid X_0 = i) = 0, \text{ i.e. } \mathbb{P}(R_i < \infty \mid X_0 = i) = 1, \quad (6.3.2)$$

(iii) the mean number of returns to $\textcircled{i} \in \mathbb{S}$ *is* finite*, i.e.*

$$\mathbb{E}[R_i \mid X_0 = i] < \infty, \quad (6.3.3)$$

(iv) we have

$$\sum_{n \geq 1} f_{i,i}^{(n)} < 1. \quad (6.3.4)$$

where $f_{i,i}^{(n)} := \mathbb{P}(T_i^r = n \mid X_0 = i)$, $n \geq 1$, *is the distribution of* T_i^r.

Proof This is a direct consequence of Proposition 6.5 and the definition (6.3.1) of transience. Regarding point (6.8) and the Condition (6.3.2) we also note that the state $\textcircled{i} \in \mathbb{S}$ is transient if and only if

$$\mathbb{P}(R_i = \infty \mid X_0 = i) < 1,$$

which, by (5.4.5) is equivalent to $\mathbb{P}(R_i = \infty \mid X_0 = i) = 0$. □

In other words, a state $\textcircled{i} \in \mathbb{S}$ is *transient* if and only if

$$\mathbb{P}(R_i < \infty \mid X_0 = i) > 0,$$

which by (5.4.4) is equivalent to

$$\mathbb{P}(R_i < \infty \mid X_0 = i) = 1,$$

i.e. the number of returns to state $i \in \mathbb{S}$ is finite with a non-zero probability which is necessarily equal to one. As a consequence of Corollary 6.6 we also have the following result.

[3] "Almost surely", i.e. with probability one.

Corollary 6.9 *A state $i \in \mathbb{S}$ is transient if and only if*

$$\sum_{n\geq 1}[P^n]_{i,i} < \infty,$$

i.e. the above series converges.

Similarly to Corollary 6.7, Corollary 6.9 admits the following consequence, which shows that any state communicating with a transient state is itself transient. Therefore, transience is also a *class property*, as all states in a given communicating class are transient as soon as one of them is transient.

Corollary 6.10 (Class property) *Let $j \in \mathbb{S}$ be a transient state. Then any state $i \in \mathbb{S}$ that communicates with state j is also transient.*

Proof If a state $i \in \mathbb{S}$ communicates with a transient state j then i is also transient (otherwise the state j would be recurrent by Corollary 6.7). □

A communicating class $A \subset \mathbb{S}$ is therefore transient if any of its states is transient. By Corollary 6.9 and the relation

$$\sum_{n\geq 0}\left[P^n\right]_{i,j} = \left[(I-P)^{-1}\right]_{i,j}, \qquad i, j \in \mathbb{S}, \tag{6.3.5}$$

see (5.4.9), we find that a chain with finite state space cannot be transient as the matrix $I - P$ is invertible in this case, cf. Problem 4.13. Indeed, 0 is clearly an eigenvalue of $I - P$ with eigenvector $[1, 1, \ldots, 1]$, therefore $I - P$ is not invertible, and a finite state space chain always admits at least one recurrent state, as noted in Theorem 6.11 below.

Clearly, any absorbing state is recurrent, and any state that leads to an absorbing state is transient.

Example

For the two-state Markov chain of Section 4.5, Relations (4.5.4)-(4.5.5) show that

$$\sum_{n\geq 1}[P^n]_{0,0} = \sum_{n\geq 1}\frac{b+a\lambda_2^n}{a+b} = \begin{cases} \infty, & \text{if } b > 0, \\ \displaystyle\sum_{n\geq 1}(1-a)^n < \infty, & \text{if } b = 0 \text{ and } a > 0, \end{cases}$$

hence state 0 is transient if $b = 0$ and $a > 0$, and recurrent otherwise. Similarly, we have

$$\sum_{n\geq 1}[P^n]_{1,1} = \sum_{n\geq 1}\frac{a+b\lambda_2^n}{a+b} = \begin{cases} \infty, & \text{if } a > 0, \\ \displaystyle\sum_{n\geq 1}(1-b)^n < \infty, & \text{if } a = 0 \text{ and } b > 0, \end{cases}$$

hence state (1) is transient if $a = 0$ and $b > 0$, and recurrent otherwise.

The above results can be recovered by a simple first step analysis for $g_i(j) = \mathbb{P}(T_i^r < \infty \mid X_0 = j)$, $i, j \in \{0, 1\}$, i.e.

$$\begin{cases} g_0(0) = a g_0(1) + 1 - a \\ g_0(1) = b + (1-b) g_0(1) \\ g_1(0) = (1-a) g_1(0) + a \\ g_1(1) = b g_1(0) + 1 - b, \end{cases}$$

which shows that $g_0(0) = 1$ if $b > 0$ and $g_1(1) = 1$ if $a > 0$.

We close this section with the following result for Markov chains with finite state space.

Theorem 6.11 *Let $(X_n)_{n\geq 0}$ be a Markov chain with* finite *state space $\mathbb{S}$. Then $(X_n)_{n\geq 0}$ has at least one recurrent state.*

Proof Recall that from (5.4.6) we have

$$\mathbb{E}[R_j \mid X_0 = i] = p_{ij}(1 - p_{jj}) \sum_{n\geq 1} n (p_{jj})^{n-1} = \frac{p_{ij}}{1 - p_{jj}},$$

for any states (i), (j) $\in \mathbb{S}$. Assuming that the state (j) $\in \mathbb{S}$ is transient we have $p_{jj} < 1$ by (6.3.1), hence by (6.2.6) we have

$$\mathbb{E}[R_j \mid X_0 = i] = \sum_{n\geq 1} [P^n]_{i,j} < \infty,$$

which is finite by (6.3.3), implying[4] that

$$\lim_{n\to\infty} [P^n]_{i,j} = 0$$

for all transient states $j \in \mathbb{S}$. In case all states in $\mathbb{S}$ were transient, since $\mathbb{S}$ is finite, by the *law of total probability* (4.2.2) we would have

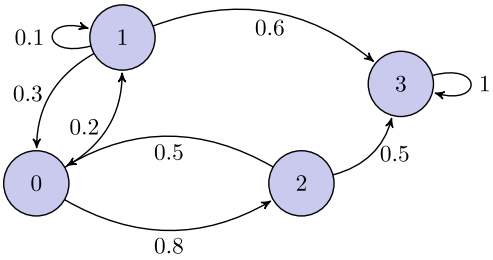
$$0 = \sum_{j\in\mathbb{S}} \lim_{n\to\infty} [P^n]_{i,j} = \lim_{n\to\infty} \sum_{j\in\mathbb{S}} [P^n]_{i,j} = \lim_{n\to\infty} 1 = 1,$$

which is a contradiction. Hence not all states can be transient, and there exists at least one recurrent state. □

[4] For any sequence $(a_n)_{n\geq 0}$ of *nonnegative* real numbers, $\sum_{n\geq 0} a_n < \infty$ implies $\lim_{n\to\infty} a_n = 0$.

Exercises:

(i) Find which states are transient and recurrent in the chain (5.4.10) represented as follows:



$$P = \begin{bmatrix} 0 & 0.2 & 0.8 & 0 \\ 0.3 & 0.1 & 0 & 0.6 \\ 0.5 & 0 & 0 & 0.5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Answer: State (3) is clearly recurrent since we have $T_3^r = 1$ with probability one when $X_0 = 3$. State (2) is transient because

$$1 - p_{2,2} = \mathbb{P}(T_2^r = \infty \mid X_0 = 2) = \frac{4}{7} \geq \mathbb{P}(X_1 = 3 \mid X_0 = 2) = 0.5 > 0, \tag{6.3.6}$$

and state (1) is transient because

$$\mathbb{P}\big(T_1^r = \infty \mid X_0 = 1\big) = 0.8 \geq \mathbb{P}(X_1 = 3 \mid X_0 = 1) = 0.6, \tag{6.3.7}$$

see the Exercise 5.4 for the computations of

$$p_{1,1} = \mathbb{P}(T_1^r < \infty \mid X_0 = 1) = 0.8$$

and

$$p_{2,2} = \mathbb{P}(T_2^r < \infty \mid X_0 = 2) = \frac{3}{7}.$$

By Corollary 6.7, the states (0) and (1) are transient because they communicate with state (2).

(ii) Which are the recurrent states in the simple random walk $(S_n)_{n\geq 0}$ of Chap. 3 on $\mathbb{S} = \mathbb{Z}$?

Answer: First, we note that this random walk is irreducible as all states communicate when $p \in (0,1)$. The simple random walk $(S_n)_{n\geq 0}$ on $\mathbb{S} = \mathbb{Z}$ has the transition matrix

$$P_{i,i+1} = p, \qquad P_{i,i-1} = q = 1-p, \qquad i \in \mathbb{Z}.$$

We have

$$[P^n]_{i,i} = \mathbb{P}(S_n = i \mid S_0 = i) = \mathbb{P}(S_n = 0 \mid S_0 = 0),$$

with

$$\mathbb{P}(S_{2n} = 0) = \binom{2n}{n} p^n q^n \quad \text{and} \quad \mathbb{P}(S_{2n+1} = 0) = 0, \quad n \geq 0.$$

Hence, we have

$$\begin{aligned}\sum_{n\geq 0} [P^n]_{0,0} &= \sum_{n\geq 0} \mathbb{P}(S_n = 0 \mid S_0 = 0) = \sum_{n\geq 0} \mathbb{P}(S_{2n} = 0 \mid S_0 = 0) \\ &= \sum_{n\geq 0} \binom{2n}{n} p^n q^n = H(1) = \frac{1}{\sqrt{1-4pq}},\end{aligned}$$

and

$$\mathbb{E}[R_0 \mid S_0 = 0] = \sum_{n\geq 1} \mathbb{P}(S_n = 0 \mid S_0 = 0) = \frac{1}{\sqrt{1-4pq}} - 1,$$

where $H(s)$ is defined in (3.4.8), cf. also Problem 3.9-(f).

Consequently, by Corollary 6.6, all states $\textcircled{i} \in \mathbb{Z}$ are recurrent when $p = q = 1/2$, whereas by Corollary 6.9 they are all transient when $p \neq q$, cf. Corollary 6.7.

Alternatively we could reach the same conclusion by directly using (3.4.14) and (6.2.1) which state that

$$\mathbb{P}(T_i^r < \infty \mid X_0 = i) = 2\min(p,q).$$

6.4 Positive Versus Null Recurrence

The expected time of return (or mean recurrence time) to a state $\textcircled{i} \in \mathbb{S}$ is given by

$$\begin{aligned}\mu_i(i) :&= \mathbb{E}[T_i^r \mid X_0 = i] \\ &= \sum_{n\geq 1} n\mathbb{P}\big(T_i^r = n \,\big|\, X_0 = i\big) \\ &= \sum_{n\geq 1} n f_{i,i}^{(n)}.\end{aligned}$$

Recall that a state $\textcircled{i}$ is recurrent when $\mathbb{P}(T_i^r < \infty \mid X_0 = i) = 1$, i.e. when the random return time T_i^r is almost surely *finite* starting from state $\textcircled{i}$. However, the recurrence property yields no information on the finiteness of its expected value $\mu_i(i) = \mathbb{E}[T_i^r \mid X_0 = i]$, cf. the example (1.6.5).

Definition 6.12 A *recurrent* state $i \in \mathbb{S}$ is said to be:

(a) *positive recurrent* if the mean return time to $\textcircled{i}$ is *finite*, i.e.

$$\mu_i(i) = \mathbb{E}[T_i^r \mid X_0 = i] < \infty,$$

(b) *null recurrent* if the mean return time to $\textcircled{i}$ is *infinite*, i.e.

$$\mu_i(i) = \mathbb{E}[T_i^r \mid X_0 = i] = \infty.$$

Exercise: Which states are positive/null recurrent in the simple random walk $(S_n)_{n\geq 0}$ of Chap. 3 on $\mathbb{S} = \mathbb{Z}$?

- From (3.4.20) and (3.4.17) we know that $\mathbb{E}[T_i^r \mid S_0 = i] = \infty$ for all values of $p \in (0, 1)$, hence all states of the random walk on $\mathbb{Z}$ are null recurrent when $p = 1/2$, while all states are transient when $p \neq 1/2$ due to (3.4.14).

The following Theorem 6.13 shows in particular that a Markov chain with finite state space cannot have any null recurrent state, cf. e.g. Corollary 2.3 in Kijima (1997), and also Corollary 3.7 in Asmussen (2003).

Theorem 6.13 *Assume that the state space $\mathbb{S}$ of a Markov chain $(X_n)_{n\geq 0}$ is* finite. *Then all recurrent states in $\mathbb{S}$ are also positive recurrent.*

As a consequence of Definition 6.3, Corollary 6.7, and Theorems 6.11 and 6.13 we have the following corollary.

Corollary 6.14 *Let $(X_n)_{n\geq 0}$ be an irreducible Markov chain with* finite *state space $\mathbb{S}$. Then* all *states of the chain $(X_n)_{n\geq 0}$ are positive recurrent.*

6.5 Periodicity and Aperiodicity

Given a state $i \in \mathbb{S}$, consider the sequence

$$\{n \geq 1 \ : \ [P^n]_{i,i} > 0\}$$

of integers which represent the possible travel times from state $\textcircled{i}$ to itself.

Definition 6.15 The *period* of the state $i \in \mathbb{S}$ is the greatest common divisor of the sequence

$$\{n \geq 1 \ : \ [P^n]_{i,i} > 0\}.$$

A state $i \in \mathbb{S}$ having period 1 is said to be *aperiodic*. This is the case in particular when $P_{ii} > 0$, i.e. when $\textcircled{i}$ admits a returning loop with nonzero probability.

In particular, any absorbing state is both aperiodic and recurrent. A *recurrent* state $i \in \mathbb{S}$ is said to be *ergodic* if it is both *positive recurrent* and *aperiodic*.

If $[P^n]_{i,i} = 0$ for all $n \geq 1$ then the set $\{n \geq 1 \ : \ [P^n]_{i,i} > 0\}$ is empty and by convention the period of state $\textcircled{i}$ is defined to be 0. In this case, state $\textcircled{i}$ is also transient.

Note also that if

$$\{n \geq 1 \ : \ [P^n]_{i,i} > 0\}$$

contains two distinct numbers that are relatively prime to each other (i.e. their greatest common divisor is 1) then state $\textcircled{i}$ aperiodic.

Proposition 6.16 shows that periodicity is a *class property*, as all states in a given communicating class have same periodicity.

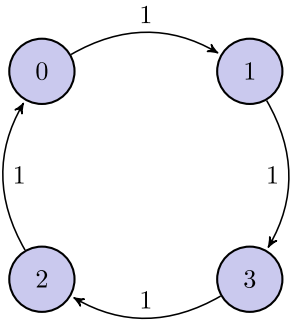
Proposition 6.16 (Class property) *All states that belong to a same communicating class have the same period.*

Proof Assume that state $\textcircled{i}$ has period d_i, that $\textcircled{j}$ communicates with $\textcircled{i}$, and let $n \in \{m \geq 1 \ : \ [P^m]_{j,j} > 0\}$. Since $\textcircled{i}$ and $\textcircled{j}$ communicate, there exists $k, l \geq 1$ such that $[P^k]_{i,j} > 0$ and $[P^l]_{j,i} > 0$, hence by (6.1.1) we have $[P^{k+l}]_{i,i} > 0$ hence $k + l$ is a multiple of d_i. Similarly by (6.1.1) we also have $[P^{n+k+l}]_{i,i} > 0$, hence $n + k + l$ and n are multiples of d_i, which implies $d_j \geq d_i$. Exchanging the roles of $\textcircled{i}$ and $\textcircled{j}$ we obtain similarly that $d_i \geq d_j$. □

A Markov chain is said to be *aperiodic* when *all* of its states are aperiodic. Note that any state that communicates with an aperiodic state becomes itself aperiodic. In particular, if a communicating class contains an aperiodic state then the whole class becomes aperiodic.

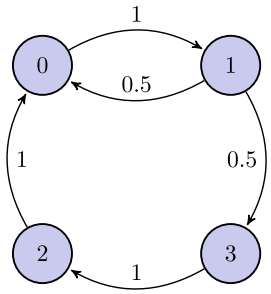
Examples

(i) The chain



clearly has periodicity equal to 4.

(ii) Consider the following chain:

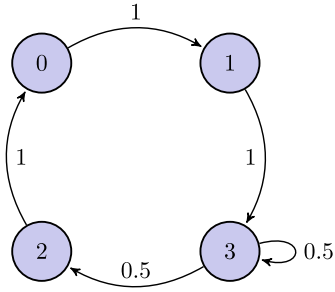


Here, we have

$$\{n \geq 1 \ : \ [P^n]_{0,0} > 0\} = \{2, 4, 6, 8, 10, \ldots\},$$
$$\{n \geq 1 \ : \ [P^n]_{1,1} > 0\} = \{2, 4, 6, 8, 10, \ldots\},$$
$$\{n \geq 1 \ : \ [P^n]_{2,2} > 0\} = \{4, 6, 8, 10, 12, \ldots\},$$
$$\{n \geq 1 \ : \ [P^n]_{3,3} > 0\} = \{4, 6, 8, 10, 12, \ldots\},$$

hence *all* states have period 2, and this is also consequence of Proposition 6.16.

(iii) Consider the following chain:

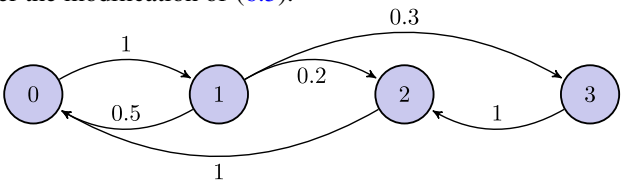


Here, we have

$$\begin{aligned}
\{n \geq 1 \ : \ [P^n]_{0,0} > 0\} &= \{4, 5, 6, 7, \ldots\}, \\
\{n \geq 1 \ : \ [P^n]_{1,1} > 0\} &= \{4, 5, 6, 7, \ldots\}, \\
\{n \geq 1 \ : \ [P^n]_{2,2} > 0\} &= \{4, 5, 6, 7, \ldots\}, \\
\{n \geq 1 \ : \ [P^n]_{3,3} > 0\} &= \{4, 5, 6, 7, \ldots\},
\end{aligned}$$

hence *all* states have period 1, see also Proposition 6.16.

(iv) Next, consider the modification of (6.5):



Here, the chain is aperiodic since we have

$$\begin{aligned}
\{n \geq 1 \ : \ [P^n]_{0,0} > 0\} &= \{2, 3, 4, 5, 6, 7, \ldots\}, \\
\{n \geq 1 \ : \ [P^n]_{1,1} > 0\} &= \{2, 3, 4, 5, 6, 7, \ldots\}, \\
\{n \geq 1 \ : \ [P^n]_{2,2} > 0\} &= \{3, 4, 5, 6, 7, 8, \ldots\}, \\
\{n \geq 1 \ : \ [P^n]_{3,3} > 0\} &= \{4, 6, 7, 8, 9, 10, \ldots\},
\end{aligned}$$

hence *all* states have period 1.

Exercises:

(i) What is the periodicity of the simple random walk $(S_n)_{n\geq 0}$ of Chap. 3 on $\mathbb{S} = \mathbb{Z}$?

Answer: By (3.3.3) We have

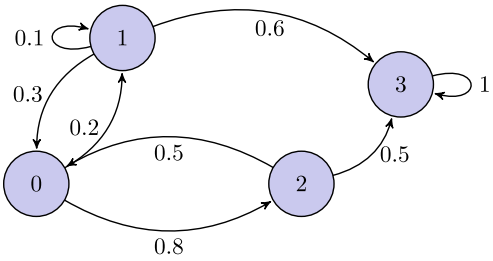
$$[P^{2n}]_{i,i} = \binom{2n}{n} p^n q^n > 0 \quad \text{and} \quad [P^{2n+1}]_{i,i} = 0, \qquad n \geq 0,$$

hence

$$\{n \geq 1 \ : \ [P^n]_{i,i} > 0\} = \{2, 4, 6, 8, \ldots\},$$

and the chain has period 2.

(ii) Find the periodicity of the chain (5.4.10).



$$\begin{bmatrix} 0 & 0.2 & 0.8 & 0 \\ 0.3 & 0.1 & 0 & 0.6 \\ 0.5 & 0 & 0 & 0.5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Answer: States ⓪, ①, ②, ③ have period 1, hence the chain is aperiodic.

(iii) The chain of Fig. 4.1 is aperiodic since it is irreducible and state ③ has a returning loop.

Exercises

Exercise 6.1 Consider the Markov chain $(X_n)_{n\geq 0}$ on the state space $\{0, 1, 2, 3\}$, with transition matrix

$$\begin{bmatrix} 1/3 & 1/3 & 1/3 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

(a) Draw the graph of this chain and find its communicating classes. Is this Markov chain reducible? Why?
(b) Find the periods of states , ⓪, ①, ②, ③.
(c) Compute $\mathbb{P}(T_0 < \infty \mid X_0 = 0)$, $\mathbb{P}(T_0 = \infty \mid X_0 = 0)$, and $\mathbb{P}(R_0 < \infty \mid X_0 = 0)$.
(d) Which state(s) is (are) absorbing, recurrent, and transient?

Exercise 6.2 Consider the Markov chain on $\{0, 1, 2\}$ with transition matrix

$$\begin{bmatrix} 1/3 & 1/3 & 1/3 \\ 1/4 & 3/4 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

(a) Is the chain irreducible? Give its communicating classes.
(b) Which states are absorbing, transient, recurrent, positive recurrent?
(c) Find the period of every state.

Exercise 6.3 Consider the Markov chain $(X_n)_{n\geq 0}$ on the state space $\{0, 1, 2, 3, 4\}$, with transition matrix

$$\begin{bmatrix} 0 & 1/4 & 1/4 & 1/4 & 1/4 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

(a) Draw the graph of this chain.
(b) Find the periods of states ⓪, ①, ② and ③.
(c) Which state(s) is (are) absorbing, recurrent, and transient?
(d) Is the Markov chain reducible? Why?

Exercise 6.4 Consider the Markov chain $(X_n)_{n\geq 0}$ with transition matrix

$$\left[P_{i,j} \right]_{0\leq i,j\leq 5} = \begin{bmatrix} 1/2 & 0 & 1/4 & 0 & 0 & 1/4 \\ 1/3 & 1/3 & 1/3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 1/6 & 1/2 & 1/6 & 0 & 0 & 1/6 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

(a) Is the chain $(X_n)_{n\geq 0}$ reducible? If yes, find its communicating classes.
(b) Determine the transient and recurrent states of $(X_n)_{n\geq 0}$.
(c) Find the period of each state.

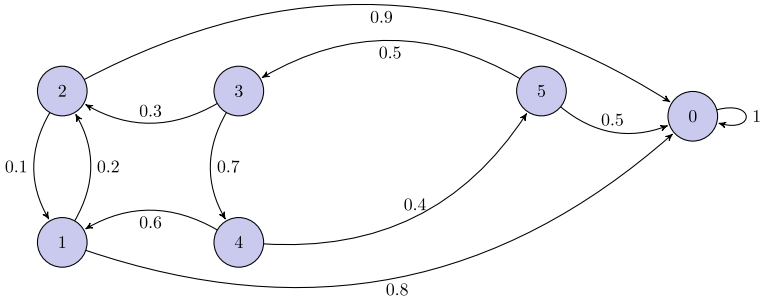
Exercise 6.5 Consider the Markov chain $(X_n)_{n\geq 0}$ with transition matrix

$$\begin{bmatrix} 0.8 & 0 & 0.2 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0.3 & 0.4 & 0 & 0.3 \end{bmatrix}.$$

(a) Is the chain $(X_n)_{n\geq 0}$ irreducible? If not, give its communicating classes.
(b) Find the period of each state. Which states are absorbing, transient, recurrent, positive recurrent?

Exercise 6.6 In the following chain, find:

(a) the communicating class(es),
(b) the transient state(s),
(c) the recurrent state(s),
(d) the positive recurrent state(s),
(e) the period of every state.



Exercise 6.7 Consider two boxes containing a total of N balls. At each unit of time one ball is chosen randomly among N and moved to the other box.

(a) Write down the transition matrix of the Markov chain $(X_n)_{n\geq 0}$ with state space $\mathbb{S} = \{0, 1, 2, \ldots, N\}$, representing the number of balls in the first box.
(b) Determine the periodicity, transience and recurrence of the Markov chain.

Exercise 6.8 (a) Is the Markov chain of Exercise 4.10-(a) recurrent? positive recurrent?
(b) Find the periodicity of every state.
(c) Same questions for the success runs Markov chain of Exercise 4.10-(b).

Problem 6.9 Let $\alpha > 0$ and consider the Markov chain with state space $\mathbb{S} = \mathbf{N}$ and transition matrix given by

$$P_{i,i-1} = \frac{1}{\alpha+1}, \quad P_{i,i+1} = \frac{\alpha}{\alpha+1}, \qquad i \geq 1.$$

and a reflecting barrier at 0, such that $P_{0,1} = 1$. Compute the mean return times $\mathbb{E}[T_k^r \mid X_0 = k]$ for $k \in \mathbf{N}$, and show that the chain is positive recurrent if and only if $\alpha < 1$.

Chapter 7
Long-Run Behavior of Markov Chains



This chapter is concerned with the behavior of Markov chains in large time, including the computation of their limiting and stationary distributions. Here, the notions of recurrence, transience, and classification of states introduced in the previous chapter play a major role.

7.1 Limiting Distributions

Definition 7.1 A Markov chain $(X_n)_{n\geq 0}$ is said to admit a *limiting probability distribution* if the following conditions are satisfied:

(i) the limits

$$\lim_{n\to\infty} \mathbb{P}(X_n = j \mid X_0 = i) \tag{7.1.1}$$

exist for all $i, j \in \mathbb{S}$, and

(ii) they form a *probability distribution* on $\mathbb{S}$, i.e.

$$\sum_{j\in\mathbb{S}} \lim_{n\to\infty} \mathbb{P}(X_n = j \mid X_0 = i) = 1, \tag{7.1.2}$$

for all $i \in \mathbb{S}$.

Note that Condition (7.1.2) is always satisfied if the limits (7.1.1) exist and the state space $\mathbb{S}$ is finite.

As remarked in (4.5.6) and (4.5.7) above, the two-state Markov chain has a limiting distribution given by

$$[\pi_0, \pi_1] = \left[\frac{b}{a+b}, \frac{a}{a+b}\right], \tag{7.1.3}$$

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provided that $(a,b) \neq (0,0)$ and $(a,b) \neq (1,1)$, while the corresponding mean return times are given from (5.3.3) by

$$(\mu_0(0), \mu_1(1)) = \left(1 + \frac{a}{b}, 1 + \frac{b}{a}\right),$$

i.e. the limiting probabilities are given by the mean return time inverses, as

$$[\pi_0, \pi_1] = \left[\frac{b}{a+b}, \frac{a}{a+b}\right] = \left[\frac{1}{\mu_0(0)}, \frac{1}{\mu_1(1)}\right] = \left[\frac{\mu_1(0)}{\mu_0(1)+\mu_1(0)}, \frac{\mu_0(1)}{\mu_0(1)+\mu_1(0)}\right].$$

This fact is not a simple coincidence, and it is actually a consequence of the following more general result, which shows that the longer it takes on average to return to a state, the smaller the probability is to find the chain in that state. Recall that a chain $(X_n)_{n\geq 0}$ is said to be recurrent, resp. aperiodic, if all its states are recurrent, resp. aperiodic.

Theorem 7.2 (Karlin and Taylor (1998), Theorem 4.4.1) *Consider a Markov chain $(X_n)_{n\geq 0}$ satisfying the following* 3 *conditions:*

(i) irreducibility,
(ii) recurrence, and
(iii) aperiodicity.

Then, the chain $(X_n)_{n\geq 0}$ admits the limiting distribution

$$\lim_{n\to\infty} \mathbb{P}(X_n = j \mid X_0 = i) = \frac{1}{\mu_j(j)}, \qquad i, j \in \mathbb{S}, \tag{7.1.4}$$

independently of the initial state $i \in \mathbb{S}$, where

$$\mu_j(j) = \mathbb{E}[T_j^r \mid X_0 = j] \in [1, \infty]$$

is the mean return time to state $j \in \mathbb{S}$.

In Theorem 7.2, Condition (i), resp. Condition (ii), is satisfied from Proposition 6.16, resp. from Proposition 6.7, provided that at least one state is aperiodic, resp. recurrent, since the chain is irreducible.

The conditions stated in Theorem 7.2 are sufficient, but they are not all necessary. For example, a Markov chain may admit a limiting distribution when the recurrence and irreducibility Conditions (i) and (iii) above are not satisfied, cf. for example Exercise 7.16-(b) below.

Note that the limiting probability (7.1.4) is independent of the initial state (i), and it vanishes whenever the state (i) is transient or null recurrent, cf. Proposition 7.4 below. In the case of the two-state Markov chain this result in consistent with (4.5.6), (4.5.7), and (7.1.3). However, it does not apply to e.g. the simple random walk of Chap. 3 which is *not* recurrent when $p \neq q$ from (6.2.5), and has period 2.

For an aperiodic chain with finite state space $\mathbb{S}$, we can show that the limit $\lim_{n\to\infty} \mathbb{P}(X_n = i \mid X_0 = j)$ exists for all $i, j \in \mathbb{S}$ by breaking the chain into communicating classes, however, it may depend on the initial state (j). This, however, does not apply to the random walk of Chap. 3 which is not aperiodic and has an infinite state space, although it can be turned into an aperiodic chain by allowing a draw as in Exercise 2.1.

The following sufficient condition for existence of a limiting distribution is a consequence of Theorem 4.1.1 in Karlin and Taylor (1998).

Proposition 7.3 *Consider a Markov chain* $(X_n)_{n\geq 0}$ *with* finite *state space* $\mathbb{S} = \{0, 1, \ldots, N\}$, *whose transition matrix* P *is* regular, *i.e. there exists* $n \geq 1$ *such that all entries of the matrix power* P^n *are non-zero. Then* $(X_n)_{n\geq 0}$ *admits a limiting probability distribution* $\pi = (\pi_i)_{i=0,1,\ldots,N}$ *given by*

$$\pi_j = \lim_{n\to\infty} \mathbb{P}(X_n = j \mid X_0 = i), \qquad 0 \leq i, j \leq N, \tag{7.1.5}$$

A chain with finite state space is regular if it is aperiodic and irreducible, cf. Proposition 1.7 in Levin et al. (2009).

We close this section with the following proposition, whose proof uses an argument similar to that of Theorem 6.11.

Proposition 7.4 *Let* $(X_n)_{n\geq 0}$ *be a Markov chain with a transient state* $(j) \in \mathbb{S}$. *Then we have*

$$\lim_{n\to\infty} \mathbb{P}(X_n = j \mid X_0 = i) = 0,$$

for all $(i) \in \mathbb{S}$.

Proof Since (j) is a transient state, the probability p_{jj} of return to (j) in finite time satisfies $p_{jj} < 1$ by definition, hence by Relation (5.4.6) Sect. 5.3, the expected number of returns to (j) starting from state (i) is finite[1]:

[1] The exchange of infinite sums and expectation is valid in particular for nonnegative series.

$$\begin{aligned}\mathbb{E}[R_j \mid X_0 = i] &= \mathbb{E}\left[\sum_{n\geq 1} \mathbb{1}_{\{X_n=j\}} \,\middle|\, X_0 = i\right] \\ &= \sum_{n\geq 1} \mathbb{E}\left[\mathbb{1}_{\{X_n=j\}} \mid X_0 = i\right] \\ &= \sum_{n\geq 1} \mathbb{P}(X_n = j \mid X_0 = i) \\ &= \frac{p_{ij}}{1-p_{jj}} < \infty.\end{aligned}$$

The convergence of the above series implies the convergence to 0 of its general term, i.e.

$$\lim_{n\to\infty} \mathbb{P}(X_n = j \mid X_0 = i) = 0$$

for all $(i) \in \mathbb{S}$, which is the expected conclusion. □

7.2 Stationary Distributions

Definition 7.5 A *probability distribution* on $\mathbb{S}$ is any a family $\pi = (\pi_i)_{i\in\mathbb{S}}$ in $[0, 1]$ such that

$$\sum_{i\in\mathbb{S}} \pi_i = 1.$$

Next, we state the definition of stationary distribution.

Definition 7.6 A *probability distribution* $\pi = (\pi_i)_{i\in\mathbb{S}}$ on $\mathbb{S}$ is said to be *stationary* if, starting X_0 at time 0 with the distribution $(\pi_i)_{i\in\mathbb{S}}$, it turns out that the distribution of X_1 is still $(\pi_i)_{i\in\mathbb{S}}$ at time 1.

In other words, $(\pi_i)_{i\in\mathbb{S}}$ is stationary for the Markov chain with transition matrix P if, letting

$$\mathbb{P}(X_0 = i) := \pi_i, \qquad i \in \mathbb{S},$$

at time 0, implies

$$\mathbb{P}(X_1 = i) = \mathbb{P}(X_0 = i) = \pi_i, \quad i \in \mathbb{S},$$

at time 1. This also means that

$$\pi_j = \mathbb{P}(X_1 = j) = \sum_{i\in\mathbb{S}} \mathbb{P}(X_1 = j \mid X_0 = i)\mathbb{P}(X_0 = i) = \sum_{i\in\mathbb{S}} \pi_i P_{i,j}, \quad j \in \mathbb{S},$$

i.e. the distribution π is stationary if and only if the vector π is *invariant* (or stationary) by the matrix P, that means

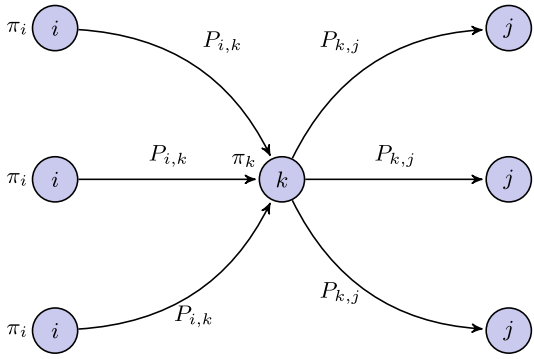


Fig. 7.1 Global balance condition (discrete time)

$$\pi = \pi P. \tag{7.2.1}$$

Note that in contrast with (5.1.4), the multiplication by P in (7.2.1) is on the *right*-hand side and *not* on the left-hand side. The relation (7.2.1) can be rewritten as the *global balance condition*

$$\sum_{i\in\mathbb{S}} \pi_i P_{i,k} = \pi_k = \pi_k \sum_{j\in\mathbb{S}} P_{k,j} = \sum_{j\in\mathbb{S}} \pi_k P_{k,j}, \tag{7.2.2}$$

which is illustrated in Fig. 7.1.

We also note that the stationarity and limiting properties of distributions are quite different concepts. If the chain is started in the stationary distribution then it will remain in that distribution at *any* subsequent time step (which is stronger than saying that the chain will reach that distribution after an *infinite* number of time steps). On the other hand, in order to reach the limiting distribution the chain can be started from any given initial distribution or even from any fixed given state, and it will converge to the limiting distribution if it exists. Nevertheless, the limiting and stationary distribution may coincide in some situations as in Theorem 7.8 below.

More generally, assuming that X_n has the invariant (or stationary) distribution π at time n, i.e. $\mathbb{P}(X_n = i) = \pi_i$, $i \in \mathbb{S}$, we have

$$\begin{aligned}\mathbb{P}(X_{n+1} = j) &= \sum_{i\in\mathbb{S}} \mathbb{P}(X_{n+1} = j \mid X_n = i)\mathbb{P}(X_n = i)\\ &= \sum_{i\in\mathbb{S}} P_{i,j}\mathbb{P}(X_n = i) = \sum_{i\in\mathbb{S}} P_{i,j}\pi_i\\ &= [\pi P]_j = \pi_j, \qquad j \in \mathbb{S},\end{aligned}$$

since the transition matrix of $(X_n)_{n\geq 0}$ is *time homogeneous*, hence

$$\mathbb{P}(X_n = j) = \pi_j, \quad j \in \mathbb{S}, \quad \Longrightarrow \quad \mathbb{P}(X_{n+1} = j) = \pi_j, \quad j \in \mathbb{S}.$$

By induction on $n \geq 0$, this yields

$$\mathbb{P}(X_n = j) = \pi_j, \qquad j \in \mathbb{S}, \quad n \geq 1,$$

i.e. the chain $(X_n)_{n\geq 0}$ remains in the same distribution π at all times $n \geq 1$, provided that it has been started with the stationary distribution π at time $n = 0$.

Proposition 7.7 *Assume that $\mathbb{S} = \{0, 1, \ldots, N\}$ is finite and that the limits*

$$\pi_j^{(i)} := \lim_{n\to\infty} \mathbb{P}(X_n = j \mid X_0 = i) = \lim_{n\to\infty} [P^n]_{i,j}$$

exist for all $j \in \mathbb{S}$ and are independent of the initial state $i \in \mathbb{S}$, i.e. we have

$$\lim_{n\to\infty} P^n = \begin{bmatrix} \pi_0^{(0)} & \pi_1^{(0)} & \cdots & \pi_N^{(0)} \\ \pi_0^{(1)} & \pi_1^{(1)} & \cdots & \pi_N^{(1)} \\ \vdots & \vdots & \ddots & \vdots \\ \pi_0^{(N)} & \pi_1^{(N)} & \cdots & \pi_N^{(N)} \end{bmatrix}.$$

Then for every $i = 0, 1, \ldots, N$, the vector $\pi^{(i)} := (\pi_j^{(i)})_{j\in\{0,1,\ldots,N\}}$ is a stationary distribution and we have

$$\pi^{(i)} = \pi^{(i)} P, \tag{7.2.3}$$

i.e. $\pi^{(i)}$ is invariant (or stationary) by P, $i = 0, 1, \ldots, N$.

Proof We have

$$\begin{aligned} \pi_j^{(i)} &:= \lim_{n\to\infty} \mathbb{P}(X_n = j \mid X_0 = i) \\ &= \lim_{n\to\infty} \mathbb{P}(X_{n+1} = j \mid X_0 = i) \\ &= \lim_{n\to\infty} \sum_{l\in\mathbb{S}} \mathbb{P}(X_{n+1} = j \mid X_n = l)\mathbb{P}(X_n = l \mid X_0 = i) \\ &= \lim_{n\to\infty} \sum_{l\in\mathbb{S}} P_{l,j}\mathbb{P}(X_n = l \mid X_0 = i) \\ &= \sum_{l\in\mathbb{S}} P_{l,j} \lim_{n\to\infty} \mathbb{P}(X_n = l \mid X_0 = i) \\ &= \sum_{l\in\mathbb{S}} \pi_l^{(i)} P_{l,j}, \qquad i, j \in \mathbb{S}, \end{aligned}$$

where we exchanged limit and summation because the state space $\mathbb{S}$ is assumed to be finite, which shows that

$$\pi^{(i)} = \pi^{(i)} P,$$

i.e. (7.2.3) holds and $\pi^{(i)}$ is a stationary distribution, $i = 0, 1, \ldots, N$. □

Proposition 7.7 can be applied in particular when the limiting distribution $\pi_j := \lim_{n\to\infty} \mathbb{P}(X_n = j \mid X_0 = i)$ does not depend on the initial state $\textcircled{i}$, i.e.

$$\lim_{n\to\infty} P^n = \begin{bmatrix} \pi_0 & \pi_1 & \cdots & \pi_N \\ \pi_0 & \pi_1 & \cdots & \pi_N \\ \vdots & \vdots & \ddots & \vdots \\ \pi_0 & \pi_1 & \cdots & \pi_N \end{bmatrix}.$$

For example, the limiting distribution (7.1.3) of the two-state Markov chain is also an invariant distribution, i.e. it satisfies (7.2.1). In particular, we have the following result which does not require the finiteness of the state space $\mathbb{S}$.

Theorem 7.8 (Karlin and Taylor (1998), Theorem 4.4.2) *Assume that the Markov chain $(X_n)_{n\geq 0}$ satisfies the following* 3 *conditions:*

(i) irreducibility,
(ii) positive recurrence, and
(iii) aperiodicity.

Then the chain $(X_n)_{n\geq 0}$ admits a limiting distribution

$$\pi_j := \lim_{n\to\infty} \mathbb{P}(X_n = j \mid X_0 = i) = \lim_{n\to\infty} [P^n]_{i,j} = \frac{1}{\mu_j(j)}, \qquad i, j \in \mathbb{S},$$

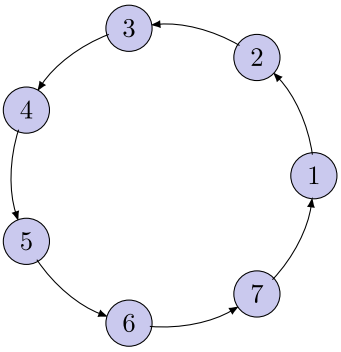
independently of the initial state $i \in \mathbb{S}$, which also forms a stationary distribution $(\pi_j)_{j\in\mathbb{S}} = (1/\mu_j(j))_{j\in\mathbb{S}}$, uniquely determined by the equation

$$\pi = \pi P.$$

In Theorem 7.8 above, Condition (ii), is satisfied from Proposition 6.16, provided that at least one state is aperiodic, since the chain is irreducible. We also note that the limiting distribution π does not depend on the initial distribution of X_0.

See Exercise 7.23 for an application of Theorem 7.8 on an infinite state space.

In the following trivial example of a finite circular chain, Theorems 7.2 and 7.8 cannot be applied since the chain is not aperiodic, and it clearly does not admit a limiting distribution. However, Theorem 7.10 below applies here, and the chain admits a stationary distribution: one can easily check that $\mu_k(k) = n$ and $\pi_k = 1/n = 1/\mu_k(k)$, $k = 1, 2, \ldots, n$, with $n = 7$.



$$P = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{7.2.4}$$

In view of Theorem 6.13, we have the following corollary of Theorem 7.8:

Corollary 7.9 *Consider an irreducible aperiodic Markov chain with finite state space* $\mathbb{S}$*. Then, the limiting probabilities*

$$\pi_i := \lim_{n\to\infty} \mathbb{P}(X_n = i \mid X_0 = j) = \frac{1}{\mu_i(i)}, \qquad i, j \in \mathbb{S},$$

exist and form a stationary distribution which is uniquely determined by the equation

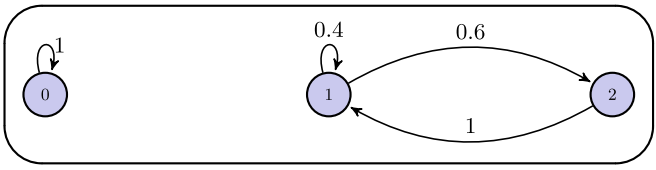
$$\pi = \pi P.$$

Corollary 7.9 can also be applied separately to derive a stationary distribution on each closed component of a reducible chain.

The convergence of the two-state chain to its stationary distribution has been illustrated in Fig. 4.4. Before proceeding further we make some comments on the assumptions of Theorems 7.2 and 7.8.

Remarks

- Irreducibility.
  The irreducibility assumption on the chain in Theorems 7.2 and 7.8 is truly required in general, as a reducible chain may have a limiting distribution that depends on the initial state as in the following trivial example on the state space {0, 1, 2}:



in which the chain is aperiodic and positive recurrent, but not irreducible. Note that the sub-chain $\{1, 2\}$ admits $[\pi_1, \pi_2] = [1/1.6, 0.6/1.6]$ as stationary and limiting distribution, however, any vector of the form $(1 - \alpha, \alpha\pi_1, \alpha\pi_2)$ is also a stationary distribution on $\mathbb{S} = \{0, 1, 2\}$ for any $\alpha \in [0, 1]$, showing the non uniqueness of the stationary distribution.

More generally, in case the chain is not irreducible we can split it into subchains and consider the subproblems separately. For example, when the state space $\mathbb{S}$ is a finite set it admits at least one communicating class $A \subset \mathbb{S}$ that leads to no other class, and it admits a stationary distribution π_A by Corollary 7.11 since it is irreducible, hence a chain with finite state space $\mathbb{S}$ admits at least one stationary distribution of the form $(0, 0, \ldots, 0, \pi_A)$.

Similarly, the constant two-state Markov chain with transition matrix $P = I$ is reducible, it admits an infinity of stationary distributions, and a limiting distribution which is dependent on the initial state.

- Aperiodicity.
  The conclusions of Theorems 7.2, 7.8 and Corollary 7.9 ensure the existence of the limiting distribution by requiring the aperiodicity of the Markov chain. Indeed, the limiting distribution may not exist when the chain is not aperiodic. For example, the two-state Markov chain with transition matrix

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

  is not aperiodic (both states have period 2) and it has no limiting distribution because[2]

$$\mathbb{P}(X_{2n} = 1 \mid X_0) = 1 \quad \text{and} \quad \mathbb{P}(X_{2n+1} = 1 \mid X_0) = 0, \quad n \geq 0. \tag{7.2.5}$$

  The chain does have an invariant (or stationary) distribution π solution of $\pi = \pi P$, and given by

$$\pi = [\pi_0, \pi_1] = \left[\frac{1}{2}, \frac{1}{2}\right].$$

- Positive recurrence.

  Theorems 7.8, 7.10 below, and Corollary 7.9 do not apply to the unrestricted random walk $(S_n)_{n\geq 0}$ of Chap. 3, because this chain is not positive recurrent, cf. Relations (3.4.17) and (3.4.20), and admits no stationary distribution.

  If a stationary distribution $\pi = (\pi_i)_{i\in\mathbb{Z}}$ existed it would satisfy the equation $\pi = \pi P$ which, according to (4.3.1), would read

$$\pi_i = p\pi_{i-1} + q\pi_{i+1}, \qquad i \in \mathbb{Z},$$

[2] This two-state chain is a particular case of the circular chain (7.2.4) for $n = 2$.

i.e.

$$(p+q)\pi_i = p\pi_{i-1} + q\pi_{i+1}, \qquad i \in \mathbb{Z},$$

or

$$\pi_{i+1} - \pi_i = \frac{p}{q}(\pi_i - \pi_{i-1}), \qquad i \in \mathbb{Z}.$$

As in the direct solution method of Sect. 2.2, this implies

$$\pi_{i+1} - \pi_i = \left(\frac{p}{q}\right)^i (\pi_1 - \pi_0), \qquad i \in \mathbb{N},$$

so that by a telescoping summation argument we have

$$\begin{aligned}
\pi_k &= \pi_0 + \sum_{i=0}^{k-1} (\pi_{i+1} - \pi_i) \\
&= \pi_0 + (\pi_1 - \pi_0) \sum_{i=0}^{k-1} \left(\frac{p}{q}\right)^i \\
&= \pi_0 + (\pi_1 - \pi_0) \frac{1 - (p/q)^k}{1 - p/q}, \qquad k \geq 0,
\end{aligned}$$

which cannot satisfy the condition $\sum_{k\in\mathbb{Z}} \pi_k = 1$, with $p \neq q$.

When $p = q = 1/2$ we similarly obtain

$$\pi_k = \pi_0 + \sum_{i=0}^{k-1} (\pi_{i+1} - \pi_i) = \pi_0 + k(\pi_1 - \pi_0), \qquad k \in \mathbb{Z},$$

and in this case as well, the sequence $(\pi_k)_{k\geq 0}$ cannot satisfy the condition $\sum_{k\in\mathbb{Z}} \pi_k = 1$. Since there is no uniform distribution on an infinite space, we conclude that the chain does not admit a stationary distribution. Hence the stationary distribution of a Markov chain may not exist at all.

In addition, it follows from (3.3.3) and the Stirling approximation formula that

$$\begin{aligned}
\lim_{n\to\infty} \mathbb{P}(S_{2n} = 2k \mid S_0 = 0) &= \lim_{n\to\infty} \frac{(2n)!}{(n+k)!(n-k)!} p^{n+k} q^{n-k} \\
&\leq \lim_{n\to\infty} \frac{(2n)!}{2^{2n} n!^2} \\
&= \lim_{n\to\infty} \frac{1}{\sqrt{\pi n}} = 0, \quad k \geq 0,
\end{aligned}$$

as in Problem 3.9-(f), so that the limiting distribution does not exist as well. Here, Theorem 7.2 cannot be applied because the chain is not aperiodic (it has period

2), however, aperiodicity and irreducibility are not sufficient in general when the state space is infinite, cf. e.g. the model of Exercise 2.1.

The following theorem gives sufficient conditions for the existence of a stationary distribution, without requiring aperiodicity or finiteness of the state space. As noted in (7.2.5) above, the limiting distribution may not exist in this case. See also Problem 6.9 and Exercise 7.23 for an example of a null recurrent chain which does not admit a stationary distribution.

Theorem 7.10 (Bosq and Nguyen (1996), Theorem 4.1) *Consider a Markov chain* $(X_n)_{n\geq 0}$ *satisfying the following two conditions:*

(i) irreducibility, and
(ii) positive recurrence.

Then, the probabilities

$$\pi_i = \frac{1}{\mu_i(i)}, \qquad i \in \mathbb{S},$$

form a stationary distribution which is uniquely determined by the equation $\pi = \pi P$.

Note that the conditions stated in Theorem 7.10 are sufficient, but they are not all necessary. For example, Condition (ii) is not necessary as the trivial constant chain, whose transition matrix $P = \mathrm{I}$ is reducible, does admit a stationary distribution.

Note that the positive recurrence assumption in Theorem 7.2 is required in general on infinite state spaces. For example, the process in Exercise 7.23 is positive recurrent for $\alpha < 1$ only, whereas no stationary distribution exists when $\alpha \geq 1$.

As a consequence of Corollary 6.14 we have the following corollary of Theorem 7.10, which does not require aperiodicity for the stationary distribution to exist.

Corollary 7.11 *Let* $(X_n)_{n\geq 0}$ *be an irreducible Markov chain with* finite *state space* $\mathbb{S}$. *Then, the probabilities*

$$\pi_k = \frac{1}{\mu_k(k)}, \qquad k \in \mathbb{S},$$

form a stationary distribution which is uniquely determined by the equation

$$\pi = \pi P.$$

Similarly to Proposition 7.3, any Markov chain $(X_n)_{n\geq 0}$ with *finite* state space and whose transition matrix P is *regular* admits a stationary distribution, as it is irreducible.

According to Corollary 7.11, the limiting distribution and stationary distribution both exist (and coincide) when the chain is irreducible aperiodic with finite state space $\mathbb{S}$, and in this case we have $\pi_k > 0$ for all $k \in \mathbb{S}$ by Corollaries 6.14 and 7.11. When the chain is irreducible it is usually easier to compute the stationary distribution, which will give us the limiting distribution.

Under the assumptions of Theorem 7.8, if the stationary and limiting distributions both exist then they are equal and in this case we only need to compute one of them. However, in some situations only the stationary distribution might exist. According to Corollary 7.11 above the stationary distribution always exists when the chain is irreducible with finite state space, nevertheless the limiting distribution may not exist if the chain is not aperiodic, consider for example the two-state switching chain with $a = b = 1$.

Finding a Limiting Distribution

In summary:

– We usually attempt first to compute the stationary distribution whenever possible, and this also gives the limiting distribution when it exists. For this, we first check whether the chain is positive recurrent, aperiodic and irreducible, in which case the limiting distribution can be found by solving $\pi = \pi P$ according to Theorem 7.8.

– In case the above properties are not satisfied we need to compute the limiting distribution by taking the limit $\lim_{n\to\infty} P^n$ of the powers P^n of the transition matrix, if possible by decomposing the state space in communicating classes as in e.g. Exercise 7.12. This can turn out to be much more complicated and done only in special cases. If the chain has period $d \geq 2$ we may need to investigate the limits $\lim_{n\to\infty} P^{nd}$ instead, see e.g. Exercise 7.4 and (7.2.6)–(7.2.7) below.

To further summarize, we note that by Theorem 7.2 we have

(a) irreducible + recurrent + aperiodic $\Longrightarrow$ existence of a limiting distribution,

by Theorem 7.8 we get

(b) irreducible + positive recurrent + aperiodic $\Longrightarrow$ existence of a limiting distribution which is also stationary,

and by Theorem 7.10 we get

(c) irreducible + positive recurrent $\Longrightarrow$ existence of a stationary distribution.

In addition, the limiting or stationary distribution $\pi = (\pi_i)_{i\in\mathbb{S}}$ satisfies

$$\pi_i = \frac{1}{\mu_i(i)}, \qquad i \in \mathbb{S},$$

in all above cases (a), (b) and (c).

The Ergodic Theorem

The *Ergodic Theorem*, cf. e.g. Theorem 1.10.2 of Norris (1998) states the following.

Theorem 7.12 *Assume that the chain $(X_n)_{n\geq 0}$ is* irreducible. *Then the sample average time spent at state i converges almost surely to $1/\mu_i(i)$, i.e.*

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \mathbb{1}_{\{X_k=i\}} = \frac{1}{\mu_i(i)}, \qquad i \in \mathbb{S}.$$

In case $(X_n)_{n\geq 0}$ is also positive recurrent, Theorem 7.10 shows that we also have

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \mathbb{1}_{\{X_k=i\}} = \pi_i, \qquad i \in \mathbb{S},$$

where $(\pi_i)_{i\in\mathbb{S}}$ is the stationary distribution of $(X_n)_{n\geq 0}$. We refer to Fig. 4.4 for an illustration of convergence in the setting of the Ergodic Theorem 7.12.

Example Consider the maze random walk (5.3.7) with transition matrix

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 0 & 0 & 1/2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 1/2 & 0 & 1/2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 0 & 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 0 & 1/3 & 0 & 1/3 & 0 & 1/3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}.$$

The equation $\pi = \pi P$ yields

$$\begin{cases} \pi_1 = \frac{1}{2}\pi_2 \\ \pi_2 = \pi_1 + \frac{1}{2}\pi_3 \\ \pi_3 = \frac{1}{2}\pi_2 + \frac{1}{2}\pi_6 \\ \pi_4 = \frac{1}{2}\pi_7 \\ \pi_5 = \frac{1}{2}\pi_6 + \frac{1}{3}\pi_8 \\ \pi_6 = \frac{1}{2}\pi_3 + \frac{1}{2}\pi_5 \\ \pi_7 = \pi_4 + \frac{1}{3}\pi_8 \\ \pi_8 = \frac{1}{2}\pi_5 + \frac{1}{2}\pi_7 + \pi_9 \\ \pi_9 = \frac{1}{3}\pi_8, \end{cases} \quad \text{hence} \quad \begin{cases} \pi_1 = \frac{1}{2}\pi_2 \\ \pi_2 = \pi_3 \\ \pi_3 = \pi_6 \\ \pi_4 = \frac{1}{2}\pi_3 \\ \frac{1}{2}\pi_7 = \frac{1}{3}\pi_8 \\ \pi_6 = \pi_5 \\ \pi_6 = \pi_7 \\ \pi_9 = \frac{1}{3}\pi_8, \end{cases}$$

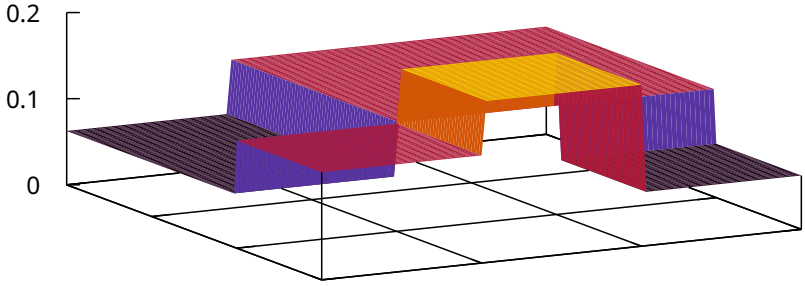


Fig. 7.2 Stationary distribution on the maze (5.3.7)

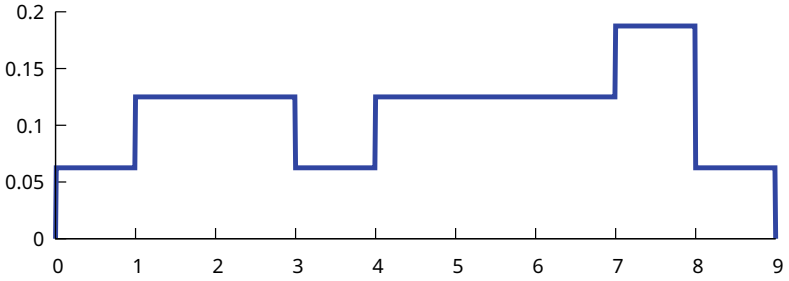


Fig. 7.3 Stationary distribution by state numbering

and

$$\begin{aligned} 1 &= \pi_1 + \pi_2 + \pi_3 + \pi_4 + \pi_5 + \pi_6 + \pi_7 + \pi_8 + \pi_9 \\ &= \pi_1 + 2\pi_1 + 2\pi_1 + \pi_1 + 2\pi_1 + 2\pi_1 + 2\pi_1 + 3\pi_1 + \pi_1 \\ &= 16\pi_1, \end{aligned}$$

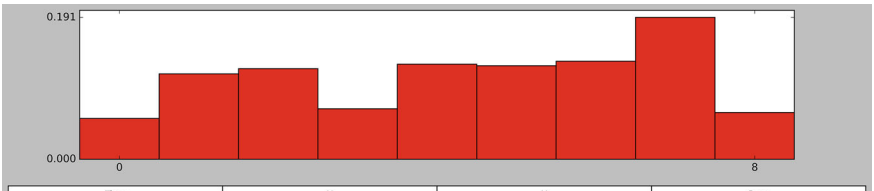
hence

$$\begin{aligned} &\pi_1 = \frac{1}{16},\ \pi_2 = \frac{2}{16},\ \pi_3 = \frac{2}{16},\ \pi_4 = \frac{1}{16}, \\ &\pi_5 = \frac{2}{16},\ \pi_6 = \frac{2}{16},\ \pi_7 = \frac{2}{16},\ \pi_8 = \frac{3}{16},\ \pi_9 = \frac{1}{16}, \end{aligned}$$

cf. Figs. 7.3 and 7.2 below, and we check that since $\mu_1(1) = 16$ by (5.3.8), we indeed have

$$\pi_1 = \frac{1}{\mu_1(1)} = \frac{1}{16},$$

according to Corollary 7.11.
The stationary probability distribution of Figs. 7.3 and 7.2 can be compared to the proportions of time spent at each state simulated in Fig. 7.4.



Time	X	X	Data
0	0	0	0.055
1	1	1	0.115
2	2	2	0.122
3	1	3	0.068
4	2	4	0.128
5	5	5	0.126
6	4	6	0.132
7	5	7	0.191
8	4	8	0.063

Fig. 7.4 Simulated stationary distribution

Note that this chain has period 2 and the matrix powers $(P^n)_{n\geq 0}$ do not converge as n tends to infinity, i.e. it does not admit a limiting distribution. In fact, using the following Matlab/Octave commands:

```
P = [0,1,0,0,0,0,0,0,0;
1/2,0,1/2,0,0,0,0,0,0;
0,1/2,0,0,0,1/2,0,0,0;
0,0,0,0,0,0,1,0,0;
0,0,0,0,0,1/2,0,1/2,0;
0,0,1/2,0,1/2,0,0,0,0;
0,0,0,1/2,0,0,0,1/2,0;
0,0,0,0,1/3,0,1/3,0,1/3;
0,0,0,0,0,0,0,1,0]
mpower(P,1000)
mpower(P,1001)
```

we see that

$$\lim_{n\to\infty} P^{2n} = \begin{bmatrix} 1/8 & 0 & 1/4 & 0 & 1/4 & 0 & 1/4 & 0 & 1/8 \\ 0 & 1/4 & 0 & 1/8 & 0 & 1/4 & 0 & 3/8 & 0 \\ 1/8 & 0 & 1/4 & 0 & 1/4 & 0 & 1/4 & 0 & 1/8 \\ 0 & 1/4 & 0 & 1/8 & 0 & 1/4 & 0 & 3/8 & 0 \\ 1/8 & 0 & 1/4 & 0 & 1/4 & 0 & 1/4 & 0 & 1/8 \\ 0 & 1/4 & 0 & 1/8 & 0 & 1/4 & 0 & 3/8 & 0 \\ 1/8 & 0 & 1/4 & 0 & 1/4 & 0 & 1/4 & 0 & 1/8 \\ 0 & 1/4 & 0 & 1/8 & 0 & 1/4 & 0 & 3/8 & 0 \\ 1/8 & 0 & 1/4 & 0 & 1/4 & 0 & 1/4 & 0 & 1/8 \end{bmatrix}, \tag{7.2.6}$$

and

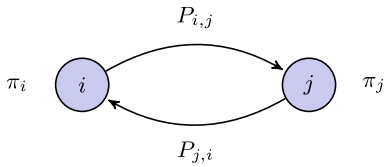


Fig. 7.5 Detailed balance condition (discrete time)

$$\lim_{n\to\infty} P^{2n+1} = \begin{bmatrix} 0 & 1/4 & 0 & 1/8 & 0 & 1/4 & 0 & 3/8 & 0 \\ 1/8 & 0 & 1/4 & 0 & 1/4 & 0 & 1/4 & 0 & 1/8 \\ 0 & 1/4 & 0 & 1/8 & 0 & 1/4 & 0 & 3/8 & 0 \\ 1/8 & 0 & 1/4 & 0 & 1/4 & 0 & 1/4 & 0 & 1/8 \\ 0 & 1/4 & 0 & 1/8 & 0 & 1/4 & 0 & 3/8 & 0 \\ 1/8 & 0 & 1/4 & 0 & 1/4 & 0 & 1/4 & 0 & 1/8 \\ 0 & 1/4 & 0 & 1/8 & 0 & 1/4 & 0 & 3/8 & 0 \\ 1/8 & 0 & 1/4 & 0 & 1/4 & 0 & 1/4 & 0 & 1/8 \\ 0 & 1/4 & 0 & 1/8 & 0 & 1/4 & 0 & 3/8 & 0 \end{bmatrix}, \tag{7.2.7}$$

which shows that, although $(P^n)_{n\geq 1}$ admits two converging subsequences, $\lim_{n\to\infty} P^n$ does not exist, therefore this Markov chain does not admit a limiting distribution.

7.3 Markov Chain Monte Carlo (MCMC)

The goal of the Markov Chain Monte Carlo (MCMC) method, or Metropolis algorithm, is to generate random samples according to a target distribution $\pi = (\pi_i)_{i\in\mathbb{S}}$ via a Markov chain that admits π as limiting and stationary distribution. It can be applied in particular in the setting of large state spaces $\mathbb{S}$, cf. e.g. Problem 7.27-(a).

A Markov chain $(X_k)_{k\geq 0}$ with transition matrix P on a state space $\mathbb{S}$ is said to satisfy the *detailed balance* (or *reversibility*) condition with respect to the probability distribution $\pi = (\pi_i)_{i\in S}$ if

$$\pi_i P_{i,j} = \pi_j P_{j,i}, \qquad i, j \in \mathbb{S}, \tag{7.3.1}$$

see Fig. 7.5.

Lemma 7.13 *The detailed balance condition* (7.3.1) *implies the global balance condition* (7.2.2).

Proof By summation over $i \in \mathbb{S}$ in (7.3.1), we have

$$\sum_{i\in\mathbb{S}} \pi_i P_{i,j} = \sum_{i\in\mathbb{S}} \pi_j P_{j,i} = \pi_j \sum_{i\in\mathbb{S}} P_{j,i} = \pi_j, \quad j \in \mathbb{S},$$

which shows that $\pi P = \pi$, i.e. π is a stationary distribution for P, cf. e.g. Problem 7.25-(c).

If the transition matrix P satisfies the detailed balance condition with respect to π then the probability distribution of X_n will naturally converge to the stationary distribution π in the long run, e.g. under the hypotheses of Theorem 7.8, i.e. when the chain $(X_k)_{k\geq 0}$ is positive recurrent, aperiodic, and irreducible .

Metropolis-Hastings Algorithm

In general, however, the detailed balance (or reversibility) condition (7.3.1) may not be satisfied by π and P. In this case, starting from a proposal matrix P, one can construct a modified transition matrix $\widetilde{P}$ that will satisfy the detailed balance condition with respect to π. This modified transition matrix $\widetilde{P}$ is defined by

$$\begin{aligned}\widetilde{P}_{i,j} &:= P_{i,j} \times \min\left(1, \frac{\pi_j P_{j,i}}{\pi_i P_{i,j}}\right)\\ &= \min\left(P_{i,j}, \frac{\pi_j P_{j,i}}{\pi_i}\right)\\ &= \begin{cases} P_{j,i}\dfrac{\pi_j}{\pi_i} & \text{if } \pi_j P_{j,i} \leq \pi_i P_{i,j},\\ P_{i,j} & \text{if } \pi_j P_{j,i} \geq \pi_i P_{i,j},\end{cases}\end{aligned} \tag{7.3.2}$$

for $i \neq j$. We note that

$$\sum_{\substack{j\in\mathbb{S}\\ j\neq i}} \widetilde{P}_{i,j} \leq \sum_{\substack{j\in\mathbb{S}\\ j\neq i}} P_{i,j} \leq 1,$$

and for $i \in \mathbb{S}$ we let

$$\begin{aligned}\widetilde{P}_{i,i} &:= 1 - \sum_{\substack{j\in\mathbb{S}\\ j\neq i}} \widetilde{P}_{i,j}\\ &= P_{i,i} + \sum_{\substack{j\in\mathbb{S}\\ j\neq i}} P_{i,j}\left(1 - \min\left(1, \frac{\pi_j P_{j,i}}{\pi_i P_{i,j}}\right)\right)\\ &= P_{i,i} + \sum_{\substack{j\in\mathbb{S}\\ j\neq i}} P_{i,j}\left(1 - \frac{\pi_j P_{j,i}}{\pi_i P_{i,j}}\right)^+\\ &= P_{i,i} + \sum_{\substack{j\in\mathbb{S}\\ j\neq i}} \left(P_{i,j} - \frac{\pi_j P_{j,i}}{\pi_i}\right)^+.\end{aligned}$$

Clearly, we have $\widetilde{P} = P$ when the detailed balance (or reversibility) condition (7.3.1) is satisfied by P. In the general case, we can check that for $i \neq j$, we have

$$\pi_i \widetilde{P}_{i,j} = \left\{ \begin{array}{l} P_{j,i}\pi_j = \pi_j \widetilde{P}_{j,i} \text{ if} \pi_j P_{j,i} \le \pi_i P_{i,j}, \\ \pi_i P_{i,j} = \pi_j \widetilde{P}_{j,i} \quad \text{if } \pi_j P_{j,i} \ge \pi_i P_{i,j}, \end{array} \right\} = \pi_j \widetilde{P}_{j,i},$$

hence $\widetilde{P}$ satisfies the detailed balance condition with respect to π (the condition is obviously satisfied when $i = j$). Therefore, the random simulation of $(\widetilde{X}_n)_{n\ge 0}$ according to the transition matrix $\widetilde{P}$ will provide samples of the distribution π in the long run as n tends to infinity, provided that the chain $(\widetilde{X}_n)_{n\ge 0}$ is positive recurrent, aperiodic, and irreducible.

In standard MCMC sampling, the transition matrix is usually symmetric with $P_{i,j} = P_{j,i} > 0$, $i, j \in \mathbb{S}$, typically $P_{i,j} = \varphi(i-j)$ with φ a Gaussian type density kernel, and the modified transition matrix $\widetilde{P}$ simplifies to

$$\widetilde{P}_{i,j} := P_{i,j} \times \min\left(1, \frac{\pi_j}{\pi_i}\right) = \min\left(P_{i,j}, \frac{\pi_j}{\pi_i} P_{i,j}\right) = \left\{ \begin{array}{ll} P_{i,j}\dfrac{\pi_j}{\pi_i} & \text{if } \pi_j < \pi_i, \\ P_{i,j} & \text{if } \pi_j \ge \pi_i, \end{array} \right.$$

for $i \neq j$, with

$$\widetilde{P}_{i,i} := 1 - \sum_{\substack{j\in\mathbb{S} \\ j\neq i}} \widetilde{P}_{i,j} = P_{i,i} + \sum_{\substack{j\in\mathbb{S}, j\neq i \\ \pi_j<\pi_i}} P_{i,j}\left(1 - \frac{\pi_j}{\pi_i}\right), \qquad i \in \mathbb{S}.$$

In other words, starting from state $\textcircled{i}$, a proposal $\textcircled{j}$ is generated with probability $P_{i,j}$. This proposal is then accepted if $\pi_j \ge \pi_i$, otherwise if $\pi_j < \pi_i$, the proposal is accepted with probability π_j/π_i, and one remains at state $\textcircled{i}$ with probability $1 - \pi_j/\pi_i$.

Generating Posterior Samples Using MCMC

In this case, $\pi = (\pi_i)_{i\in\mathbb{S}}$ represents the prior distribution of a model parameter. Given $\mathcal{O}$ a set of observations sampled according to a distribution $(p_k)_{k\in\mathcal{O}}$, we are given a likelihood function $l(k|i)$ which represents the probability of observing $k \in \mathcal{O}$ when the system parameter is $\textcircled{i}$.

The posterior probability distribution $p(i|k)$ of being in the state $\textcircled{i}$ given that we observed $k \in \mathcal{O}$ is obtained by the Bayes formula as

$$p(i|k) = l(k|i)\frac{\pi_i}{p_k}, \qquad i \in \mathbb{S},\ k \in \mathcal{O}.$$

Computing the posterior distribution $p(i|k)$ and generating the corresponding random samples may require to estimate the distribution p_k, $k \in \mathcal{O}$.

The Markov Chain Monte Carlo method provides an efficient way to generate random samples according to the posterior distribution $p(i|k)$. For this, the ratio π_j/π_i in (7.3.2) is replaced with

$$\frac{l(k|j)\pi_j}{l(k|i)\pi_i} = \frac{p(j|k)p_k}{p(i|k)p_k} = \frac{p(j|k)}{p(i|k)}, \qquad i, j \in \mathbb{S},\ k \in \mathcal{O}, \tag{7.3.3}$$

which uses the information given by the observation $\textcircled{k}$.

Relation (7.3.3) shows that the proposal $\textcircled{j}$ generated with probability $P_{i,j}$ is accepted if $p(j|k) \geq p(i|k)$, i.e. if its posterior probability $p(j|k)$ given the observation k is higher than the posterior probability $p(i|k)$ of the initial state $\textcircled{i}$. Otherwise, if $p(j|k) < p(i|k)$ the proposal $\textcircled{j}$ is accepted only with the probability given by (7.3.3).

Improved versions of the MCMC algorithms include the Hamiltonian Monte Carlo method and the No U-Turn Sampler (NUTS). See Chap. 6 in Privault (2024) for more details and an implementation of the MCMC algorithm.

In Table 7.1 we summarize the definitions introduced in this chapter and in Chap. 6.

Exercises

Exercise 7.1 Find the stationary distribution $[\pi_0, \pi_1]$ of the two-state Markov chain on $\mathbb{S} = \{0, 1\}$ with transition probability matrix

$$P = \begin{matrix} \\ 0 \\ 1 \end{matrix} \begin{matrix} \begin{matrix} 0 & 1 \end{matrix} \\ \begin{bmatrix} 1/3 & 2/3 \\ 2/3 & 1/3 \end{bmatrix} \end{matrix}.$$

Exercise 7.2 We consider the Markov chain of Exercise 4.10-(a).

(a) Is this chain irreducible, aperiodic, recurrent, positive recurrent?
(b) Does this Markov chain admit a stationary distribution?
(c) Does this Markov chain admit a limiting distribution?

Exercise 7.3 We consider the success runs Markov chain of Exercise 4.10-(b).

(a) Is the success runs chain irreducible, aperiodic, recurrent, positive recurrent?
(b) Does this Markov chain admit a stationary distribution?
(c) Does this Markov chain admit a limiting distribution?

Table 7.1 Summary of Markov chain properties

Property	Definition
Absorbing (state)	$P_{i,i} = 1$
Recurrent (state)	$\mathbb{P}(T_i^r < \infty \mid X_0 = i) = 1$
Transient (state)	$\mathbb{P}(T_i^r < \infty \mid X_0 = i) < 1$
Positive recurrent (state)	Recurrent *and* $\mathbb{E}[T_i^r \mid X_0 = i] < \infty$
Null recurrent (state)	Recurrent *and* $\mathbb{E}[T_i^r \mid X_0 = i] = \infty$
Aperiodic (state or chain)	Period(s) $= 1$
Ergodic (state or chain)	Positive recurrent *and* aperiodic
Irreducible (chain)	*All* states communicate
Regular (chain)	All coefficients of P^n are > 0 for some $n \geq 1$
Stationary distribution π	Obtained from solving $\pi = \pi P$

Exercise 7.4 We consider the Ehrenfest chain (4.3.2)–(4.3.3).

(a) Is the Ehrenfest chain irreducible, aperiodic, recurrent, positive recurrent?
(b) Does this Markov chain admit a stationary distribution? *Hint*: Try a binomial distribution.
(c) Does this Markov chain admit a limiting distribution?

Exercise 7.5 Consider the Bernoulli-Laplace chain $(X_n)_{n\geq 0}$ of Exercise 4.9 with state space $\mathbb{S} = \{0, 1, 2, \ldots, N\}$ and transition matrix

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 & \cdots & \cdots & 0 & 0 \\ 1/N^2 & 2(N-1)/N^2 & (N-1)^2/N^2 & 0 & \cdots & \cdots & 0 & 0 \\ 0 & 2^2/N^2 & 4(N-2)/N^2 & (N-2)^2/N^2 & \cdots & \cdots & 0 & 0 \\ 0 & 0 & 3^2/N^2 & 0 & \cdots & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \ddots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \ddots & \vdots & \vdots & \vdots \\ & & & \vdots & & & & \\ 0 & 0 & \cdots & 0 & 3^2/N^2 & 0 & 0 & \\ 0 & 0 & \cdots & 0 & (N-2)^2/N^2 & 4(N-2)/N^2 & 2^2/N^2 & 0 \\ 0 & 0 & \cdots & 0 & 0 & (N-1)^2/N^2 & 2(N-1)/N^2 & 1/N^2 \\ 0 & 0 & \cdots & 0 & 0 & 0 & 1 & 0 \end{bmatrix},$$

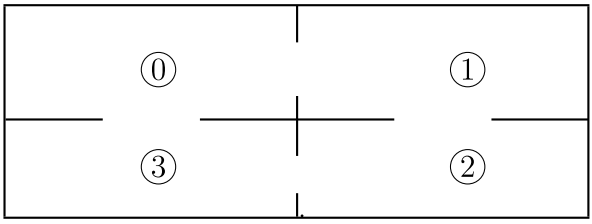
i.e.

$$P_{k,k-1} = \frac{k^2}{N^2}, \quad P_{k,k} = \frac{2k(N-k)}{N^2}, \quad P_{k,k+1} = \frac{(N-k)^2}{N^2}, \quad k = 1, 2, \ldots, N-1.$$

(a) Is the Bernoulli-Laplace chain irreducible, aperiodic, recurrent, positive recurrent?
(b) Does this Markov chain admit a stationary distribution?
(c) Does this Markov chain admit a limiting distribution?

Exercise 7.6 Consider a robot evolving in the following circular maze, moving from one room to a different room at each time step, according to a Markov chain with equal transition probabilities.

N. Privault



Let $X_n \in \{0, 1, 2, 3\}$ denote the state of the robot at time $n \in \mathbb{N}$.

(a) Write down the transition matrix P of the chain.
(b) By first step analysis, compute the mean **return** times $\mu_0(k)$ from state $k = 0, 1, 2, 3$ to state ⓪.

Hint: One may use the symmetry of the problem to simplify the calculations.
(c) Guess an invariant (or stationary) probability distribution $[\pi_0, \pi_1, \pi_2, \pi_3]$ for the chain, and show that it does satisfy the condition $\pi = \pi P$.

Exercise 7.7 A signal processor is analysing a sequence of signals that can be either distorted or non-distorted. It turns out that on average, 1 out of 4 signals following a distorted signal are distorted, while 3 out of 4 signals are non-distorted following a non-distorted signal.

(a) Let $X_n \in \{D, N\}$ denote the state of the $n\text{-}th$ signal being analysed by the processor. Show that the process $(X_n)_{n\geq 1}$ can be modeled as a Markov chain and determine its transition matrix.
(b) Compute the stationary distribution of $(X_n)_{n\geq 1}$.
(c) In the long run, what fraction of analysed signals are distorted?
(d) Given that the last observed signal was distorted, how long does it take on average until the next non-distorted signal?
(e) Given that the last observed signal was non-distorted, how long does it take on average until the next distorted signal?

Exercise 7.8 Consider the Markov chain $(X_n)_{n\geq 0}$ on the state space $\{0, 1, 2, 3\}$ with transition probability matrix P given by

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0.2 & 0 & 0.8 & 0 \\ 0.3 & 0 & 0.7 & 0 \\ 0.4 & 0.6 & 0 & 0 \end{bmatrix}.$$

(a) Draw the graph of this chain. Is the chain reducible?
(b) Find the recurrent, transient, and absorbing state(s) of this chain.
(c) Compute the fraction of time spent at state ⓪ in the long run.
(d) On the average, how long does it take to reach state ⓪ after starting from state ②?

Exercise 7.9 Consider the transition probability matrix

$$P = \left[P_{i,j} \right]_{0\leq i,j\leq 3} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0.1 & 0.4 & 0.2 & 0.3 \\ 0.2 & 0.2 & 0.5 & 0.1 \\ 0.3 & 0.3 & 0.4 & 0 \end{bmatrix}.$$

(a) Compute the limiting distribution $[\pi_0, \pi_1, \pi_2, \pi_3]$ of this Markov chain.
(b) Compute the average time $\mu_0(1)$ it takes to the chain to travel from state ① to state ⓪.

Hint: The data of the first row of the matrix P should play no role in the computation of $\mu_0(k)$, $k = 0, 1, 2, 3$.

(c) Prove by direct computation that the relation $\pi_0 = 1/\mu_0(0)$ holds, where $\mu_0(0)$ represents the mean return time to state $\textcircled{0}$ for this chain.

Exercise 7.10 Consider the Markov chain with transition probability matrix

$$\left[P_{i,j} \right]_{0 \le i,j \le 3} = \begin{bmatrix} 0 & 1/2 & 0 & 1/2 \\ 1/4 & 0 & 3/4 & 0 \\ 0 & 1/3 & 0 & 2/3 \\ 1/2 & 0 & 1/2 & 0 \end{bmatrix}.$$

(a) Show that the chain is periodic[3] and compute its period.
(b) Determine the stationary distribution of this chain.

Exercise 7.11 The lifetime of a given component of a machine is a discrete random variable T with distribution

$$\mathbb{P}(T = 1) = 0.1, \quad \mathbb{P}(T = 2) = 0.2, \quad \mathbb{P}(T = 3) = 0.3, \quad \mathbb{P}(T = 4) = 0.4.$$

The component is immediately replaced with a new component upon failure, and the machine starts functioning with a new component. Compute the long run probability of finding the machine about to fail at the next time step.

Exercise 7.12 Suppose that a Markov chain has the one-step transition probability matrix P on the state space $\{A, B, C, D, E\}$ given by

$$P = \begin{bmatrix} 0.6 & 0.4 & 0 & 0 & 0 \\ 0.3 & 0.7 & 0 & 0 & 0 \\ 0.2 & 0 & 0.4 & 0 & 0.4 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

Find $\lim_{n\to\infty} \mathbb{P}(X_n = A \mid X_0 = C)$.

Exercise 7.13 Consider the Markov chain $(X_n)_{n\ge 0}$ on the state space $\{0, 1, 2, 3, 4\}$ with transition probability matrix P given by

$$P = \begin{bmatrix} 1/3 & 2/3 & 0 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1/7 & 6/7 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

[3] A chain is periodic when all states have the same period

(a) Draw the graph of this chain.
(b) Identify the communicating class(es).
(c) Find the recurrent, transient, and absorbing state(s) of this chain.
(d) Find $\lim_{n\to\infty} \mathbb{P}(X_n = 0 \mid X_0 = 4)$.

Exercise 7.14 Three out of 4 trucks passing under a bridge are followed by a car, while only 1 out of every 5 cars passing under that same bridge is followed by a truck. Let $X_n \in \{C, T\}$ denote the nature of the n-th vehicle passing under the bridge, $n \geq 1$.

(a) Show that the process $(X_n)_{n\geq 1}$ can be modeled as a Markov chain, and write down its transition matrix.
(b) Compute the stationary distribution of $(X_n)_{n\geq 1}$.
(c) In the long run, what fraction of vehicles passing under the bridge are trucks?
(d) Given that the last vehicle seen was a truck, how long does it take on average until the next truck is seen under that same bridge?

Exercise 7.15 Consider a discrete-time Markov chain $(X_n)_{n\geq 0}$ on $\mathbb{S} = \{1, 2, \ldots, N\}$, whose transition matrix $P = (P_{i,j})_{1\leq i,j\leq N}$ is assumed to be *symmetric*, i.e. $P_{i,j} = P_{j,i}$, $1 \leq i, j \leq N$,

(a) Find an invariant (or stationary) distribution of the chain. *Hint*: The equation $\pi P = \pi$ admits an easy solution.
(b) Assume further that $P_{i,i} = 0$, $1 \leq i \leq N$, and that $P_{i,j} > 0$ for all $1 \leq i < j \leq N$. Find the period of every state.

Exercise 7.16 (Gusev 2014, Problem 5.22-(d) continued)

(a) Is the chain $(Y_k)_{k\geq 0}$ reducible? Find its communicating classes.
(b) Find the limiting distribution, and the possible stationary distributions of the chain $(Y_k)_{k\geq 0}$.

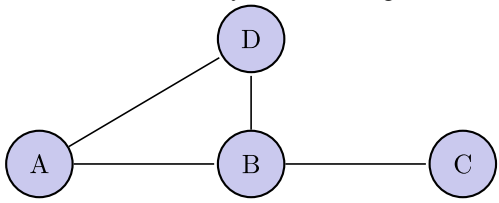
Exercise 7.17 Consider the Markov chain with transition matrix

$$\begin{bmatrix} q & p & 0 & 0 & 0 \\ q & 0 & p & 0 & 0 \\ q & 0 & 0 & p & 0 \\ q & 0 & 0 & 0 & p \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix},$$

where $p, q \in (0, 1)$ satisfy $p + q = 1$.

(a) Compute the stationary distribution $[\pi_0, \pi_1, \pi_2, \pi_3, \pi_4]$ of this chain.
(b) Compute the limiting distribution of the chain.

Exercise 7.18 Four players A, B, C, D are connected by the following network,



and play by exchanging a token.
At each step of the game, the player who holds the token chooses another player he is connected to, and sends the token to that player.

(a) Assuming that the player choices are made at random and are equally distributed, model the states of the token as a Markov chain $(X_n)_{n\geq 1}$ on $\{A, B, C, D\}$ and give its transition matrix.
(b) Compute the stationary distribution $[\pi_A, \pi_B, \pi_C, \pi_D]$ of $(X_n)_{n\geq 1}$.

Hint: To simplify the resolution, start by arguing that we have $\pi_A = \pi_D$.
(c) Compute the mean return times $\mu_D(i)$, $i \in \{A, B, C, D\}$. On average, how long does player D have to wait to recover the token ?
(d) In the long run, what is the probability that player D holds the token?

Exercise 7.19 Consider a Markov chain with transition probability matrix

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ a & b & c \end{bmatrix},$$

with $a + b + c = 1$.

(a) Compute the matrix power P^n for all $n \geq 2$.
(b) Does this Markov chain admit a limiting distribution? If yes, compute that distribution.
(c) Does this Markov chain admit a stationary distribution? Compute this distribution if it exists.

Exercise 7.20 Consider a game server that can become offline with probability p and can remain online with probability $q = 1 - p$ on any given day. Assume that the random time N it takes to fix the server has the geometric distribution

$$\mathbb{P}(N = k) = \beta(1 - \beta)^{k-1}, \qquad k \geq 1,$$

with parameter $\beta \in (0, 1)$. We let $X_n = 1$ when the server is online on day n, and $X_n = 0$ when it is offline.

(a) Show that the process $(X_n)_{n\geq 0}$ can be modeled as a discrete-time Markov chain, and write down its transition matrix.
(b) Compute the probability that the server is online in the long run, in terms of the parameters β and p.

Exercise 7.21 Let $(X_n)_{n\geq 0}$ be an irreducible aperiodic Markov chain on the finite state space $\mathbb{S} = \{1, 2, \ldots, N\}$.

(a) Show that there *exists* a state $i \in \{1, 2, \ldots, N\}$ such that the mean return time $\mu_i(i)$ from state i to itself is lower or equal to N, i.e. $\mu_i(i) \leq N$.
(b) Show that there *exists* a state $i \in \{1, 2, \ldots, N\}$ such that the mean return time $\mu_i(i)$ from state i to itself is higher or equal to N, i.e. $\mu_i(i) \geq N$.

Exercise 7.22 Consider a Markov chain on the state space $\{1, 2, \ldots, N\}$. For any $i \in \{2, \ldots, N-1\}$, the chain has probability $p \in (0, 1)$ to switch from state $\textcircled{i}$ to state $\boxed{i+1}$, and probability $q = 1 - p$ to switch from $\textcircled{i}$ to $\boxed{i-1}$. When the chain reaches state $\textcircled{i}=\ \textcircled{1}$ it rebounds to state $\textcircled{2}$ with probability p or stays at state $\textcircled{1}$ with probability q. Similarly, after reaching state $\textcircled{N}$ it rebounds to state or remains at N with probability p.

(a) Write down the transition probability matrix of this chain.
(b) Is the chain reducible?
(c) Determine the absorbing, transient, recurrent, and positive recurrent states of this chain.
(d) Compute the stationary distribution of this chain.
(e) Compute the limiting distribution of this chain.

Exercise 7.23 (Problem 6.9 continued) Let $\alpha > 0$ and consider the Markov chain with state space $\mathbb{S} = \mathbb{N}$ and transition matrix given by

$$P_{i,i-1} = \frac{1}{\alpha+1}, \quad P_{i,i+1} = \frac{\alpha}{\alpha+1}, \qquad i \geq 1.$$

and a reflecting barrier at 0, such that $P_{0,1} = 1$.

(a) Show that when $\alpha \in (0, 1)$ this chain admits a stationary distribution of the form

$$\pi_k = \alpha^{k-1}(1-\alpha^2)/2, \qquad k \geq 1,$$

where the value of π_0 has to be determined.
(b) Does this Markov chain admit a stationary distribution when $\alpha \geq 1$?
(c) Show that this Markov chain is positive recurrent when $\alpha < 1$.

Exercise 7.24 Consider two discrete-time stochastic processes $(X_n)_{n\geq 0}$ and $(Y_n)_{n\geq 0}$ on a state space $\mathbb{S}$, such that

$$X_n = Y_n, \qquad n \geq \tau,$$

where τ is a random time called the *coupling time* of $(X_n)_{n\geq 0}$ and $(Y_n)_{n\geq 0}$.

(a) Show that for all $x \in \mathbb{S}$ and $n \in \mathbb{N}$ we have

$$\mathbb{P}(X_n = x) \leq \mathbb{P}(Y_n = x) + \mathbb{P}(\tau > n) \qquad x \in \mathbb{S}, \quad n \geq 0.$$

Hint: Use the law of total probability as

$$\mathbb{P}(A) = \mathbb{P}(A \cap \{\tau \leq n\}) + \mathbb{P}(A \cap \{\tau > n\}).$$

(b) Show that for all $n \in \mathbb{N}$ we have

$$\sup_{x\in\mathbb{S}} |\mathbb{P}(X_n = x) - \mathbb{P}(Y_n = x)| \leq \mathbb{P}(\tau > n), \qquad n \geq 0.$$

Problem 7.25 *Time reversibility* is a fundamental issue in physics as it is akin to the idea of "traveling backward in time". This problem studies the reversibility of Markov chains, and applies it to the computation of stationary and limiting distributions. Given $N \geq 1$ and $(X_k)_{k=0,1,\ldots,N}$ a Markov chain with transition matrix P on a state space $\mathbb{S}$, we let

$$Y_k := X_{N-k}, \qquad k = 0, 1, \ldots, N,$$

denote the *time reversal* of $(X_k)_{k=0,1,\ldots,N}$.

(a) Assume that X_k has same distribution $\pi = (\pi_i)_{i\in\mathbb{S}}$ for every $k = 0, 1, \ldots, N$, i.e.

$$\mathbb{P}(X_k = i) = \pi_i, \qquad i \in \mathbb{S}, \quad k = 0, 1, \ldots, N.$$

Show that the process $(Y_k)_{k=0,1,\ldots,N}$ is a (forward) Markov chain (i.e. $(X_k)_{k=0,1,\ldots,N}$ has the backward Markov property), and find the transition matrix

$$[\mathbb{P}(Y_{n+1} = j \mid Y_n = i)]_{i,j\in\mathbb{S}}$$

in terms of P and π.

Hint: Use the basic definition of conditional probabilities to compute

$$[\mathbb{P}(Y_{n+1} = j \mid Y_n = i)]_{i,j\in\mathbb{S}},$$

and then show that $(Y_n)_{n=0,1,\ldots,N}$ has the Markov property

$$\mathbb{P}(Y_{n+1} = j \mid Y_n = i_n, \ldots, Y_0 = i_0) = \mathbb{P}(Y_{n+1} = j \mid Y_n = i_n).$$

(b) We say that $(X_k)_{k=0,1,\ldots,N}$ is *reversible* for π when $(X_k)_{k=0,1,\ldots,N}$ and $(Y_k)_{k=0,1,\ldots,N}$ have same transition probabilities.

Write down this reversibility condition in terms of P and π. From now on, we refer to that condition as the *detailed balance condition*, which can be stated independently of N.

Hint: By "same transition probabilities" we mean

$$\mathbb{P}(X_{n+1} = j \mid X_n = i) = \mathbb{P}(Y_{n+1} = j \mid Y_n = i).$$

(c) Show that if $(X_k)_{k\geq 0}$ is *reversible* for π, then π is also a stationary distribution for $(X_k)_{k\geq 0}$.

Hint: The fact that $\sum_i P_{j,i} = 1$ plays a role here.

(d) Show that if an irreducible positive recurrent aperiodic Markov chain is reversible for its stationary distribution π then we have

$$P_{k_1,k_2} P_{k_2,k_3} \cdots P_{k_{n-1},k_n} P_{k_n,k_1} = P_{k_1,k_n} P_{k_n,k_{n-1}} \cdots P_{k_3,k_2} P_{k_2,k_1} \tag{7.3.4}$$

for all sequences $\{k_1, k_2, \ldots, k_n\}$ of states and $n \geq 2$.

Hint: This is a standard algebraic manipulation.

(e) Show that conversely, if an irreducible positive recurrent aperiodic Markov chain satisfies Condition (7.3.4) for all sequences $\{k_1, k_2, \ldots, k_n\}$ of states, $n \geq 2$, then it is reversible for its stationary distribution π.

Hint: This question is more difficult and here you need to apply Theorem 7.8.

(f) From now on, we assume that $\mathbb{S} = \{0, 1, \ldots, M\}$ and that P is the transition matrix

$$P_{i,i+1} = \frac{1}{2} - \frac{i}{2M}, \qquad P_{i,i} = \frac{1}{2}, \qquad P_{i,i-1} = \frac{i}{2M}, \qquad 1 \leq i \leq M-1,$$

of the *modified* Ehrenfest chain, with $P_{0,0} = P_{0,1} = P_{M,M-1} = P_{M,M} = 1/2$.

Find a probability distribution π for which the chain $(X_k)_{k\geq 0}$ is reversible.

Hint: The reversibility condition will yield a relation that can be used to compute π by induction. Remember to make use of the condition $\sum_i \pi_i = 1$.

(g) Confirm that the distribution of Question (f) is invariant (or stationary) by checking explicitly that the equality

$$\pi = \pi P$$

does hold.

(h) Show, by quoting the relevant theorem, that π is also the limiting distribution of the *modified* Ehrenfest chain $(X_k)_{k\geq 0}$.

(i) Show, by the result of Question (h), that

$$\lim_{n\to\infty} \mathbb{E}[X_n \mid X_0 = i] = \frac{M}{2},$$

for all $i = 0, 1, \ldots, M$.

(j) Show, using first step analysis and induction on $n \geq 0$, that we have

$$\mathbb{E}\left[X_n - \frac{M}{2} \,\middle|\, X_0 = i\right] = \left(i - \frac{M}{2}\right)\left(1 - \frac{1}{M}\right)^n, \qquad n \geq 0,$$

for all $i = 0, 1, \ldots, M$, and that this relation can be used to recover the result of Question (i).

Hint: Letting

$$h_n(i) = \mathbb{E}\left[X_n - \frac{M}{2} \,\middle|\, X_0 = i\right], \qquad n \geq 0,$$

in order to prove the formula by induction one has to

(i) show that the formula holds for $h_0(i)$ when $n = 0$;

(ii) show that assuming that the formula holds for $h_n(i)$, it then holds for $h_{n+1}(i)$.

It can help to start by proving the formula for $h_1(i)$ when $n = 1$ by first step analysis.

Problem 7.26 Meta search engines (Schalekamp and van Zuylen 2009). A **meta search engine** tries to provide a single optimized ranking of search results $\{a, b, c, d, e\}$ based on the outputs of *4* different search engines denoted $\mathcal{S}_1$, $\mathcal{S}_2$, $\mathcal{S}_3$, $\mathcal{S}_4$, a technique known as *rank aggregation*.

Five possible search results a, b, c, d, e have been respectively ranked as

Rank	$\mathcal{S}_1$	$\mathcal{S}_2$	$\mathcal{S}_3$	$\mathcal{S}_4$
1	b	c	d	e
2	c	b	e	a
3	d	d	a	d
4	a	e	b	b
5	e	a	c	c

by the Search Engines $\mathcal{S}_1$, $\mathcal{S}_2$, $\mathcal{S}_3$, $\mathcal{S}_4$.

(a) A state $y \in \{a, b, c, d, e\}$ is said to be better ranked than another state $x \in \{a, b, c, d, e\}$, and we write $x \preceq y$ and $y \succeq x$ if y ranks higher than x for **at least three** of the four search engines. We also write "$x \not\preceq y$" when neither "$x \preceq y$" nor "$x \succeq y$" is satisfied.

Complete the ranking table

$\preceq$	a	b	c	d	e
a	=				
b		=			
c			=		
d				=	
e					=

with "$x \preceq y$", "$x \succeq y$" or "$x \not\preceq y$" at the positions (x, y), $x, y \in \{a, b, c, d, e\}$. The diagonal entries are not relevant.

(b) The meta search engine works by constructing a self-improving random sequence $(X_n)_{n\geq 0}$ which is supposed to "converge" to the best possible search result based on the data of the four rankings.
Given a search result $X_n = x \in \{a, b, c, d, e\}$, we choose the next search result X_{n+1} by assigning probability $1/5$ to each of the search results that are *better ranked* than x. If no search result is better than x, then we keep $X_{n+1} = x$.
What is the state space of the sequence $(X_n)_{n\geq 0}$?
Model the process $(X_n)_{n\geq 0}$ as a Markov chain, and complete the transition matrix

$$P = \begin{bmatrix} 3/5 & \square & \square & \square & \square \\ 0 & \square & \square & \square & \square \\ \square & 1/5 & \square & \square & \square \\ \square & \square & \square & \square & 0 \\ \square & \square & \square & 1/5 & \square \end{bmatrix}$$

(c) Draw the graph of the chain $(X_n)_{n\geq 0}$.
Is the chain $(X_n)_{n\geq 0}$ reducible?

(d) Does the Markov chain $(X_n)_{n\geq 0}$ admit a limiting distribution independent of its initial state?

(e) Does the Markov chain $(X_n)_{n\geq 0}$ admit a stationary distribution? Find all possible stationary distribution(s) of the chain $(X_n)_{n\geq 0}$

(f) In PageRank™-type algorithms, one typically chooses to perturb the transition matrix P into the new matrix

$$\widetilde{P} := \frac{\varepsilon}{n} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix} + (1 - \varepsilon) P,$$

with $n = 5$ here, and $\varepsilon \in (0, 1)$. Show that $\widetilde{P}$ is a Markov transition matrix and that the corresponding chain $(\widetilde{X}_n)_{n\geq 1}$ is irreducible and aperiodic.

(g) Show that $\widetilde{P}$ admits a stationary distribution $\tilde{\pi}$ that satisfies

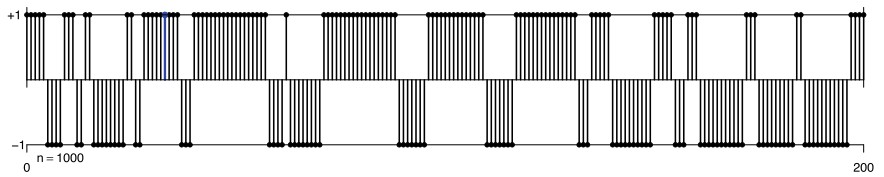


Fig. 7.6 Simulation of the voter model with $N = 199$, $p = 0.98$, and $z_0 = z_{N+1} = +1$

$$\tilde{\pi} = \left[\frac{\varepsilon}{5}, \frac{\varepsilon}{5}, \frac{\varepsilon}{5}, \frac{\varepsilon}{5}, \frac{\varepsilon}{5}\right] + (1-\varepsilon)\tilde{\pi}P,$$

and that all probabilities in $\tilde{\pi}$ are greater than $\varepsilon/5$.

(h) Compute the stationary distribution of $\widetilde{P}$.

(i) Provide a ranking of the states $\{a, b, c, d, e\}$ based on the stationary distribution $\tilde{\pi}$.

(j) Compute the mean return times for $\widetilde{P}$, and show that they are all below $5/\epsilon$.

Problem 7.27 (Agapie and Höns 2007) The voter modelis a particular case of the Ising model. This model has applications to spatial statistics, image analysis and segmentation, opinion studies, urban segregation, language change, metal alloys, magnetic materials, liquid/gas coexistence, phase transitions, plasmas, cell membranes in biophysics, ...

In dimension one, the model is built on the state space $\mathbb{S} := \{-1, +1\}^N$ made of elements $z = (z_k)_{1\le k\le N} \in \mathbb{S}$ whose components $z_k \in \{-1, 1\}$, $k = 1, 2, \ldots N$, are called *spins*.

We consider a Markov chain $(Z_n)_{n\ge 0}$ *on the state space* $\mathbb{S} = \{-1, +1\}^N$, whose transitions from an initial configuration $Z_0 = z = (z_k)_{1\le k\le N}$ to a new configuration $Z_1 = \tilde{z} = (\tilde{z}_k)_{1\le k\le N}$ are defined as follows (Fig. 7.6):

First, choose one component z_k in $z = (z_k)_{1\le k\le N}$ with probability $1/N$, $k = 1, 2, \ldots, N$, and then consider the following cases:

(i) if $(z_{k-1}, z_{k+1}) = (-1, +1)$ or $(z_{k-1}, z_{k+1}) = (+1, -1)$: $\Rightarrow$ flip the sign of z_k, i.e. set $\tilde{z}_k := \pm z_k$ with probability $1/2$,

(ii) if $(z_{k-1}, z_{k+1}) = (+1, +1)$:
$\Rightarrow$ set $\tilde{z}_k := +1$ with probability $p > 0$, and $\tilde{z}_k := -1$ with probability $q > 0$.

(iii) if $(z_{k-1}, z_{k+1}) = (-1, -1)$:
$\Rightarrow$ set $\tilde{z}_k := -1$ with probability $p > 0$, and $\tilde{z}_k := +1$ with probability $q > 0$,

where $p + q = 1$. The probabilities p and q can be respectively viewed as the probabilities of "agreeing", resp. "disagreeing" with two neighbors sharing the same opinion. The boundary conditions z_0 and z_{N+1} are arbitrarily fixed, and the corresponding instructions can be coded in R as follows:

```
if (z[k-1]!=z[k+1]) z[k]=sample(c(-1,1), 1,prob=c(0.5,0.5))
if (z[k-1]==1 && z[k+1]==1) z[k]=sample(c(-1,1), 1, prob=c(q,p))
if (z[k-1]==-1 && z[k+1]==-1) z[k]=sample(c(-1,1), 1, prob=c(p,q))
```

(a) Find the cardinality of the state space $\mathbb{S}$.

(b) Replace the question marks (?) below with the corresponding transition probabilities given that $Z_0 = z = (z_k)_{1 \le k \le N}$:

(i) if $(z_{k-1}, z_k, z_{k+1}) = (-1, \pm 1, +1)$ or $(z_{k-1}, z_k, z_{k+1}) = (+1, \pm 1, -1)$,

$$\mathbb{P}(Z_1 = (z_1, \ldots, z_{k-1}, -z_k, z_{k+1}, \ldots, z_N) \mid Z_0 = z) := (?),$$

$k = 1, 2, \ldots, N$,

(ii) if $(z_{k-1}, z_k, z_{k+1}) = (+1, +1, +1)$ or $(z_{k-1}, z_k, z_{k+1}) = (-1, -1, -1)$,

$$\mathbb{P}(Z_1 = (z_1, \ldots, z_{k-1}, -z_k, z_{k+1}, \ldots, z_N) \mid Z_0 = z) := (?),$$

$k = 1, 2, \ldots, N$,

(iii) if $(z_{k-1}, z_k, z_{k+1}) = (+, -1, +1)$ or $(z_{k-1}, z_k, z_{k+1}) = (-1, +1, -1)$,

$$\mathbb{P}(Z_1 = (z_1, \ldots, z_{k-1}, -z_k, z_{k+1}, \ldots, z_N) \mid Z_0 = z) := (?),$$

$k = 1, 2, \ldots, N$.

(c) Show that, based on the three above cases (i)–(ii)–(iii), the transition probabilities of $(Z_n)_{n \ge 0}$ take the general form

$$\mathbb{P}(Z_1 = \bar{z}^k \mid Z_0 = z) = \frac{1}{N(1 + (p/q)^{z_k(z_{k-1}+z_{k+1})/2})}, \qquad k = 1, 2, \ldots, N, \tag{7.3.5}$$

where $z = (z_1, \ldots, z_N)$ and

$$\bar{z}^k := (z_1, \ldots, z_{k-1}, -z_k, z_{k+1}, \ldots, z_N) \tag{7.3.6}$$

denotes the state $z \in \mathbb{S}$ after flipping its kth component z_k.

(d) Compute $\mathbb{P}(Z_1 = z \mid Z_0 = z)$ for all $z \in \mathbb{S}$ using the complement rule, Relation (7.3.5), and the law of total probability.

(e) Taking $N = 3$ and setting $z_0 = z_4 = -1$, i.e. $(z_0, z_1, z_2, z_3, z_4)$ takes the form $(-, \pm, \pm, \pm, -)$, complete the missing entries following transition probability matrix P of $(Z_n)_{n \ge 0}$ on the state space $\mathbb{S} = \{- - -, - - +, - + -, - - +, + - -, + - +, + + -, + + +\}$:

$$
\begin{array}{c}
 \\
---\\
--+\\
-+-\\
-++\\
+--\\
+-+\\
++-\\
+++
\end{array}
\begin{array}{c}
\begin{array}{cccccccc}
--- & --+ & -+- & -++ & +-- & +-+ & ++- & +++
\end{array}\\
\left[\begin{array}{cccccccc}
(?) & (?) & q/3 & 0 & q/3 & 0 & 0 & 0\\
(?) & 1/2 & 0 & (?) & 0 & q/3 & 0 & 0\\
(?) & 0 & (1+q)/3 & (?) & 0 & 0 & 1/6 & 0\\
0 & 1/6 & (?) & (?) & 0 & 0 & 0 & 1/6\\
p/3 & 0 & 0 & 0 & 1/2 & (?) & (?) & 0\\
0 & p/3 & 0 & 0 & p/3 & q & 0 & p/3\\
0 & 0 & 1/6 & 0 & (?) & 0 & 1/2 & (?)\\
0 & 0 & 0 & 1/6 & 0 & q/3 & 1/6 & (?)
\end{array}\right]
\end{array}.
$$

Is the chain $(Z_n)_{n\geq 0}$ reducible? Why?

(f) Does the Markov chain $(Z_n)_{n\geq 0}$ admit a limiting distribution? A stationary distribution? Why?

(g) Show that if a probability distribution $(\pi_z)_{z\in\mathbb{S}}$ on $\mathbb{S}$ satisfies the relation

$$
\frac{\pi_{\bar{z}^k}}{\pi_z} = \frac{\mathbb{P}(Z_1 = \bar{z}^k \mid Z_0 = z)}{\mathbb{P}(Z_1 = z \mid Z_0 = \bar{z}^k)}, \qquad k = 1, 2, \ldots, N, \quad z \in \mathbb{S}, \tag{7.3.7}
$$

where $\bar{z}^k$ is defined in (7.3.6), then $(\pi_z)_{z\in\mathbb{S}}$ is a *stationary distribution* for the chain $(Z_n)_{n\geq 0}$, i.e. we have

$$
\Big(\mathbb{P}(Z_0 = z) = \pi_z,\ \forall\, z \in \mathbb{S}\Big) \quad \Longrightarrow \quad \Big(\mathbb{P}(Z_1 = z) = \pi_z,\ \forall\, z \in \mathbb{S}\Big).
$$

Hint: Start from the law of total probability

$$
\begin{aligned}
\mathbb{P}(Z_1 = z) &= \mathbb{P}(Z_1 = z \mid Z_0 = z)\mathbb{P}(Z_0 = z)\\
&\quad + \sum_{k=1}^{N} \mathbb{P}(Z_1 = z \mid Z_0 = \bar{z}^k)\mathbb{P}(Z_0 = \bar{z}^k),
\end{aligned}
$$

and show, using (7.3.7), that the above equals π_z if $\mathbb{P}(Z_0 = z) = \pi_z$ for all $z \in \mathbb{S}$.

(h) Show that

$$
\frac{\mathbb{P}(Z_1 = \bar{z}^k \mid Z_0 = z)}{\mathbb{P}(Z_1 = z \mid Z_0 = \bar{z}^k)} = \left(\frac{q}{p}\right)^{z_k(z_{k-1}+z_{k+1})/2}, \qquad k = 1, 2, \ldots, N, \quad z \in \mathbb{S}. \tag{7.3.8}
$$

Hint: Use Eq. (7.3.5) and the relations

$$
\frac{1 + (q/p)}{1 + (p/q)} = \frac{q}{p} \quad \text{and} \quad \frac{1 + (p/q)}{1 + (q/p)} = \frac{p}{q}.
$$

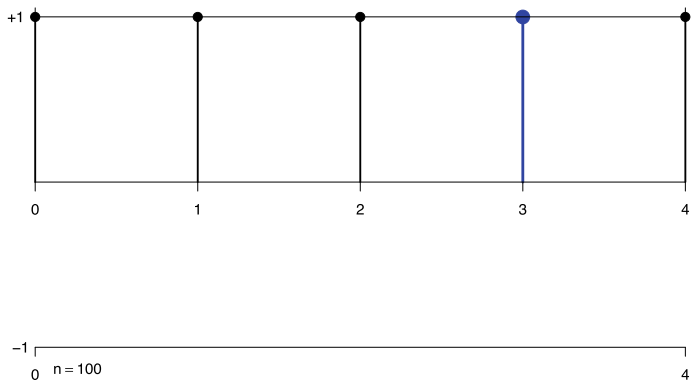


Fig. 7.7 Simulation of the voter model with $N = 3$, $p = \sqrt{0.75}$, and $z_0 = z_4 = +1$

(i) Show that the probability distribution $(\pi_z)_{z\in\mathbb{S}}$ defined as

$$\pi_z := C_\beta \exp\left(\beta \sum_{l=0}^{N} z_l z_{l+1}\right), \qquad z \in \mathbb{S}, \tag{7.3.9}$$

is the stationary and limiting distribution of $(Z_n)_{n\geq 0}$, where C_β is a normalization constant[4] and β is to be given in terms of p and q.
Hint: Using (7.3.8), show that $(\pi_z)_{z\in\mathbb{S}}$ defined in (7.3.9) satisfies (7.3.7).

(j) Taking $N = 3$ and $z_0 = z_4 = +1$, i.e. $(z_0, z_1, z_2, z_3, z_4)$ takes the form $(+, \pm, \pm, \pm, +)$, compute the limiting distribution of each of the 8 configurations in

$$\mathbb{S} = \{---, --+, -+-, --+, +--, +-+, ++-, +++\},$$

and find the value of C_β (Fig. 7.7).

[4] The normalization constant C_β does not have to be computed here.

Chapter 8
Branching Processes



Branching processes are used as a tool for modeling in genetics, biomolecular reproduction, population growth, genealogy, disease spread, photomultiplier cascades, nuclear fission, earthquake triggering, queueing models, viral phenomena, social networks, neuroscience, etc. This chapter mainly deals with the computation of probabilities of extinction and explosion in finite time for branching processes.

8.1 Construction and Examples

Consider a time-dependent population made of a number X_n of individuals at generation $n \geq 0$. In the branching process model, each of these X_n individuals may have a random number of descendants born at time $n+1$.

For each $k = 1, 2, \ldots, X_n$ we let Y_k denote the number of descendants of individual n^o k. That means, we have $X_0 = 1$, $X_1 = Y_1$, and at time $n+1$, the new population size X_{n+1} will be given by

$$X_{n+1} = Y_1 + \cdots + Y_{X_n} = \sum_{k=1}^{X_n} Y_k, \tag{8.1.1}$$

where the $(Y_k)_{k\geq 1}$ form a sequence of independent, identically distributed, nonnegative integer valued random variables which are assumed to be almost surely finite, i.e.

$$\mathbb{P}(Y_k < \infty) = \sum_{n\geq 0} \mathbb{P}(Y_k = n) = 1.$$

Note that new independent samples of $(Y_k)_{k\geq 1}$ are generated at each generation, and the X_n individuals of generation n "die" at time $n+1$ as they are not considered in the sum (8.1.1). In order to keep them at the next generation we would have to

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N. Privault, *Understanding Markov Chains*, Springer Undergraduate Mathematics Series,
https://doi.org/10.1007/978-981-95-4057-0_8

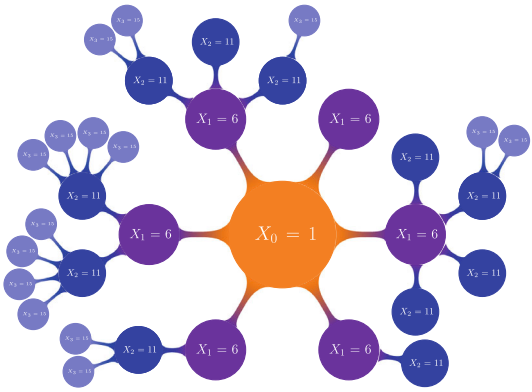


Fig. 8.1 Sample of a branching process

modify (8.1.1) into the *progeny*

$$X_{n+1} = X_n + Y_1 + \cdots + Y_{X_n},$$

however, we will *not* adopt this convention, and we will rely on (8.1.1) instead.

As a consequence of (8.1.1), the branching process $(X_n)_{n\geq 0}$ is a Markov process with state space $\mathbb{S} = \mathbb{N}$ and transition matrix given by

$$P = \big[P_{i,j}\big]_{i,j\geq 0} = \begin{bmatrix} 1 & 0 & 0 & \cdots \\ \mathbb{P}(Y_1 = 0) & \mathbb{P}(Y_1 = 1) & \mathbb{P}(Y_1 = 2) & \cdots \\ P_{2,0} & P_{2,1} & P_{2,2} & \cdots \\ P_{3,0} & P_{3,1} & P_{3,2} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}. \tag{8.1.2}$$

Note that state $\textcircled{0}$ is absorbing since by construction we always have

$$P_{0,0} = \mathbb{P}(X_{n+1} = 0 \mid X_n = 0) = 1, \qquad n \geq 0.$$

Figure 8.1 represents an example of branching process with $X_0 = 1$ and $Y_1 = 6$, hence

$$X_1 = Y_{X_0} = Y_1 = 6,$$

then successively

$$(Y_k)_{k=1,2,\ldots,X_1} = (Y_1, Y_2, Y_3, Y_4, Y_5, Y_6) = (0, 4, 1, 1, 2, 3)$$

and

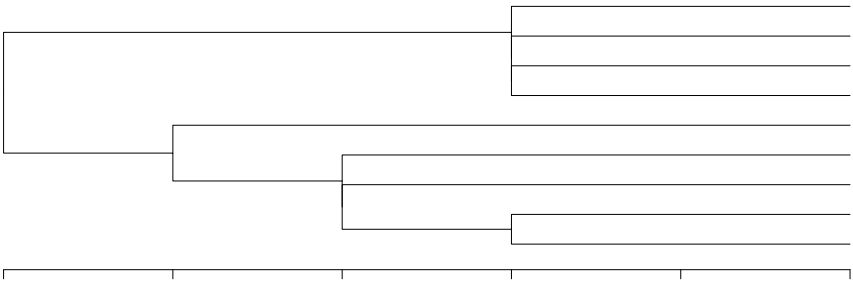


Fig. 8.2 Sample graph of a branching process

$$\begin{aligned} X_2 &= Y_1 + \cdots + Y_{X_1} \\ &= Y_1 + Y_2 + Y_3 + Y_4 + Y_5 + Y_6 \\ &= 0 + 4 + 1 + 1 + 2 + 3 \\ &= 11, \end{aligned}$$

then

$$\begin{aligned} (Y_k)_{k=1,2,\ldots,X_2} &= (Y_1, Y_2, Y_3, Y_4, Y_5, Y_6, Y_7, Y_8, Y_9, Y_{10}, Y_{11}) \\ &= (0, 2, 0, 0, 0, 2, 4, 4, 2, 0, 1), \end{aligned}$$

and

$$\begin{aligned} X_3 &= Y_1 + \cdots + Y_{X_2} \\ &= Y_1 + Y_2 + Y_3 + Y_4 + Y_5 + Y_6 + Y_7 + Y_8 + Y_9 + Y_{10} + Y_{11} \\ &= 0 + 2 + 0 + 0 + 0 + 2 + 4 + 4 + 2 + 0 + 1 \\ &= 15. \end{aligned}$$

Figure 8.2 presents another sample tree for the path of a branching process.

In Fig. 8.2 the branching process starts from $X_0 = 2$, with $X_1 = 3$, $X_2 = 5$, $X_3 = 9$, $X_4 = 9$, $X_5 = 9$. However, in the sequel and except if otherwise specified, all branching processes will start from $X_0 = 1$.

See Sellke et al. (2008) and Iribarren and Moro (2011) for results on the modeling of the offspring distribution of Y_1 based on social network and internet data and the use of power tail distributions. The use of power tail distributions leads to probability generating functions of polylogarithmic form.

8.2 Probability Generating Functions

Let now $G_1(s)$ denote the probability generating function of $X_1 = Y_1$, defined as

$$G_1(s) := \mathbb{E}\left[s^{X_1} \middle| X_0 = 1\right] = \mathbb{E}\left[s^{Y_1}\right] = \sum_{k\geq 0} s^k \mathbb{P}(Y_1 = k), \quad -1 \leq s \leq 1,$$

cf. (1.7.3), denote the probability generating function of the (almost surely finite) random variable $X_1 = Y_1$, with

$$\begin{cases} G_1(0) = \mathbb{P}(Y_1 = 0), \\ G_1(1) = \sum_{n\geq 0} \mathbb{P}(Y_1 = n) = \mathbb{P}(Y_1 < \infty) = 1, \\ \mu := G_1'(1^-) = \sum_{k\geq 0} k\mathbb{P}(Y_1 = k) = \mathbb{E}[X_1 \mid X_0 = 1] = \mathbb{E}[Y_1]. \end{cases} \quad (8.2.1a)$$

More generally, letting $G_n(s)$ denote the probability generating function of X_n, defined as

$$G_n(s) := \mathbb{E}\left[s^{X_n} \mid X_0 = 1\right] = \sum_{k\geq 0} s^k \mathbb{P}(X_n = k \mid X_0 = 1), \quad -1 \leq s \leq 1,$$

$n \in \mathbb{N}$, we have

$$\begin{cases} G_0(s) = s, \qquad -1 \leq s \leq 1, \\ G_n(0) = \mathbb{P}(X_n = 0 \mid X_0 = 1), \qquad n \geq 0, & (8.2.2a) \\ \mu_n := \mathbb{E}[X_n \mid X_0 = 1] = G_n'(1^-) = \sum_{k\geq 0} k\mathbb{P}(X_n = k \mid X_0 = 1), & (8.2.2b) \end{cases}$$

cf. (1.7.5). When $X_0 = k$ we can view the branching tree as the union of k independent trees started from $X_0 = 1$ and we can write X_n as the sum of independent random variables

$$X_n = \sum_{l=1}^{k} X_n^{(l)}, \qquad n \geq 0,$$

where $X_n^{(l)}$ denotes the size of the tree $n^o l$ at time n, with $X_n^{(l)} = 1$, $l = 1, 2, \ldots, k$. In this case, we have

$$\begin{aligned}\mathbb{E}\left[s^{X_n} \mid X_0 = k\right] &= \mathbb{E}\left[s^{X_n^{(1)}+\cdots+X_n^{(k)}} \;\middle|\; X_0^{(1)} = 1, \ldots, X_0^{(k)} = 1\right] \\ &= \prod_{l=1}^{k} \mathbb{E}\left[s^{X_n^{(l)}} \mid X_0^{(l)} = 1\right] \\ &= \left(\mathbb{E}\left[s^{X_n} \mid X_0 = 1\right]\right)^k \\ &= (G_n(s))^k, \quad -1 \le s \le 1, \quad n \in \mathbb{N}.\end{aligned}$$

The next proposition provides an algorithm for the computation of the probability generating function G_n.

Proposition 8.1 *We have the recurrence relation*

$$G_{n+1}(s) = G_n(G_1(s)) = G_1(G_n(s)), \quad -1 \le s \le 1, \qquad n \ge 0. \tag{8.2.3}$$

Proof By the identity (1.6.11) on random products we have

$$\begin{aligned}G_{n+1}(s) &= \mathbb{E}\left[s^{X_{n+1}} \mid X_0 = 1\right] \\ &= \mathbb{E}\left[s^{Y_1+\cdots+Y_{X_n}} \mid X_0 = 1\right] \\ &= \mathbb{E}\left[\prod_{l=1}^{X_n} s^{Y_l} \;\middle|\; X_0 = 1\right] \\ &= \sum_{k\ge 0} \mathbb{E}\left[\prod_{l=1}^{X_n} s^{Y_l} \;\middle|\; X_n = k\right] \mathbb{P}(X_n = k \mid X_0 = 1) \\ &= \sum_{k\ge 0} \mathbb{E}\left[\prod_{l=1}^{k} s^{Y_l} \;\middle|\; X_n = k\right] \mathbb{P}(X_n = k \mid X_0 = 1) \\ &= \sum_{k\ge 0} \mathbb{E}\left[\prod_{l=1}^{k} s^{Y_l}\right] \mathbb{P}(X_n = k \mid X_0 = 1) \\ &= \sum_{k\ge 0} \left(\prod_{l=1}^{k} \mathbb{E}\left[s^{Y_l}\right]\right) \mathbb{P}(X_n = k \mid X_0 = 1) \\ &= \sum_{k\ge 0} \left(\mathbb{E}\left[s^{Y_1}\right]\right)^k \mathbb{P}(X_n = k \mid X_0 = 1) \\ &= G_n\left(\mathbb{E}\left[s^{Y_1}\right]\right) \\ &= G_n(G_1(s)), \qquad -1 \le s \le 1.\end{aligned}$$

□

Instead of (8.2.3) we may also write

$$G_n(s) = G_1(G_1(\cdots(G_1(s), \cdots)), \qquad -1 \le s \le 1, \tag{8.2.4}$$

and

$$G_n(s) = G_1(G_{n-1}(s)) = G_{n-1}(G_1(s)), \qquad -1 \leq s \leq 1.$$

Mean Population Size

In case the random variable Y_k is equal to a deterministic constant $\mu \in \mathbb{N}$, the population size at generation $n \geq 0$ will clearly be equal to μ^n. The next proposition shows that for branching processes, this property admits a natural extension to the random case.

Proposition 8.2 *The mean population size μ_n at generation $n \geq 0$ is given by*

$$\mu_n = \mathbb{E}[X_n \mid X_0 = 1] = (\mathbb{E}[X_1 \mid X_0 = 1])^n = \mu^n, \quad n \geq 0, \tag{8.2.5}$$

where $\mu = \mathbb{E}[Y_1]$ is given by (8.2.1a).

Proof By (8.2.4), (8.2.2b) and the chain rule of derivation, we have

$$\begin{aligned}
\mu_n &= G_n'(1^-) \\
&= \frac{d}{ds} G_1(G_{n-1}(s))_{|s=1} \\
&= G_{n-1}'(1^-) G_1'(G_{n-1}(1^-)) \\
&= G_{n-1}'(1^-) G_1'(1^-) \\
&= \mu \times \mu_{n-1},
\end{aligned}$$

hence $\mu_1 = \mu$, $\mu_2 = \mu \times \mu_1 = \mu^2$, $\mu_3 = \mu \times \mu_2 = \mu^3$, and by induction on $n \geq 1$ we obtain (8.2.5). □

Similarly, we find

$$\mathbb{E}[X_n \mid X_0 = k] = k\, \mathbb{E}[X_n \mid X_0 = 1] = k\mu^n, \quad n \geq 1,$$

hence starting from $X_0 = k \geq 1$, the average of X_n goes to infinity when $\mu > 1$. On the other hand, μ_n converges to 0 when $\mu < 1$, and in this case the *total* mean population size, including all ancestors, is finite and equal to

$$\sum_{n \geq 0} \mathbb{E}[X_n \mid X_0 = 1] = \sum_{n \geq 0} \mu^n = \frac{1}{1-\mu}.$$

Examples

(i) Supercritical case. When $\mu > 1$ the average population size $\mu_n = \mu^n$ grows to infinity as n tends to infinity, and we say that the branching process $(X_n)_{n \geq 0}$ is *supercritical*.

This condition holds in particular when $\mathbb{P}(Y_1 \geq 1) = 1$ and Y_1 is not almost surely equal to 1, i.e. $\mathbb{P}(Y_1 = 1) < 1$. Indeed, under those conditions we have $\mathbb{P}(Y_1 \geq 2) > 0$ and

$$\begin{aligned}\mu = \mathbb{E}[Y_1] &= \sum_{n\geq 1} n\mathbb{P}(Y_1 = n)\\ &\geq \mathbb{P}(Y_1 = 1) + 2\sum_{n\geq 2} \mathbb{P}(Y_1 = n)\\ &= \mathbb{P}(Y_1 = 1) + 2\mathbb{P}(Y_1 \geq 2)\\ &> \mathbb{P}(Y_1 \geq 1)\\ &= 1,\end{aligned}$$

hence $\mu > 1$.

(ii) Critical case. When $\mu = 1$ we have $\mu_n = (\mu)^n = 1$ for all $n \in \mathbb{N}$, and we say that the branching process $(X_n)_{n\geq 0}$ is *critical*.

(iii) Subcritical case. In case $\mu < 1$, the average population size $\mu_n = \mu^n$ tends to 0 as n tends to infinity and we say that the branching process $(X_n)_{n\geq 0}$ is *subcritical*. In this case, we necessarily have $\mathbb{P}(Y_1 = 0) > 0$ as

$$\begin{aligned}1 > \mu = \mathbb{E}[Y_1] &= \sum_{n\geq 1} n\mathbb{P}(Y_1 = n)\\ &\geq \sum_{n\geq 1} \mathbb{P}(Y_1 = n)\\ &= \mathbb{P}(Y_1 \geq 1)\\ &= 1 - \mathbb{P}(Y_0 = 0),\end{aligned}$$

although the converse is not true in general.

The variance $\sigma_n^2 = \mathrm{Var}[X_n \mid X_0 = 1]$ of X_n given that $X_0 = 1$ can be shown in a similar way to satisfy the recurrence relation

$$\sigma_{n+1}^2 = \sigma^2\mu^n + \mu\sigma_n^2,$$

cf. also Relation (1.7.8), where $\sigma^2 = \mathrm{Var}[Y_1]$, which shows by induction that

$$\sigma_n^2 = \mathrm{Var}[X_n \mid X_0 = 1] = \begin{cases} n\sigma^2, & \mu = 1,\\ \sigma^2\mu^{n-1}\dfrac{1-\mu^n}{1-\mu} = \sigma^2\displaystyle\sum_{k=0}^{n-1}\mu^{n+k-1}, & \mu \neq 1,\end{cases}$$

$n \geq 1$ cf. e.g. pages 180–181 of Karlin and Taylor (1998), and Exercise 8.3-(a) for an application. We also have

$$\mathrm{Var}[X_n \mid X_0 = k] = k\mathrm{Var}[X_n \mid X_0 = 1] = k\sigma_n^2, \qquad k, n \geq 0,$$

due to Relation (1.6.10) for the variance of a sum of independent random variables.

8.3 Extinction Probabilities

Here we are interested in the time to extinction[1]

$$T_0 := \inf\{n \geq 0 \ : \ X_n = 0\},$$

and in the extinction probability

$$\alpha_k := \mathbb{P}(T_0 < \infty \mid X_0 = k)$$

within a finite time, after starting from $X_0 = k$. Note that the word "extinction" can have negative as well as positive meaning, for example when the branching process is used to model the spread of an infection.

Proposition 8.3 *The probability distribution of T_0 can be expressed using the probability generating function G_n as*

$$\mathbb{P}(T_0 = n \mid X_0 = 1) = G_n(0) - G_{n-1}(0) = G_1(G_{n-1}(0)) - G_{n-1}(0), \quad n \geq 1,$$

with $\mathbb{P}(T_0 = 0 \mid X_0 = 1) = 0$.

Proof By the relation $\{X_{n-1} = 0\} \subset \{X_n = 0\}$, we have

$$\big\{T_0 = n\big\} = \big\{X_n = 0\big\} \cap \big\{X_{n-1} \geq 1\big\} = \big\{X_n = 0\big\} \setminus \big\{X_{n-1} = 0\big\}$$

and

$$\begin{aligned}
\mathbb{P}(T_0 = n \mid X_0 = 1) &= \mathbb{P}(\{X_n = 0\} \cap \{X_{n-1} \geq 1\} \mid X_0 = 1) \\
&= \mathbb{P}(\{X_n = 0\} \setminus \{X_{n-1} = 0\} \mid X_0 = 1) \\
&= \mathbb{P}(X_n = 0 \mid X_0 = 1)) - \mathbb{P}(X_{n-1} = 0 \mid X_0 = 1) \\
&= G_n(0) - G_{n-1}(0) \\
&= G_1(G_{n-1}(0)) - G_{n-1}(0), \quad n \geq 1,
\end{aligned}$$

where we applied Proposition 8.1. □

First, we note that by the independence assumption, starting from $X_0 = k \geq 2$ independent individuals, we have

$$\alpha_k = \mathbb{P}(T_0 < \infty \mid X_0 = k) = (\mathbb{P}(T_0 < \infty \mid X_0 = 1))^k = (\alpha_1)^k, \quad k \geq 1. \quad (8.3.1)$$

Indeed, given k individuals at generation 0, each of them will start independently a new branch of offsprings, and in order to have extinction of the whole population, all of k branches should become extinct. Since the k branches behave independently, α_k

[1] We normally start from $X_0 \geq 1$.

is the product of the extinction probabilities for each branch, which yields $\alpha_k = (\alpha_1)^k$ since these extinction probabilities are all equal to α_1 and there are k of them.

The next proposition is a consequence of Lemma 5.2.

Proposition 8.4 *The extinction probability within a finite time after starting from $X_0 = k$ satisfies*

$$\alpha_1 = \mathbb{P}(T_0 < \infty \mid X_0 = 1) = \lim_{n\to\infty} G_n(0).$$

Proof Since state ⓪ is absorbing, by Lemma 5.2 with $j = 0$ and $i = 1$ we find

$$\alpha_1 = \mathbb{P}(T_0 < \infty \mid X_0 = 1) = \lim_{n\to\infty} \mathbb{P}(X_n = 0 \mid X_0 = 1) = \lim_{n\to\infty} G_n(0).$$

□

The next proposition shows that the extinction probability α_1 can be computed as the solution of an equation.

Proposition 8.5 *The extinction probability*

$$\alpha_1 := \mathbb{P}(T_0 < \infty \mid X_0 = 1)$$

is a solution of the equation

$$\alpha = G_1(\alpha). \tag{8.3.2}$$

Proof By first step analysis, we have

$$\begin{aligned}
\alpha_1 &= \mathbb{P}(T_0 < \infty \mid X_0 = 1) \\
&= \mathbb{P}(X_1 = 0 \mid X_0 = 1) + \sum_{k\geq 1} \mathbb{P}(T_0 < \infty \mid X_1 = k)\mathbb{P}(X_1 = k \mid X_0 = 1) \\
&= \mathbb{P}(Y_1 = 0) + \sum_{k\geq 1} \mathbb{P}(T_0 < \infty \mid X_0 = k)\mathbb{P}(Y_1 = k) \\
&= \sum_{k\geq 0} (\alpha_1)^k \mathbb{P}(Y_1 = k) \\
&= G_1(\alpha_1),
\end{aligned}$$

hence the extinction probability α_1 solves (8.3.2). □

Note that from the above proof we find

$$\alpha_1 \geq \mathbb{P}(X_1 = 0 \mid X_0 = 1) = \mathbb{P}(Y_1 = 0), \tag{8.3.3}$$

which shows that the extinction probability is non-zero whenever $\mathbb{P}(Y_1 = 0) > 0$. On the other hand, any solution α of (8.3.2) also satisfies

$$\alpha = G_1(G_1(\alpha)), \quad \alpha = G_1(G_1(G_1(\alpha))),$$

and more generally

$$\alpha = G_n(\alpha), \qquad n \geq 1, \tag{8.3.4}$$

by Proposition 8.1. On the other hand the solution of (8.3.2) may not be unique, for example $\alpha = 1$ is always solution of (8.3.2) since $G_1(1) = 1$, and it may not be equal to the extinction probability. The next proposition clarifies this point.

Proposition 8.6 *The extinction probability*

$$\alpha_1 := \mathbb{P}(T_0 < \infty \mid X_0 = 1)$$

is the smallest nonnegative solution of the equation $\alpha = G_1(\alpha)$.

Proof By Lemma 8.4 we have $\alpha_1 = \lim_{n\to\infty} G_n(0)$. Next, we note that the function $s \longmapsto G_1(s)$ is increasing because

$$G_1'(s) = \mathbb{E}[Y_1 s^{Y_1 - 1}] > 0, \qquad s \in [0, 1).$$

Hence $s \longmapsto G_n(s)$ is also increasing by Proposition 8.1, and for any solution $\alpha \geq 0$ of (8.3.2) we have, by (8.3.4),

$$0 \leq G_n(0) \leq G_n(\alpha) = \alpha, \qquad n \geq 1,$$

and taking the limit in this inequality as n goes to infinity we get

$$0 \leq \alpha_1 = \lim_{n\to\infty} G_n(0) \leq \alpha,$$

by Proposition 8.4, hence the extinction probability α_1 is always smaller than any solution α of (8.3.2). This fact can also be recovered from Proposition 8.4 and

$$\begin{aligned}
\alpha &= \lim_{n\to\infty} G_n(\alpha) \\
&= \lim_{n\to\infty} \left(G_n(0) + \sum_{k\geq 1} \alpha^k \mathbb{P}(X_n = k \mid X_0 = 1) \right) \\
&\geq \lim_{n\to\infty} G_n(0) \\
&= \alpha_1.
\end{aligned}$$

Therefore α_1 is the smallest solution of (8.3.2). □

Since $G_1(0) = \mathbb{P}(Y_1 = 0)$ we have

$$\mathbb{P}(Y_1 = 0) = G_1(0) \leq G_1(\alpha_1) = \alpha_1,$$

which recovers (8.3.3).

On the other hand, if $\mathbb{P}(Y_1 \geq 1) = 1$ then we have $G_1(0) = 0$, which implies $\alpha_1 = 0$ by Proposition 8.6.

Note that from Lemma 5.2, Proposition 8.4, and (8.3.1), the transition matrix (8.1.2) satisfies

$$\lim_{n\to\infty} \left([P^n]_{i,0}\right)_{i\geq 0} = \begin{bmatrix} 1 \\ \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \vdots \end{bmatrix} = \begin{bmatrix} 1 \\ \alpha_1 \\ (\alpha_1)^2 \\ (\alpha_1)^3 \\ \vdots \end{bmatrix}.$$

Examples

(i) Assume that Y_1 has a Bernoulli distribution with parameter $p \in (0, 1)$, i.e.

$$\mathbb{P}(Y_1 = 1) = p, \qquad \mathbb{P}(Y_1 = 0) = 1 - p.$$

Compute the extinction probability of the associated branching process.

In this case, the branching process is actually a two-state Markov chain with transition matrix

$$P = \begin{bmatrix} 1 & 0 \\ 1-p & p \end{bmatrix},$$

and we have

$$G_n(0) = \mathbb{P}(X_n = 0 \mid X_0 = 1) = (1-p)\sum_{k=0}^{n-1} p^k = 1 - p^n, \tag{8.3.5}$$

where we used the geometric series (A.2), hence as in (5.1.6) the extinction probability α_1 is given by

$$\begin{aligned} \alpha_1 &= \mathbb{P}(T_0 < \infty \mid X_0 = 1) = \mathbb{P}\left(\bigcup_{n\geq 1}\{X_n = 0\} \,\middle|\, X_0 = 1\right) \\ &= \lim_{n\to\infty} \mathbb{P}(X_n = 0 \mid X_0 = 1) \\ &= \lim_{n\to\infty} G_n(0) = 1, \end{aligned}$$

provided that $p = \mathbb{E}[Y_1] < 1$, otherwise we have $\alpha_1 = 0$ when $p = 1$. The value of α_1 can be recovered using the generating function

$$G_1(s) = \mathbb{E}\left[s^{Y_1}\right] = \sum_{k\geq 0} s^k \mathbb{P}(Y_1 = k) = 1 - p + ps, \tag{8.3.6}$$

for which the unique solution of $G_1(\alpha) = \alpha$ is the extinction probability $\alpha_1 = 1$, as shown in Fig. 8.3.

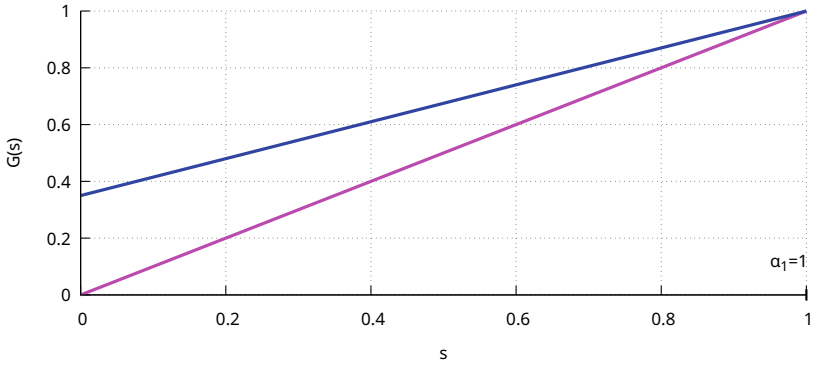


Fig. 8.3 Generating function of Y_1 with $p = 0.65$

From (8.2.4) and (8.3.6) we can also show by induction on $n \geq 1$ as in Exercise 8.2 that

$$G_n(s) = p^n s + (1-p) \sum_{k=0}^{n-1} p^k = 1 - p^n + p^n s,$$

which recovers (8.3.5) from (1.7.5) or (8.2.2a) as

$$\mathbb{P}(X_n = 0 \mid X_0 = 1) = G_n(0) = 1 - p^n.$$

We also have $\mathbb{E}[X_n] = p^n$, $n \geq 0$.

(ii) Same question as in (i) above for

$$\mathbb{P}(Y_1 = 2) = p, \qquad \mathbb{P}(Y_1 = 0) = q = 1 - p.$$

Here, we will directly use the probability generating function

$$\begin{aligned} G_1(s) = \mathbb{E}\left[s^{Y_1}\right] &= \sum_{k \geq 0} s^k \mathbb{P}(Y_1 = k) \\ &= s^0 \mathbb{P}(Y_1 = 0) + s^2 \mathbb{P}(Y_1 = 2) = 1 - p + ps^2. \end{aligned}$$

We check that the solutions of

$$G_1(\alpha) = 1 - p + p\alpha^2 = \alpha,$$

i.e.[2]

$$p\alpha^2 - \alpha + q = p(\alpha - 1)(\alpha - q/p) = 0, \tag{8.3.7}$$

with $q = 1 - p$, are given by

$$\left\{ \frac{1 + \sqrt{1 - -4pq}}{2p}, \frac{1 - \sqrt{1 - -4pq}}{2p} \right\} = \left\{ 1, \frac{q}{p} \right\}, \qquad p \in (0, 1]. \tag{8.3.8}$$

Hence the extinction probability is $\alpha_1 = 1$ if $q \geq p$, and it is equal to $\alpha_1 = q/p < 1$ if $q < p$, or equivalently if $\mathbb{E}[Y_1] > 1$, due to the relation $\mathbb{E}[Y_1] = 2p$.

(iii) Assume that Y_1 has the geometric distribution with parameter $p \in (0, 1)$, i.e.

$$\mathbb{P}(Y_1 = n) = (1 - p)p^n, \qquad n \geq 0,$$

with $\mu = \mathbb{E}[Y_1] = p/q$. We have

$$G_1(s) = \mathbb{E}\left[s^{Y_1}\right] = \sum_{n \geq 0} s^n \mathbb{P}(Y_1 = n) = (1 - p) \sum_{n \geq 0} p^n s^n = \frac{1 - p}{1 - ps}. \tag{8.3.9}$$

The equation $G_1(\alpha) = \alpha$ reads

$$\frac{1 - p}{1 - p\alpha} = \alpha,$$

i.e.

$$p\alpha^2 - \alpha + q = p(\alpha - 1)(\alpha - q/p) = 0,$$

which is identical to (2.2.16) and (8.3.7) with $q = 1 - p$, and has for solutions (8.3.8). Hence the finite time extinction probability is

$$\begin{aligned} \alpha_1 &= \mathbb{P}(T_0 < \infty \mid X_0 = 1) \\ &= \min\left(1, \frac{q}{p}\right) = \begin{cases} \dfrac{q}{p}, & p \geq 1/2, \ \text{(super)critical case,} \\ 1, & p \leq 1/2, \ \text{(sub)critical case.} \end{cases} \end{aligned}$$

Note that we have $\alpha_1 < 1$ if and only if $\mathbb{E}[Y_1] > 1$, due to the equality $\mathbb{E}[Y_1] = p/q$. As can be seen from Figs. 8.4 and 8.5, the extinction probability α_1 is equal to 1 when $p \leq 1/2$, meaning that extinction within a finite time is certain in that case. Note that we also find

[2] Remark that (8.3.7) is identical to the characteristic equation (2.2.16).

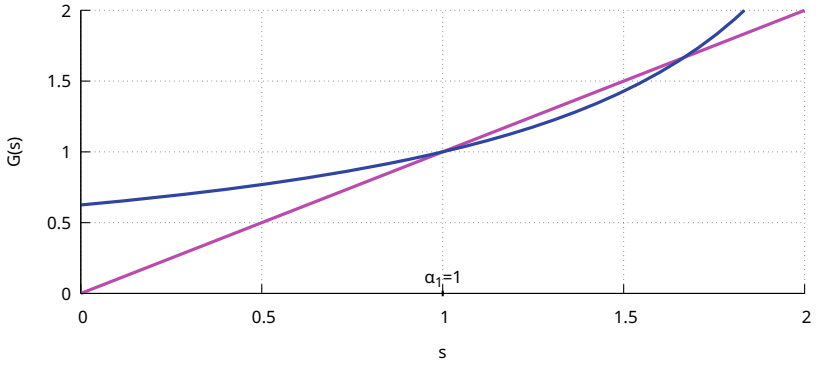


Fig. 8.4 Generating function of Y_1 with $p = 3/8 < 1/2$ and $\alpha_1 = 1$

$$\mathbb{P}(T_0 < \infty \mid X_0 = k) = \min\left(1, \left(\frac{q}{p}\right)^k\right)$$
$$= \begin{cases} \left(\frac{q}{p}\right)^k, & p \geq 1/2, \text{ (super)critical case,} \\ 1, & p \leq 1/2, \text{ (sub)critical case,} \end{cases}$$

which incidentally coincides with the finite time hitting probability found in (3.4.16) for the simple random walk started from $k \geq 1$.
Next, in Fig. 8.5 is a graph of the generating function $s \longmapsto G_1(s)$ for $p = 1/2$.
The generating function graph in Fig. 8.6 corresponds to $p = 3/4$.
We also have $\mu_n = (\mathbb{E}[Y_1])^n = (p/q)^n$, $n \geq 1$.

(iv) Assume now that Y_1 is the sum of two independent geometric variables with parameter p, i.e. it has the negative binomial distribution

$$\mathbb{P}(Y_1 = n) = \binom{n+r-1}{r-1} q^r p^n = (n+1) q^r p^n = (n+1) q^2 p^n, \qquad n \geq 0,$$

with $r = 2$, cf. (1.5.12).
In this case, we have[3,4]

$$G_1(s) = \mathbb{E}\left[s^{Y_1}\right] = \sum_{n \geq 0} s^n \mathbb{P}(Y_1 = n)$$
$$= q^2 \sum_{n \geq 0} (n+1) p^n s^n = \left(\frac{1-p}{1-ps}\right)^2, \qquad -1 \leq s \leq 1.$$

[3] Here, Y_1 is the sum of two independent geometrically distributed random variables, and G_1 is the square of the generating function (8.3.9) of the geometric distribution.
[4] We used the identity $\sum_{n \geq 0} (n+1) r^n = (1-r)^{-2}$, cf. (A.4).

When $p = 1/2$ we check that $G_1(\alpha) = \alpha$ reads

$$s^3 - 4\,s^2 + 4\,s - 1 = 0,$$

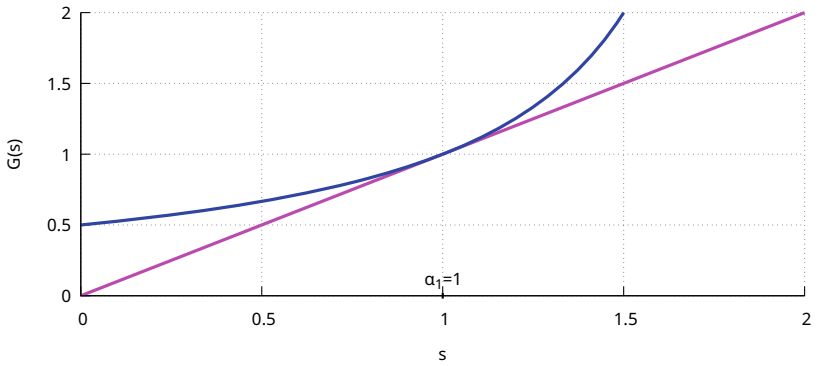


Fig. 8.5 Generating function of Y_1 with $p = 1/2$ and $\alpha_1 = 1$

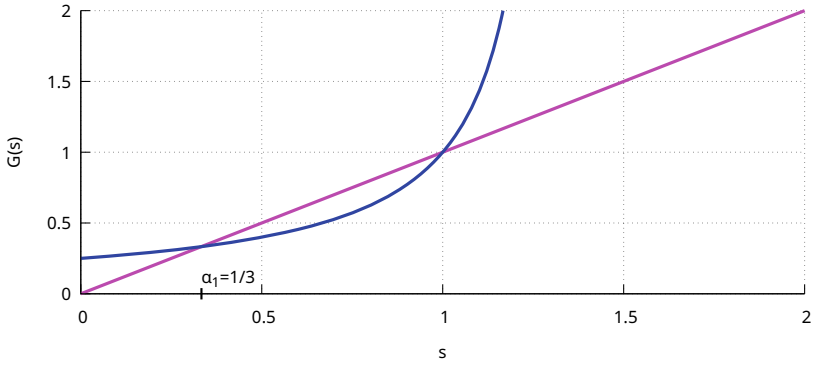


Fig. 8.6 Generating function of Y_1 with $p = 3/4 > 1/2$ and $\alpha_1 = q/p = 1/3$

which is an equation of degree 3 in the unknown s. Now, since $\alpha = 1$ is solution of this equation we can factorise it as follows:

$$(s - 1)(s^2 - 3\,s + 1) = 0,$$

and we check that the smallest nonnegative solution of this equation is given by

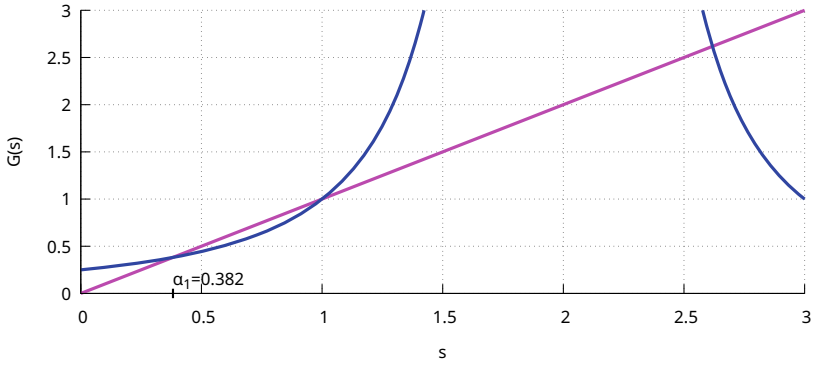


Fig. 8.7 Probability generating function of Y_1

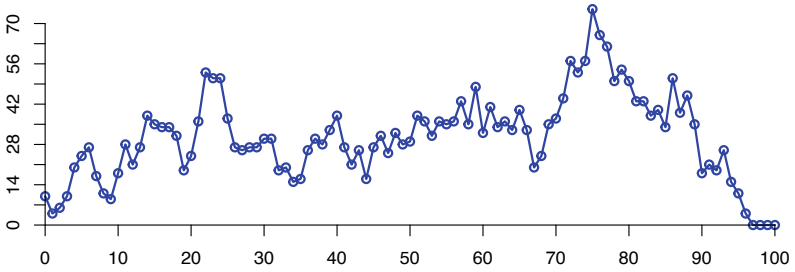


Fig. 8.8 Sample path of a branching process $(X_n)_{n\geq 0}$

$$\alpha_1 = \frac{1}{2}(3-\sqrt{5}) \simeq 0.382$$

which is the extinction probability, as illustrated in Fig. 8.7. Here, we have $\mathbb{E}[Y_1] = 2$.

Figure 8.8 illustrates the extinction of a branching process in finite time when Y_1 has the geometric distribution with $p = 1/2$, in which case there is extinction within finite time with probability 1.

In Table 8.1 we summarize some questions and their associated solution methods introduced in this chapter and the previous ones.

Exercises

Exercise 8.1 A parent particle can be divided into 0, 1 or 2 particles with probabilities $1/5$, $3/5$, and $1/5$, respectively. It disappears after splitting. Starting with one

Table 8.1 Summary of computing methods

How to compute	Method
Expected value $\mathbb{E}[X]$	Sum the values of X weighted by their probabilities
Uses of $G_X(s)$	$G_X(0) = \mathbb{P}(X = 0)$, $G_X(1) = \mathbb{P}(X < \infty)$, $G'_X(1^-) = \mathbb{E}[X]$
Hitting probabilities $g(k)$	Solve* $g = Pg$ for $g(k)$
Mean hitting times $h(k)$	Solve* $h = \mathbb{1} + Ph$ for $h(k)$
Stationary distribution π	Solve† $\pi = \pi P$ for π
Extinction probability α_1	Solve $G_1(\alpha) = \alpha$ for α, and choose the smallest solution
$\lim_{n\to\infty} \left(\begin{bmatrix} 1-a & a \\ b & 1-b \end{bmatrix} \right)^n$	$\begin{bmatrix} \frac{b}{a+b} & \frac{a}{a+b} \\ \frac{b}{a+b} & \frac{a}{a+b} \end{bmatrix}$

*Be sure to write only the relevant rows of the system under the appropriate boundary conditions.
†Remember that the values of $\pi(k)$ have to add up to 1

particle, the ancestor, let us denote by X_n the size of the corresponding branching process at the n-th generation.

(a) Find $P(X_2 > 0)$.
(b) Find $P(X_2 = 1)$.
(c) Find the probability that $X_1 = 2$ given that $X_2 = 1$.

Exercise 8.2 Each individual in a population has a random number Y of offsprings, with

$$\mathbb{P}(Y = 0) = 1/2, \qquad \mathbb{P}(Y = 1) = 1/2.$$

We let X_n denote the size of the population at time $n \geq 0$, starting from $X_0 = 1$.

(a) Compute the generating function $G_1(s) = \mathbb{E}[s^Y]$ of Y for $s \in \mathbb{R}_+$.
(b) Let $G_n(s) := \mathbb{E}\left[s^{X_n}\right]$ denote the generating function of X_n. Show that

$$G_n(s) = 1 - \frac{1}{2^n} + \frac{s}{2^n}, \qquad s \in \mathbb{R}. \tag{8.3.10}$$

(c) Compute the probability $\mathbb{P}(X_n = 0 \mid X_0 = 1)$ that the population is extinct at time n.
(d) Compute the average size $\mathbb{E}[X_n \mid X_0 = 1]$ of the population at step n.
(e) Compute the extinction probability of the population after starting from $X_0 = 1$.

Exercise 8.3 Each individual in a population has a random number ξ of offsprings, with distribution

$$\mathbb{P}(\xi = 0) = 0.2, \qquad \mathbb{P}(\xi = 1) = 0.5, \qquad \mathbb{P}(\xi = 2) = 0.3.$$

Let X_n denote the number of individuals in the population at the n-th generation, with $X_0 = 1$.

(a) Compute the mean and variance of X_2.
(b) Give the probability distribution of the random variable X_2.
(c) Compute the probability that the population is extinct by the fourth generation.
(d) Compute the expected number of offsprings at the tenth generation.
(e) What is the probability of extinction of this population?

Exercise 8.4 Each individual in a population has a random number Y of offsprings, with

$$\mathbb{P}(Y = 0) = c, \qquad \mathbb{P}(Y = 1) = b, \qquad \mathbb{P}(Y = 2) = a,$$

where $a + b + c = 1$.

(a) Compute the generating function $G_1(s)$ of Y for $s \in [-1, 1]$.
(b) Compute the probability that the population is extinct at time 2, starting from 1 individual at time 0.
(c) Compute the probability that the population is extinct at time 2, starting from 2 individuals at time 0.
(d) Show that when $0 < c \leq a$ the probability of eventual extinction of the population, starting from 2 individuals at time 0, is $(c/a)^2$.
(e) What is this probability equal to when $0 < a < c$?

Exercise 8.5 Consider a branching process $(Z_n)_{n\geq 0}$ in which the offspring distribution at each generation is binomial with parameter $(2, p)$, i.e.

$$\mathbb{P}(Y = 0) = q^2, \quad \mathbb{P}(Y = 1) = 2pq, \quad \mathbb{P}(Y = 2) = p^2,$$

with $q := 1 - p$.

(a) Compute the probability generating function G_Y of Y.
(b) Compute the extinction probability of this process, starting from $Z_0 = 1$.
(c) Compute the probability that the population becomes extinct for the first time in the second generation ($n = 2$), starting from $Z_0 = 1$.
(d) Suppose that the initial population size Z_0 is a Poisson random variable with parameter $\lambda > 0$. Compute the extinction probability in this case.

Exercise 8.6 A cell culture is started with one red cell at time 0. After one minute the red cell dies and two new cells are born according to the following probability distribution:

Color configuration	Probability
2 red cells	1/4
1 red cell + 1 white cell	2/3
2 white cells	1/12

The above procedure is repeated minute after minute for any red cell present in the culture. On the other hand, the white cells can only live for one minute, and disappear after that without reproducing. We assume that the cells behave independently.

(a) What is the probability that no white cells have been generated until time n included?
(b) Compute the extinction probability of the whole cell culture.
(c) Same questions as above for the following probability distribution:

Color configuration	Probability
2 red cells	1/3
1 red cell + 1 white cell	1/2
2 white cells	1/6

Exercise 8.7 Using first step analysis, show that if $(X_n)_{n\geq 0}$ is a subcritical branching process, i.e. $\mu = \mathbb{E}[Y_1] < 1$, the time to extinction $T_0 := \inf\{n \geq 0 \ : \ X_n = 0\}$ satisfies $\mathbb{E}[T_0 \mid X_0 = 1] < \infty$.

Exercise 8.8 Consider a branching process $(X_n)_{n\geq 0}$ started at $X_0 = 1$, in which the numbers Y_k of descendants of individual $n^o k$ form an *i.i.d.* sequence with the negative binomial distribution

$$\mathbb{P}(Y_k = n) = (n+1)q^2 p^n, \qquad n \geq 0, \quad k \geq 1,$$

where $0 < q = 1 - p < 1$.

(a) Compute the probability generating function

$$G_1(s) := \mathbb{E}\left[s^{Y_k}\right] = \sum_{n\geq 0} s^n \mathbb{P}(Y_1 = n)$$

of Y_k, $k \geq 1$.
(b) Compute the extinction probability $\alpha_1 := \mathbb{P}(T_0 < \infty \mid X_0 = 1)$ of the branching process $(X_n)_{n\geq 0}$ in finite time.

Exercise 8.9 Families in a given society have children until the birth of the first girl, after which the family stops having children. Let X denote the number of male children of a given husband.

(a) Assuming that girls and boys are equally likely to be born, compute the probability distribution of X.
(b) Compute the probability generating function $G_X(s)$ of X.
(c) What is the probability that a given man has no male descendant (patrilineality) by the time of the third generation?
(d) Suppose now that one fourth of the married couples have no children at all while the others continue to have children until the first girl, and then cease childbearing. What is the probability that the wife's female line of descent (matrilineality) will cease to exist by the third generation?

Exercise 8.10 Consider a branching process $(Z_k)_{k\geq 0}$ with $Z_0 = 1$ and offspring distribution given by

$$\mathbb{P}(Z_1 = 0) = \frac{1-p-q}{1-p} \quad \text{and} \quad \mathbb{P}(Z_1 = k) = qp^{k-1}, \qquad k = 1, 2, 3, \ldots,$$

where $0 \leq p < 1$ and $0 \leq q \leq 1 - p$.

(a) Find the probability generating function of Z_1.
(b) Compute $\mathbb{E}[Z_1]$.
(c) Find the value of q for which $\mathbb{E}[Z_1] = 1$, known as the *critical value*.
(d) Using the critical value of q, show by induction that the probability generating function of Z_k is given by

$$G_{Z_k}(s) = \frac{kp - (kp + p - 1)s}{1 - p + kp - kps}, \qquad -1 < s < 1,$$

for all $k \geq 1$.

Exercise 8.11 Borel distribution. Consider a branching process started from one individual, in which every individual gives rise to a Poisson number N of offsprings with the Poisson distribution with mean $\mu > 0$, i.e.

$$\mathbb{P}(N = n) = \mathrm{e}^{-\mu}\frac{\mu^n}{n!}, \qquad n \geq 0.$$

We let X denote the total number of descendants of a given ancestor, including this ancestor and all subsequent ancestors.

(a) Show that the Probability Generating Function (PGF) of X satisfies the induction relation

$$G(s) = sG_\mu(G(s)), \qquad -1 \leq s \leq 1,$$

where

$$G_\mu(s) = \mathrm{e}^{-\mu}\sum_{n=0}^{\infty}\frac{\mu^n}{n!}s^n = \mathrm{e}^{\mu(s-1)}$$

denotes the PGF of the Poisson distribution with mean $\mu > 0$.

(b) Show that the mean count of all descendants including all ancestors is given by

$$\mathbb{E}[X] = \frac{1}{1-\mu}.$$

(c) Show that the variance of the count of all descendants including all ancestors is given by

$$\mathrm{Var}[X] = \frac{\mu}{(1-\mu)^3}.$$

Problem 8.12 Consider a branching process with *i.i.d.* offspring sequence $(Y_k)_{k\geq 1}$. The number of individuals in the population at generation $n+1$ is given by the relation $X_{n+1} = Y_1 + \cdots + Y_{X_n}$, with $X_0 = 1$.

(a) Let

$$Z_n = \sum_{k=1}^{n} X_k,$$

denote the total number of individuals generated from time 1 to n. Compute $\mathbb{E}[Z_n]$ as a function of $\mu = \mathbb{E}[Y_1]$.

(b) Let $Z = \sum_{k\geq 1} X_k$. denote the total number of individuals generated from time 1 to infinity. Compute $\mathbb{E}[Z]$ and show that it is finite when $\mu < 1$.
In what follows we work under the condition $\mu < 1$.

(c) Let

$$H(s) = \mathbb{E}\left[s^Z\right], \qquad -1 \leq s \leq 1,$$

denote the generating function of Z.
Show, by first step analysis, that the relation

$$H(s) = G_1(sH(s)), \qquad 0 \leq s \leq 1,$$

holds, where $G_1(x)$ is the probability generating function of Y_1.

(d) In what follows we assume that Y_1 has the geometric distribution $\mathbb{P}(Y_1 = k) = qp^k$, $k \in \mathbb{N}$, with $p \in (0,1)$ and $q = 1-p$. Compute $H(s)$ for $s \in [0,1]$.

(e) Using the expression of the generating function $H(s)$ computed in Question (d), check that we have $H(0) = \lim_{s \searrow 0} H(s)$, where $H(0) = \mathbb{P}(Z = 0) = \mathbb{P}(Y_1 = 0) = G_1(0)$.

(f) Using the generating function $H(s)$ computed in Question (d), recover the value of $\mathbb{E}[Z]$ found in Question (b).

(g) Assume that each of the Z individuals earns an income U_k, $k = 1, 2, \ldots, Z$, where $(U_k)_{k\geq 1}$ is an *i.i.d.* sequence of random variables with finite expectation $\mathbb{E}[U]$ and distribution function $F(x) = \mathbb{P}(U \leq x)$.
Compute the expected value of the sum of gains of all the individuals in the population.

(h) Compute the probability that none of the individuals earns an income higher than $x > 0$.
(i) Evaluate the results of Questions (g) and (h) when U_k has the exponential distribution with $F(x) = 1 - \mathrm{e}^{-x}$, $x \geq 0$.

Hints and comments on Problem 8.12.

(a) Use the expression (8.2.5) of $\mathbb{E}[X_k]$.
(b) $\mathbb{E}[Z] < \infty$ implies that $Z < \infty$ almost surely.
(c) Given that $X_1 = k$, Z can be decomposed into the sum of k independent population sizes, each of them started with 1 individual.
(d) Compute $G_1(s)$ and μ in this model, and check the condition $\mu < 1$. When computing $H(s)$ you should have to solve a quadratic equation, and to choose the relevant solution out of two possibilities.
(e) The goal of this question is to confirm the result of Question (d) by checking the value of $H(0) = \lim_{s \searrow 0} H(s)$. For this, find the Taylor expansion of $\sqrt{1 - 4pqs}$ as s tends to 0.
(f) The identity $1 - 4pq = (q - p)^2$ can be useful, and the sign of $q - p$ has to be taken into account when computing $\sqrt{1 - 4pq}$.
(g) Use conditioning on $Z = n$, $n \in \mathbb{N}$.
(h) The answer will use both H and F.
(i) Compute the value of $\mathbb{E}[U_1]$.

Chapter 9
Continuous-Time Markov Chains



In this chapter we start the study of *continuous-time* stochastic processes, which are families $(X_t)_{t\in\mathbf{R}_+}$ of random variables indexed by $\mathbf{R}_+$. Our aim is to make the transition from discrete to continuous-time Markov chains, the main difference between the two settings being the replacement of the transition matrix with the continuous-time *infinitesimal generator* of the process. We will start with the two fundamental examples of the Poisson and birth and death processes, followed by the construction of continuous-time Markov chains and their generators in more generality. From the point of view of simulations, the use of continuous-time Markov chains does not bring any special difficulty as any continuous-time simulation is actually based on discrete-time samples. From a theoretical point of view, however, the rigorous treatment of the continuous-time Markov property is much more demanding than its discrete-time counterpart, notably due to the use of the strong Markov property. Here, we focus on the understanding of the continuous-time case by simple calculations, and we will refer to the literature for the use of the strong Markov property.

9.1 The Poisson Process

The *standard Poisson process* $(N_t)_{t\in\mathbf{R}_+}$ is a continuous-time *counting* process, i.e. $(N_t)_{t\in\mathbf{R}_+}$ has jumps of size $+1$ only, and its paths are constant (and right-continuous) in between jumps. Figure 9.1 represents a sample path of a Poisson process.

We denote by $(T_k)_{k\geq 1}$ the increasing sequence of jump times of $(N_t)_{t\in\mathbf{R}_+}$, which can be defined from the (right-continuous) Poisson process path $(N_t)_{t\in\mathbf{R}_+}$ by noting that T_k is the first hitting time of state k, i.e.

$$T_k = \inf\{t \in \mathbf{R}_+ \ : \ N_t = k\}, \qquad k \geq 1,$$

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N. Privault, *Understanding Markov Chains*, Springer Undergraduate Mathematics Series,
https://doi.org/10.1007/978-981-95-4057-0_9

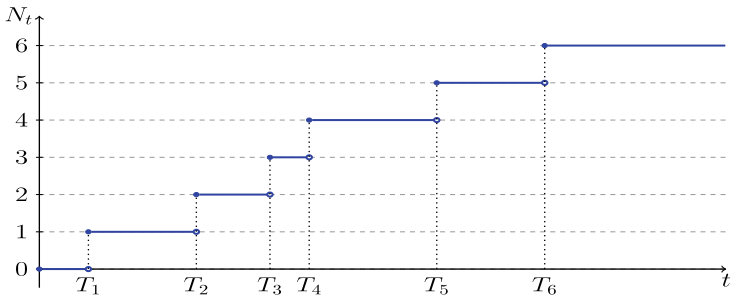


Fig. 9.1 Sample path of a Poisson process $(N_t)_{t\in\mathbf{R}_+}$

with

$$\lim_{k\to\infty} T_k = \infty.$$

The value N_t at time t of the Poisson process can be recovered from its jump times $(T_k)_{k\geq 1}$ as

$$N_t = \sum_{k\geq 1} k\mathbb{1}_{[T_k,T_{k+1})}(t) = \sum_{k\geq 1} \mathbb{1}_{[T_k,\infty)}(t), \qquad t \geq 0,$$

where

$$\mathbb{1}_{[T_k,\infty)}(t) = \begin{cases} 1 \text{ if } t \geq T_k, \\ 0 \text{ if } 0 \leq t < T_k, \end{cases}$$

and

$$\mathbb{1}_{[T_k,T_{k+1})}(t) = \begin{cases} 1 \text{ if } T_k \leq t < T_{k+1}, & k \geq 0, \\ 0 \text{ if } 0 \leq t < T_k \text{ or } t \geq T_{k+1}, & k \geq 0. \end{cases}$$

with $T_0 = 0$.

In addition, $(N_t)_{t\in\mathbf{R}_+}$ is assumed to satisfy the following conditions:

(i) Independence of increments: for all $0 \leq t_0 < t_1 < \cdots < t_n$ and $n \geq 1$ the increments

$$N_{t_1} - N_{t_0}, \ldots, \ N_{t_n} - N_{t_{n-1}},$$

over the disjoint time intervals $[t_0, t_1)$, $[t_1, t_2)$, $\ldots$, $[t_{n-2}, t_{n-1})$, $[t_{n-1}, t_n]$ are mutually independent random variables.

(ii) Stationarity of increments: $N_{t+h} - N_{s+h}$ has the same distribution as $N_t - N_s$ for all $h > 0$ and $0 \leq s \leq t$.

The meaning of the above stationarity condition is that for all fixed $k \in \mathbf{N}$ we have

$$\mathbb{P}(N_{t+h} - N_{s+h} = k) = \mathbb{P}(N_t - N_s = k),$$

for all $h > 0$ and $0 \leq s \leq t$.

The stationarity of increments means that for all $k \in \mathbf{N}$, the probability $\mathbb{P}(N_{t+h} - N_{s+h} = k)$ does not depend on $h > 0$.

Based on the above assumption, a natural question arises:

what is the distribution of N_t at time t?

We already know that N_t takes values in $\mathbf{N}$ and therefore it has a discrete distribution for all $t \in \mathbf{R}_+$. It is a remarkable fact that the distribution of the increments of $(N_t)_{t\in\mathbf{R}_+}$, can be completely determined from the above conditions, as shown in the following theorem.

As seen in the next result, the random variable $N_t - N_s$ has the Poisson distribution with parameter $\lambda(t-s)$.

Theorem 9.1 *Assume that the counting process $(N_t)_{t\in\mathbf{R}_+}$ satisfies the independence and stationarity Conditions (i) and (ii) above. Then we have*

$$\mathbb{P}(N_t - N_s = k) = \mathrm{e}^{-\lambda(t-s)} \frac{(\lambda(t-s))^k}{k!}, \qquad k \in \mathbf{N}, \quad 0 \leq s \leq t,$$

for some constant $\lambda > 0$.

Theorem 9.1 shows in particular that

$$\mathbb{E}[N_t - N_s] = \lambda(t-s) \quad \text{and} \quad \mathrm{Var}[N_t - N_s] = \lambda(t-s),$$

$0 \leq s \leq t$, cf. the solution of Exercise 1.4-(a).

The parameter $\lambda > 0$ is called the *intensity* of the process and it can be recovered given from $\mathbb{P}(N_h = 1) = \lambda h \mathrm{e}^{-\lambda h}$ as the limit

$$\lambda = \lim_{h \searrow 0} \frac{1}{h} \mathbb{P}(N_h = 1). \tag{9.1.1}$$

Proof of Theorem 9.1 We only quote the main steps of the proof and we refer to Bosq and Nguyen (1996) for the complete argument. Using the independence and stationarity of increments, we show that the probability generating function

$$G_t(u) := \mathbb{E}[u^{N_t}], \qquad -1 \leq u \leq 1,$$

satisfies

$$G_t(u) := (G_1(u))^t, \qquad -1 \leq u \leq 1,$$

which implies that

$$G_t(u) := \mathrm{e}^{-tf(u)}, \qquad -1 \leq u \leq 1,$$

for some function $f(u)$ of u. Still relying on the independence and stationarity of increments, it can be shown that $f(u)$ takes the form

$$f(u) = \lambda \times (1-u), \qquad -1 \le u \le 1,$$

where $\lambda > 0$ is given by (9.1.1). □

In particular, given that $N_0 = 0$, the random variable N_t has a Poisson distribution with parameter λt:

$$\mathbb{P}(N_t = k) = \frac{(\lambda t)^k}{k!} \mathrm{e}^{-\lambda t}, \qquad t \ge 0.$$

From (9.1.1) above we see that[1]

$$\begin{cases} \mathbb{P}(N_h = 0) = \mathrm{e}^{-\lambda h} = 1 - \lambda h + o(h), & h \searrow 0, \\ \mathbb{P}(N_h = 1) = h\lambda \mathrm{e}^{-\lambda h} \simeq \lambda h, & h \searrow 0, \end{cases}$$

and more generally that

$$\begin{cases} \mathbb{P}(N_{t+h} - N_t = 0) = \mathrm{e}^{-\lambda h} = 1 - \lambda h + o(h), & h \searrow 0, \quad (9.1.2a) \\ \mathbb{P}(N_{t+h} - N_t = 1) = \lambda h \mathrm{e}^{-\lambda h} \simeq \lambda h, & h \searrow 0, \quad (9.1.2b) \\ \mathbb{P}(N_{t+h} - N_t = 2) = h^2 \dfrac{\lambda^2}{2} \mathrm{e}^{-\lambda h} \simeq h^2 \dfrac{\lambda^2}{2}, & h \searrow 0, \quad (9.1.2c) \end{cases}$$

for all $t \ge 0$. This means that within "short" time intervals $[kh, (k+1)h]$ of length $h = t/n > 0$, the increments $N_{(k+1)h} - N_{kh}$ can be approximated by independent Bernoulli random variables X_{kh} with parameter λh, whose sum

$$\sum_{k=0}^{n-1} X_{kh} \simeq \sum_{k=0}^{n-1} (N_{(k+1)h} - N_{kh}) = N_t - N_0 = N_t$$

converges in distribution as n goes to infinity to the Poisson random variable N_t with parameter λt. This remark can be used for the random simulation of Poisson process paths.

More generally, we have

$$\mathbb{P}(N_{t+h} - N_t = k) \simeq h^k \frac{\lambda^k}{k!}, \qquad h \searrow 0, \qquad t > 0.$$

In order to determine the distribution of the first jump time T_1 we note that we have the equivalence

$$\{T_1 > t\} \Longleftrightarrow \{N_t = 0\},$$

[1] The notation $f(h) \simeq h^k$ means $\lim_{h\to 0} f(h)/h^k = 1$, and $f(h) = o(h)$ means $\lim_{h\to 0} f(h)/h = 0$.

which implies

$$\mathbb{P}(T_1 > t) = \mathbb{P}(N_t = 0) = \mathrm{e}^{-\lambda t}, \qquad t \geq 0,$$

i.e. T_1 has an exponential distribution with parameter $\lambda > 0$.

In order to prove the next proposition we note that more generally, we have the equivalence

$$\{T_n > t\} = \{N_t < n\}, \qquad t \geq 0, \quad n \geq 1.$$

Indeed, stating that the n-th jump time T_n is strictly larger than t is equivalent to saying that at most $n-1$ jumps of the Poisson process have occurred over the interval $[0, t]$, i.e. $N_t \leq n-1$. The next proposition shows that T_n has a gamma distribution with parameter (λ, n) for $n \geq 1$, also called the Erlang distribution in queueing theory.

Proposition 9.2 *The random variable T_n has the gamma probability density function*

$$x \longmapsto \lambda^n \mathrm{e}^{-\lambda x} \frac{x^{n-1}}{(n-1)!}$$

$x \geq 0$, $n \geq 1$.

Proof For $n = 1$ we have

$$\mathbb{P}(T_1 > t) = \mathbb{P}(N_t = 0) = \mathrm{e}^{-\lambda t}, \qquad t \geq 0,$$

and by induction on $n \geq 1$, assuming that

$$\mathbb{P}(T_{n-1} > t) = \lambda \int_t^\infty \mathrm{e}^{-\lambda s} \frac{(\lambda s)^{n-2}}{(n-2)!} ds,$$

at the rank $n-1$ with $n \geq 2$, we obtain

$$\begin{aligned}
\mathbb{P}(T_n > t) &= \mathbb{P}(T_n > t \geq T_{n-1}) + \mathbb{P}(T_{n-1} > t) \\
&= \mathbb{P}(N_t = n-1) + \mathbb{P}(T_{n-1} > t) \\
&= \mathrm{e}^{-\lambda t} \frac{(\lambda t)^{n-1}}{(n-1)!} + \lambda \int_t^\infty \mathrm{e}^{-\lambda s} \frac{(\lambda s)^{n-2}}{(n-2)!} ds \\
&= \lambda \int_t^\infty \mathrm{e}^{-\lambda s} \frac{(\lambda s)^{n-1}}{(n-1)!} ds, \qquad t \geq 0,
\end{aligned}$$

which proves the desired relation at the rank n, where we applied an integration by parts on $\mathbf{R}_+$ to derive the last line. □

Let now

$$\tau_k = T_{k+1} - T_k, \qquad k \geq 1,$$

denote the time spent in state $k \in \mathbf{N}$, with $T_0 = 0$. In addition to Proposition 9.2 we could show the following proposition which is based on the *strong Markov property*, see e.g. Theorem 6.5.4 of Norris (1998), (9.2.4) below and Exercise 5.8 in discrete time.

Proposition 9.3 *The random inter-jump times*

$$\tau_k := T_{k+1} - T_k$$

spent in state $k \in \mathbf{N}$ form a sequence of independent identically distributed random variables having the exponential distribution with parameter $\lambda > 0$, i.e.

$$\mathbb{P}(\tau_0 > t_0, \tau_1 > t_1, \ldots, \tau_n > t_n) = \mathrm{e}^{-\lambda(t_0+t_1+\cdots+t_n)}, \qquad t_0, t_1, \ldots, t_n \geq 0.$$

Random samples of Poisson process jump times can be generated using the following R code.

```
lambda = 2.0;n = 10
taun <- rexp(n)/lambda; Tn <- cumsum(taun)
taun
Tn
```

Similarly, random samples of Poisson process paths can be generated using the following code.

```
n<-100
x<-cumsum(rexp(50,rate=0.5))
y<-cumsum(c(0,rep(1,50)))
plot(stepfun(x,y),xlim = c(0,10),do.points = F,main="L=0.5")
```

In other words, we have

$$\begin{aligned} \mathbb{P}(\tau_0 > t_0, \tau_1 > t_1, \ldots, \tau_n > t_n) &= \mathbb{P}(\tau_0 > t_0) \times \cdots \times \mathbb{P}(\tau_n > t_n) \\ &= \prod_{k=0}^{n} \mathrm{e}^{-\lambda t_k} \\ &= \mathrm{e}^{-\lambda(t_0+\cdots+t_n)}, \end{aligned} \tag{9.1.3}$$

for all $t_0, t_1, \ldots, t_n \geq 0$. In addition, from Proposition 9.2 the sum

$$T_k = \tau_0 + \tau_1 + \cdots + \tau_{k-1}, \qquad k \geq 1,$$

has a gamma distribution with parameter (λ, k), cf. also Exercise 9.13 for a proof in the particular case $k = 2$.

As the expectation of the exponentially distributed random variable τ_k with parameter $\lambda > 0$ is given by

$$\mathbb{E}[\tau_k] = \lambda \int_0^\infty x \mathrm{e}^{-\lambda x} dx = \frac{1}{\lambda},$$

we can check that the higher the intensity λ (i.e. the higher the probability of having a jump within a small interval), the smaller is the time spent in each state $k \in \mathbf{N}$ on average. Poisson random samples on arbitrary spaces will be considered in Chap. 11.

9.2 Continuous-Time Markov Chains

A $\mathbb{S}$-valued continuous-time stochastic process $(X_t)_{t\in\mathbf{R}_+}$ is said to be *Markov*, or to have the *Markov property* if, for all $t \in [s, \infty)$, the probability distribution of X_t given the past of the process up to time s is determined by the state X_s of the process at time s, and does not depend on the past values of X_u for $u < s$. In other words, for all

$$0 < s_1 < \cdots < s_{n-1} < s < t$$

we have

$$\mathbb{P}(X_t = j \mid X_s = i_n,\ X_{s_{n-1}} = i_{n-1}, \ldots, X_{s_1} = i_0) = \mathbb{P}(X_t = j \mid X_s = i_n). \tag{9.2.1}$$

In particular, we have

$$\mathbb{P}(X_t = j \mid X_s = i_n \text{ and } X_{s_{n-1}} = i_{n-1}) = \mathbb{P}(X_t = j \mid X_s = i_n).$$

Example

The Poisson process $(N_t)_{t\in\mathbf{R}_+}$ considered in Sect. 9.1 is a continuous-time Markov chain because it has independent increments by Condition (i) Sect. 9.1. The birth and death processes discussed below are also continuous-time Markov chains, although they may not have independent increments.

More generally, any continuous-time process $(X_t)_{t\in\mathbf{R}_+}$ with independent increments has the Markov property. Indeed, for all $j, i_n, \ldots, i_1 \in \mathbb{S}$ we have (note that $X_0 = 0$ here)

$$\begin{aligned}
&\mathbb{P}(X_t = j \mid X_s = i_n,\ X_{s_{n-1}} = i_{n-1}, \ldots, X_{s_1} = i_1) \\
&= \frac{\mathbb{P}(X_t = j, X_s = i_n, X_{s_{n-1}} = i_{n-1}, \ldots, X_{s_1} = i_1)}{\mathbb{P}(X_s = i_n,\ X_{s_{n-1}} = i_{n-1}, \ldots, X_{s_1} = i_1)} \\
&= \frac{\mathbb{P}(X_t - X_s = j - i_n, X_s = i_n, \ldots, X_{s_2} = i_2, X_{s_1} = i_1)}{\mathbb{P}(X_s = i_n, , \ldots, X_{s_2} = i_2, X_{s_1} = i_1)} \\
&= \frac{\mathbb{P}(X_t - X_s = j - i_n)\mathbb{P}(X_s = i_n, X_{s_{n-1}} = i_{n-1}, \ldots, X_{s_2} = i_2, X_{s_1} = i_1)}{\mathbb{P}(X_s = i_n, X_{s_{n-1}} = i_{n-1}, \ldots, X_{s_2} = i_2, X_{s_1} = i_1)}
\end{aligned}$$

$$= \mathbb{P}(X_t - X_s = j - i_n) = \frac{\mathbb{P}(X_t - X_s = j - i_n)\mathbb{P}(X_s = i_n)}{\mathbb{P}(X_s = i_n)}$$
$$= \frac{\mathbb{P}(X_t - X_s = j - i_n \text{ and } X_s = i_n)}{\mathbb{P}(X_s = i_n)}$$
$$= \frac{\mathbb{P}(X_t = j \text{ and } X_s = i_n)}{\mathbb{P}(X_s = i_n)} = \mathbb{P}(X_t = j \mid X_s = i_n),$$

cf. (4.1.6) for the discrete-time version of this argument. Hence, continuous-time processes with *independent increments* are Markov chains. However, *not all continuous-time Markov chains have independent increments*, and in fact the continuous-time Markov chains of interest in this chapter will not have independent increments.

Birth Process

The pure birth process behaves similarly to the Poisson process, by making the parameter of every exponential inter-jump time dependent on the current state of the process.

In other words, a continuous-time Markov chain $(X_t^b)_{t\in\mathbb{R}_+}$ such that[2]

$$\mathbb{P}(X_{t+h}^b = i+1 \mid X_t^b = i) = \mathbb{P}(X_{t+h}^b - X_t^b = 1 \mid X_t^b = i) \simeq \lambda_i h, \quad h \searrow 0, \quad i \in \mathbb{S},$$

and

$$\mathbb{P}(X_{t+h}^b = X_t^b \mid X_t^b = i) = \mathbb{P}(X_{t+h}^b - X_t^b = 0 \mid X_t^b = i) = 1 - \lambda_i h + o(h), \quad h \searrow 0, \quad i \in \mathbb{S}, \tag{9.2.2}$$

is called a *pure birth process* with (possibly) state-dependent birth rates $\lambda_i \geq 0, i \in \mathbb{S}$, see Fig. 9.2. Its inter-jump times $(\tau_k)_{k\geq 0}$ form a sequence of exponential independent random variables with state-dependent parameters.

This process is stationary in time because the rates λ_i, $i \in \mathbf{N}$, are independent of time t. The Poisson process $(N_t)_{t\in\mathbb{R}_+}$ is a pure birth process with state-independent birth rates $\lambda_i = \lambda > 0$, $i \in \mathbf{N}$.

As a consequence of (9.2.2) we can recover the fact that the time $\tau_{i,i+1}$ spent in state $\boxed{i}$ by the pure birth process $(X_t^b)_{t\in\mathbb{R}_+}$ started at state $\boxed{i}$ at time 0 before it moves to state $\boxed{i+1}$ has an exponential distribution with parameter λ_i. Indeed we have, using the Markov property in continuous time,

[2] Recall that by definition $f(h) \simeq g(h)$, $h \to 0$, if and only if $\lim_{h\to 0} f(h)/g(h) = 1$.

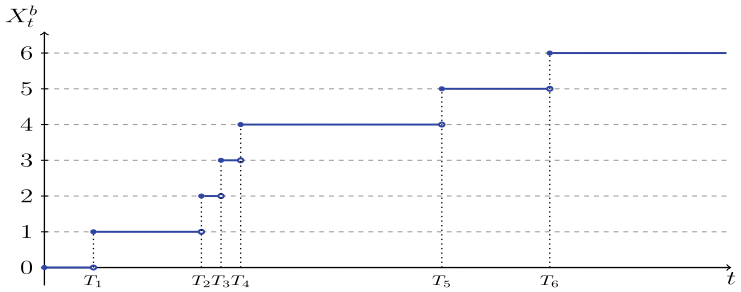


Fig. 9.2 Sample path of a birth process $(X_t^b)_{t\in\mathbb{R}_+}$

$$\begin{aligned}
\mathbb{P}(\tau_{i,i+1} > t+h \mid \tau_{i,i+1} > t \text{ and } X_0^b = i) &= \frac{\mathbb{P}(\tau_{i,i+1} > t+h \mid X_0^b = i)}{\mathbb{P}(\tau_{i,i+1} > t \mid X_0^b = i)} \\
&= \frac{\mathbb{P}(X_{t+h}^b = i \mid X_0^b = i)}{\mathbb{P}(X_t^b = i \mid X_0^b = i)} \\
&= \frac{\mathbb{P}(X_{t+h}^b = i \text{ and } X_0^b = i)\mathbb{P}(X_0^b = i)}{\mathbb{P}(X_t^b = i \text{ and } X_0^b = i)\mathbb{P}(X_0^b = i)} \\
&= \frac{\mathbb{P}(X_{t+h}^b = i \text{ and } X_0^b = i)}{\mathbb{P}(X_t^b = i \text{ and } X_0^b = i)} \\
&= \frac{\mathbb{P}(X_{t+h}^b = i,\ X_t^b = i,\ X_0^b = i)}{\mathbb{P}(X_t^b = i \text{ and } X_0^b = i)} \\
&= \mathbb{P}(X_{t+h}^b = i \mid X_t^b = i \text{ and } X_0^b = i) \\
&= \mathbb{P}(X_{t+h}^b = i \mid X_t^b = i) \\
&= \mathbb{P}(X_h^b = i \mid X_0^b = i) \\
&= \mathbb{P}(\tau_{i,i+1} > h \mid X_0^b = i) \\
&= 1 - \lambda_i h + o(h), \qquad (9.2.3)
\end{aligned}$$

which is often referred to as the *memoryless property* of Markov processes. In other words, since the above ratio is independent of $t > 0$ we get

$$\mathbb{P}(\tau_{i,i+1} > t+h \mid \tau_{i,i+1} > t \text{ and } X_0^b = i) = \mathbb{P}(\tau_{i,i+1} > h \mid X_0^b = i),$$

which means that the distribution of the waiting time after time t does not depend on t, cf. (12.1.1) in Chap. 12 for a similar argument.

From (9.2.3) we have

$$\begin{aligned}
&\frac{\mathbb{P}(\tau_{i,i+1} > t+h \mid X_0^b = i) - \mathbb{P}(\tau_{i,i+1} > t \mid X_0^b = i)}{h\mathbb{P}(\tau_{i,i+1} > t \mid X_0^b = i)} \\
&= \frac{1}{h}\left(\frac{\mathbb{P}(\tau_{i,i+1} > t+h \mid X_0^b = i)}{\mathbb{P}(\tau_{i,i+1} > t \mid X_0^b = i)} - 1\right) \\
&= \frac{\mathbb{P}(\tau_{i,i+1} > t+h \mid \tau_{i,i+1} > t \text{ and } X_0^b = i) - 1}{h} \\
&\simeq -\lambda_i, \qquad h \to 0,
\end{aligned}$$

which can be read as the differential equation

$$\frac{d}{dt}\log \mathbb{P}(\tau_{i,i+1} > t \mid X_0^b = i) = -\lambda_i,$$

where "log" denotes the *natural logarithm* "ln", with solution

$$\mathbb{P}(\tau_{i,i+1} > t \mid X_0^b = i) = \mathrm{e}^{-\lambda_i t}, \qquad t \geq 0, \tag{9.2.4}$$

i.e. $\tau_{i,i+1}$ is an exponentially distributed random variable with parameter λ_i, and the mean time spent at state $\boxed{i}$ before switching to state $\boxed{i+1}$ is given by

$$\mathbb{E}[\tau_{i,i+1} \mid X_0^b = i] = \frac{1}{\lambda_i}, \qquad i \in \mathbb{S},$$

see (9.3.12) below for the general case of continuous-time Markov chains. More generally, and similarly to (9.1.3) it can also be shown as a consequence of the *strong Markov property* that the sequence $(\tau_{j,j+1})_{j \geq i}$ is made of independent random variables which are respectively exponentially distributed with parameters λ_j, $j \geq i$.

Letting $T_{i,j}^b = \tau_{i,i+1} + \cdots + \tau_{j-1,j}$ denote the hitting time of state $\boxed{j}$ starting from state $\boxed{i}$ by the birth process $(X_t^b)_{t \in \mathbb{R}_+}$, we have the representation

$$X_t^b = i + \sum_{i<j<\infty} \mathbb{1}_{[T_{i,j}^b, \infty)}(t), \qquad t \geq 0.$$

Note that since the pure birth process has stationary increments, by Theorem 9.1 it has independent increments when the rates $\lambda_i = \lambda$ are state independent, i.e. when $(X_t^b)_{t \in \mathbb{R}_+}$ is a standard Poisson process with intensity $\lambda > 0$.

Death Process

A continuous-time Markov chain $(X_t^d)_{t \in \mathbb{R}_+}$ such that

$$\begin{cases}
\mathbb{P}(X_{t+h}^d - X_t^d = -1 \mid X_t^d = i) \simeq \mu_i h, & h \searrow 0, \quad i \in \mathbb{S}, \\
\mathbb{P}(X_{t+h}^d - X_t^d = 0 \mid X_t^d = i) = 1 - \mu_i h + o(h), & h \searrow 0, \quad i \in \mathbb{S},
\end{cases}$$

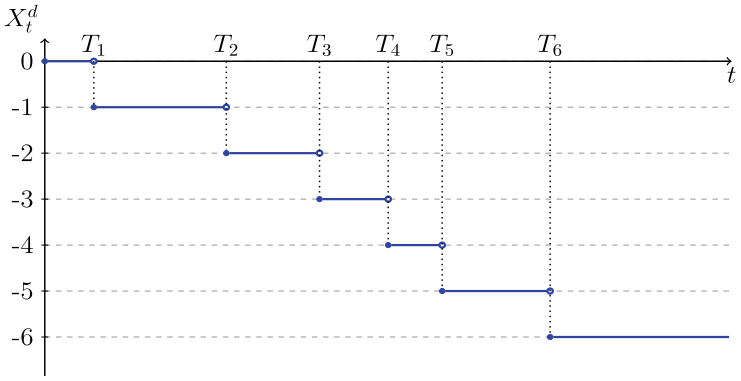


Fig. 9.3 Sample path of a death process $(X_t^d)_{t\in\mathbb{R}_+}$

is called a *pure death process* with (possibly) state-dependent death rates $\mu_i \geq 0, i \in \mathbb{S}$. Its inter-jump times $(\tau_k)_{k\geq 0}$ form a sequence of exponential independent random variables with state-dependent parameters, see Fig. 9.3.

In the case of a pure death process $(X_t^d)_{t\in\mathbb{R}_+}$ we denote by $\tau_{i,i-1}$ the time spent in state $\textcircled{i}$ by $(X_t^d)_{t\in\mathbb{R}_+}$ before it moves to state $\boxed{i-1}$. Similarly to the pure birth process, that the sequence $(\tau_{j,j-1})_{j\leq i}$ is made of independent random variables which are exponentially distributed with parameter μ_j, $j \leq i$, with

$$\mathbb{P}(\tau_{j,j-1} > t) = \mathrm{e}^{-\mu_j t}, \qquad t \geq 0,$$

and

$$\mathbb{E}[\tau_{i,i-1}] = \frac{1}{\mu_i}, \qquad i \in \mathbb{S}.$$

Letting $T_{i,j}^d = \tau_{i,i-1} + \cdots + \tau_{j+1,j}$ denote the hitting time of state $\textcircled{j}$ starting from state $\textcircled{i}$ by the death process $(X_t^d)_{t\in\mathbb{R}_+}$ we have the representation

$$X_t^d = i - \sum_{-\infty<j<i} \mathbb{1}_{[T_{i,j}^d,\infty)}(t), \qquad t \geq 0.$$

When $(N_t)_{t\in\mathbb{R}_+}$ is a Poisson process, the process $(-N_t)_{t\in\mathbb{R}_+}$ is a pure death process with state-independent death rates $\mu_n = \lambda > 0, n \in \mathbf{N}$.

Birth and Death Process

A continuous-time Markov chain $(X_t)_{t\in\mathbb{R}_+}$ such that, for all $i \in \mathbb{S}$,

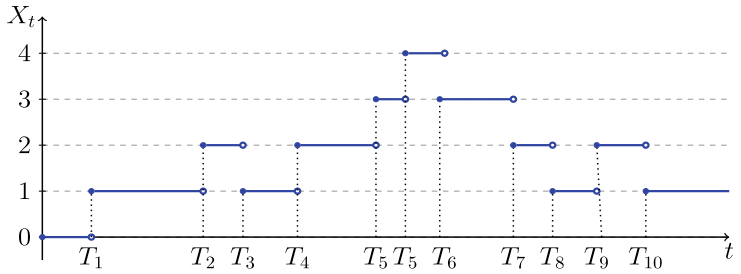


Fig. 9.4 Sample path of a birth and death process $(X_t)_{t\in\mathbb{R}_+}$

$$\begin{cases} \mathbb{P}(X_{t+h} - X_t = 1 \mid X_t = i) \simeq \lambda_i h, \quad h \searrow 0, & \text{(9.2.5a)} \\ \mathbb{P}(X_{t+h} - X_t = -1 \mid X_t = i) \simeq \mu_i h, \quad h \searrow 0, \text{ and} \\ \mathbb{P}(X_{t+h} - X_t = 0 \mid X_t = i) = 1 - (\lambda_i + \mu_i)h + o(h), \quad h \searrow 0, & \text{(9.2.5b)} \end{cases}$$

is called a *birth and death process* with (possibly) state-dependent birth rates $\lambda_i \geq 0$ and death rates $\mu_i \geq 0$, $i \in \mathbb{S}$, see Fig. 9.4.

The birth and death process $(X_t)_{t\in\mathbb{R}_+}$ can be built as

$$X_t = X_t^b + X_t^d, \qquad t \geq 0,$$

in which case the time τ_i spent in state $\textcircled{i}$ by $(X_t)_{t\in\mathbb{R}_+}$ satisfies the identity in distribution

$$\tau_i = \min(\tau_{i,i+1}, \tau_{i,i-1})$$

i.e. τ_i is an exponentially distributed random variable with parameter $\lambda_i + \mu_i$ and

$$\mathbb{E}[\tau_i] = \frac{1}{\lambda_i + \mu_i}.$$

Indeed, since $\tau_{i,i+1}$ and $\tau_{i,i-1}$ are two independent exponentially distributed random variables with parameters λ_i and μ_i, we have

$$\begin{aligned} \mathbb{P}(\min(\tau_{i,i+1} \text{ and } \tau_{i,i-1}) > t) &= \mathbb{P}(\tau_{i,i+1} > t \text{ and } \tau_{i,i-1} > t) \\ &= \mathbb{P}(\tau_{i,i+1} > t)\mathbb{P}(\tau_{i,i-1} > t) \\ &= \mathrm{e}^{-t(\lambda_i+\mu_i)}, \qquad t \geq 0, \end{aligned}$$

hence $\tau_i = \min(\tau_{i,i+1}, \tau_{i,i-1})$ is an exponentially distributed random variable with parameter $\lambda_i + \mu_i$, cf. also (1.5.8) in Chap. 1.

9.3 Transition Semigroup and Infinitesimal Generator

The transition semigroup of the continuous time Markov process $(X_t)_{t\in\mathbf{R}_+}$ is the family $(P(t))_{t\in\mathbf{R}_+}$ of matrices determined by

$$P_{i,j}(t) := \mathbb{P}(X_{t+s} = j \mid X_s = i), \qquad i, j \in \mathbb{S}, \quad s, t \geq 0,$$

where we assume that the probability $\mathbb{P}(X_{t+s} = j \mid X_s = i)$ does not depend on $s \in \mathbf{R}_+$. In this case, the Markov process $(X_t)_{t\in\mathbf{R}_+}$ is said to be *time homogeneous*.

Definition 9.4 A continuous-time Markov chain $(X_t)_{t\in\mathbf{R}_+}$ is *irreducible* if for all $t > 0$, $P(t)$ is the transition matrix of an irreducible discrete-time chain.

Note that we always have

$$P(0) = \mathrm{I}.$$

These data can be recorded as a time-dependent matrix indexed by $\mathbb{S}^2 = \mathbb{S} \times \mathbb{S}$, called the *transition semigroup* of the Markov process:

$$\left[\, P_{i,j}(t) \,\right]_{i,j\in\mathbb{S}} = \left[\, \mathbb{P}(X_{t+s} = j \mid X_s = i) \,\right]_{i,j\in\mathbb{S}},$$

also written as

$$\left[\, P_{i,j}(t) \,\right]_{i,j\in\mathbb{S}} = \begin{bmatrix} \ddots & \vdots & \vdots & \vdots & \vdots & \vdots & \iddots \\ \cdots & P_{-2,-2}(t) & P_{-2,-1}(t) & P_{-2,0}(t) & P_{-2,1}(t) & P_{-2,2}(t) & \cdots \\ \cdots & P_{-1,-2}(t) & P_{-1,-1}(t) & P_{-1,0}(t) & P_{-1,1}(t) & P_{-1,2}(t) & \cdots \\ \cdots & P_{0,-2}(t) & P_{0,-1}(t) & P_{0,0}(t) & P_{0,1}(t) & P_{0,2}(t) & \cdots \\ \cdots & P_{1,-2}(t) & P_{1,-1}(t) & P_{1,0}(t) & P_{1,1}(t) & P_{1,2}(t) & \cdots \\ \cdots & P_{2,-2}(t) & P_{2,-1}(t) & P_{2,0}(t) & P_{2,1}(t) & P_{2,2}(t) & \cdots \\ \iddots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}.$$

As in the discrete-time case, note the inversion of the order of indices (i, j) between $\mathbb{P}(X_{t+s} = j \mid X_s = i)$ and $P_{i,j}(t)$. In particular, the initial state $\textcircled{i}$ correspond to a *row number* in the matrix $P(t)$, while the final state $\textcircled{j}$ corresponds to a *column number*.

Due to the relation

$$\sum_{j\in\mathbb{S}} \mathbb{P}(X_{t+s} = j \mid X_s = i) = 1, \qquad i \in \mathbb{S}, \tag{9.3.1}$$

all *rows* of the transition matrix semigroup $(P(t))_{t\in\mathbf{R}_+}$ satisfy the condition

$$\sum_{j\in\mathbb{S}} P_{i,j}(t) = 1,$$

for $i \in \mathbb{S}$. In the sequel we will only consider $\mathbf{N}$-valued Markov process, and in this case the *transition semigroup* $(P(t))_{t\in\mathbf{R}_+}$ of the Markov process is written as

$$P(t) = \left[\, P_{i,j}(t) \,\right]_{i,j\in\mathbf{N}} = \begin{bmatrix} P_{0,0}(t) & P_{0,1}(t) & P_{0,2}(t) & \cdots \\ P_{1,0}(t) & P_{1,1}(t) & P_{1,2}(t) & \cdots \\ P_{2,0}(t) & P_{2,1}(t) & P_{2,2}(t) & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}.$$

From (9.3.1) we have

$$\sum_{j\geq 0} P_{i,j}(t) = 1,$$

for all $i \in \mathbf{N}$ and $t \geq 0$.
Exercise: Write down the transition semigroup $\left[\, P_{i,j}(t) \,\right]_{i,j\in\mathbf{N}}$ of the Poisson process $(N_t)_{t\in\mathbf{R}_+}$.

We can show that

$$\left[\, P_{i,j}(t) \,\right]_{i,j\in\mathbf{N}} = \begin{bmatrix} \mathrm{e}^{-\lambda t} & \lambda t\mathrm{e}^{-\lambda t} & \dfrac{\lambda^2 t^2}{2}\mathrm{e}^{-\lambda t} & \cdots \\ 0 & \mathrm{e}^{-\lambda t} & \lambda t\mathrm{e}^{-\lambda t} & \cdots \\ 0 & 0 & \mathrm{e}^{-\lambda t} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}.$$

Indeed we have

$$\begin{aligned} P_{i,j}(t) &= \mathbb{P}(N_{s+t} = j \mid N_s = i) = \frac{\mathbb{P}(N_{s+t} = j \text{ and } N_s = i)}{\mathbb{P}(N_s = i)} \\ &= \frac{\mathbb{P}(N_{s+t} - N_s = j - i \text{ and } N_s = i)}{\mathbb{P}(N_s = i)} = \frac{\mathbb{P}(N_{s+t} - N_s = j - i)\mathbb{P}(N_s = i)}{\mathbb{P}(N_s = i)} \\ &= \mathbb{P}(N_{s+t} - N_s = j - i) = \begin{cases} \mathrm{e}^{-\lambda t}\dfrac{(\lambda t)^{j-i}}{(j-i)!} & \text{if } j \geq i, \\ 0 & \text{if } j < i. \end{cases} \end{aligned}$$

In case the Markov process $(X_t)_{t\in\mathbf{R}_+}$ takes values in the finite state space $\mathbb{S} = \{0, 1, \ldots, N\}$ its transition semigroup will simply have the form

$$P(t) = \left[\, P_{i,j}(t) \,\right]_{0\leq i,j\leq N} = \begin{bmatrix} P_{0,0}(t) & P_{0,1}(t) & P_{0,2}(t) & \cdots & P_{0,N}(t) \\ P_{1,0}(t) & P_{1,1}(t) & P_{1,2}(t) & \cdots & P_{1,N}(t) \\ P_{2,0}(t) & P_{2,1}(t) & P_{2,2}(t) & \cdots & P_{2,N}(t) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ P_{N,0}(t) & P_{N,1}(t) & P_{N,2}(t) & \cdots & P_{N,N}(t) \end{bmatrix}.$$

As noted above, the semigroup matrix $P(t)$ is a convenient way to record the values of $\mathbb{P}(X_{t+s} = j \mid X_s = i)$ in a table.

Proposition 9.5 *The family* $(P(t))_{t\in\mathbf{R}_+}$ *satisfies the relation*

$$P(s+t) = P(s)P(t) = P(t)P(s), \tag{9.3.2}$$

which is called the semigroup property.

Proof Using the Markov property and denoting by $\mathbb{S}$ the state space of the process, by standard arguments based on the *law of total probability* (1.3.1) for the probability measure $\mathbb{P}(\cdot \mid X_0 = i)$ and the Markov property (9.2.1), we have

$$\begin{aligned}
&P_{i,j}(t+s) = \mathbb{P}(X_{t+s} = j \mid X_0 = i) \\
&= \sum_{l\in\mathbb{S}} \mathbb{P}(X_{t+s} = j \text{ and } X_s = l \mid X_0 = i) = \sum_{l\in\mathbb{S}} \frac{\mathbb{P}(X_{t+s} = j,\ X_s = l,\ X_0 = i)}{\mathbb{P}(X_0 = i)} \\
&= \sum_{l\in\mathbb{S}} \frac{\mathbb{P}(X_{t+s} = j,\ X_s = l,\ X_0 = i)}{\mathbb{P}(X_s = l \text{ and } X_0 = i)} \frac{\mathbb{P}(X_s = l \text{ and } X_0 = i)}{\mathbb{P}(X_0 = i)} \\
&= \sum_{l\in\mathbb{S}} \mathbb{P}(X_{t+s} = j \mid X_s = l \text{ and } X_0 = i)\mathbb{P}(X_s = l \mid X_0 = i) \\
&= \sum_{l\in\mathbb{S}} \mathbb{P}(X_{t+s} = j \mid X_s = l)\mathbb{P}(X_s = l \mid X_0 = i) = \sum_{l\in\mathbb{S}} P_{i,l}(s)P_{l,j}(t) \\
&= [P(s)P(t)]_{i,j},
\end{aligned}$$

$i, j \in \mathbb{S}$, $s, t \geq 0$. We have shown the relation

$$P_{i,j}(s+t) = \sum_{l\in\mathbb{S}} P_{i,l}(s)P_{l,j}(t),$$

which leads to (9.3.2). □

From (9.3.2) property one can check in particular that the matrices $P(s)$ and $P(t)$ *commute*, i.e. we have

$$P(s)P(t) = P(t)P(s), \qquad s, t \geq 0.$$

Example

For the transition semigroup $(P(t))_{t\in\mathbf{R}_+}$ of the Poisson process we can check by hand computation that

$$P(s)P(t) = \begin{bmatrix} \mathrm{e}^{-\lambda s} & \lambda s\mathrm{e}^{-\lambda s} & \frac{\lambda^2}{2}s^2\mathrm{e}^{-\lambda s} & \cdots \\ 0 & \mathrm{e}^{-\lambda s} & \lambda s\mathrm{e}^{-\lambda s} & \cdots \\ 0 & 0 & \mathrm{e}^{-\lambda s} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} \times \begin{bmatrix} \mathrm{e}^{-\lambda t} & \lambda t\mathrm{e}^{-\lambda t} & \frac{\lambda^2}{2}t^2\mathrm{e}^{-\lambda t} & \cdots \\ 0 & \mathrm{e}^{-\lambda t} & \lambda t\mathrm{e}^{-\lambda t} & \cdots \\ 0 & 0 & \mathrm{e}^{-\lambda t} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

$$= \begin{bmatrix} \mathrm{e}^{-\lambda(s+t)} & \lambda(s+t)\mathrm{e}^{-\lambda(s+t)} & \frac{\lambda^2}{2}(s+t)^2\mathrm{e}^{-\lambda(s+t)} & \cdots \\ 0 & \mathrm{e}^{-\lambda(s+t)} & \lambda(s+t)\,\mathrm{e}^{-\lambda(s+t)} & \cdots \\ 0 & 0 & \mathrm{e}^{-\lambda(s+t)} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

$$= P(s+t).$$

The above identity can be recovered by the following calculation, for all $0 \leq i \leq j$, which amounts to saying that the sum of two independent Poisson random variables with parameters s and t has a Poisson distribution with parameter $s+t$, cf. the solution of Exercise 1.7-(a). We have $P_{i,j}(s) = 0$, $i > j$, and $P_{l,j}(t) = 0$, $l > j$, hence

$$[P(s)P(t)]_{i,j} = \sum_{l\geq 0} P_{i,l}(s)P_{l,j}(t) = \sum_{l=i}^{j} P_{i,l}(s)P_{l,j}(t)$$

$$= \mathrm{e}^{-\lambda s-\lambda t}\sum_{l=i}^{j}\frac{(\lambda s)^{l-i}}{(l-i)!}\frac{(\lambda t)^{j-l}}{(j-l)!} = \mathrm{e}^{-\lambda s-\lambda t}\frac{1}{(j-i)!}\sum_{l=i}^{j}\binom{j-i}{l-i}(\lambda s)^{l-i}(\lambda t)^{j-l}$$

$$= \mathrm{e}^{-\lambda s-\lambda t}\frac{1}{(j-i)!}\sum_{l=0}^{j-i}\binom{j-i}{l}(\lambda s)^{l}(\lambda t)^{j-i-l} = \mathrm{e}^{-\lambda(s+t)}\frac{1}{(j-i)!}(\lambda s+\lambda t)^{j-i}$$

$$= P_{i,j}(s+t), \qquad s,t \geq 0.$$

Infinitesimal Generator

The infinitesimal generator of a continuous-time Markov process allows us to encode all properties of the process $(X_t)_{t\in\mathbb{R}_+}$ in a single matrix.

By differentiating the semigroup relation (9.3.2) with respect to t we get, by componentwise differentiation and assuming a finite state space $\mathbb{S}$,

$$\begin{aligned} P'(t) &= \lim_{h\searrow 0}\frac{P(t+h)-P(t)}{h} = \lim_{h\searrow 0}\frac{P(t)P(h)-P(t)}{h} \\ &= P(t)\lim_{h\searrow 0}\frac{P(h)-P(0)}{h} \\ &= P(t)Q, \end{aligned} \tag{9.3.3}$$

where

$$Q := P'(0) = \lim_{h \searrow 0} \frac{P(h) - P(0)}{h}$$

is called the *infinitesimal generator* of $(X_t)_{t \in \mathbf{R}_+}$.

When $\mathbb{S} = \{0, 1, \ldots, N\}$ we will denote by $\lambda_{i,j}$, $i, j \in \mathbb{S}$, the entries of the infinitesimal generator matrix $Q = (\lambda_{i,j})_{i,j \in \mathbb{S}}$, i.e.

$$Q = \frac{dP(t)}{dt}_{|t=0} = \left[\lambda_{i,j} \right]_{0 \leq i,j \leq N} = \begin{bmatrix} \lambda_{0,0} & \lambda_{0,1} & \lambda_{0,2} & \cdots & \lambda_{0,N} \\ \lambda_{1,0} & \lambda_{1,1} & \lambda_{1,2} & \cdots & \lambda_{1,N} \\ \lambda_{2,0} & \lambda_{2,1} & \lambda_{2,2} & \cdots & \lambda_{2,N} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \lambda_{N,0} & \lambda_{N,1} & \lambda_{N,2} & \cdots & \lambda_{N,N} \end{bmatrix}. \tag{9.3.4}$$

Denoting $Q = [\lambda_{i,j}]_{i,j \in \mathbb{S}}$, for all $i \in \mathbb{S}$ we have

$$\sum_{j \in \mathbb{S}} \lambda_{i,j} = \sum_{j \in \mathbb{S}} P'_{i,j}(0) = \frac{d}{dt} \sum_{j \in \mathbb{S}} P_{i,j}(t)_{|t=0} = \frac{d}{dt} \mathbb{1}_{|t=0} = 0,$$

hence the *rows* of the infinitesimal generator matrix $Q = [\lambda_{i,j}]_{i,j \in \mathbb{S}}$ always add up to 0, i.e.

$$\sum_{l \in \mathbb{S}} \lambda_{i,l} = \lambda_{i,i} + \sum_{l \neq i} \lambda_{i,l} = 0,$$

or

$$\lambda_{i,i} = -\sum_{l \neq i} \lambda_{i,l}. \tag{9.3.5}$$

Note that a state (i) such that $\lambda_{i,j} = 0$ for all $j \in \mathbb{S}$ is absorbing.

Equation (9.3.3), i.e.

$$P'(t) = P(t)Q, \qquad t > 0, \tag{9.3.6}$$

is called the *forward Kolmogorov equation*, cf. (1.2.4). In a similar way we can show the relation

$$P'(t) = QP(t), \qquad t > 0, \tag{9.3.7}$$

which is called the *backward Kolmogorov equation*.

The forward and backward Kolmogorov equations (9.3.6)–(9.3.7) can be solved either using the matrix exponential e^{tQ} defined as

$$\exp(tQ) := \sum_{n\geq 0} \frac{t^n}{n!} Q^n = \mathrm{I} + \sum_{n\geq 1} \frac{t^n}{n!} Q^n, \tag{9.3.8}$$

or by viewing the Kolmogorov equations (9.3.6)–(9.3.7) component by component as systems of differential equations.

In (9.3.8) above, $Q^0 = \mathrm{I}$ is the identity matrix, written as

$$\mathrm{I} = \begin{bmatrix} 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix}$$

when the state space is $\mathbb{S} = \{0, 1, \ldots, N\}$. Using matrix exponentials, the solution of (9.3.6) is given by

$$P(t) = P(0)\exp(tQ) = \exp(tQ), \quad t \geq 0.$$

We will often use the first order approximation in $h \to 0$ of

$$P(h) = \exp(hQ) = \mathrm{I} + \sum_{n\geq 1} \frac{h^n}{n!} Q^n = \mathrm{I} + hQ + \frac{h^2}{2!} Q^2 + \frac{h^3}{3!} Q^3 + \frac{h^4}{4!} Q^4 + \cdots,$$

given by the first order Taylor expansion

$$P(h) = \mathrm{I} + hQ + o(h), \qquad h \searrow 0, \tag{9.3.9}$$

where $o(h)$ is a function such that $\lim_{h\to 0} o(h)/h = 0$, i.e.

$$P(h) = \begin{bmatrix} 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix} + h \begin{bmatrix} \lambda_{0,0} & \lambda_{0,1} & \lambda_{0,2} & \cdots & \lambda_{0,N} \\ \lambda_{1,0} & \lambda_{1,1} & \lambda_{1,2} & \cdots & \lambda_{1,N} \\ \lambda_{2,0} & \lambda_{2,1} & \lambda_{2,2} & \cdots & \lambda_{2,N} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \lambda_{N,0} & \lambda_{N,1} & \lambda_{N,2} & \cdots & \lambda_{N,N} \end{bmatrix} + o(h), \quad h \searrow 0.$$

Relation (9.3.9) yields the transition probabilities over a small time interval of length $h > 0$, as:

$$\mathbb{P}(X_{t+h} = j \mid X_t = i) = P_{i,j}(h) = \begin{cases} \lambda_{i,j}h + o(h), & i \neq j, \quad h \searrow 0, \\ 1 + \lambda_{i,i}h + o(h), & i = j, \quad h \searrow 0, \end{cases}$$

and by (9.3.5) we also have

$$\mathbb{P}(X_{t+h} = j \mid X_t = i) = P_{i,j}(h) = \begin{cases} \lambda_{i,j}h + o(h), & i \neq j, \quad h \searrow 0, \\ 1 - h\sum_{l \neq i} \lambda_{i,l} + o(h), & i = j, \quad h \searrow 0. \end{cases} \tag{9.3.10}$$

For example, in the case of a two-state continuous-time Markov chain we have

$$Q = \begin{bmatrix} -\alpha & \alpha \\ \beta & -\beta \end{bmatrix},$$

with $\alpha, \beta \geq 0$, and

$$\begin{aligned} P(h) &= \mathrm{I} + hQ + o(h) \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + h \begin{bmatrix} -\alpha & \alpha \\ \beta & -\beta \end{bmatrix} + o(h) \\ &= \begin{bmatrix} 1 - \alpha h & \alpha h \\ \beta h & 1 - \beta h \end{bmatrix} + o(h), \end{aligned} \tag{9.3.11}$$

as $h \searrow 0$. In this case, $P(h)$ above has the same form as the transition matrix (4.5.1) of a *discrete-time* Markov chain with "small" time step $h > 0$ and "small" transition probabilities $h\alpha$ and $h\beta$, namely $h\alpha$ is the probability of switching from state $\textcircled{0}$ to state $\textcircled{1}$, and $h\beta$ is the probability of switching from state $\textcircled{1}$ to state $\textcircled{0}$ within a short period of time $h > 0$.

We note that since

$$\mathbb{P}(X_{t+h} = j \mid X_t = i) \simeq \lambda_{i,j}h, \qquad h \searrow 0, \quad i \neq j,$$

and

$$\mathbb{P}(X_{t+h} \neq j \mid X_t = i) = 1 - \lambda_{i,j}h + o(h), \qquad h \searrow 0, \quad i \neq j,$$

the transition of the process $(X_t)_{t \in \mathbf{R}_+}$ from state $\textcircled{i}$ to state $\textcircled{j}$ behaves identically to that of a Poisson process with intensity $\lambda_{i,j}$, cf. (9.1.2)–(9.1.2b) above. Similarly to the Poisson, birth and death processes, the relation

$$\mathbb{P}(X_{t+h} = j \mid X_t = i) = \lambda_{i,j}h + o(h), \qquad h \searrow 0, \quad i \neq j,$$

shows that the time $\tau_{i,j}$ spent in state $\textcircled{i}$ "before moving to state $\textcircled{j} \neq \textcircled{i}$", i.e. given the first jump is to state $\textcircled{j}$, is an *exponentially distributed random variable* with parameter $\lambda_{i,j}$, i.e.

$$\mathbb{P}(\tau_{i,j} > t) = \mathrm{e}^{-\lambda_{i,j}t}, \qquad t \geq 0, \tag{9.3.12}$$

and we have

$$\mathbb{E}[\tau_{i,j}] = \lambda_{i,j} \int_0^\infty t \mathrm{e}^{-t\lambda_{i,j}} dt = \frac{1}{\lambda_{i,j}}, \qquad i \neq j.$$

When $i = j$ we have

$$\mathbb{P}(X_{t+h} \neq i \mid X_t = i) \simeq h \sum_{l \neq i} \lambda_{i,l} = -\lambda_{i,i} h, \qquad h \searrow 0,$$

and

$$\mathbb{P}(X_{t+h} = i \mid X_t = i) = 1 - h \sum_{l \neq i} \lambda_{i,l} + o(h) = 1 + \lambda_{i,i} h + o(h), \quad h \searrow 0,$$

hence, by the same Poisson process analogy, the time τ_i spent in state $\textcircled{i}$ before the next transition to a different state is an exponentially distributed random variable with parameter $\sum_{j \neq i} \lambda_{i,j}$, i.e.

$$\mathbb{P}(\tau_i > t) = \exp\left(-t \sum_{j \neq i} \lambda_{i,j}\right) = \mathrm{e}^{t\lambda_{i,i}}, \qquad t \geq 0.$$

In other words, we can also write the time τ_i spent in state $\textcircled{i}$ as

$$\tau_i = \min_{\substack{j \in \mathbb{S} \\ j \neq i}} \tau_{i,j},$$

and this recovers the fact that τ_i is an exponential random variable with parameter $\sum_{j \neq i} \lambda_{i,j}$, since

$$\begin{aligned} \mathbb{P}(\tau_i > t) &= \mathbb{P}\left(\min_{\substack{j \in \mathbb{S} \\ j \neq i}} \tau_{i,j} > t\right) \\ &= \prod_{\substack{j \in \mathbb{S} \\ j \neq i}} \mathbb{P}(\tau_{i,j} > t) \\ &= \exp\left(-t \sum_{j \neq i} \lambda_{i,j}\right) = \mathrm{e}^{t\lambda_{i,i}}, \qquad t \geq 0. \end{aligned}$$

cf. (1.5.8) in Chap. 1. In addition we have

$$\mathbb{E}[\tau_i] = \sum_{l\neq i} \lambda_{i,l} \int_0^\infty t \exp\left(-t \sum_{l\neq i} \lambda_{i,l}\right) dt = \frac{1}{\sum_{l\neq i} \lambda_{i,l}} = -\frac{1}{\lambda_{i,i}},$$

and the times $(\tau_k)_{k\in\mathbb{S}}$ spent in each state $k \in \mathbb{S}$ form a sequence of independent random variables.

Examples

(i) Two-state continuous-time Markov chain.

For the two-state continuous-time Markov chain with generator

$$Q = \begin{bmatrix} -\alpha & \alpha \\ \beta & -\beta \end{bmatrix},$$

the mean time spent at state $\textcircled{0}$ is $1/\alpha$, whereas the mean time spent at state $\textcircled{1}$ is $1/\beta$. We will come back to this example in more detail in the following Sect. 9.4.

(ii) Poisson process.

The generator of the Poisson process is given by $\lambda_{i,j} = \mathbb{1}_{\{j=i+1\}}\lambda$, $i \neq j$, i.e.

$$Q = \left[\ \lambda_{i,j}\ \right]_{i,j\in\mathbf{N}} = \begin{bmatrix} -\lambda & \lambda & 0 & \cdots \\ 0 & -\lambda & \lambda & \cdots \\ 0 & 0 & -\lambda & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}.$$

From the relation $P(h) = \mathrm{I} + hQ + o(h)$ we recover the infinitesimal transition probabilities of the Poisson process as

$$\mathbb{P}(N_{t+h} - N_t = 1) = \mathbb{P}(N_{t+h} = i+1 \mid N_t = i) \simeq \lambda h,$$

$h \searrow 0$, $i \in \mathbf{N}$, and

$$\mathbb{P}(N_{t+h} - N_t = 0) = \mathbb{P}(N_{t+h} = i \mid N_t = i) = 1 - \lambda h + o(h),$$

$h \searrow 0$, $i \in \mathbf{N}$.

(iii) Pure birth process.

The generator of the pure birth process on $\mathbf{N} = \{0, 1, 2, \ldots\}$ is

$$Q = \left[\lambda_{i,j} \right]_{i,j\geq 0} = \begin{bmatrix} -\lambda_0 & \lambda_0 & 0 & \cdots \\ 0 & -\lambda_1 & \lambda_1 & \cdots \\ 0 & 0 & -\lambda_2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix},$$

in which the rate λ_i is (possibly) state-dependent. From the relation

$$P(h) = \mathrm{I} + hQ + o(h), \qquad h \searrow 0,$$

i.e.

$$P(h) = \begin{bmatrix} 1 & 0 & 0 & \cdots \\ 0 & 1 & 0 & \cdots \\ 0 & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} + h \begin{bmatrix} -\lambda_0 & \lambda_0 & 0 & \cdots \\ 0 & -\lambda_1 & \lambda_1 & \cdots \\ 0 & 0 & -\lambda_2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} + o(h), \quad h \searrow 0,$$

we recover the infinitesimal transition probabilities of the pure birth process as

$$\mathbb{P}(X_{t+h} - X_t = 1 \mid X_t = i) = \mathbb{P}(X_{t+h} = i+1 \mid X_t = i) \simeq \lambda_i h,$$

$h \searrow 0$, $i \in \mathbf{N}$, and

$$\mathbb{P}(X_{t+h} - X_t = 0 \mid X_t = i) = \mathbb{P}(X_{t+h} = i \mid X_t = i) = 1 - \lambda_i h + o(h),$$

$h \searrow 0$, $i \in \mathbf{N}$.

(iv) Pure death process.

The generator of the pure death process on $-\mathbf{N} = \{\ldots, -2, -1, 0\}$ is

$$Q = \left[\lambda_{i,j} \right]_{i,j\leq 0} = \begin{bmatrix} \cdots & 0 & \mu_0 & -\mu_0 \\ \cdots & \mu_1 & -\mu_1 & 0 \\ \cdots & -\mu_2 & 0 & 0 \\ \iddots & \vdots & \vdots & \vdots \end{bmatrix}.$$

From the relation

$$P(h) = \begin{bmatrix} \cdots & 0 & 0 & 1 \\ \cdots & 0 & 1 & 0 \\ \cdots & 1 & 0 & 0 \\ \iddots & \vdots & \vdots & \vdots \end{bmatrix} + h \begin{bmatrix} \cdots & 0 & \mu_0 & -\mu_0 \\ \cdots & \mu_1 & -\mu_1 & 0 \\ \cdots & -\mu_2 & 0 & 0 \\ \iddots & \vdots & \vdots & \vdots \end{bmatrix} + o(h), \quad h \searrow 0,$$

we recover the infinitesimal transition probabilities

$$\mathbb{P}(X_{t+h} - X_t = -1 \mid X_t = i) = \mathbb{P}(X_{t+h} = i-1 \mid X_t = i) \simeq \mu_i h, \quad h \searrow 0,$$

$i \in \mathbb{S}$, and

$$\mathbb{P}(X_{t+h} = i \mid X_t = i) = \mathbb{P}(X_{t+h} - X_t = 0 \mid X_t = i) = 1 - \mu_i h + o(h), \quad h \searrow 0,$$

$i \in \mathbb{S}$, of the pure death process.

(v) Birth and death process on $\{0, 1, \ldots, N\}$.

By (9.2.5a)–(9.2.5b) and (9.3.10), the generator of the birth and death process on $\{0, 1, \ldots, N\}$ is

$$\left[\lambda_{i,j} \right]_{0 \le i,j \le N} = \begin{bmatrix} -\lambda_0 & \lambda_0 & 0 & 0 & \cdots & \cdots & \cdots & 0 & 0 \\ \mu_1 & -\lambda_1 - \mu_1 & \lambda_1 & 0 & \cdots & \cdots & \cdots & 0 & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \ddots & \ddots & \ddots & \ddots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \ddots & \ddots & \ddots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 0 & \mu_{N-1} & -\lambda_{N-1} - \mu_{N-1} & \lambda_{N-1} \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & \mu_N & -\mu_N \end{bmatrix},$$

with $\mu_0 = \lambda_N = 0$.
From the Relation (9.3.9) we have

$$P(h) = \mathrm{I} + hQ + o(h), \qquad h \searrow 0,$$

and we recover the infinitesimal transition probabilities

$$\mathbb{P}(X_{t+h} - X_t = 1 \mid X_t = i) \simeq \lambda_i h, \qquad h \searrow 0, \quad i = 0, 1, \ldots, N,$$

and

$$\mathbb{P}(X_{t+h} - X_t = -1 \mid X_t = i) \simeq \mu_i h, \qquad h \searrow 0, \quad i = 0, 1, \ldots, N,$$

and

$$\mathbb{P}(X_{t+h} - X_t = 0 \mid X_t = i) = 1 - (\lambda_i + \mu_i)h + o(h), \qquad h \searrow 0, \quad i = 0, 1, \ldots, N,$$

of the birth and death process on $\{0, 1, \ldots, N\}$, with $\mu_0 = \lambda_N = 0$.
Recall that the time τ_i spent in state $\textcircled{i}$ is an exponentially distributed random variable with parameter $\lambda_i + \mu_i$ and we have

$$P_{i,i}(t) \ge \mathbb{P}(\tau_i > t) = \mathrm{e}^{-t(\lambda_i + \mu_i)}, \qquad t \ge 0,$$

and

$$\mathbb{E}[\tau_i] = \frac{1}{\lambda_i + \mu_i}.$$

In the case of a pure birth process we find

$$P_{i,i}(t) = \mathbb{P}(\tau_i > t) = \mathrm{e}^{-t\lambda_i}, \qquad t \geq 0,$$

and similarly for a pure death process. This allows us in particular to compute the diagonal entries in the matrix exponential $P(t) = \exp(tQ)$, $t \geq 0$.

When $\mathbb{S} = \{0, 1, \ldots, N\}$ with $\lambda_i = \lambda$ and $\mu_i = \mu$, $i = 1, 2, \ldots, N-1$, and $\lambda_0 = \mu_N = 0$, the above birth and death process becomes a continuous-time analog of the discrete-time gambling process.

9.4 Two-State Continuous-Time Markov Chain

In this section we consider a continuous-time Markov process with state space $\mathbb{S} = \{0, 1\}$, in the same way as in Sect. 4.5 which covered the two-state Markov chain in discrete time. This continuous-time Markov chain is also called a *telegraph process*.

In this case, the infinitesimal generator Q of $(X_t)_{t\in\mathbf{R}_+}$ has the form

$$Q = \begin{bmatrix} -\alpha & \alpha \\ \beta & -\beta \end{bmatrix}, \tag{9.4.1}$$

with $\alpha, \beta \geq 0$. The forward Kolmogorov equation (9.3.6) reads

$$P'(t) = P(t) \times \begin{bmatrix} -\alpha & \alpha \\ \beta & -\beta \end{bmatrix}, \qquad t > 0, \tag{9.4.2}$$

i.e.

$$\begin{bmatrix} P'_{0,0}(t) & P'_{0,1}(t) \\ P'_{1,0}(t) & P'_{1,1}(t) \end{bmatrix} = \begin{bmatrix} P_{0,0}(t) & P_{0,1}(t) \\ P_{1,0}(t) & P_{1,1}(t) \end{bmatrix} \times \begin{bmatrix} -\alpha & \alpha \\ \beta & -\beta \end{bmatrix}, \qquad t > 0,$$

or

$$\begin{cases} P'_{0,0}(t) = -\alpha P_{0,0}(t) + \beta P_{0,1}(t), & P'_{0,1}(t) = \alpha P_{0,0}(t) - \beta P_{0,1}(t), \\ P'_{1,0}(t) = -\alpha P_{1,0}(t) + \beta P_{1,1}(t), & P'_{1,1}(t) = \alpha P_{1,0}(t) - \beta P_{1,1}(t), \end{cases}$$

$t > 0$, which is a system of four differential equations with initial condition

$$P(0) = \begin{bmatrix} P_{0,0}(0) & P_{0,1}(0) \\ P_{1,0}(0) & P_{1,1}(0) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathrm{I}.$$

The solution of the forward Kolmogorov equation (9.4.2) is given by the matrix exponential

$$P(t) = P(0)\exp(tQ) = \exp(tQ) = \exp\left(t\begin{bmatrix} -\alpha & \alpha \\ \beta & -\beta \end{bmatrix}\right),$$

which is computed in the next Proposition 9.6.

Proposition 9.6 *The solution $P(t)$ of the forward Kolmogorov equation (9.4.2) is given by*

$$\begin{aligned} P(t) &= \begin{bmatrix} P_{0,0}(t) & P_{0,1}(t) \\ P_{1,0}(t) & P_{1,1}(t) \end{bmatrix} \\ &= \begin{bmatrix} \dfrac{\beta}{\alpha+\beta} + \dfrac{\alpha}{\alpha+\beta}\mathrm{e}^{-t(\alpha+\beta)} & \dfrac{\alpha}{\alpha+\beta} - \dfrac{\alpha}{\alpha+\beta}\mathrm{e}^{-t(\alpha+\beta)} \\ \dfrac{\beta}{\alpha+\beta} - \dfrac{\beta}{\alpha+\beta}\mathrm{e}^{-t(\alpha+\beta)} & \dfrac{\alpha}{\alpha+\beta} + \dfrac{\beta}{\alpha+\beta}\mathrm{e}^{-t(\alpha+\beta)} \end{bmatrix}, \end{aligned} \tag{9.4.3}$$

$t \geq 0$.

Proof We will compute the matrix exponential e^{tQ} by the diagonalization technique. The matrix Q has two eigenvectors

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} -\alpha \\ \beta \end{bmatrix},$$

with respective eigenvalues $\lambda_1 = 0$ and $\lambda_2 = -\alpha - \beta$. Hence it can be transformed into the *diagonal form*

$$Q = M \times D \times M^{-1}$$

as follows:

$$Q = \begin{bmatrix} 1 & -\alpha \\ 1 & \beta \end{bmatrix} \times \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \times \begin{bmatrix} \dfrac{\beta}{\alpha+\beta} & \dfrac{\alpha}{\alpha+\beta} \\ -\dfrac{1}{\alpha+\beta} & \dfrac{1}{\alpha+\beta} \end{bmatrix}.$$

Consequently we have

$$\begin{aligned} P(t) = \exp(tQ) &= \sum_{n\geq 0} \frac{t^n}{n!} Q^n = \sum_{n\geq 0} \frac{t^n}{n!} (M \times D \times M^{-1})^n \\ &= \sum_{n\geq 0} \frac{t^n}{n!} M \times D^n \times M^{-1} = M \times \left(\sum_{n\geq 0} \frac{t^n}{n!} D^n\right) \times M^{-1} \\ &= M \times \exp(tD) \times M^{-1} \\ &= \begin{bmatrix} 1 & -\alpha \\ 1 & \beta \end{bmatrix} \times \begin{bmatrix} \mathrm{e}^{t\lambda_1} & 0 \\ 0 & \mathrm{e}^{t\lambda_2} \end{bmatrix} \times \begin{bmatrix} \dfrac{\beta}{\alpha+\beta} & \dfrac{\alpha}{\alpha+\beta} \\ -\dfrac{1}{\alpha+\beta} & \dfrac{1}{\alpha+\beta} \end{bmatrix} \end{aligned}$$

$$
\begin{aligned}
&= \begin{bmatrix} 1 & -\alpha \\ 1 & \beta \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 0 & \mathrm{e}^{-t(\alpha+\beta)} \end{bmatrix} \times \begin{bmatrix} \dfrac{\beta}{\alpha+\beta} & \dfrac{\alpha}{\alpha+\beta} \\ -\dfrac{1}{\alpha+\beta} & \dfrac{1}{\alpha+\beta} \end{bmatrix} \\
&= \frac{1}{\alpha+\beta} \begin{bmatrix} \beta & \alpha \\ \beta & \alpha \end{bmatrix} + \frac{\mathrm{e}^{-t(\alpha+\beta)}}{\alpha+\beta} \begin{bmatrix} \alpha & -\alpha \\ -\beta & \beta \end{bmatrix} \\
&= \begin{bmatrix} \dfrac{\beta}{\alpha+\beta} + \dfrac{\alpha}{\alpha+\beta}\mathrm{e}^{-t(\alpha+\beta)} & \dfrac{\alpha}{\alpha+\beta} - \dfrac{\alpha}{\alpha+\beta}\mathrm{e}^{-t(\alpha+\beta)} \\ \dfrac{\beta}{\alpha+\beta} - \dfrac{\beta}{\alpha+\beta}\mathrm{e}^{-t(\alpha+\beta)} & \dfrac{\alpha}{\alpha+\beta} + \dfrac{\beta}{\alpha+\beta}\mathrm{e}^{-t(\alpha+\beta)} \end{bmatrix},
\end{aligned}
$$

$t > 0$. □

From Proposition 9.6 we obtain the probabilities

$$
\begin{cases}
\mathbb{P}(X_h = 0 \mid X_0 = 0) = \dfrac{\beta + \alpha \mathrm{e}^{-(\alpha+\beta)h}}{\alpha+\beta}, \\
\mathbb{P}(X_h = 1 \mid X_0 = 0) = \dfrac{\alpha}{\alpha+\beta}(1 - \mathrm{e}^{-(\alpha+\beta)h}), \\
\mathbb{P}(X_h = 0 \mid X_0 = 1) = \dfrac{\beta}{\alpha+\beta}(1 - e^{-h(\alpha+\beta)h}), \\
\mathbb{P}(X_h = 1 \mid X_0 = 1) = \dfrac{\alpha + \beta \mathrm{e}^{-(\alpha+\beta)h}}{\alpha+\beta}, \qquad h \geq 0.
\end{cases}
$$

In other words, (9.4.3) can be rewritten as

$$
P(h) = \begin{bmatrix} 1 - \dfrac{\alpha}{\alpha+\beta}(1 - \mathrm{e}^{-(\alpha+\beta)h}) & \dfrac{\alpha}{\alpha+\beta}(1 - \mathrm{e}^{-(\alpha+\beta)h}) \\ \dfrac{\beta}{\alpha+\beta}(1 - \mathrm{e}^{-(\alpha+\beta)h}) & 1 - \dfrac{\beta}{\alpha+\beta}(1 - \mathrm{e}^{-(\alpha+\beta)h}) \end{bmatrix}, \qquad h > 0, \tag{9.4.4}
$$

hence, since

$$
1 - \mathrm{e}^{-(\alpha+\beta)h} \simeq h(\alpha+\beta), \qquad h \searrow 0,
$$

the expression (9.4.4) above recovers (9.3.11) as $h \searrow 0$, i.e. we have

$$
P(h) = \begin{bmatrix} 1 - h\alpha & h\alpha \\ h\beta & 1 - h\beta \end{bmatrix} + o(h) = \mathrm{I} + hQ + o(h), \qquad h \searrow 0,
$$

which recovers (9.4.1).

From these expressions we can determine the large time behavior of the continuous-time Markov chain by taking limits as t goes to infinity:

$$
\lim_{t\to\infty} P(t) = \lim_{t\to\infty} \begin{bmatrix} \mathbb{P}(X_t = 0 \mid X_0 = 0) & \mathbb{P}(X_t = 1 \mid X_0 = 0) \\ \mathbb{P}(X_t = 0 \mid X_0 = 1) & \mathbb{P}(X_t = 1 \mid X_0 = 1) \end{bmatrix} = \frac{1}{\alpha+\beta} \begin{bmatrix} \beta & \alpha \\ \beta & \alpha \end{bmatrix},
$$

whenever $\alpha > 0$ or $\beta > 0$, whereas if $\alpha = \beta = 0$ we simply have

$$P(t) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathrm{I}, \qquad t \geq 0,$$

and the chain is constant. Note that in continuous time the limiting distribution of the two-state chain always exists (unlike in the discrete-time case), and convergence will be faster when $\alpha + \beta$ is larger. Hence we have

$$\lim_{t\to\infty} \mathbb{P}(X_t = 1 \mid X_0 = 0) = \lim_{t\to\infty} \mathbb{P}(X_t = 1 \mid X_0 = 1) = \frac{\alpha}{\alpha+\beta}$$

and

$$\lim_{t\to\infty} \mathbb{P}(X_t = 0 \mid X_0 = 0) = \lim_{t\to\infty} \mathbb{P}(X_t = 0 \mid X_0 = 1) = \frac{\beta}{\alpha+\beta}$$

and

$$\pi = [\pi_0, \pi_1] = \left(\frac{\beta}{\alpha+\beta}, \frac{\alpha}{\alpha+\beta}\right) = \left(\frac{1/\alpha}{1/\alpha+1/\beta}, \frac{1/\beta}{1/\alpha+1/\beta}\right)$$

appears as a *limiting distribution* as t goes to infinity, provided that $(\alpha, \beta) \neq (0, 0)$. This means that whatever the starting point X_0, the probability of being at $\textcircled{1}$ after a "large" time is close to $\alpha/(\alpha+\beta)$, while the probability of being at $\textcircled{0}$ is close to $\beta/(\alpha+\beta)$.

Next, we consider a simulation of the two-state continuous Markov chain with infinitesimal generator

$$Q = \begin{bmatrix} -20 & 20 \\ 40 & -40 \end{bmatrix},$$

i.e. $\alpha = 20$ and $\beta = 40$. Figure 9.5 represents a sample path $(x_t)_{t\in\mathbb{R}_+}$ of the continuous-time chain, while Fig. 9.6 represents the sample average

$$y_t = \frac{1}{t}\int_0^t x_s ds, \qquad t \in [0, 1],$$

which counts the proportion of values of the chain in the state $\textcircled{1}$. This proportion is found to converge to $\alpha/(\alpha+\beta) = 1/3$, as a consequence of the Ergodic Theorem in continuous time, see (9.5.4) below.

9.5 Limiting and Stationary Distributions

A probability distribution $\pi = (\pi_i)_{i\in\mathbb{S}}$ on $\mathbb{S}$ is said to be *stationary* for $P(t)$ if it satisfies the equation

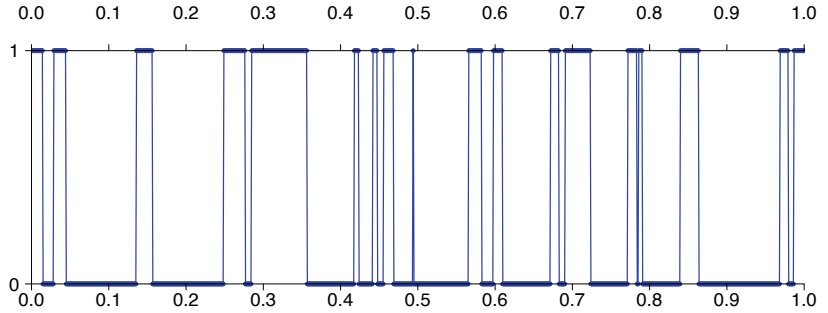


Fig. 9.5 Sample path of a continuous-time two-state chain with $\alpha = 20$ and $\beta = 40$

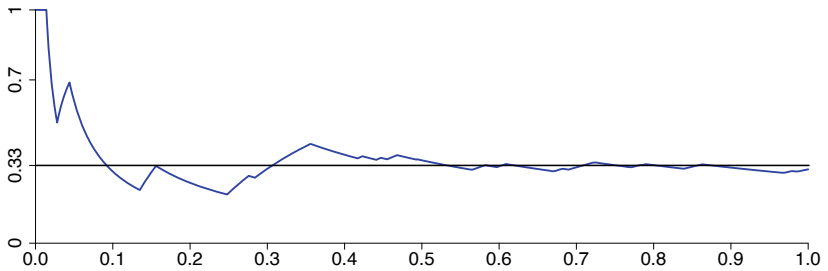


Fig. 9.6 The proportion of process values in the state 1 converges to $1/3 = \alpha/(\alpha + \beta)$

$$\pi P(t) = \pi, \qquad t \geq 0.$$

In Proposition 9.7, we show that the notion of stationary distribution admits a simpler characterization.

Proposition 9.7 *The probability distribution $\pi = (\pi_i)_{i \in \mathbb{S}}$ is stationary if and only if it satisfies the equation*

$$\pi Q = 0.$$

Proof Assuming that $\pi Q = 0$, we have

$$\begin{aligned} \pi P(t) &= \pi \exp(tQ) = \pi \sum F_{n \geq 0} \frac{t^n}{n!} Q^n \\ &= \pi \left(\mathrm{I} + \sum_{n \geq 1} \frac{t^n}{n!} Q^n \right) \\ &= \pi + \sum_{n \geq 1} \frac{t^n}{n!} \pi Q^n \\ &= \pi, \end{aligned}$$

since $\pi Q^n = \pi Q Q^{n-1} = 0$, $n \geq 1$. Conversely, the relation $\pi = \pi P(t)$ shows, by differentiation at $t = 0$, that

$$0 = \pi P'(0) = \pi Q. \qquad \square$$

When $\mathbb{S} = \{0, 1, \ldots, N\}$ and the generator Q has the form (9.3.4) the relation $\pi Q = 0$ reads

$$\pi_0 \lambda_{0,j} + \pi_1 \lambda_{1,j} + \cdots + \pi_N \lambda_{N,j} = 0, \qquad j = 0, 1, \ldots, N,$$

i.e.,

$$\sum_{\substack{i=0 \\ i \neq j}}^{N} \pi_i \lambda_{j,i} = -\pi_j \lambda_{j,j}, \qquad j = 0, 1, \ldots, N,$$

hence from (9.3.5) we find the balance condition

$$\sum_{\substack{i=0 \\ i \neq j}}^{N} \pi_i \lambda_{i,j} = \sum_{\substack{k=0 \\ k \neq j}}^{N} \pi_j \lambda_{j,k},$$

which can be interpreted by stating the equality between incoming and outgoing "flows" into and from state $\textcircled{j}$ are equal for all $j = 0, 1, \ldots, N$.

Next is the continuous-time analog of Proposition 7.7 in Sect. 7.2 (Fig. 9.7).

Proposition 9.8 *Consider a continuous-time Markov chain* $(X_t)_{t \in \mathbb{R}_+}$ *on a* finite state space $\mathbb{S}$*, which admits a limiting distribution given by*

$$\pi_j := \lim_{t \to \infty} \mathbb{P}(X_t = j \mid X_0 = i) = \lim_{t \to \infty} P_{i,j}(t), \qquad j \in \mathbb{S}, \tag{9.5.1}$$

independent of the initial state $i \in \mathbb{S}$*. Then we have*

$$\pi Q = 0, \tag{9.5.2}$$

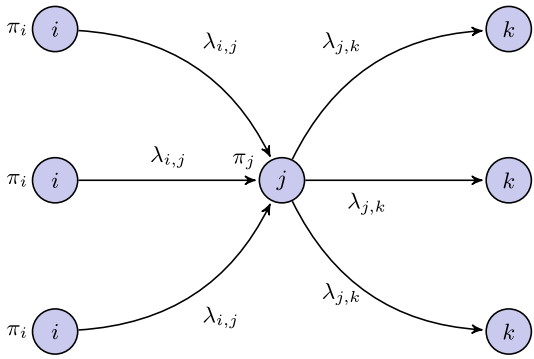


Fig. 9.7 Global balance condition (continuous time)

i.e. π *is a stationary distribution for the chain* $(X_t)_{t\in\mathbf{R}_+}$.

Proof Taking $\mathbb{S} = \{0, 1, \ldots, N\}$ we note that by (9.5.1) since the limiting distribution is independent of the initial state it satisfies

$$\lim_{t\to\infty} P(t) = \begin{bmatrix} \lim\limits_{t\to\infty} P_{0,0}(t) & \cdots & \lim\limits_{t\to\infty} P_{0,N}(t) \\ \vdots & \ddots & \vdots \\ \lim\limits_{t\to\infty} P_{N,0}(t) & \cdots & \lim\limits_{t\to\infty} P_{N,N}(t) \end{bmatrix}$$
$$= \begin{bmatrix} \pi_0 & \pi_1 & \cdots & \pi_N \\ \vdots & \vdots & \ddots & \vdots \\ \pi_0 & \pi_1 & \cdots & \pi_N \end{bmatrix} = \begin{bmatrix} \pi \\ \vdots \\ \pi \end{bmatrix},$$

where π is the row vector

$$\pi = [\pi_0, \pi_1, \ldots, \pi_n].$$

By the forward Kolmogorov equation (9.3.6) and (9.5.1) we find that the limit of $P'(t)$ exists as $t \to \infty$ since

$$\lim_{t\to\infty} P'(t) = \lim_{t\to\infty} P(t)Q = \begin{bmatrix} \pi \\ \vdots \\ \pi \end{bmatrix} Q.$$

On the other hand, since $P'(t)$ converges as $t \to \infty$ we should have

$$\lim_{t\to\infty} P'(t) = 0,$$

for the integral

$$P(t) = P(0) + \int_0^t P'(s)ds \tag{9.5.3}$$

to converge as $t \to \infty$. This shows that

$$\begin{bmatrix} \pi Q \\ \vdots \\ \pi Q \end{bmatrix} = \begin{bmatrix} \pi \\ \vdots \\ \pi \end{bmatrix} Q = 0$$

by (9.3.6), hence we have $\pi Q = 0$, or $\sum\limits_{i\in\mathbb{S}} \pi_i \lambda_{i,j} = 0$, $j \in \mathbb{S}$. □

Equation (9.5.2) is actually equivalent to

$$\pi = \pi(\mathrm{I} + hQ), \qquad h > 0,$$

which yields the stationary distribution of a discrete-time Markov chain with transition matrix $P(h) = \mathrm{I} + hQ + o(h)$ on "small" discrete intervals of length $h \searrow 0$.

Proposition 9.8 admits the following partial converse. More generally it can be shown, cf. Corollary 6.2 of Lalley (2013), that an irreducible continuous-time Markov chain admits its stationary distribution π as limiting distribution, similarly to the discrete-time Theorem 7.8, cf. also Proposition 1.1 Sigman (2014) in the positive recurrent case.

Proposition 9.9 *Consider an irreducible, continuous-time Markov chain* $(X_t)_{t\in\mathbb{R}_+}$ *on a* finite state space $\mathbb{S}$*, with stationary distribution* π*, i.e.*

$$\pi Q = 0,$$

and assume that the matrix Q *is diagonalizable. Then* $(X_t)_{t\in\mathbb{R}_+}$ *admits* π *as limiting distribution, i.e.*

$$\lim_{t\to\infty} P_{i,j}(t) = \lim_{t\to\infty} \mathbb{P}(X_t = j \mid X_0 = i) = \pi_j, \quad j \in \mathbb{S},$$

which is independent of the initial state $i \in \mathbb{S}$.

Proof By e.g. Theorem 2.1 in Chap. 10 of Karlin and Taylor (1981), since the chain is irreducible, $\lambda_1 = 0$ is an eigenvalue of Q with multiplicity one and eigenvector $u^{(1)} = (1, 1, \ldots, 1)$. In addition, the remaining eigenvectors $u^{(2)}, \ldots, u^{(n)} \in \mathbf{R}^n$ with eigenvalues $\lambda_2, \ldots, \lambda_n$ are orthogonal to the invariant (or stationary) distribution $[\pi_1, \pi_2, \ldots, \pi_n]$ of $(X_t)_{t\in\mathbb{R}_+}$ as we have $\lambda_k \langle u^{(k)}, \pi\rangle_{\mathbf{R}^n} = \langle Qu^{(k)}, \pi\rangle_{\mathbf{R}^n} = \langle u^{(k)}, Q^\top \pi\rangle_{\mathbf{R}^n} = \pi^\top Q u^{(k)} = 0$, $k = 2, \ldots, n$. Hence by diagonalization we have $Q = M^{-1} D M$ where the matrices M and M^{-1} take the form

$$M = \begin{bmatrix} \pi_1 & \cdots & \pi_n \\ M_{2,1} & \cdots & M_{2,n} \\ \vdots & \ddots & \vdots \\ M_{n,1} & \cdots & M_{n,n} \end{bmatrix} \quad \text{and} \quad M^{-1} = \begin{bmatrix} 1 & u_1^{(2)} & \cdots & u_1^{(n)} \\ 1 & u_2^{(2)} & \cdots & u_2^{(n)} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & u_n^{(2)} & \cdots & u_n^{(n)} \end{bmatrix},$$

and D is the diagonal matrix $D = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$. This allows us to compute the transition probabilities of $(X_t)_{t\in\mathbb{R}_+}$ as

$$P_{i,j}(t) = \mathbb{P}(X_t = j \mid X_0 = i) = [\exp(tQ)]_{i,j} = [M^{-1} \exp(tD) M]_{i,j}$$

where $\exp(tD)$ is the diagonal matrix

$$\exp(tD) = \mathrm{diag}(1, \mathrm{e}^{t\lambda_2}, \ldots, \mathrm{e}^{t\lambda_n}),$$

and the eigenvalues $\lambda_2, \ldots, \lambda_n$ have to be strictly negative, hence we have

$$\lim_{t\to\infty}[\mathbb{P}(X_t = j \mid X_0 = i)]_{1\le i,j\le n} = \lim_{t\to\infty}[M^{-1}\exp(tD)M]_{1\le i,j\le n}$$

$$= \begin{bmatrix} 1 & u_1^{(2)} & \cdots & u_1^{(n)} \\ 1 & u_2^{(2)} & \cdots & u_2^{(n)} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & u_n^{(2)} & \cdots & u_n^{(n)} \end{bmatrix} \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} \pi_1 & \cdots & \pi_n \\ M_{2,1} & \cdots & M_{2,n} \\ \vdots & \ddots & \vdots \\ M_{n,1} & \cdots & M_{n,n} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} \pi_1 & \cdots & \pi_n \\ M_{2,1} & \cdots & M_{2,n} \\ \vdots & \ddots & \vdots \\ M_{n,1} & \cdots & M_{n,n} \end{bmatrix}$$

$$= \begin{bmatrix} \pi_1 & \cdots & \pi_n \\ \pi_1 & \cdots & \pi_n \\ \vdots & \ddots & \vdots \\ \pi_1 & \cdots & \pi_n \end{bmatrix}.$$

□

The discrete-time Ergodic Theorem 7.12 also admits a continuous-time version with a similar proof, stating that if the chain $(X_t)_{t\in\mathbf{R}_+}$ is *irreducible* then the sample average of the number of visits to state $\textcircled{i}$ converges almost surely to π_i, i.e.,

$$\lim_{t\to\infty}\frac{1}{t}\int_0^t \mathbb{1}_{\{X_t=i\}}dt = \pi_i, \qquad i \in \mathbb{S}. \tag{9.5.4}$$

Examples

(i) Two-state Markov chain.
Consider a two-state Markov chain with infinitesimal generator

$$Q = \begin{bmatrix} -\alpha & \alpha \\ \beta & -\beta \end{bmatrix},$$

the limiting distribution solves $\pi Q = 0$, i.e.

$$\begin{cases} 0 = -\alpha\pi_0 + \beta\pi_1 \\ 0 = \alpha\pi_0 - \beta\pi_1, \end{cases}$$

with $\pi_0 + \pi_1 = 1$, i.e.

$$\pi = [\pi_0, \pi_1] = \left[\frac{\beta}{\alpha+\beta}, \frac{\alpha}{\alpha+\beta}\right]. \tag{9.5.5}$$

(ii) Birth and death process on $\mathbf{N}$.
Next, consider the birth and death process on $\mathbf{N}$ with infinitesimal generator

$$Q = \left[\ \lambda_{i,j}\ \right]_{i,j\in\mathbf{N}} = \begin{bmatrix} -\lambda_0 & \lambda_0 & 0 & 0 & 0 & \cdots \\ \mu_1 & -\lambda_1-\mu_1 & \lambda_1 & 0 & 0 & \cdots \\ 0 & \mu_2 & -\lambda_2-\mu_2 & \lambda_2 & 0 & \cdots \\ 0 & 0 & \mu_3 & -\lambda_3-\mu_3 & \lambda_3 & \cdots \\ \vdots & \vdots & \vdots & \ddots & \ddots & \ddots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \ddots \end{bmatrix},$$

the stationary distribution solves $\pi Q = 0$, i.e.

$$\begin{cases} 0 = & -\lambda_0\pi_0 + \mu_1\pi_1 \\ 0 = & \lambda_0\pi_0 - (\lambda_1+\mu_1)\pi_1 + \mu_2\pi_2 \\ 0 = & \lambda_1\pi_1 - (\lambda_2+\mu_2)\pi_2 + \mu_3\pi_3 \\ & \vdots \\ 0 = & \lambda_{j-1}\pi_{j-1} - (\lambda_j+\mu_j)\pi_j + \mu_{j+1}\pi_{j+1}, \\ & \vdots \end{cases}$$

i.e.

$$\begin{cases} \pi_1 = \dfrac{\lambda_0}{\mu_1}\pi_0 \\ \pi_2 = -\dfrac{\lambda_0}{\mu_2}\pi_0 + \dfrac{\lambda_1+\mu_1}{\mu_2}\pi_1 = -\dfrac{\lambda_0}{\mu_2}\pi_0 + \dfrac{\lambda_1+\mu_1}{\mu_2}\dfrac{\lambda_0}{\mu_1}\pi_0 = \dfrac{\lambda_1}{\mu_2}\dfrac{\lambda_0}{\mu_1}\pi_0 \\ \pi_3 = -\dfrac{\lambda_1}{\mu_3}\pi_1 + \dfrac{\lambda_2+\mu_2}{\mu_3}\pi_2 = -\dfrac{\lambda_1}{\mu_3}\dfrac{\lambda_0}{\mu_1}\pi_0 + \dfrac{\lambda_2+\mu_2}{\mu_3}\dfrac{\lambda_1}{\mu_2}\dfrac{\lambda_0}{\mu_1}\pi_0 = \dfrac{\lambda_2}{\mu_3}\dfrac{\lambda_1}{\mu_2}\dfrac{\lambda_0}{\mu_1}\pi_0 \\ \quad\vdots \\ \pi_{j+1} = \dfrac{\lambda_j\cdots\lambda_0}{\mu_{j+1}\cdots\mu_1}\pi_0. \\ \quad\vdots \end{cases}$$

Using the convention

$$\lambda_{j-1}\cdots\lambda_0 = \prod_{l=0}^{j-1}\lambda_l = 1 \quad\text{and}\quad \mu_j\cdots\mu_1 = \prod_{l=1}^{j}\mu_l = 1$$

in the case $j = 0$, we have

$$\begin{aligned} 1 = \sum_{j\geq 0}\pi_j &= \pi_0 + \pi_0\sum_{j\geq 0}\frac{\lambda_j\cdots\lambda_0}{\mu_{j+1}\cdots\mu_1} = \pi_0 + \pi_0\sum_{j\geq 1}\frac{\lambda_{j-1}\cdots\lambda_0}{\mu_j\cdots\mu_1} \\ &= \pi_0\sum_{i\geq 0}\frac{\lambda_0\cdots\lambda_{i-1}}{\mu_1\cdots\mu_i}, \end{aligned}$$

hence

$$\pi_0 = \frac{1}{\displaystyle\sum_{i\geq 0} \frac{\lambda_0 \cdots \lambda_{i-1}}{\mu_1 \cdots \mu_i}},$$

and

$$\pi_j = \frac{\lambda_0 \cdots \lambda_{j-1}}{\mu_1 \cdots \mu_j \displaystyle\sum_{i\geq 0} \frac{\lambda_0 \cdots \lambda_{i-1}}{\mu_1 \cdots \mu_i}}, \qquad j \in \mathbf{N}.$$

When $\lambda_i = \lambda$, $i \in \mathbf{N}$, and $\mu_i = \mu$, $i \geq 1$, this gives

$$\pi_j = \frac{\lambda^j}{\mu^j \displaystyle\sum_{i\geq 0} (\lambda/\mu)^i} = \left(1 - \frac{\lambda}{\mu}\right)\left(\frac{\lambda}{\mu}\right)^j, \qquad j \in \mathbf{N}.$$

provided that $\lambda < \mu$, hence in this case the stationary distribution is the geometric distribution with parameter λ/μ, otherwise the stationary distribution does not exist .

(iii) Birth and death process on $\mathbb{S} = \{0, 1, \ldots, N\}$.
The birth and death process on $\mathbb{S} = \{0, 1, \ldots, N\}$ has the infinitesimal generator

$$Q = [\,\lambda_{i,j}\,]_{0\leq i,j\leq N} = \begin{bmatrix} -\lambda_0 & \lambda_0 & 0 & \cdots & \cdots & 0 & 0 & 0 & 0 \\ \mu_1 & -\lambda_1 - \mu_1 & \lambda_1 & \cdots & \cdots & 0 & 0 & 0 & 0 \\ 0 & \ddots & \ddots & \ddots & \vdots & \vdots & 0 & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \ddots & \ddots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \vdots & \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & 0 & 0 & \cdots & \cdots & 0 & \mu_{N-1} & -\lambda_{N-1} - \mu_{N-1} & \lambda_{N-1} \\ 0 & 0 & 0 & \cdots & \cdots & 0 & 0 & \mu_N & -\mu_N \end{bmatrix},$$

we can apply (1.6.6) with $\lambda_j = 0$, $j \geq N$, and $\mu_j = 0$, $j \geq N+1$, which yields

$$\pi_j = \frac{\lambda_{j-1} \cdots \lambda_0}{\mu_j \cdots \mu_1 \displaystyle\sum_{i=0}^{N} \frac{\lambda_{i-1} \cdots \lambda_0}{\mu_i \cdots \mu_1}}, \qquad j \in \{0, 1, \ldots, N\},$$

and coincides with (9.5.5) when $N = 1$.

When $\lambda_i = \lambda$, $i \in \mathbf{N}$, and $\mu_i = \mu$, $i \geq 1$, this gives

$$\pi_j = \frac{1 - \lambda/\mu}{1 - (\lambda/\mu)^{N+1}} \left(\frac{\lambda}{\mu}\right)^j, \qquad j = 0, 1, \ldots, N,$$

which is a truncated geometric distribution since $\pi_j = 0$ for all $j > N$ and any $\lambda, \mu > 0$.

9.6 Discrete-Time Embedded Chain

Consider the sequence $(T_n)_{n\geq 0}$ the sequence of jump times of the continuous-time Markov process $(X_t)_{t\in\mathbf{R}_+}$, defined recursively by $T_0 = 0$, then

$$T_1 = \inf\{t > 0 \ : \ X_t \neq X_0\},$$

and

$$T_{n+1} = \inf\{t > T_n \ : \ X_t \neq X_{T_n}\}, \qquad n \geq 0.$$

The *embedded chain* of $(X_t)_{t\in\mathbf{R}_+}$ is the *discrete-time* Markov chain $(Z_n)_{n\geq 0}$ defined by $Z_0 = X_0$ and

$$Z_n := X_{T_n}, \qquad n \geq 1.$$

Figure 9.8 shows the graph of the embedded chain of a birth and death process.

The results of Chaps. 2–8 can now be applied to the discrete-time embedded chain. Figure 9.9 represents the discrete-time embedded chain associated to the path of Fig. 9.8, in which we have $Z_0 = 0$, $Z_1 = 1$, $Z_2 = 2$, $Z_3 = 3$, $Z_4 = 4$, $Z_5 = 3$, $Z_6 = 4$, $Z_7 = 3$, ...

For example, if $\lambda_0 > 0$ and $\mu_1 > 0$, the embedded chain of the two-state continuous-time Markov chain has the transition matrix

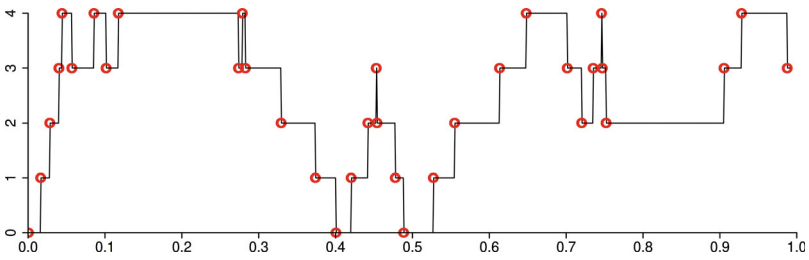


Fig. 9.8 Birth and death process $(X_t)_{t\in\mathbf{R}_+}$ with its embedded chain $(Z_n)_{n\geq 0}$

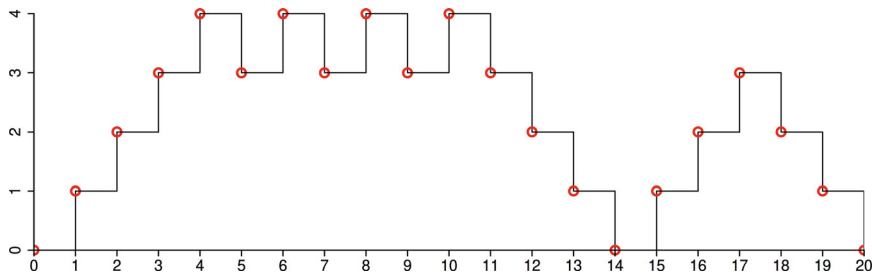


Fig. 9.9 Discrete-time embedded chain $(Z_n)_{n\geq 0}$ based on the path of Fig. 9.8

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \tag{9.6.1}$$

which switches permanently between the states (0) and (1).

In case one of the states $\{0, 1\}$ is absorbing, the transition matrix becomes

$$P = \begin{cases} \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, & \text{if } \lambda_0 = 0,\ \mu_1 > 0, \\ \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}, & \text{if } \lambda_0 > 0,\ \mu_1 = 0, \\ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, & \text{if } \lambda_0 = 0,\ \mu_1 = 0. \end{cases}$$

Birth and Death Embedded Chain

More generally, consider now the birth and death process with infinitesimal generator

$$Q = [\,\lambda_{i,j}\,]_{0\leq i,j\leq N} = \begin{bmatrix} -\lambda_0 & \lambda_0 & 0 & \cdots & \cdots & 0 & 0 & 0 & 0 \\ \mu_1 & -\lambda_1-\mu_1 & \lambda_1 & \cdots & \cdots & 0 & 0 & 0 & 0 \\ 0 & \ddots & \ddots & \ddots & \vdots & \vdots & 0 & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \ddots & \ddots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \vdots & \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & 0 & 0 & \cdots & \cdots & 0 & \mu_{N-1} & -\lambda_{N-1}-\mu_{N-1} & \lambda_{N-1} \\ 0 & 0 & 0 & \cdots & \cdots & 0 & 0 & \mu_N & -\mu_N \end{bmatrix}.$$

Given that a transition occurs from state (i) in a "short" time interval $[t, t+h]$, the probability that the chain switches to state $\boxed{i+1}$ is given by

$$\begin{aligned}\mathbb{P}(X_{t+h} = i+1 \mid \ X_t = i \text{ and } X_{t+h} - X_t \neq 0) &= \frac{\mathbb{P}(X_{t+h} = i+1 \text{ and } X_t = i)}{\mathbb{P}(X_{t+h} - X_t \neq 0 \text{ and } X_t = i)} \\ &= \frac{\mathbb{P}(X_{t+h} = i+1 \text{ and } X_t = i)}{\mathbb{P}(X_{t+h} - X_t \neq 0 \mid X_t = i)\mathbb{P}(X_t = i)} \\ &= \frac{\mathbb{P}(X_{t+h} = i+1 \mid X_t = i)}{\mathbb{P}(X_{t+h} - X_t \neq 0 \mid X_t = i)} \\ &= \frac{\mathbb{P}(X_{t+h} - X_t = 1 \mid X_t = i)}{\mathbb{P}(X_{t+h} - X_t \neq 0 \mid X_t = i)} \\ &\simeq \frac{\lambda_i h}{\lambda_i h + \mu_i h} \\ &= \frac{\lambda_i}{\lambda_i + \mu_i}, \qquad h \searrow 0, \quad i \in \mathbb{S},\end{aligned}$$

where we applied (9.3.10), hence the transition matrix of the embedded chain satisfies

$$P_{i,i+1} = \lim_{h \searrow 0} \mathbb{P}(X_{t+h} = i+1 \mid X_t = i \text{ and } X_{t+h} - X_t \neq 0) = \frac{\lambda_i}{\lambda_i + \mu_i}, \quad i \in \mathbb{S}. \tag{9.6.2}$$

This result can also be obtained from (1.5.9) which states that

$$\mathbb{P}(\tau_{i,i+1} < \tau_{i,i-1}) = \frac{\lambda_i}{\lambda_i + \mu_i}. \tag{9.6.3}$$

Similarly the probability that a given transition occurs from $\textcircled{i}$ to $\boxed{i-1}$ is

$$\mathbb{P}(X_{t+h} = i-1 \mid X_t = i \text{ and } X_{t+h} - X_t \neq 0) = \frac{\mu_i}{\lambda_i + \mu_i}, \qquad h \searrow 0, \quad i \in \mathbb{S},$$

which can also be obtained from (1.5.9) which states that

$$\mathbb{P}(\tau_{i,i-1} < \tau_{i,i+1}) = \frac{\mu_i}{\lambda_i + \mu_i}.$$

Hence we have

$$P_{i,i-1} = \lim_{h \searrow 0} \mathbb{P}(X_{t+h} = i-1 \mid X_t = i \text{ and } X_{t+h} - X_t \neq 0) = \frac{\mu_i}{\lambda_i + \mu_i}, \quad i \in \mathbb{S},$$

and the embedded chain $(Z_n)_{n \geq 0}$ has the transition matrix

$$P = \left[P_{i,j} \right]_{i,j \in \mathbb{S}}$$

$$= \begin{bmatrix} 0 & 1 & 0 & 0 & \cdots & 0 & 0 & 0 & 0 \\ \frac{\mu_1}{\lambda_1+\mu_1} & 0 & \frac{\lambda_1}{\lambda_1+\mu_1} & 0 & \cdots & 0 & 0 & 0 & 0 \\ 0 & \ddots & \ddots & \ddots & \vdots & \vdots & 0 & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \ddots & \ddots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \vdots & \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 & \frac{\mu_{N-1}}{\lambda_{N-1}+\mu_{N-1}} & 0 & \frac{\lambda_{N-1}}{\lambda_{N-1}+\mu_{N-1}} \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 1 & 0 \end{bmatrix},$$

provided that $\lambda_0 > 0$ and $\mu_N > 0$. When $N = 1$, this coincides with (9.6.1). In case $\lambda_0 = \mu_N = 0$, states $\textcircled{0}$ and $\textcircled{N}$ are both absorbing since the birth rate starting from $\textcircled{0}$ and the death rate starting from $\textcircled{N}$ are both 0, hence the transition matrix of the embedded chain can be written as

$$P = \left[P_{i,j} \right]_{0 \le i,j \le N}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & 0 \\ \frac{\mu_1}{\lambda_1+\mu_1} & 0 & \frac{\lambda_1}{\lambda_1+\mu_1} & 0 & \cdots & 0 & 0 & 0 & 0 \\ 0 & \ddots & \ddots & \ddots & \vdots & \vdots & 0 & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \ddots & \ddots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \vdots & \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 & \frac{\mu_{N-1}}{\lambda_{N-1}+\mu_{N-1}} & 0 & \frac{\lambda_{N-1}}{\lambda_{N-1}+\mu_{N-1}} \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & 1 \end{bmatrix},$$

which is the transition matrix of a gambling type process on $\{0, 1, \ldots, N\}$. When $N = 1$ this yields $P = \mathrm{I}$, which is consistent with the fact that a two-state Markov chain with two absorbing states is constant.

For example, for a continuous-time chain with infinitesimal generator

$$Q = \left[\lambda_{i,j} \right]_{0 \le i,j \le 4} = \begin{bmatrix} -10 & 10 & 0 & 0 & 0 \\ 10 & -20 & 10 & 0 & 0 \\ 0 & 10 & -30 & 20 & 0 \\ 0 & 0 & 10 & -40 & 30 \\ 0 & 0 & 0 & 20 & -20 \end{bmatrix},$$

the transition matrix of the embedded chain is

$$P = \left[P_{i,j} \right]_{0 \le i,j \le 4} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 1/3 & 0 & 2/3 & 0 \\ 0 & 0 & 1/4 & 0 & 3/4 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}.$$

In case the states ⓪ and ④ are absorbing, i.e.

$$Q = \left[\lambda_{i,j} \right]_{0 \le i,j \le 4} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 10 & -20 & 10 & 0 & 0 \\ 0 & 10 & -30 & 20 & 0 \\ 0 & 0 & 10 & -40 & 30 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

the transition matrix of the embedded chain becomes

$$P = \left[P_{i,j} \right]_{0 \le i,j \le 4} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 1/3 & 0 & 2/3 & 0 \\ 0 & 0 & 1/4 & 0 & 3/4 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

More generally, by (9.3.10) and (9.3.5) we could also show that the embedded chain of a continuous-time Markov chain with generator Q of the form (9.3.4) has the transition matrix

$$P = \left[P_{i,j} \right]_{i,j \in \mathbb{S}}$$

$$= \begin{bmatrix} 0 & -\frac{\lambda_{0,1}}{\lambda_{0,0}} & -\frac{\lambda_{0,2}}{\lambda_{0,0}} & \cdots\cdots & -\frac{\lambda_{0,N-1}}{\lambda_{0,0}} & -\frac{\lambda_{0,N}}{\lambda_{0,0}} \\ -\frac{\lambda_{1,0}}{\lambda_{1,1}} & 0 & -\frac{\lambda_{1,2}}{\lambda_{1,1}} & \cdots\cdots & -\frac{\lambda_{1,N-1}}{\lambda_{1,1}} & -\frac{\lambda_{1,N}}{\lambda_{1,1}} \\ \vdots & \ddots & \vdots & \vdots\ \vdots & \ddots & \vdots \\ \vdots & \ddots & \vdots & \vdots\ \vdots & \ddots & \vdots \\ \vdots & \ddots & \vdots & \vdots\ \vdots & \ddots & \vdots \\ -\frac{\lambda_{N-1,0}}{\lambda_{N-1,N-1}} & -\frac{\lambda_{N-1,1}}{\lambda_{N-1,N-1}} & \cdots & \cdots\cdots & 0 & -\frac{\lambda_{N-1,N}}{\lambda_{N-1,N-1}} \\ -\frac{\lambda_{N,0}}{\lambda_{N,N}} & -\frac{\lambda_{N,1}}{\lambda_{N,N}} & \cdots & \cdots\cdots & -\frac{\lambda_{N,N-1}}{\lambda_{N,N}} & 0 \end{bmatrix}$$

see (9.3.10), provided that $\lambda_{i,i} > 0$, $i = 0, 1, \ldots, N$.

9.7 Mean Absorption Time and Probabilities

Absorption Probabilities

The absorption probabilities of the continuous-time process $(X_t)_{t\in\mathbb{R}_+}$ can be computed based on the behaviour of the embedded chain $(Z_n)_{n\geq 0}$. In fact the continuous waiting time between two jumps has no influence on the absorption probabilities. Here, we consider only the simple example of birth and death processes, which can be easily generalized to more complex situations.

The basic idea is to perform a first step analysis on the underlying discrete-time embedded chain. Assume that state $\textcircled{0}$ is absorbing, i.e. $\lambda_0 = 0$, and let

$$T_0 = \inf\{t \geq 0 \ : \ X_t = 0\}$$

denote the absorption time of the chain into state $\textcircled{0}$. Let now

$$g_0(k) = \mathbb{P}(T_0 < \infty \mid X_0 = k), \qquad k = 0, 1, \ldots, N,$$

denote the probability of absorption in $\textcircled{0}$ starting from state $k \in \{0, 1, \ldots, N\}$. We have the boundary condition $g_0(0) = 1$, and by first step analysis on the chain $(Z_n)_{n\geq 1}$ we get

$$g_0(k) = \frac{\lambda_k}{\lambda_k + \mu_k} g_0(k+1) + \frac{\mu_k}{\lambda_k + \mu_k} g_0(k-1), \qquad k = 1, 2, \ldots, N-1.$$

When the rates $\lambda_k = \lambda$ and $\mu_k = \mu$ are independent of $k \in \{1, 2, \ldots, N-1\}$, this equation becomes

$$g_0(k) = p g_0(k+1) + q g_0(k-1), \qquad k = 1, 2, \ldots, N-1,$$

which is precisely Eq. (2.2.6) for the gambling process with

$$p = \frac{\lambda}{\lambda + \mu} \quad \text{and} \quad q = \frac{\mu}{\lambda + \mu}.$$

When $\lambda_0 = \mu_N = 0$ we have the boundary conditions

$$g_0(0) = 1 \quad \text{and} \quad g_0(N) = 0$$

since the state $\textcircled{N}$ becomes absorbing, and the solution becomes

$$g_0(k) = \frac{(\mu/\lambda)^k - (\mu/\lambda)^N}{1 - (\mu/\lambda)^N}, \qquad k = 0, 1, \ldots, N,$$

when $\lambda \neq \mu$, according to (2.2.12). When $\lambda = \mu$, Relation (2.2.13) shows that

$$g_0(k) = \frac{N-k}{N} = 1 - \frac{k}{N}, \qquad k = 0, 1, \ldots, N.$$

Mean Absorption Time

We may still use the embedded chain $(Z_n)_{n\geq 0}$ to compute the mean absorption time, using the mean inter-jump times. Here, unlike in the case of absorption probabilities, the random time spent by the continuous-time process $(X_t)_{t\in\mathbb{R}_+}$ should be taken into account in the calculation.

Next, we consider a birth and death process on the state space $\mathbb{S} = \{0, 1, \ldots, N\}$ with absorbing states $\textcircled{0}$ and $\textcircled{N}$. Recall that the the mean time spent at state $\textcircled{i}$, given that the next transition is from $\textcircled{i}$ to $\boxed{i+1}$, is equal to

$$\mathbb{E}[\tau_{i,i+1}] = \frac{1}{\lambda_i}, \qquad i = 1, 2, \ldots, N-1,$$

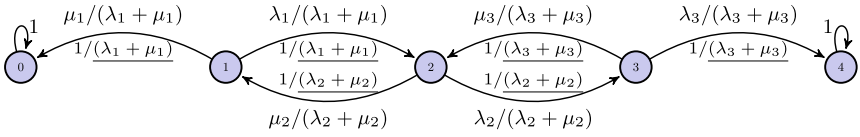
and the mean time spent at state $\textcircled{i}$, given that the next transition is from $\textcircled{i}$ to $\boxed{i-1}$, is equal to

$$\mathbb{E}[\tau_{i,i-1}] = \frac{1}{\mu_i}, \qquad i = 1, 2, \ldots, N-1.$$

We associate a *weighted* graph to the Markov chain $(Z_n)_{n\geq 0}$ that includes the average

$$\mathbb{E}[\tau_i] = \frac{1}{\lambda_i + \mu_i}$$

of the time $\tau_i = \min(\tau_{i,i-1}, \tau_{i,i+1})$ spent in state $\textcircled{i}$ before the next transition, $i = 1, 2, \ldots, N-1$. In the next graph, which is drawn for $N = 4$, the weights are *underlined*:



with $\lambda_0 = \mu_4 = 0$.

Proposition 9.10 *The mean absorption times*

$$h_{0,N}(i) = \mathbb{E}\left[T_{0,N} \mid X_0 = i\right], \qquad i = 0, 1, \ldots, N,$$

into states $\{0, N\}$ *starting from state* $i \in \{0, 1, \ldots, N\}$ *satisfy the boundary conditions* $h_{0,N}(0) = h_{0,N}(N) = 0$ *the first step analysis equation*

$$h_{0,N}(i) = \frac{1}{\lambda_i + \mu_i} + \frac{\mu_i}{\lambda_i + \mu_i} h_{0,N}(i-1) + \frac{\lambda_i}{\lambda_i + \mu_i} h_{0,N}(i+1),$$

$i = 1, 2, \ldots, N-1$.

Proof By first step analysis on the discrete-time embedded chain $(Z_n)_{n \geq 1}$ with transition matrix

$$P = \big[P_{i,j} \big]_{i,j \in \mathbb{S}} = \begin{bmatrix} 0 & 1 & 0 & 0 & \cdots & 0 & 0 & 0 \\ \frac{\mu_1}{\lambda_1 + \mu_1} & 0 & \frac{\lambda_1}{\lambda_1 + \mu_1} & 0 & \cdots & 0 & 0 & 0 \\ 0 & \ddots & \ddots & \ddots & \vdots & \vdots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \vdots & \vdots & \ddots & \ddots & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & \frac{\lambda_{N-1}}{\lambda_{N-1} + \mu_{N-1}} \\ 0 & 0 & 0 & 0 & \cdots & 0 & 1 & 0 \end{bmatrix},$$

we have

$$\begin{aligned} h_{0,N}(i) &= \frac{\mu_i}{\lambda_i + \mu_i} \big(\mathbb{E}[\tau_i] + h_{0,N}(i-1) \big) + \frac{\lambda_i}{\lambda_i + \mu_i} \big(\mathbb{E}[\tau_i] + h_{0,N}(i+1) \big) \\ &= \frac{\mu_i}{\lambda_i + \mu_i} \left(\frac{1}{\lambda_i + \mu_i} + h_{0,N}(i-1) \right) + \frac{\lambda_i}{\lambda_i + \mu_i} \left(\frac{1}{\lambda_i + \mu_i} + h_{0,N}(i+1) \right), \end{aligned} \tag{9.7.1}$$

$i = 1, 2, \ldots, N-1$.

Note that by conditioning on the independent exponential random variables $\tau_{i,i-1}$ and $\tau_{i,i+1}$ we can also show that

$$\mathbb{E}[\tau_i \mid \tau_{i,i-1} < \tau_{i,i+1}] = \mathbb{E}[\tau_i \mid \tau_{i,i+1} < \tau_{i,i-1}] = \mathbb{E}[\tau_i] = \frac{1}{\lambda_i + \mu_i},$$

$i = 1, 2, \ldots, N-1$, cf. (1.8.10) in Exercise 1.5, hence (9.7.1) can be rewritten as

$$\begin{aligned} h_{0,N}(i) = \ & \frac{\mu_i}{\lambda_i + \mu_i} \big(\mathbb{E}[\tau_i \mid \tau_{i,i-1} < \tau_{i,i+1}] + h_{0,N}(i-1) \big) \\ & + \frac{\lambda_i}{\lambda_i + \mu_i} \big(\mathbb{E}[\tau_i \mid \tau_{i,i+1} < \tau_{i,i-1}] + h_{0,N}(i+1) \big). \end{aligned}$$

When the rates $\lambda_i = \lambda$ and $\mu_i = \mu$ are independent of $i \in \{1, 2, \ldots, N-1\}$, this equation becomes

$$h_{0,N}(i) = \frac{1}{\lambda+\mu} + \frac{\lambda}{\lambda+\mu} h_{0,N}(i+1) + \frac{\mu}{\lambda+\mu} h_{0,N}(i-1),$$

$i = 1, 2, \ldots, N-1$, which is a modification of Eq. (2.3.7a), by replacing the discrete time step by the average time $1/(\lambda+\mu)$ spent at any state. Rewriting the equation as

$$h_{0,N}(i) = \frac{1}{\lambda+\mu} + p h_{0,N}(i+1) + q h_{0,N}(i-1),$$

$i = 1, 2, \ldots, N-1$, or

$$(\lambda+\mu) h_{0,N}(i) = 1 + p(\lambda+\mu) h_{0,N}(i+1) + q(\lambda+\mu) h_{0,N}(i-1),$$

$i = 1, 2, \ldots, N-1$, with

$$p = \frac{\lambda}{\lambda+\mu} \quad \text{and} \quad q = \frac{\mu}{\lambda+\mu},$$

we find from (2.3.12) that, with $r = q/p = \mu/\lambda$,

$$(\lambda+\mu) h_{0,N}(k) = \frac{1}{q-p} \left(k - N \frac{1-r^k}{1-r^N} \right),$$

i.e.

$$h_{0,N}(k) = \frac{1}{\mu-\lambda} \left(k - N \frac{1-(\mu/\lambda)^k}{1-(\mu/\lambda)^N} \right), \qquad k = 0, 1, \ldots, N, \tag{9.7.2}$$

when $\lambda \neq \mu$. In the limit $\lambda \to \mu$ we find by (2.3.19) that

$$h_{0,N}(k) = \frac{1}{2\mu} k(N-k), \qquad k = 0, 1, \ldots, N.$$

This solution is similar to that of the gambling problem with draw Exercise 2.1 as we multiply the solution of the gambling problem in the fair case by the average time $1/(2\mu)$ spent in any state in $\{1, 2, \ldots, N-1\}$.

The mean absorption time for the embedded chain $(Z_n)_{n\geq 0}$ can be recovered by dividing (9.7.2) by the mean time $\mathbb{E}[\tau_i] = 1/(\lambda+\mu)$ between two jumps, as

$$\frac{\lambda+\mu}{\mu-\lambda} \left(k - N \frac{1-(\mu/\lambda)^k}{1-(\mu/\lambda)^N} \right), \qquad k = 0, 1, \ldots, N, \tag{9.7.3}$$

Table 9.1 Summary of computing methods

How to compute	Method
Infinitesimal generator $Q = (\lambda_{i,j})_{i,j\in\mathbb{S}}$	$Q = \dfrac{dP(t)}{dt}_{\|t=0} = P'(0)$
Semigroup $(P(t))_{t\in\mathbb{R}_+}$	$P(t) = \exp(tQ)$, $t \geq 0$, $P(h) = \mathrm{I} + hQ + o(h)$, $h \searrow 0$
Stationary distribution π	Solve[a] $\pi Q = 0$ for π
Probability distribution of the time $\tau_{i,j}$ spent in $i \to j$	Exponential distribution $(\lambda_{i,j})$
Probability distribution of the time τ_i spent at state $\textcircled{i}$	Exponential distribution $\left(\sum_{l\neq i} \lambda_{i,l}\right)$
$\lim_{t\to\infty} \exp\left(t \begin{bmatrix} -\alpha & \alpha \\ \beta & -\beta \end{bmatrix}\right)$	$\begin{bmatrix} \frac{\beta}{\alpha+\beta} & \frac{\alpha}{\alpha+\beta} \\ \frac{\beta}{\alpha+\beta} & \frac{\alpha}{\alpha+\beta} \end{bmatrix}$
Hitting probabilities	Solve[b] $g = Pg$ for the embedded chain
Mean hitting times	Use the embedded chain with weighted links using mean inter-jump times

[a] Remember that the values of $\pi(k)$ have to add up to 1.
[b] Be sure to write only the relevant rows of the system under the appropriate boundary conditions

which coincides with (2.3.12) in the non-symmetric case with $p = \lambda/(\lambda+\mu)$ and $p = \mu/(\lambda+\mu)$, and recovers (2.3.19), i.e.

$$k(N-k), \qquad k = 0, 1, \ldots, N,$$

in the symmetric case $\lambda = \mu$.

In Table 9.1 we gather some frequent questions with their corresponding solution methods.

Exercises

Exercise 9.1 A workshop has five machines and one repairman. Each machine functions until it fails at an exponentially distributed random time with rate $\mu = 0.20$ per hour. On the other hand, it takes a exponentially distributed random time with (rate) $\lambda = 0.50$ per hour to repair a given machine. We assume that the machines behave independently of one another, and that

(i) up to five machines can operate at any given time,
(ii) at most one can be under repair at any time.

Compute the proportion of time the repairman is idle in the long run.

Exercise 9.2 Two types of consultations occur at a database according to two independent Poisson processes: "read" consultations arrive at the rate (or intensity) λ_R and "write" consultations arrive at the rate (or intensity) λ_W.

(a) What is the probability that the time interval between two consecutive "read" consultations is larger than $t > 0$?
(b) What is the probability that during the time interval $[0, t]$, at most three "write" consultations arrive?
(c) What is the probability that the next arriving consultation is a "read" consultation?
(d) Determine the distribution of the number of arrived "read" consultations during $[0, t]$, given that in this interval a total number of n consultations occurred.

Exercise 9.3 Consider two machines, operating simultaneously and independently, where both machines have an exponentially distributed time to failure with mean $1/\mu$. There is a single repair facility, and the repair times are exponentially distributed with rate λ.

(a) In the long run, what is the probability that no machines are operating when $\lambda = \mu = 1$?
(b) We now assume that at most one machine can operate at any time. Namely, while one machine is working, the other one may be either under repair or already fixed and waiting to take over. How does this modify your answer to question (a)?

Exercise 9.4 Passengers arrive at a cable car station according to a Poisson process with intensity $\lambda > 0$. Each car contains at most 4 passengers, and a cable car arrives immediately and leaves with 4 passengers as soon as there are at least 4 people in the queue. We let $(X_t)_{t\in\mathbf{R}_+}$ denote the number of passengers in the waiting queue at time $t \geq 0$.

(a) Explain why $(X_t)_{t\in\mathbb{R}_+}$ is a continuous-time Markov chain with state space $\mathbb{S} = \{0, 1, 2, 3\}$, and give its matrix infinitesimal generator Q.
(b) Compute the limiting distribution $\pi = [\pi_0, \pi_1, \pi_2, \pi_3]$ of $(X_t)_{t\in\mathbb{R}_+}$.
(c) Compute the mean time between two departures of cable cars.

Exercise 9.5 (Muni Toke (2015)) We consider a stock whose prices can only belong to the following five ticks:

$$\$10.01; \quad \$10.02; \quad \$10.03; \quad \$10.04; \quad \$10.05,$$

numbered $k = 1, 2, 3, 4, 5$.
At time t, the order book for this stock contains exactly $N_t^{(k)}$ sell orders at the price tick n$^o k$, $k = 1, 2, 3, 4, 5$, where $(N_t^{(k)})_{t\in\mathbf{R}_+}$ are independent Poisson processes with same intensity $\lambda > 0$. In addition,

- Any *sell* order can be cancelled after an exponentially distributed random time with parameter $\mu > 0$,

- *Buy* market orders are submitted according to another Poisson process with intensity $\theta > 0$, and are filled instantly at the lowest order price present in the book.

Order cancellations can occur as a result of various trading algorithms such as, e.g., "spoofing", "layering", or "front running".

(a) Show that the *total* number of sell orders L_t in the order book at time t forms a continuous-time Markov chain, and write down its infinitesimal generator Q.
(b) It is estimated that 95% percent of high-frequency trader orders are later cancelled. What relation does this imply between μ and λ?

Exercise 9.6 The size of a fracture in a rock formation is modeled by a continuous-time pure birth process with parameters

$$\lambda_k = (1+k)^\rho, \qquad k \geq 1,$$

i.e. the growth rate of the fracture is a power of $1+k$, where k is the current fracture length. Show that when $\rho > 1$, the mean time for the fracture length to grow to infinity is finite. Conclude that the time to failure of the rock formation is almost-surely finite.[3]

Exercise 9.7 Customers arrive at a processing station according to a Poisson process with rate $\lambda = 0.1$, i.e. on average one customer per ten minutes. Processing of customer queries starts as soon as the third customer enters the queue.

(a) Compute the expected time until the start of the customer service.
(b) Compute the probability that no customer service occurs within the first hour.

Exercise 9.8 Online user logins are modeled according to a standard Poisson process $(N_t)_{t\in\mathbb{R}_+}$ with intensity $\lambda > 0$. Each user spends the same amount of time $\theta > 0$ checking the site, and then quits. We assume that the process started at time $t = 0$.

(a) Compute the average number of users checking the website at any given time t.
(b) Compute the probability that no user is checking the website at a given time $t > 0$.

Exercise 9.9 Suppose that customers arrive at a facility according to a Poisson process having rate $\lambda = 3$. Let N_t be the number of customers that have arrived up to time t and let T_n be the arrival time of the n-th customer, $n = 1, 2, \ldots$ Determine the following (conditional) probabilities and (conditional) expectations, where $0 < t_1 < t_2 < t_3 < t_4$.

(a) $\mathbb{P}(N_{t_3} = 5 \mid N_{t_1} = 1)$.
(b) $\mathbb{E}[N_{t_1} N_{t_4}(N_{t_3} - N_{t_2})]$.
(c) $\mathbb{E}[N_{t_2} \mid T_2 > t_1]$.

[3] Recall that a finite-valued random variable may have an infinite mean.

Exercise 9.10 Let $(X_t)_{t\in\mathbb{R}_+}$ be a birth and death process on $\{0, 1, 2\}$ with birth and death parameters $\lambda_0 = 2\alpha$, $\lambda_1 = \alpha$, $\lambda_2 = 0$, and $\mu_0 = 0$, $\mu_1 = \beta$, $\mu_2 = 2\beta$. Determine the stationary distribution of $(X_t)_{t\in\mathbf{R}_+}$.

Exercise 9.11 Let $(X_t)_{t\in\mathbb{R}_+}$ be a birth and death process on $\{0, 1, \ldots, N\}$ with birth and death parameters $\lambda_n = \alpha(N - n)$ and $\mu_n = \beta n$, respectively. Determine the stationary distribution of $(X_t)_{t\in\mathbf{R}_+}$.

Exercise 9.12 Consider the pure birth process with birth rates $\lambda_0 = 1$, $\lambda_1 = 3$, $\lambda_2 = 2$, $\lambda_3 = 5$. Compute $P_{0,n}(t)$ for $n = 0, 1, 2, 3$.

Exercise 9.13 Consider a pure birth process $(X_t)_{t\in\mathbf{R}_+}$ started at $X_0 = 0$, and let T_k denote the time until the k-th birth. Show that

$$\mathbb{P}(T_1 > t \text{ and } T_2 > t + s) = P_{0,0}(t)(P_{0,0}(s) + P_{0,1}(s)).$$

Determine the joint probability density function of (T_1, T_2), and then the joint density of $(\tau_0, \tau_1) := (T_1, T_2 - T_1)$.

Exercise 9.14 Cars pass a certain street location with identical speeds, according to a Poisson process with rate $\lambda > 0$. A woman at that location needs T units of time to cross the street, i.e. she waits until it appears that no car will cross that location within the next T time units.

(a) Find the probability that her waiting time is 0.
(b) Find her expected waiting time.
(c) Find the total average time it takes to cross the street.
(d) Assume that, due to other factors, the crossing time in the absence of cars is an independent exponentially distributed random variable with parameter $\mu > 0$. Find the total average time it takes to cross the street in this case.
(e) Assume now that the street is split in two lanes, the crossing of each lane requiring $T/2$ units of time. Is it better to cross the two lanes at once, or to make a stop in between the two lanes?

Exercise 9.15 A machine is maintained at random times, such that the inter-service times $(\tau_k)_{k\geq 0}$ are *i.i.d.* with exponential distribution of parameter $\mu > 0$. The machine breaks down if it has not received maintenance for more than T units of time. After breaking down it is automatically repaired.

(a) Compute the probability that the machine breaks down before its first maintenance after it is started.
(b) Find the expected time until the machine breaks down.
(c) Assuming that the repair time is exponentially distributed with parameter $\lambda > 0$, find the proportion of time the machine is working.

Exercise 9.16 A system consists of two machines and two repairmen. Each machine can work until failure at an exponentially distributed random time with parameter 0.2. A failed machine can be repaired only by one repairman, within an exponentially distributed random time with parameter 0.25. We model the number X_t of working machines at time $t \in \mathbb{R}_+$ as a continuous-time Markov process.

(a) Complete the missing entries in the matrix

$$Q = \begin{bmatrix} \square & 0.5 & 0 \\ 0.2 & \square & \square \\ 0 & \square & -0.4 \end{bmatrix}$$

of its generator.
(b) Calculate the long-run probability distribution $[\pi_0, \pi_1, \pi_2]$ of X_t.
(c) Compute the average number of working machines in the long run.
(d) Given that a working machine can produce 100 units every hour, how many units can the system produce per hour in the long run?
(e) Assume now that in case a single machine is under failure then both repairmen can work on it, therefore dividing the mean repair time by a factor 2. Complete the missing entries in the matrix

$$Q = \begin{bmatrix} -0.5 & \square & \square \\ \square & -0.7 & \square \\ \square & \square & -0.4 \end{bmatrix}$$

of the modified generator and calculate the long run probability distribution $[\pi_0, \pi_1, \pi_2]$ for X_t.

Exercise 9.17 Let $X_1(t)$ and $X_2(t)$ be two independent two-state Markov chains on $\{0, 1\}$ and having the same infinitesimal matrix

$$\begin{bmatrix} -\lambda & \lambda \\ \mu & -\mu \end{bmatrix}.$$

Argue that $Z(t) := X_1(t) + X_2(t)$ is a Markov chain on the state space $\mathbb{S} = \{0, 1, 2\}$ and determine the transition semigroup $P(t)$ of $Z(t)$.

Exercise 9.18 Consider a two-state discrete-time Markov chain $(\xi_n)_{n\geq 0}$ on $\{0, 1\}$ with transition matrix

$$\begin{bmatrix} 0 & 1 \\ 1-\alpha & \alpha \end{bmatrix}. \tag{9.7.4}$$

Let $(N_t)_{t\in\mathbb{R}_+}$ be a Poisson process with parameter $\lambda > 0$, and let the

$$X_t = \xi_{N_t}, \qquad t \in \mathbb{R}_+,$$

i.e. $(X_t)_{t\in\mathbb{R}_+}$ is a two-state birth and death process.

(a) Compute the mean return time $\mathbb{E}[T_0^r \mid X_0 = 0]$ of X_t to state $\textcircled{0}$, where T_0^r is defined as

$$T_0^r = \inf\{t > T_1 \ : \ X_t = 0\}$$

and

$$T_1 = \inf\{t > 0 \ : \ X_t = 1\}$$

is the first hitting time of state $\textcircled{1}$.

(b) Compute the mean return time $\mathbb{E}[T_1^r \mid X_0 = 1]$ of X_t to state $\textcircled{1}$, where T_1^r is defined as

$$T_1^r = \inf\{t > T_0 \ : \ X_t = 1\}$$

and

$$T_0 = \inf\{t > 0 \ : \ X_t = 0\}$$

is the first hitting time of state $\textcircled{0}$. The return time T_1^r to $\textcircled{1}$ starting from $\textcircled{1}$ is evaluated by switching first to state $\textcircled{0}$ before returning to state $\textcircled{1}$.

(c) Show that $(X_t)_{t\in\mathbb{R}_+}$ is a two-state birth and death process and determine its generator matrix Q in terms of α and λ.

Problem 9.19 Let $(N_t^1)_{t\in\mathbb{R}_+}$ and $(N_t^2)_{t\in\mathbb{R}_+}$ be two independent Poisson processes with intensities $\lambda_1 > 0$ and $\lambda_2 > 0$.

(a) Show that $(N_t^1 + N_t^2)_{t\in\mathbb{R}_+}$ is a Poisson process and find its intensity.

(b) Consider the difference

$$M_t = N_t^1 - N_t^2, \qquad t \geq 0,$$

and that $(M_t)_{t\in\mathbb{R}_+}$ has stationary independent increments.

(c) Find the distribution of $M_t - M_s$, $0 < s < t$.

(d) Compute

$$\lim_{t\to\infty} \mathbb{P}(|M_t| \leq c)$$

for any $c > 0$.

(e) Suppose that N_t^1 denotes the number of clients arriving at a taxi station during the time interval $[0, t]$, and that N_t^2 denotes the number of taxis arriving at that same station during the same time interval $[0, t]$.
How do you interpret the value of M_t depending on its sign?
How do you interpret the result of Question (d)?

Problem 9.20 We consider a birth and death process $(X_t)_{t\in\mathbb{R}_+}$ on $\{0, 1, \ldots, N\}$ with transition semigroup $(P(t))_{t\in\mathbb{R}}$ and birth and death rates

$$\lambda_n = (N - n)\lambda, \qquad \mu_n = n\mu, \qquad n = 0, 1, \ldots, N.$$

This process is used for the modeling of membrane channels in neuroscience.

(a) Write down the infinitesimal generator Q of $(X_t)_{t\in\mathbb{R}_+}$.

(b) From the forward Kolmogorov equation $P'(t) = P(t)Q$, show that for all $n = 0, 1, \ldots, N$ we have

$$\begin{cases} P'_{n,0}(t) = -\lambda_0 P_{n,0}(t) + \mu_1 P_{n,1}(t), \\ P'_{n,k}(t) = \lambda_{k-1} P_{n,k-1}(t) - (\lambda_k + \mu_k) P_{n,k}(t) + \mu_{k+1} P_{n,k+1}(t), \\ P'_{n,N}(t) = \lambda_{N-1} P_{n,N-1}(t) - \mu_N P_{n,N}(t), \end{cases}$$

$k = 1, 2, \ldots, N-1$.

(c) Let

$$G_k(s,t) = \mathbb{E}\left[s^{X_t} \mid X_0 = k\right] = \sum_{n=0}^{N} s^n \mathbb{P}(X_t = n \mid X_0 = k) = \sum_{n=0}^{N} s^n P_{k,n}(t)$$

denote the generating function of X_t given that $X_0 = k \in \{0, 1, \ldots, N\}$. From the result of Question (b), show that $G_k(s,t)$ satisfies the *partial differential equation*

$$\frac{\partial G_k}{\partial t}(s,t) = \lambda N(s-1) G_k(s,t) + (\mu + (\lambda - \mu)s - \lambda s^2) \frac{\partial G_k}{\partial s}(s,t), \quad (9.7.5)$$

with $G_k(s,0) = s^k$, $k = 0, 1, \ldots, N$.

(d) Verify that the solution of (9.7.5) is given by

$$G_k(s,t) = \frac{1}{(\lambda+\mu)^N} (\mu + \lambda s + \mu(s-1)\mathrm{e}^{-(\lambda+\mu)t})^k (\mu + \lambda s - \lambda(s-1)\mathrm{e}^{-(\lambda+\mu)t})^{N-k},$$

$k = 0, 1, \ldots, N$.

(e) Show that

$$\begin{aligned} \mathbb{E}[X_t \mid X_0 = k] = {} & \frac{k}{(\lambda+\mu)^N} (\lambda + \mu \mathrm{e}^{-(\lambda+\mu)t}) (\mu+\lambda)^{k-1} (\mu+\lambda)^{N-k} \\ & + \frac{N-k}{(\lambda+\mu)^N} (\mu+\lambda)^k (\lambda - \lambda \mathrm{e}^{-(\lambda+\mu)t}) (\mu+\lambda)^{N-k-1}. \end{aligned}$$

(f) Compute

$$\lim_{t \to \infty} \mathbb{E}[X_t \mid X_0 = k]$$

and show that it does not depend on $k \in \{0, 1, \ldots, N\}$.

Problem 9.21 Let $T_1, T_2, \ldots$ be the first two jump times of a Poisson process $(N_t)_{t \in \mathbb{R}_+}$ with intensity λ. Given $f : \mathbf{R} \longrightarrow \mathbf{R}$ an integrable function, show that

$$\mathbb{E}\left[\sum_{k=1}^{N_t} f(T_i)\right] = \lambda \int_0^t f(s) ds. \quad (9.7.6)$$

Chapter 10
Discrete-Time Martingales



As mentioned in the introduction, stochastic processes can be classified into two main families, namely *Markov processes* on the one hand, and *martingales* on the other hand. Markov processes have been our main focus of attention so far, and in this chapter we turn to the notion of martingale. We will give a precise mathematical meaning to the description of martingales, which says that when $(X_n)_{n\geq 0}$ is a martingale, the best possible estimate at time $n \geq 0$ of the future value X_m at time $m > n$ is X_n itself. In this chapter, the main application of martingales will be to recover in an elegant way the previous results on gambling processes of Chap. 2. The concept of martingale has also many applications in stochastic modeling, for example in financial mathematics, where martingales are used to characterize the fairness and equilibrium in a market model.

10.1 Filtrations and Conditional Expectations

Before dealing with martingales we need to introduce the important notion of *filtration* generated by a discrete-time stochastic process $(S_n)_{n\geq 0}$. In what follows, we work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition 10.1 The *filtration* $(\mathcal{F}_n)_{n\geq 0}$ generated by a stochastic process $(S_n)_{n\geq 0}$ taking its values in a state space $\mathbb{S}$, is the family of σ-algebras

$$\mathcal{F}_n := \sigma(S_0, S_1, \ldots, S_n), \qquad n \geq 0,$$

where $\sigma(S_0, S_1, \ldots, S_n)$ denotes the collections of events generated by $S_0, S_1, \ldots, S_n, n \geq 0$.

Examples of events in $\mathcal{F}_n$ include:

$$\{S_0 \leq a_0,\ S_1 \leq a_1,\ \ldots,\ S_n \leq a_n\}$$

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N. Privault, *Understanding Markov Chains*, Springer Undergraduate Mathematics Series,
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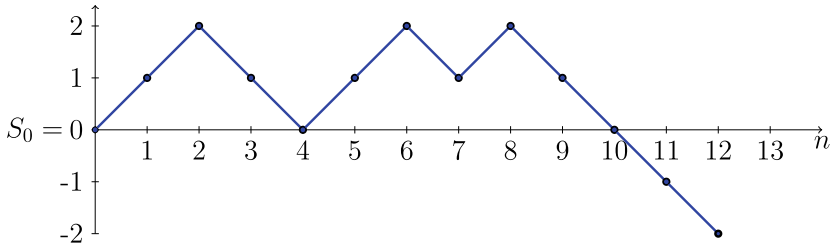


Fig. 10.1 Sample path of the random walk $(S_n)_{n\geq 0}$

for $a_0, a_1, \ldots, a_n$ a given fixed sequence of real numbers. Note that we have the inclusion $\mathcal{F}_n \subset \mathcal{F}_{n+1}$, $n \geq 0$, i.e. $(\mathcal{F}_n)_{n\geq 0}$ is *non-decreasing*.

One refers to $\mathcal{F}_n$ as the *information* generated by $(S_k)_{k\geq 0}$ up to time n, and to $(\mathcal{F}_n)_{n\geq 0}$ as the *information flow* generated by $(S_n)_{n\geq 0}$. We say that a random variable is $\mathcal{F}_n$-measurable whenever F can be written as a function $F = f(S_0, S_1, \ldots, S_n)$ of $(S_0, S_1, \ldots, S_n)$ (Fig. 10.1).

Example

Consider the simple random walk

$$S_n := X_1 + X_2 + \cdots + X_n, \qquad n \geq 0,$$

where $(X_k)_{k\geq 1}$ is a sequence of independent, identically distributed $\{-1, 1\}$-valued random variables, and $S_0 := 0$.
The filtration (or information flow) $(\mathcal{F}_n)_{n\geq 0}$ generated by $(S_n)_{n\geq 0}$ satisfies

$$\mathcal{F}_0 = \big\{\{S_0 \neq 0\}, \{S_0 = 0\}\big\} = \big\{\emptyset, \Omega\big\}, \quad \mathcal{F}_1 = \big\{\emptyset, \{X_1 = 1\}, \{X_1 = -1\}, \Omega\big\}, \tag{10.1.1}$$

and

$$\mathcal{F}_2 = \sigma\big(\big\{\emptyset, \{X_1 = 1, X_2 = 1\}, \{X_1 = 1, X_2 = -1\}, \{X_1 = -1, X_2 = 1\}, \{X_1 = -1, X_2 = -1\}, \Omega\big\}\big).$$

The notation $\mathcal{F}_n$ is useful to represent a quantity of information available at time n, and various sub σ-algebras of $\mathcal{F}_n$ can be defined. For example, the σ-algebra $\mathcal{G}$ generated by S_2 satisfies

$$\begin{aligned}\mathcal{G} &= \sigma\big(\big\{\emptyset, \{S_2 = -2\}, \{S_2 = 2\}, \{S_2 = 0\}, \Omega\big\}\big)\\ &= \sigma\big(\big\{\emptyset, \{X_1 = -1, X_2 = -1\}, \{X_1 = 1, X_2 = 1\}\\ &\qquad \{X_1 = 1, X_2 = -1\} \cup \{X_1 = -1, X_2 = 1\}, \Omega\big\}\big),\end{aligned}$$

which contains less information than $\mathcal{F}_2$, as it only tells whether the increments X_1, X_2 are both equal to 1 or to -1.

Definition 10.2 A stochastic process $(Z_n)_{n\geq 0}$ is said to be $(\mathcal{F}_n)_{n\geq 0}$-*adapted* if the value of Z_n depends on no more than the information available up to time n in $\mathcal{F}_n$.

In other words, the value of an $(\mathcal{F}_n)_{n\geq 0}$-adapted process Z_n is determined by a function of $X_0, X_1, \ldots, X_n$ for all $n \geq 0$.

We now review the definition of *conditional expectation*, see Sects. 1.6 and 1.8 for details. Given X a random variable with finite mean, the conditional expectation $\mathbb{E}[X \mid \mathcal{F}_n]$ refers to

$$\mathbb{E}[X \mid X_0, X_1, \ldots, X_n] = \mathbb{E}[X \mid X_0 = k_0, \ldots, X_n = k_n]_{k_0=X_0,\ldots,k_n=X_n},$$

given that $X_0, X_1, \ldots, X_n$ are respectively equal to $k_0, k_1, \ldots, k_n \in \mathbb{S}$.

The conditional expectation $\mathbb{E}[X \mid \mathcal{F}_n]$ is itself a random variable that depends only on the values of $X_0, X_1, \ldots, X_n$, i.e. on the history of the process up to time $n \in \mathbb{N}$. It can also be interpreted as the best possible estimate of X in mean-square sense, given the values of $X_0, X_1, \ldots, X_n$, see Propositions 1.8.2 and 1.19.

By point (ii) Sect. 1.8, any integrable $\mathcal{F}_n$-*adapted* process $(Z_n)_{n\geq 0}$ satisfies

$$\mathbb{E}[Z_n \mid \mathcal{F}_n] = Z_n, \qquad n \geq 0.$$

10.2 Martingales–Definition and Properties

We now turn to the definition of *martingale*.

Definition 10.3 An integrable,[1] discrete-time stochastic process $(M_n)_{n\geq 0}$ is a *martingale* with respect to $(\mathcal{F}_n)_{n\geq 0}$ if $(M_n)_{n\geq 0}$ is $(\mathcal{F}_n)_{n\geq 0}$-adapted and satisfies the property

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n, \qquad n \geq 0. \tag{10.2.1}$$

The process $(M_n)_{n\geq 0}$ is a martingale with respect to $(\mathcal{F}_n)_{n\geq 0}$ if, given the information $\mathcal{F}_n$ known up to time n, the best possible estimate of M_{n+1} is simply M_n.

Remark 10.4 Definition 10.3 can be equivalently stated by saying that

$$\mathbb{E}[M_n \mid \mathcal{F}_k] = M_k, \qquad k = 0, 1, \ldots, n.$$

[1] Integrable means $\mathbb{E}[|M_n|] < \infty$ for all $n \geq 0$.

Proof Let $k \geq 0$. We prove the statement by induction on $n \geq k$. For $n = k$ we have $M_k = \mathbb{E}[M_k \mid \mathcal{F}_k]$. Next, assuming that $M_k = \mathbb{E}[M_n \mid \mathcal{F}_k]$ for some $n \geq k$, by the *tower property* of conditional expectations, see point (iii) Sect. 1.8, we have

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_k] = \mathbb{E}[\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] \mid \mathcal{F}_k] = \mathbb{E}[M_n \mid \mathcal{F}_k] = M_k.$$

□

A particular property of martingales is that their expectation is constant over time.

Proposition 10.5 *Let $(M_n)_{n\geq 0}$ be an $(\mathcal{F}_n)_{n\geq 0}$-martingale. We have*

$$\mathbb{E}[M_n] = \mathbb{E}[M_0], \qquad n \geq 0.$$

Proof By (1.8.6) and the *tower property* of conditional expectations, see point (iii) Sect. 1.8, we have

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_0] = \mathbb{E}[\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] \mid \mathcal{F}_0] = \mathbb{E}[M_n \mid \mathcal{F}_0], \qquad n \geq 0.$$

Hence, by induction on $n \geq 0$, we have

$$\begin{aligned}
\mathbb{E}[M_{n+1} \mid \mathcal{F}_0] &= \mathbb{E}[M_n \mid \mathcal{F}_0] \\
&= \mathbb{E}[M_{n-1} \mid \mathcal{F}_0] \\
&= \cdots \\
&= \mathbb{E}[M_1 \mid \mathcal{F}_0] \\
&= \mathbb{E}[M_0 \mid \mathcal{F}_0], \quad n \geq 0,
\end{aligned}$$

hence

$$\mathbb{E}[M_n \mid \mathcal{F}_0] = \mathbb{E}[M_0 \mid \mathcal{F}_0], \qquad n \geq 0,$$

and the conclusion follows from (1.8.5). □

Examples of Martingales

1. Any *centered*[2] integrable process $(S_n)_{n\geq 0}$ with mutually *independent* increments is a martingale with respect to the filtration $(\mathcal{F}_n)_{n\geq 0}$ generated by $(S_n)_{n\geq 0}$, see the example of Sect. 10.1.
   Indeed, in this case we have

[2] A process $(S_n)_{n\geq 0}$ is said to be *centered* if $\mathbb{E}[S_n] = 0$ for all $n \geq 0$.

$$\begin{aligned}\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}[S_n + S_{n+1} - S_n \mid \mathcal{F}_n] \\ &= \mathbb{E}[S_n \mid \mathcal{F}_n] + \mathbb{E}[S_{n+1} - S_n \mid \mathcal{F}_n] \\ &= \mathbb{E}[S_n \mid \mathcal{F}_n] + \underbrace{\mathbb{E}[S_{n+1} - S_n]}_{=0} \\ &= \mathbb{E}[S_n \mid \mathcal{F}_n] = S_n, \qquad n \geq 0.\end{aligned}$$

In addition to being a martingale, a stochastic process $(S_n)_{n\geq 0}$ with centered independent increments is also a Markov process, cf. Sect. 4.1.
However, not all martingales have the Markov property, and not all Markov processes are martingales. In addition, there are martingales and Markov processes which do not have independent increments.

2. Given $X \in L^2(\Omega)$ a square-integrable random variable and $(\mathcal{F}_n)_{n\geq 0}$ a filtration, the process $(X_n)_{n\geq 0}$ defined by

$$X_n := \mathbb{E}[X \mid \mathcal{F}_n], \qquad n \geq 0,$$

is an $(\mathcal{F}_n)_{n\geq 0}$-martingale under the probability measure $\mathbb{P}$, as follows from the tower property (1.6.8):

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_{n+1}] \mid \mathcal{F}_n] = \mathbb{E}[X \mid \mathcal{F}_n] = X_n, \quad n \geq 0. \tag{10.2.2}$$

10.3 Stopping Times

Next, we turn to the definition of stopping time. If an event occurs at a (random) *stopping time*, it should be possible, at any time $n \in \mathbb{N}$, to determine whether the event has already occurred, based on the information available up to time n. This idea is formalized in the next definition.

Definition 10.6 A random variable $\tau : \Omega \longrightarrow \mathbb{N} \cup \{+\infty\}$ is a *stopping time* with respect to $(\mathcal{F}_n)_{n\geq 0}$ if

$$\{\tau > n\} \in \mathcal{F}_n, \qquad n \geq 0. \tag{10.3.1}$$

The meaning of Relation (10.3.1) is that the knowledge of $\{\tau > n\}$ depends only on the information present in $\mathcal{F}_n$ up to time n, i.e. on the knowledge of $X_0, X_1, \ldots, X_n$.

Note that condition (10.3.1) is equivalent to the condition

$$\{\tau \leq n\} \in \mathcal{F}_n, \qquad n \geq 0,$$

since $\{\tau \leq n\} = \{\tau > n\}^c$ and $\mathcal{F}_n$ is *stable by complement*.

Not every $\mathbb{N}$-valued random variable is a stopping time, however, hitting times provide natural examples of stopping times. We refer to Definition 10.1 for the construction of the filtration generated by a discrete-time stochastic process.

Proposition 10.7 *Let $(X_n)_{n\geq 0}$ denote a stochastic process taking its values in $\mathbb{S}$, and generating a filtration $(\mathcal{F}_n)_{n\geq 0}$ The first hitting time*

$$\tau_x := \inf\{k \geq 0 \ : \ X_k = x\}$$

of $x \in \mathbb{S}$ by $(X_n)_{n\geq 0}$ is a stopping time *with respect to the filtration $(\mathcal{F}_n)_{n\geq 0}$.*

Proof We have

$$\begin{aligned}\{\tau_x > n\} &= \{X_0 \neq x, \ X_1 \neq x, \ \ldots, X_n \neq x\} \\ &= \{X_0 \neq x\} \cap \{X_1 \neq x\} \cap \cdots \cap \{X_n \neq x\} \in \mathcal{F}_n, \quad n \geq 0,\end{aligned}$$

since

$$\{X_0 \neq x\} \in \mathcal{F}_0 \subset \mathcal{F}_n, \ \{X_1 \neq x\} \in \mathcal{F}_1 \subset \mathcal{F}_n, \ldots, \ \{X_n \neq x\} \in \mathcal{F}_n, \quad n \geq 0.$$

□

Hitting times can be used to trigger "buy limit" or "sell stop" orders in finance. On the other hand, the first time

$$\tau := \inf\left\{k \geq 0 \ : \ X_k = \max_{l=0,1,\ldots,N} X_l\right\}$$

the process $(X_k)_{k\geq 0}$ reaches its maximum over $\{0, 1, \ldots, N\}$ is not a stopping time. Indeed, it is not possible to decide whether $\{\tau \leq n\}$, i.e. the maximum has been reached before time n, based on the information available up to time n.

Exercise: Show from Definition 10.6 that the minimum $\tau \wedge \nu := \min(\tau, \nu)$ and the maximum $\tau \vee \nu := \max(\tau, \nu)$ of two stopping times are themselves stopping times.

Definition 10.8 Given $(Z_n)_{n\geq 0}$ a stochastic process and $\tau : \Omega \longrightarrow \mathbb{N}$ a stopping time, the *stopped process*

$$(Z_{\tau\wedge n})_{n\geq 0} = (Z_{\min(\tau,n)})_{n\geq 0}$$

is defined as

$$Z_{\tau\wedge n} = Z_{\min(\tau,n)} = \begin{cases} Z_n & \text{if } n < \tau, \\ Z_\tau & \text{if } n \geq \tau, \end{cases}$$

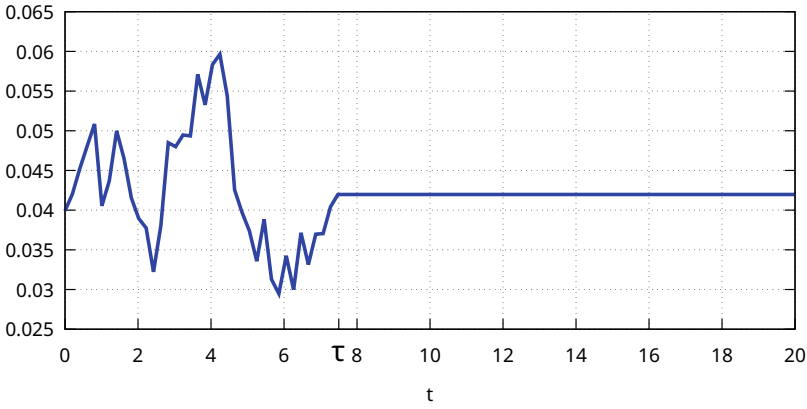


Fig. 10.2 Stopped process

Using indicator functions, we may also write

$$Z_{\tau\wedge n} = Z_n \mathbb{1}_{\{n<\tau\}} + Z_\tau \mathbb{1}_{\{n\geq\tau\}}, \qquad n \geq 0.$$

Figure 10.2 presents a sample path of a stopped process.

The following Theorem 10.9 is called the *Stopping Time Theorem*, and is due to J.L. Doob (1910-2004).

Theorem 10.9 *Assume that* $(M_n)_{n\geq 0}$ *is a martingale with respect to* $(\mathcal{F}_n)_{n\geq 0}$ *and that* $\tau : \Omega \longrightarrow \mathbb{N} \cup \{+\infty\}$ *is a stopping time with respect to* $(\mathcal{F}_n)_{n\geq 0}$. *Then the* stopped process $(M_{\tau\wedge n})_{n\geq 0}$ *is also a martingale with respect to* $(\mathcal{F}_n)_{n\geq 0}$.

Proof Writing

$$M_n = M_0 + \sum_{l=1}^{n}(M_l - M_{l-1}) = M_0 + \sum_{l\geq 1} \mathbb{1}_{\{l\leq n\}}(M_l - M_{l-1}),$$

we have

$$M_{\tau\wedge n} = M_0 + \sum_{l=1}^{\tau\wedge n}(M_l - M_{l-1}) = M_0 + \sum_{l=1}^{n} \mathbb{1}_{\{l\leq\tau\}}(M_l - M_{l-1}),$$

and for $k \leq n$ we find

$$
\begin{aligned}
\mathbb{E}[M_{\tau\wedge n} \mid \mathcal{F}_k] &= \mathbb{E}\left[M_0 + \sum_{l=1}^{n} \mathbb{1}_{\{l\le\tau\}}(M_l - M_{l-1}) \,\middle|\, \mathcal{F}_k \right] \\
&= M_0 + \sum_{l=1}^{n} \mathbb{E}\left[\mathbb{1}_{\{l\le\tau\}}(M_l - M_{l-1}) \,\middle|\, \mathcal{F}_k \right] \\
&= M_0 + \sum_{l=1}^{k} \mathbb{E}\left[\mathbb{1}_{\{l\le\tau\}}(M_l - M_{l-1}) \,\middle|\, \mathcal{F}_k \right] + \sum_{l=k+1}^{n} \mathbb{E}\left[\mathbb{1}_{\{l\le\tau\}}(M_l - M_{l-1}) \,\middle|\, \mathcal{F}_k \right] \\
&= M_0 + \sum_{l=1}^{k} (M_l - M_{l-1})\, \mathbb{E}\left[\mathbb{1}_{\{\tau>l-1\}} \,\middle|\, \mathcal{F}_k \right] \\
&\quad + \sum_{l=k+1}^{n} \mathbb{E}\left[\mathbb{E}\left[(M_l - M_{l-1}) \mathbb{1}_{\{l-1<\tau\}} \,\middle|\, \mathcal{F}_{l-1} \right] \middle|\, \mathcal{F}_k \right] \\
&= M_0 + \sum_{l=1}^{k} (M_l - M_{l-1}) \mathbb{1}_{\{l\le\tau\}} \\
&\quad + \sum_{l=k+1}^{n} \mathbb{E}\left[\mathbb{1}_{\{l-1<\tau\}} \underbrace{\mathbb{E}\left[M_l - M_{l-1} \,\middle|\, \mathcal{F}_{l-1} \right]}_{=0} \,\middle|\, \mathcal{F}_k \right] \\
&= M_0 + \sum_{l=1}^{\tau\wedge k} (M_l - M_{l-1}) \\
&= M_{\tau\wedge k},
\end{aligned}
$$

$k = 0, 1, \ldots, n$, where we used the tower property (1.6.8), the martingale property

$$
\begin{aligned}
\mathbb{E}\left[M_l - M_{l-1} \,\middle|\, \mathcal{F}_{l-1} \right] &= \mathbb{E}\left[M_l \,\middle|\, \mathcal{F}_{l-1} \right] - \mathbb{E}\left[M_{l-1} \,\middle|\, \mathcal{F}_{l-1} \right] \\
&= M_{l-1} - M_{l-1} = 0
\end{aligned}
$$

of $(M_l)_{l\ge 0}$, and the fact that

$$
\{\tau \ge l\} = \{\tau > l-1\} \in \mathcal{F}_{l-1} \supset \mathcal{F}_k, \qquad l \ge k+1.
$$

□

By the Stopping Time Theorem 10.9 we know that the stopped process $(M_{\tau\wedge n})_{n\ge 0}$ is a martingale, hence its expectation is constant over time by Proposition 10.5.

(i) If τ is a stopping time bounded by a constant $N \ge 1$, i.e. $\tau \le N$, by Proposition 10.5 we have

$$
\mathbb{E}[M_\tau] = \mathbb{E}[M_{\tau\wedge N}] = \mathbb{E}[M_{\tau\wedge 0}] = \mathbb{E}[M_0]. \tag{10.3.2}
$$

(ii) As a consequence of (10.3.2), if $(M_n)_{n\geq 0}$ is a martingale and $\tau \leq N$ and $\nu \leq N$ are two *bounded* stopping times bounded by a constant $N \geq 1$, we have

$$\mathbb{E}[M_\tau] = \mathbb{E}[M_\nu] = \mathbb{E}[M_0]. \tag{10.3.3}$$

(iii) In case τ is only *a.s.* finite, i.e. $\mathbb{P}(\tau < \infty) = 1$, we may also write

$$\mathbb{E}[M_\tau] = \mathbb{E}\left[\lim_{n\to\infty} M_{\tau\wedge n}\right] = \lim_{n\to\infty} \mathbb{E}[M_{\tau\wedge n}] = \mathbb{E}[M_0],$$

provided that the limit and expectation signs can be exchanged, however, this may not be always the case.

In some situations the exchange of limit and expectation signs may not be valid.[3] Nevertheless, the exchange is possible when the stopped process $(M_{\tau\wedge n})_{n\geq 0}$ is bounded in absolute value, i.e. $|M_{\tau\wedge n}| \leq K$ *a.s.*, $n \geq 0$, for some constant $K > 0$, as a consequence of the *dominated convergence theorem*.

Analog statements can be proved for *sub*martingales, see e.g. Exercise 10.6 for this notion.

10.4 Ruin Probabilities

In the sequel we will show that, as an application of the Stopping Time Theorem 10.9, the ruin probabilities computed for random walks in Chap. 2 can be recovered in a simple and elegant way.

Consider the standard random walk (or gambling process) $(S_n)_{n\geq 0}$ on $\{0, 1, \ldots, B\}$ with independent $\{-1, 1\}$-valued increments, and

$$\mathbb{P}(S_{n+1} - S_n = +1) = p \quad \text{and} \quad \mathbb{P}(S_{n+1} - S_n = -1) = q, \quad n \geq 0,$$

as introduced in Sect. 2.1. Let

$$\tau_{0,B} : \Omega \longrightarrow \mathbb{N}$$

be the first hitting time of the boundary $\{0, B\}$, defined by (Fig. 10.3)

$$\tau = \tau_{0,B} := \inf\{n \geq 0 \ : \ S_n = B \text{ or } S_n = 0\}. \tag{10.4.1}$$

One can easily check that the event $\{\tau > n\}$ depends only on the history of $(S_k)_{k\geq 0}$ up to time n since

$$\{\tau > n\} = \{0 < S_0 < B\} \cap \{0 < S_1 < B\} \cap \cdots \cap \{0 < S_n < B\} \subset \mathcal{F}_n,$$

hence τ is a stopping time.

[3] Consider for example the sequence $M_n := n\mathbb{1}_{\{U<1/n\}}$, $n \geq 1$, where $U \simeq U(0, 1]$ is a uniformly distributed random variable on $(0, 1]$, see also Exercise 10.5.

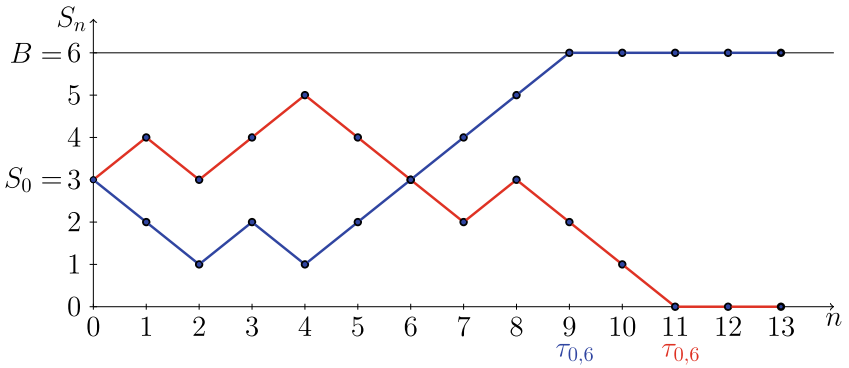


Fig. 10.3 Two sample paths of the random walk $(S_n)_{n\geq 0}$

We will recover the ruin probabilities

$$\mathbb{P}(S_\tau = 0 \mid S_0 = k), \qquad k = 0, 1, \ldots, B,$$

computed in Chap. 2 in three steps, first in the unbiased case $p = q = 1/2$ (note that the hitting time τ can be shown to be *a.s.* finite, i.e. $\mathbb{P}(\tau < \infty) = 1$, cf. e.g. the identity (2.2.30)).

Unbiased case $p = q = 1/2$

Step 1. The process $(S_n)_{n\geq 0}$ is a martingale.

We note that the process $(S_n)_{n\geq 0}$ has independent increments, and in the unbiased case $p = q = 1/2$ those increments are centered:

$$\mathbb{E}[S_{n+1} - S_n] = 1 \times p + (-1) \times q = 1 \times \frac{1}{2} + (-1) \times \frac{1}{2} = 0, \qquad (10.4.2)$$

hence $(S_n)_{n\geq 0}$ is a *martingale* by Point 1 Sect. 10.2.

Step 2. The stopped process $(S_{\tau\wedge n})_{n\geq 0}$ is also a martingale, as a consequence of the Stopping Time Theorem 10.9.

Step 3. Since the stopped process $(S_{\tau\wedge n})_{n\geq 0}$ is a martingale by the Stopping Time Theorem 10.9, we find that its expectation $\mathbb{E}[S_{\tau\wedge n} \mid S_0 = k]$ is constant in $n \geq 0$ by Proposition 10.5, which gives

$$k = \mathbb{E}[S_0 \mid S_0 = k] = \mathbb{E}[S_{\tau\wedge n} \mid S_0 = k], \qquad k = 0, 1, \ldots, B.$$

Letting n go to infinity and noting that $|S_{\tau\wedge n}| \leq B, n \geq 0$, by dominated convergence we get

$$\begin{aligned}\mathbb{E}[S_\tau \mid S_0 = k] &= \mathbb{E}\Big[\lim_{n\to\infty} S_{\tau\wedge n} \Big| S_0 = k\Big] \\ &= \lim_{n\to\infty} \mathbb{E}[S_{\tau\wedge n} \mid S_0 = k] \\ &= \lim_{n\to\infty} \mathbb{E}[S_{\tau\wedge 0} \mid S_0 = k] \\ &= \lim_{n\to\infty} \mathbb{E}[S_0 \mid S_0 = k] \\ &= k,\end{aligned}$$

provided that $\mathbb{P}(\tau < \infty) = 1$, see Exercise 10.1, where the exchange between limit and expectation is justified by the boundedness $|S_{\tau\wedge n}| \leq B$ *a.s.*, $n \geq 0$. Hence we have

$$\begin{cases} 0 \times \mathbb{P}(S_\tau = 0 \mid S_0 = k) + B \times \mathbb{P}(S_\tau = B \mid S_0 = k) = \mathbb{E}[S_\tau \mid S_0 = k] = k \\ \mathbb{P}(S_\tau = 0 \mid S_0 = k) + \mathbb{P}(S_\tau = B \mid S_0 = k) = \mathbb{P}(\tau < \infty) = 1, \end{cases}$$

which shows that

$$\mathbb{P}(S_\tau = B \mid S_0 = k) = \frac{k}{B} \quad \text{and} \quad \mathbb{P}(S_\tau = 0 \mid S_0 = k) = 1 - \frac{k}{B},$$

$k = 0, 1, \ldots, B$, which recovers (2.2.22) without relying on boundary conditions, and with shorter calculations. Namely, the solution has been obtained in a simple way without solving any finite difference equation, demonstrating the power of the martingale approach.

Biased case $p \neq q$

Next, we turn to the biased case where $p \neq q$. In this case, the process $(S_n)_{n\geq 0}$ is no longer a martingale because its increments are not centered:

$$\mathbb{E}[S_{n+1} - S_n] = 1 \times p + (-1) \times q = p - q \neq 0. \tag{10.4.3}$$

In order to apply the Stopping Time Theorem 10.9, we need to construct a martingale of a different type. Here, we note that the process

$$M_n := \left(\frac{q}{p}\right)^{S_n}, \qquad n \geq 0,$$

is a martingale with respect to $(\mathcal{F}_n)_{n\geq 0}$.

Step 1. The process $(M_n)_{n\geq 0}$ is a martingale.

Indeed, we have

$$
\begin{aligned}
\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}\left[\left(\frac{q}{p}\right)^{S_{n+1}} \middle| \mathcal{F}_n\right] = \mathbb{E}\left[\left(\frac{q}{p}\right)^{S_{n+1}-S_n}\left(\frac{q}{p}\right)^{S_n} \middle| \mathcal{F}_n\right] \\
&= \left(\frac{q}{p}\right)^{S_n} \mathbb{E}\left[\left(\frac{q}{p}\right)^{S_{n+1}-S_n} \middle| \mathcal{F}_n\right] \\
&= \left(\frac{q}{p}\right)^{S_n} \mathbb{E}\left[\left(\frac{q}{p}\right)^{S_{n+1}-S_n}\right] \\
&= \left(\frac{q}{p}\right)^{S_n} \left(\frac{q}{p}\mathbb{P}(S_{n+1}-S_n=1) + \left(\frac{q}{p}\right)^{-1}\mathbb{P}(S_{n+1}-S_n=-1)\right) \\
&= \left(\frac{q}{p}\right)^{S_n} \left(p\frac{q}{p} + q\left(\frac{q}{p}\right)^{-1}\right) \\
&= \left(\frac{q}{p}\right)^{S_n} \left(\frac{pq^2+p^2q}{pq}\right) \\
&= \left(\frac{q}{p}\right)^{S_n} (q+p) = \left(\frac{q}{p}\right)^{S_n} = M_n, \quad n \geq 0.
\end{aligned}
$$

In particular, the expectation of $(M_n)_{n\geq 0}$ is constant over time by Proposition 10.5 since it is a martingale, i.e. we have

$$
\left(\frac{q}{p}\right)^k = \mathbb{E}[M_0 \mid S_0 = k] = \mathbb{E}[M_n \mid S_0 = k], \qquad k = 0, 1, \ldots, B, \quad n \geq 0.
$$

Step 2. The stopped process $(M_{\tau\wedge n})_{n\geq 0}$ is also a martingale, as a consequence of the Stopping Time Theorem 10.9.

Step 3. Since the stopped process $(M_{\tau\wedge n})_{n\geq 0}$ remains a martingale by the Stopping Time Theorem 10.9, its expected value $\mathbb{E}[M_{\tau\wedge n} \mid S_0 = k]$ is constant in $n \geq 0$ by Proposition 10.5. This gives

$$
\left(\frac{q}{p}\right)^k = \mathbb{E}[M_0 \mid S_0 = k] = \mathbb{E}[M_{\tau\wedge n} \mid S_0 = k].
$$

Next, letting n go to infinity, by dominated convergence we find[4]

[4] Since $\mathbb{P}(\tau < \infty) = 1$, see Exercise 10.1.

$$\begin{aligned}\mathbb{E}[M_\tau \mid S_0 = k] &= \mathbb{E}\left[\lim_{n\to\infty} M_{\tau\wedge n} \,\middle|\, S_0 = k\right] \\ &= \lim_{n\to\infty} \mathbb{E}[M_{\tau\wedge n} \mid S_0 = k] \\ &= \mathbb{E}[M_0 \mid S_0 = k] \\ &= \left(\frac{q}{p}\right)^k,\end{aligned}$$

hence

$$\begin{aligned}\left(\frac{q}{p}\right)^k &= \mathbb{E}[M_\tau \mid S_0 = k] \\ &= \left(\frac{q}{p}\right)^B \mathbb{P}\left(M_\tau = \left(\frac{q}{p}\right)^B \,\middle|\, S_0 = k\right) + \left(\frac{q}{p}\right)^0 \mathbb{P}\left(M_\tau = \left(\frac{q}{p}\right)^0 \,\middle|\, S_0 = k\right) \\ &= \left(\frac{q}{p}\right)^B \mathbb{P}\left(M_\tau = \left(\frac{q}{p}\right)^B \,\middle|\, S_0 = k\right) + \mathbb{P}(M_\tau = 1 \mid S_0 = k).\end{aligned}$$

Solving the system of equations

$$\begin{cases} \left(\frac{q}{p}\right)^k = \left(\frac{q}{p}\right)^B \mathbb{P}\left(M_\tau = \left(\frac{q}{p}\right)^B \,\middle|\, S_0 = k\right) + \mathbb{P}(M_\tau = 1 \mid S_0 = k) \\ \mathbb{P}\left(M_\tau = \left(\frac{q}{p}\right)^B \,\middle|\, S_0 = k\right) + \mathbb{P}(M_\tau = 1 \mid S_0 = k) = 1, \end{cases}$$

gives

$$\begin{aligned}\mathbb{P}(S_\tau = B \mid S_0 = k) &= \mathbb{P}\left(M_\tau = \left(\frac{q}{p}\right)^B \,\middle|\, S_0 = k\right) \\ &= \frac{(q/p)^k - 1}{(q/p)^B - 1}, \qquad k = 0, 1, \ldots, B,\end{aligned} \tag{10.4.4}$$

and

$$\begin{aligned}\mathbb{P}(S_\tau = 0 \mid S_0 = k) &= \mathbb{P}(M_\tau = 1 \mid S_0 = k) \\ &= 1 - \frac{(q/p)^k - 1}{(q/p)^B - 1}, \\ &= \frac{(q/p)^B - (q/p)^k}{(q/p)^B - 1},\end{aligned}$$

$k = 0, 1, \ldots, B$, which recovers (2.2.12).

10.5 Mean Game Duration

In this section we show that the mean game durations $\mathbb{E}[\tau \mid S_0 = k]$ computed in Sect. 2.3 can also be recovered as a second application of the Stopping Time Theorem 10.9.

Unbiased case $p = q = 1/2$

In the unbiased case of a fair game with $p = q = 1/2$, the martingale method can be used by noting that $\left(S_n^2 - n\right)_{n\geq 0}$ is also a martingale.

Step 1. The process $\left(S_n^2 - n\right)_{n\geq 0}$ is a martingale.

We have

$$\begin{aligned}
&\mathbb{E}[S_{n+1}^2 - (n+1) \mid \mathcal{F}_n] = \mathbb{E}[(S_{n+1} - S_n + S_n)^2 - (n+1) \mid \mathcal{F}_n] \\
&= \mathbb{E}[S_n^2 + (S_{n+1} - S_n)^2 + 2S_n(S_{n+1} - S_n) - (n+1) \mid \mathcal{F}_n] \\
&= \mathbb{E}[S_n^2 - n - 1 \mid \mathcal{F}_n] + \mathbb{E}[(S_{n+1} - S_n)^2 \mid \mathcal{F}_n] + 2\,\mathbb{E}[S_n(S_{n+1} - S_n) \mid \mathcal{F}_n] \\
&= S_n^2 - n - 1 + \mathbb{E}[(S_{n+1} - S_n)^2 \mid \mathcal{F}_n] + 2S_n\,\mathbb{E}[S_{n+1} - S_n \mid \mathcal{F}_n] \\
&= S_n^2 - n - 1 + \underbrace{\mathbb{E}[(S_{n+1} - S_n)^2]}_{=1} + 2S_n \underbrace{\mathbb{E}[S_{n+1} - S_n]}_{=0} \\
&= S_n^2 - n, \quad n \geq 0,
\end{aligned}$$

since $\mathbb{E}[S_{n+1} - S_n] = 0$ and $\mathbb{E}[(S_{n+1} - S_n)^2] = 1$.

Step 2. The stopped process $\left(S_{\tau\wedge n}^2 - \tau \wedge n\right)_{n\geq 0}$ is also a martingale, as a consequence of the Stopping Time Theorem 10.9.

Step 3. Since the stopped process $\left(S_{\tau\wedge n}^2 - \tau \wedge n\right)_{n\geq 0}$ is also a martingale by the Stopping Time Theorem 10.9, its expectation $\mathbb{E}[S_{\tau\wedge n}^2 - \tau \wedge n \mid S_0 = k]$ is constant in $n \geq 0$ by Proposition 10.5, hence we have

$$k^2 = \mathbb{E}[S_0^2 - 0 \mid S_0 = k] = \mathbb{E}\left[S_{\tau\wedge n}^2 - \tau \wedge n \,\middle|\, S_0 = k\right],$$

and since $\mathbb{P}(\tau < \infty) = 1$, after taking the limit as n tends to infinity and using dominated convergence, we find

$$\begin{aligned}
\mathbb{E}\left[S_\tau^2 - \tau \mid S_0 = k\right] &= \mathbb{E}\left[\lim_{n\to\infty} S_{\tau\wedge n}^2 - \lim_{n\to\infty} \tau \wedge n \,\middle|\, S_0 = k\right] \\
&= \mathbb{E}\left[\lim_{n\to\infty} S_{\tau\wedge n}^2 \mid S_0 = k\right] - \mathbb{E}\left[\lim_{n\to\infty} \tau \wedge n \mid S_0 = k\right]
\end{aligned}$$

$$
\begin{aligned}
&= \lim_{n\to\infty} \mathbb{E}\left[S^2_{\tau\wedge n} \mid S_0 = k\right] - \lim_{n\to\infty} \mathbb{E}[\tau\wedge n \mid S_0 = k] \\
&= \lim_{n\to\infty} \mathbb{E}\left[S^2_{\tau\wedge n} - \tau\wedge n \,\middle|\, S_0 = k\right] \\
&= k^2,
\end{aligned}
$$

by application of the *dominated* and *monotone* convergence theorems, since $S^2_{\tau\wedge n} \in [0, B^2]$ for all $n \geq 0$ and the mapping $n \mapsto \tau \wedge n$ is non-decreasing in $n \geq 0$. This now yields

$$
\begin{aligned}
k^2 &= \mathbb{E}\left[S^2_\tau - \tau \mid S_0 = k\right] \\
&= \mathbb{E}\left[S^2_\tau \mid S_0 = k\right] - \mathbb{E}[\tau \mid S_0 = k] \\
&= B^2\mathbb{P}(S_\tau = B \mid S_0 = k) + 0^2 \times \mathbb{P}(S_\tau = 0 \mid S_0 = k) - \mathbb{E}[\tau \mid S_0 = k],
\end{aligned}
$$

i.e.

$$
\begin{aligned}
\mathbb{E}[\tau \mid S_0 = k] &= B^2\mathbb{P}(S_\tau = B \mid S_0 = k) - k^2 \\
&= B^2\frac{k}{B} - k^2 \\
&= k(B-k),
\end{aligned}
$$

$k = 0, 1, \ldots, B$, which recovers (2.3.19).

Biased case $p \neq q$

Finally, we show how to recover the value of the mean game duration, i.e. the mean hitting time of the boundaries $\{0, B\}$, in the biased non-symmetric case $p \neq q$.

Step 1. The process $S_n - (p-q)n$ is a martingale.

In this case, we note that although $(S_n)_{n\geq 0}$ does not have centered increments and is not a martingale, the compensated process

$$
S_n - (p-q)n, \qquad n \geq 0,
$$

is a martingale because, in addition to being independent, its increments are centered random variables:

$$
\mathbb{E}[S_{n+1} - S_n - (p-q)] = \mathbb{E}[S_{n+1} - S_n] - (p-q) = 0, \qquad n \geq 0,
$$

by (10.4.3).

Step 2. The stopped process $(S_{\tau\wedge n} - (p-q)(\tau\wedge n))_{n\geq 0}$ is also a martingale, as a consequence of the Stopping Time Theorem 10.9.

Step 3. The expectation $\mathbb{E}[S_{\tau\wedge n} - (p-q)(\tau\wedge n) \mid S_0 = k]$ is constant in $n \geq 0$.

Table 10.1 List of martingales

Problem \ Probabilities	Unbiased	Biased
Ruin probability	S_n	$\left(\frac{q}{p}\right)^{S_n}$
Mean game duration	$S_n^2 - n$	$S_n - (p-q)n$

Step 4. Since the stopped process $(S_{\tau\wedge n} - (p-q)(\tau\wedge n))_{n\geq 0}$ is a martingale by the Stopping Time Theorem 10.9, we have

$$k = \mathbb{E}[S_0 - 0 \mid S_0 = k] = \mathbb{E}[S_{\tau\wedge n} - (p-q)(\tau\wedge n) \mid S_0 = k],$$

and, since $\mathbb{P}(\tau < \infty) = 1$, after taking the limit as n goes to infinity we find

$$\begin{aligned}\mathbb{E}[S_\tau - (p-q)\tau \mid S_0 = k] &= \lim_{n\to\infty} \mathbb{E}[S_{\tau\wedge n} - (p-q)(\tau\wedge n) \mid S_0 = k] \\ &= \mathbb{E}\Big[\lim_{n\to\infty} S_{\tau\wedge n} - (p-q)\lim_{n\to\infty} \tau\wedge n \,\Big|\, S_0 = k\Big] \\ &= k,\end{aligned}$$

which gives

$$\begin{aligned}k &= \mathbb{E}[S_\tau - (p-q)\tau \mid S_0 = k] \\ &= \mathbb{E}[S_\tau \mid S_0 = k] - (p-q)\,\mathbb{E}[\tau \mid S_0 = k] \\ &= B \times \mathbb{P}(S_\tau = B \mid S_0 = k) + 0 \times \mathbb{P}(S_\tau = 0 \mid S_0 = k) - (p-q)\,\mathbb{E}[\tau \mid S_0 = k],\end{aligned}$$

i.e.

$$\begin{aligned}(p-q)\,\mathbb{E}[\tau \mid S_0 = k] &= B \times \mathbb{P}(S_\tau = B \mid S_0 = k) - k \\ &= B\frac{(q/p)^k - 1}{(q/p)^B - 1} - k,\end{aligned}$$

from (10.4.4), hence

$$\mathbb{E}[\tau \mid S_0 = k] = \frac{1}{p-q}\left(B\frac{(q/p)^k - 1}{(q/p)^B - 1} - k\right), \qquad k = 0, 1, \ldots, B,$$

which recovers (2.3.12). In Table 10.1, we summarize the family of martingales used to treat the above problems.

The above examples can be summarized in Fig. 10.4, which can be generated using the following R code.

```
nsim <- 100;a=-2;b=6;x=1;mu=0;N=21; T<-2.0; t <- 0:(N-1); dt <- T/N;
     prob=0; time=0;
dev.new(width=16,height=8); for (i in 1:nsim){signal=0;colour="blue";Z <-
     2*(rbinom(N,1,0.5)-0.5);
tau=N+1; X <- c(1,N);X[1]=x;for (j in 2:N){X[j]=X[j-1]+Z[j]+mu*dt
if (X[j]<=a && signal==0)
     {xtau=a;tau=j-1;signal=-1;colour="purple";time=time+j}
if (X[j]>=b && signal==0)
     {xtau=b;tau=j-1;signal=1;colour="blue";prob=prob+1;time=time+j}}
plot(t, X, xlab = "n", ylab = "", type = "l",lwd=3, ylim = c(a-2,b+2), col
     = "blue", xaxs="i",yaxs="i", xaxt="n",
     yaxt="n",cex.axis=1.8,cex.lab=1.8,cex.main=2)
if (N>tau) {lines(t, c(X[1:tau],rep(xtau,N-tau))-signal*0.07, xlab = "n",
     ylab = "", type = "l",lwd=3, ylim = c(a-2,b+2), col = "red",
     xaxs="i",yaxs="i", xaxt="n",
     yaxt="n",cex.axis=1.8,cex.lab=1.8,cex.main=2)}
points(t, X, xlab = "t", ylab = "", type = "p", pch = 16, cex=1.6, ylim =
     c(-6,12), col = "blue", xaxs="i",yaxs="i", xaxt="n",
     yaxt="n",cex.axis=1.8,cex.lab=1.8,cex.main=2); yticks<-c(a,x,b);
     axis(side=2, mgp=c(1,1,0), at=yticks,labels = c("a=
     ",as.expression(bquote('M'['0']~"= ")),"b= "), las = 2,cex.axis=1.3)
axis(side=2, at = seq(a-2, b+2, by = 1), las = 2,cex.axis=1.3)
axis(side=1, at = seq(0, N, by = 1), las = 2,cex.axis=1.3)
axis(side=1, at = c(tau), labels = expression(paste(tau,"= ")),las =
     2,cex.axis=1.3)
abline(h=x,lw=2); abline(h=a,col="black",lwd=3);
     abline(h=b,col="black",lwd=3)
abline(v=tau,col="purple",lwd=3); abline(v=tau,col="purple",lwd=3);
text(1.17, 5.5, as.expression(bquote('M'[tau~"\u2227 10"]~"= ")),cex=2)
text(2.2, 5.5, X[min(10,tau)+1],cex=2)
text(0.75, 4.5, as.expression(bquote('M'[tau]~"= ")),cex=2); text(1.3, 4.5,
     X[tau+1],cex=2)
text(0.75, 3.5, as.expression(bquote('M'['5']~"= ")),cex=2); text(1.3, 3.5,
     X[6],cex=2)
text(0.75, 2.5, as.expression(bquote('M'['2']~"= ")),cex=2); text(1.3, 2.5,
     X[3],cex=2)
legend(-0.3, 8.35, legend = c(expression(Process ~ M[n]),
     expression(Stopped ~ process ~ M[tau~"\u2227 n"])), col = c("blue",
     "red"), lty = 1, lwd = 3, box.lty = 0, cex = 2)
grid(N-1,b-a+4,col="black"); grid(N-1,b-a+4,col="black")
readline(prompt = "Pause. Press <Enter> to continue...")}
```

Exercises

Exercise 10.1 (Steele (2001)) Recover (2.2.30) by showing that for all $k = 0, 1, \ldots, B$ we have $\mathbb{P}(\tau_{0,B} < \infty \mid S_0 = k) = 1$, i.e. the stopping time $\tau_{0,B}$ defined in (10.4.1) is finite almost surely.

Exercise 10.2 Doubling down. Consider a sequence $(X_n)_{n\geq 1}$ of independent Bernoulli random variables, with

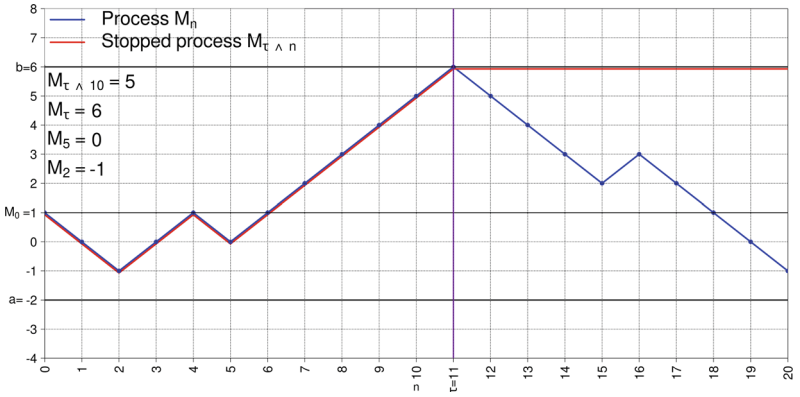


Fig. 10.4 Stopped process

$$\mathbb{P}(X_n = 1) = \mathbb{P}(X_n = -1) = \frac{1}{2}, \qquad n \geq 1,$$

and the process $(M_n)_{n\geq 0}$ defined by $M_0 := 0$ and

$$M_n := \sum_{k=1}^{n} 2^{k-1} X_k, \qquad n \geq 1,$$

with in particular

$$\begin{cases} M_1 = X_1, \\ M_2 = X_1 + 2X_2, \\ M_3 = X_1 + 2X_2 + 4X_3, \\ \vdots \end{cases}$$

see Fig. 10.5.

Note that when $X_1 = X_2 = \cdots = X_{n-1} = -1$ and $X_n = 1$, we have

$$M_n = -\sum_{k=1}^{n-1} 2^{k-1} + 2^{n-1} = -\frac{1-2^{n-1}}{1-2} + 2^{n-1} = 1, \qquad n \geq 1,$$

while when $X_1 = X_2 = \cdots = X_{n-1} = X_n = -1$, we have

$$M_n = -\sum_{k=1}^{n} 2^{k-1} = -\frac{1-2^n}{1-2} = 1 - 2^n, \qquad n \geq 1.$$

(a) Show that the process $(M_n)_{n\geq 0}$ is a martingale.

(b) Is the random time

$$\tau := \inf\{n \geq 1 \ : \ M_n = 1\}$$

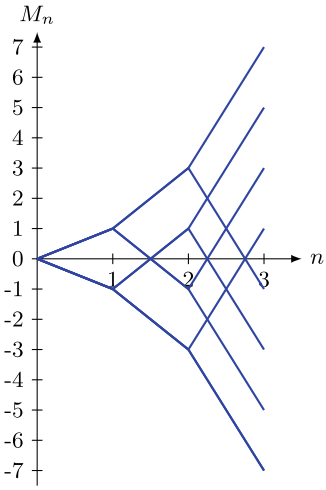


Fig. 10.5 Possible paths of the process $(M_n)_{n\geq 0}$

a stopping time?

(c) Consider the stopped process

$$M_{\tau\wedge n} := M_n \mathbb{1}_{\{n<\tau\}} + \mathbb{1}_{\{\tau\leq n\}} = \begin{cases} M_n = 1-2^n & \text{if } n < \tau, \\ M_\tau = 1 & \text{if } n \geq \tau, \end{cases}$$

$n \geq 0$, see Fig. 10.6. Give an interpretation of $(M_{n\wedge\tau})_{n\geq 0}$ in terms of betting strategy for a gambler starting a game at $M_0 = 0$.

(d) Determine the two possible values of $M_{\tau\wedge n}$ and the probability distribution of $M_{\tau\wedge n}$ at any time $n \geq 1$.

(e) Show, using the result of Question (d), that we have

$$\mathbb{E}[M_{\tau\wedge n}] = 0, \qquad n \geq 0.$$

(f) Show that the result of Question (e) can be recovered using the Stopping Time Theorem 10.9.

(g) Show that $\mathbb{P}(\tau < \infty) = 1$.

(h) Do we have $\mathbb{E}[M_\tau] = \lim_{n\to\infty} \mathbb{E}[M_{\tau\wedge n}]$?

Exercise 10.3 Let $(S_n)_{n\geq 0}$ be the random walk defined by $S_0 := 0$ and

$$S_n := \sum_{k=1}^{n} X_k = X_1 + \cdots + X_n, \qquad n \geq 1,$$

where $(X_n)_{n\geq 1}$ is an i.i.d. Bernoulli sequence of $\{-1, 1\}$-valued random variables with $\mathbb{P}(X_n = 1) = \mathbb{P}(X_n = -1) = 1/2$, $n \geq 1$. We consider the random time

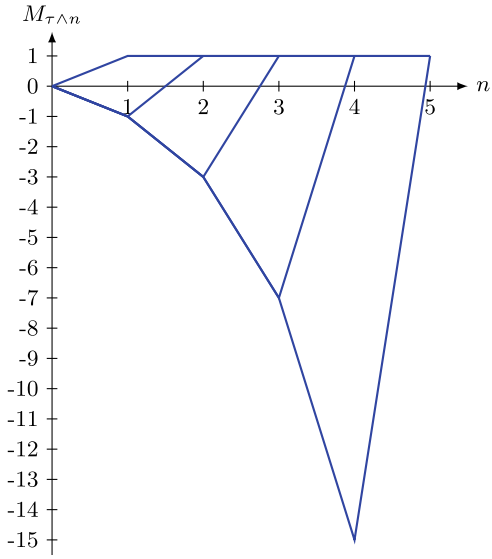


Fig. 10.6 Possible paths of the stopped process $(M_{\tau\wedge n})_{n\geq 0}$

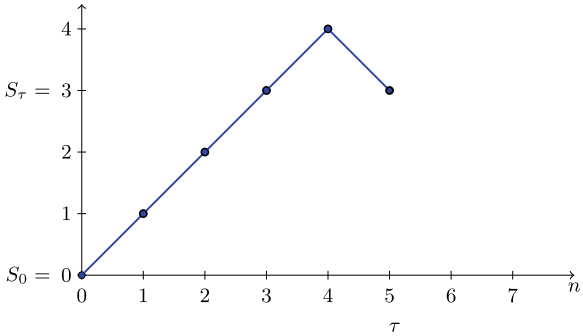


Fig. 10.7 Sample path of the random walk $(S_n)_{n\geq 0}$

$$\tau := \inf\{n \geq 1 \ : \ S_n - S_{n-1} = -1\}$$

at which $(S_n)_{n\geq 0}$ drops for the first time, and decide to use τ as an exit strategy (Fig. 10.7).

(a) Show that τ is a stopping time.
(b) Find the range of possible values of S_τ.
(c) Give the value of $\mathbb{E}[S_\tau]$ according to the stopping time theorem.
(d) Find the probability distribution of S_τ, i.e. find $\mathbb{P}(S_\tau = k)$ for all $k \in \mathbb{Z}$.
(e) Compute

$$\mathbb{E}[S_\tau] = \sum_{k\in\mathbb{Z}} k\mathbb{P}(S_\tau = k),$$

and recover the conclusion of part (c).

Exercise 10.4 Show that, as the discrete-time filtration $(\mathcal{F}_n)_{n\geq 0}$ satisfies $\mathcal{F}_{n-1} \subset \mathcal{F}_n$, $n \geq 1$, Condition (10.3.1) is equivalent to

$$\{\tau = n\} \in \mathcal{F}_n, \qquad n \geq 0. \tag{10.5.1}$$

Exercise 10.5 Give an example of an a.s. converging and unbounded sequence $(X_n)_{n\geq 0}$ of random variables for which expectation and limit cannot be exchanged, i.e.

$$\lim_{n\to\infty} \mathbb{E}[X_n] \neq \mathbb{E}[\lim_{n\to\infty} X_n].$$

Exercise 10.6 Let $(M_n)_{n\geq 0}$ be a discrete-time *sub*martingale with respect to a filtration $(\mathcal{F}_n)_{n\geq 0}$, with $\mathcal{F}_0 = \{\emptyset, \Omega\}$, i.e. we have

$$M_n \leq \mathbb{E}[M_{n+1} \mid \mathcal{F}_n], \qquad n \geq 0.$$

(a) Show that for all $n \geq 0$, we have $\mathbb{E}[M_n] \leq \mathbb{E}[M_{n+1}]$, i.e. *sub*martingales have a *non-decreasing* expected value.

(b) Show that independent increment processes whose increments have nonnegative expectation are examples of *sub*martingales.

(c) (Doob-Meyer decomposition). Show that there exists two processes $(N_n)_{n\geq 0}$ and $(A_n)_{n\geq 0}$ such that

(i) $(N_n)_{n\geq 0}$ is a martingale with respect to $(\mathcal{F}_n)_{n\geq 0}$,

(ii) $(A_n)_{n\geq 0}$ is *non-decreasing*, i.e. $A_n \leq A_{n+1}$, *a.s.*, $n \geq 0$,

(iii) $(A_n)_{n\geq 0}$ is predictable in the sense that A_n is $\mathcal{F}_{n-1}$-measurable, $n \geq 1$, and

(iv) $M_n = N_n + A_n$, $n \in \mathbb{N}$.
Hint: Let $A_0 := 0$, $A_1 := A_0 + \mathbb{E}[M_1 - M_0 \mid \mathcal{F}_0]$, and

$$A_{n+1} := A_n + \mathbb{E}[M_{n+1} - M_n \mid \mathcal{F}_n], \quad n \geq 0,$$

and define $(N_n)_{n\geq 0}$ in such a way that it satisfies the four required properties.

(d) Show that for all bounded stopping times σ and τ such that $\sigma \leq \tau$ *a.s.*, we have

$$\mathbb{E}[M_\sigma] \leq \mathbb{E}[M_\tau].$$

Hint: Use the Doob Stopping Time Theorem 10.9 for martingales and (10.3.3).

Exercise 10.7 We say that a function $\phi : \mathbb{R} \to \mathbb{R}$ is *convex* if

$$\phi(px + qy) \leq p\phi(x) + q\phi(y), \qquad x, y \in \mathbb{R},$$

for all $p, q \in [0, 1]$ such that $p + q = 1$ (Fig. 10.8).

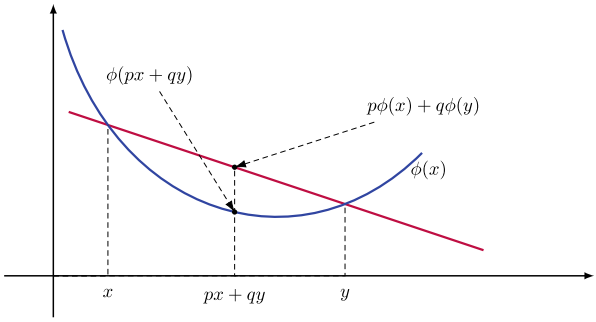


Fig. 10.8 Convex function

(a) Show that for any convex function $\phi : \mathbb{R} \to \mathbb{R}$ we have the inequality

$$\phi\left(\frac{x_1+x_2+\cdots+x_n}{n}\right) \leq \frac{\phi(x_1)+\phi(x_2)+\cdots+\phi(x_n)}{n}, \tag{10.5.2}$$

$x_1, \ldots, x_n \in \mathbb{R}$, $n \geq 1$.

(b) Consider a martingale $(S_n)_{n=0,1,\ldots,N}$ under the probability measure $\mathbb{P}$, with respect to the filtration $(\mathcal{F}_n)_{n\geq 0}$, and let ϕ be a convex function. Show an inequality between the quantities

$$\mathbb{E}\left[\phi\left(\frac{S_1+S_2+\cdots+S_N}{N}\right)\right] \quad \text{and} \quad \mathbb{E}[\phi(S_N)].$$

Hint: Use in the following order:

(i) the convexity inequality (10.5.2),
(ii) the martingale property of $(S_k)_{k\geq 0}$,
(iii) the conditional Jensen inequality $\phi(\mathbb{E}[F \mid \mathcal{G}]) \leq \mathbb{E}[\phi(F) \mid \mathcal{G}]$,
(iv) the tower property of conditional expectations.

(c) Given the (convex) function $\phi(x) := (x-K)^+$, show that the price

$$\mathbb{E}\left[\left(\frac{S_1+S_2+\cdots+S_N}{N}-K\right)^+\right]$$

of a discrete-time arithmetic average option with payoff

$$\left(\frac{S_1+S_2+\cdots+S_N}{N}-K\right)^+$$

is upper bounded by the price of the European *call* option with payoff $(S_N-K)^+$ and maturity N.

Exercise 10.8 A stochastic process $(M_n)_{n\geq 0}$ is a *sub*martingale if it satisfies

$$M_k \leq \mathbb{E}[M_n \mid \mathcal{F}_k], \qquad k = 0, 1, \ldots, n.$$

(a) Show that the expected value $\mathbb{E}[M_n]$ of the *sub*martingale $(M_n)_{n\geq 0}$ is non-decreasing in time $n \geq 0$.
(b) Consider the random walk given by $S_0 := 0$ and

$$S_n := \sum_{k=1}^{n} X_k = X_1 + X_2 + \cdots + X_n, \qquad n \geq 1,$$

where $(X_n)_{n\geq 1}$ is an *i.i.d.* Bernoulli sequence of $\{0, 1\}$-valued random variables with $\mathbb{P}(X_n = 1) = p$, $n \geq 1$. Under which condition on $\alpha \in \mathbb{R}$ is the process $(S_n - \alpha n)_{n\geq 0}$ a *sub*martingale?

Exercise 10.9 Recall that a discrete-time stochastic process $(M_n)_{n\geq 0}$ is a *sub*martingale with respect to a filtration $(\mathcal{F}_n)_{n\geq 0}$ if it satisfies

$$M_k \leq \mathbb{E}[M_n \mid \mathcal{F}_k], \qquad k = 0, 1, \ldots, n.$$

(a) Show that any convex function $(\phi(M_n))_{n\geq 0}$ of a martingale $(M_n)_{n\geq 0}$ is itself a *sub*martingale. *Hint*: Use Jensen's inequality.
(b) Show that any convex non-decreasing function $\phi(M_n)$ of a *sub*martingale $(M_n)_{n\geq 0}$ remains a *sub*martingale.

Problem 10.10 (a) Consider $(M_n)_{n\geq 0}$ a *nonnegative martingale* generating the filtration $(\mathcal{F}_n)_{n\geq 0}$. For any $x > 0$, let

$$\tau_x := \inf\{n \geq 0 \ : \ M_n \geq x\}.$$

Show that the random time τ_x is a stopping time.
(b) Show that for all $n \geq 0$, we have

$$\mathbb{P}\Big(\max_{k=0,1,\ldots,n} M_k \geq x\Big) \leq \frac{\mathbb{E}[M_n]}{x}, \qquad x > 0. \tag{10.5.3}$$

Hint: Proceed as in the proof of the classical Markov inequality and use the Doob Stopping Time Theorem 10.9 for the stopping time τ_x.
(c) Show that (10.5.3) remains valid for $(M_n)_{n\geq 0}$ a nonnegative *sub*martingale.

Hint: Use the Doob Stopping Time Theorem 10.9 for *sub*martingales as in Exercise 10.6-(d).

(d) Show that if $(M_n)_{n\geq 0}$ a nonnegative *sub*martingale, then for any $n \geq 0$ we have

$$\mathbb{P}\Big(\max_{k=0,1,\ldots,n} M_k \geq x\Big) \leq \frac{\mathbb{E}[(M_n)^2]}{x^2}, \qquad x > 0.$$

(e) Show that more generally, we have

$$\mathbb{P}\Big(\max_{k=0,1,\ldots,n} M_k \geq x\Big) \leq \frac{\mathbb{E}[(M_n)^p]}{x^p}, \qquad x > 0,$$

for all $n \geq 0$ and $p \geq 1$.

(f) Given $(Y_n)_{n\geq 1}$ a sequence of centered independent random variables with same mean $\mathbb{E}[Y_n] = 0$ and variance $\sigma^2 = \mathrm{Var}[Y_n]$, $n \geq 1$, consider the random walk $S_n = Y_1 + Y_2 + \cdots + Y_n$, $n \geq 1$, with $S_0 = 0$.

Show that for all $n \geq 0$, we have

$$\mathbb{P}\Big(\max_{k=0,1,\ldots,n} |S_k| \geq x\Big) \leq \frac{n\sigma^2}{x^2}, \qquad x > 0.$$

(g) Show that for any (not necessarily nonnegative) *sub*martingale, we have

$$\mathbb{P}\Big(\max_{k=0,1,\ldots,n} M_k \geq x\Big) \leq \frac{\mathbb{E}[(M_n)^+]}{x}, \qquad x > 0,$$

where $(z)^+ := \max(z, 0)$, $z \in \mathbb{R}$.

(h) A stochastic process $(M_n)_{n\geq 0}$ is a *super*martingale[5] if it satisfies

$$\mathbb{E}[M_n \mid \mathcal{F}_k] \leq M_k, \qquad k = 0, 1, \ldots, n.$$

Show that for any *nonnegative super*martingale we have

$$\mathbb{P}\Big(\max_{k=0,1,\ldots,n} M_k \geq x\Big) \leq \frac{\mathbb{E}[M_0]}{x}, \qquad x > 0.$$

(i) Show that for any *nonnegative submartingale* $(M_n)_{n\geq 0}$ and any convex nondecreasing nonnegative function ϕ we have

$$\mathbb{P}\Big(\max_{k=0,1,\ldots,n} \phi(M_k) \geq x\Big) \leq \frac{\mathbb{E}[\phi(M_n)]}{x}, \qquad x > 0.$$

Hint: Consider the stopping time

[5] "This obviously inappropriate nomenclature was chosen under the malign influence of the noise level of radio's SUPERman program, a favorite supper-time program of Doob's son during the writing of Doob (1953)", cf. Doob (1984), historical notes, page 808.

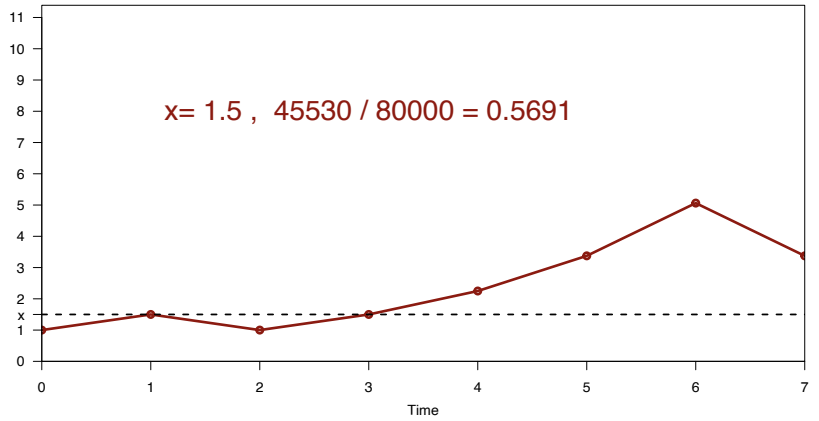


Fig. 10.9 Random walk supremum above $x = 1.5$

$$\tau_x^{\phi} := \inf\{n \geq 0 \ : \ \phi(M_n) \geq x\},$$

and use the result of Exercise 10.9-(b).

(j) Give an example of a nonnegative *super*martingale which is *not* a martingale.

Exercise 10.11 Consider the random walk $(S_n)_{n\geq 0}$ on $\{0, 1, \ldots, B\}$ with $S_0 := 0$ and independent $\{-1, 1\}$-valued increments $(S_{n+1} - S_n)_{n\geq 0}$ such that

$$\mathbb{P}(S_{n+1} - S_n = +1) = p \quad \text{and} \quad \mathbb{P}(S_{n+1} - S_n = -1) = q, \quad n \geq 0,$$

and the martingale $(M_n)_{n\geq 0}$ defined as

$$M_n := \left(\frac{q}{p}\right)^{S_n}, \qquad n \geq 0,$$

where $p, q \in (0, 1)$ are such that $p + q = 1$. Show that for all $n \geq 0$ and $r \geq 1$, we have (Fig. 10.9)[6]

$$\mathbb{P}\Big(\max_{k=0,1,\ldots,n} M_k \geq x\Big) \leq \frac{(p(q/p)^r + q(p/q)^r)^n}{x^r}, \qquad x > 0.$$

[6] The animation works in Acrobat Reader on the entire pdf file.

References

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Chapter 11
Spatial Poisson Processes



Spatial Poisson point processes are typically used to model the random scattering of configuration points within a plane or a three-dimensional space $\mathbb{X}$. In case $\mathbb{X} = \mathbb{R}_+$ is the real half line, these random points can be identified with the jump times $(T_k)_{k\geq 1}$ of the standard Poisson process $(N_t)_{t\in\mathbb{R}_+}$ introduced in Sect. 9.1. However, in contrast with the previous chapter, no time ordering is a priori imposed here on the index set $\mathbb{X}$. Sections 11.4 and 11.5 contain some more advanced results on moments and deviation inequalities for Poisson stochastic integrals.

11.1 Spatial Poisson Processes

In this section, we present the construction of spatial Poisson point processes on the space

$$\Omega^{\mathbb{X}} := \left\{\omega := (x_i)_{i=1}^N \subset \mathbb{X},\ N \in \mathbb{N} \cup \{\infty\}\right\}$$

of subsets of $\mathbb{X} \subset \mathbb{R}^d$ called *configurations*, for $d \geq 1$. Figure 11.1 illustrates a given configuration $\omega \in \Omega^{\mathbb{X}}$.

Definition 11.1 Given a (measurable) subset A of $\mathbb{X}$, we let

$$\omega(A) = \#\{x \in \omega\ :\ x \in A\} = \sum_{x\in\omega} \mathbb{1}_A(x)$$

denote the number of configuration points in ω that are contained in the set A.

We consider an intensity measure $\sigma(dx)$ on $\mathbb{X}$, possibly given from a nonnegative density function $\rho : \mathbb{X} \longrightarrow \mathbb{R}_+$ as $\sigma(dx) = \rho(x)dx$, i.e. for any (measurable) subset A of $\mathbb{X}$ we have

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N. Privault, *Understanding Markov Chains*, Springer Undergraduate Mathematics Series,
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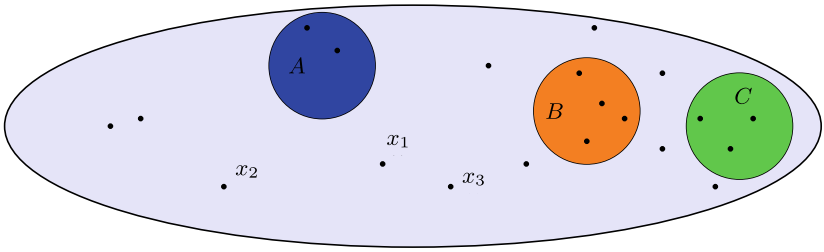


Fig. 11.1 Poisson random samples with $\omega(A) = 2$, $\omega(B) = 4$, $\omega(C) = 3$

$$\begin{aligned}\sigma(A) &= \int_A \sigma(dx) \\ &= \int_A \rho(x)dx \\ &= \int_{\mathbb{X}} \mathbb{1}_A(x)\rho(x)dx.\end{aligned}$$

When $\sigma(\mathbb{X}) < \infty$, the Poisson point process with intensity $\sigma(dx)$ can be constructed in three steps:

1. First, choose the number $\omega(\mathbb{X})$ of points in $\mathbb{X}$ according to a standard Poisson distribution with mean $\sigma(\mathbb{X})$:

$$\mathbb{P}_\sigma^{\mathbb{X}}(\omega(\mathbb{X}) = n) = \mathrm{e}^{-\sigma(\mathbb{X})}\frac{(\sigma(\mathbb{X}))^n}{n!}, \qquad n \geq 0.$$

2. Second, scatter $n = \omega(\mathbb{X})$ points $(X_1 \ldots, X_n)$ over $\mathbb{X}$ independently, each of them with the probability distribution $\sigma(dx)/\sigma(\mathbb{X})$, i.e.

$$\mathbb{P}_\sigma^{\mathbb{X}}((X_1, \ldots, X_n) \in A_1 \times \cdots \times A_n \mid \omega(\mathbb{X}) = n) = \frac{\sigma(A_1)}{\sigma(\mathbb{X})} \cdots \frac{\sigma(A_n)}{\sigma(\mathbb{X})}, \tag{11.1.1}$$

for $A_1, \ldots, A_n$ measurable subsets of $\mathbb{X}$ with finite σ-measure.

```
library(spatstat)
lambda = 10000
bellcurve <- function(x,y,s){return(exp(-s*((x-0.5)**2+(y-0.5)**2)))}
rho <- function(x,y){lambda*bellcurve(x+0.2,y+0.2,70)+lambda*bellcurve(x-0.2,y-0.1,40)}
X <- rpoispp(rho)
plot(X, cols="blue", pch=16, cex=0.7, main = '')
```

Figure 11.2 presents another Poisson point process sample together with the density of its intensity measure (Fig. 11.3).

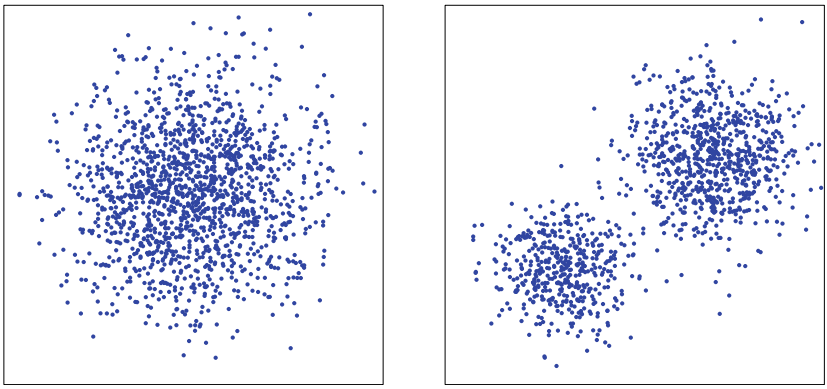
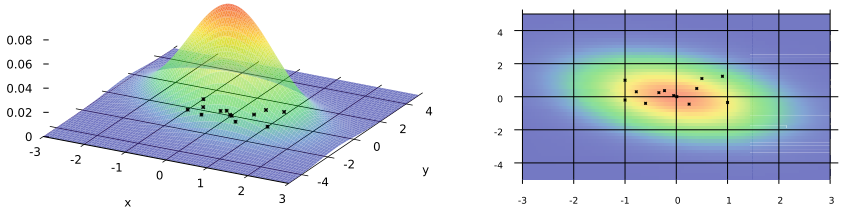


Fig. 11.2 Two Poisson point process samples



(a) Sample points *vs.* intensity $\rho(x)$. (b) View from above (heat map).

Fig. 11.3 Poisson point process sample on the plane

The Poisson probability measure $\mathbb{P}_\sigma^{\mathbb{X}}$ with intensity $\sigma(dx) = \rho(x)dx$ on $\mathbb{X}$ satisfying the above points 1 and 2 can be characterized in the next theorem, see Proposition I.6 in Neveu (1977).

Theorem 11.2 *Given $\rho : \mathbb{X} \longrightarrow \mathbb{R}_+$ a nonnegative function, the Poisson probability measure $\mathbb{P}_\sigma^{\mathbb{X}}$ with intensity $\sigma(dx) = \rho(x)dx$ on $\mathbb{X}$ is the only probability measure on $\Omega^{\mathbb{X}}$ satisfying the following two properties:*

(i) For any (measurable) subset A of $\mathbb{X}$ such that $\sigma(A) < \infty$, the number $\omega(A)$ of configuration points contained in A is a Poisson random variable with intensity $\sigma(A)$, i.e.

$$\mathbb{P}_\sigma^{\mathbb{X}}(\omega \in \Omega^{\mathbb{X}} \ : \ \omega(A) = n) = \mathrm{e}^{-\sigma(A)} \frac{(\sigma(A))^n}{n!}, \qquad n \geq 0.$$

(ii) For any sequence $A_1, A_2, \ldots, A_n$ are disjoint measurable subsets of $\mathbb{X}$ with $\sigma(A_k) < \infty$, $k = 1, 2, \ldots, n$, the $\mathbb{N}^n$-valued random vector

$$\omega \longmapsto (\omega(A_1), \ldots, \omega(A_n)), \qquad \omega \in \Omega^{\mathbb{X}},$$

is made of independent random variables for all $n \geq 1$.

In the remaining of this chapter, we will assume for simplicity that $\sigma(\mathbb{X}) < \infty$. The Poisson measure $\mathbb{P}_\sigma^{\mathbb{X}}$ can also be defined from

$$\mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}[F] = \mathrm{e}^{-\sigma(\mathbb{X})} \sum_{n\geq 0} \frac{1}{n!} \int_{\mathbb{X}^n} f_n(x_1, x_2, \ldots, x_n)\sigma(dx_1)\cdots\sigma(dx_n) \tag{11.1.2}$$

for F written as

$$F(\omega) = f_0 \mathbb{1}_{\{\omega(\mathbb{X})=0\}} + \sum_{n\geq 1} \mathbb{1}_{\{\omega(\mathbb{X})=n\}} f_n(x_1, x_2, \ldots, x_n)$$

where f_n is a symmetric integrable function of $\omega = \{x_1, x_2, \ldots, x_n\}$ when $\omega(\mathbb{X}) = n$, $n \geq 1$, cf. e.g. Proposition 6.1.3 and Sect. 6.1 in Privault (2009).

In many applications, the intensity function $\rho(x)$ can be constant, i.e. $\rho(x) = \lambda > 0$, $x \in \mathbb{X}$, where $\lambda > 0$ is called the intensity parameter, and

$$\sigma(A) = \lambda \int_A dx = \lambda \int_{\mathbb{X}} \mathbb{1}_A(x)dx$$

represents the surface or volume of A in $\mathbb{R}^d$. In this case, (11.1.1) can be used to show that the random points $\{x_1, \ldots, x_n\}$ are uniformly distributed on A^n given that $\{\omega(A) = n\}$.

11.2 Poisson Stochastic Integrals

In Proposition 11.3, we consider the Poisson stochastic integral defined as

$$\int_{\mathbb{X}} f(x)\omega(dx) := \sum_{x\in\omega} f(x), \tag{11.2.1}$$

for $f \in L^1(\mathbb{X}, \sigma)$ an integrable function on $\mathbb{X}$, and we compute its first and second order moments and cumulants via its characteristic function.

Proposition 11.3 *Let $f \in L^1(\mathbb{X}, \sigma)$ be an integrable function on $\mathbb{X}$. We have*

$$\mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}\left[\exp\left(i\int_{\mathbb{X}} f(x)\omega(dx)\right)\right] = \exp\left(\int_{\mathbb{X}} (\mathrm{e}^{if(x)} - 1)\sigma(dx)\right).$$

Proof We assume that $\sigma(\mathbb{X}) < \infty$. By (11.1.2) and the definition (11.2.1) of the Poisson stochastic integral, we have

$$\begin{aligned}
&\mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}\left[\exp\left(i\int_{\mathbb{X}} f(x)\omega(dx)\right)\right] \\
&\quad = \mathrm{e}^{-\sigma(\mathbb{X})}\sum_{n\geq 0}\frac{1}{n!}\int_{\mathbb{X}}\cdots\int_{\mathbb{X}}\mathrm{e}^{i(f(x_1)+\cdots+f(x_n))}\sigma(dx_1)\cdots\sigma(dx_n). \\
&\quad = \mathrm{e}^{-\sigma(\mathbb{X})}\sum_{n\geq 0}\frac{1}{n!}\left(\int_{\mathbb{X}}\mathrm{e}^{if(x)}\sigma(dx)\right)^n \\
&\quad = \exp\left(\int_{\mathbb{X}}(\mathrm{e}^{if(x)}-1)\sigma(dx)\right).
\end{aligned}$$

□

The characteristic function also allows us to compute the expectation of $\int_{\mathbb{X}} f(x)\omega(dx)$ using the relation $i^2 = -1$, as

$$\begin{aligned}
\mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}\left[\int_{\mathbb{X}} f(x)\omega(dx)\right] &= -i\frac{d}{d\varepsilon}\,\mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}\left[\exp\left(i\varepsilon\int_{\mathbb{X}} f(x)\omega(dx)\right)\right]_{|\varepsilon=0} \\
&= -i\frac{d}{d\varepsilon}\exp\left(\int_{\mathbb{X}}(\mathrm{e}^{i\varepsilon f(x)}-1)\sigma(dx)\right)_{|\varepsilon=0} \\
&= \int_{\mathbb{X}} f(x)\sigma(dx),
\end{aligned}$$

for f an integrable function on $\mathbb{X}$. As a consequence, the *compensated* Poisson stochastic integral

$$\int_{\mathbb{X}} f(x)\omega(dx) - \int_{\mathbb{X}} f(x)\sigma(dx)$$

is a *centered* random variable, i.e. we have

$$\mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}\left[\int_{\mathbb{X}} f(x)\omega(dx) - \int_{\mathbb{X}} f(x)\sigma(dx)\right] = 0.$$

The variance can be similarly computed as

$$\mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}\left[\left(\int_{\mathbb{X}} f(x)(\omega(dx)-\sigma(dx))\right)^2\right] = \int_{\mathbb{X}}|f(x)|^2\sigma(dx),$$

for all f in the space $L^2(\mathbb{X},\sigma)$ of functions which are square-integrable on $\mathbb{X}$ with respect to $\sigma(dx)$.

More generally, the logarithmic generating function

$$\begin{aligned}\log \mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}\left[\exp\left(\int_{\mathbb{X}} f(x)\omega(dx)\right)\right] &= \int_{\mathbb{X}}(\mathrm{e}^{f(x)}-1)\sigma(dx)\\ &= \sum_{n\geq 1}\frac{1}{n!}\int_{\mathbb{X}} f^n(x)\sigma(dx),\end{aligned}$$

shows that the *cumulants* of $\int_{\mathbb{X}} f(x)\omega(dx)$, see (1.7.11), are given by

$$\kappa_n = \int_{\mathbb{X}} f^n(x)\sigma(dx), \qquad n \geq 1. \tag{11.2.2}$$

Example-Gamma Distribution

When $\mathbb{X} = \mathbb{R}_+$ and $\rho(x) = \lambda \mathrm{e}^{-xt}/x$, $\lambda, t > 0$, i.e. σ is given by $\sigma(dx) = \rho(x)dx = \lambda \mathrm{e}^{-xt}dx/x$, the Poisson stochastic integral $\displaystyle\int_0^\infty x\omega(dx) = \sum_{x\in\omega} x$ has the Laplace transform

$$\begin{aligned}\mathbb{E}_\sigma\left[\exp\left(-s\int_{\mathbb{X}} x\omega(dx)\right)\right] &= \exp\left(\int_{\mathbb{X}}(\mathrm{e}^{-sx}-1)\sigma(dx)\right)\\ &= \exp\left(\lambda\int_{\mathbb{X}}(\mathrm{e}^{-sx}-1)\mathrm{e}^{-xt}\frac{dx}{x}\right)\\ &= \exp\left(-\lambda\log\left(1+\frac{s}{t}\right)\right)\\ &= \left(1+\frac{s}{t}\right)^{-\lambda}\\ &= \frac{t^\lambda}{\Gamma(\lambda)}\int_0^\infty \mathrm{e}^{-sy}y^{\lambda-1}\mathrm{e}^{-yt}dy, \qquad s > -t,\end{aligned}$$

where we used Frullani's identity

$$\log\left(1+\frac{s}{t}\right) = \int_0^\infty (1-\mathrm{e}^{-sx})\mathrm{e}^{-xt}\frac{dx}{x}, \qquad s,t>0.$$

This shows that the random variable $\int_0^\infty x\omega(dx) = \sum_{x\in\omega} x$ has the gamma distribution with probability density function (Fig. 11.4)

$$y \mapsto \frac{t^\lambda}{\Gamma(\lambda)}y^{\lambda-1}\mathrm{e}^{-yt}, \qquad y > 0,$$

shape parameter λ, scaling parameter $1/t$, and mean λ/t.

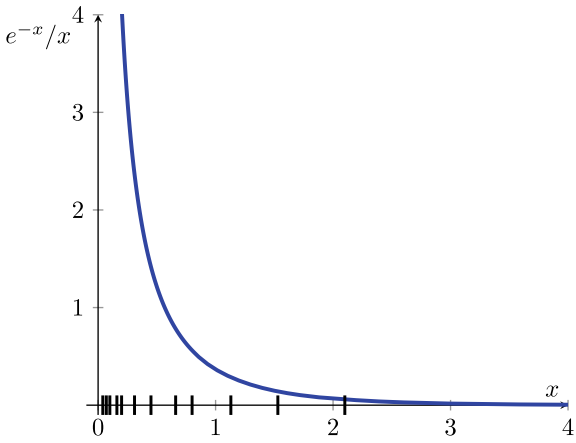


Fig. 11.4 Gamma Lévy density $\rho(x) = \lambda e^{-xt}/x$

11.3 Transformations of Poisson Measures

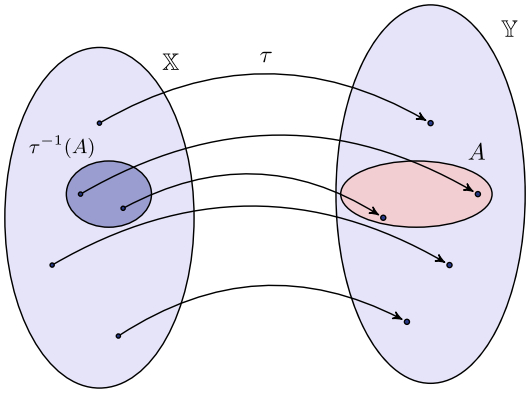
Consider a mapping $\tau : (\mathbb{X}, \sigma) \longrightarrow \mathbb{Y}$, and let

$$\tau_* : \Omega^{\mathbb{X}} \longrightarrow \Omega^{\mathbb{Y}}$$

be the transformed configuration defined by

$$\tau_*\omega = \{\tau(x_1), \tau(x_2), \tau(x_3), \ldots\} := \{\tau(x) \ : \ x \in \omega\}, \qquad \omega \in \Omega^{\mathbb{X}},$$

as illustrated in the following figure.



Proposition 11.4 *Assume that $\tau : \mathbb{X} \to \mathbb{Y}$ is a one-to-one mapping. The random configuration*

$$\begin{aligned} \Omega^{\mathbb{X}} : &\longrightarrow \Omega^{\mathbb{Y}} \\ \omega &\longmapsto \tau_*(\omega) \end{aligned}$$

has the Poisson distribution $\mathbb{P}_\mu^{\mathbb{Y}}$ *with intensity* μ *on* $\mathbb{Y}$*, where* $\mu := \tau_*\sigma$ *is the push-forward of the measure* σ *by* τ*.*

Proof For any set $A \subset \mathbb{Y}$ of finite μ-measure, we have

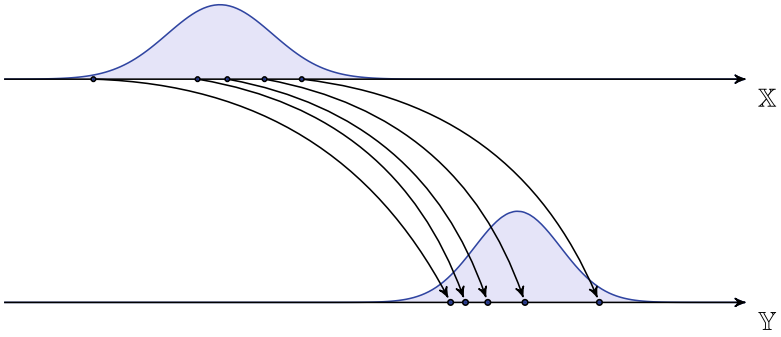
$$\begin{aligned} \mathbb{P}_\sigma^{\mathbb{X}}(\{\omega \in \mathbb{X} : \tau_*\omega(A) = n\}) &= \mathbb{P}_\sigma^{\mathbb{X}}(\{\omega \in \mathbb{X} : \omega(\tau^{-1}(A)) = n\}) \\ &= \mathrm{e}^{-\sigma(\tau^{-1}(A))} \frac{(\sigma(\tau^{-1}(A)))^n}{n!} \\ &= \mathrm{e}^{-\tau_*\sigma(A)} \frac{(\tau_*\sigma(A))^n}{n!} \\ &= \mathrm{e}^{-\mu(A)} \frac{(\mu(A))^n}{n!}. \end{aligned}$$

More generally, we can check that for all families $A_1, A_2, \ldots, A_n$ of disjoint subsets of $\mathbb{Y}$ and $k_1, k_2, \ldots, k_n \in \mathbb{N}$, we have

$$\begin{aligned} &\mathbb{P}_\sigma^{\mathbb{X}}\big(\{\omega \in \Omega^{\mathbb{X}} : \tau_*\omega(A_1) = k_1, \ldots, \tau_*\omega(A_n) = k_n\}\big) \\ &\quad = \mathbb{P}_\sigma^{\mathbb{X}}\big(\{\omega \in \Omega^{\mathbb{X}} : \omega(\tau^{-1}(A_1)) = k_1, \ldots, \tau_*\omega(\tau^{-1}(A_n)) = k_n\}\big) \\ &\quad = \prod_{i=1}^n \mathbb{P}_\sigma^{\mathbb{X}}(\{\omega \in \Omega^{\mathbb{X}} : \omega(\tau^{-1}(A_i)) = k_i\}) \\ &\quad = \prod_{i=1}^n \mathbb{P}_\sigma^{\mathbb{X}}\big(\{\omega \in \Omega^{\mathbb{X}} : \tau_*\omega(A_i) = k_i\}\big). \end{aligned}$$

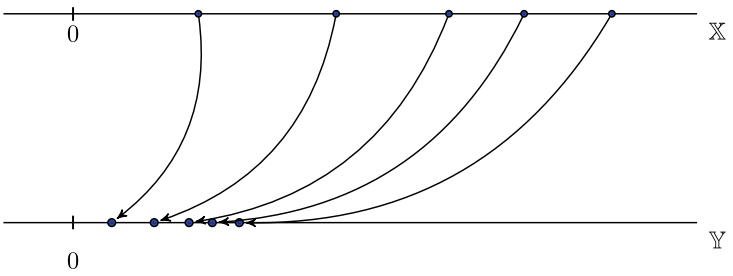
□

The next figure illustrates the transport of measure in the case of Gaussian intensities on $\mathbb{X} = \mathbb{R}$.



For example, in the case of a flat intensity $\rho(x) = \lambda$ on $\mathbb{X} = \mathbb{R}_+$ the intensity of the original Poisson point process becomes doubled under the mapping $\tau(x) = x/2$, since

$$\begin{aligned}\mathbb{P}_\sigma^{\mathbb{X}}(\tau_*\omega([0,t]) = n) &= \mathbb{P}_\sigma^{\mathbb{X}}(\omega(\tau^{-1}([0,t])) = n)\\ &= \mathrm{e}^{-\sigma(\tau^{-1}([0,t]))}\frac{(\sigma(\tau^{-1}([0,t])))^n}{n!}\\ &= \mathrm{e}^{-\sigma([0,2t])}\frac{(\sigma([0,2t]))^n}{n!}\\ &= \mathrm{e}^{-2\lambda t}(2\lambda t)^n/n!.\end{aligned}$$



11.4 Moments of Poisson Stochastic Integrals

As a consequence of (11.2.2) and the relation (1.7.11) between cumulants and moments we find that the *n-th* moment of $\int_{\mathbb{X}} f(x)\omega(dx)$ can be written as

$$\begin{aligned}&\mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}\left[\left(\int_{\mathbb{X}} f(x)\omega(dx)\right)^n\right]\\ &= \sum_{k=0}^n \frac{1}{k!}\sum_{d_1+\cdots+d_k=n}\frac{n!}{d_1!\cdots d_k!}\int_{\mathbb{X}} f^{d_1}(x)\sigma(dx)\cdots\int_{\mathbb{X}} f^{d_k}(x)\sigma(dx)\\ &= \sum_{k=1}^n \sum_{B_1^n,B_2^n,\ldots,B_k^n}\int_{\mathbb{X}} f^{|B_1^n|}(x)\sigma(dx)\cdots\int_{\mathbb{X}} f^{|B_k^n|}(x)\sigma(dx),\end{aligned}$$

where the sum runs over the partitions $B_1^n, B_2^n, \ldots, B_k^n$ of $\{1,\ldots,n\}$ with cardinality $|B_i^n|$. Similarly, we find that the moments of the centered stochastic integral

$$\int_{\mathbb{X}} f(x)\omega(dx) - \int_{\mathbb{X}} f(x)\sigma(dx) = \int_{\mathbb{X}} f(x)(\omega(dx) - \sigma(dx))$$

satisfy

$$\mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}\left[\left(\int_{\mathbb{X}} f(x)\omega(dx) - \int_{\mathbb{X}} f(x)\sigma(dx)\right)^n\right]$$
$$= \sum_{k=1}^{n} \sum_{\substack{B_1^n, B_2^n, \ldots, B_k^n \\ |B_1^n|\geq 2, \ldots, |B_k^n|\geq 2}} \int_{\mathbb{X}} f^{|B_1^n|}(x)\sigma(dx) \cdots \int_{\mathbb{X}} f^{|B_k^n|}(x)\sigma(dx),$$

where the sum runs over the partitions $B_1^n, B_2^n, \ldots, B_k^n$ of $\{1, \ldots, n\}$ of size $|B_k^n| \geq 2$.

By taking $f = \mathbb{1}_A$ we can specialize the above results to the computation of the moments and central moments of the Poisson random variable $\omega(A)$ with parameter $\lambda := \sigma(A)$. For the moments we recover the well-known identity

$$\mathbb{E}\left[(\omega(A))^n\right] = \sum_{k=1}^{n} \lambda^n S(n,k),$$

where $S(n,k)$ is the Stirling number of the second kind, i.e. the number of partitions of $\{1, 2, \ldots, n\}$ made of k subsets, $k = 0, 1, \ldots, n$. Similarly, for the central moments we have

$$\mathbb{E}[(\omega(A) - \sigma(A))^n] = \sum_{k=1}^{n} \lambda^n S_2(n,k), \tag{11.4.1}$$

where $S_2(n,k)$ is the number of partitions of $\{1, 2, \ldots, n\}$ made of k subsets of size at least 2, $k = 0, 1, \ldots, n$, cf. Privault (2011).

Slivnyak-Mecke Identity

The following version of the Slivnyak-Mecke identity (Slivnyak 1962; Mecke 1967) allows us to compute the first moment of the first order stochastic integral of a random integrand.

Proposition 11.5 *For $u : \mathbb{X} \times \Omega^{\mathbb{X}} \longrightarrow \mathbb{R}$ a measurable process, we have*

$$\mathbb{E}_\sigma\left[\sum_{x\in\omega} u(x,\omega)\right] = \mathbb{E}_\sigma\left[\int_{\mathbb{X}} u(x, \omega\cup\{x\})\sigma(\mathrm{d}x)\right], \tag{11.4.2}$$

provided that

$$\mathbb{E}_\sigma\left[\int_{\mathbb{X}} |u(x, \omega\cup\{x\})|\sigma(\mathrm{d}x)\right] < \infty.$$

Proof The proof is done when $\sigma(\mathbb{X}) < \infty$. We consider $u(x,\omega)$ written as

$$u(x,\omega) = \sum_{n\geq 0} \mathbb{1}_{\{\omega(\mathbb{X})=n\}} f_n(x; X_1, \ldots, X_n),$$

where for every $x \in \mathbb{X}$, $(x_1, \ldots, x_n) \longmapsto f_n(x; x_1, \ldots, x_n)$ is a symmetric integrable function of $\omega = \{X_1, \ldots, X_n\}$ when $\omega(\mathbb{X}) = n$, for each $n \geq 1$. We have

$$\begin{aligned}
&\mathbb{E}_\sigma\left[\sum_{x\in\omega} u(x,\omega)\right] \\
&= \mathrm{e}^{-\sigma(\mathbb{X})} \sum_{n\geq 1} \frac{1}{n!} \sum_{k=1}^{n} \int_{\mathbb{X}^n} f_n(x_k; x_1, \ldots, x_n)\sigma(\mathrm{d}x_1)\cdots\sigma(\mathrm{d}x_n) \\
&= \mathrm{e}^{-\sigma(\mathbb{X})} \sum_{n\geq 0} \frac{1}{(n+1)!} \sum_{k=1}^{n+1} \int_{\mathbb{X}^{n+1}} f_{n+1}(x_k; x_1, \ldots, x_{n+1})\sigma(\mathrm{d}x_1)\cdots\sigma(\mathrm{d}x_{n+1}) \\
&= \mathrm{e}^{-\sigma(\mathbb{X})} \sum_{n\geq 0} \frac{1}{n!} \int_{\mathbb{X}^n}\int_{\mathbb{X}} f_{n+1}(x; x, x_1, \ldots, x_n)\sigma(\mathrm{d}x)\sigma(\mathrm{d}x_1)\cdots\sigma(\mathrm{d}x_n) \\
&= \mathbb{E}_\sigma\left[\int_{\mathbb{X}} u(x,\omega\cup\{x\})\sigma(\mathrm{d}x)\right].
\end{aligned}$$

□

Proposition 11.5 can actually be extended to a moment identity, as follows.

Proposition 11.6 (Privault (2012)) *Let $u : \mathbb{X}\times\Omega^{\mathbb{X}} \longrightarrow \mathbb{R}$ be a (measurable) process. We have*

$$\begin{aligned}
&\mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}\left[\left(\int_{\mathbb{X}} u(x,\omega)\omega(dx)\right)^n\right] \\
&= \sum_{k=1}^{n} \sum_{B_1^n, B_2^n, \ldots, B_k^n} \mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}\left[\int_{\mathbb{X}^k} u^{|B_1^n|}(x_1, \omega\cup\{x_1, x_2, \ldots, x_k\})\cdots \right. \\
&\quad \left. \cdots u^{|B_k^n|}(x_k, \omega\cup\{x_1, x_2, \ldots, x_k\})\sigma(dx_1)\cdots\sigma(dx_k)\right],
\end{aligned}$$

where the sum runs over the partitions $B_1^n, B_2^n, \ldots, B_k^n$ of $\{1, \ldots, n\}$, for any $n \geq 1$ such that all terms are integrable.

Proof We will prove the following slightly more general formula

$$\begin{aligned}
&\mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}\left[F\left(\int_{\mathbb{X}} u(x,\omega)\omega(dx)\right)^n\right] \\
&= \sum_{k=1}^{n} \sum_{B_1^n, B_2^n, \ldots, B_k^n} \mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}\left[\int_{\mathbb{X}^k} F(\omega\cup\{x_1, x_2, \ldots, x_k\})u^{|B_1^n|}(x_1, \omega\cup\{x_1, x_2, \ldots, x_k\})\cdots \right. \\
&\quad \left. \cdots u^{|B_k^n|}(x_k, \omega\cup\{x_1, x_2, \ldots, x_k\})\sigma(dx_1)\cdots\sigma(dx_k)\right],
\end{aligned}$$

by induction on $n \geq 1$, for F a sufficiently integrable random variable. Clearly, the formula holds for $n = 0$ and $n = 1$. Assuming that it holds at the rank n and using Proposition 11.5 we get

$$
\begin{aligned}
&\mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}\left[F\left(\int_{\mathbb{X}} u(x,\omega)\omega(dx)\right)^{n+1}\right] \\
&=\sum_{k=1}^{n}\sum_{B_1^n,B_2^n,\ldots,B_k^n} \\
&\quad\mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}\left[\int_{\mathbb{X}^k} u^{|B_1^n|}(x_1,\omega\cup\{x_1,x_2\ldots,x_k\})\cdots u^{|B_k^n|}(x_k,\omega\cup\{x_1,x_2,\ldots,x_k\})\right. \\
&\quad\left.\times F(\omega\cup\{x_1,x_2,\ldots,x_k\})\sum_{i=1}^{k} u(x_i,\omega\cup\{x_1,x_2,\ldots,x_k\})\sigma(dx_1)\cdots\sigma(dx_k)\right] \\
&\quad+\sum_{k=1}^{n}\sum_{B_1^n,B_2^n,\ldots,B_k^n} \\
&\quad\mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}\left[\int_{\mathbb{X}^k} u^{|B_1^n|}(x_1,\omega\cup\{x_1,x_2,\ldots,x_k\})\cdots u^{|B_k^n|}(x_k,\omega\cup\{x_1,x_2,\ldots,x_k\})\right. \\
&\quad\left.\times F(\omega\cup\{x_1,x_2,\ldots,x_k\})\int_{\mathbb{X}} u(y,\omega\cup\{x_1,x_2,\ldots,x_k\})\omega(dy)\sigma(dx_1)\cdots\sigma(dx_k)\right] \\
&=\sum_{k=1}^{n}\sum_{B_1^n,B_2^n\ldots,B_k^n} \\
&\quad\mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}\left[\int_{\mathbb{X}^k} u^{|B_1^n|}(x_1,\omega\cup\{x_1,x_2,\ldots,x_k\})\cdots u^{|B_k^n|}(x_k,\omega\cup\{x_1,x_2,\ldots,x_k\})\right. \\
&\quad\left.\times F(\omega\cup\{x_1,x_2,\ldots,x_k\})\sum_{i=1}^{k} u(x_i,\omega\cup\{x_1,x_2,\ldots,x_k\})\sigma(dx_1)\cdots\sigma(dx_k)\right] \\
&\quad+\sum_{k=1}^{n}\sum_{B_1^n,B_2^n,\ldots,B_k^n} \\
&\quad\mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}\left[\int_{\mathbb{X}^{k+1}} u^{|B_1^n|}(x_1,\omega\cup\{x_1,x_2,\ldots,x_k,y\})\cdots u^{|B_k^n|}(x_k,\omega\cup\{x_1,x_2,\ldots,x_k,y\})\right. \\
&\quad\left.\times F(\omega\cup\{x_1,x_2,\ldots,x_k,y\})u(y,\omega\cup\{x_1,x_2,\ldots,x_k,y\})\sigma(dy)\sigma(dx_1)\cdots\sigma(dx_k)\right] \\
&=\sum_{k=1}^{n+1}\sum_{B_1^{n+1},B_2^{n+1},\ldots,B_k^{n+1}} \\
&\quad\mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}\left[\int_{\mathbb{X}^k} F(\omega\cup\{x_1,x_2,\ldots,x_k\})u^{|B_1^{n+1}|}(x_1,\omega\cup\{x_1,x_2,\ldots,x_k\})\times\cdots\right. \\
&\quad\left.\cdots\times u^{|B_k^{n+1}|}(x_k,\omega\cup\{x_1,x_2,\ldots,x_k\})\sigma(dx_1)\cdots\sigma(dx_k)\right],
\end{aligned}
$$

where the sum runs over the partitions $B_1^{n+1}, B_2^{n+1}, \ldots, B_k^{n+1}$ of $\{1,\ldots,n+1\}$. □

11.5 Deviation Inequalities

The next proposition is a particular case of results proved for general Poisson functionals in Wu (2000), Houdré and Privault (2002), cf. also Corollary 6.9.3 of Privault (2009). Such results are useful for the accurate estimation of loss (or outage) probabilities in a wireless network whose users are randomly located within the space $\mathbb{X}$, cf. e.g. Decreusefond et al. (2009).

Proposition 11.7 *Let $f : \mathbb{X} \longrightarrow \mathbb{R}$ be a (measurable) function such that*

(i) $f(y) \leq K$, $\sigma(dy)$*-a.e., for some* $K > 0$, *and*

(ii) $\displaystyle\int_{\mathbb{X}} |f(y)|^2 \sigma(dy) \leq \alpha^2$, *for some* $\alpha > 0$.

Then we have

$$\mathbb{P}_\sigma^{\mathbb{X}} \left(\int_{\mathbb{X}} f(y)\omega(dy) - \int_{\mathbb{X}} f(y)\sigma(dy) \geq x \right) \leq \mathrm{e}^{x/K} \left(1 + \frac{xK}{\alpha^2} \right)^{-\frac{x}{K} - \frac{\alpha^2}{K^2}}, \quad (11.5.1)$$

for all $x > 0$.

Proof We let

$$\int_{\mathbb{X}} f(y)\tilde{\omega}(dy) := \int_{\mathbb{X}} f(y)(\omega(dy) - \sigma(dy))$$

denote the compensated Poisson stochastic integral of f, and let

$$M(s) := \mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}} \left[\exp\left(s \int_{\mathbb{X}} f(y)\tilde{\omega}(dy) \right) \right], \qquad s \geq 0,$$

denote the moment generating function of $\int_{\mathbb{X}} f(y)\tilde{\omega}(dy)$. By Proposition 11.5 we have

$$\begin{aligned}
M'(s) &= \mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}} \left[\int_{\mathbb{X}} f(x)\tilde{\omega}(dx) \exp\left(s \int_{\mathbb{X}} f(y)\tilde{\omega}(dy) \right) \right] \\
&= \mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}} \left[\int_{\mathbb{X}} \exp\left(s \int_{\mathbb{X}} f(y)\tilde{\omega}(dy) \right) f(x)\tilde{\omega}(dx) \right. \\
&\quad \left. - \int_{\mathbb{X}} \exp\left(s \int_{\mathbb{X}} f(y)\tilde{\omega}(dy) \right) f(x)\sigma(dx) \right] \\
&= \mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}} \left[\int_{\mathbb{X}} \exp\left(sf(x) + s \int_{\mathbb{X}} f(y)\tilde{\omega}(dy) \right) f(x)\sigma(dx) \right. \\
&\quad \left. - \int_{\mathbb{X}} \exp\left(s \int_{\mathbb{X}} f(y)\tilde{\omega}(dy) \right) f(x)\sigma(dx) \right] \\
&= \mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}} \left[\int_{\mathbb{X}} f(x)(\mathrm{e}^{sf(x)} - 1)\sigma(dx) \exp\left(s \int_{\mathbb{X}} f(y)\tilde{\omega}(dy) \right) \right]
\end{aligned}$$

$$
\begin{aligned}
&= \int_{\mathbb{X}} f(x)(\mathrm{e}^{sf(x)}-1)\sigma(dx)\,\mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}\left[\exp\left(s\int_{\mathbb{X}} f(y)\tilde{\omega}(dy)\right)\right] \\
&= s\int_{\mathbb{X}} |f(x)|^2 \frac{\mathrm{e}^{sf(x)}-1}{sf(x)}\sigma(dx)\,\mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}\left[\exp\left(s\int_{\mathbb{X}} f(y)\tilde{\omega}(dy)\right)\right] \\
&\le \frac{\mathrm{e}^{sK}-1}{K}\int_{\mathbb{X}} |f(x)|^2\sigma(dx)\,\mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}\left[\exp\left(s\int_{\mathbb{X}} f(y)\tilde{\omega}(dy)\right)\right] \\
&= \alpha^2\frac{\mathrm{e}^{sK}-1}{K}\,\mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}\left[\exp\left(s\int_{\mathbb{X}} f(y)\tilde{\omega}(dy)\right)\right] = \alpha^2\frac{\mathrm{e}^{sK}-1}{K}M(s),
\end{aligned}
$$

which shows that

$$
\frac{M'(s)}{M(s)} \le h(s) := \alpha^2\frac{\mathrm{e}^{sK}-1}{K}, \qquad s \ge 0,
$$

hence

$$
M(t) \le \exp\left(\int_0^t h(s)ds\right) = \exp\left(\alpha^2\int_0^t \frac{\mathrm{e}^{sK}-1}{K}ds\right), \qquad t \ge 0.
$$

Consequently we have

$$
\mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}[F\mathrm{e}^{tF}] \le h(t)\,\mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}[\mathrm{e}^{tF}], \qquad t \ge 0.
$$

Next, the Markov inequality shows that

$$
\begin{aligned}
\mathbb{P}_\sigma^{\mathbb{X}}\left(\int_{\mathbb{X}} f(y)(\omega(dy)-\sigma(dy)) \ge x\right) &= \mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}\left[\mathbb{1}_{\{\int_{\mathbb{X}} f(y)(\omega(dy)-\sigma(dy))\ge x\}}\right] \\
&\le \mathrm{e}^{-tx}\,\mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}\left[\mathbb{1}_{\{\int_{\mathbb{X}} f(y)(\omega(dy)-\sigma(dy))\ge x\}}\exp\left(t\int_{\mathbb{X}} f(y)\omega(dy)\right)\right] \\
&\le \mathrm{e}^{-tx}\,\mathbb{E}_{\mathbb{P}_\sigma^{\mathbb{X}}}\left[\exp\left(t\int_{\mathbb{X}} f(y)\omega(dy)\right)\right] \\
&\le \exp\left(-tx+\int_0^t h(s)ds\right) \\
&\le \exp\left(-tx+\alpha^2\int_0^t \frac{\mathrm{e}^{sK}-1}{K}ds\right) \\
&= \exp\left(-tx+\frac{\alpha^2}{K^2}(\mathrm{e}^{tK}-tK-1)\right).
\end{aligned}
$$

Minimizing the above term in t with $t = K^{-1}\log(1 + Kx/\alpha^2)$ shows that

$$\mathbb{P}_\sigma^{\mathbb{X}}\left(\int_{\mathbb{X}} f(y)(\omega(dy) - \sigma(dy)) \geq x\right) \leq \exp\left(\frac{x}{K} - \left(\frac{x}{K} + \frac{\alpha^2}{K^2}\right)\log\left(1 + \frac{xK}{\alpha^2}\right)\right),$$

which yields (11.5.1). □

The bound (11.5.1) also shows that

$$\mathbb{P}_\sigma^{\mathbb{X}}\left(\int_{\mathbb{X}} f(y)\omega(dy) - \int_{\mathbb{X}} f(y)\sigma(dy) \geq x\right)$$
$$\leq \exp\left(-\frac{x}{2K}\log\left(1 + \frac{xK}{\alpha^2}\right)\right) = \left(1 + \frac{xK}{\alpha^2}\right)^{-x/2K},$$

for all $x > 0$.
In case the function $f : \mathbb{X} \longrightarrow \mathbb{R}$ is such that

(i) $f(y) \leq 0$, $\sigma(dy)$-a.e., for some $K \in \mathbb{R}$, and
(ii) $\displaystyle\int_{\mathbb{X}} |f(y)|^2\sigma(dy) \leq \alpha^2$,

the above argument can be applied with $K = 0$, and also shows that

$$\mathbb{P}_\sigma^{\mathbb{X}}\left(\int_{\mathbb{X}} f(y)\omega(dy) - \int_{\mathbb{X}} f(y)\sigma(dy) \geq x\right) \leq \exp\left(-\frac{x^2}{2\alpha^2}\right),$$

for all $x > 0$.

Exercises

Exercise 11.1 Consider the standard Poisson process $(N_t)_{t\in\mathbb{R}_+}$ on $\mathbb{R}_+$ with intensity $\lambda = 2$ and jump times $(T_k)_{k\geq 1}$. Compute

$$\mathbb{E}[T_1 + T_2 + T_3 \mid N_2 = 2].$$

Exercise 11.2 Consider the spatial Poisson point process on $\mathbb{R}^2$ with intensity $\lambda = 0.5$ per square meter. What is the probability that there are 10 events within a circle of radius 3 meters.

Exercise 11.3 Some living organisms are distributed in space according to a Poisson point process of intensity $\theta = 0.6$ units per mm^3. Compute the probability that more than two living organisms are found within a 10 mm^3 volume.

Exercise 11.4 Defects are present over a piece of fabric according to a Poisson point process with intensity of one defect per piece of fabric. Both halves of the piece is checked separately. What is the probability that both inspections record at least one defect?

Exercise 11.5 Let $\lambda > 0$ and suppose that N points are independently and uniformly distributed over the interval $[0, N]$. Determine the probability distribution for the number of points in the interval $[0, \lambda]$ as $N \to \infty$.

Exercise 11.6 Suppose that $X(A)$ is a spatial Poisson point process of discrete items scattered on the plane $\mathbb{R}^2$ with intensity $\lambda = 0.5$ per square meter. We let

$$D((x, y), r) = \{(u, v) \in \mathbb{R}^2 \ : \ (x-u)^2 + (y-v)^2 \leq r^2\}$$

denote the disc with radius r centered at (x, y) in $\mathbb{R}^2$. No evaluation of numerical expressions is required in this exercise.

(a) What is the probability that 10 items are found within the disk $D((0, 0), 3)$ with radius 3 meters centered at the origin?
(b) What is the probability that 5 items are found within the disk $D((0, 0), 3)$ *and* 3 items are found within the disk $D((x, y), 3)$ with $(x, y) = (7, 0)$?
(c) What is the probability that 8 items are found anywhere within $D((0, 0), 3) \bigcup D((x, y), 3)$ with $(x, y) = (7, 0)$?
(d) Given that 5 items are found within the disk $D((0, 0), 1)$, what is the probability that 3 of them are located within the disk $D((1/2, 0), 1/2)$ centered at $(1/2, 0)$ with radius $1/2$?

Exercise 11.7 Let S_n be a Poisson random variable with parameter λn for all $n \geq 1$, with $\lambda > 0$. Show that the moments of order p of $(S_n - \lambda n)/\sqrt{n}$ satisfy the bound

$$\sup_{n \geq 1} \mathbb{E}\left[\left|\frac{S_n - \lambda n}{\sqrt{n}}\right|^p\right] < C_p$$

where $C_p > 0$ is a finite constant for all $p \geq 1$. *Hint*: Use Relation (11.4.1).

Exercise 11.8 Let $(N_t)_{t \in \mathbb{R}_+}$ denote a standard Poisson process on $\mathbb{R}_+$. Given a bounded function $f \in L^1(\mathbb{R}_+)$ we let

$$\int_0^\infty f(y)(dN_y - dy)$$

denote the compensated Poisson stochastic integral of f, and let

$$M(s) := \mathbb{E}\left[\exp\left(s \int_0^\infty f(y)(dN_y - dy)\right)\right] = \exp\left(\int_0^\infty (\mathrm{e}^{sf(y)} - sf(y) - 1)dy\right),$$

$s \geq 0$, denote the moment generating function of $\int_0^\infty f(y)(dN_y - dy)$.

(a) Show that we have

$$\frac{M'(s)}{M(s)} \leq h(s) := \alpha^2 \frac{e^{sK} - 1}{K}, \qquad s \geq 0,$$

provided that $f(t) \leq K$, dt-a.e., for some $K > 0$ and provided in addition that $\int_0^\infty |f(y)|^2 dy \leq \alpha^2$, for some $\alpha > 0$.

(b) Show that

$$M(t) \leq \exp\left(\int_0^t h(s)ds\right) = \exp\left(\alpha^2 \int_0^t \frac{e^{sK} - 1}{K} ds\right), \qquad t \geq 0.$$

(c) Show, using Markov's inequality, that

$$\mathbb{P}\left(\int_0^\infty f(y)(dN_y - dy) \geq x\right) \leq \mathrm{e}^{-tx}\, \mathbb{E}\left[\exp\left(t \int_0^\infty f(y) dN_y\right)\right],$$

and that

$$\mathbb{P}\left(\int_0^\infty f(y)(dN_y - dy) \geq x\right) \leq \exp\left(-tx + \alpha^2 \int_0^t \frac{e^{sK} - 1}{K} ds\right).$$

(d) By minimization in t, show that

$$\mathbb{P}\left(\int_0^\infty f(y) dN_y - \int_0^\infty f(y) dy \geq x\right) \leq \mathrm{e}^{x/K} \left(1 + \frac{xK}{\alpha^2}\right)^{-x/K - \alpha^2/K^2},$$

for all $x > 0$, and that

$$\mathbb{P}\left(\int_0^\infty f(y) dN_y - \int_0^\infty f(y) dy \geq x\right) \leq \left(1 + \frac{xK}{\alpha^2}\right)^{-x/2K},$$

for all $x > 0$.

Exercise 11.9 Let Φ be a Poisson point process with finite intensity measure $\sigma(dx)$ on $\mathbb{X} := \mathbb{R}^d \times \mathbb{R}_+$, given by $\sigma(dx) = \lambda dy \rho(r) dr$, $x = (y, r) \in \mathbb{R}^d \times \mathbb{R}_+$, where $\lambda > 0$ and $\rho(r)$ is a probability density function on $\mathbb{R}_+$. We also consider the Probability Generating Functional (PGFl) $\mathcal{G}_\Phi(f) := \mathbb{E}\left[\prod_{x \in \Phi} f(x)\right]$.

(a) Show that $\mathcal{G}_\Phi(f) = \exp\left(\int_{\mathbb{X}} (f(x) - 1)\sigma(dx)\right)$, where $f - 1 \in L^1(\mathbb{X}, \sigma)$.

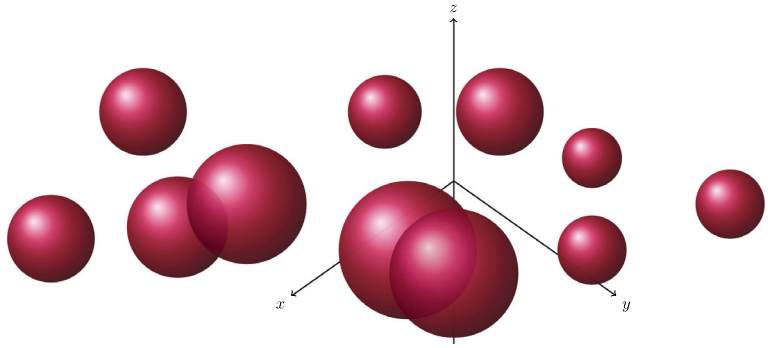


Fig. 11.5 Boolean model in dimension three

(b) Using the PGFl $\mathcal{G}_\Phi$, recover the probability $\mathbb{P}(\Phi \cap A = \emptyset)$ that *no* process points can be found within a given subset A of $\mathbb{X}$.
(c) Consider the Boolean model Ξ on $\mathbb{R}^d$ made of the union of balls constructed by associating every point (y, r) of Φ to a ball of radius r centered at $y \in \mathbb{R}^d$. Using the volume $v_d = \pi^{d/2}/\Gamma(1 + d/2)$ of the unit ball in $\mathbb{R}^d$, find the probability that the union of balls contains the point 0.
(d) Find the probability that this Boolean model covers the whole space $\mathbb{R}^d$ (Fig. 11.5).

Exercise 11.10 Consider a Poisson point process ω with intensity $\sigma(dy)\mathrm{e}^{-r}dr$ on $[0, 1]^d \times [0, \infty)$, given by $\omega := \{(Y_k, R_k)\}_k$. Each point $(x, r) \in \omega$ models a location $x \in [0, 1]^d$ along with a radius $r \in [0, \infty)$ corresponding to the radius of the ball centered around it. The spherical Boolean model Ξ with ω as its driving Poisson point process can now be constructed as

$$\Xi = \bigcup_{(x,r)\in\omega} B(x, r),$$

which consists in the random subset of points in $[0, 1]^d$ which are covered by at least one Euclidean ball centered around the points of the Poisson point process ω.

(a) Give the probability of not observing any ball of radius smaller than $1/2$.
(b) Give the mean number of balls which have radius less than $1/2$.

Chapter 12
Reliability Theory



This chapter consists in a short review of survival probabilities based on failure rate and reliability functions, in connection with Poisson processes having a time-dependent intensity.

12.1 Survival Probabilities

Let $\tau : \Omega \longrightarrow \mathbb{R}_+$ denote (random) the lifetime of an entity, and let $\mathbb{P}(\tau \geq t)$ denote its probability of surviving at least t years, $t > 0$. The probability of surviving up to a (deterministic) time T, given that the entity has already survived up to time t, is given by

$$\begin{aligned}\mathbb{P}(\tau > T \mid \tau > t) &= \frac{\mathbb{P}(\tau > T \text{ and } \tau > t)}{\mathbb{P}(\tau > t)} \\ &= \frac{\mathbb{P}(\tau > T)}{\mathbb{P}(\tau > t)}, \qquad 0 \leq t \leq T.\end{aligned}$$

Let now

$$\lambda(t) := \lim_{h \searrow 0} \frac{\mathbb{P}(\tau < t + h \mid \tau > t)}{h}, \qquad t \geq 0,$$

denote the *failure rate function* associated to τ. Letting $A = \{\tau < t + h\}$ and $B = \{\tau > t\}$ we note that $(\Omega \setminus A) \subset B$, hence $A \cap B = B \setminus A^c$, and

$$\begin{aligned}\lambda(t) &= \lim_{h \searrow 0} \frac{\mathbb{P}(\tau < t + h \mid \tau > t)}{h} \\ &= \frac{1}{\mathbb{P}(\tau > t)} \lim_{h \searrow 0} \frac{\mathbb{P}(\tau < t + h \text{ and } \tau > t)}{h}\end{aligned}$$

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N. Privault, *Understanding Markov Chains*, Springer Undergraduate Mathematics Series,
https://doi.org/10.1007/978-981-95-4057-0_12

$$\begin{aligned} &= \frac{1}{\mathbb{P}(\tau > t)} \lim_{h \searrow 0} \frac{\mathbb{P}(\tau > t) - \mathbb{P}(\tau > t + h)}{h} \\ &= -\frac{d}{dt} \log \mathbb{P}(\tau > t) \\ &= -\frac{1}{\mathbb{P}(\tau > t)} \frac{d}{dt} \mathbb{P}(\tau > t) \\ &= -\frac{1}{R(t)} \frac{d}{dt} R(t), \end{aligned} \tag{12.1.1}$$

where the *reliability function* $R(t)$ is defined by

$$R(t) := \mathbb{P}(\tau > t), \qquad t \geq 0.$$

This yields

$$R'(t) = -\lambda(t) R(t),$$

with $R(0) = \mathbb{P}(\tau > 0) = 1$, which has for solution

$$R(t) = \mathbb{P}(\tau > t) = R(0) \exp\left(- \int_0^t \lambda(u) du \right) = \exp\left(- \int_0^t \lambda(u) du \right), \tag{12.1.2}$$

$t \geq 0$. Hence we have

$$\mathbb{P}(\tau > T \mid \tau > t) = \frac{R(T)}{R(t)} = \exp\left(- \int_t^T \lambda(u) du \right), \qquad t \in [0, T]. \tag{12.1.3}$$

In case the failure rate function $\lambda(t) = c$ is constant, this recovers the memoryless property of the exponential distribution with parameter $c > 0$, cf. (9.2.3).

Relation (12.1.2) can be recovered informally as

$$\mathbb{P}(\tau > T) = \prod_{0 < t < T} \mathbb{P}(\tau > t + dt \mid \tau > t) = \prod_{0 < t < T} \exp\left(-\lambda(t) dt \right),$$

which yields

$$\mathbb{P}(\tau > t) = \exp\left(- \int_0^t \lambda(s) ds \right), \qquad t \geq 0,$$

in the limit.

12.2 Poisson Process with Time-Dependent Intensity

Recall that the random variable τ has the exponential distribution with parameter $\lambda > 0$ if

$$\mathbb{P}(\tau > t) = \mathrm{e}^{-\lambda t}, \qquad t \geq 0,$$

cf. (1.5.3). Given $(\tau_n)_{n\geq 0}$ a sequence of *i.i.d.* exponentially distributed random variables, letting

$$T_n = \tau_0 + \cdots + \tau_{n-1}, \qquad n \geq 1,$$

and

$$N_t = \sum_{n\geq 1} \mathbb{1}_{[T_n,\infty)}(t), \qquad t \geq 0,$$

defines the standard Poisson process with intensity $\lambda > 0$ of Sect. 9.1 and we have

$$\mathbb{P}(N_t - N_s = k) = \mathrm{e}^{-\lambda(t-s)} \frac{(\lambda(t-s))^k}{k!}, \qquad k \geq 0.$$

The intensity of the Poisson process can in fact made time-dependent. For example under the time change

$$X_t = N_{\int_0^t \lambda(s)ds}$$

where $(\lambda(u))_{u\in\mathbb{R}_+}$ is a deterministic function of time, we have

$$\mathbb{P}(X_t - X_s = k) = \frac{\left(\int_s^t \lambda(u)du\right)^k}{k!} \exp\left(-\int_s^t \lambda(u)du\right), \qquad k \geq 0.$$

In this case, we have

$$\mathbb{P}(X_{t+h} - X_t = 0) = \mathrm{e}^{-\lambda(t)h} + o(h) = 1 - \lambda(t)h + o(h), \qquad h \searrow 0, \quad (12.2.1)$$

and

$$\mathbb{P}(X_{t+h} - X_t = 1) = 1 - \mathrm{e}^{-\lambda(t)h} + o(h) \simeq \lambda(t)h, \qquad h \searrow 0, \quad (12.2.2)$$

which can also viewed as a pure birth process with time-dependent intensity. Letting τ_0 denote the first jump time of $(X_t)_{t\in\mathbb{R}_+}$, we have

$$R(t) = \mathbb{P}(\tau_0 > t) = \mathbb{P}(X_t = 0) = \exp\left(-\int_0^t \lambda(u)du\right), \qquad t \geq 0, \quad (12.2.3)$$

hence by (12.1.3) we find

$$\begin{aligned}\mathbb{P}(X_{t+h} = 0 \mid X_t = 0) &= \mathbb{P}(\tau_0 > t+h \mid \tau_0 > t) \\ &= \mathrm{e}^{-\lambda(t)h} + o(h) = 1 - \lambda(t)h + o(h), \qquad h \searrow 0,\end{aligned}$$

and

$$\begin{aligned}\mathbb{P}(X_{t+h} \geq 1 \mid X_t = 0) &= \mathbb{P}(\tau_0 < t+h \mid \tau_0 > t) = 1 - \mathbb{P}(\tau_0 > t+h \mid \tau_0 > t) \\ &= 1 - \mathrm{e}^{-\lambda h} \simeq \lambda(t)h + o(h), \qquad h \searrow 0,\end{aligned}$$

which coincide respectively with $\mathbb{P}(X_{t+h} - X_t = 0)$ and $\mathbb{P}(X_{t+h} - X_t = 1)$ in (12.2.1) and (12.2.2) above, as $(X_t)_{t\in\mathbb{R}_+}$ has independent increments.

Cox Processes

The intensity process $\lambda(s)$ can also be made random. In this case, $(X_t)_{t\in\mathbb{R}_+}$ is called a *Cox process* and it may not have independent increments. For example, assume that $(\lambda_u)_{u\in\mathbb{R}_+}$ is a two-state Markov chain on $\{0, \lambda\}$, with transitions

$$\mathbb{P}(\lambda_{t+h} = \lambda \mid \lambda_t = 0) = \alpha h, \qquad h \searrow 0,$$

and

$$\mathbb{P}(\lambda_{t+h} = 0 \mid \lambda_t = \lambda) = \beta h, \qquad h \searrow 0.$$

In this case the probability distribution of N_t can be explicitly computed, cf. Chaps. 6–7 in Karlin and Taylor (1998).

Renewal Processes

A *renewal process* is a counting process $(N_t)_{t\in\mathbb{R}_+}$ given by

$$N_t = \sum_{k\geq 1} k \mathbb{1}_{[T_k, T_{k+1})}(t) = \sum_{k\geq 1} \mathbb{1}_{[T_k,\infty)}(t), \qquad t \geq 0,$$

in which $\tau_k = T_{k+1} - T_k$, $k \in \mathbb{N}$, is a sequence of independent identically distributed random variables. In particular, Poisson processes are renewal processes.

12.3 Mean Time to Failure

The mean time to failure is given, from (12.1.1), by

$$\begin{aligned}\mathbb{E}[\tau] &= \int_0^\infty t \frac{d}{dt}\mathbb{P}(\tau < t)dt = -\int_0^\infty t \frac{d}{dt}\mathbb{P}(\tau > t)dt \\ &= -\int_0^\infty t R'(t)dt = \int_0^\infty R(t)dt, \end{aligned} \tag{12.3.1}$$

provided that $\lim_{t \nearrow \infty}(t R(t)) = 0$. For example, when τ has the distribution function (12.2.3), we get

$$\mathbb{E}[\tau] = \int_0^\infty R(t)dt = \int_0^\infty \exp\left(-\int_0^t \lambda(u)du\right) dt.$$

In case the function $\lambda(t) = \lambda > 0$ is constant we recover the mean value

$$\mathbb{E}[\tau] = \int_0^\infty \mathrm{e}^{-\lambda t} dt = \frac{1}{\lambda}$$

of the exponential distribution with parameter $\lambda > 0$.

Exercises

Exercise 12.1 Assume that the random time τ has the *Weibull* distribution with probability density

$$f_\beta(x) = \beta \mathbb{1}_{[0,\infty)} x^{\beta-1} \mathrm{e}^{-x^\beta}, \qquad x \in \mathbb{R},$$

where $\beta > 0$ is called the shape parameter.

(a) Compute the distribution function F_β of the Weibull distribution.
(b) Compute the reliability function $R(t) = \mathbb{P}(\tau > t)$.
(c) Compute the failure rate function $\lambda(t)$.
(d) Compute the mean time to failure.

Exercise 12.2 Assume that the random time τ has the *Fréchet* distribution with probability density function

$$g_\beta(x) = \mathbb{1}_{[0,\infty)} \frac{\beta}{x^{\beta+1}} \mathrm{e}^{-x^{-\beta}}, \qquad x \in \mathbb{R},$$

where $\beta > 0$ is called the shape parameter.

(a) Compute the distribution function F_β of the Fréchet distribution.
(b) Compute the reliability function $R(t) = \mathbb{P}(\tau > t)$.
(c) Compute the failure rate function $\lambda(t)$.
(d) Compute the mean time to failure when $\beta > 1$.

Exercise 12.3 Consider the default time τ defined as

$$\tau := \inf\left\{t \in \mathbb{R}_+ \, : \, \int_0^t X_u du \geq L\right\},$$

where

- $(X_t)_{t\in\mathbb{R}_+}$ is an integrable stochastic process generating the filtration $(\mathcal{F}_t)_{t\in\mathbb{R}_+}$,
- L is an exponentially distributed random variable with parameter $\mu > 0$ and distribution function $\mathbb{P}(L > x) = \mathrm{e}^{-\mu x}$, $x \geq 0$, independent of $(\mathcal{F}_t)_{t\in\mathbb{R}_+}$.

Show that for the failure rate process $(\lambda_t)_{t\in\mathbb{R}_+}$ associated to τ conditionally to $(\mathcal{F}_t)_{t\in\mathbb{R}_+}$ is given by $\lambda_t = \mu X_t$, $t \in \mathbb{R}_+$.

Some Useful Identities

Here, we present a summary of algebraic identities that are used in this text.
Indicator functions

$$\mathbb{1}_A(x) = \begin{cases} 1 & \text{if } x \in A, \\ 0 & \text{if } x \notin A. \end{cases} \quad \mathbb{1}_{[a,b]}(x) = \begin{cases} 1 & \text{if } a \le x \le b, \\ 0 & \text{otherwise.} \end{cases}$$

Binomial coefficients

$$\binom{n}{k} := \frac{n!}{(n-k)!k!}, \qquad k = 0, 1, \ldots, n.$$

Exponential series

$$\mathrm{e}^x = \sum_{n \ge 0} \frac{x^n}{n!}, \qquad x \in \mathbb{R}. \tag{A.1}$$

Geometric sum

$$\sum_{k=0}^{n} r^k = \frac{1 - r^{n+1}}{1 - r}, \qquad r \ne 1. \tag{A.2}$$

Geometric series

$$\sum_{k \ge 0} r^k = \frac{1}{1 - r}, \qquad -1 < r < 1. \tag{A.3}$$

Differentiation of geometric series

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$$\sum_{k\geq 1} kr^{k-1} = \frac{\partial}{\partial r}\sum_{k\geq 0} r^k = \frac{\partial}{\partial r}\frac{1}{1-r} = \frac{1}{(1-r)^2}, \qquad -1 < r < 1. \tag{A.4}$$

Binomial identities

$$\sum_{k=0}^{n}\binom{n}{k}a^k b^{n-k} = (a+b)^n.$$

$$\sum_{k=0}^{n}\binom{n}{k} = 2^n. \tag{A.5}$$

$$\sum_{k=1}^{n} k\binom{n}{k} = n2^{n-1}.$$

$$\begin{aligned}\sum_{k=0}^{n} k\binom{n}{k}a^k b^{n-k} &= \sum_{k=1}^{n}\frac{n!}{(n-k)!(k-1)!}a^k b^{n-k}\\ &= \sum_{k=0}^{n-1}\frac{n!}{(n-1-k)!k!}a^{k+1}b^{n-1-k}\\ &= n\sum_{k=0}^{n-1}\binom{n-1}{k}a^{k+1}b^{n-1-k}\\ &= na(a+b)^{n-1}, \qquad n\geq 1,\end{aligned}$$

$$\begin{aligned}\sum_{k=0}^{n} k\binom{n}{k}a^k b^{n-k} &= a\frac{\partial}{\partial a}\sum_{k=0}^{n}\binom{n}{k}a^k b^{n-k}\\ &= a\frac{\partial}{\partial a}(a+b)^n\\ &= na(a+b)^{n-1}, \qquad n\geq 1.\end{aligned}$$

Sums of integers

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}. \tag{A.6}$$

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}. \tag{A.7}$$

Taylor expansion

$$(1+x)^{\alpha} \sum_{k\geq 0} \frac{x^k}{k!} \alpha(\alpha-1) \times \cdots \times (\alpha-(k-1)). \tag{A.8}$$

Differential equation

$$\text{The solution of } f'(t) = cf(t) \text{ is given by } f(t) = f(0)\mathrm{e}^{ct}, \quad t \geq 0. \tag{A.9}$$

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Subject Index

Symbols
0 − 1 law, 159
Q-learning, 171
σ-algebra, 2
σ-field, 2
code, 22, 175, 179, 225, 256, 317, 328

A
Absorbing
  set, 136
  state, 107, 136, 188, 213
Absorption
  probability, 290
  time, 143, 291
Accepting state, 169
Accessible state, 177
Action-value function, 172
Adjacency matrix, 104
Algorithm
  Baum-Welch, 132
  Expectation-Maximization (EM), 132
Aperiodic
  chain, 188
  state, 188, 196, 201, 203, 213
Aperiodicity, 188, 203

B
Backward Kolmogorov equation, 267
Balance
  detailed, 210
  global, 199
Balance condition, 199, 279
Baum-Welch algorithm, 132
Bellman equation, 167, 173
Bernoulli
  distribution, 41
  random walk, 75
Bernoulli distribution, 16
Bernoulli-Laplace chain, 126, 215
Binary classification, 132
Binomial
  coefficient, 351
  distribution, 16, 78
  identity, 351
Birth and death process, 261, 284
Birth process, 258
Boltzmann distribution, 228
Boolean model, 343, 344
Borel-Cantelli Lemma, 6
Borel distribution, 248
Boundary condition, 47, 59
Branching process, 229
  progeny, 229
  subcritical, 235
  supercritical, 234
Buy limit, 306

C
Cauchy
  distribution, 13
Centered random variable, 304
Central moments, 336
Cesàro mean, 94
Chain rule, 102
Characteristic

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N. Privault, *Understanding Markov Chains*, Springer Undergraduate Mathematics Series, https://doi.org/10.1007/978-981-95-4057-0

equation, 50, 72
function, 28, 330
Class
communicating, 178
property, 181, 188
recurrent, 180, 181
transient, 183
Class property, 183
Classification of states, 177
Code
Matlab, 209
Communicating
class, 178
state, 177, 178
Compensated Poisson stochastic integral, 331
Complement rule, 5
Conditional
expectation, 20, 35, 303
probability, 6, 7
Conditioning, 6
Continuous-time Markov chain, 251, 257
Convolution equation, 80, 95, 156
Cookie, 97
Counting process, 251
Coupling, 221
Cox process, 348
Critical (branching process), 235
Cumulants, 34, 332
Cumulative distribution function, 12
joint, 14

D
Death process, 260
Density
function, 10
marginal, 15
Detailed balance, 210, 222, 227
Deviation inequalities, 339
Diagonalization, 115, 275
Dirichlet problem, 138
Discrete distribution, 16
Discrete-time
Markov chain, 101
martingale, 303
Distribution
Bernoulli, 16
binomial, 16, 78
Boltzmann, 228
Borel, 248
Cauchy, 13
discrete, 16
Erlang, 255
exponential, 12
Fréchet, 349
gamma, 13
Gaussian, 12
geometric, 17
zero-modified, 157
invariant, 106, 199, 201, 203
limiting, 117, 195, 277
lognormal, 13
marginal, 24
negative binomial, 17, 128, 242, 247
Pascal, 17
phase-type, 128
Poisson, 17
power tail, 231
stationary, 198, 210, 222, 277, 294
uniform, 12
Weibull, 349
Dominated convergence, 315
Dominated convergence theorem, 309
Doob-Meyer decomposition, 321
Dynamic programming, 166, 171

E
Ehrenfest chain, 108, 214, 222
Elephant random walk, 124
EM algorithm, 132
Embedded chain, 285
Emission probabilities, 122, 130
Equation
characteristic, 50, 72
convolution, 80, 95, 156
homogeneous, 62, 67
Kolmogorov (backward), 267
Kolmogorov (forward), 267, 274, 280, 299
partial differential, 300
Equivalence relation, 178
Ergodic
state, 188, 213
theorem
continuous time, 277, 282
discrete time, 120, 206
Erlang
distribution, 255
ERW, 124
Event, 2
Excited random walk, 97
Expectation, 18, 244
conditional, 20, 35
Expectation-Maximization algorithm, 132

Exponential distribution, 12, 250, 256, 269, 270, 346
Exponential series, 351
Extinction probability, 236, 244

F
Factorial moment, 33
Failure rate function, 345
Failure time, 348
Filtration, 301
First step analysis, 46, 60, 67, 111, 135, 149, 167, 216, 237, 249, 290
Forward Kolmogorov equation, 267, 274, 280, 299
Fréchet distribution, 349
Front running, 296
Frullani's identity, 332

G
Gambling process, 43, 107, 113
Gamma
  distribution, 13
  function, 13
Gaussian
  distribution, 12
  random variable, 29
Generating function, 28, 30, 81, 239, 300
Geometric
  distribution, 17
    zero-modified, 157
  series, 45, 152, 351
  sum, 351
Global balance, 199
GridWorld, 171

H
Hidden Markov, 122, 130
High-frequency trading, 296
Hitting
  probability, 135, 244
  time, 143
Hidden Markov Model (HMM), 132
Homogeneous equation, 62, 67

I
Inclusion-exclusion principle, 5
Increment
  independent, 103, 252, 257, 258, 304, 321
  stationary, 252
Independence, 6, 8, 10, 15, 17, 22, 29, 33, 38
Independent increments, 103, 252, 257, 258, 304, 321
Indicator function, 9, 351
Infimum, 17, 58
Infinitesimal generator, 267, 294
Information flow, 302
Intensity parameter, 253
Invariant distribution, 106, 199, 201, 203
Inverse
  temperature, 228
Irreducible, 178, 196, 201, 205, 213, 263
Ising model, 225

J
Joint
  cumulative distribution function, 14
  probability density function, 14

K
Kernel
  potential, 159
Kolmogorov equation
  backward, 267
  forward, 267, 274, 280, 299

L
Law
  of total expectation, 24, 143
  of total probability, 4, 8, 24, 102, 104, 111, 136, 184, 265
Layering, 296
Lazy random walk, 66
Limiting distribution, 117, 195, 277
  Bernoulli-Laplace chain, 215
  Ehrenfest chain, 214
  success runs chain, 213
Lognormal
  distribution, 13

M
Marginal
  density, 15
  distribution, 24
Markov
  decision process, 166, 171
  hidden, 122, 130
  property, 101, 143, 257
    strong, 163, 251, 256, 260

Markov chain, 101
  Bernoulli-Laplace, 126, 215
  continuous time, 251, 257
  discrete time, 101
  embedded, 285
  irreducible, 178, 196, 201, 205, 213
  Monte Carlo, 210
  recurrent, 180, 196
  reducible, 178
  regular, 179, 197, 205, 213
  reversible, 210, 222
  success runs, 126, 193, 213
  two-state, 114, 140, 145, 151, 195, 201, 239, 267, 274, 277, 285, 298, 348
Markov Chain Monte Carlo (MCMC), 210
Markov Decision Process (MDP), 166, 171
  toolbox, 171
Martingale, 317
  discrete time, 303
  submartingale, 309, 321
Matlab, 209
Matrilineality, 248
Maze, 152, 164
Mean
  game duration, 58, 314
  hitting time, 244, 294
  number of returns, 83, 155
  recurrence time, 186
  return time, 149
  time to failure, 348
Mean-square distance, 37
Memoryless property, 259, 346
Metropolis algorithm, 211
Metropolis-Hastings algorithm, 211
Model
  Boolean, 343, 344
  hidden Markov, 122, 130
Moment, 34, 332
  generating function, 28, 330
Monotone convergence, 315

N
Natural logarithm, 98, 260
Negative
  binomial distribution, 17, 128, 242, 247
Null recurrent, 187, 213
Number of returns, 83, 155

O
Octave, 209
Order book, 295

P
PageRank™, 224
Partial differential equation, 300
Partition, 8, 35, 335
Pascal
  distribution, 17
Patrilineality, 248
Periodicity, 188
Phase-type distribution, 128
Poisson
  central moments, 336
  cumulants, 332
  distribution, 17
  moments, 332
  point process, 327
  process, 251, 264, 296, 327, 346
    transformation, 333
  stochastic integral, 330
Poisson moments, 336
Policy, 167, 171
Polylogarithmic, 231
Population genetics, 167
Positive recurrence, 187, 201, 205, 213
  Bernoulli-Laplace chain, 215
  Ehrenfest chain, 214
  success runs chain, 213
Potential kernel, 159
Power tail distribution, 231
Probabilistic automata, 169
Probability
  absorption, 290
  conditional, 6, 7
  density function, 10
    joint, 14
  distribution, 10, 198
  extinction, 236
  generating function, 30, 232
  measure, 4
  ruin, 45, 309
  sample space, 1
  space, 5
Problem
  Dirichlet, 138
Process
  birth, 258
  birth and death, 261, 284
  branching, 229
  counting, 251
  Cox, 348
  death, 260
  gambling, 43, 107, 113
  Poisson, 251, 264, 296, 346
  Poisson point, 327

renewal, 348
spatial Poisson, 327
stopped, 306
telegraph, 274
Progeny, 229
Python code, 85, 87

R
Random
product, 26
sum, 25
variable, 8
walk, 75, 95, 103
Random walk
elephant, 124
excited, 97
two-dimensional, 96, 124
Range process, 94
Rank aggregation, 223
Rebound, 69
Recurrence, 95
Recurrent, 180
class, 180, 181
null, 187
positive, 187
random walk, 95
state, 180, 196, 213
Reducible, 178
Reflected path, 92
Reflection principle, 92
Reflexive (relation), 178
Regular transition matrix, 179, 197, 205, 213

Reinforcement learning, 171
Relation
equivalence, 178
reflexive, 178
symmetric, 178
transitive, 178
Reliability function, 346
Renewal process, 348
Resolvent, 159
Return
probabilities, 155
time, 149
Reversibility, 221
condition, 210
Reversible Markov chain, 210, 222
Reward function, 167
Rlogo code, 120, 132
R package
Hidden Markov Model (HMM), 132
markovchain, 179
MDPtoolbox, 171
Ruin probability, 45, 309

S
Search engine, 223
Second chance, 69
Sell stop, 306
Semigroup, 263
property, 265
Sink state, 169
Snake
and ladders, 109, 159
game, 124
Spatial Poisson point process, 327
Spin, 225
Spoofing, 296
State
absorbing, 136, 188, 213
accepting, 169
accessible, 177
aperiodic, 188, 196, 201, 203, 213
communicating, 177, 178
ergodic, 188, 213
null recurrent, 187, 213
positive recurrent, 187, 201, 205, 213
recurrent, 180, 196, 213
sink, 169
transient, 181, 213
Stationary
distribution, 198, 210, 277, 294
increments, 252
Stationary distribution, 222
Bernoulli-Laplace chain, 215
Ehrenfest chain, 214
success runs chain, 213
Stickiness, 119
Stirling
approximation, 96, 204
number, 336
Stochastic dynamic programming, 166, 171

Stopped process, 306
Stopping time, 305
theorem, 307
St. Petersburg paradox, 20, 64, 87
Strongly connected (graph theory), 178
Strong Markov property, 163, 251, 256, 260

Subcritical, 235
Success runs Markov chain, 126, 193, 213
Supercritical, 234

Survival probability, 345
Symmetric (relation), 178
Synchronizing automata, 169

T
Taylor expansion, 82, 268
Telegraph process, 274
Telescoping sum, 53, 93, 94, 204
Thinning, 41
Time homogeneous, 44, 104, 199, 263
Tower property, 23, 27, 38, 304, 305
Trading, 296
Transience, 181
Transient
  class, 183
Transient state, 181, 213
Transition
  matrix, 104
  semigroup, 263
Transitive (relation), 178
Two-dimensional random walk, 96, 124
Two-state Markov chain, 114, 140, 145, 151, 195, 201, 239, 267, 274, 277, 285, 298, 348

U
Uniform distribution, 12
Unsupervised learning, 130
Utility function, 146

V
Value function, 173
Variance, 25, 33
Voter model, 225

W
Weibull distribution, 349
Wright-Fisher model, 167

Z
Zero-modified distribution, 157

Author Index

A
Abiola, S., 127
Agapie, A., 225
Althoen, S.C., 159
Anděl, J., 72
Antal, T., 97
Asmussen, S., 187

B
Bagchi, S., 231
Benjamini, I., 97
Bosq, D., 205, 253

C
Çınlar, E., vi

D
Decreusefond, L., 339
Devore, J.L., 1
Doob, J.L., 307, 321

E
Eleyoowo, I.O., 127

F
Ferraz, E., 339

G
Goldberg, S., 137
Gusev, V.V., 169, 218

H
Höns, R., 225
Houdré, C., 339
Hudecová, Š., 72

I
Iribarren, J.L., 231

J
Jacod, J. ,1
Jones, P.W., vi

K
Kallenberg, O., 39
Karlin, S., 137, 150, 196, 197, 201, 235, 281, 348
Kijima, M., 187
King, L., 159

L
Lalley, S.P., 281
Levin, D.A., 197

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N. Privault, *Understanding Markov Chains*, Springer Undergraduate Mathematics Series,
https://doi.org/10.1007/978-981-95-4057-0

M
Martins, P., 339
Medhi, J., vi
Meyer, P.A., 321
Moro, E., 231
Muni Toke, I., 295

N
Neveu, J., 329
Nguyen, H.T., 205, 253
Norris, J.R., 116, 206, 256

O
Olaleye, T., 127

P
Peres, Y., 197
Pitman, J., 1
Protter, P., 1

R
Redner, S., 97
Ross, S., vi

S
Salako, M.O., 127
Schalekamp, F., 223
Schilling, K., 159
Schütz, G.M., 124
Sellke, S.H., 231
Shroff, N.B., 231
Sigman, K., 281
Smith, P., vi
Sowunmi, F.A., 127
Steele, J., v
Stroock, D.W., 37

T
Taylor, H.M., 137, 150, 196, 197, 201, 235, 281, 348
Trimper, S., 124

V
van Zuylen, A., 223

W
Wilmer, E.L., 197
Wilson, D.B., 97
Wu, L.M., 339